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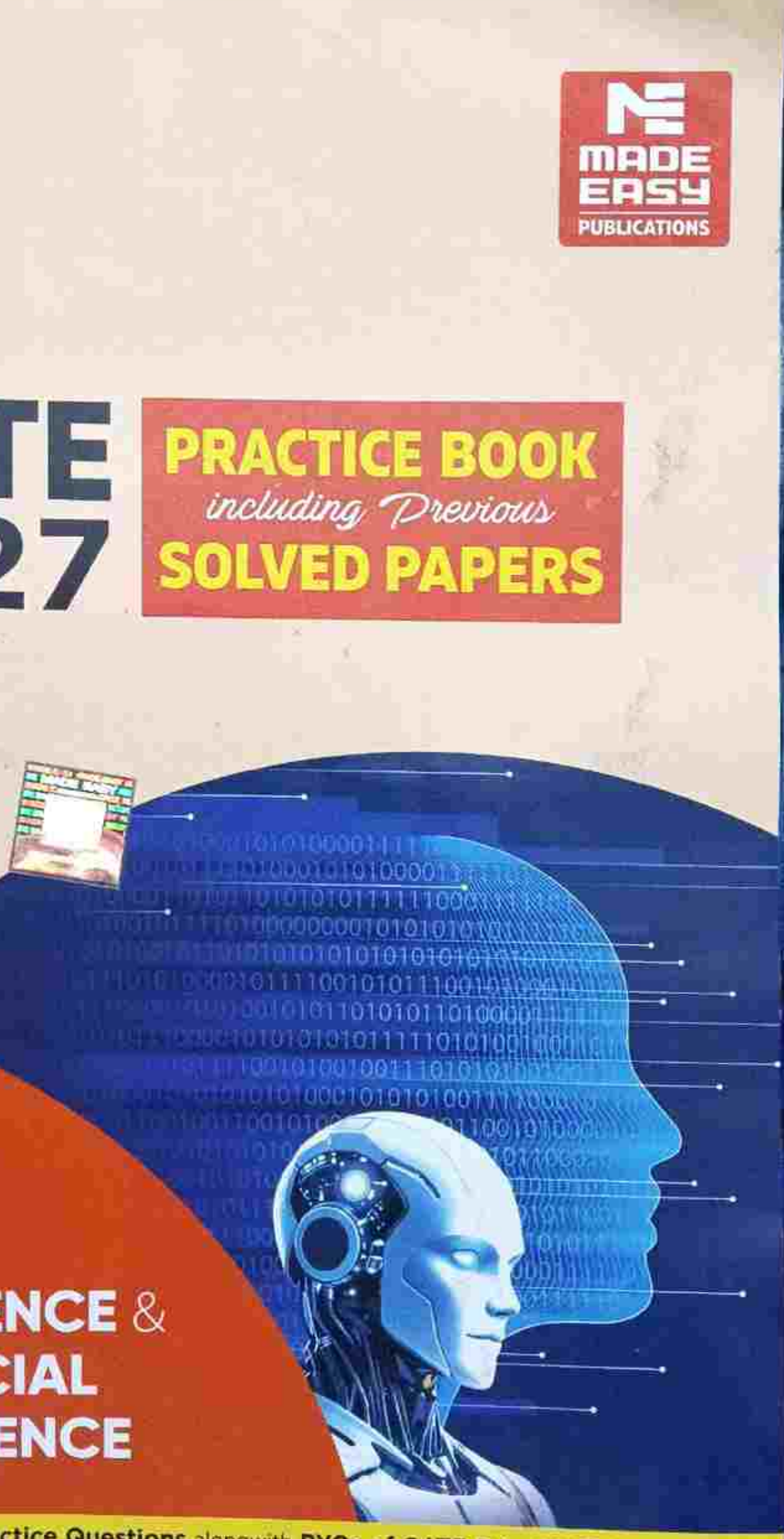

DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Step by Step Solved Practice Questions alongwith **PYQs of GATE-DA : 2025 & 2026**

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UNIT 1

Linear Algebra

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Knowledge Resource

1. Vector Space and Subspace

Q.1 A and B are real non-zero 3×3 matrices and satisfy the equation $(AB)^T + B^{-1} A = 0$. If B is orthogonal, then A is

(a) Symmetric (b) Antisymmetric
(c) Hermitian (d) Anti-Hermitian

Q.2 Let C[0, 1] be the space of continuous real valued functions on [0, 1].

Define $\langle f,g\rangle = \int_0^1 f(t)(g(t))^2 dt$ for all $f, g \in$ C[0, 1]

Then which of the following statements is true?

(a) $\langle\ ,\rangle$ is an inner product on C[0, 1].
(b) $\langle\ ,\rangle$ is a bilinear form on C[0, 1] but is not an inner product on C[0, 1].
(c) $\langle\ ,\rangle$ is not a bilinear form on C[0, 1].
(d) $\langle f, f\rangle \geq 0$ for all $f \in$ C[0, 1].

Q.3 Consider the subspaces W_1 and W_2 of $\mathbb{R}^3$ given by

$W_1 = \{(x, y, z)\} \in \mathbb{R}^3 : x + y + z = 0\}$
$W_2 = \{(x, y, z)\} \in \mathbb{R}^3 : x - y + z = 0\}$

If W is subspace of $\mathbb{R}^3$ such that

(i) $W \cap W_2$ = span $\{(0, 1, 1)\}$
(ii) $W \cap W_1$ is orthogonal to span $W \cap W_2$ with respect to the usual inner product of $\mathbb{R}^3$, then

(a) W = span $\{(0, 1, -1), (0, 1, 1)\}$
(b) W = span $\{(1, 0, -1), (0, 1, -1)\}$
(c) W = span $\{(1, 0, -1), (0, 1, 1)\}$
(d) W = span $\{(1, 0, -1), (1, 0, 1)\}$

Q.4 Consider the set of (column) vectors defined by $X = \{x \in R^3 | x_1 + x_2 + x_3 = 0$, where $x^T = [x_1, x_2, x_3]^T\}$. Which of the following is TRUE?

(a) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X.
(b) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X
(c) X is not a subspace for R^3
(d) None of the above

Q.5 For a quadratic form in 3 variables over $\mathbb{R}$, let r be the rank and s be the signature. The number of possible pairs (r, s) is

(a) 13 (b) 9
(c) 10 (d) 16

Q.6 The L_2 norm of $g(x) = 5x^2 - 3x - 1$ over [0, 1] is

(a) $\sqrt{\frac{5}{6}}$ (b) $\sqrt{\frac{25}{6}}$
(c) $\sqrt{\frac{7}{6}}$ (d) $\frac{5}{\sqrt{6}}$

Q.7 Let $P \neq 0$ be a 3×3 real matrix. There exist linearly independent vectors x and y such that $PX = 0$ and $PY = 0$. The dimension to the range space of P is

(a) 0 (b) 1
(c) 2 (d) 3

Q.8 Let $n > 1$ be a fixed natural number. Which of the following is an inner product on the vector space of $n \times n$ real symmetric matrices?

(a) $\langle A, B\rangle$ = (trace(A)) (trace(B))
(b) $\langle A, B\rangle$ = trace(AB)
(c) $\langle A, B\rangle$ = determinant(AB)
(d) $\langle A, B\rangle$ = trace(A) + trace(B)

Q.9 Let V be the real vector space of a real polynomials in one variable with degree less than or equal to 10 (including the zero polynomial). Let $T : V \to V$ be the linear map defined by $T(p) = p'$, where p' denotes the derivative of p. Which of the following statements are correct?

(a) rank(T) = 10
(b) determinant(T) = 1
(c) trace(T) = 0
(d) All the eigenvalues of T are equal to 0

Q.10 The linear operation $L(x)$ is defined by the cross product $L(x) = b \times X$, where $b = [0\ 1\ 0]^T$ and $X = [x_1\ x_2\ x_3]^T$ are three dimensional vectors. The 3×3 matrix M of this operation satisfies

$$L(x) = M\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then the eigen values of M are

(a) $0, +1, -1$ (b) $1, -1, 1$

(c) $i, -i, 1$ (d) $i, -i, 0$

Q.11 Let V be a vector space of dimension 3 over $\mathbb{R}$. Let $T: V \to V$ be a linear transformation, given by the matrix $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -4 & 3 \\ -2 & 5 & -3 \end{pmatrix}$ with respect to an ordered basis $[v_1, v_2, v_3]$ of V. Then which of the following statements is true?

(a) $T(v_3) = 0$

(b) $T(v_1 + v_2) = 0$

(c) $T(v_1 + v_2 + v_3) = 0$

(d) $T(v_1 + v_3) = T(v_2)$

Q.12 $P = \begin{bmatrix} -10 \\ -1 \\ 3 \end{bmatrix}^T$, $Q = \begin{bmatrix} -2 \\ -5 \\ 9 \end{bmatrix}^T$ and $R = \begin{bmatrix} 2 \\ -7 \\ 12 \end{bmatrix}^T$ are three vectors. An orthogonal set of vectors having a span that contains P, Q, R is

(a) $\begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix}$ $\begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} -4 \\ 2 \\ 4 \end{bmatrix}$ $\begin{bmatrix} 5 \\ 7 \\ -11 \end{bmatrix}$ $\begin{bmatrix} 8 \\ 2 \\ -3 \end{bmatrix}$

(c) $\begin{bmatrix} 6 \\ 7 \\ -1 \end{bmatrix}$ $\begin{bmatrix} -3 \\ 2 \\ -2 \end{bmatrix}$ $\begin{bmatrix} 3 \\ 9 \\ -4 \end{bmatrix}$

(d) $\begin{bmatrix} 4 \\ 3 \\ 11 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 31 \\ 3 \end{bmatrix}$ $\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix}$

Q.13 Consider the quadratic form $Q(x, y, z)$ associated to the matrix $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$.

Let $S = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in R^3 \mid Q(a,b,c) = 0 \right\}$.

Which of the following statements is FALSE?

(a) The intersection of S with the xy-plane is a line

(b) The intersection of S with the xz-plane is a ellipse

(c) S is the union of two planes

(d) Q is a degenerate quadratic form

Q.14 Let A be a 3×3 real matrix whose characteristic polynomial $p(T)$ is divisible by T^2. Which of the following statements is true?

(a) The eigenspace of A for the eigenvalue 0 is two-dimensional

(b) All the eigenvalues of A are real

(c) $A^3 = 0$

(d) A is diagonalizable

Q.15 It is given that $X_1, X_2, \ldots X_M$ are M non-zero orthogonal vectors. The dimension of the vector space spanned by the 2M vectors $X_1, X_2, \ldots X_M, -X_1, -X_2, \ldots -X_M$ is

(a) $2M$

(b) $M + 1$

(c) M

(d) Dependent on the choice of $X_1, X_2, \ldots X_M$

Q.16 The two vectors $[1, 1, 1]$ and $[1, a, a^2]$, where $a = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$, are

(a) orthonormal (b) orthogonal

(c) parallel (d) collinear

Q.17 Let P_3 be the vector space of polynomials with real coefficients and of degree at most 3. Consider the linear map $T: P_3 \to P_3$ defined by $T(p(x)) = p(x+1) + p(x-1))$ Which of the following properties does the matrix of T (with respect to the standard basis $B = [1, x, x^2, x^3]$ of P_3) satisfy?

(a) $\det T = 0$

(b) $(T - 2I)^4 = 0$ but $(T - 2I)^3 \neq 0$

(c) $(T - 2I)^3 = 0$ but $(T - 2I)^2 \neq 0$

(d) 2 is an eigenvalue with multiplicity 4

Q.18 A scalar valued function is defined as

$$f(X) = X^T AX + b^T X + c,$$

where A is a symmetric positive definite matrix with dimension $n \times n$; b and x are vectors of dimension $n \times 1$. The minimum value of $f(X)$ will occur when X equals

(a) $(A^T A)^{-1} b$ (b) $-(A^T A)^{-1} b$

(c) $-\left(\frac{A^{-1} b}{2}\right)$ (d) $\frac{A^{-1} b}{2}$

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Q.19 Let $M_4(\mathbb{R})$ be the space of all (4×4) matrices over $\mathbb{R}$. Let

$$W = \left\{(a_{ij}) \in M_4(\mathbb{R}) \sum_{i+j=k} a_{ij} = 0, \text{ for } k = 2,3,4,5,6,7,8\right\}$$

Then dim(W) is

(a) 7 (b) 8
(c) 9 (d) 10

Q.20 If the vectors $e_1 = (1, 0, 2)$, $e_2 = (0, 1, 0)$ and $e_3 = (-2, 0, 1)$ form an orthogonal basis of the three-dimensional real space R^3, then the vector $u = (4, 3, -3) \in R^3$ can be expressed as

(a) $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

(b) $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

(c) $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

(d) $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

Q.21 Let T be a linear operator on $\mathbb{R}^3$. Let $f(X) \in \mathbb{R}[X]$ denote its characteristic polynomial.

Consider the following statements:

A. Suppose T is non-zero and 0 is an eigen value of T. If we write $f(X) = Xg(X)$ in $\mathbb{R}[X]$, then the linear operator $g(T)$ is zero.

B. Suppose 0 is an eigenvalue of T with at least two linearly independent eigenvectors. If we write $f(X) = Xg(X)$ in $\mathbb{R}[X]$, then the linear operator $g(T)$ is zero.

Which of the following is true?

(a) Both (A) and (B) are true
(b) Both (A) and (B) are false
(c) (A) is true and (B) is false
(d) (A) is false and (B) is true

Q.22 Consider a 2×2 matrix $M = [v_1 \ v_2]$, where v_1 and v_2 are the column vectors.

Suppose $M^{-1} = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$, where u_1^T and u_2^T are the row vectors.

Consider the following statements:

Statement-1 : $u_1^T v_1 = 1$ and $u_2^T v_2 = 1$

Statement-2 : $u_1^T v_2 = 0$ and $u_2^T v_1 = 0$

Which of the following options is correct?

(a) Statement 2 is true and statement 1 is false.
(b) Both the statements are false.
(c) Both the statements are true.
(d) Statement 1 is true and statement 2 is false.

Q.23 If $v_1, v_2, \ldots, v_6$ are six vectors in R^4, which one of the following statements is false?

(a) If $\{v_1, v_3, v_5, v_6\}$ spans R^4, then it forms a basis for R^4.
(b) These vectors are not linearly independent.
(c) It is not necessary that these vectors span R^4.
(d) Any four of these vectors form a basis for R^4.

Q.24 If the vectors (1.0, −1.0, 2.0), (7.0, 3.0, x) and (2.0, 3.0, 1.0) and R^3 are linearly dependent, the value of x is _____.

Q.25 For a given vector $w = [1\ 2\ 3]^T$, the vector normal to the plane defined by $w^T x = 1$ is

(a) $[-2\ \ -2\ \ 2]^T$ (b) $[3\ \ 0\ \ -1]^T$
(c) $[3\ \ 2\ \ 1]^T$ (d) $[1\ \ 2\ \ 3]^T$

Q.26 Two vectors $[2\ \ 1\ \ 0\ \ 3]^T$ and $[1\ \ 0\ \ 1\ \ 2]^T$ belong to the null space of a 4×4 matrix of rank 2. Which one of the following vectors also belongs to the null space?

(a) $[1\ \ 1\ \ -1\ \ 1]^T$ (b) $[2\ \ 0\ \ 1\ \ 2]^T$
(c) $[0\ \ -2\ \ 1\ \ -1]^T$ (d) $[3\ \ 1\ \ 1\ \ 2]^T$

Q.27 Suppose A is a 5×5 block diagonal real matrix with diagonal blocks $\begin{pmatrix} e & 1 \\ 0 & e \end{pmatrix}$, $\begin{pmatrix} e & 1 & 0 \\ 0 & e & 0 \\ 0 & 0 & e \end{pmatrix}$.

Which of the following statements are true?

(a) The algebraic multiplicity of e in A is 5
(b) A is not diagonalisable
(c) The geometric multiplicity of e in A is 3
(d) The geometric multiplicity of e in A is 2

Q.28 If u and v are orthogonal vectors in the inner product space v, such that $\|u\| = 6$ and $\|u+v\| = 10$, then what is the value of $\|v\|$ and $\|u-v\|$?

(a) 4, 2 (b) 8, 10
(c) 6, 12 (d) 8, $\sqrt{28}$

Q.29 Find the dimension of the plane $2x - 3y = 0$ in $\mathbb{R}^3$.

Q.30 Let $W_1 = \{(u, v, w, x) \in \mathbb{R}^4 \mid u + v + w = 0, 2v + x = 0, 2u + 2w - x = 0\}$ and $W_2 = \{(u, v, w, x) \in \mathbb{R}^4 \mid u + w + x = 0, u + w - 2x = 0, v - x = 0\}$.
Then which among the following is true?
(a) $\dim(W_1) = 1$
(b) $\dim(W_2) = 2$
(c) $\dim(W_1 \cap W_2) = 1$
(d) $\dim(W_1 + W_2) = 3$

Q.31 Let $S = \{u_1, u_2, u_3\}$ be a linearly independent set in $\mathbb{R}^n$, which of the following are true?
(a) The set $\{u_1 + u_2, u_2 + u_3, u_3 + u_1\}$ is linearly independent.
(b) The set $\{u_1 - u_2, u_2 - u_3, u_3 - u_1\}$ is linearly independent.
(c) The set $\{u_1, u_1 + u_2, u_1 + u_2 + u_3\}$ is linearly independent.
(d) The set $\{u_1 - 2u_2, u_2 - 2u_3, u_3 - 2u_1\}$ is linearly independent.

Q.32 What is the maximum value of $x - 2y + z$ subject to the constraint $x^2 + 2y^2 + z^2 = 1$?
(a) 0 (b) 1
(c) 2 (d) 3

Q.33 Consider the quadratic forms:
$Q_1(x, y) = xy$
$Q_2(x, y) = x^2 + 2xy + y^2$
$Q_3(x, y) = x^2 + 3xy + 2y^2$
on $\mathbb{R}^2$. Which of the following is correct?
(a) Q_1 and Q_2 are equivalent
(b) Q_1 and Q_3 are equivalent
(c) Q_2 and Q_3 are equivalent
(d) All are equivalent

Q.34 Consider the two statements given below:
I. There exists a matrix $N \in \mathbb{M}_4(\mathbb{R})$ such that $\{(1, 1, 1, -1), (1, -1, 1, 1)^T\}$ is a basis of Row(N) and $(1, 2, 1, 4) \in$ Null(N).
II. There exists a matrix $M \in \mathbb{M}_4(\mathbb{R})$ such that $\{(1, 1, 1, 0)^T, (1, 0, 1, 1)^T\}$ is a basis of Col(M) and $(1, 1, 1, 1)^T, (1, 0, 1, 0)^T \in$ Null(M).
Which of the following statements is true?
(a) Statement I is FALSE and Statement II is TRUE.
(b) Statement I is TRUE and Statement II is FALSE.
(c) Both Statement I and Statement II are FALSE.
(d) Both Statement I and Statement II are TRUE.

Q.35 Let $v = \{A \in M_{3\times 3}(\mathbb{R}): A^t + A \in \mathbb{R} \times I\}$, where I is the identity matrix. Consider the quadratic form defined as $q(A) = \text{Trace}(A)^2 - \text{Trace}(A)^2$. What is the signature of this quadratic form?
(a) (+ + + +) (b) (+ 0 0 0)
(c) (+ − − −) (d) (− − − 0)

2. Matrix and Determinants

Q.36 Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then B^{-1} is
(a) $D^{-1} C^{-1} A^{-1}$
(b) CDA
(c) ADC
(d) Does not necessarily exist

Q.37 Which of the following matrices is/are Skew Hermitian?

(a) $\begin{bmatrix} 0 & -i & -2i \\ i & 0 & -i \\ 2i & i & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 3 & 1 \\ -3 & 0 & 3 \\ -1 & -3 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -i-4 & 1+i \\ i+4 & 0 & -i \\ -(1+i) & i & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & -(1+i) & \frac{i}{2} \\ 1-i & 2i & 2 \\ \frac{i}{2} & -2 & -i \end{bmatrix}$

Q.38 If matrix $A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ and $A^2 = kA$, then what is the value of k? _____

Q.39 Given an orthogonal matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \left[AA^T\right]^{-1} \text{ is}$$

(a) $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ (b) $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

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(c) $\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$ (d) $\begin{bmatrix}\frac{1}{4}&0&0&0\\0&\frac{1}{4}&0&0\\0&0&\frac{1}{4}&0\\0&0&0&\frac{1}{4}\end{bmatrix}$

Q.40 Which of the following statement is/are true?

(a) For any square matrix A, $(A^T - A)^T$ is symmetric matrix.

(b) For any square matrix A, $(A - A^T)$ is skew-symmetric matrix.

(c) If A and B are two symmetric matrices then $A^T + B^T$ is also symmetric matrix.

(d) If A and B are two skew symmetric matrix and AB^T exists then A^TB is also symmetric matrix.

Q.41 Let, $A = \begin{bmatrix}2&-0.1\\0&3\end{bmatrix}$ and $A^{-1} = \begin{bmatrix}\frac{1}{2}&a\\0&b\end{bmatrix}$.

Then $(a+b) =$

(a) $\frac{7}{20}$ (b) $\frac{3}{20}$

(c) $\frac{19}{60}$ (d) $\frac{11}{20}$

Q.42 The inverse of the 2×2 matrix $\begin{bmatrix}1&2\\5&7\end{bmatrix}$ is

(a) $\frac{1}{3}\begin{bmatrix}-7&2\\5&-1\end{bmatrix}$ (b) $\frac{1}{3}\begin{bmatrix}7&2\\5&1\end{bmatrix}$

(c) $\frac{1}{3}\begin{bmatrix}7&-2\\-5&1\end{bmatrix}$ (d) $\frac{1}{3}\begin{bmatrix}-7&-2\\-5&-1\end{bmatrix}$

Q.43 Let A be an $n \times n$ invertible matrix with real entries whose row sum are all equal to C.

Consider the following statements:

1. Every row in the matrix $2A$ sums to $2C$.
2. Every row in the matrix A^2 sums to C^2.
3. Every row in the matrix A^{-1} sums to C^{-1}.

Which of the following is true?

(a) None of statement 1, 2, 3 is correct

(b) Statement 1 is correct but not necessarily (2) or (3)

(c) Statement 2 is correct but not necessarily (1) or (3)

(d) All the statement (1) (2) (3) are correct.

Q.44 If A is Skew-Hermitian then iA is

(a) hermitian

(b) symmetric

(c) unitary

(d) skew-hermitian

Q.45 Which of the following statement is/are true about square matrix A of order n?

(a) If all the elements of the matrix A are non-zero then A can be diagonal matrix.

(b) If A is scalar matrix then A is diagonal matrix.

(c) If A is orthogonal matrix then A is non-singular matrix.

(d) None of these

Q.46 $X = [x_1, x_2, \ldots x_n]^T$ is an n-tuple non-zero vector. The $n \times n$ matrix $V = XX^T$

(a) has rank zero

(b) has rank 1

(c) is orthogonal

(d) has rank n

Q.47 If $R = \begin{bmatrix}1&0&-1\\2&1&-1\\2&3&2\end{bmatrix}$, then top row of R^{-1} is

(a) [5 6 4] (b) [5 −3 1]

(c) [2 0 −1] (d) $\begin{bmatrix}2&-1&\frac{1}{2}\end{bmatrix}$

Q.48 The product of matrices $(PQ)^{-1}P$ is

(a) P^{-1} (b) Q^{-1}

(c) $P^{-1}Q^{-1}P$ (d) PQP^{-1}

Q.49 The determinant $\begin{vmatrix}1+b&b&1\\b&1+b&1\\1&2b&1\end{vmatrix}$ equals to

(a) 0 (b) $2b(b-1)$

(c) $2(1-b)(1+2b)$ (d) $3b(1+b)$

Q.50 X and Y are non-zero square matrices of size $n \times n$. If $XY = 0_{n \times n}$ then

(a) $|X| = 0$ and $|Y| \neq 0$

(b) $|X| \neq 0$ and $|Y| = 0$

(c) $|X| = 0$ and $|Y| = 0$

(d) $|X| \neq 0$ and $|Y| \neq 0$

Q.51 A square matrix B is skew-symmetric if

(a) $B^T = -B$ (b) $B^T = B$

(c) $B^{-1} = B$ (d) $B^{-1} = B^T$

Q.52 The characteristic equation of a (3 × 3) matrix P is defined as

$$a(\lambda) = |P - \lambda I| = \lambda^3 + \lambda^2 + 2\lambda + 1 = 0$$

If I denotes identity matrix, then the inverse of matrix P will be

(a) $(P^2 + P + 2I)$
(b) $(P^2 + P + 1)$
(c) $-(P^2 + P + 1)$
(d) $-(P^2 + P + 2I)$

Q.53 The inverse of the matrix $\begin{bmatrix} 3+2i & i \\ -i & 3-2i \end{bmatrix}$ is

(a) $\frac{1}{12}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$

(b) $\frac{1}{12}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

(c) $\frac{1}{14}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$

(d) $\frac{1}{14}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

Q.54 Let $A = [a_{ij}]$, $1 \le i, j \le n$, with $n \ge 3$ and $a_{ij} = i.j$. Then the rank of A is

(a) 0 (b) 1
(c) $n-1$ (d) n

Q.55 A is $m \times n$ full rank matrix with $m > n$ and I is an identity matrix. Let matrix $A' = (A^TA)^{-1}A^T$ then, which one of the following statement is TRUE?

(a) $AA'A = A$ (b) $(AA')^2 = A$
(c) $AA'A = I$ (d) $AA'A = A'$

Q.56 If the rank of a (5 × 6) matrix Q is 4, then which one of the following statements is correct?

(a) Q will have four linearly independent rows and four linearly independent columns
(b) Q will have four linearly independent rows and five linearly independent columns
(c) QQ^T will be invertible
(d) Q^TQ will be invertible

Q.57 The eigen values of a symmetric matrix are all

(a) complex with non-zero positive imaginary part
(b) complex with non-zero negative imaginary part
(c) real
(d) pure imaginary

Q.58 Given the matrices $J = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix}$ and $K = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, the product $K^T JK$ is ____

Q.59 Which one of the following statements is true for all real symmetric matrices?

(a) All the eigen values are real.
(b) All the eigen values are positive.
(c) All the eigen values are distinct.
(d) Sum of all the eigen values is zero.

Q.60 For matrices of same dimension M, N and scalar c, which one of these properties DOES NOT ALWAYS hold?

(a) $(M^T)^T = M$ (b) $(cM)^T = c(M)^T$
(c) $(M + N)^T = M^T + N^T$ (d) $MN = NM$

Q.61 The determinant of matrix $\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{bmatrix}$ is ____

Q.62 If A is periodic matrix with period 3 then which of the following statement is true?

(a) $A^4 = A$ (b) $A^2 = A^7$
(c) $A^5 = A^{25}$ (d) $A^4 = A^{100}$

Q.63 Two matrices A and B are given below:

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}; \quad B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}$$

If the rank of matrix A is N, then the rank of matrix B is

(a) $\frac{N}{2}$ (b) $N - 1$
(c) N (d) $2N$

Q.64 Perform the following operations on the matrix

$$\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$$

1. Add the third row to the second row.
2. Subtract the third column from the first column.

The determinant of the resultant matrix is ____

Q.65 Let $M^4 = I$, (where I denotes the identity matrix) and $M \neq I$, $M^2 \neq I$ and $M^3 \neq I$. Then, for any natural number k, M^{-1} equals:

(a) M^{4k+1} (b) M^{4k+2}

(c) M^{4k+3} (d) M^{4k}

Q.66 The rank of the following matrix is $\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix}$

(a) 1 (b) 2

(c) 3 (d) 4

Q.67 Let X be a square matrix. Consider the following two statements on X.

I. X is invertible.

II. Determinant of X is non-zero.

Which one of the following is TRUE?

(a) I implies II; II does not imply I.

(b) II implies I; I does not imply II.

(c) I and II are equivalent statements.

(d) I does not imply II; II does not imply I.

Q.68 The transformation matrix for mirroring a point in x-y plane about the line $y = x$ is given by

(a) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Q.69 If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, then AA^* will be

(where, A^* is the conjugate transpose of A)

(a) Unitary matrix

(b) Orthogonal matrix

(c) Hermitian matrix

(d) Skew Hermitian matrix

Q.70 If A is a square matrix then orthogonality property mandates

(a) $AA^T = A^{-1}$

(b) $AA^T = 0$

(c) $AA^T = I$

(d) $AA^T = A^2$

Q.71 Let X and Y are invertible matrices, then product of the matrices $(X(Y^{-1}X)^{-1}\ Y^{-1})^{-1}$ is/are equivalent to .

(a) $(X)^{-1}$ (b) $(I)^{-1}$

(c) $(I\ I^{-1})^{-1}$ (d) $(Y)^{-1}$

Q.72 The rank of the matrix; every element of which is unity is =

(a) 1

(b) 0

(c) Can not to the order of the matrix

(d) None of these

Q.73 Let X^T represents the transpose of X, and $\bar{X}$ represents the conjugate of X. The transpose conjugate of X is denoted by X^θ. Then which of following statement is/are true?

(a) $(XY)^T = (Y^T X^T)$ (b) $(XY)^\theta = (YX)^\theta$

(c) $(\overline{YX}) = \bar{Y}\bar{X}$ (d) $(\overline{YX})^T = (\bar{X})^T(\bar{Y})^T$

Q.74 If $A = \begin{bmatrix} i & i-1 & 3 \\ 2+i & 0 & 2i \\ -i+1 & -i & 1-i \end{bmatrix}$ then conjugate of a matrix is equal to

(a) $\begin{bmatrix} -i & -2i-1 & 3 \\ 2+i & 0 & -2i \\ -i+1 & -2i & 1-i \end{bmatrix}$

(b) $\begin{bmatrix} -i & -2i-1 & 3 \\ 2-i & 0 & -2i \\ i+1 & -2i & 1+i \end{bmatrix}$

(c) $\begin{bmatrix} -i & i+1 & -1 \\ 2-i & -0 & -2i \\ -i-1 & i & 1+i \end{bmatrix}$

(d) None of these

Q.75 Let A and B be two matrices then which of the following statement is/are true?

(a) $AB = (B^T A^T)^T$

(b) $(AB = AC) \Rightarrow (B \neq C)$

(c) If $AB = O$ (null matrix) then $A \neq O$ and $B \neq O$ is possible

(d) $A(B - C) = AB - AC$

Q.76 Let A is a square matrix of order n then which of the following statement is true?

(a) If $A^2 = A$ then A is, orthogonal matrix.

(b) If $A^T = -A$ then A is symmetric matrix.

(c) If $A^T = A^{-1}$ then A is idempotent matrix.

(d) None of these

Q.77 If P and Q are any two matrices then which of the following is false?

(a) $\rho(PQ) < \rho(P)$ (b) $\rho(PQ) > \rho(Q)$

(c) $\rho(PQ) < \rho(Q)$ (d) $\rho(PQ) \leq \rho(P)$

Q.78 If $M_{n\times n}$ is a non-singular matrix then which of the following is correct?
(a) $\rho(M) = \rho(M^1)$
(b) $\rho(M) = \rho(M^{-1})$
(c) $\rho(M) = \rho(I_n)$
(d) None of these

Q.79 Which of the following matrices is invertible matrix?

(a) $\begin{bmatrix} 2 & -4 & -8 \\ -2 & 0 & 8 \\ 4 & 4 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} -2 & -3 & -3 \\ 1 & 0 & 1 \\ 2 & 4 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 2 & 4 \\ 0 & 3 & 0 \\ 0 & -2 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 5 & 8 & 0 \\ -3 & -5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$

Q.80 If 'M' is a non-singular matrix and
$(I - M + M^2 - \ldots + (-1)^n M^n) = 0$
then $M^{-1} =$ _____.
(a) M^n (b) $-M^{(n-1)}$
(c) $(-1)^n M^{n-1}$ (d) $(-1)^{(n+1)} \cdot M^n$

Q.81 Given matrix $p\begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & 4 \\ 6 & 0 & 7 \end{pmatrix}$ and if it's inverse $p^{-1} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, determine the value of $c =$ _____.

Q.82 Rank of the matrix $\begin{pmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{pmatrix}$ is
(i) 1 if $a = -6$
(ii) 2 if $a \neq -6$
Which of the following is true?
(a) (i) is true and (ii) is false
(b) (i) is false and (ii) is true
(c) Both (i) and (ii) are true
(d) Both (i) and (ii) are false

Q.83 If A is a matrix of order $m \times n$ then which of the following statement is/are true?
(a) If $n = m$ then A is called square matrix.
(b) If $n < m$ then A is called horizontal matrix.
(c) If $n > m$ then A is called horizontal matrix.
(d) The value of $m \times n$ is prime or composite number.

Q.84 If 'M' is 3×3 matrix with characteristics equation $\lambda^3 - 6\lambda^2 + 8\lambda - 5 = 0$ then which of the following is true?
(a) Determinant of matrix = 5
(b) Determinant of matrix = –5
(c) Trace of matrix = 6
(d) Trace of matrix = –6

Q.85 If $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then $|\text{adj}(A^{-1})| =$ _____.

Q.86 Let M be a non-singular matrix and
$I + M + M^2 + \ldots + M^n = 0$
then $M^{-1} =$
(a) M (b) M^{n-1}
(c) I (d) M^n

Q.87 Let A be a non-singular matrix of order n and k is some number then which of the following is simplified expression for Adj (kA^{-1})?
(a) $k^{(n-1)} |A|^{1-n}$
(b) $k^{n(n-1)} |A|^{n-1}$
(c) $k^{n(n-1)} |A|^{1-n}$
(d) $k^n |A|^{n-1}$

Q.88 If $A = \begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix}$ then what is the value of $|A^T A^{-1}|$?

Q.89 Let A is a symmetric matrix of order n, and the matrix.
$B = (I + A)^{-1} (I + A)^T$; where I is an identity matrix of order n. Then B is
(a) Idempotent matrix
(b) Symmetric matrix
(c) Orthogonal matrix
(d) None of these

Q.90 If A is $A^{-1} = \dfrac{\text{Adj } A}{|A|}$
(a) Orthogonal
(b) Symmetric
(c) Hermitian
(d) Invertible

Q.91 If $A_{2\times 3}$, $B_{3\times 4}$, and $C_{4\times 2}$ are multiplied to get ABC then minimum number of multiplications to get ABC is
(a) 32 (b) 36
(c) 40 (d) 44

Q.92 Consider the two matrices $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ and

$B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Which of the following statement is/are true?

(a) $(2A + 3B)$ is a diagonal matrix

(b) AB is a symmetric matrix

(c) AB is a singular matrix

(d) $AB = BA$

Q.93 For which value of x will the matrix given below become singular? _____

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & x \\ 3 & 6 & 0 \end{bmatrix}$$

Q.94 Matrix A is expressed as $A = B + C$ where B and C are respectively symmetric and skew symmetric matrix then matrix B is

Given, $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 5 \end{bmatrix}$

(a) $\begin{bmatrix} 1 & \frac{5}{2} & \frac{5}{2} \\ 5 & 8 & 2 \\ 5 & 2 & 10 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & \frac{5}{2} & \frac{5}{2} \\ \frac{5}{2} & 4 & 1 \\ \frac{5}{2} & 1 & 5 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 5 & 5 \\ 5 & 8 & 2 \\ 5 & 2 & 10 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & \frac{5}{2} & \frac{5}{2} \\ \frac{5}{2} & 4 & 1 \\ \frac{5}{2} & 1 & 10 \end{bmatrix}$

Q.95 If $\rho(A_{m \times n}) = r$ then which of the following is correct?

(a) $r \le m, r \le n$ (b) $r > m, r \le n$

(c) $r < m, r \ge n$ (d) $r \le m, r > n$

Q.96 Consider the following matrix A:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 5 & x & 6 \end{bmatrix}$$

For which value of x is the matrix A not invertible?

Q.97 Let A be an invertible 10×10 matrix with real entries such that the sum of each row is 1. Then,

(a) The sum of the entries of each row of the inverse of A is 1.

(b) The sum of the entries of each column of the inverse of A is 1.

(c) The trace of inverse of A is non zero.

(d) None of the above

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Q.98 If $D_1 = \begin{vmatrix} p & q & r \\ r & p & q \\ q & r & p \end{vmatrix}$ and

$D_2 = \begin{vmatrix} p+r & r+q & p+q \\ p+q & q+r & r+p \\ r+p & p+q & q+r \end{vmatrix}$

then what is the relation between D_1 and D_2?

(a) $D_2 = D_1$ (b) $D_2 = -D_1$

(c) $D_2 = 2D_1$ (d) $D_2 = -2D_1$

Q.99 If M is a square matrix with a zero determinant, which of the following assertion(s) is (are) correct?

S_1: Each row of M can be represented as a linear combination of the other rows.

S_2: Each column of M can be represented as a linear combination of the other columns.

S_3: $MX = O$ has a nontrivial solution.

S_4: M has an inverse.

(a) S_3 and S_2 only

(b) S_1 and S_4 only

(c) S_1 and S_3 only

(d) S_1, S_2 and S_3 only

Q.100 Which of the following is a scalar matrix?

(a) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Q.101 If the adjoint of a 3×3 matrix A is $\begin{bmatrix} -1 & 0 & 2 \\ 3 & 5 & -1 \\ -2 & 2 & 3 \end{bmatrix}$,

then the possible value(s) of the determinant of A is (are)

(a) 16 (b) 4

(c) −16 (d) −4

Q.102 If $A=\begin{bmatrix}2 & -1\\ 3 & -1\end{bmatrix}$ then adjoint of A is equivalent to

(a) adj $A=\begin{bmatrix}-1 & 1\\ 3 & 2\end{bmatrix}$ (b) adj $A=\begin{bmatrix}-1 & 1\\ -3 & 2\end{bmatrix}$

(c) adj $A=\begin{bmatrix}1 & 1\\ 3 & 2\end{bmatrix}$ (d) Does not exist

Q.103 Consider the following statements:

S_1 : The sum of two singular $n\times n$ matrices may be non-singular

S_2 : The sum of two $n\times n$ non-singular matrices may be singular

Which of the following statements is correct?

(a) S_1 and S_2 are both true

(b) S_1 is true, S_2 is false

(c) S_1 is false, S_2 is true

(d) S_1 and S_2 are both false

Q.104 Let $A=\begin{bmatrix}-1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix}_{3\times2}$. If $AB=I$ then element $f=$ _____.

(a) a (b) $a+1$

(c) b (d) $1-b$

Q.105 A is 4-square matrix and A ^ 5 (a raised to the power of 5) = 0 then $|A|=$ _____.

(a) 1 (b) −1

(c) 0 (d) None of these

Q.106 If A, B, C are any three matrices, then $A^T+B^T+C^T$ is equal to

(a) A null matrix (b) $A+B+C$

(c) $(A+B+C)$ (d) $-(A+B+C)$

Q.107 If A is a 2×2 matrix where $A^TA=I$ and $|A^{-1}|=1$, then what is the value of $|A-I|$? _____

Q.108 Consider the matrix $A=\begin{bmatrix}0 & i\\ -i & 0\end{bmatrix}$. Consider the following statements:

I. A is Hermitian

II. A is Skew-Hermitian

III. A is Unitary

Which of the above statements are true?

(a) I only

(b) II only

(c) III only

(d) I and III only

Q.109 If rank of a 3×4 matrix A is 2 and P is a non-singular matrix of order 4 then rank of matrix AP is _____.

(a) 2 (b) <2

(c) ≤2 (d) >2

Q.110 If $A=\begin{bmatrix}2 & i\\ i & 0\end{bmatrix}$, the trace of A ^ 10 is

(a) 2 (b) $2(1+i)$

(c) 0 (d) 2^{10}

Q.111 Which of the following matrix can be invertible?

(a) Idempotent matrix

(b) Involutory matrix

(c) Nilpotent matrix

(d) Orthogonal matrix

Q.112 Let $A=\begin{bmatrix}1 & 1 & 0\\ 2 & 1 & 2\\ 2 & 0 & 3\end{bmatrix}$. Find A^{-1}?

(a) $\begin{bmatrix}3 & 3 & -2\\ -2 & 3 & 2\\ -2 & 2 & 1\end{bmatrix}$ (b) $\begin{bmatrix}3 & -3 & 2\\ -2 & 3 & -2\\ 2 & 2 & 1\end{bmatrix}$

(c) $\begin{bmatrix}3 & -3 & 2\\ -2 & 3 & -2\\ -2 & 2 & -1\end{bmatrix}$ (d) $\begin{bmatrix}3 & 3 & -2\\ -2 & 3 & -2\\ -2 & 2 & -1\end{bmatrix}$

Q.113 Let $A=\begin{bmatrix}1 & 1 & 0\\ 0 & a & b\\ 1 & 0 & 1\end{bmatrix}$ then A^{-1} does not exist if (a, b) is equal to

(a) $(1, -1)$ (b) $(1, 0)$

(c) $(-1, -1)$ (d) $(0, 1)$

Q.114 Consider the following matrix $A=\begin{bmatrix}-a & 0 & 1\\ 0 & b & -e\\ 0 & d & c\end{bmatrix}$.

Which of the following statement is/are true?

(a) If $d=e=0$ then A is singular matrix.

(b) If $d=e=0$ and $a\neq b\neq c$ then A is scalar matrix.

(c) If $d=0$ and $e\neq0$, then A is lower triangular matrix.

(d) If $b=0$ and $a, c\neq0$, then A is invertible matrix.

Q.115 If P is a Idempotent Matrix. Then possible value(s) of $|P|$ is/are _____.

(a) 1 (b) −1

(c) 0 (d) None of these

Q.116 The inverse of the 2×2 matrix $\begin{bmatrix} -1 & 0 \\ 3 & 1 \end{bmatrix}$ is

(a) $\begin{bmatrix} -1 & 0 \\ 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ -3 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 \\ -3 & 1 \end{bmatrix}$ (d) Inverse does not exist

3. System of Linear Equations and Their Solution

Q.117 The system of equations $x + y + z = 6$, $2x + y + z = 7$, $x + 2y + z = 8$ has

(a) A unique solution
(b) No solution
(c) An infinite number of solutions
(d) None of these

Q.118 For the system

$x + y + z = 6$
$x + 2y + 3z = 10$
$x + 2y + \lambda z = \mu$

Which of the following is true?

(a) The system has unique solution if $\lambda = 3$ and $\mu = 10$
(b) The system has unique solution if $\lambda = 3$ and $\mu \neq 10$
(c) The system has unique solution if $\lambda \neq 3$
(d) The system has unique solution if $\lambda = 3$

Q.119 Consider the following system of equations:

$x + 2y + z = 3$
$x + 2z = 1$
$y + z = 0$

The system has

(a) Unique solution (b) Infinite many solution
(c) Finite many solution
(d) No solution

Q.120 Consider the following system of equations.

$x + y = 2$
$x + py = 2$

Where p is constant. Find the value of 'p' such that the system has more than two solutions.

Q.121 Consider the system:

$x_1 + 2x_2 = a$
$4x_1 + bx_2 = 5$

If the above system has infinitely many solutions, find the values of 'a' and 'b'?

(a) $a = \frac{5}{4}, b \neq 8$ (b) $a \neq \frac{5}{4}, b = 8$

(c) $a = \frac{5}{4}, b = 8$ (d) $a \neq \frac{5}{4}, b \neq 8$

Q.122 Consider the following system of equation:

$x + y + z = 0$
$x - y - 6z = 0$
$3x + 7y - 10z = 0$
$x + 2y = 0$

The number of solutions for this systems is

(a) Unique (b) No solution
(c) Infinite many (d) Finite many

Q.123 If the system of equations:

$x - ky - z = 0$, $kx - y - z = 0$, $x + y - z = 0$

has a non-zero solution, then the possible values of k are: '$-a$' and 'b' then $|a + b| =$ _____.

Q.124 Consider the following set of equations:

$x + 2y = 5$
$4x + 8y = 12$
$3x + 6y + 3z = 15$

This set

(a) Has a unique solution
(b) Has no solution
(c) Has finite number of solutions
(d) Has infinite number of solutions

Q.125 The value of η for which the following systems of equations:

$x + y + z = 1$
$x + 2y + 4z = \eta$
$x + 4y + 10z = \eta^2$

Has a solution are _____.

(a) $\eta = 1, -2$ (b) $\eta = -1, -2$
(c) $\eta = 3, -3$ (d) $\eta = 1, 2$

Q.126 Let A is singular matrix of order 50 and $AX = 0$ is a system of equation then system has

(a) Only trivial solution
(b) Infinite solutions
(c) Two linearly independent non-trivial solutions
(d) No solution

Q.127 The system $AX = B$ has a unique solution if

(a) $\rho(A)$ = number of variables in the system $(n) = \rho\left(\frac{A}{B}\right)$

(b) $\rho(A) < \rho(A|B)$ (c) $\rho(A) > n$

(d) $\rho(A) = \rho\left(\frac{A}{B}\right) < n$

Q.128 The equation $\begin{bmatrix}2 & -2\\ 1 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$ has

(a) no solution

(b) only one solution $\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$

(c) non-zero unique solution

(d) multiple solutions

Q.129 Consider the following system of equation:

$x + 3z = 1$

$3y + 1z = 2$

$x + y + kz = 0$

If the system of equation has unique solution then the value of k would be

(a) $\frac{5}{3}$ (b) $\frac{20}{5}$

(c) $\frac{10}{3}$ (d) All of these

Q.130 Find the real value of λ for which the following system of equations has a non-zero solution

$x + 2y + 3z = \lambda x$

$3x + y + 2z = \lambda y$

$2x + 3y + z = \lambda z$

(a) 6 (b) 4

(c) –4 (d) –6

Q.131 The system of equation, given below, has

$x + 2y + 4z = 2$

$4x + 3y + z = 5$

$3x + 2y + 3z = 1$

(a) Unique solution

(b) Two solutions

(c) No solutions

(d) More than two solutions

Q.132 The system of equations,

$x + y + z = 6$

$x + 4y + 6z = 20$

$x + 4y + \lambda z = \mu$

has **NO** solution for values of λ and μ given by

(a) $\lambda = 6, \mu = 20$ (b) $\lambda = 6, \mu \neq 20$

(c) $\lambda \neq 6, \mu = 20$ (d) $\lambda \neq 6, \mu \neq 20$

Q.133 $x + 2y + z = 4$

$2x + y + 2z = 5$

$x - y + z = 1$

The system of algebraic given below has

(a) A unique solution of $x = 1$, $y = 1$ and $z = 1$

(b) only the two solutions of $(x = 1, y = 1, z = 1)$ and $(x = 2, y = 1, z = 0)$

(c) infinite number of solutions

(d) no feasible solution

Q.134 For the set of equations:

$x_1 + 2x_2 + x_3 + 4x_4 = 2$

$3x_1 + 6x_2 + 3x_3 + 12x_4 = 6$

The following statement is true:

(a) Only the trivial solution $x_1 = x_2 = x_3 = x_4 = 0$ exists

(b) There are no solution

(c) A unique non-trivial solution exists

(d) Multiple non-trivial solutions exist

Q.135 The value of x_3 obtained by solving following system of linear equation is

$x_1 + 2x_2 - 2x_3 = 4$

$2x_1 + x_2 + x_3 = -2$

$-x_1 + x_2 - x_3 = 2$

(a) –12 (b) –2

(c) 0 (d) 12

Q.136 The number of linearly independent solutions of the system of equations

$\begin{bmatrix}1 & 2 & 0\\ 1 & 0 & -1\\ 2 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = 0$ is equal to

(a) 1 (b) 2

(c) 3 (d) 0

Q.137 The following system of equations:

$x_1 + x_2 + 2x_3 = 1$

$x_1 + 2x_3 + 3x_3 = 2$

$x_1 + 4x_2 + ax_3 = 4$

has a unique solution. The only possible value(s) for a is/are

(a) 0

(b) either 0 or 1

(c) one of 0, 1 or –1

(d) any real number other than 5

Q.138 For what values of α and β, the following simultaneous equations have an infinite number of solutions?

$x + y + z = 5$

$x + 3y + 3z = 9$

$x + 2y + \alpha z = \beta$

(a) 2, 7 (b) 3, 8

(c) 8, 3 (d) 7, 2

Q.139 Consider the system of simultaneous equations

$$x + 2y + z = 6$$
$$2x + y + 2z = 6$$
$$x + y + z = 5$$

This system has

(a) unique solution
(b) infinite number of solutions
(c) no solution
(d) exactly two solutions

Q.140 Consider a non-homogeneous system of linear equations representing mathematically an over-determined system. Such a system will be

(a) consistent having a unique solution
(b) consistent having many solutions
(c) inconsistent having a unique solution
(d) inconsistent having no solution

Q.141 Consider the following system of equations:

$3x + 2y = 1$; $\quad 4x + 7z = 1$

$x + y + z = 3$; $\quad x - 2y + 7z = 0$

The number of solutions for this system is _____.

Q.142 Given a system of equations:

$x + 2y + 2z = b_1$

$5x + y + 3z = b_2$

Which of the following is true regarding its solution?

(a) The system has a unique solution for any given b_1 and b_2
(b) The system will have infinitely many solutions for any given b_1 and b_2
(c) Whether or not a solution exists depends on the given b_1 and b_2
(d) The system would have no solution for any values of b_1 and b_2

Q.143 For what value of k the following set of equations will have no solution? _____

$$-x + 2y = 3$$
$$3x + ky = 4$$

Q.144 Consider the system of equations:

$$x + y + 3z = 2$$
$$x + 2y + 4z = 3$$
$$2x + 3y + 7z = 5$$

Which of the following is correct?

(a) No solution
(b) Infinitely many solution
(c) On trivial solution
(d) None of these

Q.145 A is 4×5 real matrix and $AX = B$ is an inconsistent system of equations. The highest possible rank of A is _____.

Q.146 Consider the following linear system:

$$x + 2y - 3z = a$$
$$2x + 3y + 3z = b$$
$$5x + 9y - 6z = c$$

This system is consistent if a, b and c satisfy the equation

(a) $7a - b - c = 0$ (b) $3a + b - c = 0$
(c) $3a - b + c = 0$ (d) $7a - b + c = 0$

Q.147 Consider the systems, each consisting of m linear equations in n variables.

I. If $m < n$, then all such systems have a solution.
II. If $m > n$, then none of these systems has a solution.
III. If $m = n$, then there exists a system which has a solution

Which one of the following is CORRECT?

(a) I, II and III are true
(b) Only II and III are true
(c) Only III is true
(d) None of them is true

Q.148 If the system

$$2x - y + 3z = 2$$
$$x + y + 2z = 2$$
$$5x - y + 8z = b$$

has infinitely many solutions, then the values of a and b, respectively, are

(a) -8 and 6 (b) 8 and 6
(c) -8 and -6 (d) 8 and -6

Q.149 The system $AX = B$ in n variables has no solutions if

(a) $\rho(A) = n$ (b) $\rho(A) > n$
(c) $\rho(A) < \rho(A|B)$ (d) $\rho(A) \geq \rho(A|B)$

Q.150 The system of linear equations

$$p - x + 2y - z = 1$$
$$p + y + z = 0$$
$$-x + y - 2z = -1$$

has _____

(a) $\rho(A) = \rho(A|B) = 3$
(b) $\rho(A) = \rho(A|B) = 2$
(c) $\rho(A) = 2, \rho(A|B) = 3$
(d) $\rho(A) = 3, \rho(A|B) = 2$

Q.151 Consider the following system of equations:

$5y + 7z = 0$

$x + z = 0$

$2x + y - 8z = 0$

Then system has

(a) Only trivial solution.

(b) Infinite many solutions.

(c) One trivial solution and two linearly independent non-trivial infinite solutions.

(d) No solution.

Q.152 If A is a square matrix then the system $AX = 0$ has non-zero solutions when

(a) $|A| \neq 0$

(b) $|A| = 0$

(c) $\rho(A)$ = number of variables

(d) $\rho(A) \geq$ number of variables

Q.153 Consider the systems, each consisting of m linear equations in n variables.

Which of the following statement is/are true?

(i) If $m = n$, then there exists a system which has infinite solutions.

(ii) If $m \neq n$, then none of these systems has a solution.

(iii) If $m = n$, then all such system have a solution.

(a) (i) only (b) (ii) only

(c) (i) and (iii) only (d) Data insufficient

4. Eigenvalues, Eigenvectors and SVD

Q.154 Which of the following statement is true for all real symmetric matrices?

(a) All the eigenvalues are non-zero.

(b) All the eigenvalues are real.

(c) Determinant of A can be imaginary.

(d) Trace is always positive.

Q.155 What are the eigenvalues of the following 2×2 matrix?

$$\begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$$

(a) -1 and 1 (b) 1 and 6

(c) 2 and 5 (d) 4 and -1

Q.156 Which of the following vector(s) is/are eigen-vectors of the following matrix?

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

(a) $t\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (b) $t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(c) $t\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ (d) $t\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

Q.157 Find an Eigen vector corresponding to largest Eigen value of matrix $A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$

(a) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ (d) $\begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}$

Q.158 The Eigen values of matrix are $\begin{bmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

(a) $\pm\cos\alpha$

(b) $\tan\alpha$ and $\cot\alpha$

(c) $\pm\sin\alpha$

(d) $\cos\alpha + \sin\alpha$ and $\cos\alpha - \sin\alpha$

Q.159 Consider the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. The sum of its Eigen values is _____.

Q.160 If the eigenvalues of A are 2, 3, 6 then what is the value of trace of A^2? _____

(a) 11 (b) 49

(c) 40 (d) None of these

Q.161 Let A be a 3×3 real matrix. Suppose 1 and -1 are two of the three Eigen values of matrix A and 18 is one of the Eigen values of $A^2 + 3A$, then

(a) Both A and $A^2 + 3A$ are invertible

(b) $A^2 + 3A$ is invertible but A is not invertible

(c) A is invertible but $A^2 + 3A$ is not invertible

(d) Both A and $A^2 + 3A$ are not invertible

Q.162 Let A be 2×2 matrix. Suppose $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ are Eigen vectors corresponding to eigen values 1 and 2 of A respectively. Then A is _____.

(a) $\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix}$

Q.163 If 2, – 4 are the Eigen values of a non-singular matrix A and $|A| = 4$, then the Eigen values of Adj A are x and $-y$ then the value of $x + y$ is _____.

Q.164 Find the absolute sum of squares of Eigen values for the matrix

$$A=\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Q.165 Which of the following statement is true?

(a) The eigen values of unitary matrix are of zero modulus.

(b) The eigen values of orthogonal matrix are of unit modulus.

(c) One of the eigen value of square nilpotent matrix is always non-zero.

(d) One of the eigen value of idempotent matrix is either 0 or –1.

Q.166 The Eigen values of $A = \begin{bmatrix} a & 1 & 0 \\ 1 & a & 1 \\ a & 1 & a \end{bmatrix}$ are _____.

(a) a, a, a

(b) $0, a, 2a$

(c) $-a, 2a, 2a$

(d) $a, a+\sqrt{2}, a-\sqrt{2}$

Q.167 Consider the matrix:

$$A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

What are the Eigen values of above matrix?

(a) 1, 1, 4 (b) 1, 0, 4

(c) 2, 1, 4 (d) 1, 4, 4

Q.168 $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, the α is equal to _____.

(a) 2 (b) 5

(c) –3 (d) ± 2

Q.169 Given the matrix $\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$, the eigen vector is

(a) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ (b) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ (d) $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Q.170 The number of linearly independent eigenvectors of $\begin{bmatrix} 1 & -3 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ is _____.

Q.171 What are the eigen values of the following 2 × 2 matrix?

$$\begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$$

(a) –1 and 1 (b) 1 and 6

(c) 2 and 5 (d) 4 and –1

Q.172 For the matrix $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ the eigen values are

(a) 3 and – 3 (b) –3 and – 5

(c) 3 and 5 (d) 5 and 0

Q.173 For the matrix $A = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, one of the eigen values is equal to –2. Which of the following is an eigen vector?

(a) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. (d) $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

Q.174 The trace and determinant of a 2 × 2 matrix are known to be 5, –14 respectively. Its eigenvalues are

(a) 7, –2 (b) 14, –1

(c) –7, 2 (d) –14, 1

Q.175 The number of linearly independent eigen vectors of $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is

(a) 0 (b) 1

(c) 2 (d) Infinite

Q.176 Eigen values of a real symmetric matrix are always
(a) Positive
(b) Negative
(c) Real
(d) Complex

Q.177 The eigen values of a skew-symmetric matrix are
(a) always zero
(b) always pure imaginary
(c) either zero or pure imaginary
(d) always real.

Q.178 An eigen vector of $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ is
(a) $[-1 \quad 1 \quad 1]^T$ (b) $[1 \quad 2 \quad 1]^T$
(c) $[1 \quad -1 \quad 2]^T$ (d) $[2 \quad 1 \quad -1]^T$

Q.179 The trace and determinant of a 2 × 2 matrix are known to be –2 and –35 respectively. It eigen values are
(a) –30 and –5 (b) –37 and –1
(c) –7 and 5 (d) 17.5 and –2

Q.180 If $[1, 0, -1]^T$ is an eigen vector of the following matrix $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ then corresponding eigen value is
(a) 1 (b) 2
(c) 3 (d) 5

Q.181 The matrix $M = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ has eigen values –3, –3, 5. An eigen vector corresponding to the eigen value 5 is $[1 \quad 2 \quad -1]^T$. One of the eigen vectors of the matrix M^3 is
(a) $[1 \quad 8 \quad -1]^T$ (b) $[1 \quad 2 \quad -1]^T$
(c) $\left[1 \quad \sqrt[3]{2} \quad -1\right]^T$ (d) $[1 \quad 1 \quad -1]^T$

Q.182 A real (4 × 4) matrix A satisfies the equation $A^2 = I$, where I is the (4 × 4) identity matrix. The positive eigen value of A is _____.

Q.183 The eigen values of a symmetric matrix are all
(a) complex with non-zero positive imaginary part
(b) complex with non-zero negative imaginary part
(c) real
(d) pure imaginary

Q.184 The minimum eigen value of the following matrix is
$$\begin{bmatrix} 3 & 5 & 2 \\ 5 & 12 & 7 \\ 2 & 7 & 5 \end{bmatrix}$$
(a) 0 (b) 1
(c) 2 (d) 3

Q.185 Consider the following 2 × 2 matrix A where two elements are unknown and are marked by a and b. The eigen values of this matrix are –1 and 7. What are the values of a and b?
$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$
(a) $a = 6, b = 4$
(b) $a = 4, b = 6$
(c) $a = 3, b = 5$
(d) $a = 5, b = 3$

Q.186 The matrix $\begin{pmatrix} 2 & -4 \\ 4 & -2 \end{pmatrix}$ has
(a) real eigenvalues and eigenvectors
(b) real eigenvalues but complex eigenvectors
(c) complex eigenvalues but real eigenvectors
(d) complex eigenvalues and eigenvectors

Q.187 Let M be a real 4 × 4 matrix. Consider the following statements:
S_1 : M has 4 linearly independent eigenvectors.
S_2 : M has 4 distinct eigenvalues.
S_3 : M is non-singular (invertible).
Which one among the following is TRUE?
(a) S_1 implies S_2
(b) S_1 implies S_3
(c) S_2 implies S_1
(d) S_3 implies S_2

Q.188 If the eigen values of A are 2, –1, 3 then what is the value of trace of 5A? _____

Q.189 If a square matrix of order 100 has exactly 15 distinct eigen values, the degree of the minimal polynomial is
(a) At least 15 (b) At most 15
(c) Always 15 (d) Exactly 100

Q.190 A 3 × 3 matrix has eigen values 1, 2 and 5. The determinant of the matrix is _____.

Q.191 M is a 2 × 2 matrix with eigenvalues 4 and 9. The eigenvalues of M^2 are
(a) –2 and –3 (b) 2 and 3
(c) 4 and 9 (d) 16 and 81

Q.192 If λ is eigenvalues of A, and A is idempotent matrix, then
(a) $\lambda \neq 0$
(b) $\lambda \neq 1$
(c) Either $\lambda = 0$ or $\lambda = 1$
(d) $\lambda \neq 0$ and $\lambda = 1$

Q.193 The highest Eigen value of the 2 × 2 matrix $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ is
(a) –1 (b) –5
(c) 5 (d) 1

Q.194 If a matrix is squared, then
(a) both eigenvalues and eigenvectors are retained.
(b) eigenvalues get squared but eigenvectors are retained.
(c) both eigenvalues and eigenvectors must change.
(d) eigenvalues are retained but eigenvectors change.

Q.195 If the sum and product of eigen values of a 2 × 2 matrix $\begin{bmatrix} 3 & p \\ p & q \end{bmatrix}$ are 4 and –1 respectively, then $|p|$ is _____ (in integer).

Q.196 If the eigenvalues of A are 1, 2, 1 then what is the value of $|\text{adj}(2A)|$? _____
(a) 256 (b) 286
(c) 64 (d) 128

■■■■

Answers Linear Algebra

1.	(b)	**2.**	(c)	**3.**	(a)	**4.**	(a)	**5.**	(c)	**6.**	(c)	**7.**	(b)
8.	(b)	**9.**	(a, c, d)	**10.**	(d)	**11.**	(c)	**12.**	(a)	**13.**	(b)	**14.**	(b)
15.	(c)	**16.**	(b)	**17.**	(d)	**18.**	(c)	**19.**	(c)	**20.**	(d)	**21.**	(d)
22.	(c)	**23.**	(d)	**24.**	(8)	**25.**	(d)	**26.**	(a)	**27.**	(a, b, c)	**28.**	(b)
29.	(2)	**30.**	(c)	**31.**	(a, c, d)	**32.**	(c)	**33.**	(b)	**34.**	(a)	**35.**	(a)
36.	(b)	**37.**	(b, d)	**38.**	(–4)	**39.**	(c)	**40.**	(b, c)	**41.**	(a)	**42.**	(a)
43.	(d)	**44.**	(d)	**45.**	(b, c)	**46.**	(b)	**47.**	(b)	**48.**	(b)	**49.**	(a)
50.	(c)	**51.**	(a)	**52.**	(d)	**53.**	(b)	**54.**	(b)	**55.**	(a)	**56.**	(a)
57.	(c)	**58.**	(23)	**59.**	(a)	**60.**	(d)	**61.**	(88)	**62.**	(d)	**63.**	(c)
64.	(0)	**65.**	(c)	**66.**	(b)	**67.**	(c)	**68.**	(b)	**69.**	(c)	**70.**	(c)
71.	(b, c)	**72.**	(a)	**73.**	(c, d)	**74.**	(d)	**75.**	(a, c, d)	**76.**	(d)	**77.**	(b)
78.	(a, b, c)	**79.**	(d)	**80.**	(d)	**81.**	(–1)	**82.**	(c)	**83.**	(a, d)	**84.**	(a, c)
85.	(1)	**86.**	(d)	**87.**	(c)	**88.**	(1)	**89.**	(a, b, c)	**90.**	(d)	**91.**	(b)
92.	(a, b, d)	**93.**	(0)	**94.**	(b)	**95.**	(a)	**96.**	(5.5)	**97.**	(a)	**98.**	(c)
99.	(d)	**100.**	(c)	**101.**	(b, d)	**102.**	(b)	**103.**	(a)	**104.**	(c)	**105.**	(c)
106.	(c)	**107.**	(0)	**108.**	(d)	**109.**	(a)	**110.**	(a)	**111.**	(a, b, d)	**112.**	(c)
113.	(a)	**114.**	(b)	**115.**	(a, c)	**116.**	(a)	**117.**	(a)	**118.**	(c)	**119.**	(a)

120. (1)	**121.** (c)	**122.** (a)	**123.** (2)	**124.** (b)	**125.** (d)	**126.** (b)
127. (a)	**128.** (d)	**129.** (a, b)	**130.** (a)	**131.** (a)	**132.** (b)	**133.** (c)
134. (d)	**135.** (b)	**136.** (a)	**137.** (d)	**138.** (a)	**139.** (c)	**140.** (b)
141. (1)	**142.** (b)	**143.** (–6)	**144.** (b)	**145.** (3)	**146.** (b)	**147.** (c)
148. (b)	**149.** (c)	**150.** (c)	**151.** (a)	**152.** (b)	**153.** (a)	**154.** (b)
155. (b)	**156.** (a, c)	**157.** (a)	**158.** (d)	**159.** (6)	**160.** (b)	**161.** (a)
162. (c)	**163.** (3)	**164.** (2)	**165.** (b)	**166.** (d)	**167.** (a)	**168.** (c)
169. (c)	**170.** (2)	**171.** (b)	**172.** (c)	**173.** (d)	**174.** (a)	**175.** (b)
176. (c)	**177.** (c)	**178.** (b)	**179.** (c)	**180.** (a)	**181.** (b)	**182.** (1)
183. (c)	**184.** (a)	**185.** (d)	**186.** (d)	**187.** (c)	**188.** (20)	**189.** (a)
190. (10)	**191.** (d)	**192.** (c)	**193.** (c)	**194.** (b)	**195.** (2)	**196.** (a)

Explanations Linear Algebra

1. (b)

Given $(AB)^T + B^{-1}A = 0$ and $BB^T = I$

Pre-multiply by B on both sides

$\Rightarrow B(B^TA^T) + B(B^{-1}A) = 0$

$\Rightarrow (BB^T)A^T + (BB^{-1})A = 0$

$\Rightarrow IA^T + IA = 0$

$\Rightarrow A^T = -A$

$\therefore A$ is skew-symmetric.

2. (c)

For bilinear form

$$B(f, (g+h)) = B(f, g) + B(f+h)$$

$$\Rightarrow B(f, (g+h)) = \int_0^1 f(x)\cdot g(x) + (h(x))^2 dx$$

$$\neq \int_0^1 f(x)\cdot (g(x))^2 dx + \int_0^1 f(x)(h(x))^2 dx$$

$\Rightarrow$ Not a bilinear form

So, also not inner product.

3. (a)

Given, W_1 and W_2 = subspace of $\mathbb{R}^3$

$W_1 = \{(x, y, z)\} \in \mathbb{R}^3 : x + y + z = 0\}$ and

$W_2 = \{(x, y, z)\} \in \mathbb{R}^3 : x - y + z = 0\}$

If W is a subspace of $\mathbb{R}^3$ such that

$W \cap W_2$ = span $\{(0, 1, 1)\}$ ...(i)

$W \cap W_1$ is orthogonal to $W \cap W_2$ ...(ii)

From (i)

$\because W \cap W_2$ = span $\{(0, 1, 1)\}$

i.e., $(0, 1, 1)$ is in $W \cap W_2$

or $(0, 1, 1)$ is also in W

Option (b) and (d) are incorrect.

From (ii)

Let, $\langle (x, y, z) \in (W \cap W_2)$

$\therefore \langle (x, y, z) \cdot (0, 1, 1) \rangle = 0$

$\Rightarrow y + z = 0$ ...(•)

Option (a) $1 - 1 = 0$ (equal to (•))

$\Rightarrow 1 \Rightarrow$ option (a) is correct.

Option (c) $0 - 1 \neq 0$ option (c) is incorrect

4. (a)

To be basis for subspace X, two conditions are to be satisfied

1. The vectors have to be linearly independent.
2. They must span X.

Here, $X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0\}$

$x^T = [x_1, x_2, x_3]^T$

Step 1: Now, $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set because one cannot be obtained from another by scalar multiplication. The fact that it is independent can also be established by seeing that rank of $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ is 2.

Step 2: Next, we need to check if the set spans X.

Here, $X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0\}$

The general infinite solution of $X = \begin{bmatrix} -k_1 - k_2 \\ k_1 \\ k_2 \end{bmatrix}$

Choosing k_1, k_2 as $\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 0 \\ k \end{bmatrix}$ and $\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} k \\ 0 \end{bmatrix}$,

we get 2 linearly independent solutions, for X.

$$X = \begin{bmatrix} -k \\ 0 \\ k \end{bmatrix} \text{ or } \begin{bmatrix} -k \\ k \\ 0 \end{bmatrix}$$

Now since both of these can be generated by linear combinations of $[1, -1, 0]^T$ and $[1, 0, -1]^T$, the set spans X. Since we have shown that the set is not only linearly independent but also spans X, therefore by definition it is a basis for the subspace X.

5. (c)

Quadratic form:

Let, p = Number of positive term (+ve)

n = Number of negative term (–ve)

r = Rank

s = Signature

$r = p + n, s = p - n$

rank(r) = rank(A) = 3

Here 4 possibility.

Case I:

If $r = 0\ (0, 0)$

$\Rightarrow$ $s = 0$

Case II: If $r = 1$

$r = p + n$	$s = p - n$	
$1 = 1 + 0 = 1$	$= 1 - 0 = 1$	(1,1)(1,–1)
$0 + 1 = 1$	$= 0 - 1 = -1$	

Case III: If $r = 2$

$r = p + n$	$s = p - n$	
$2 = 2 + 0 = 2$	$2 - 0 = 2$	(2,2)(2,–2),(2,0)
$0 + 2 = 2$	$0 - 2 = -2$	
$1 + 1 = 2$	$1 - 1 = 0$	

Case III: If $r = 3$

$r = p + n$	$s = p - n$	
$3 = 1 + 2 = 3$	$1 - 2 = -1$	$\rightarrow$ (3,–1)
$2 + 1 = 3$	$2 - 1 = 1$	(3,1)
$3 + 0 = 3$	$3 - 0 = 3$	(3,3)
$0 + 3 = 3$	$0 - 3 = -3$	(3,–3)

Total number of pairs of $(r, s) = 1 + 2 + 3 + 4 = 10$

6. (c)

$g(x) = 5x^2 - 3x - 1$

$$\|g\|_2 = \left(\int_0^1 |g(x)|^2 dx \right)^{1/2}$$

$$\|g\|_2 = \left(\int_0^1 \left| (5x^2 - 3x - 1) \right|^2 dx \right)^{1/2}$$

$$\|g\|_2 = \left(\int_0^1 (25x^4 - 30x^3 - x^2 + 6x + 1) dx \right)^{1/2}$$

$$= \left(\left[\frac{25x^5}{5} - \frac{30x^5}{4} - \frac{x^3}{3} + \frac{6x^2}{2} + 1 \right]_0^1 \right)^{1/2}$$

$$= \left(25 \times \frac{1}{5} - 30 \times \frac{1}{4} - \frac{1}{3} + 6 \times \frac{1}{2} + 1 \right)^{1/2}$$

$$= \left(5 - \frac{15}{2} - \frac{1}{3} + 3 + 1 \right)^{1/2}$$

$$= \left(9 - \frac{15}{2} - \frac{1}{3} \right)^{1/2} = \left(\frac{54 - 45 - 2}{6} \right)^{1/2}$$

$$= \left(\frac{7}{6} \right)^{1/2}$$

$$\|g\|_2 = \sqrt{\frac{7}{6}}$$

7. (b)

Given that, $P_{3\times 3} \neq 0$

X, Y such that $PX = 0$, $PY = 0$ and X, Y are L.I.

$\Rightarrow X, Y$ are null space of P

$\Rightarrow$ Dimension $(N(p)) = 2$

By Rank – Nullity theorem

Rank + Nullity = Dimension of P

Rank + 2 = 3

Rank = 1

Therefore, dimension of the range space

= Rank = 1

8. (b)

By using the property of inner product

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a real symmetric matrix

trace(A) = 1 – 1 = 0

Option (a) $\langle A, B \rangle$ = (trace(A)) (trace(B))

By using properly

$\langle A, A \rangle$ = (trace(A)) (trace(B))

= 0 but $A \neq 0$

It does not satisfy properly of inner product.

Option (c)

Example: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is a real symmetric matrix

$$\langle A, A\rangle = \text{determinant}(AA)$$

$$= \text{determinant}\left(\begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix}\right)$$

$$= 0 \text{ but } A \neq 0$$

It does not satisfied properly.

It is not an inner product.

Option (d)

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a real symmetric matrix

$$\text{trace}(A) = 1 - 1 = 0$$

$$\langle A, B\rangle = \text{trace}(A) + \text{trace}(B)$$

$$\langle A, A\rangle = \text{trace}(A) + \text{trace}(A)$$

$$= 2\,\text{trace}(A) = 0 \text{ but } A \neq 0$$

It does not satisfy properly.

9. (a, c, d)

$$\dim V = 11$$

$$B = [1, x, x^2, \ldots, x^{10}] \text{ ve basis of } V$$

$$T(1) = 1,\ T(n) = 1$$

$$T(n^2) = 2n$$

$$\vdots$$

$$T(x^{10}) = 10x^9$$

$$T = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & \\ & & 0 & 3 & & \\ & & & 0 & & \\ & & & & & 10 \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}_{11 \times 11}$$

$$\text{rank}(T) = 10$$

$$\det(T) = 0$$

10. (d)

The cross product of $b = [0\ 1\ 0]^T$ and $X = [x_1\ x_2\ x_3]^T$ can be written as

$$b \times X = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ x_1 & x_2 & x_3 \end{vmatrix}$$

$$= x_3\hat{i} + 0\hat{j} - x_1\hat{k}$$

$$= [x_3 \quad 0 \quad -x_1]$$

Now $L(x) = b \times X = M\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

where M is a 3×3 matrix

Let $M = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$

Now $M\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = b \times X$

$$\Rightarrow \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ 0 \\ -x_1 \end{bmatrix}$$

By matching LHS and RHS we get

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ 0 \\ -x_1 \end{bmatrix}$$

So, $M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

Now we have to find the eigen values of M

$$|M - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} -\lambda & 0 & 1 \\ 0 & -\lambda & 0 \\ -1 & 0 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda(\lambda^2 - 0) + 1(0 - \lambda) = 0$$

$$\Rightarrow \lambda^3 + \lambda = 0$$

$$\Rightarrow \lambda(\lambda^2 + 1) = 0$$

$$\Rightarrow \lambda = 0,\ \lambda = \pm i$$

So, the eigen values of M are i, $-i$ and 0.

11. (c)

Given, V is a vector space of dimension 3 over $\mathbb{R}$ and $\{v_1, v_2, v_3\}$ is a ordered basis of V

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -4 & 3 \\ -2 & 5 & -3 \end{pmatrix}$$

$$T(v) = Av$$

$$T(v) = 0$$

$$\Rightarrow Av = 0$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 1 & -4 & 3 \\ -2 & 5 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By using Cramer's rule

$$\frac{x}{\begin{vmatrix}-4 & 3\\ 5 & -3\end{vmatrix}} = \frac{-y}{\begin{vmatrix}1 & 3\\ -2 & -3\end{vmatrix}} = \frac{z}{\begin{vmatrix}1 & -4\\ -2 & 5\end{vmatrix}}$$

$$\frac{x}{12-15} = \frac{-y}{-3+6} = \frac{z}{5-8}$$

$$\frac{x}{-3} = \frac{y}{-3} = \frac{z}{-3}$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{1}$$

$$V = \begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$\Rightarrow \quad V = (1)\,v_1 + (1)\,v_2 + (1)\,v_3$$

$$= v_1 + v_2 + v_3$$

Since, $T(v) = 0$

$T(v_1 + v_2 + v_3) = 0$

12. (a)

We are looking for orthogonal vectors having a span that contain P, Q and R.

Take choice (a) $\begin{bmatrix}-6\\-3\\6\end{bmatrix}$ and $\begin{bmatrix}4\\-2\\3\end{bmatrix}$

Firstly these are orthogonal, as can be seen by taking their dot product

$$= -6 \times 4 + -3 \times -2 + 6 \times 3 = 0$$

The space spanned by these two vectors is

$$k_1\begin{bmatrix}-6\\-3\\6\end{bmatrix} + k_2\begin{bmatrix}4\\-2\\3\end{bmatrix} \qquad \ldots (i)$$

The span of $\begin{bmatrix}-6\\-3\\6\end{bmatrix}$ and $\begin{bmatrix}4\\-2\\3\end{bmatrix}$ contains P, Q and R.

We can show this by successively setting equation (i) to P, Q and R one by one and solving for k_1 and k_2 uniquely.

Notice also that choices (b), (c) and (d) are wrong since none of them are orthogonal as can be seen by taking pairwise dot products.

13. (b)

$$B = \begin{bmatrix}1 & 1 & 0\\ 1 & 1 & 0\\ 0 & 0 & -2\end{bmatrix}$$

$$Q(a, b, c) = (a,b,c)\begin{bmatrix}1 & 1 & 0\\ 1 & 1 & 0\\ 0 & 0 & -2\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix}$$

$$= (a+b,\, a+b-2c)\begin{bmatrix}a\\b\\c\end{bmatrix}$$

$$= a^2 + ab + ab + b^2 - 2c^2$$

$$(a+b)^2 - 2c^2$$

$$(a+b-\sqrt{2}c)(a+b+\sqrt{2}c)$$

$Q(a, b, c) = 0$

$\Rightarrow \quad a+b-\sqrt{2}c = 0$ or $a+b+\sqrt{2}c = 0$

S is union of two planes.

In $x-2$ plane $y = 0$, $b = 0$

$a-\sqrt{2}c = 0$ or $a+\sqrt{2}c = 0$

$a = \sqrt{2}c$ and $a = -\sqrt{2}c$ pair of straight line

14. (b)

$$p(T) = 3$$

$T^2 \mid p(T)$

$$\Rightarrow \quad p(T) = T^2(T-\lambda)$$

Take, $T = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & \lambda\end{bmatrix}$

dimension eigen space for eigen value 0 = 1

T is non-diagonalizable also

$$T^3 \neq 0$$

15. (c)

Since $(X_1, X_2, \ldots X_M)$ are orthogonal, they span a vector space of dimension M.

Since $(-X_1, -X_2, \ldots X_M)$ are linearly dependent on $X_1, X_2, \ldots X_M$, the set $(X_1, X_2, X_3, \ldots X_M, -X_1, -X_2, \ldots -X_M)$ will also span a vector space of dimension M only.

16. (b)

Given $[1, 1, 1]$ and $[1, a, a^2]$

Hence $a = \omega = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$

$$a^2 = \omega^2$$

So the vectors we

$$u = [1, 1, 1]$$

and $v = [1, \omega, \omega^2]$

Now $u \cdot v = 1 \cdot 1 + 1 \cdot \omega + 1 \cdot \omega^2$

$$= 1 + \omega + \omega^2 = 0$$

So u and v are orthogonal.

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17. (d)

$T: P_3 \to P_3$

$T(p(x)) = p(x+1) + p(x-1)$

$B = [1, x, x^2, x^3]$

$p(x) = 1$

$\Rightarrow T(1) = 1 + 1 = 2 = (2 \times 1 + 0 \times x^2 + 0 \times x^3)$

$p(x) = x$

$\Rightarrow T(x) = x + 1 + x - 1 = 2x$

$= (0 \times 1 + 2 \times x + 0 \times x^2 + 0 \times x^3)$

$p(x) = x^2$

$\Rightarrow T(x^2) = (x+1)^2 + (x-1)^2 = 2 + 2x^2$

$= 2 \times 1 + 0 \times x + 2 \times x^2 + 0 \times x^3)$

$p(x) = x^3$

$\Rightarrow T(x^3) = (x+1)^3 + (x-1)^3 = 6x + 2x^3$

$= (0 \times 1 + 6 \times x + 0 \times x^2 + 2 \times x^3)$

We can form the matrix.

$$T = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 6 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Above matrix is upper triangular matrix.

So, eigen value is diagonal element.

Eigen value = 2, 2, 2, 2

$A \cdot m(2) = 4$

$\Rightarrow |T| \neq 0$

Now, $(T - 2I) = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$(T-2I)^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$(T-2I)^3 = 0$

and $(T-2I)^4 = 0$

18. (c)

Let $n = 2$ and $A = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

then, $f(x) = X^T AX + B^T X + C$

$= (px^2 + 2qxy + ry^2) + (b_1 x + b_2 y) + C$

$$\frac{\partial f}{\partial x} = 2px + 2qy + b_1 = 0 \qquad ...(i)$$

$$\frac{\partial f}{\partial y} = 2qx + 2ry + b_2 = 0 \qquad ...(ii)$$

From equations (i) and (ii)

$$2\begin{bmatrix} p & q \\ q & r \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = -\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

i.e., $2AX = -B$

$$\Rightarrow \quad AX = -\frac{B}{2}$$

$$\therefore X = -\frac{A^{-1}B}{2} \quad [\because A \text{ is +ve definitely} \Rightarrow |A| \geq 0]$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2p \; ; \; s = \frac{\partial^2 f}{\partial x \partial y} = 2q$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2r$$

$rt - s^2 = 4(pr - q^2) > 0$

$r = 2p > 0 \quad [\because A \text{ is +ve definitely}]$

$$\therefore \quad \text{At } X = -\frac{A^{-1}B}{2}$$

We get minimum of $f(x)$.

Similarly we can verify for $n > 2$ also.

19. (c)

Given, $M_4(\mathbb{R})$ = Space of all (4×4) matrices over $\mathbb{R}$

$$W = \left\{(a_{ij}) \in M_4(\mathbb{R}) \sum_{i+j=k} a_{ij} = 0, \text{ for } k = 2,3,4,5,6,7,8\right\}$$

dimension$(M_{4\times4}(\mathbb{R})) = (4)^2 = 16$

$\Rightarrow$ dimension(W) = dimension$(M_{4\times4}(\mathbb{R}))$ – Number of condition i.i $(k = 7)$

$= 16 - 7 = 9$

20. (d)

$$\begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$a - 2c = 4$

$b = 3$

$2a + c = -3$

From here $a = -\frac{2}{5}$

$b = 3$

$c = -\frac{11}{5}$

$$u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$$

21. (d)

Let, $f(X) = X^3$

$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$g(T) = T^2 \neq 0$

So, Statement I is false.

G.M $\geq$ 2

We know that,

A.M $\geq$ G.M $\geq$ 2

A.M = 2 or 3

If A.M = 3, $f(X) = X^3$

Since,

G.M = 2 or 3

min poly$X$ or X^2

$\Rightarrow$ $g(T) = T^2 = 0$

Again,

If A.M = 2, $f(X) = X^2(X - \lambda)$

Since,

A.M = G.M

$g(T) = 0$

So, Statement II is true.

22. (c)

$$M^{-1}M = I$$

$$\Rightarrow \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix} [v_1 \;\; v_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} u_1^T v_1 & u_1^T v_2 \\ u_2^T v_1 & u_2^T v_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$u_1^T v_1 = 1 \qquad u_1^T v_2 = 0$

$u_2^T v_1 = 0 \qquad u_2^T v_2 = 1$

Statement (1) and (2) are both correct.

Option (c) is correct.

24. (8)

$\because$ (1, −1, 2), (7, 3, x) and (2, 3, 1) are linearly dependent.

Then,

$$\begin{vmatrix} 1 & -1 & 2 \\ 7 & 3 & x \\ 2 & 3 & 1 \end{vmatrix} = 0$$

$\Rightarrow 1(3 - 3x) + 1(7 - 2x) + 2(15) = 0$

$\Rightarrow \quad -5x = -40$

$x = 8$

25. (d)

$$w = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\phi = w^T x = 1$

Let $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\phi = [1 \;\; 2 \;\; 3] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 1$$

$\phi = x + 2y + 3z = 1$

Normal vector to ϕ is given by $\vec{\nabla}\phi$

$$N = \vec{\nabla}\phi = i\frac{\partial \phi}{\partial x} + j\frac{\partial \phi}{\partial y} + k\frac{\partial \phi}{\partial z}$$

$$N = \vec{\nabla}\phi = i(1) + j(2) + k(3)$$

$$N = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

26. (a)

$\rho(A_{4 \times 4}) = 2$

$N(A)$ = Number of column − Rank

$= 4 - 2 = 2$

I.e. Null space of A will consist only two linearly independent vectors which is given as x and y.

Eigen vectors of matrix A,

$$X = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 3 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

As these are linearly independent eigen vectors so remaining eigen vectors of null space must be linearly dependent.

Hence, $X - Y = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$

27. (a, b, c)

Given, 5 × 5 matrix

Diagonal block of matrix $A = \begin{pmatrix} e & 1 \\ 0 & e \end{pmatrix}, \begin{pmatrix} e & 1 & 0 \\ 0 & e & 0 \\ 0 & 0 & e \end{pmatrix}$

We can write,

$$A = \begin{bmatrix} \begin{pmatrix} e & 1 \\ 0 & e \end{pmatrix} & 0 & 0 & 0 \\ 0 & 0 & \begin{pmatrix} e & 1 \\ 0 & e \end{pmatrix} & 0 & 0 \\ 0 & 0 & 0 & 0 & (e) \end{bmatrix}$$

Then, eigen values of A are $\lambda_1, \lambda_2, \lambda_3 \ldots\ldots, \lambda_5 = e$

$\therefore$ Algebraic multiplicity of e in A is = 5

So, option (a) is correct.

Number of Jordan block corresponding $(\lambda = e) = 3$

The geometric multiplicity of $e = 3$

So, option (c) is correct.

Algebraic multiplicity $(\lambda = e)$ = Geometric multiplicity $(\lambda = e)$

$$\text{A.M}(\lambda = e) = 5 \neq \text{B.M}(\lambda = e) = 3$$

So, A is not diagonalisable.

So, option (b) is also correct.

28. (b)

$$u + v = 10$$
$$v = 10 - u$$

$u \perp v$, and the space has inner product.

We know that,

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2$$
$$10^2 = 6^2 + \|v\|^2$$
$$100 = 36 + \|v\|^2$$
$$\|v\|^2 = 100 - 36 = 64$$
$$\|v\| = \sqrt{64} = 8$$
$$\|u-v\|^2 = \|u\|^2 + \|v\|^2$$
$$= 36 + 64 = 100$$
$$\|u-v\| = \sqrt{100} = 10$$

29. (2)

$$2x - 3y = 0$$
$$2x = 3y$$
$$x = \frac{3}{2}y$$

Put, $y = t$

$z = s$

$$x = \frac{3}{2}t$$

So, the general vector

$$(x, y, z) = \left(\frac{3}{2}t, t, s\right) = t\left(\frac{3}{2}, 1, 0\right) + s(0,0,1)$$

This is a linear combination of two linearly independent vectors

$$\left(\frac{3}{2}, 1, 0\right) \text{ and } (0, 0, 1)$$

30. (c)

Given,

$W_1 = \{(u, v, w, x) \in \mathbb{R}^4 \mid u + v + w = 0,$
$2v + x = 0, 2u + 2w - x = 0\}$

$W_2 = \{(u, v, w, x) \in \mathbb{R}^4 \mid u + w + x = 0,$
$u + w - 2x = 0, v - x = 0\}$

For W_1 (Dimension)

$$u + v + w = 0 \Rightarrow w = -u - v \quad \ldots(i)$$
$$2v + x = 0 \Rightarrow x = -2v \quad \ldots(ii)$$

Substitute in W_1 (i) and (ii)

$$W_1 = \{u, v, -u - v, -2u\}$$
$$= \{(1, 0, -1, 0), (0, 1, -1, -2)\}$$

So, number of free variable in $W_1 = 2$

Hence, Dimension of $W_1 = \dim(W_1) = 2$

For W_2 (Dimension)

$$u + w + x = 0 \quad \ldots(iii)$$
$$u + w - 2x = 0 \quad \ldots(iv)$$

By solving (ii) and (iv) we get

$$x = 0$$
$$v - x = 0$$
$$\Rightarrow \quad v = x = 0$$

and $u = -w$ or $w = -u$

Substitute in W_2

$$W_2 = \{(u, 0, -v, 0), (1, 0, -1, 0)\}$$

So, number of free variable = 1

$$\dim(W_2) = 1$$

33. (b)

Given,

$$Q_1(x, y) = xy \quad \ldots(i)$$
$$Q_2(x, y) = x^2 + 2xy + y^2 \quad \ldots(ii)$$
$$Q_3(x, y) = x^2 + 3xy + 2y^2 \quad \ldots(iii)$$

From (i)

$$Q_1(x, y) = xy$$

$$\Rightarrow \quad M_1 = \begin{vmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{vmatrix}$$

$$\rho(M_1) = 2$$

$$\Rightarrow \quad |M_1 - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 0-\lambda & \frac{1}{2} \\ \frac{1}{2} & 0-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - \frac{1}{4} = 0$$

$$\lambda = \pm\frac{1}{2}$$

Number of +ve eigen value = 1
Number of –ve eigen value = 1

$$\text{Signature} = 1 - 1 = 0$$

$$S(M_1) = 0$$

From (ii)

$$Q_2(x, y) = x^2 + 2xy + y^2$$

$$M_2 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

$$\Rightarrow \quad \rho(M_2) = 1$$

$$|M_3 - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 - 2\lambda - 1 = 0$$

$$\lambda^2 - 2\lambda = 0$$

$$\lambda(\lambda - 2) = 0$$

$$\lambda = 0, 2$$

Number of +ve eigen value = 2
Number of –ve eigen value = 0

$$S(M_2) = 2 - 0 = 2$$

From (iii)

$$Q_3(x, y) = x^2 + 3xy + 2y^2$$

$$M_3 = \begin{vmatrix} 1 & \frac{3}{2} \\ \frac{3}{2} & 2 \end{vmatrix}$$

$$\rho(M_3) = 2$$

$$|M_3 - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & \frac{3}{2} \\ \frac{3}{2} & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(2-\lambda) - \frac{9}{4} = 0$$

$$\lambda = \frac{3 \pm \sqrt{10}}{2}$$

Number of +ve eigen value = 1
Number of –ve eigen value = 1

$$\Rightarrow \quad S(M_3) = 0$$

$\because$ Q_1 and Q_3 have same rank and signature.
So, Q_1 and Q_3 are equivalent.

34. (a)

For statement I:

Given, $N = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$

If, $V = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 4 \end{bmatrix} \in \text{Null}(N) = \{X \mid NX = 0\}$

We know that,

$$NV = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1+2+1-4 \\ 1-2+1+4 \\ * \\ * \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Which is not possible, therefore matrix N does not exist. So, Statement I is false.

For statement II:

Given, $M = \begin{bmatrix} 1 & 1 & a & x \\ 1 & 0 & b & y \\ 1 & 1 & c & z \\ 0 & 1 & d & w \end{bmatrix}$

If $V = (1, 1, 1, 0)^T$,

$U = (1, 0, 1, 1)^T \in \text{Null}(M) = \{X \mid MX = 0\}$

We know that,

$$MV = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & a & x \\ 1 & 0 & b & y \\ 1 & 1 & c & z \\ 0 & 1 & d & w \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1+1+a+x \\ 1+0+b+y \\ 1+1+c+z \\ 0+1+d+w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix}2+a+x\\1+b+y\\2+c+z\\1+d+w\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$2T\,a+x=0,\ 1+b+y=0$$
$$2T\,c+2=0,\ 1+d+w=0$$
$$MU=0$$

$$\Rightarrow \begin{bmatrix}1&1&a&x\\1&0&b&y\\1&1&c&z\\0&1&d&w\end{bmatrix}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}1+0+a+0\\1+0+b+0\\1+0+c+0\\0+0+d+0\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$1+a=0,\ 1+b=0,\ 1+c=0,\ d=0$$
$$a=-1,\ b=-1,\ c=-1,\ d=0$$

Since,

$2+a+x=0,\ 1+b+y=0,\ 2+c+z=0,$
$1+d+w=0$

$$x=-1,\ z=-1,\ w=-1$$
$$a=-1,\ b=-1,\ c=-1,\ d=0$$
$$x=-1,\ y=0,\ z=-1,\ w=-1$$

$$M=\begin{bmatrix}1&1&-1&-1\\1&0&-1&0\\1&1&-1&-1\\0&1&0&-1\end{bmatrix}$$

Therefore matrix M exists.

So, Statement II is true.

35. (a)

$$\mathbb{R}\cdot I=\left\{k\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix};\ k\in\mathbb{R}\right\}$$

$$=\left\{\begin{bmatrix}k&0&0\\0&k&0\\0&0&k\end{bmatrix};\ k\in\mathbb{R}\right\}$$

Given that, $A^t+A\in\mathbb{R}\cdot I$

$$A^t+A=\begin{bmatrix}k&0&0\\0&k&0\\0&0&k\end{bmatrix};\ \text{for some } k\in\mathbb{R}$$

$$A=\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix},\ A^t=\begin{bmatrix}a&d&g\\b&e&h\\c&f&i\end{bmatrix}$$

$$\Rightarrow A^t+A\begin{bmatrix}2a&b+d&c+g\\b+d&2c&f+h\\c+g&f+h&2i\end{bmatrix}$$

Since, $A^t+A=\begin{bmatrix}k&0&0\\0&k&0\\0&0&k\end{bmatrix}$

$$\begin{bmatrix}2a&b+d&c+g\\b+d&2c&f+h\\c+g&f+h&2i\end{bmatrix}=\begin{bmatrix}k&0&0\\0&k&0\\0&0&k\end{bmatrix}$$

$$a=\frac{k}{2},\ e=\frac{k}{2},\ i=\frac{k}{2},\ d=-b,\ g=-c,\ h=-f$$

Since, $A=\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix}$

$$\Rightarrow A=\begin{bmatrix}\frac{k}{2}&b&c\\-b&\frac{k}{2}&f\\-c&-f&\frac{k}{2}\end{bmatrix}$$

$$\text{Trace}(A)=\frac{k}{2}+\frac{k}{2}+\frac{k}{2}=\frac{3k}{2}$$

$$A^2=\begin{bmatrix}\frac{k}{2}&b&c\\-b&\frac{k}{2}&f\\-c&-f&\frac{k}{2}\end{bmatrix}\begin{bmatrix}\frac{k}{2}&b&c\\-b&\frac{k}{2}&f\\-c&-f&\frac{k}{2}\end{bmatrix}$$

$$\begin{bmatrix}\frac{k^2}{4}-b^2-c^2&-&-\\-&\frac{k^2}{4}-b^2-f^2&-\\-&-&\frac{k^2}{4}-c^2-f^2\end{bmatrix}$$

$$\text{Trace}(A)^2=\frac{3k^2}{4}-2b^2-2c^2-2f^2$$

$$q(A)=\text{Trace}(A)^2-\text{Trace}(A)^2$$
$$=\left(\frac{3k^2}{2}\right)-\left(\frac{3k^2}{4}-2b^2-2c^2-2f^2\right)$$
$$=\frac{9k^2}{2}-\frac{3k^2}{4}+2b^2+2c^2+2f^2$$
$$=\frac{6k^2}{4}+2b^2+2c^2+2f^2$$

$$= \frac{3k^2}{2} + 2b^2 + 2c^2 + 2f^2$$

Matrix associated with quadratic form

$$M = \begin{bmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Eigen vector of $M = \frac{3}{2}, 2, 2, 2$

All eigen value of matrix M is +ve

So, signature = + + + +

36. (b)

A, B, C, D is $n \times n$ matrix.

Given $ABCD = I$

$\Rightarrow ABCDD^{-1}C^{-1} = D^{-1}C^{-1}$

$\Rightarrow AB = D^{-1}C^{-1}$

$\Rightarrow A^{-1}AB = A^{-1}D^{-1}C^{-1}$

$\Rightarrow B = A^{-1}D^{-1}C^{-1}$

$B^{-1} = (A^{-1}D^{-1}C^{-1})^{-1}$

$= (C^{-1})^{-1} \cdot (D^{-1})^{-1} \cdot (A^{-1})^{-1}$

$= CDA$

37. (b, d)

For Skew Hermitian matrix

$$A = A^{\theta}$$

$\therefore$ θ is conjugate transverse of A.

Option (b) and (d) satisfies it.

38. (–4)

Given, $A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$$A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} = kA$$

$$\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} = k\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$-8\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = 2k\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$-8 = 2k$$

$$k = -4$$

39. (c)

For orthogonal matrix

$AA^T = I$ i.e. Identity matrix.

$\therefore$ $(AA^T)^{-1} = I^{-1} = 0$

40. (b, c)

Option (a) false

$P = (A^T - A)^T = (A - A^T)$

$P^T = (A^T - A) = -P$

P is skew symmetric matrix.

Option (b) true

$P = A - A^T$

$P^T = (A - A^T)^T$

$= A^T - A = -P$

P is skew symmetric matrix.

Option (c) true

Given, $A = A^T, B = B^T$

(A, B are symmetric matrix)

$P = A^T + B^T = A + B$

$P^T = (A^T + B^T)^T = A + B$

Hence P is symmetric matrix.

Option (d) false

Given, $A = -A^T, B = -B^T$

(A, B are skew symmetric matrix)

$P = AB^T = -AB$

$P^T = (-AB)^T = -B^TA^T$

$= -(-B)(-A) = -BA$

P is neither symmetric, the skew symmetric matrix.

41. (a)

$$[AA^{-1}] = I$$

$$\Rightarrow \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2a - 0.1b \\ 0 & 3b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow 2a - 0.1b = 0 \Rightarrow a = \frac{0.1b}{2} \quad \ldots(i)$$

$$3b = 1 \Rightarrow b = \frac{1}{3}$$

Now substitute b in equation (i), we get

$$a = \frac{1}{60}$$

So, $a + b = \frac{1}{60} + \frac{1}{3} = \frac{1+20}{60} = \frac{21}{60} = \frac{7}{20}$

42. **(a)**

Inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad-bc)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\therefore \quad \begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix}^{-1} = \frac{1}{(7-10)}\begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix}$$

$$= \frac{1}{-3}\begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -7 & 2 \\ 5 & -1 \end{bmatrix}$$

43. **(d)**

Take a matrix : $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

Sum of each row = $a + b$ [matrix is invertible hence determine is not 0 i.e., a and b are different]

Now, lets prove

$$\Rightarrow \quad 2\times\begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} 2a & 2b \\ 2b & 2a \end{bmatrix}:$$

sum of each row = $2(a + b)$

$$\Rightarrow \begin{bmatrix} a & b \\ b & a \end{bmatrix}\times\begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} a^2+b^2 & ab+ba \\ ab+ba & a^2+b^2 \end{bmatrix}:$$

sum of each rows = $(a^2 + b^2 + 2ab) = (a + b)^2$

$$\Rightarrow \frac{1}{a^2-b^2}\begin{bmatrix} a & -b \\ -b & a \end{bmatrix} = \frac{(a-b)}{(a^2-b^2)} = \frac{a-b}{(a+b)(a-b)}$$

Hence, all are true.

So, correct option is (d).

45. **(b, c)**

- Scalar matrix is always diagonal matrix.
- Orthogonal matrix is always non-singular.

46. **(b)**

If $X = (x_1, x_2, \ldots x_n)^T$

Rank $X = 1$, since it is non-zero n-tuple.

Rank X^T = Rank $X = 1$

Now Rank $(X^T) \le \min(\text{Rank } X, \text{Rank } X^T)$

$\Rightarrow$ Rank $(XX^T) \le \min(1, 1)$

$\Rightarrow$ Rank $(XX^T) \le 1$.

So XX^T has a rank of a either 0 or 1.

But since both X and X^T are non-zero vectors, so neither of their ranks can be zero.

So XX^T has a rank 1.

47. **(b)**

$$R = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$$

$$R^{-1} = \frac{\text{adj}(R)}{|R|} = \frac{[\text{cofactor }(R)]^T}{|R|}$$

$$|R| = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{vmatrix}$$

$$= 1(2 + 3) - 0(4 + 2) - 1(6 - 2)$$

$$= 5 - 4 = 1$$

Since we need only the top row of R^{-1}, we need to find only first column of cof (R) which after transpose will become first row of adj (R).

$$\text{cof. } (1, 1) = +\begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} = 2 + 3 = 5$$

$$\text{cof. } (2, 1) = -\begin{vmatrix} 0 & -1 \\ 3 & 2 \end{vmatrix} = -3$$

$$\text{cof. } (3, 1) = +\begin{vmatrix} 0 & -1 \\ 1 & -1 \end{vmatrix} = +1$$

$$\therefore \quad \text{cof. } (A) = \begin{bmatrix} 5 & - & - \\ -3 & - & - \\ 1 & - & - \end{bmatrix}$$

$$\text{Adj } (A) = [\text{cof. } (A)]^T = \begin{bmatrix} 5 & -3 & 1 \\ - & - & - \\ - & - & - \end{bmatrix}$$

Dividing by $|R| = 1$ gives

$$R^{-1} = \begin{bmatrix} 5 & -3 & 1 \\ - & - & - \\ - & - & - \end{bmatrix}$$

$\therefore$ Top row of $R^{-1} = [5 \ -3 \ 1]$

48. **(b)**

$$(PQ)^{-1} P = (Q^{-1} P^{-1})P$$

$$= (Q^{-1})(P^{-1} P) = (Q^{-1})(I) = Q^{-1}$$

49. **(a)**

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix}$$

$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} 2(1+b) & b & 1 \\ 2(1+b) & 1+b & 1 \\ 2(1+b) & 2b & 1 \end{vmatrix}$$

$$= 2(1+b)\begin{vmatrix}1 & b & 1\\ 1 & 1+b & 1\\ 1 & 2b & 1\end{vmatrix}$$

$= 0$ [Since C_1 and C_3 are identical)]

50. (c)

$[X]_{n\times n}[Y]_{n\times n} = [0]_{n\times n}$

$\det([X][Y]) = \det([Y][X]) = \det[X]\det[Y]$

$\therefore \det[X]\det[Y] = 0$

$\Rightarrow \det[X] = 0$ or $\det[Y] = 0$

or both are zero but it is not necessary that one of them is non-zero.

51. (a)

A square matrix B is defined as skew-symmetric if and only if $B^T = -B$, by definition.

52. (d)

If characteristic equation is

$\lambda^3 + \lambda^2 + 2\lambda + 1 = 0$

Then by Cayley-Hamilton theorem,

$P^3 + P^2 + 2P + I = 0$

$I = -P^3 - P^2 - 2P$

Multiplying by P^{-1} on both sides,

$P^{-1} = -P^2 - P - 2I = -(P^2 + P + 2I)$

53. (b)

$$\begin{bmatrix}a & b\\ c & d\end{bmatrix}^{-1} = \frac{1}{(ad-bc)}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$\therefore \begin{bmatrix}3+2i & i\\ -i & 3-2i\end{bmatrix}^{-1} = \frac{1}{\left[(3+2i)(3-2i)+i^2\right]}\begin{bmatrix}3-2i & -i\\ i & 3+2i\end{bmatrix}$$

$$= \frac{1}{12}\begin{bmatrix}3-2i & -i\\ i & 3+2i\end{bmatrix}$$

54. (b)

$$A = \begin{bmatrix}1\cdot1 & 1\cdot2 & 1\cdot3 & \dots & 1\cdot n\\ 2\cdot1 & 2\cdot2 & 2\cdot3 & \dots & 2\cdot n\\ 3\cdot1 & 3\cdot2 & 3\cdot3 & \dots & 3\cdot n\\ \vdots & \vdots & \vdots & \vdots & \vdots\\ n\cdot1 & n\cdot2 & n\cdot3 & \dots & n\cdot n\end{bmatrix}$$

All row are the multiple of first row

$$\therefore \quad A = \begin{bmatrix}1\cdot1 & 1\cdot2 & 1\cdot3 & \dots & 1\cdot n\\ 0 & 0 & 0 & \dots & 0\\ 0 & 0 & 0 & \dots & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots\\ 0 & 0 & 0 & \dots & 0\end{bmatrix}$$

Rank of $A = 1$.

55. (a)

Choice (a) $AA'A = A$ is correct

Since, $AA'A = A[(A^TA)^{-1}A^T]A$

$= A[(A^TA)^{-1}A^TA]$

Let, $A^TA = P$

Then, $= A[P^{-1}P] = A \cdot I = A$

56. (a)

If rank of (5 x 6) matrix is 4, then surely it must have exactly 4 linearly independent rows as well as 4 linearly independent columns, since rank = row rank = column rank.

57. (c)

(i) The Eigen values of symmetric matrix $[A^T = A]$ are purely real.

(ii) The Eigen value of skew-symmetric matrix $[A^T = -A]$ are either purely imaginary or zeros.

58. (23)

$$J = \begin{bmatrix}3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 6\end{bmatrix}$$

$$K = \begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$$

$$K^T JK = \begin{bmatrix}1 & 2 & -1\end{bmatrix}\begin{bmatrix}3 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 6\end{bmatrix}\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$$

$$= \begin{bmatrix}6 & 8 & -1\end{bmatrix}\begin{bmatrix}1\\ 2\\ -1\end{bmatrix} = 6+16+1 = 23$$

60. (d)

Matrix multiplication is not commutative.

61. (88)

$$\Delta = \begin{vmatrix}0 & 1 & 2 & 3\\ 1 & 0 & 3 & 0\\ 2 & 3 & 0 & 1\\ 3 & 0 & 1 & 2\end{vmatrix}$$

$R_4 \to R_4 - R_2 - R_3$

$$\Delta = \begin{vmatrix}0 & 1 & 2 & 3\\ 1 & 0 & 3 & 0\\ 2 & 3 & 0 & 1\\ 0 & -3 & -2 & 1\end{vmatrix}$$

$R_4 \to R_4 + 3R_1$

$$\Delta = \begin{vmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 0 & 0 & 4 & 10 \end{vmatrix}$$

$R_3 \to R_3 - 3R_1$

$$\Delta = \begin{vmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 0 \\ 2 & 0 & -6 & -8 \\ 0 & 0 & 4 & 10 \end{vmatrix}$$

Interchanging column 1 and column 2 and taking transpose

$$\Delta = -\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 2 & 3 & -6 & 4 \\ 3 & 0 & -8 & 10 \end{vmatrix} = -1\times\begin{vmatrix} 1 & 2 & 0 \\ 3 & -6 & 4 \\ 0 & -8 & 10 \end{vmatrix}$$

$= -1 \times |1(-60+32)+2(0-30)|$

$= -(-28 - 60) = 88$

62. (d)

Given, period of matrix $A = 3$

$A^3 = I$

$A^4 = A^{100}$

$A^3 \cdot A = A \cdot A^{99}$

$I \cdot A = A \cdot I$

$A = A$

63. (c)

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$$

$$B = \begin{bmatrix} p^2+q^2 & pr+qs \\ pr+qs & r^2+s^2 \end{bmatrix} = AA^t$$

There are three cases for the rank of A

Case I:

rank $(A) = 0$

$\Rightarrow$ A is null. So $B = AA^T$ also has to be null and hence rank (B) is also equal to 0. Therefore in this case rank (A) = rank (B).

Case II:

rank $(A) = 1$

$\Rightarrow$ A cannot be null. So B also cannot be null, since $B = AA^T$

and $|B| = |AA^T| = |A| \cdot |A^T| = |A|^2$

So rank $(B) \neq 0$. Now since rank $(A) \neq 2$ in this case, $|A| = 0$, which means that $|B| = |A|^2 = 0$

So rank (B) is also $\neq 2$. Now since rank $(B) \neq 0$ and $\neq 2$, therefore rank (B) must be equal to 1. Therefore in this case also rank (A) = rank (B).

Case III:

rank $(A) = 2$

So A has to be non-singular. i.e. $|A| \neq 0$.

Therefore, $|B| = |A|^2$ is also $\neq 0$. So rank $(B) = 2$

Therefore in this case also rank (A) = rank (B).

Therefore, in all three cases rank (A) = rank (B).

So rank of A is N, then the rank of matrix B is also N.

64. (0)

Since operations 1 and 2 are elementary operations of the type of $R_i \pm kR_j$ and $C_i \pm kC_j$ respectively, the determinant will be unchanged from the original determinant.

So the required determinant $= \begin{vmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{vmatrix}$

$$\begin{vmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{vmatrix} \xrightarrow{C_3 - 15C_1} \begin{vmatrix} 3 & 4 & 0 \\ 7 & 9 & 0 \\ 13 & 2 & 0 \end{vmatrix} = 0$$

So the required determinant = 0.

65. (c)

Given that $M^4 = I$ or $M^{4k} = I$ or $M^{4(k+1)} = I$

$\therefore$ $M^{-1} \times I = M^{4(k+1)} \times M^{-1}$

$\therefore$ $M^{-1} = M^{4k+3}$

66. (b)

$$A = \begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 4R_1$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix}$$

$R_3 \to R_3 - \frac{3}{2}R_2$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Number of non zero rows = 2

rank of $A = 2$

67. (c)

Both I and II are equivalent statements.

68. (b)

The transformation matrix for mirroring a point in x-y plane about the line $y = x$ is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

69. (c)

$$A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2-i & 3 & -1-3i \\ -5 & -i & 4+2i \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix}$$

$$A \times A^* = A \times (\bar{A})^T$$

$$= \begin{bmatrix} 5+9+10 & -10-5i-3i-4 \\ & -2i+12i-6 \\ -10+5i+3i-4 & \\ -12i+2i-6 & 25+1+16+4 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & -20+2i \\ -20-2i & 46 \end{bmatrix}$$

Which is Hermitian,

$a_{ij} = \bar{a}_{ji}$ for all i, j and a_{ii} are real

70. (c)

If, $AA^T = I$ or $A^{-1} = A^T$

Then the matrix is orthogonal.

71. (b, c)

Given, $P = (X(Y^{-1}X)^{-1}\, Y^{-1})^{-1}$

$= (X(X^{-1}(Y^{-1})^{-1})\, Y^{-1})^{-1}$

$= ((X \cdot X^{-1})\,(Y)\,(Y^{-1}))^{-1}$

$= (I \cdot (Y \cdot Y^{-1}))^{-1}$

[I is identity matrix]

$= (I \cdot I)^{-1} = (I)^{-1} = I$

72. (a)

If A is matrix with each element as 1, then A has only one independence row.

$\therefore \quad \rho(A) = 1$

73. (c, d)

Option (a) false

$(XY)^T = Y^T X^T$

Option (b) false

$(XY)^\theta = Y^\theta X^\theta$

Option (c) true

$(\overline{YX}) = \bar{Y}\,\bar{X}$

Option (d) true

$(\overline{YX})^T = (\bar{X})^T (\bar{Y})^T$

74. (d)

Given, $A = \begin{bmatrix} i & i-1 & 3 \\ 2+i & 0 & 2i \\ -i+1 & -i & 1-i \end{bmatrix}$

$$\text{Conjugate } (A) = \begin{bmatrix} -i & -i-1 & 3 \\ 2-i & 0 & -2i \\ i+1 & i & 1+i \end{bmatrix}$$

75. (a, c, d)

Option (a) true

$AB = (B^T A^T)^T$

$= B(A^T)^T (B^T)^T = AB$

Option (b) false

$AB = AC \Rightarrow B = C$ or $B \neq C$

Option (c) true

$AB = 0$

e.g.: $A = [1 \;\; 0],\ B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

Option (d) true

$A(B - A) = AB - AC$

76. (d)

Option (a) false

$A^2 = A \Rightarrow A$ is idempotent matrix

Option (b) false

$A^T = -A \Rightarrow A$ is skew-symmetric matrix

Option (c) false

$A^T = A^{-1}$

$A^T A = I \Rightarrow A$ is orthogonal matrix

77. (b)

$\rho(PQ) \leq$ minimum of $[\rho(P), \rho(Q)]$

78. (a, b, c)

If $M_{n \times n}$ is non-singular, then $\rho(M) = n$.

79. (d)

For invertible matrix, $|A| \neq 0$

$$A = \begin{bmatrix} 5 & 8 & 0 \\ -3 & -5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$$

$|A| = 5(-5) = -25 \neq 0$

80. (d)

Given that,

$(I - M + M^2 - ... + (-1)^n M^n) = 0$ ...(i)

Multiplying by M^{-1} we get

$M^{-1} - I + M - M^2 + ... + (-1)^{n-1} M^{n-1} = 0$ ...(ii)

Adding (i) and (ii), we get

$M^{-1} + (-1)^n M^n = 0$

$\therefore \quad M^{-1} = (-1)^{n+1} \cdot M^n$

81. (-1)

Here, determinant of $p = -8$

$$\therefore \quad p^{-1} = \frac{\text{adj}\, p}{|p|}$$

$$\Rightarrow \quad c = \frac{-1}{8} \text{ cofactor of the element 6 in } p$$

$$\Rightarrow \quad c = \frac{-1}{8}\left[(-1)^{3+1}\begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix}\right]$$

$$\Rightarrow \quad c = \frac{-1}{8}\left[1\begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix}\right]$$

$$\Rightarrow \quad c = \frac{-1}{8}[(2\times 4) - (0\times 3)]$$

$$= \frac{-1}{8}(8 - 0)$$

$$= \frac{-1}{8} \times 8 = -1$$

83. (a, d)

A = $m \times n$ order

For $m = n$, A is square matrix

For $n < m$, A is vertical matrix

For $n > m$, A is horizontal matrix

$m \times n$ is either prime or composite number

84. (a, c)

The given characteristic equation is

$$\lambda^3 - 6\lambda^2 + 8\lambda - 5 = 0$$

$|M|$ = Product of the roots of characteristic equation = 5 and

Trace of M = Sum of the roots of the characteristic equation = 6

85. (1)

$$|A| = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix} = 1$$

$\because \quad |\text{adj}(A_{n\times n})| = |A|^{n-1}$

$\Rightarrow \quad |\text{adj}(A_{3\times 3})| = |A|^{3-1} = |A|^2$

$\Rightarrow \quad |\text{adj}(A^{-1})| = |A^{-1}|^2 = |A|^{-2} = \frac{1}{|A|^2}$

$\therefore \quad |\text{adj}(A^{-1})| = \frac{1}{1} = 1$

86. (d)

Given that,

$I + M + M^2 + M^3 + ... + M^n = 0$ ...(i)

Multiplying by M^{-1} we get

$M^{-1} + I + M + M^2 + ... + M^{n-1} = 0$ ...(ii)

Subtracting (ii) from (i), we get

$M^n - M^{-1} = 0$

$\Rightarrow \quad M^{-1} = M^n$

87. (c)

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$\text{adj}(A) = A^{-1}|A|$$

$$\text{adj}(A^{-1}) = (A^{-1})^{-1}|A^{-1}|$$

$$\text{adj}(A^{-1}) = A|A^{-1}|$$

$$\text{adj}(kA^{-1}) = k^{n-1}\,\text{adj}(A^{-1})$$

$$|\text{adj}(kA^{-1})| = |k^{n-1}\,\text{adj}(A^{-1})|$$

$$= k^{n(n-1)}|\text{adj}(A^{-1})|$$

$$= k^{n(n-1)}|A|A^{-1}||$$

$$= k^{n(n-1)}|A^{-1}|^n|A|$$

$$= k^{n(n-1)}|A|^{1-n}$$

88. (1)

Given, $A = \begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix}$

$$A^T = \begin{bmatrix} 5 & 4 \\ 5 & 5 \end{bmatrix}$$

$$A^{-1} = -\frac{1}{10}\begin{bmatrix} 2 & -5 \\ -4 & 2 \end{bmatrix}$$

$$A^TA^{-1} = -\frac{1}{10}\begin{bmatrix} 5 & 4 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 2 & -5 \\ -4 & 5 \end{bmatrix}$$

$$|A^TA^{-1}| = \left|-\frac{1}{10}\begin{pmatrix} -6 & -5 \\ 2 & -15 \end{pmatrix}\right|$$

$$= \left(-\frac{1}{10}\right)^2\begin{vmatrix} -6 & -5 \\ 2 & -15 \end{vmatrix}$$

$$= \frac{1}{100} \times 100 = 1$$

Short method:

$$|A^TA^{-1}| = |A^T||A^{-1}|$$
$$= \frac{|A|^T}{|A|} = \frac{|A|}{|A|} = 1$$

89. (a, b, c)

Given A is a symmetric matrix

$$A = A^T$$
$$B = (I+A)^{-1}(I-A)^{-1}$$
$$B \cdot B = B^2 = (I+A)^{-1}(I+A)^T \cdot (I+A)^{-1}(I+A)^T$$
$$= (I+A)^{-1} \cdot I\,(I+A)$$
$$= (I+A)^{-1} \cdot (I+A)$$
$$= I$$

Hence B is identity matrix.

90. (d)

$$A^{-1} = \frac{\text{Adj}\,(A)}{|A|} \text{ is invertible}$$

91. (b)

To get ABC their are two ways,

(i) $(AB)C$

Now, number of multiplications in AB

$$= 2 \times 3 \times 4 = 24$$

Now, $ABC = (AB)_{2\times4}\, C_{4\times2}$

Number of multiplication for $(AB)C$

$$= 2 \times 4 \times 2 = 16$$

Total multiplication = 24 + 16 = 40

(ii) $A(BC)$

Number of multiplication operations in BC

$$= 3 \times 4 \times 2 = 24$$

Now, $ABC = A_{2\times3}(BC)_{3\times2}$

Number of multiplication for $A(BC)$

$$= 2 \times 3 \times 2 = 12$$

Total multiplication = 24 + 12 = 36

So, Minimum number = 36

92. (a, b, d)

$$A = \begin{bmatrix}2&0&0\\0&1&0\\0&0&5\end{bmatrix},\ B = \begin{bmatrix}5&0&0\\0&1&0\\0&0&2\end{bmatrix}$$

A, B are symmetric and diagonal matrix.

Option (a) true

$2A + 3B$ is a diagonal matrix

Option (b) true, as AB is diagonal and symmetric matrix.

Option (c) false

$$AB = \begin{bmatrix}10&0&0\\0&1&0\\0&0&10\end{bmatrix} \text{ is non-singular matrix}$$

Option (d) true

$$BA = \begin{bmatrix}10&0&0\\0&1&0\\0&0&10\end{bmatrix} \text{ hence } AB = BA$$

93. (0)

$$A = \begin{bmatrix}2&1&0\\1&0&x\\3&6&0\end{bmatrix}$$

For singular matrix $|A| = 0$

$$|A| = 2(-6x) - (-3x)$$
$$= -12x + 3x = -9x$$
$$-|A| = 0$$
$$-9x = 0$$
$$x = 0$$

94. (b)

$$A = \frac{A+A^T}{2} + \frac{A-A^T}{2}$$

where $B = \frac{A+A^T}{2}$ is symmetric matrix.

$$C = \frac{A-A^T}{2} \text{ is skew symmetric matrix}$$

Now, $A^T = \begin{bmatrix}1&2&3\\2&4&1\\2&1&5\end{bmatrix}$

$$\Rightarrow \quad B = \frac{A+A^T}{2} = \frac{1}{2}\begin{bmatrix}2&5&5\\5&8&2\\5&2&10\end{bmatrix} = \begin{bmatrix}1&\frac{5}{2}&\frac{5}{2}\\\frac{5}{2}&4&1\\\frac{5}{2}&1&5\end{bmatrix}$$

96. (5.5)

$$A = \begin{bmatrix}1&1&1\\0&1&2\\5&x&6\end{bmatrix}$$
$$|A| = |(6-2x) - 1\,(0-10) + 1\,(0-5)|$$
$$= 11 - 2x$$

If $|A| = 0 \Rightarrow A$ is not invertible

$$\therefore\ 11 - 2x = 0 \Rightarrow x = \frac{11}{2}$$

97. (a)

Let we first define a vector e, which is vector of all 1's (vector is nothing but a column matrix)

So, $$e = \begin{bmatrix}1\\1\\1\end{bmatrix}_{10\times 1}$$

Now, what will be Ae = ?

It turns out, that since the sum of each row of A is 1, Ae will be equal to e.

So, $$Ae = e$$

Multiply both side by A^{-1}.

$$\Rightarrow \quad A^{-1}Ae = A^{-1}e$$

$$\Rightarrow \quad e = A^{-1}e$$

$$\Rightarrow \quad A^{-1}e = e$$

The above equation means that when we multiply A^{-1} with e, we get e.

This means that sum of each row is A^{-1} is 1.

So, option (a) is correct.

98. (c)

$$|D_1| = p^3 + q^3 + r^3 - 3pqr$$

$$|D_2| = 2(p+q+r)(p^2+q^2+r^2-pq+qr+pr)$$
$$= 2(p^3+q^3+r^3-3pqr)$$
$$= 2|D_1|$$

99. (d)

S_1 and S_2: Since M has zero determinant, its rank is not full i.e. if M is of size 3 × 3, then its rank is not 3. So there is a linear combination of rows which evaluates to 0 i.e.

$$k_1R_1 + k_2R_2 + \ldots + k_nR_n = 0$$

and there is a linear combination of columns which evaluates to 0 i.e.

$$k_1C_1 + k_2C_2 + \ldots + k_nC_n = 0$$

Now any row R_i can be written as linear combination of other rows as:

$$R_i = \frac{k_1}{k_i}R_1 - \frac{k_2}{k_i}R_2 - \ldots - \frac{k_{i-1}}{k_i}R_{i-1} - \frac{k_{i+1}}{k_i}R_{i+1} - \ldots - \frac{k_n}{k_i}R_n$$

Similar is the case for columns.

So S_1 and S_2 are true.

S_3: MX = O is a homogenous equation and such an equation when $|M| = 0$ has non-trivial solution.

S_4: Since M is singular, M has no inverse. So S_4 is false.

So S_1, S_2 and S_3 only are true.

100. (c)

If all diagonal elements of a square matrix are equal then such matrix is scalar matrix.

$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ is scalar matrix.

101. (b, d)

Given, $$\text{adj}(A) = \begin{bmatrix}-1&0&2\\1&5&-1\\-3&1&3\end{bmatrix}$$

$$\text{adj}(A) = |A|A^{-1}$$

$\therefore$ A^{-1} is inverse of matrix A.

$$|\text{adj}(A)| = |A|^{(n-1)}$$

$\therefore$ n is the order of matrix A.

$$|\text{adj}(A)| = |A|^{(3-1)},\ n = 3$$
$$|\text{adj}(A)| = |A|^2$$
$$-(15-(-1)) + 2(6+10) = |A|^2$$
$$-(16) + 2(16) = |A|^2$$
$$|A|^2 = 16$$
$$A = \pm 4$$

102. (b)

$$A = \begin{bmatrix}a&b\\c&d\end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix}d&-b\\-c&a\end{bmatrix}$$

$$A = \begin{bmatrix}2&-1\\3&-1\end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix}-1&-(-1)\\-3&2\end{bmatrix} \Rightarrow \begin{bmatrix}-1&1\\-3&2\end{bmatrix}$$

103. (a)

S_1 is true

Consider two singular matrices

$$A = \begin{bmatrix}1&0\\0&0\end{bmatrix} \text{ and } B = \begin{bmatrix}0&0\\0&1\end{bmatrix}$$

Sum of A and B is given as

However $(A + B)$ is a non-singular matrix $\begin{bmatrix}1&0\\0&1\end{bmatrix}$

So, S_1 is true.

Now, consider two non-singular matrices

$$C = \begin{bmatrix}1&0\\0&1\end{bmatrix} \text{ and } D = \begin{bmatrix}-1&0\\0&-1\end{bmatrix}$$

$$C+D=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

However $(C+D)$ is a singular matrix.
So, S_2 is also true.
Therefore, both S_1 and S_2 are true.

104. (c)

$$A=\begin{bmatrix}-1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix} \text{ and } B=\begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix}$$

Given, $AB = I$

$$\therefore \begin{bmatrix}-1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}-a+e & -b+f\\ c & d\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$-b+f=0$$
$$\Rightarrow \quad f=b$$

105. (c)

$$A \wedge 5 = A \wedge 4 * A = 0$$

So one of A and $A \wedge 4$ determinant must be zero.

Suppose $|A| \neq 0$

Then, $A \wedge 4 = 0$

$$A \wedge 4 = A \wedge 3 * A = 0$$

$|A \wedge 3|$ should be 0

$$A \wedge 3 = A \wedge 2 * A = 0$$

$|A \wedge 2|$ should be 0.

$$A \wedge 2 = A * A = 0$$

$|A| = 0$ but that make our initial assumption wrong. So $|A| = 0$

106. (c)

We know that transpose with respect to additional
$A' + B' + C' = (A + B + C)'$

107. (0)

Given, $A^TA = I \Rightarrow A^T = A^{-1}$

$$|A^{-1}| = 1 \Leftrightarrow |A^T| = 1$$
$$|A| = 1$$

Now, $(A-I)^T = A^T - I^T$

$$(A-I)^T = A^T-(A^TA)^T$$
$$(A-I)^T = A^T-A^T(A^T)^T$$
$$(A-I)^T = A^T-A^TA$$
$$(A-I)^T = A^T(I-A)$$

Taking determinant on both sides.

$$|(A-I)^T| = |A^T(I-A)|$$
$$|A-I| = |A^T|\,|I-A|$$
$$|A-I| = 1\cdot(-1)^3|A-I|$$
$$|A-I| = -|A-I|$$
$$2|A-I| = 0$$
$$|A-I| = 0$$

108. (d)

In this matrix

$$A^Q = A^{-1} = \begin{bmatrix}0 & i\\ -i & 0\end{bmatrix}$$

$$A^{-1} = \frac{1}{i^2}\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} = \begin{bmatrix}0 & i\\ -i & 0\end{bmatrix}$$

Since here $A^\theta = A$, the matrix is Hermitian
Since, also $A^\theta = A^{-1}$, the matrix is Unitary

109. (a)

We know that the 'the rank of a matrix does not alter by post-multiplication with a non-singular matrix'.

$\therefore$ if $\rho(A) = 2$ then $\rho(AP)$ is also 2.

So, correct option is (a).

110. (a)

$$\begin{vmatrix}2-\lambda & i\\ i & -\lambda\end{vmatrix} = 0$$

$$\Rightarrow \quad \frac{-(2-\lambda)}{\lambda+1} = 0$$

$$\Rightarrow \quad (\lambda-1)^2 = 0$$

$$\Rightarrow \quad \lambda = 1, 1$$

Eigen values of A is 1, 1.

Eigen values of A^{10} will be 1^{10}, $1^{10} = 1, 1$

Trace of A^{10} = Sum of Eigen values = 1 + 1 = 2

So, option (a) is correct.

111. (a, b, d)

For Nilpotent matrix, A

$$|A| = 0$$

and inverse $(A^{-1}) = \frac{\text{adj}(A)}{|A|}$, $|A| \neq 0$

for invertible matrix

Hence Nilpotent matrix is not invertible.

112. (c)

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 2 & 0 & 3 \end{bmatrix}$$

$$\left[\frac{A}{I}\right] = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 0 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 2 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & -2 & 2 & -1 \end{array}\right]$$

$$\Rightarrow \quad A^{-1} = \begin{bmatrix} 3 & -3 & 2 \\ -2 & 3 & -2 \\ -2 & 2 & -1 \end{bmatrix}$$

113. (a)

A matrix is not invertible if it's determinant is 0.

$|A| = 1(a-0) - 1(0-b) = a+b$

So, option (a) is correct.

114. (b)

Given, $A = \begin{bmatrix} -a & 0 & 1 \\ 0 & b & -e \\ 0 & d & c \end{bmatrix}$

Option (a) false

$|A| = abc$, so A may be singular or not.

Option (b) true

Option (c) false

A is upper triangular matrix.

Option (d) false

$|A| = 0$, hence not invertible.

115. (a, c)

Given, P is Idempotent matrix

$$P^2 = P$$
$$P^2 - P = 0$$
$$P(P-I) = 0$$
$$|P|\,|P-I| = 0$$
$$|P| = 0 \text{ or } |P| = 1$$

116. (a)

Given, $A = \begin{bmatrix} -1 & 0 \\ 3 & 1 \end{bmatrix}$

$|A| = (-1)(1) - 0(3)$

$= -1 \neq 0$, hence inverse exist

$$\text{adj}(A) = \begin{bmatrix} 1 & 0 \\ -3 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$= -\begin{bmatrix} 1 & 0 \\ -3 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 3 & 1 \end{bmatrix}$$

117. (a)

$$\begin{array}{c} \overset{\overset{C}{\longleftarrow}}{\overset{\overset{A}{\longleftarrow}}{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 7 \\ 1 & 2 & 1 & 8 \end{array}\right]}} \end{array}$$

$R_2 \leftarrow R_2 - 2R_1$

$R_3 \leftarrow R_3 - R_1$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -5 \\ 0 & 1 & 0 & 2 \end{array}\right]$$

$R_3 \leftarrow R_3 + R_1$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -5 \\ 0 & 1 & 0 & 2 \end{array}\right]$$

$R(A) = 3, R(C) = 3$

n (unknown) $= 3$

So, unique solution so, option (a) is correct.

118. (c)

Let the given system be $AX = B$

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

$R_2 - R_1, R_3 - R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right]$$

$R_3 - R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

$\therefore$ The system has a unique solution if $\lambda \neq 3$.

119. (a)

$$x + 2y + z = 3$$
$$x + 2z = 1$$
$$y + z = 0$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$AX = B$$
$$\text{Rank } \rho(A) = 3$$
$$\text{Rank } \rho(A:B) = 3 = \text{Number of unknowns}$$
$\therefore$ Hence unique solution

120. (1)

$$\left.\begin{matrix} x+y=2 \\ x+py=2 \end{matrix}\right\} Ax = B$$

where, $A = \begin{bmatrix} 1 & 1 \\ 1 & p \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$$(A|B) = \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & p & 2 \end{array}\right)$$

$$R_2 \leftarrow R_2 - R_1$$

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & p-1 & 0 \end{array}\right)$$

If $p = 1$ rank of A = rank of B and rank is less than 2.
$\therefore$ The system has multiple solutions when $p = 1$

121. (c)

$$\left[\begin{array}{cc|c} 1 & 2 & a \\ 4 & b & 5 \end{array}\right] = \left[\begin{array}{cc|c} 1 & 2 & a \\ 0 & b-8 & 5-4a \end{array}\right] R_2 \leftarrow R_2 - 4R_1$$

If $b = 8$ and $a = \frac{5}{4}$, the last row is reduced to zero.

$\therefore$ System has infinite solutions.

122. (a)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -6 \\ 3 & 7 & -10 \\ 1 & 2 & 0 \end{bmatrix}$$

Rank, $\rho(A) = 3 = \text{Number of unknowns}$

Hence trivial/unique solutions.

123. (2)

For the given homogeneous system to have non zero solution, determinant of coefficient matrix should be zero; i.e.,

$$= \begin{vmatrix} 1 & -k & -1 \\ k & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 1(1+1) + k(-k+1) - 1(k+1) = 0$$

$$\Rightarrow 2 - k^2 + k - k - 1 = 0 \Rightarrow k^2 = 1$$
$$\Rightarrow k = +1, -1$$
$$a = 1, b = 1$$
$$\Rightarrow \quad a + b = 2$$

124. (b)

Set of equations is

$$x + 2y = 5$$
$$4x + 8y = 12$$
$$3x + 6y + 3z = 15$$

Above set of equations can be written as

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 8 & 0 \\ 3 & 6 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ x \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \\ 15 \end{bmatrix}$$

Augmented matrix $[AB]$ is given as

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 4 & 8 & 0 & 12 \\ 3 & 6 & 3 & 15 \end{array}\right]$$

Performing gauss-Elimination on the above matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 4 & 8 & 0 & 12 \\ 3 & 6 & 3 & 15 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 4R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 0 & -8 \\ 0 & 0 & 3 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -8 \end{array}\right]$$

The rank of matrix A is 2 and rank of matrix $[AB]$ is 3.
Therefore, the system is inconsistent and has no solutions.

125. (d)

Now, using Gaussian elimination method on the systems of equations:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & \eta \\ 1 & 4 & 10 & \eta^2 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \eta - 1 \\ 0 & 3 & 9 & \eta^2 - 1 \end{array}\right];$$

$[r_2 \leftarrow r_2 - r_1,\ r_3 \leftarrow r_3 - r_1]$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \eta - 1 \\ 0 & 0 & 0 & \eta^2 - 1 - 3(\eta - 1) \end{array}\right]; [r_3 \leftarrow r_3 - 3 \times r_2]$$

To have a solution, as the LHS of the final row are all zeros, the RHS of this final row must be zero.

$$\therefore \ \eta^2 - 1 - 3(\eta - 1) = 0$$
$$\Rightarrow (\eta - 1)(\eta + 1 - 3) = 0$$
$$\Rightarrow \quad (\eta - 1)(\eta - 2) = 0$$
$$\Rightarrow \quad \eta = 1, 2$$

So, option (d) is correct.

126. (b)

Given $|A_{50\times 50}| = 0$ and $AX = 0$, homogeneous system of equations.

$\therefore$ Infinite solutions exists.

127. (a)

In the system $AX = 0$, if $|A| \neq 0$ then the system has only zero solution.

128. (d)

$$\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x_1 - 2x_2 = 0$$

$$x_1 - x_2 = 0$$

$\Rightarrow$ $x_1 = x_2$

i.e. x_1 and x_2 are having infinite number of solutions.

$\Rightarrow$ Multiple solutions are these.

129. (a, b)

Given, system of equations has unique solution

$$|A| \neq 0$$

$$\begin{vmatrix} 1 & 0 & 3 \\ 0 & 3 & 1 \\ 1 & 1 & k \end{vmatrix} \neq 0$$

$$(3k - 1) + 3[-3] \neq 0$$

$$3k - 1 - 9 \neq 0$$

$$k \neq \frac{10}{3}$$

130. (a)

If the system has a non-zero solution then determinant of coefficient matrix is zero.

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (6-\lambda)(\lambda^2 + 3\lambda + 3) = 0$$

$$\Rightarrow \lambda = 6$$

131. (a)

Augmented matrix, $[A : B]$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 2 \\ 4 & 3 & 1 & 5 \\ 3 & 2 & 3 & 1 \end{array}\right]$$

$R_2 - (R_1 + R_3)$; $R_3 - 3R_1$

$$= \left[\begin{array}{ccc:c} 1 & 2 & 4 & 2 \\ 0 & -1 & -6 & 2 \\ 0 & -4 & -9 & -5 \end{array}\right] \xrightarrow{R_3 - 4R_2} \left[\begin{array}{ccc:c} 1 & 2 & 4 & 2 \\ 0 & -1 & -6 & 2 \\ 0 & 0 & 15 & -13 \end{array}\right]$$

$\Rightarrow$ $\rho(A) = \rho(A : B) = 3$ = Number of variables

$\Rightarrow$ Unique solution

132. (b)

The augmented matrix for the system of equations is

$$[A \mid B] = \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & 4 & 6 & 20 \\ 1 & 4 & \lambda & \mu \end{array}\right]$$

$$[A \mid B] = \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & 4 & 6 & 20 \\ 0 & 0 & \lambda - 6 & \mu - 20 \end{array}\right] \quad [R_3 \to R_3 - R_2]$$

If $\lambda = 6$ and $\mu \neq 20$ then

Rank $(A \mid B) = 3$ and Rank $(A) = 2$

$\because$ Rank $(A \mid B) \neq$ Rank (A)

$\therefore$ Given system of equations has no solution for $\lambda = 6$ and $\mu \neq 20$.

133. (c)

Method 1: Since in options values of x, y and z are given. So, quickest method to check from options.

$x = 1$; $y = 1$ and $z = 1$

$1 + 2 + 1 = 4$ (satisfied)

$2 + 1 + 2 = 5$ (satisfied)

$1 - 1 + 1 = 1$ (satisfied)

$x = 2$; $y = 1$ and $z = 0$

$2 + 2 \neq 0 = 4$ (satisfied)

$2 \times 2 + 1 + 0 = 5$ (satisfied)

$2 - 1 + 0 = 1$ (satisfied)

$\Rightarrow$ Simultaneous equations have two solutions

But set of equations has either

(i) No solution $\rho(A) \neq \rho(A : B)$

(ii) Infinite solutions ($\rho(A) = \rho(A : B) <$ Number of variables)

(iii) Unique solutions ($\rho(A) = \rho(A : B) =$ Number of variables)

So given set of equation has infinite solutions. [Out of which two we have calculated using options].

Methods 2:

Augmented matrix $[A : B]$

$$= \left[\begin{array}{ccc:c} 1 & 2 & 1 & 4 \\ 2 & 1 & 2 & 5 \\ 1 & -1 & 1 & 1 \end{array}\right]$$

$R_2 - 2R_1;\ R_3 - R_1$

$$= \left[\begin{array}{ccc:c} 1 & 2 & 1 & 4 \\ 0 & -3 & 0 & -3 \\ 0 & -3 & 0 & -3 \end{array}\right]$$

$R_3 - R_1$

$$= \left[\begin{array}{ccc:c} 1 & 2 & 1 & 4 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow \rho(A) = \rho(A:B) = 2 <$ Number of variables $= 3$

$\Rightarrow$ Infinitely many solutions

134. (d)

$x_1 + 2x_2 + x_3 + 4x_4 = 2$

$3x_1 + 6x_2 + 3x_3 + 12x_4 = 6$

The augmented matrix is $\left[\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 2 \\ 3 & 6 & 3 & 12 & 6 \end{array}\right]$

Performing gauss-elimination on this we get

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 2 \\ 3 & 6 & 3 & 12 & 6 \end{array}\right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

rank (A) = rank $(A \mid B) = 1$

So, system is consistent.

Since, system's rank = 1 is less than the number of variables, only infinite (multiple) non-trivial solution exists.

135. (b)

Method-1: Writing given non-homogenous equations in matrix form,

$$\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix}$$

i.e., $AX = B$

$\Rightarrow X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 - 2R_1;\ \ R_3 + R_1$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 5 \\ 0 & 3 & -3 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$R_3 + R_2$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 5 \\ 0 & 0 & 2 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$R_2 - \frac{5}{2}R_3$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & -1.5 & -2.5 \\ -1 & 1 & 1 \end{bmatrix}$$

$R_1 + R_3$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix} A^{-1} = \begin{bmatrix} 0 & 1 & 1 \\ 0.5 & -1.5 & -2.5 \\ -1 & 1 & 1 \end{bmatrix}$$

$R_1 + \frac{2}{3}R_2;\ \ \frac{R_2}{-3};\ \ \frac{R_3}{2}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & +1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ -\frac{1}{6} & \frac{1}{2} & \frac{5}{6} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Now, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = A^{-1}B$

$$= \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ -\frac{1}{6} & \frac{1}{2} & \frac{5}{6} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$$

$\Rightarrow \quad x_3 = -2$

Method 2: $[A:B] = \left[\begin{array}{ccc:c} 1 & 2 & -2 & 4 \\ 2 & 1 & 1 & -2 \\ -1 & 1 & 1 & 2 \end{array}\right]$

$R_2 - 2R_1;\ R_3 + R_1$

$$\left[\begin{array}{ccc:c} 1 & 2 & -2 & 4 \\ 0 & -3 & 5 & -10 \\ 0 & 3 & -3 & 6 \end{array}\right] \xrightarrow{R_3 + R_2} \left[\begin{array}{ccc:c} 1 & 2 & -2 & 4 \\ 0 & -3 & 5 & -10 \\ 0 & 0 & 2 & 6 \end{array}\right]$$

Also $AX = B$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 5 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ -4 \end{bmatrix}$$

$\Rightarrow \quad x_1 + 2x_2 - 2x_3 = 4$

$-3x_2 + 5x_3 = -10$

$-2x_3 = -4$

$\Rightarrow \quad x_3 = -2$

Method 3: (Quickest method)

Given: $x_1 + 2x_2 - 2x_3 = 4$ ...(1)

$2x_1 + x_2 + x_3 = -2$ ...(2)

$-x_1 + x_2 - x_3 = 2$ ...(3)

We will try to eliminate either x_1 or x_2 and form 2 equations with x_3 and x_1 or x_2 as variables.

(1) + (3), $2 \times (1) - (2)$ [i.e., eliminating x_1]

$$3x_2 - 3x_3 = 6$$
$$\Rightarrow \quad x_2 - x_3 = 2$$
$$\Rightarrow \quad x_2 = 2 + x_3 \qquad ...(4)$$
$$3x_2 - 5x_3 = 10 \qquad ...(5)$$
$$\Rightarrow \quad 6 + 3x_2 - 5x_3 = 10$$
$$\Rightarrow \quad -2x_3 = 4$$
$$\Rightarrow \quad x_3 = -2$$

136. (a)

Given, $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 0 & -1 \\ 2 & 0 & -2 \end{bmatrix}$ and $n = 3$

$R_2 \to R_2 - R_1$; $R_3 \to R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & -1 \\ 0 & -4 & -2 \end{bmatrix}$$

$R_3 \to R_3 - 2R_2$

$$\sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Here, $r = \rho(A_{3\times 3}) = 2$

and n = Number of variables = 3

$\therefore$ The number of linearly independent solutions of $AX = 0$ is given by

$$p = n - r = 3 - 2 = 1$$

137. (d)

The augmented matrix for above system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 4 & a & 4 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & a-2 & 3 \end{array}\right]$$

$$\xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & 0 \end{array}\right]$$

Now as long as $a - 5 \neq 0$, rank (A) = rank $(A|B) = 3$

$\therefore$ A can take any real value except 5. Closest correct answer is (d).

138. (a)

The augmented matrix for this system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 3 & 3 & 9 \\ 1 & 2 & \alpha & \beta \end{array}\right]$$

Using Gauss-elimination method we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 3 & 3 & 9 \\ 1 & 2 & \alpha & \beta \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 2 & 2 & 4 \\ 0 & 1 & \alpha - 1 & \beta - 5 \end{array}\right]$$

$$\xrightarrow{R_3 - \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & \alpha - 2 & \beta - 7 \end{array}\right]$$

Now, for infinite solution last row must be completely zero i.e., $\alpha - 2$ and $\beta - 7 = 0$

$$\Rightarrow \quad \alpha = 2 \text{ and } \beta = 7$$

139. (c)

Given equation are

$$x + 2y + z = 6$$
$$2x + y + 2z = 6$$
$$x + y + z = 5$$

Given system can be written as

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 5 \end{bmatrix}$$

Augmented matrix is $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 6 \\ 2 & 1 & 2 & 6 \\ 1 & 1 & 1 & 5 \end{array}\right]$

By gauss elimination

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 6 \\ 2 & 1 & 2 & 6 \\ 1 & 1 & 1 & 5 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 0 & -1 & 0 & -1 \end{array}\right]$$

$$\xrightarrow{R_3 - \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$r(A) = 2$$
$$r(A \mid B) = 3$$

Since the rank of coefficient matrix is 2 and rank of argument matrix is 3, which is not equal. Hence system has no solution i.e. system is inconsistent.

140. (b)

$$r(A_{m\times n}) \leq \min(m, n)$$

So, Highest possible rank = Least value of 3 and 4.

i.e. highest possible rank (based on size of A) = 3

However if the rank of $A = 3$ then rank of $[A|B]$ also would be 3, which means the system would become consistent. But it is given that the system is inconsistent.

So the maximum rank of A could only be 2.

141. (1)

Given:
$$3x + 2y = 1$$
$$4x + 7z = 1$$
$$x + y + z = 3$$
$$x - 2y + 7z = 0$$
$$x + y + z = 3$$
$$-x - 2y + 7z = 0$$
$$3y - 6z = 3$$
$$\Rightarrow \quad y - 2z = 1$$
$$\Rightarrow \quad 2y - 4z = 2$$
$$2y + 3x = 1$$
$$3x + 7z = -1$$
$$4x + 7z = 1$$
$$x = 2$$
$$3x + 2y = 1$$
$$\Rightarrow \quad y = -\frac{5}{2}$$
$$\Rightarrow \quad 4x + 7z = 1 \qquad (\text{Put } x = 2)$$
$$8 + 7z = 1$$
$$z = -1$$

∴ The number of solutions for this system is one.

$x = 2$, $y = -\frac{5}{2}$ and $z = -1$ is the only solution.

142. (b)

The augmented matrix for this system is

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & b_1 \\ 5 & 1 & 3 & b_2 \end{array}\right] \xrightarrow{R_2 - 5R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & b_1 \\ 0 & -9 & -7 & b_2 - 5b_1 \end{array}\right]$$

Now Gauss elimination is completed. We can see that the Rank(A) = 2.

Rank$[A \mid B]$ is also = 2 (does not depend on value of b_1 and b_2)

Rank(A) = Rank$[A \mid B]$ < Number of variables = 3

Therefore the system is consistent and as infinitely many solutions.

143. (–6)

Given, system of equation has no solution

$$A = \begin{bmatrix} -1 & 2 \\ 3 & k \end{bmatrix}, B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Rank $\rho(A) \neq$ Rank $\rho(A : B)$

$$A : B = \left[\begin{array}{cc|c} -1 & 2 & 3 \\ 3 & k & 4 \end{array}\right] R_2 \to R_2 + 3 \times R_1$$

$$= \left[\begin{array}{cc|c} -1 & 2 & 3 \\ 0 & k+6 & 13 \end{array}\right]$$

For, $k = -6$

$\rho(A) = 1$, $\rho(A : B) = 2$

144. (b)

$$(A \mid B) = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 7 & 5 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Rank $\rho(A) = 2$

$= \rho(A \mid B) = 2 < 3$

∴ Infinitely many solution

145. (3)

For inconsistent system of equations

$$|A_{4\times 5}| = 0$$

∴ Highest possible rank of $A = 3$

146. (b)

We can represent the system of equation in matrix form as

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 3 \\ 5 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$[A : B] = \left[\begin{array}{ccc:c} 1 & 2 & -3 & a \\ 2 & 3 & 3 & b \\ 5 & 9 & -6 & c \end{array}\right]$$

By elementary operation $R_3 \to R_3 - (3R_1 - R_2)$.

$$[A : B] = \left[\begin{array}{ccc:c} 1 & 2 & -3 & a \\ 2 & 3 & 3 & b \\ 0 & 0 & 0 & c - 3a - b \end{array}\right]$$

For consisting of system, $c - 3a - b = 0$

147. (c)

I. $m < n$ (system may still be inconsistent so incorrect)

II. $m > n$ (rank may still be equal to n of hence solution may exist so incorrect).

III. $m = n$ (some system rank may be equal to n and hence may have solution so correct).

So only III is correct.

148. **(b)**

$$\text{Consider } (AB) = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & -1 & 3 & 2 \\ 5 & -1 & a & b \end{bmatrix}$$

$$R_2 - 2R_1,\ R_3 - 5R_1$$

$$= 2\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & -6 & a-10 & b-10 \end{bmatrix}$$

$$R_3 - 2R_2$$

$$= \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & 0 & a-8 & b-6 \end{bmatrix}$$

$$a = 8,\ b = 6$$

∴ Infinite many solutions.

149. **(c)**

The system $AX = B$ has no solution if

$$\rho(A) < \rho(A|B)$$

150. **(c)**

The augmented matrix of the given system is

$$[A|B] = \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -2 & -1 \end{array}\right]$$

$R_2 - R_1$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 1 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & -1 & 1 & -2 & -1 \end{array}\right]$$

$R_3 + R_1$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & 1 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -2 \end{array}\right]$$

Here, rank of coefficient matrix $A = 2$ and rank of $[A|B] = 3$

$$\Rightarrow \quad \rho(A) \neq \rho(A|B)$$

∴ The system has no solution.

151. **(a)**

$$A = \begin{bmatrix} 0 & 5 & 7 \\ 1 & 0 & 1 \\ 2 & 1 & -8 \end{bmatrix}$$

$|A| \neq 0$ and $AX = 0$ is homogeneous system of equations.

∴ Unique solutions exists

152. **(b)**

The system $AX = 0$ has non-zero solution

iff $\quad \rho(A) < n$

∴ $\quad |A| = 0$

153. **(a)**

(i) e.g. $A = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix}_{2\times 2}$ and $AX = 0$

$$|A| = 0$$

∴ Infinite solutions

Statement (i) is true.

(ii) e.g. $A = \begin{bmatrix} 2 & 4 \\ 4 & 8 \\ 8 & 16 \end{bmatrix}_{3\times 2}$ $AX = 0$

Rank, $\rho(A) = 1 < n$

∴ Infinite solutions

Statement (ii) is false.

154. **(b)**

Eigen values in real symmetric matrix is always real either negative, positive or zero.

So determinant and trace of A is always real.

155. **(b)**

$$A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$$

The characteristic equation of this matrix is given by

$$|A - \lambda| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 \\ -4 & 5-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(5-\lambda) - 4 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\lambda = 1, 6$$

∴ The Eigen values of A are 1 and 6.

156. **(a, c)**

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

[λ is eigen value, X is corresponding eigen vector]

$$AX = \lambda X$$

$$(A - \lambda I)X = 0$$

$$|A - \lambda I| = 0$$

where λ is the eigen value of A

$$\begin{vmatrix} 1-\lambda & -2 & 0 \\ 0 & -5-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$(1-\lambda)(-5-\lambda)(2-\lambda) = 0$

$\lambda = 1, -5, 2$

For $\lambda = 1$, eigen vector is $t\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

For $\lambda = -5$, eigen vector is $t\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$

For $\lambda = 2$, eigen vector is $t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

157. (a)

$$|\lambda - A| = (1-\lambda)(\lambda^2 - 2) + (2-\lambda) - \lambda$$
$$= -\lambda^3 + \lambda^2$$
$$\Rightarrow \quad -\lambda^3 + \lambda^2 = 0$$
$$\Rightarrow \quad -\lambda^2(\lambda - 1) = 0$$
$$\lambda = 0, \lambda = 1$$

The largest Eigen value is 1

$$A - I = \begin{bmatrix} 0 & -1 & 1 \\ 1 & -2 & 1 \\ -1 & 1 & 0 \end{bmatrix}_{R_1 \leftrightarrow R_2}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}_{R_3 \leftarrow R_2 + R_1}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix}_{R_3 \leftarrow R_3 - R_2}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{R_1 \leftarrow R_1 - 2R_2}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$[A - I]\vec{x} = 0$

$x_1 - x_3 = 0 \Rightarrow x_1 = x_3$

$-x_2 + x_3 = 0 \Rightarrow x_2 = x_3$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} x^3$$

$\therefore \quad x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is an Eigen vector.

158. (d)

"Sum of diagonal element = Sum of Eigen value"

$$\begin{bmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

Sum of diagonals = $\cos\alpha + \cos\alpha = 2\cos\alpha$

Now, lets take option one by one, and check

Sum of diagonals = sum of Eigen value

Option (i): $\lambda_1 = \cos\alpha, \lambda_2 = -\cos\alpha$

$\lambda_2 + \lambda_2 = 0$

Option (ii): $\lambda_1 = \sin\alpha, \lambda_2 = -\sin\alpha$

$\lambda_2 + \lambda_2 = 0$

Option (iii): $\lambda_1 = \tan\alpha, \lambda_2 = \cot\alpha$

$\lambda_2 + \lambda_2 = \tan\alpha + \cot\alpha$

Option (iv): $\lambda_1 = \cos\alpha + \sin\alpha$

$\lambda_2 = \cos\alpha - \sin\alpha$

$\lambda_1 + \lambda_2 = 2\cos\alpha$

Which is equal to sum of eigen value.

Hence, option (d) is correct.

159. (6)

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix}$$
$$= -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

Characteristic equation

$\lambda^3 - 6\lambda + 9\lambda - 4 = 0$

$\lambda = 1, 1, 4$

160. (b)

trace of $(A^2) = 2^2 + 3^2 + 6^2$

$= 4 + 9 + 36$

$= 40 + 9 = 49$

161. (a)

$\lambda, 1, -1$ are Eigen values of A then

$\lambda + 3, 4, 2$ are Eigen of $A + 3I$

$\lambda^2 + 3\lambda, 4, -2$ are Eigen of $A^2 + 3A$

Since $\lambda^2 + 3\lambda = 18$

$\therefore \quad |A^2 + 3A| \neq 0$

Since $\lambda \neq 0$

$\therefore \quad |A| \neq 0$

162. (c)

Let, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$AX_2 = \lambda_2 X_2$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a+0 \\ c+0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$AX_1 = \lambda_1 X_1$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = 1\begin{bmatrix} 2 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2a-b \\ 2c-d \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow \quad a = 2 \quad b = +2$$
$$c = 0 \quad d = 1$$

$$\Rightarrow \quad A = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}$$

163. (3)

If $\lambda_1, \lambda_2, \ldots \lambda_n$ are the Eigen value of A then the Eigen value of adj A are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \ldots \frac{|A|}{\lambda_n}$; $|A| \neq 0$.

Thus Eigen values of adj A are $\frac{4}{2}, \frac{-4}{4}$ i.e. 2 and -1, so, $x = 2$, $y = 1$ then sum $x + y = 3$.

164. (2)

$$\lambda I - A = \begin{bmatrix} \lambda & 0 & 0 \\ -1 & \lambda & 1 \\ 0 & -1 & \lambda \end{bmatrix}$$

Note: det $(\lambda I - A) = 0$ is same as det $(A - \lambda I) = 0$.

det $(\lambda I - A) = \lambda . (\lambda^2 + 1)$

$\lambda_1 = 0, \lambda_2 = +i, \lambda_3 = -i$

Sum of squares of Eigen values

$$= \lambda_1^2 + \lambda_2^2 + \lambda_3^3 = (0)^2 + (i)^2 + (-i)^2$$
$$= 0 + (-1) + (-1) = -2$$

$\therefore$ 2 is absolute value.

165. (b)

(a) Eigen values of unitary matrix are of unit modulus.

(b) Eigen values of orthogonal matrix are of unit modulus.

(c) Atleast one of the eigen value of square nilpotent matrix is zero.

(d) Eigen values of idempotent matrix is 0 or 1.

166. (d)

Characteristic equation of A is given by,

$$|A - \lambda I| = 0$$

Here, $|A - \lambda I|$ is a characteristic polynomial in terms of λ and the solution of $|A - \lambda I| = 0$ gives Eigen values of matrix A.

$$\Rightarrow \begin{vmatrix} a-\lambda & 1 & 0 \\ 1 & a-\lambda & 1 \\ 0 & 1 & a-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a-\lambda & 0 & -(a-\lambda) \\ 1 & a-\lambda & 1 \\ 0 & 1 & a-\lambda \end{vmatrix} = 0 \quad [R_1' = R_1 - R_3]$$

$$\Rightarrow (a-\lambda)\begin{vmatrix} 1 & 0 & -1 \\ 0 & a-\lambda & 2 \\ 0 & 1 & a-\lambda \end{vmatrix} = 0$$

[Taking $(a - \lambda)$ common from R_1]

$R_2' = R_2 - R_1$

$$\Rightarrow (a-\lambda)\begin{vmatrix} 1 & 0 & -1 \\ 0 & a-\lambda & 2 \\ 0 & 1 & a-\lambda \end{vmatrix} = 0$$

$\Rightarrow (a-\lambda)[(a-\lambda)^2 - 2] = 0$

[Expanding using e_1]

$\therefore$ Either $(a - \lambda) = 0$ or $(a - \lambda)^2 - 2 = 0$

$\because \quad (a - \lambda) = 0$

$\Rightarrow \quad \lambda = a$

$(a - \lambda)^2 - 2 = 0$

$\Rightarrow \{(a-\lambda) + \sqrt{2}\}\{(a-\lambda) - \sqrt{2}\} = 0$

$\Rightarrow \lambda = a + \sqrt{2}, a - \sqrt{2}$

$\therefore$ Eigen values of A are a, $(a + \sqrt{2}), (a - \sqrt{2})$

So, correct option is (d).

167. (a)

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix}$$

$$= -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

Characteristic equation

$\lambda^3 - 6\lambda + 9\lambda - 4 = 0$

$\lambda = 1, 1, 4$

168. (c)

$$|A^3| = |A|^3$$
$$|A| = 5$$
$$\alpha^2 - 4 = 5$$
$$\alpha^2 = 9$$
$$\alpha = \pm 3$$

–3 is most appropriate option.
So, option (c) is correct.

169. (c)

First, find the eigen values of $A = \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-4-\lambda)(3-\lambda) - 8 = 0$$
$$\Rightarrow \lambda^2 + \lambda - 20 = 0$$
$$\Rightarrow (\lambda + 5)(\lambda - 4) = 0$$
$$\Rightarrow \lambda_1 = -5 \text{ and } \lambda_2 = 4$$

Corresponding to $\lambda_1 = -5$ we need to find eigen vector:
The eigen value problem is $[A - \lambda I]X = 0$

$$\Rightarrow \begin{bmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix} = 0$$

Putting $\lambda = -5$

We get, $\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$x_1 + 2x_2 = 0 \quad \text{... (i)}$$
$$4x_1 + 8x_2 = 0 \quad \text{... (ii)}$$

Since (i) and (ii) are the same equation we take

$$x_1 + 2x_2 = 0$$
$$x_1 = -2x_2$$
$$x_1 : x_2 = -2 : 1$$

$$\Rightarrow \frac{x_1}{x_2} = -2$$

Now from the answers given, we look for any vector in this ratio and we find choice (c) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ is.

in this ratio $\frac{x_1}{x_2} = \frac{2}{-1} = -2$.

So choice (c) is an eigen vector corresponding to $\lambda = -5$.

Since we already got an answer, there is no need to find the second eigen vector corresponding to $\lambda = 4$.

170. (2)

Let λ is the eigen value of A.

$$|A - \lambda| = 0$$

$$\begin{vmatrix} 1-\lambda & -3 & 1 \\ 0 & -2-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$(1-\lambda)(-2-\lambda)(1-\lambda) = 0$

$$\lambda = 1, -2, 1$$

$\therefore$ 2 distinct eigen values
$\therefore$ 2 linearly independent eigen vector

171. (b)

$$A = \begin{bmatrix} 2 & -1 \\ -4 & 5 \end{bmatrix}$$

The characteristic equation of this matrix is given by

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 \\ -4 & 5-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(5-\lambda) - 4 = 0$$
$$\lambda^2 - 7\lambda + 6 = 0$$
$$\lambda = 1, 6$$

$\therefore$ The eigen values of A are 1 and 6.

172. (c)

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

Now, $A - \lambda I = 0$
Where λ = eigen value

$$\therefore \begin{bmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{bmatrix} = 0$$

$$(4-\lambda)^2 - 1 = 0$$
or, $(4-\lambda)^2 - (1)^2 = 0$
or, $(4-\lambda+1)(4-\lambda-1) = 0$
or, $(5-\lambda)(3-\lambda) = 0$
$\therefore$ $\lambda = 3, \lambda = 5$

173. (d)

Since matrix is triangular, the eigen values are the diagonal elements themselves namely $\lambda = 3, -2$ and 1. Corresponding to eigen value, $\lambda = -2$ let us find the eigen vector

$$[A-\lambda I]x = 0$$

$$\begin{bmatrix} 3-\lambda & -2 & 2 \\ 0 & -2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Putting $\lambda = -2$ in above equation we get,

$$\begin{bmatrix} 5 & -2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Which gives the equations,

$5x_1 - 2x_2 + 2x_3 = 0$ ... (i)

$x_3 = 0$ ... (ii)

$3x_3 = 0$ ... (iii)

Since eq. (ii) and (iii) are same we have

$5x_1 - 2x_2 + 2x_3 = 0$ ... (i)

$x_3 = 0$ ... (ii)

Putting $x_2 = k$, in eq. (i) we get

$5x_1 - 2k + 2 \times 0 = 0$

$$\Rightarrow \quad x_1 = \frac{2}{5}k$$

∴ Eigen vectors are of the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{2}{5}k \\ k \\ 0 \end{bmatrix}$$

i.e. $x_1 : x_2 : x_3 = \frac{2}{5}k : k : 0 = \frac{2}{5} : 1 : 0 = 2 : 5 : 0$

$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$ is an eigen vector of matrix A.

174. (a)

Sum of eigen values = Trace of matrix

Multiplication of eigen values = Determinant of matrix

Let eigen values be x and y

Then, $x + y = 5$ ...(i)

$xy = -14$ ...(ii)

Solving equation (i) and (ii)

$x = 7, \ y = -2$

175. (b)

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$[a-\lambda I] = 0$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix} = 0$$

$$\Rightarrow \quad (2-\lambda)^2 = 0$$

$$\Rightarrow \quad \lambda = 2$$

Now, consider the eigen value problem

$$[A-\lambda I]X = 0$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Put $\lambda = 2$, we get,

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_2 = 0$ ... (i)

$0 = 0$ ... (ii)

The solution is therefore $x_2 = 0$, x_1 = anything

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} k \\ 0 \end{bmatrix} = k\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

176. (c)

Eigen values of symmetric matrix are always real.

177. (c)

Eigen values of a skew symmetric matrix are either zero or pure imaginary.

178. (b)

Given, $$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

P is triangular. So eigen values are the diagonal elements themselves. Eigen values are therefore, $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$.

Now, the eigen value problem is $[A-\lambda I]\hat{x} = 0$

$$\begin{bmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 2 \\ 0 & 0 & 3-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Putting $\lambda_1 = 1$, we get the eigen vector corresponding to this eigen value,

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Which gives the equations

$x_2 = 0$

$x_2 + 2x_3 = 0$

$2x_3 = 0$

The solution is $x_2 = 0, x_3 = 0, x_1 = k$

So, one eigen vector is $\hat{x}_1 = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix}$

i.e., $x_1 : x_2 : x_3 = k : 0 : 0$

Since, none of the eigen vectors given in choices matches with this, ratio we need to proceed further and find the other eigen vectors corresponding to the other Eigen values.

Now, corresponding to $\lambda_2 = 2$, we get by substituting $\lambda = 2$, in the eigen value problem, the following set of equations,

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Which gives the equations,

$$-x_1 + x_2 = 0$$
$$2x_3 = 0$$
$$x_3 = 0$$

Solution is $x_3 = 0, x_1 = k, x_2 = k$

$$\therefore\ X_2 = \begin{bmatrix} k \\ k \\ 0 \end{bmatrix} \text{ i.e., } x_1 : x_2 : x_3 = 1 : 1 : 0$$

Since none of the eigen vectors given in the choices is of this ratio, we need to proceed further and find 3rd eigen vector also.

By putting $\lambda = 3$ in the eigen valve problem, we get

$$\begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 + x_2 = 0$$
$$-x_2 + 2x_3 = 0$$

Putting $x_1 = k$, we get, $x_2 = 2k$ and $x_3 = \frac{x_2}{2} = k$

$$\therefore \qquad \hat{X}_3 = \begin{bmatrix} k \\ 2k \\ k \end{bmatrix}$$

i.e., $\qquad x_1 : x_2 : x_3 = 1 : 2 : 1$

Only the eigen vector given in choice (b) $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, is in this ratio.

So, the correct answer is choice (b).

179. (c)

$$\Sigma\lambda_i = \text{Trace } (A) = -2$$
$$\Rightarrow \quad \lambda_1 + \lambda_2 = -2 \qquad \ldots (i)$$
$$\Pi\lambda_i = |A| = -35$$
$$\Rightarrow \quad \lambda_1 \lambda_2 = -35 \qquad \ldots (ii)$$

Solving (i) and (ii) we get λ_1 and $\lambda_2 = 5, -7$.

180. (a)

Every eigen vector X_i of matrix A corresponding to eigen value λ satisfy,

$$AX = \lambda X$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \qquad \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow \lambda = 1$$

181. (b)

If $\lambda_1, \lambda_2, \lambda_3 \ldots\ldots \lambda_n$ are the eigen values of matrix A, then $\lambda_1^k, \lambda_2^k, \ldots \lambda_n^k$ will be the eigen values of A^k. But the A and A^k will have same eigen vector.

Example:

Let a matrix

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 22 & 27 \\ 18 & 31 \end{bmatrix}$$

eigen values of $A = 2, 7$

eigen values of $A^2 = 4, 49$

eigen vector of matrix A corresponding to eigen value 2 is $[3, \ -2]^T$

eigen vector of matrix A^2 corresponding to eigen value 4 is $[3, \ -2]^T$.

182. (1)

Since, $A^2 = I$, $\text{eig}(A^2) = \text{eig}(I) = 1$

$$\Rightarrow \qquad \text{eig}(A)^2 = 1$$
$$\Rightarrow \qquad \text{eig}(A) = \pm 1$$

Therefore, the positive eigen value of A is $+1$.

183. (c)

(i) The Eigen values of symmetric matrix $[A^T = A]$ are purely real.

(ii) The Eigen value of skew-symmetric matrix $[A^T = -A]$ are either purely imaginary or zeros.

184. (a)

$$|A-\lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 5 & 2 \\ 5 & 12-\lambda & 7 \\ 2 & 7 & 5-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)[(12-\lambda)(5-\lambda)-49]-5[5(5-\lambda)-14]$$
$$+2[35-2(12-\lambda)]=0$$

$$(3-\lambda)\left[60-17\lambda+\lambda^2-49\right]-5(25-5\lambda-14)$$
$$+2(35-24+2\lambda)=0$$

$$(3-\lambda)(\lambda^2-17\lambda+11)-5(11-5\lambda)+2(11+2\lambda)=0$$

$$3\lambda^2-51\lambda+33-\lambda^3+17\lambda^2-11\lambda-55+25\lambda+22+4\lambda=0$$

$$-\lambda^3+20\lambda^2-33\lambda=0$$

$$\lambda^3-20\lambda^2+33\lambda=0$$

$$\lambda(\lambda^2-20\lambda+33)=0$$

$$\lambda = 0,\ 1.82,\ 18.2$$

So minimum eigen value is 0.

185. (d)

Trace = Sum of eigen values

$$1 + a = 6$$

$$\Rightarrow \quad a = 5$$

Determinant = Product of eigen values

$$(a - 4b) = -7$$

$$5 - 4b = -7$$

$$-4b = -12$$

$$\Rightarrow \quad b = 3$$

$$\therefore \quad a = 5,\ b = 3$$

186. (d)

$$A = \begin{bmatrix} 2 & -4 \\ 4 & -2 \end{bmatrix}$$

$$|A-\lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -4 \\ 4 & -2-\lambda \end{vmatrix} = 0$$

$$-4-2\lambda+2\lambda+\lambda^2+16=0$$

$$\lambda^2+12=0$$

$$\lambda = \pm 2\sqrt{3}i \text{ (Complex eigen values)}$$

(1) $\lambda = 2\sqrt{3}i$

Consider $(A-\lambda I)X=0$

$$\begin{bmatrix} 2-2\sqrt{3}i & -4 \\ 4 & -2-2\sqrt{3}i \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2-2\sqrt{3}i\,x_1 = 4x_2$$

$$\frac{x_1}{4} = \frac{x_2}{2-2\sqrt{3}i}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2-2\sqrt{3}i \end{bmatrix}$$

(2) $\lambda = -2\sqrt{3}i$

Consider $(A-\lambda I)X=0$

$$\begin{bmatrix} 2+2\sqrt{3}i & -4 \\ 4 & -2+2\sqrt{3}i \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2+2\sqrt{3}i\,x_1 = 4x_2$$

$$\frac{x_1}{4} = \frac{x_2}{2+2\sqrt{3}i}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2+2\sqrt{3}i \end{bmatrix}$$

$\therefore$ Complex eigen values and complex eigen vectors.

187. (c)

Eigen vectors corresponding to distinct eigen values are linearly independent.

Hence, "S_2 implies S_1".

188. (20)

$$\text{trace}(5A) = 5 \text{ trace}(A)$$
$$= 5 \times (2 + (-1) + 3)$$
$$= 5 \times (1 + 3)$$
$$= 5 \times (4) = 20$$

189. (a)

If A is square matrix of order 100 has 15 distinct eigen values. The of minimal polynomial is a divisor of the characteristic polynomial. Every root of minimal polynomial is also a root of characteristic equation.

190. (10)

Product of eigen values = $|A|$

$$= (1)\ (2)\ (5) = 10$$

191. (d)

M is a 2×2 matrix with eigen values 4 and 9.

The eigen values of M^2 are 16 and 81.

192. (c)

A is idempotent $\Rightarrow A^2 = A$

λ is eigen value of A.

$$\Rightarrow \quad \lambda^2 = \lambda$$
$$\Rightarrow \quad \lambda^2 - \lambda = 0$$
$$\lambda(\lambda - 1) = 0$$
$$\lambda = 0 \text{ (or) } 1$$

193. (c)

Characteristic equation

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$
$$\Rightarrow \quad \lambda^2 - 4\lambda - 5 = 0$$
$$\Rightarrow \quad \lambda = 5, -1$$
$$\therefore \quad \lambda_{max} = 5$$

194. (b)

If X is an eigen vector of A for eigen value λ, then X is also an eigen vector of A^2 for eigen value λ^2.

195. (2)

From properties of eigen values,

1. Sum of eigen values = Trace of matrix

$$\Rightarrow \quad 4 = 3 + q$$
$$\text{or} \quad q = 1$$

2. Product of eigen values = Determinant

$$\Rightarrow \quad -1 = \begin{vmatrix} 3 & p \\ p & q \end{vmatrix}$$
$$\Rightarrow \quad 3q - p^2 = -1$$
$$\Rightarrow \quad 3 - p^2 = -1$$
$$\Rightarrow \quad p^2 = 4$$
$$\Rightarrow \quad p = \pm 2$$
$$\text{Thus,} \quad |p| = 2$$

196. (a)

$$|\text{adj}(kA)| = |k^{n-1}\text{adj}(A)|$$

where n is the order of matrix A

$$= (k^{n-1})^n |A^{-1}||A||$$
$$= |\text{adj}(A) = A^{-1}|A||$$
$$= (k^{n-1})^n |A|^n |A|^{-1}$$
$$= k^{n(n-1)} |A|^{n-1}$$
$$= n = 3$$
$$= k^{3\times 2} |A|^2 = k^6 |A|^2$$
$$\therefore \quad k = 2, \ |A| = 1 \times 2 \times 1 = 2$$
$$= 2^6 2^2 = 2^8$$
$$= 256$$

■■■■

UNIT 2

Calculus and Optimization

1. Limit, Continuity and Differentiability

Q.1 $\lim_{x\to\infty}\frac{x-\sin x}{x+\cos x}$ equals

(a) 1 (b) -1
(c) ∞ (d) $-\infty$

Q.2 Which of the following statement is/are false?

(a) $\lim_{x\to 2}\frac{\sin(x-2)}{3x-6}=\frac{1}{3}$

(b) $\lim_{x\to 2}\frac{\tan(x-2)}{x-2}=\frac{1}{2}$

(c) $\lim_{x\to\infty}\left(2+\frac{1}{x}\right)^{x}=e^2$

(d) $\lim_{x\to 0}\frac{\log_e(1+2x)}{2x}=\frac{1}{2}$

Q.3 $\lim_{x\to\infty} x^{1/x}$ is

(a) ∞ (b) 0
(c) 1 (d) Not defined

Q.4 Evaluate the limit $\lim_{n\to\infty}\left[\cos\frac{\pi}{2^2}\cos\frac{\pi}{2^3}\ldots\cos\frac{\pi}{2^n}\right]$

(a) π (b) $\frac{1}{\pi}$
(c) $\frac{2}{\pi}$ (d) $\frac{\pi}{e}$

Q.5 The value of $\lim_{x\to 1}(1-x^2)^{[\log(1-x)]^{-1}}$ is equal to

(a) π (b) e
(c) 0 (d) ∞

Q.6 What is the value of $\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{2n}$?

(a) 0 (b) e^{-2}
(c) $e^{-1/2}$ (d) 1

Q.7 $\lim_{x\to 0}\left(\frac{7x^2-5x+3}{-2x+3}\right)^{\frac{1}{x}}$ is equal to

(a) e^{-1} (b) $e^{2/3}$
(c) $e^{-3/2}$ (d) $e^{-2/3}$

Q.8 If $f(x)$ is defined as follows, what is the minimum value of $f(x)$ for $x \in (0, 2]$?

$$f(x)=\begin{cases}\frac{25}{8x} \text{ when } x\le\frac{3}{2}\\ x+\frac{1}{x} \text{ otherwise}\end{cases}$$

(a) 2 (b) $2\left(\frac{1}{12}\right)$
(c) $2\left(\frac{1}{6}\right)$ (d) $2\left(\frac{1}{2}\right)$

Q.9 What is the value of the $L=\lim_{x\to 0}\frac{x^2-4x}{5x}$?

(a) $-\frac{5}{4}$ (b) $\frac{4}{5}$
(c) $-\frac{4}{5}$ (d) Doesn't exist

Q.10 The value of $\lim_{x\to\infty}(1+x^2)^{e^{-x^2}}$ is

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) ∞

Q.11 Consider the following two statements about the function $f(x)=|x|$

P : $f(x)$ is continuous for all real values of x.
Q : $f(x)$ is differentiable for all real values of x.
Which of the following is TRUE?

(a) P is true and Q is false
(b) P is false and Q is true
(c) Both P and Q are true
(d) Both P and Q are false

Q.12 $\lim_{x\to 0}\left(e^{-\frac{1}{x}}-2\right)\left(7+\frac{\sin\sqrt{x}}{2\sqrt{x}}\right)=$ _____.

Q.13 $\lim_{\theta \to 0} \frac{\sin\left(\frac{\theta}{2}\right)}{\theta}$ is

(a) 0.5 (b) 1
(c) 2 (d) not defined

Q.14 The value of the function $f(x) = \lim_{x \to 0} \frac{x^3 + x^2}{2x^3 - 7x^2}$ is

(a) 0 (b) $-\frac{1}{7}$
(c) $\frac{1}{7}$ (d) ∞

Q.15 Consider the function $f(x) = |x|^3$, where x is real.
Then the function $f(x)$ at $x = 0$ is
(a) continuous but not differentiable
(b) once differentiable but not twice
(c) twice differentiable but not thrice
(d) thrice differentiable

Q.16 The $\lim_{x \to 0} \frac{\sin\left[\frac{2}{3}x\right]}{x}$ is

(a) $\frac{2}{3}$ (b) 1
(c) $\frac{3}{2}$ (d) ∞

Q.17 The values of p and q for which is function

$$f(x) = \begin{cases} 2x+1, & \text{if } x \le 2 \\ px^2 + q, & \text{if } 2 < x < 4 \\ 5x + 2p, & \text{if } x \ge 4 \end{cases}$$

is continuous every where

(a) $p = \frac{2}{3}, q = -1$ (b) $p = 1, q = \frac{2}{3}$
(c) $p = \frac{3}{2}, q = -1$ (d) $p = \frac{2}{3}, q = +1$

Q.18 The function $y = |2 - 3x|$
(a) is continuous $\forall x \in R$ and differentiable $\forall x \in R$
(b) is continuous $\forall x \in R$ and differentiable $\forall x \in R$ except at $x = \frac{3}{2}$
(c) is continuous $\forall x \in R$ and differentiable $\forall x \in R$ except at $x = \frac{2}{3}$
(d) is continuous $\forall x \in R$ except $x = 3$ and differentiable $\forall x \in R$

Q.19 $\lim_{x \to 0} \frac{\sin^2 x}{x}$ is equal to

(a) 0 (b) ∞
(c) 1 (d) -1

Q.20 If $f(x) = \frac{2x^2 - 7x + 3}{5x^2 - 12x - 9}$, then $\lim_{x \to 3} f(x)$ will be

(a) $-\frac{1}{3}$ (b) $\frac{5}{18}$
(c) 0 (d) $\frac{2}{5}$

Q.21 $\lim_{x \to 0} \frac{e^x - \left(1 + x + \frac{x^2}{2}\right)}{x^3} =$

(a) 0 (b) $\frac{1}{6}$
(c) $\frac{1}{3}$ (d) 1

Q.22 $\lim_{x \to 0} \frac{\left(\sqrt{x+2} - \sqrt{2-x}\right)}{\left(\sqrt{x+3} - \sqrt{3-x}\right)}$ is equal to

(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{3}{4}}$
(c) $\sqrt{\frac{4}{3}}$ (d) $\sqrt{\frac{3}{2}}$

Q.23 Given the following statements about a function $f : R \to R$, select the right option:
P : If $f(x)$ is continuous at $x = x_0$, then it is differential at $x = x_0$.
Q : If $f(x)$ is continuous at $x = x_0$, then it may not be differentiable at $x = x_0$.
R : If $f(x)$ is differentiable at $x = x_0$, then it is also continuous at $x = x_0$.
(a) P is true, Q is false, R is false
(b) P is false, Q is true, R is true
(c) P is false, Q is true, R is false
(d) P is true, Q is false, R is true

Q.24 The values of a and b such that

$$\lim_{x \to 0} \frac{a^2 \sin x + b \log(\cos x)}{x^2} = -1$$

(a) $-1, +2$ (b) $0, -2$
(c) $0, 2$ (d) $1, -2$

Q.25 The value of $\lim_{x\to 0}\frac{x^3-\sin(x)}{x}$ is

(a) 0 (b) 3
(c) 1 (d) -1

Q.26 $\lim_{n\to\infty}\left(\sqrt{n^2+n}-\sqrt{n^2+1}\right)$ is _____.

Q.27 $\lim_{x\to 4}\frac{\sin(x-4)}{x-4}=$ _____.

Q.28 $\lim_{x\to 0}\frac{\log_e(1+4x)}{e^{3x}-1}$ is equal to

(a) 0 (b) $\frac{1}{12}$
(c) $\frac{4}{3}$ (d) 1

Q.29 $\lim_{x\to\infty} x^{1/x}$ is

(a) ∞ (b) 0
(c) 1 (d) Not defined

Q.30 If a function is continuous at a point,

(a) the limit of the function may not exist at the point.
(b) the function must be derivable at the point.
(c) the limit of the function at the point tends to infinity.
(d) the limit must exist at the point and the value of limit should be same as the value of the function at that point.

Q.31 $\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)$ is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

Q.32 Which of the following option(s) is/are true?

(a) $\lim_{x\to 0}\frac{\log_a(1+x)}{x}=\log_e a$

(b) $\lim_{x\to 0}\frac{a^x+1}{x}=1$

(c) $\lim_{x\to 0}\frac{e^x-1}{x}=1$

(d) $\lim_{x\to 0}\frac{\sin^{-1}x}{x^2}=1$

Q.33 $\lim_{x\to\infty}\sqrt{x^2+x-1}-x$ is

(a) 0 (b) ∞
(c) $\frac{1}{2}$ (d) $-\infty$

Q.34 The value of $\lim_{x\to 1}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

(a) is 0 (b) is -1
(c) is 1 (d) Does not exist

Q.35 Let f be a real-valued function of a real variable defined as $f(x)=x^2$ for $x\geq 0$, and $f(x)=-x^2$ for $x<0$. Which one of the following statements is true?

(a) $f(x)$ is discontinuous at $x=0$.
(b) $f(x)$ is continuous but not differentiable at $x=0$.
(c) $f(x)$ is differentiable but its first derivative is not continuous at $x=0$.
(d) $f(x)$ is differentiable but its first derivative is not differentiable at $x=0$.

Q.36 Compute $\lim_{x\to 3}\frac{x^4-81}{2x^2-5x-3}$

(a) $\frac{108}{7}$ (b) $\frac{53}{12}$
(c) 1 (d) Limit does not exists

Q.37 Which of the following option(s) is true?

(a) $\lim_{x\to 0}\frac{\sin x}{x}=0$ (b) $\lim_{x\to\infty}\frac{\frac{\pi}{2}-\tan\left(\frac{1}{x}\right)}{\frac{1}{x}}=1$

(c) $\lim_{x\to 0}\frac{\tan x}{x}=1$ (d) $\lim_{x\to 0}\frac{\cos x}{x}=1$

Q.38 Consider the following function:

$f(x)=\frac{-|x|}{2x}$; where x is real number.

Which of the following statement is/are true?

(a) The limit of $f(x)$ exists at $x=1$ and the value of limit is 1 at $x=\frac{1}{2}$.
(b) The limit of $f(x)$ exists at $x=0$ and the value of limit is 1 at $x=0$.
(c) The limit of $f(x)$ exists at $x=0$ but not continuous at $x=0$.
(d) $\lim_{x\to 0} f(x)$ does not exist.

Q.39 Consider the following two functions:

$$f(x)=\begin{Bmatrix}-3x; & x<-1\\ 3; & -1\le x\le 1\\ 3x; & x>1\end{Bmatrix}$$

Which of the following statement is/are true?

(a) $f(x)$ is continuous at $x=1$.

(b) $f(x)$ is not continuous at $x=-1$.

(c) $f(x)$ is differentiable at $x=1$.

(d) $f(x)$ is not differentiable at $x=-1$.

Q.40 Function $f(x)=|\cos x|$ is

(a) Continuous only in $\left[0,\frac{\pi}{2}\right]$

(b) Continuous only in $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

(c) Continuous only in $[-\pi,\pi]$

(d) None of these

Q.41 $\lim\limits_{x\to\infty}\left(\frac{x^2+2x+3}{x^2-1x+1}\right)^{x/2}=$ _____

(a) $e^{3/2}$ (b) $e^{1/2}$

(c) $e^{-3/2}$ (d) e^{3}

Q.42 If $\lim\limits_{x\to 4}\frac{(x^{3/2}-k^3)}{x-4}=3$, what is the value of k?

Q.43 If $f(x)=\frac{\sin[x]}{[x]}$, $[x]\neq 0$

$=0$, $[x]=0$

where $[x]$ is the greatest integer less than or equal to x, then $k\lim\limits_{x\to 0}f(x)$ is

(a) 1

(b) Does not exist

(c) 0

(d) -1

Q.44 Let the function $f(x)$ be defined as $f(x)=|x-1|+|x-2|$. Then which of the following statements is true?

(a) $f(x)$ is differentiable at $x=1$

(b) $f(x)$ is differentiable at $x=2$

(c) $f(x)$ is differentiable at $x=1$ but not at $x=2$

(d) None of the above

Q.45 If the function $f(x)$ defined by

$$\begin{cases}\frac{\log(1+ax)-\log(1-bx)}{x}, & \text{if } x\neq 0\\ k & \text{if } x=0\end{cases}$$

is continuous at $x=0$, then value of k is:

(a) $b-a$ (b) $a-b$

(c) $a+b$ (d) $-a-b$

Q.46 $\lim\limits_{x\to 0}\frac{e^x(\cos x)\sin^{1/2}x}{\sqrt{x}}=$

(a) 1 (b) $e^{-1/2}$

(c) -1 (d) Limit does not exist

Q.47 Which one of the following functions is continuous at $x=3$?

(a) $f(x)=\begin{cases}2, & \text{if } x=3\\ x-1, & \text{if } x>3\\ \frac{x+3}{3}, & \text{if } x<3\end{cases}$

(b) $f(x)=\begin{cases}4, & \text{if } x=3\\ 8-x & \text{if } x\neq 3\end{cases}$

(c) $f(x)=\begin{cases}x+3, & \text{if } x\le 3\\ x-4 & \text{if } x>3\end{cases}$

(d) $f(x)=\frac{1}{x^3-27}$, if $x\neq 3$

Q.48 $\lim\limits_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}=$ _____

(a) e^5 (b) e^6

(c) e^2 (d) e^{10}

Q.49 $\lim\limits_{x\to 0}\frac{x(e^{2x}-1)+5(\cos x^2-1)}{x(2-\cos x)}=$ _____

Q.50 If a continuous function $f(x)$ does not have a root in the interval $[a, b]$, then which one of the following statements is TRUE?

(a) $f(a)\cdot f(b)=0$

(b) $f(a)\cdot f(b)<0$

(c) $f(a)\cdot f(b)>0$

(d) $f(a)/f(b)\le 0$

Q.51 $\lim\limits_{x\to\infty}\left(\frac{1}{1-x^2}+\frac{2}{1-x^2}+\ldots+\frac{x}{1-x^2}\right)$ is equal to

(a) 0 (b) $\frac{1}{2}$

(c) $-\frac{1}{2}$ (d) None of the above

Q.52 $\lim_{x\to 0}\frac{4x^4 - x + 5}{x^3 - 5x^2 + 20}$ is equal to _____.

Q.53 Let the function

$$f(\theta) = \begin{vmatrix} \sin\theta & \cos\theta & \tan\theta \\ \sin\left(\frac{\pi}{6}\right) & \cos\left(\frac{\pi}{6}\right) & \tan\left(\frac{\pi}{6}\right) \\ \sin\left(\frac{\pi}{3}\right) & \cos\left(\frac{\pi}{3}\right) & \tan\left(\frac{\pi}{3}\right) \end{vmatrix}$$

where $\theta \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ and $f'(\theta)$ denote the derivative of f with respect to θ. Which of the following statements is/are TRUE?

I. There exists $\theta \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ such that $f'(\theta) = 0$.

II. There exists $\theta \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ such that $f'(\theta) \neq 0$.

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

Q.54 At the point $x = 1$, the function

$$f(x) = \begin{cases} x^3 - 1; & 1 < x < \infty \\ x - 1; & -\infty < x \le 1 \end{cases}$$ is

(a) Continuous and differentiable
(b) Continuous and not differentiable
(c) Discontinuous and differentiable
(d) Discontinuous and not differentiable

Q.55 $\lim_{x\to\infty} e^{2x}\sin\left(\frac{2d}{e^x}\right)^2$ is equal to

(a) $4d^2$ (b) $2d^2$
(c) $4d$ (d) None of these

Q.56 Consider the function:

$$f(x) = \begin{cases} x^4, & x^2 < 1 \\ x, & x^2 \ge 1 \end{cases}$$

Then which of the following is true.

(a) $f(x)$ is continuous but not differentiable at $x = -1$.
(b) $f(x)$ is neither continuous nor differentiable at $x = -1$.
(c) $f(x)$ is continuous and differentiable at $x = -1$.
(d) $f(x)$ is not continuous but differentiable at $x = -1$.

Q.57 $\lim_{x\to 0}\frac{2\left(a^{\frac{kx}{2}} - 2\right)}{3x}$ is equal to

(a) $\frac{1\log_e a}{3}$
(b) $\frac{2k\cdot\log_e a}{3}$
(c) $\frac{k\log_e a}{3}$
(d) $\frac{3}{2}\log_e a$

Q.58 If $\lim_{x\to 1}\frac{x^x - x}{x - 1 - \log x} = A$, then A is _____.

(a) 0 (b) 1
(c) 2 (d) Limit does not exists

Q.59 The value of $\lim_{x\to\frac{\pi}{4}}\frac{\sec^2 x - 2\tan x}{1 + \cos 4x}$ is _____

(a) 0.5 (b) 0.25
(c) 1 (d) 2

Q.60 $\lim_{x\to\infty}\sqrt{\frac{x - \sin\left(\frac{x}{2}\right)}{\frac{x}{3} + \cos^2 x}}$ is equal to

(a) 0 (b) $\sqrt{3}$
(c) $\sqrt{6}$ (d) None of these

Q.61 Let $f(y) = \begin{cases} \frac{\sin y}{y} & \text{if } y \neq 0 \\ 1 & \text{if } y = 0 \end{cases}$

Then f is

(a) discontinuous at $y = 0$.
(b) continuous but not differentiable at $y = 0$.
(c) differentiable but not continuous at at $y = 0$.
(d) differentiable and continuous at all points.

Q.62 The function $f: \mathbb{R} \to \mathbb{R}$ is given by

$$f(x) = \min\left(e^x, |x|\right)$$

Which of the following is true about the function f?

(a) It is not differentiable exactly at two points of $\mathbb{R}$.
(b) It is differentiable at $x = 0$.
(c) It is not differentiable at $x = -2$.
(d) It is strictly decreasing for $x < -1$.

Q.63 $\lim_{x\to 0}\dfrac{2^{-\sin x}-1}{x\left(x-\dfrac{\pi}{2}\right)}=$

(a) $\dfrac{2\ln 2}{\pi}$ (b) $\dfrac{\ln 2}{\pi}$

(c) $\dfrac{\ln 2}{2\pi}$ (d) Does not exists

2. Mean Value Theorem, Maxima, Minima, Taylor's Series and Optimization

Q.64 Consider the function $f(x) = x + \ln x$ and f is differentiable on $(1, e)$ and $f(x)$ is continuous on $[1, e]$. Determine the c value using mean value theorem. [By computing $f'(c) = \dfrac{f(b)-f(a)}{b-a}$]

(a) e (b) $e-1$

(c) $\dfrac{e}{e-1}$ (d) $\dfrac{e-1}{e}$

Q.65 For which value of $c \in (0, 4)$, Rolle's theorem is applicable for the function $f(x) = x^3 - 6x^2 + x$ in the interval $[0, 4]$?

(a) 0.085 (b) 2.71

(c) 3.91 (d) 0.85

Q.66 Consider the following function:

$$f(x) = x^4 - 2x^2 + 4x + 16$$

Which of the following statement is/are true?

(a) $f(x)$ has maxima at $x = 0$

(b) $f(x)$ has minima at $x = -1$

(c) $f(x)$ has maxima $x = 1$

(d) $f(x)$ has maxima $x = 1$, maximum value is 19

Q.67 Consider the function $y = x^2 - 6x + 9$. The maximum value of y obtained when x varies over the interval 2 to 5 will be at _____.

Q.68 Determine all the number(s) C which satisfy the conclusion of Rolle's Theorem for $g(t) = 2t - t^2 - t^3$ on $[-2, 1]$.

Choose correct option:

(a) 0.1234, 0.5486 (b) 1.2153, 0.5486

(c) −1.2153, 0.5486 (d) 1.2153, −0.5486

Q.69 Determine all the number(s) C which satisfy the conclusion of Mean Value Theorem for

$$A(t) = 8t + e^{-3t} \text{ on } [-2, 3]$$

Choose the correct option:

(a) −1.0973 (b) −1.0973, 3.4734

(c) 3.4734, 1.0973 (d) −3.4734

Q.70 If for $c \in (-1, 1)$, Lagrange's Mean value theorem is applicable for the function $f(x) = \sqrt{16 + x^2}$ in the interval $[0, 3]$ then possible values of c is

(a) 0.57 (b) 0.67

(c) −0.67 (d) −0.57

Q.71 A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 4x^2 + 37$ is

(a) 0 (b) 1

(c) 2 (d) 3

Q.72 What is the maximum value of $\dfrac{e^{\sin x}}{e^{\cos x}}$ where x is a real number _____.

(a) $e^{\sqrt{2}}$ (b) $e^{\sqrt{4}}$

(c) 1 (d) 0

Q.73 At $x = 2$, the function $f(x) = |x - 3|$ has

(a) a minimum at $x = 0$

(b) a maximum at $x = 3$

(c) a point of inflection at $x = 3$

(d) neither a maximum nor minimum

Q.74 Find the maximum and minimum values of $3x^4 - 2x^3 - 6x^2 + 6x + 1$ in the interval $(0, 2)$.

(a) 2.43, 2

(b) 2.43, −6

(c) 21, 2

(d) 21, −6

Q.75 The function $f(x) = -2x - x^2 + 15$ has

(a) a maxima at $x = -1$ and minima at 0

(b) a maxima at $x = -1$ and maximum value 17

(c) only a maxima with maximum value 16

(d) only a minima at $x = -1$

Q.76 A function is defined by $f(x) = 2x^3 - 3x^2 - 12x + 5$ for $-2 \le x \le 3$. Which one of the following statements is true about this function?

(a) function is decreasing for $(-2, -1)$.

(b) function has a minima for $x = -1$.

(c) function has a maxima for $x = 2$.

(d) function is decreasing for $(-1, 2)$.

Q.77 In the mean value theorem $f(b) - f(a) = (b - a) f'(c)$ determining c lying between a and b, if $f(x) = x(x - 1)(x - 2)$, $a = 0$ and $b = \dfrac{1}{2}$.

Q.78 Consider $f(x) = 2x^3 - 15x^2 + 36x + 1$, then which of the following represents the function is increasing in the interval?

(a) $]2, 3[$

(b) $]-\infty, 3[$

(c) $]-\infty, 2[\cup]3, \infty[$

(d) None of these

Q.79 Consider the function $y = x^2 - 6x + 9$. The maximum value of y obtained when x varies over the interval 2 to 5 will be at _____.

Q.80 Consider the function:

$$f(x) = \begin{cases} x^4, & x^2 < 1 \\ x, & x^2 \geq 1 \end{cases}$$

Then which of the following is true.

(a) $f(x)$ is continuous but not differentiable at $x = -1$.

(b) $f(x)$ is neither continuous nor differentiable at $x = -1$.

(c) $f(x)$ is continuous and differentiable at $x = -1$.

(d) $f(x)$ is not continuous but differentiable at $x = -1$.

Q.81 Consider the function $f(x) = x + \ln x$ and f is differentiable on $(1, e)$ and $f(x)$ is continuous on $[1, e]$. Determine the c value using mean value theorem.

[By computing $f'(c) = \dfrac{f(b) - f(a)}{b - a}$]

(a) e (b) $e - 1$

(c) $\dfrac{e}{e-1}$ (d) $\dfrac{e-1}{e}$

Q.82 Minimum of the real valued function $f(x) = (x-1)^{2/3}$ occurs at x equal to

(a) $-\infty$ (b) 0

(c) 1 (d) ∞

Q.83 For real x the maximum value of $\dfrac{e^{\sin x}}{e^{\cos x}}$ is

(a) 1 (b) e

(c) $e^{\sqrt{2}}$ (d) ∞

Q.84 For which value of $c \in (0, 3)$ Cauchy Mean Value theorem is applicable for the functions $p(x) = e^{x/2}$ and $q(x) = e^{-x/2}$ defined in $[a, b]$?

(a) $c = \dfrac{a-b}{2}$ (b) $c = \dfrac{ab}{a+b}$

(c) $c = \dfrac{(a+b)}{2}$ (d) $c = \sqrt{ab}$

Q.85 Consider the following function:

$$f(x) = \begin{cases} x - 2; & x < 2 \\ -x + 2; & x > 2 \\ 0; & x = 2 \end{cases}$$

The function has (local)

(a) Minima at $x = 2$

(b) Maxima at $x = 2$

(c) Neither maxima nor minima

(d) None of these

Q.86 Consider the function $y = x^2 - 8x + 16$. The minimum value of y obtained when x varies over the interval 2 to 5 is

(a) 1 (b) 3

(c) 4 (d) 9

Q.87 The maximum value of

$$f(x) = 2x^3 - 9x^2 + 12x - 3$$

in the interval $0 \leq x \leq 3$ is _____.

Q.88 The maximum value of the function $f(x) = \ln(1 + x) - x$ (where $x > -1$) occur at $x =$ _____.

Q.89 At $x = 0$, the function $f(x) = x^3 + 1$ has

(a) a maximum value

(b) a minimum value

(c) a singularity

(d) a point of inflection

Q.90 At $t = 0$, the function $f(t) = \dfrac{\sin t}{t}$ has

(a) a minimum

(b) a discontinuity

(c) a point of inflection

(d) a maximum

Q.91 If x is real, the maximum value of $\dfrac{2x^2 + x + 9}{2x^2 + x + 1}$ is

(a) 8 (b) $\dfrac{70}{7}$

(c) $\dfrac{71}{7}$ (d) None of these

Q.92 The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is
(a) 20 (b) 28
(c) 16 (d) 32

Q.93 The function $f(x) = 2x^3 - 3x^2 - 36x + 20$ has its minima at
(a) $x = -2$ only
(b) $x = 0$ only
(c) $x = 3$ only
(d) Both $x = -2$ and $x = 3$

Q.94 Let $f(x) = 3x^3 - 7x^2 + 5x + 6$. The maximum value of $f(x)$ over the interval $[0, 2]$ is _____.

Q.95 The global minimum of $x^3e^{-|x|}$ for $x \in (-\infty + \infty)$ occurs at $x =$ _____.

Q.96 The mean value of $y = 3x^2 + 4x + 1$ between $x = -1$ and $x = 2$ is
(a) 2 (b) 4
(c) 6 (d) 8

Q.97 The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is
(a) 21 (b) 25
(c) 41 (d) 46

Q.98 Let $f(x) = x\,e^{-x}$. The maximum value of the function in the interval $(0, \infty)$ is
(a) e^{-1} (b) e
(c) $1 - e^{-1}$ (d) $1 + e^{-1}$

Q.99 Consider the function $f(x) = (x^2 - 2)^4$; where x is a real number. Then the function has
(a) One minima (b) One maxima
(c) Two minima (c) Two maxima

Q.100 At $x = 0$, the function $f(x) = |x|$ has
(a) a minimum
(b) a maximum
(c) a point of inflection
(d) neither a maximum nor minimum

Q.101 According to the Mean Value Theorem, for a continuous function $f(x)$ in the interval $[a, b]$, there exists a value x in this interval such that

$$\int_a^b f(x)dx =$$

(a) $f(\xi)(b - a)$ (b) $f(b)(\xi - a)$
(c) $f(a)(b - \xi)$ (d) 0

Q.102 The maximum value $x^3 - 27x + 3$ in the interval $(-3, 3)$ is
(a) 81 (b) -75
(c) 3 (d) None of these

Q.103 At the point $x = 0$, the function $f(x) = x^3$ has
(a) local maximum
(b) local minimum
(c) both local maximum and minimum
(d) neither local maximum nor local minimum

Q.104 Consider function $f(x) = (x^2 - 4)^2$ where x is a real number. Then the function has
(a) Only one minimum
(b) Only two minima
(c) Three minima
(d) Three maxima

Q.105 The function $f(x) = 2x - x^2 + 3$ has
(a) a maxima at $x = 1$ and a minima at $x = 5$
(b) a maxima at $x = 1$ and a minima at $x = -5$
(c) only a maxima at $x = 1$
(d) only a minima at $x = 1$

Q.106 Given, function $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is
(a) $f(x)$ has one maximum
(b) $f(x)$ has one minimum
(c) $f(x)$ has two maximum
(d) $f(x)$ has two minimum

Q.107 Minimum of the real valued function $f(x) = (x - 1)^{2/3}$ occurs at x equal to
(a) $-\infty$ (b) 0
(c) 1 (d) ∞

Q.108 The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is
(a) 20 (b) 28
(c) 16 (d) 32

Q.109 $\lim\limits_{x \to 0} \dfrac{\sin(8x^2)}{\cos(2x)} =$
(a) 4 (b) 0
(c) 2 (d) -4

Q.110 If $f(x) = x^{2P} \log_e x$ and $f(0) = 0$, then the value(s) of P for which Rolle's theorem can be applied in $[0, 1]$ is
(a) 3 (b) -1
(c) $\dfrac{1}{3}$ (d) $\dfrac{1}{2}$

3. Definite Integrals

Q.111 What is value of $\int_0^{\pi/2} \frac{\cos x}{2\sqrt{1-\sin x}} dx$ is _____?

Q.112 $\int_0^{3\pi/2} |x \sin x| dx =$ _____.

Q.113 $\int_0^{\pi/2} \sin x \cos x \, dx$ is _____.

Q.114 Find the value of integral

$\int_0^{\pi/2} \cos^4 3\theta \sin^3 6\theta \, d\theta$

(a) 0 (b) $\frac{1}{15}$

(c) $\frac{8}{3}$ (d) 1

Q.115 $\int \frac{\sin(\ln x)}{x} dx = -\frac{1}{a} \cos(\ln x) + c$ then value of a is _____.

Q.116 What is value of $\int_0^{\pi} \log(1+\cos x) dx$

(a) $\pi \log 2$ (b) $-\pi \log 2$

(c) $\pi^2 \log 2$ (d) $-\pi^2 \log 2$

Q.117 $\int_0^{3\pi/2} \frac{(\cos x - \sin x)}{(\cos x + \sin x + 2)} dx$ is equal to

(a) ln (3) (b) –ln (4)

(c) –ln (5) (d) –ln (3)

Q.118 What is the value of $\int_0^{16} \frac{x^{1/4} dx}{1+\sqrt{x}}$?

(a) $\frac{4}{3} + \tan^{-1} 2$ (b) $\frac{8}{3} + 4\tan^{-1} 2$

(c) $\frac{2}{3} + \tan^{-1} 2$ (d) $\frac{5}{3} + \tan^{-1} 2$

Q.119 The following definite integral evaluates to

$\int_{-\infty}^{0} e^{-\left(\frac{x^2}{2}\right)} dx$

(a) $\frac{1}{2}$ (b) $\pi\sqrt{10}$

(c) $\sqrt{\frac{\pi}{2}}$ (d) π

Q.120 $\int_0^{\pi/3} \cos^4 3\phi \sin^2 6\phi \, d\phi$ is equal to

(a) $\frac{\pi}{32}$ (b) $\frac{7\pi}{96}$

(c) $\frac{5\pi}{96}$ (d) None of these

Q.121 The integral $\int_0^1 \frac{dx}{\sqrt{(1-x)}}$ is equal to _____

Q.122 The value of the integral given below is

$\int_0^{\pi} x^2 \cos x \, dx$

(a) -2π (b) π

(c) $-\pi$ (d) 2π

Q.123 The value of the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ is

(a) $-\pi$ (b) $-\frac{\pi}{2}$

(c) $\frac{\pi}{2}$ (d) π

Q.124 What is the value of the definite integral,

$\int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}} dx$?

(a) 0 (b) $\frac{a}{2}$

(c) a (d) $2a$

Q.125 If $f(x)$ is an even function and a is a positive real number, then $\int_{-a}^{a} f(x) dx$ equals

(a) 0 (b) a

(c) $2a$ (d) $2\int_0^a f(x) dx$

Q.126 $\int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx$

(a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{2}$

(c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{4}$

Q.127 The value of the quantity P, where $P = \int_0^1 xe^x dx$, is equal to

(a) 0 (b) 1

(c) e (d) $\frac{1}{e}$

Q.128 The infinite series

$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \ldots \infty$ converges to

(a) $\cos(x)$ (b) $\sin(x)$

(c) $\sinh(x)$ (d) e^x

Q.129 The value of integral $\int_{-\pi/2}^{\pi/2} (x\cos x)dx$ is

(a) 0 (b) $\pi - 2$

(c) π (d) $\pi + 2$

Q.130 If $S = \int_1^\infty x^{-3}\, dx$, then S has the value

(a) $-\frac{1}{3}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) 1

Q.131 The integral $\int_0^\pi \sin^3\theta d\theta$ is given by

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$

(c) $\frac{4}{3}$ (d) $\frac{8}{3}$

Q.132 If $\int_0^{2\pi} |x\sin x| dx = k\pi$, then the value of k is equal to _____.

Q.133 If $\int_0^{2\pi} x\sin\left(\frac{x}{2}\right)dx = k\pi^2$ then k is equal to _____.

(a) 4π (b) -4π

(c) $\frac{4}{\pi}$ (d) $-\frac{4}{\pi}$

Q.134 The value of the integral $\int_0^2 \frac{(x-1)^2\sin(x-1)}{(x-1)^2 + \cos(x-1)} dx$ is

(a) 3 (b) 0

(c) -1 (d) -2

Q.135 For the integral

$$I = \int_{-1}^{1} \frac{1}{x^2} dx$$

which of the following statements is TRUE?

(a) $I = 0$

(b) $I = 2$

(c) $I = -1$

(d) The integral does not converge

Q.136 The value of $\int_0^4 \sqrt{(16 - x^2)}\, dx$ is

(a) π (b) 2π

(c) 3π (d) 4π

Q.137 The value of $\int_0^2 \frac{3x}{\sqrt{(2x^2+1)}} dx$ is (take positive values of square roots only)

(a) 1 (b) 2

(c) 3 (d) 4

Q.138 The value of the integral $\int_0^{2\pi} \left(\frac{3}{9 + \sin^2\theta}\right) d\theta$ is

(a) $\frac{2\pi}{\sqrt{10}}$ (b) $2\sqrt{10}\,\pi$

(c) $\sqrt{10}\,\pi$ (d) 2π

Q.139 Consider the following definite integral:

$$I = \int_0^1 \frac{(\sin^{-1}x)^2}{\sqrt{1-x^2}} dx$$

The value of the integral is

(a) $\frac{\pi^3}{24}$ (b) $\frac{\pi^3}{12}$

(c) $\frac{\pi^3}{48}$ (d) $\frac{\pi^3}{64}$

Q.140 The value of $\int_0^{\pi/2} \frac{dx}{1 + \sqrt{\cot x}}$ is equal to

(a) 0.53 (b) 0.66

(c) 0.78 (d) 0.84

Q.141 $\int_{-\pi}^{\pi/2} \cos(x)\cos(\sin(x))dx$

(a) 1 (b) 0

(c) cos 1 (d) sin 1

Q.142 The value of integral $\int_1^3 \frac{|x-2|}{x}dx$ is _____.

(a) $2\ln\frac{4}{3}$ (b) $2\ln\frac{3}{4}$

(c) $\ln\frac{4}{3}$ (d) $4\ln\frac{2}{3}$

Q.143 Evaluate $\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx$

(a) $-\cos\sqrt{x}+c$

(b) $-2\cos(x)^{3/2}+c$

(c) $-2\sin\sqrt{x}+c$

(d) $-2\cos\sqrt{x}+c$

Q.144 What is the value of $\int_0^{\pi/2} \log(\tan x)dx$?

(a) $-2\pi\log 2$ (b) $-\pi\log 2$

(c) 1 (d) 0

Q.145 $\int \frac{\sin(\ln x)}{x}dx = -\frac{1}{a}\cos(\ln x)+c$ then value of a is _____.

Q.146 $\int_{-\pi}^{\pi/2} \cos(x)\cos(\sin(x))dx$

(a) 1 (b) 0

(c) cos 1 (d) sin 1

Q.147 $\int_{-3}^{4} (|x+2|+|x-1|)dx =$ _____.

Answers Calculus and Optimization

1.	(a)	**2.**	(b, c, d)	**3.**	(c)	**4.**	(c)	**5.**	(b)	**6.**	(b)	**7.**	(a)
8.	(b)	**9.**	(c)	**10.**	(c)	**11.**	(a)	**12.**	(−15)	**13.**	(a)	**14.**	(b)
15.	(c)	**16.**	(a)	**17.**	(c)	**18.**	(c)	**19.**	(a)	**20.**	(b)	**21.**	(b)
22.	(d)	**23.**	(b)	**24.**	(c)	**25.**	(d)	**26.**	(0.5)	**27.**	(1)	**28.**	(c)
29.	(c)	**30.**	(d)	**31.**	(b)	**32.**	(c)	**33.**	(c)	**34.**	(c)	**35.**	(d)
36.	(a)	**37.**	(c)	**38.**	(d)	**39.**	(a, d)	**40.**	(d)	**41.**	(a)	**42.**	(2)
43.	(b)	**44.**	(d)	**45.**	(c)	**46.**	(a)	**47.**	(a)	**48.**	(a)	**49.**	(0)
50.	(c)	**51.**	(c)	**52.**	(0.25)	**53.**	(c)	**54.**	(b)	**55.**	(a)	**56.**	(b)
57.	(c)	**58.**	(c)	**59.**	(a)	**60.**	(b)	**61.**	(d)	**62.**	(a, d)	**63.**	(a)
64.	(b)	**65.**	(a, c)	**66.**	(a, b)	**67.**	(5)	**68.**	(c)	**69.**	(a)	**70.**	(b, c)
71.	(b)	**72.**	(a)	**73.**	(d)	**74.**	(a)	**75.**	(c)	**76.**	(d)	**77.**	(0.23)
78.	(c)	**79.**	(5)	**80.**	(b)	**81.**	(b)	**82.**	(c)	**83.**	(c)	**84.**	(c)
85.	(b)	**86.**	(c)	**87.**	(6)	**88.**	(0)	**89.**	(d)	**90.**	(d)	**91.**	(c)
92.	(b)	**93.**	(c)	**94.**	(12)	**95.**	(−3.0)	**96.**	(c)	**97.**	(c)	**98.**	(a)
99.	(a, d)	**100.**	(a)	**101.**	(a)	**102.**	(d)	**103.**	(d)	**104.**	(b)	**105.**	(c)
106.	(a, b)	**107.**	(c)	**108.**	(b)	**109.**	(b)	**110.**	(a, c)	**111.**	(1)	**112.**	(7.28)
113.	(0.5)	**114.**	(b)	**115.**	(1)	**116.**	(b)	**117.**	(d)	**118.**	(b)	**119.**	(c)
120.	(c)	**121.**	(2)	**122.**	(a)	**123.**	(d)	**124.**	(b)	**125.**	(d)	**126.**	(a)
127.	(b)	**128.**	(b)	**129.**	(a)	**130.**	(c)	**131.**	(c)	**132.**	(4)	**133.**	(d)
134.	(b)	**135.**	(a)	**136.**	(d)	**137.**	(c)	**138.**	(a)	**139.**	(a)	**140.**	(c)
141.	(d)	**142.**	(a)	**143.**	(d)	**144.**	(d)	**145.**	(1)	**146.**	(d)	**147.**	(30)

Explanations Calculus and Optimization

1. (a)

$$\lim_{x\to\infty}\frac{x-\sin x}{x+\cos x} = \lim_{x\to\infty}\frac{1-\sin\frac{x}{x}}{1+\cos\frac{x}{x}}$$

$$= \frac{\lim\limits_{x\to\infty}\left(1-\sin\frac{x}{x}\right)}{\lim\limits_{x\to\infty}\left(1+\cos\frac{x}{x}\right)}$$

$$= \frac{1-\lim\limits_{x\to\infty}\frac{\sin x}{x}}{1+\lim\limits_{x\to\infty}\frac{\cos x}{x}} = \frac{1-0}{1+0} = 1$$

2. (b, c, d)

(a) $\lim\limits_{x\to2}\frac{\sin(x-2)}{3x-6}$ $\left\{\frac{0}{0}\text{ Limit case}\right\}$

$$= \lim_{x\to2}\frac{\sin(x-2)}{3(x-2)}$$

$$= \frac{1}{3} \to \text{True}$$

(b) $\lim\limits_{x\to2}\frac{\tan(x-2)}{(x-2)}$ $\left\{\frac{0}{0}\text{ Limit case}\right\}$

$$= 1 \to \text{False}$$

(c) $\lim\limits_{x\to\infty}\left(2+\frac{1}{x}\right)^x$ {No standard limit case}

$$= 2^\infty$$

= Infinity false

(d) $\lim\limits_{x\to0}\frac{\log_e(1+2x)}{2x}$ $\left\{\frac{0}{0}\text{ Limit case}\right\}$

$$= \lim_{x\to0}\left(\frac{\log_e(1+2x)}{2x}\right)$$

$$= 1 \to \text{False}$$

3. (c)

$$y = \lim_{x\to\infty} x^{1/x}$$

$$\log y = \lim_{x\to\infty}\log x^{1/x}$$

$$\log y = \lim_{x\to\infty}\frac{\log x}{x}$$

∞/∞ form, use L'Hopital's rule

$$\log y = \lim_{x\to\infty}\frac{1/x}{1}$$

$$\log y = 0$$

$$\Rightarrow \quad y = e^0 = 1$$

4. (c)

$$A = \cos\frac{\pi}{2^2}\cos\frac{\pi}{2^3}\ldots\cos\frac{\pi}{2^n}$$

$$B = \sin\frac{\pi}{2^2}\sin\frac{\pi}{2^3}\ldots\sin\frac{\pi}{2^n}$$

$$A\times B = \sin\frac{\pi}{2^2}\cos\frac{\pi}{2^2}\ldots\sin\frac{\pi}{2^n}\cos\frac{\pi}{2^n}$$

$$2^{n-1}AB = \left(2\sin\frac{\pi}{2^2}\cos\frac{\pi}{2^2}\right)\ldots\left(2\sin\frac{\pi}{2^n}\cos\frac{\pi}{2^n}\right)$$

$$= \sin\frac{\pi}{2^2}\sin\frac{\pi}{2^2}\ldots\sin\frac{\pi}{2^{n-1}}$$

$$2^{n-1}AB = (1)\sin\frac{\pi}{2^2}\sin\frac{\pi}{2^2}\ldots\sin\frac{\pi}{2^{n-1}} = \frac{B}{\sin\frac{\pi}{2^n}}$$

$$2^{n-1}AB = \frac{B}{\sin\frac{\pi}{2^n}} \Rightarrow A = \frac{1}{2^{n-1}\sin\frac{\pi}{2^n}}$$

$$\lim_{n\to\infty} A = \lim_{n\to\infty}\frac{\pi/2^n}{2^{n-1}\sin\frac{\pi}{2^n}}\frac{2^n}{\pi}$$

$$= \lim_{n\to\infty}\frac{\pi/2^n}{\sin\frac{\pi}{2^n}}\times\frac{2}{\pi}1\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$$

5. (b)

$$y = \lim_{x\to1}(1-x^2)^{[\log(1-x)^{-1}]}$$

$$\log y = \lim_{x\to1}[\log(1-x)]^{-1}\log(1-x^2)$$

$$= \lim_{x\to1}\frac{\log(1-x)^2}{\log(1-x)}$$

$$= \lim_{x\to1}\frac{\frac{1}{1-x^2}(-2x)}{\frac{1}{1-x}(-1)}$$

$$= \lim_{x\to1}\frac{1}{(1-x)(-1)}(-2x)\times\frac{1-x}{-1}$$

$$= \lim_{x\to 1}\frac{2x}{1+x} = \frac{2(1)}{2} = 1$$

$$\log y = 1$$

$$\Rightarrow \quad y = e^1 = e$$

6. (b)

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{2n} = \lim_{n\to\infty}\left[\left(1-\frac{1}{n}\right)^n\right]^2$$

$$= \left[\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n\right]^2$$

Now use the standared limit:

$$\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^x = e^a$$

Required limit $= (e^{-1})^2 = e^{-2}$

7. (a)

$$f(x) = \lim_{x\to 0}\left(\frac{7x^2-5x+3}{-2x+3}\right)^{\frac{1}{x}} \qquad [1^\infty \text{ limit case}]$$

$$= e^{\lim_{x\to 0}\left(\frac{7x^2-5x+3+2x-3}{-2x+3}\right)\frac{1}{x}}$$

$$= e^{\lim_{x\to 0}\left(\frac{7x^2-3x}{-2x+3}\right)\frac{1}{x}}$$

$$= e^{\lim_{x\to 0}\left(\frac{7x-3}{-2x+3}\right)} \qquad \text{Put } x = 0$$

$$= e^{-1}$$

8. (b)

For the function $\frac{25}{8x}$ the minimum value will come when x is maximum since it is a decreasing function.

The maximum value of x is $\frac{3}{2}$.

At $x = \frac{3}{2}$ the function has the value

$$f(x) = \frac{25}{\left(8\times\frac{3}{2}\right)} = 2.0833 = 2 + \left(\frac{1}{12}\right)$$

So, $\infty \geq f(x) \geq 2\left(\frac{1}{12}\right)$ for $x \leq \frac{3}{2}$

$$x + \frac{1}{x} \geq 2$$

But since for this function $x \geq \frac{3}{2}$, putting $\frac{3}{2}$ in this function we get the minimum of this function which is

$$\frac{3}{2} + \frac{1}{\left(\frac{3}{2}\right)} = \frac{13}{6} = 2.166$$

Now comparing the minimum value 2.0833 of the first function with minimum value 2.166 of the second function, we get the overall minimum of this function to be $2.0833 = 2\left(\frac{1}{12}\right)$ which is option (b).

9. (c)

$$L = \lim_{x\to 0}\frac{x^2-4x}{5x} \qquad \left\{\frac{0}{0} \text{ Limit case}\right\}$$

$$L = \lim_{x\to 0}\frac{x(x-4)}{5x}$$

$$= \lim_{x\to 0}\frac{x-4)}{5} \qquad \text{(Put } x = 0\text{)}$$

$$L = -\frac{4}{5}$$

10. (c)

Let, $\quad y = \lim_{x\to\infty}(1+x^2)^{e^{-x}}$

$$\log y = \lim_{x\to\infty}\log(1+x^2)^{e^{-x}}$$

$$= \lim_{x\to\infty}\left[\frac{\log(1+x^2)}{e^x}\right]$$

∞/∞ form apply L' Hopital's rule

$$\Rightarrow \quad \log y = \lim_{x\to\infty}\frac{\frac{1}{1+x^2}(2x)}{e^x}$$

$$\Rightarrow \quad \log y = \lim_{x\to\infty}\frac{2x}{(1+x^2)e^x}$$

Again we are getting ∞/∞ form apply L' Hopital's rule

$$\log y = \lim_{x\to\infty}\frac{2}{(1+x^2)e^x + e^x\cdot 2x}$$

$$\log y = \frac{2}{\infty} = 0$$

$$\Rightarrow \quad y = 1$$

11. **(a)**

$|x|$ is continuous and differentiable every where except at $x = 0$, where it is continuous but not differentiable.

Since 0 is a real number, for all real values it is continuous but not differentiable.

12. **(-15)**

$$f(x) = \lim_{x\to 0}\left(e^{\frac{-1}{x}} - 2\right)\left(7 + \frac{\sin\sqrt{x}}{2\sqrt{x}}\right)$$

$$= \left(e^{\frac{1}{0}} - 2\right)\left(7 + \frac{1}{2}\left(\frac{\sin\sqrt{x}}{\sqrt{x}}\right)\right) \quad (\text{Put } x = 0)$$

$$\left\{\lim_{y\to 0}\frac{\sin y}{y} = 1\right\}$$

$$= (0-2)\left(7+\frac{1}{2}\right)$$

$$= (-2)\left(\frac{15}{2}\right) = -15$$

13. **(a)**

$$\lim_{\theta\to 0}\frac{\frac{1}{2}\times\sin\left(\frac{\theta}{2}\right)}{\theta\times\frac{1}{2}} = \frac{1}{2}\lim_{\theta\to 0}\frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}} = \frac{1}{2} = 0.5$$

14. **(b)**

$$f(x) = \lim_{x\to 0}\left[\frac{x^3+x^2}{2x^3-7x^2}\right]$$

Since this has $\frac{0}{0}$ form, limit can be found by repeated application of L' Hopitals rule.

$$f(x) = \lim_{x\to 0}\left[\frac{3x^2+2x}{6x^2-14x}\right]$$

$$= \lim_{x\to 0}\left[\frac{6x+2}{12x-14}\right]$$

$$= \left[\frac{6\times 0+2}{12\times 0-14}\right] = -\frac{1}{7}$$

15. **(c)**

It is differentiable once.

For checking 2nd order differentiation

$$\text{R.H.D.} = \lim_{h\to 0}\frac{f'(0+h)-f'(0)}{h}$$

$$= \lim_{h\to 0}\frac{3h^2}{h} = 0$$

$$\text{L.H.D.} = \lim_{h\to 0}\frac{f'(0-h)-f'(0)}{-h}$$

$$= \lim_{h\to 0}\frac{3h^2}{-h} = 0$$

So, second order differentiation exist and

$$\frac{d^2f}{dx^2} = 6|x| \text{ at } x = 0$$

for checking 3rd order differentiation

$$\text{R.H.D.} = \lim_{h\to 0}\frac{f''(0+h)-f''(0)}{h}$$

$$= \lim_{h\to 0}\frac{6h}{h} = 6$$

$$\text{L.H.D.} = \lim_{h\to 0}\frac{f''(0-h)-f''(0)}{-h}$$

$$= \lim_{h\to 0}\frac{6h}{-h} = -6$$

$\because$ R.H.D. $\neq$ L.H.D., so not differentiable thrice.

16. **(a)**

$$\lim_{x\to 0}\frac{\sin\left(\frac{2}{3}x\right)}{x} = \lim_{\frac{2}{3}x\to 0}\frac{\sin\left(\frac{2}{3}x\right)}{\frac{2}{3}x}\cdot\frac{2}{3}$$

$$= (1)\left(\frac{2}{3}\right) = \frac{2}{3}$$

17. **(c)**

$$f(x) = \begin{cases} 2x+1, & x\le 2 \\ px^2+q, & 2<x<4 \\ 5x+2p, & x\ge 4 \end{cases}$$

At : $x = 2$

LHS : $2(2) + 1 = 5$

RHS: $p(2)^2 + q = 4p + q$

$$\text{LHS} = \text{RHS} = 5 = 4p + q \quad \ldots(i)$$

At : $x = 4$

LHS : $p(4)^2 + q$

RHS: $5(4) + 2p$

$$\text{LHS} = \text{RHS}$$

$$\Rightarrow \quad 16p + q = 20 + 2p$$

$$\Rightarrow \quad 14p + q = 20 \quad \ldots(ii)$$

From equation (i) and (ii)

$$p = \frac{3}{2},\ q = -1$$

18. (c)

$$y = |2-3x| = 2-3x \quad 2-3x \geq 0$$
$$= 3x-2 \quad 2-3x < 0$$

Therefore, $y = 2-3x \quad x \leq \frac{2}{3}$

$$= 3x-2 \quad x > \frac{2}{3}$$

Since $2-3x$ and $3x-2$ are polynomials, these are continuous at all points.

The only concern is at $x = \frac{2}{3}$

Left limit at $x = \frac{2}{3}$ is $2-3\times\frac{2}{3} = 0$.

Right limit at $x = \frac{2}{3}$ is $3\times\frac{2}{3} - 2 = 0$.

$$f\left(\frac{2}{3}\right) = 2-3\times\frac{2}{3} = 0$$

Since, Left limit = Right limit = $f\left(\frac{2}{3}\right)$,

Function is continuous at $\frac{2}{3}$.

y is therefore continuous $\forall x \in R$

Now since $2-3x$ and $3x-2$ are polynomials, they are differentiable.

Only concern is at $x = \frac{2}{3}$.

Now, at $x = \frac{2}{3}$, LD = Left derivative $= -3$

RD = Right derivative $= +3$

LD $\neq$ RD

$\therefore$ The function y is not differentiable at $x = \frac{2}{3}$

So, we can say that y is differentiable $\forall x \in R$, except at $x = \frac{2}{3}$.

19. (a)

$$\lim_{x\to 0}\frac{\sin^2 x}{x} = \lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2 x = 1\times 0 = 0$$

20. (b)

$$\lim_{x\to 3} f(x) = \text{Lt}_{x\to 3}\left(\frac{2x^2-7x+3}{5x^2-12x-9}\right)$$

Here this is of the form of $\left(\frac{0}{0}\right)$

So, applying L-Hopital's rule

$$\lim_{x\to 3}\left(\frac{4x-7}{10x-12}\right) = \frac{5}{18}$$

21. (b)

$$\lim_{x\to 0}\frac{e^x-\left(1+x+\frac{x^2}{2}\right)}{x^3}$$

This is of the form of $\left(\frac{0}{0}\right)$

Applying L'Hopital rule

$$\lim_{x\to 0}\frac{e^x-\left(1+x+\frac{x^2}{2}\right)}{x^3}\left(\frac{0}{0}\right)$$

$$\lim_{x\to 0}\frac{e^x-1}{6x}$$

$$\lim_{x\to 0}\frac{e^x}{6} = \frac{1}{6}$$

22. (d)

$$\lim_{x\to 0}\frac{\left(\sqrt{x+2}-\sqrt{2-x}\right)}{\left(\sqrt{x+3}-\sqrt{3-x}\right)}$$

At $x = 0$, $\frac{0}{0}$ limit case

Using L'Hopital rule

$$= \lim_{x\to 0}\left(\frac{\frac{1}{\sqrt{x+2}}-\frac{(-1)}{\sqrt{2-x}}}{\frac{1}{\sqrt{x+3}}-\frac{(-1)}{\sqrt{3-x}}}\right) \text{ (Put } x = 0)$$

$$= \left(\frac{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}}}\right) = \sqrt{\frac{3}{2}}$$

23. (b)

P: If $f(x)$ is continuous at $x = x_0$, then it is also differentiable at $x = x_0$

Q: If $f(x)$ is continuous at $x = x_0$, then it may or may not be derivable at $x = x_0$

R: If $f(x)$ is differentiable at $x = x_0$, then it is also continuous at $x = x_0$

P is false

Q is true

R is true

Option (b) is correct.

24. (c)

$$L = \lim_{x\to 0}\frac{a^2\sin x + b\log(\cos x)}{x^2} = -1$$

$$L = \lim_{x\to 0}\frac{a^2\cos x + \dfrac{b(-\sin x)}{\cos x}}{2x} = -1$$

[Using L'Hopital's rule]

$$L = \lim_{x\to 0}\frac{a^2 + b(-\sin x)}{2x} = -1$$

$\therefore \quad a^2 = 0$

$a = 0$, to make $\frac{0}{0}$, limit case

$$L = \lim_{x\to 0}\frac{b(-\sin x)}{2(x)} = -1 \quad \left\{\lim_{x\to 0}\frac{\sin x}{x} = 1\right\}$$

$$= \lim_{x\to 0} -\frac{b}{2}\left(\frac{\sin x}{x}\right)^{1} = -1$$

$$= \frac{-b}{2} = -1$$

$b = 2$

25. (d)

$$\lim_{x\to 0}\frac{x^3 - \sin x}{x} \quad \left(\frac{0}{0}\text{form}\right)$$

$$\lim_{x\to 0}\frac{3x^2 - \cos x}{1} = 0 - \cos 0$$

$$= 0 - 1 = -1$$

26. (0.5)

$$\lim_{n\to\infty}\sqrt{n^2+n} - \sqrt{n^2+1}$$

$$= \lim_{n\to\infty}\frac{\sqrt{n^2+n} - \sqrt{n^2+1}}{\left(\sqrt{n^2+n} + \sqrt{n^2+1}\right)}\left(\sqrt{n^2+n} + \sqrt{n^2+1}\right)$$

$$= \lim_{n\to\infty}\frac{n^2+n-n^2-1}{\sqrt{n^2+n} + \sqrt{n^2+1}}$$

$$= \lim_{n\to\infty}\frac{n-1}{n\sqrt{1+\frac{1}{n}} + n\sqrt{1+\frac{1}{n^2}}}$$

$$= \lim_{n\to\infty}\frac{n\left(1-\frac{1}{n}\right)}{n\left(\sqrt{1+\frac{1}{n}} + \sqrt{1+\frac{1}{n^2}}\right)} = \frac{1}{1+1} = \frac{1}{2}$$

27. (1)

$$\lim_{x\to 4}\frac{\sin(x-4)}{x-4}$$

Let $x - 4 = t$ not as $x \to 4$

So the requires limit is $\lim_{t\to 0}\frac{\sin(t)}{t} = 1$

28. (c)

$$\lim_{x\to 0}\frac{\ln(1+4x)}{e^{3x}-1} \quad \frac{0}{0}\text{form}$$

$$\lim_{x\to 0}\frac{\frac{1}{1+4x}\cdot 4}{3e^{3x}} = \frac{4}{3}$$

29. (c)

$$y = \lim_{x\to\infty} x^{1/x}$$

$$\log y = \lim_{x\to\infty}\log x^{1/x}$$

$$\log y = \lim_{x\to\infty}\frac{\log x}{x}$$

∞/∞ form, use L'Hopital's rule

$$\log y = \lim_{x\to\infty}\frac{\frac{1}{x}}{1} = 0$$

$$\Rightarrow \quad y = 1$$

30. (d)

$f(x)$ is continuous at any point

If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$

31. (b)

$$\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right) = \frac{1-\cos 0}{0^2} = \frac{0}{0}$$

So use L'Hopital rule

$$\lim_{x\to 0}\frac{(1-\cos x)}{x^2} = \lim_{x\to 0}\frac{\sin x}{2x} = \frac{0}{0}$$

So use L'Hopital rule again

$$= \lim_{x\to 0}\left[\frac{\cos x}{2}\right] = \frac{1}{2}$$

32. (c)

(a) $\lim_{x\to 0}\frac{\log_a(1+x)}{x}$

$$= \lim_{x\to 0}\frac{1}{\log_e a}\left(\frac{\log_e(1+x)}{x}\right)^{1} \quad \left\{\frac{0}{0}\text{ Limit case}\right\}$$

$$= \frac{1}{\log_e a} \Rightarrow \log_a e \rightarrow \text{False}$$

(b) $\lim_{x\to 0} \frac{a^x + 1}{x}$

$= \log_e a \rightarrow$ False

(c) $\lim_{x\to 0} \frac{e^x - 1}{x}$ $\left\{\frac{0}{0} \text{ Limit case}\right\}$

$= 1 \rightarrow$ True

(d) $\lim_{x\to 0} \frac{\sin^{-1} x}{x^2}$ $\left\{\frac{0}{0} \text{ Limit case}\right\}$

$$= 1 \cdot \left(\frac{\sin^{-1}(x)}{x}\right)^1 \frac{1}{x}$$

$$= \frac{1}{x} \text{ Put } x = 0$$

$$= \frac{1}{0}$$

$=$ Undefined $\rightarrow$ False

33. (c)

$$\lim_{x\to\infty} \sqrt{x^2 + x - 1} - x$$

$$\Rightarrow \lim_{x\to\infty} \frac{\left(\sqrt{x^2 + x - 1} - x\right)\left(\sqrt{x^2 + x - 1} + x\right)}{\sqrt{x^2 + x - 1} + x}$$

$$\Rightarrow \lim_{x\to\infty} \frac{x^2 + x - 1 - x^2}{\sqrt{x^2 + x + 1} + x}$$

$$\Rightarrow \lim_{x\to\infty} \frac{x - 1}{x\sqrt{1 + \frac{1}{x} + \frac{1}{x^2}} + x}$$

$$\Rightarrow \lim_{x\to\infty} \frac{1 - \frac{1}{x}}{\sqrt{1 + \frac{1}{x} + \frac{1}{x^2}} + 1} = \frac{1}{1+1} = \frac{1}{2}$$

34. (c)

$$\lim_{x\to 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2} = \frac{1 - 2 + 1}{1 - 3 + 2} = \frac{0}{0} \text{ form}$$

So, use L'Hopitals rule

$$= \lim_{x\to 1} \frac{7x^6 - 10x^4}{3x^2 - 6x}$$

$$= \frac{7 - 10}{3 - 6} = \frac{-3}{-3} = 1$$

35. (d)

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 2x & x \geq 0 \\ -2x & x < 0 \end{cases} \quad f''(x) = \begin{cases} 2 & x \geq 0 \\ -2 & x < 0 \end{cases}$$

The first derivation of f (i.e.) $f'(x)$ is not derivable at $x = 0$.

36. (a)

$$\lim_{x\to 3} \frac{x^4 - 81}{2x^2 - 5x - 3} = \frac{0}{0} \text{ form}$$

So apply L'Hopitals rule

$$\lim_{x\to 3} \frac{4x^3}{4x - 5} = \frac{108}{7}$$

37. (c)

(a) $\lim_{x\to 0} \frac{\sin x}{x}$ $\left\{\frac{0}{0} \text{ Limit case}\right\}$

$\lim_{x\to 0} \frac{\cos x}{1}$ Put $x = 0$

$= 1 \rightarrow$ False

(b) $\lim_{x\to\infty} \frac{\frac{\pi}{2} - \tan\left(\frac{1}{x}\right)}{\frac{1}{x}}$

$$= \lim_{x\to 0} \frac{\pi}{2}(x) - \frac{\tan\left(\frac{1}{x}\right)^1}{\left(\frac{1}{x}\right)}$$

$= \lim_{x\to\infty} \frac{\pi}{2}(x) - 1$ (Put $x = \infty$)

$= \infty \rightarrow$ False

(c) $\lim_{x\to 0} \frac{\tan x}{x}$ $\left\{\frac{0}{0} \text{ Limit case}\right\}$

$\lim_{x\to 0} \frac{\sec^2 x}{1}$ Put $x = 0$

$= 1 \rightarrow$ True

(d) $\lim_{x\to 0} \frac{\cos x}{x}$ Put $x = 0$

$$= \frac{1}{0}$$

$=$ Undefined $\rightarrow$ False

38. (d)

$$f(x) = \begin{cases} \dfrac{-x}{2x}, & x \geq 0 \\ \dfrac{-(-x)}{2x}, & x < 0 \end{cases}$$

Checking $f(x)$ continuity at $x = 0$

$$f(0) = -\frac{1}{2} = \text{RHS} \lim_{x \to 0^+}$$

LHS limit: $\lim_{x \to 0^-} f(x) = \frac{1}{2}$

Limit does not exist, hence not continuous.

39. (a, d)

$$f(x) = \begin{cases} -3x & x < -1 \\ 3 & -1 \leq x \leq 1 \\ 3x & x > 1 \end{cases}$$

→ Checking continuity at $x = -1$

$$f(-1) = 3 = \text{RHS} \lim_{x \to -1^+}$$

LHS limit: $\lim_{x \to -1^-} f(x) = -3x$

$$\lim_{h \to 0} f(1-h) = -3(-1-h) \text{ [Put } h = 0\text{]}$$

$$= 3$$

LHS limit: $f(-1)$ = RHS limit, hence $f(x)$ continuous at $x = -1$.

→ Checking continuity at $x = 1$

$$f(1) = 3 = \text{LHS} \lim_{x \to 1^-}$$

RHS limit: $\lim_{x \to 1^+} f(x) = 3x$

$$\lim_{h \to 0} f(1+0) = 3(1+0) \text{ [Put } h = 0\text{]}$$

$$= 3$$

LHS limit: $f(1)$ = RHS limit, hence $f(x)$ continuous at $x = 1$.

Checking differentiability

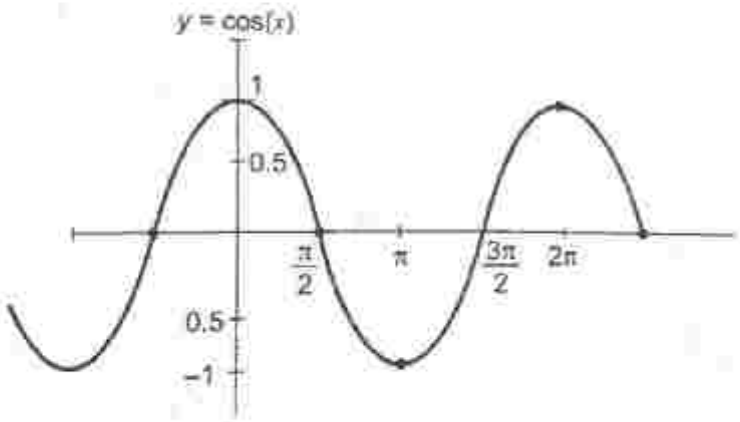
$$f'(x) = \begin{cases} -3 & x < -1 \\ 0 & -1 \leq x \leq 1 \\ 3 & x > 1 \end{cases}$$

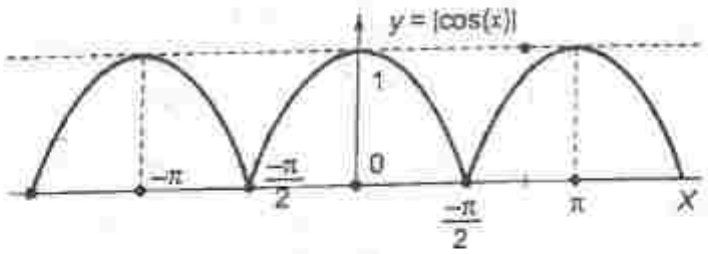
$$f'(-1) \neq \lim_{x \to -1^-} f(x) \text{ and } f'(-1) \neq \lim_{x \to 1^+} f(x)$$

Hence $f(x)$ is not differentiable at $x = 1, -1$.

40. (d)

Below are the graphs for $f(x) = \cos x$ and $f(x) = |\cos x|$. Clearly, both are continuous everywhere.





If we do it manually,

$$f(x) = |\cos x|$$

$f(x) = \cos x$ when $\cos x > 0$ i.e., when x is in between $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$f(x) = \cos x$ when $\cos x < 0$ i.e., when x is in between $\left[\frac{\pi}{2}, -\frac{\pi}{2}\right]$.

Now, do (a), (b), (c)

We will see all are continuous.

Answer is (d) because of the word "ONLY".

It is continuous everywhere.

41. (a)

$$\lim_{x \to \infty} \left(\frac{x^2+2x+3}{x^2-1x+1}\right)^{x/2} \quad [1^\infty, \text{ limit case}]$$

$$= \lim_{x \to \infty} \left(\frac{1+\frac{2}{x}+\frac{3}{x^2}}{1-\frac{1}{x}+\frac{1}{x^2}}\right)^{x/2}$$

$$= e^{\lim_{x \to \infty}} \left(\frac{1+\frac{2}{x}+\frac{3}{x^2}}{1-\frac{1}{x}+\frac{1}{x^2}} - 1\right) \times \frac{x}{2}$$

$$= e^{\lim_{x \to \infty}} \left(\frac{\frac{3}{x}+\frac{1}{x^2}}{1-\frac{1}{x}+\frac{1}{x^2}}\right) \times \frac{x}{2}$$

$$= e^{\lim_{x\to\infty}}\left(\frac{3+\frac{2}{x}}{1-\frac{1}{x}+\frac{1}{x^2}}\right)\times\frac{1}{2} \quad [\text{Put } x=0]$$

$$= e^{\lim_{x\to\infty}}\left(\frac{3}{2}\right) = e^{3/2}$$

42. (2)

$$f(x) = \lim_{x\to 4}\frac{(x^{3/2}-k^3)}{x-4} = 3$$

At $x = 4$, to form $\frac{0}{0}$, limit case

$$x^{3/2} - k^3 = 0$$
$$x^{3/2} = k^3, \text{ at } x = 4$$
$$4^{3/2} = k^3$$
$$2^{2\times 3/2} = k^3$$
$$2^3 = k^3$$
$$k = 2$$

43. (b)

Left limit $= \lim_{h\to 0} f(0-h)$

Since $[0-h] \neq 0$

$$f(0-h) = \frac{\sin[0-h]}{[0-h]}$$

$$\text{Left limit} = \lim_{h\to 0}\frac{\sin[0-h]}{[0-h]} = \frac{\sin(-1)}{-1} = \sin(1)$$

Right limit $= \lim_{h\to 0} f(0+h)$

Since $[0+h] = 0$

$f(0+h) = 0$ (by definition given)

So right limit $= \lim_{h\to 0} 0 = 0$

Since left limit ≠ right limit

The limit at $x = 0$ doesn't exist.

44. (d)

The $f(x)$ is neither differentiable at $x = 1$ nor at $x = 2$ because $f(x)$ is pointed at both $x = 1$ and $x = 2$, through it is continuous in $(-\infty, +\infty)$.

So, option (d) is correct.

45. (c)

Condition for continuity is LHL = RHL = $f(0)$

Here, $f(0) = k$

We have to find limit of this at $x = 0$

$$\frac{\log(1+ax)-\log(1-bx)}{x}$$

So, if we put $x = 0$, we will get indeterminate form of $\frac{0}{0}$ L-Hopital rule.

After differentiating $= \frac{a}{1+ax} - \frac{-b}{1-bx}$

Now, put again $x = 0$, we get $a-(-b) = a+b$

$$\text{LHL} = \text{RHL} = f(0)$$
$$a+b = k$$

So, option (c) is correct.

46. (a)

$$\lim_{x\to 0}\frac{e^x(\cos x)\sin^{1/2}x}{\sqrt{x}}$$

$$\lim_{x\to 0} e^x \cos x\left(\frac{\sin x}{x}\right)^{1/2} \quad \left\{\lim_{x\to 0}\left(\frac{\sin x}{x}\right)=1\right\}$$

$$\lim_{x\to 0} e^x \cos x \quad [\text{Put } x = 0]$$

$$= 1$$

47. (a)

$$\begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$$

$$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-}\frac{x+3}{3} = \frac{3+3}{3} = 2$$

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} x-1 = 3-1 = 2$$

Also, $f(3) = 2$

So, $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = f(3)$

So it is continuous at $x = 3$.

48. (a)

$$\underset{x\to\infty}{Lt}\left(\frac{x+6}{x+1}\right)^{x+4} = \underset{x\to\infty}{Lt}\left\{\left[1+\frac{5}{x+1}\right]^{\frac{x+1}{5}}\right\}^{5\left(\frac{x+4}{x+1}\right)}$$

[Using $\underset{x\to\infty}{Lt}\left(1+\frac{1}{x}\right)^x = e$]

$$\underset{x\to\infty}{Lt}\, e^{5\left(\frac{x+4}{x+1}\right)} = e^{5\lim_{x\to\infty}\left(\frac{1+4/x}{1+1/x}\right)} = e^5$$

49. **(0)**

$$\lim_{x\to 0}\frac{x(e^{2x}-1)+5(\cos x^2-1)}{x(2-\cos x)} \quad \left\{\frac{0}{0}, \text{ limit case}\right\}$$

Using L'Hopital's rule

$$\lim_{x\to 0}\frac{x2e^{2x}+(e^{2x}-1)+5x(-\sin x^2)(2x)}{x\sin x+(2-\cos x)}$$

$$= \frac{0}{1} \quad (\text{Put } x=0)$$

$$= 0$$

50. **(c)**

Intermediate value theorem states that if a function is continuous and $f(a)\cdot f(b)<0$, then surely there is a root in (a, b). The contrapositive of this theorem is that if a function is continuous and has no root in (a, b) then surely $f(a)\cdot f(b)\geq 0$. But since it is given that there is no root in the closed interval $[a, b]$ it means $f(a)\cdot f(b)\neq 0$.

So surely $f(a)\cdot f(b)>0$.

51. **(c)**

$$\lim_{x\to\infty}\left(\frac{1}{1-x^2}+\frac{2}{1-x^2}+\ldots+\frac{x}{1-x^2}\right)$$

$$\Rightarrow \lim_{x\to\infty}\frac{1}{x^2}(1+2+3+\ldots+x)$$

$$\Rightarrow \lim_{x\to\infty}\frac{1}{1-x^2}\cdot\frac{x(x+1)}{2}$$

$$\Rightarrow \lim_{x\to\infty}\frac{1}{(1-x)(1+x)}\cdot\frac{x(x+1)}{2}$$

$$\Rightarrow \frac{1}{2}\times\lim_{x\to\infty}\frac{1}{\left(\frac{1}{x}-1\right)}$$

Put, $x=\infty$

$$\Rightarrow -\frac{1}{2}$$

52. **(0.25)**

$$f(x) = \lim_{x\to 0}\frac{4x^4-x+5}{x^3-5x^2+20}$$

$\therefore$ Put $x = 0$

$$= \frac{4(0)-0+5}{0-5(0)+20}$$

$$= \frac{5}{20}=0.25$$

53. **(c)**

$$f(\theta)=\begin{vmatrix}\sin\theta & \cos\theta & \tan\theta\\ \sin\left(\frac{\pi}{6}\right) & \cos\left(\frac{\pi}{6}\right) & \tan\left(\frac{\pi}{6}\right)\\ \sin\left(\frac{\pi}{3}\right) & \cos\left(\frac{\pi}{3}\right) & \tan\left(\frac{\pi}{3}\right)\end{vmatrix}$$

$$f\left(\frac{\pi}{6}\right)=0$$

Since if we put $\theta=\frac{\pi}{6}$ in above determinant it will evaluate to zero, since I and II row will become same.

$$f\left(\frac{\pi}{3}\right)=0$$

Since if we put $\theta=\frac{\pi}{3}$ in above determinant it will evaluate to zero, since I and III row will become same.

So $f\left(\frac{\pi}{6}\right)=f\left(\frac{\pi}{3}\right)$ Also in the interval $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ the function $f(\theta)$ is continuous and differentiable (Note that the given interval doesn't contain any odd multiple of $\frac{\pi}{2}$ where $\tan\theta$ is neither continuous nor differentiable).

Since all the three conditions of **Roll's theorem** are satisfied the conclusion of Rolls theorem is true i.e.

I. $\exists\theta\in\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ such that $f'(\theta)=0$ is true

Now the statement

II. $\exists\theta\in\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ such that $f'(\theta)\neq 0$

is also true, since the only way it can be false is if $f'(\theta)=0$ for all values of θ, which is possible only if $f(\theta)$ is a constant which is untrue.

Therefore, both (I) and (II) are correct.

54. **(b)**

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-}(x-1)=0$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+}(x^3-1)=0$$

Also $f(1) = 0$

Thus $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)=f(1)$

$\Rightarrow$ f is continuous at $x = 1$

And $Lf'(1) = 2$, $Rf'(1) = 1$

$\Rightarrow$ f is not differentiable at $x = 1$

55. (a)

$$f(x) = \lim_{x\to\infty} e^{2x}\sin\left(\frac{2d}{e^x}\right)^2$$

$$= \lim_{x\to\infty} e^{2x}\sin\left(\frac{4d^2}{e^{2x}}\right)$$

$$= \lim_{x\to\infty} e^{2x}\frac{\sin\left(\frac{4d^2}{e^{2x}}\right)}{\left(\frac{4d^2}{e^{2x}}\right)}\times\left(\frac{4d^2}{e^{2x}}\right) \quad \left\{\lim_{x\to 0}\frac{\sin x}{x} = 1\right\}$$

$$= \lim_{x\to\infty} e^{2x}\times\frac{4d^2}{e^{2x}} = 4d^2$$

56. (b)

$$f(x) = \begin{cases} x^4 & x^2 < 1 \quad \text{or} \quad -1 < x < 1 \\ x & x^2 \geq 1 \quad \text{or} \quad x \geq 1 \text{ or } x \leq -1 \end{cases}$$

$$f(-1) = x|_{x=-1} = -1$$

$$f(-1^-) = x|_{x=-1} = -1$$

$$f(-1^+) = x^4|_{x=-1} = 1$$

At $x = -1$

Limit does not exists $\Rightarrow$ $f(x)$ is not continuous

$\Rightarrow$ $f'(-1)$ does not exist.

57. (c)

$$f(x) = \lim_{x\to 0}\left(\frac{2a^{\frac{kx}{2}} - 2}{3x}\right) \quad \left\{\frac{0}{0} \text{ Limit case}\right\}$$

Using L'Hopital rule

$$= \lim_{x\to 0}\frac{2\left(a^{\frac{kx}{2}}\frac{k\log a}{2}\right)}{3} \quad \text{Put } x = 0$$

$$= \frac{2}{3}\frac{k\log a}{2} = \frac{k\log_a a}{3}$$

58. (c)

$$\lim_{x\to 1}\frac{x^x - x}{x - 1 - \log x} \quad \left(\frac{0}{0}\text{ form}\right)$$

$$= \lim_{x\to 1}\frac{\frac{d}{dx}(x^x) - 1}{1 - 0 - \frac{1}{x}}$$

Let, $y = x^x$

$\log y = x\log x$

$\therefore$ $\frac{1}{y}\frac{dy}{dx} = x\cdot\frac{1}{x} + 1\cdot\log x$

or $\frac{d}{dx}(x^x) = x^x(1 + \log x)$

$$= \lim_{x\to 1}\frac{x^x(1+\log x) - 1}{1 - \frac{1}{x}} \quad \left(\frac{0}{0}\text{ form}\right)$$

$$= \lim_{x\to 1}\frac{\frac{d}{dx}(x^x)\cdot(1+\log x) + x^x\left(\frac{1}{x}\right) - 0}{\frac{1}{x^2}}$$

$$= \lim_{x\to 1}\frac{x^x(1+\log x)^2 + x^x\left(\frac{1}{x}\right)}{x^{-2}}$$

$$= \frac{1(1+0)^2 + 1\cdot 1}{1} = 2$$

59. (a)

Given,

$$\lim_{x\to\frac{\pi}{4}}\frac{\sec^2 x - 2\tan x}{1 + \cos 4x} : \frac{0}{0} \text{ Form}$$

Applying L' hopital rule

$$= \lim_{x\to\frac{\pi}{4}}\frac{\frac{d}{dx}(\sec^2 x - 2\tan x)}{\frac{d}{dx}(1 + \cos 4x)}$$

$$= \lim_{x\to\frac{\pi}{4}}\frac{(2\sec x\cdot\sec x\tan x - 2\sec^2 x)}{-4\sin 4x}$$

$$= \lim_{x\to\frac{\pi}{4}}\frac{\sec^2 x(\tan x - 1)}{-2\sin 4x} : \frac{0}{0} \text{ Form}$$

Applying L' hopital's rule

$$= \lim_{x\to\frac{\pi}{4}}\frac{2\sec x\cdot\sec x\cdot\tan x(\tan x - 1) + \sec^2 x\sec^2 x}{-8\cos 4x}$$

$$= \frac{2.2.1(1-1) + 2.2}{-8(-1)} = \frac{1}{2}$$

60. (b)

$$f(x) = \lim_{x\to\infty}\sqrt{\frac{x-\sin\left(\frac{x}{2}\right)}{\frac{x}{3}+\cos^2 x}}$$

$$= \lim_{x\to\infty}\sqrt{\frac{1-\frac{\sin\left(\frac{x}{2}\right)}{x}}{\frac{1}{3}+\frac{\cos^2 x}{x}}} \quad \text{Put } x=\infty$$

$$= \sqrt{\frac{1-\frac{\sin\left(\frac{x}{2}\right)}{\infty}\to 0}{\frac{1}{3}+\frac{\cos^2 x}{\infty}\to 0}}$$

$$= \sqrt{3}$$

61. (d)

$$f(y) = \begin{cases}\frac{\sin y}{y} & \text{if } y\neq 0\\ 1 & \text{if } y=0\end{cases}$$

Checking continuity:

$$f(0) = 1$$

LHS limit:

$$\lim_{h\to 0} f(0-0) = \frac{\sin(0)}{0} = 1 \quad \left\{\lim_{x\to 0}\frac{\sin(x)}{x}=1\right\}$$

RHS limit:

$$\lim_{h\to 0} f(0+0) = \frac{\sin(0)}{0} = 1 \quad \left\{\lim_{x\to 0}\frac{\sin(x)}{x}=1\right\}$$

Hence continuous at $y=0$

Checking differentiability:

$$f'(y) = \begin{cases}\frac{y\cos(y)-\sin(y)}{y^2} & y\neq 0\\ 0 & y=0\end{cases}$$

$$f'(y) = \lim_{y\to 0}\frac{y\cos y-\sin y}{y^2}, \quad \left\{\text{limit } \frac{0}{0} \text{ case}\right\}$$

$$= \lim_{y\to 0}\frac{y(-\sin y)+\cos y-\cos y}{2y}$$

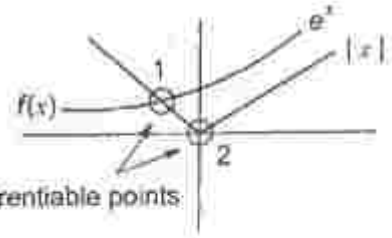
[L'Hopital's rule]

$$= \lim_{y\to 0}\frac{-\sin y}{2}$$

$= 0$, hence $f(y)$ differential at $y=0$

62. (a, d)

$$f(x) = \min\left(e^x, |x|\right)$$



Hence $f(x)$ is continuous everywhere and not differential at two points.

63. (a)

$$\lim_{x\to 0}\frac{2^{-\sin x}-1}{x\left(x-\frac{\pi}{2}\right)} \quad \left\{\frac{0}{0} \text{ Limit case}\right\}$$

Using L-Hopital's rule

$$= \lim_{x\to 0}\frac{2^{-\sin x}(-\cos x)\ln 2}{2x-\frac{\pi}{2}} \quad [\text{Put } x=0]$$

$$= \frac{2^0(-1)\ln 2}{-\frac{\pi}{2}} \Rightarrow \frac{2\ln 2}{\pi}$$

64. (b)

$$f(x) = x+\ln x$$

$$f'(x) = 1+\frac{1}{x} \qquad \ldots(1)$$

$$f'(c) = \frac{f(b)-f(a)}{b-a} = \frac{f(e)-f(1)}{e-1}$$

$$= \frac{e+\ln e-(1+\ln 1)}{e-1} = \frac{e+1-1+0}{e-1} = \frac{e}{e-1}$$

$$\Rightarrow \quad f'(c) = \frac{e}{e-1} \qquad [\text{from equation (1)}]$$

$$\Rightarrow \quad 1+\frac{1}{c} = \frac{e}{e-1}$$

$$\Rightarrow \quad \frac{1}{c} = \frac{e}{e-1}-1 = \frac{e-e+1}{e-1}$$

$$\Rightarrow \quad c = e-1$$

65. (a, c)

$$f(x) = x^3-6x^2+x$$

$$f'(c) = 0$$

by Rolle's theorem, at $x=c\in[0,4]$

$$3c^2-12c+1 = 0$$

$$c = \frac{12\pm 2\sqrt{33}}{6} = \frac{6\pm\sqrt{33}}{3} = 0.085, 3.91$$

66. (a, b)

$$f(x) = x^4 - 2x^2 + 4x + 16$$

Critical points,

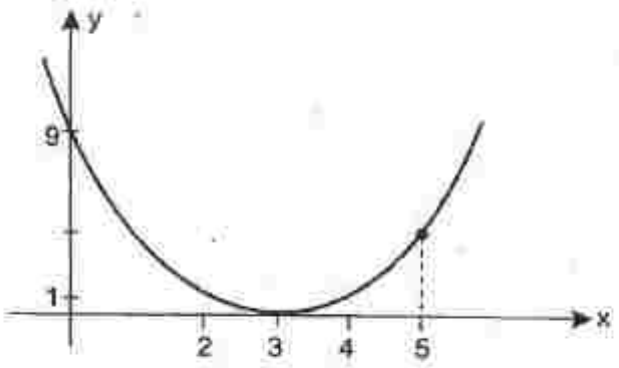
$$f'(x) = 0$$
$$4x^3 - 4x + 4 = 0$$
$$x = 0, 1, -1$$

Second derivative test

$$f''(x) = 4(3x^2 - 1)$$
$$f''(0) = -4 < 0 \Rightarrow \text{maxima}$$
$$f''(-1) = 4(3 - 1)$$
$$= 8 > 0 \Rightarrow \text{minima}$$
$$f''(1) = 4(3 - 1)$$
$$= 8 > 0 \text{ minima}$$
$$f(0) = 16 \text{ (maximum value)}$$
$$f(-1) = 1 - 2 - 4 + 16$$
$$= 11 \text{ (global minima)}$$
$$f(1) = 1 - 2 + 4 + 16$$
$$= 19 \text{ (local minima)}$$

67. (5)

$$y = x^2 - 6x + 9 = (x - 3)^2$$
$$y(2) = 1$$
$$y(5) = 4$$



∴ Maximum value of y over the interval 2 to 5 will be at $x = 5$

68. (c)

The function is a polynomial which is continuous and differentiable everywhere and should be continuous on [–2, 1] and differentiable on [–2, 1] that $g(-2) = g(1) = 0$.

Therefore, the conditions for Rolle's theorem are met and so we can actually do the problem. Now, according to Rolle's theorem, we need to do take the derivative,

$$g'(t) = 2 - 2t - 3t^2$$

and then solve $g'(c) = 0$

$$-3c^2 - 2c + 2 = 0$$

$$\Rightarrow \quad c = \frac{1 \pm \sqrt{7}}{-3} = -1.2153, 0.5486$$

So, we found two values and, in this case, they are both in the interval so, the values we want are

$$c = 0.5486, -1.2153$$

69. (a)

The function is a sum of polynomial and an exponential function both of which are continuous and differentiable everywhere. This is turn means that the sum is also continuous and differentiable everywhere and so the function will be continuous on [–2, 3] and differentiable [–2, 3].

We know that the mean value theorem can be used there really is not much to do. All we need to do some function evaluations and take the derivative.

$$A(-2) = -16 + e^6$$
$$A(3) = 24 + e^{-9}$$
$$A'(t) = 8 - 3e^{-3t}$$

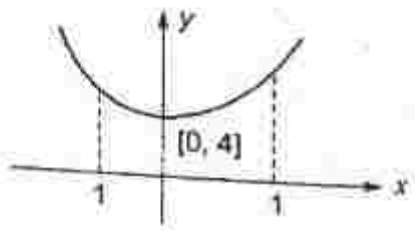
The final step is to then plug into the formula from the Mean Value Theorem and solve for C.

$$8 - 3e^{-3c} = \frac{24 + e^{-9} - (-16 + e^6)}{3 - (-2)} = -72.6857$$
$$3e^{-3c} = 80.6857$$
$$e^{-3c} = 26.8952$$
$$-3c = \ln(26.8952) = 3.29195$$
$$c = -1.0973$$

So, we found a single value and it is in the interval and so the value we wants

$$c = -1.0973$$

70. (b, c)



$$f'(c) = \frac{f(3) - f(0)}{3 - 0}$$

$$\frac{2c}{\sqrt{16 + c^2}} = \frac{\sqrt{16 + 3^2} - \sqrt{16}}{3}$$

$$\frac{2c}{\sqrt{16 + c^2}} = \frac{\sqrt{25} - \sqrt{16}}{3} = \frac{5 - 4}{3} = \frac{1}{3}$$

$$6c = \sqrt{16 + c^2}$$
$$36c^2 = 16 + c^2$$
$$35c^2 = 16$$
$$c = \pm\frac{4}{\sqrt{35}} = \pm 0.67$$

71. (b)

$$y = 3x^4 - 16x^3 + 24x^2 + 37$$

$$\frac{dy}{dx} = 12x^3 - 48x^2 + 48x = 0$$

$$x(12x^2 - 48x + 48) = 0$$

$$x = 0$$

or $12x^2 - 48x + 48 = 0$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$\frac{d^2y}{dx^2} = 36x^2 - 96x + 48$$

Now at $x = 0$

$$\frac{d^2y}{dx^2} = 48 > 0$$

$\therefore$ $f(x)$ has a minimum at $x = 0$

$$\left[\frac{d^2y}{dx^2}\right]_{x=2} = 0 \text{ and } \left[\frac{d^3y}{dx^2}\right]_{x=2} = 48 \neq 0$$

So $x = 2$ is a saddle point (point of inflection)

$\therefore$ $f(x)$ has no extremum at $x = 2$.

So $f(x)$ has only one point of extremum (at $x = 0$).

72. (a)

$\frac{e^{\sin x}}{e^{\cos x}}$ can be written as $e^{\sin x - \cos x}$. e is a famous irrational constant i.e. used as base of natural logarithms, and is approximate value is 2.7183, (which is of course greater than 1).

So, now we have to just maximize $(\sin x - \cos x)$ to find maximum value of given expression.

Clearly, $\sin x$ and $\cos x$ are differentiable functions hence, their difference $(\sin x - \cos x)$ is also differentiable.

So, we can differentiate $\sin x - \cos x$ to find its maximum value.

Solution of equation $\frac{d}{dx}(\sin x - \cos x) = 0$ will give us the points where $\sin x - \cos x$ will attain its maximum value.

On differentiating $(\sin x - \cos x)$ with respect to x we get $\cos x + \sin x$.

Putting $\cos x + \sin x = 0$ and simplifying we get $\tan x = -1$.

This equation has infinity many solutions.

One of them is $x = \frac{3\pi}{4}$ $x = \frac{3\pi}{4}$

$$\sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}} \text{ and } \cos\left(\frac{3\pi}{4}\right) = \frac{-1}{\sqrt{2}}$$

Putting these values in $e^{\sin x - \cos x}$, we get $e^{\sqrt{2}}$.

73. (d)

$$f(x) = \begin{cases} (x-3) & x > 3 \\ -(x-3) & x < 3 \\ 0 & x = 3 \end{cases}$$

Critical points,

$f'(x) = 0$, test fails

Checking limit around "$x = 2$"

$f(2) = 1$

LHS limit:

$$\lim_{h \to 0}(2-h) = -2 + h + 3 = 1 + h$$

= Increasing function

RHS limit:

$$\lim_{h \to 0}(3+h) = 2 + h - 2 = h$$

= Decreasing function

Hence at $x = 2$, $f(x)$ not maximum and minimum.

74. (a)

Let, $f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$

Then, $f'(x) = 12x^3 - 6x^2 - 12x + 6$

$f'(x) = 0$, $x = 1$ satisfied and hence 1 factor is $(x-1)$, by polynomial division we get other 2 factor are $(x+1)(2x-1)$

$= 6(x^2-1)(2x-1)$

$\therefore$ $f'(x) = 0$ when $x = \pm 1, \frac{1}{2}$

So in the interval $(0, 2)$, $f(x)$ can have maximum or minimum at $x = \frac{1}{2}$ or 1.

Now, $f''(x) = 36x^2 - 12x - 12$

$= 12(3x^2 - x - 1)$

So that $f''\left(\frac{1}{2}\right) = -9$ and $f''(1) = 12$

$\therefore$ $f(x)$ has a maximum at $x = \frac{1}{2}$ and a minimum at $x = 1$

Thus the maximum value

$$f\left(\frac{1}{2}\right) = 3\left(\frac{1}{2}\right)^4 - 2\left(\frac{1}{2}\right)^3 - 6\left(\frac{1}{2}\right)^2 + 6\left(\frac{1}{2}\right) + 1 = 2\frac{7}{16}$$

and the minimum value

$$f(1) = 3(1)^4 - 2(1)^3 - 6(1)^2 + 6(1) + 1 = 2$$

75. **(c)**

$$f(x) = -2x - x^2 + 15$$

Critical points,

$$f'(x) = 0$$
$$-2 - 2x = 0$$
$$x = -1$$

Second derivative test

$$f''(x) = -2 < 0$$

$\Rightarrow$ Maxima at $x = -1$

$$f(-1) = -2(-1) - (-1)^2 + 15$$
$$= 2 - 1 + 15$$
$$= 16 \text{ (maximum value)}$$

76. **(d)**

$$f(x) = 2x^3 - 3x^2 - 12x + 5$$
$$f'(x) = 6x^2 - 6x - 12$$

For minima/maxima, $f'(x) = 0$

$$6x^2 - 6x - 12 = 0$$
$$x^2 - x - 2 = 0$$
$$(x + 1)(x - 2) = 0$$
$$x = -1, 2$$
$$f''(x) = 12x - 6$$
$$f''(-1) = -12 - 6 = -18 < 0$$

$\Rightarrow$ maxima

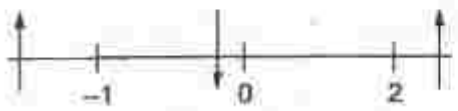
$$f''(2) = 24 - 6 = 18 > 0$$

$\Rightarrow$ minima

The function has maxima at $x = -1$ and minima at $x = 2$.

Critical point $(-1, 2)$ draw plot on line graph:

Since $0 \in (-1, 2)$ and $f'(0) = 6 \times 0^2 - 6 \times 0 - 12$
$= -12 < 0$



The function is decreasing between -1 and 2.

77. **(0.23)**

$$f(a) = 0$$
$$f(b) = \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right) = \frac{3}{8}$$
$$f'(x) = 3x^2 - 6x + 2$$
$$f'(c) = 3c^2 - 6c + 2$$

Substituting in, $f(b) - f(a)$

$= (b - a) f'(c)$

$$\frac{3}{8} - 0 = \left(\frac{1}{2} - 0\right)(3c^2 - 6c + 2)$$

or $12c^2 - 24c + 5 = 0$

whence $$c = \frac{24 \pm \sqrt{(24)^2 - 12 \times 5 \times 4}}{24}$$
$$= 1 \pm 0.764 = 1.764, 0.236$$

Hence, $c = 0.236$, since it only lies between 0 and $\frac{1}{2}$

78. **(c)**

Given function, $f(x) = 2x^3 - 15x^2 + 36x + 1$

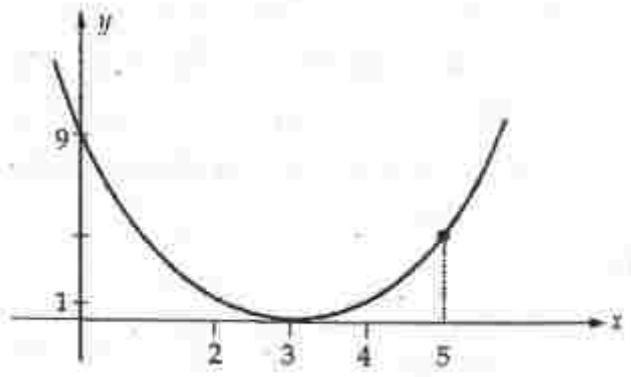
$$f'(x) = 6x^2 - 30x + 36$$
$$= 6(x - 2)(x - 3)$$

Hence, $f'(x) > 0$ when $x < 2$ and $x > 3$.

So $f(x)$ is increasing in $]-\infty, 2[\cup]3, \infty[$

79. **(5)**

$$y = x^2 - 6x + 9 = (x - 3)^2$$
$$y(2) = 1$$
$$y(5) = 4$$



$\therefore$ Maximum value of y over the interval 2 to 5 will be at $x = 5$.

80. **(b)**

$$f(x) = \begin{cases} x^4 & x^2 < 1 \quad \text{or} \quad -1 < x < 1 \\ x & x^2 \geq 1 \quad \text{or} \quad x \geq 1 \text{ or } x \leq -1 \end{cases}$$

$$f(-1) = x|_{x=-1} = -1$$
$$f(-1^-) = x|_{x=-1} = -1$$
$$f(-1^+) = x^4|_{x=-1} = 1$$

At $x = -1$

Limit does not exists $\Rightarrow f(x)$ is not continuous

$\Rightarrow f'(-1)$ does not exist.

81. **(b)**

$$f(x) = x + \ln x$$
$$f'(x) = 1 + \frac{1}{x} \quad \ldots(i)$$
$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(e) - f(1)}{e - 1}$$

$$= \frac{e+\ln e-(1+\ln 1)}{e-1}$$

$$= \frac{e+1-1+0}{e-1} = \frac{e}{e-1}$$

$\Rightarrow$ $f'(c) = \frac{e}{e-1}$ [from equation (i)]

$\Rightarrow$ $1+\frac{1}{c} = \frac{e}{e-1}$

$\Rightarrow$ $\frac{1}{c} = \frac{e}{e-1}-1 = \frac{e-e+1}{e-1}$

$\Rightarrow$ $c = e-1$

82. (c)

$$f(x) = (x-1)^{2/3} = (\sqrt[3]{x-1})^2$$

As $f(x)$ is square of $\sqrt[3]{x-1}$ hence its minimum value be 0 where at $x = 1$.

83. (c)

Let, $y = \frac{e^{\sin x}}{e^{\cos x}}$

$\Rightarrow$ $y = e^{(\sin x - \cos x)}$

$\because$ $(\sin x - \cos x) = \sqrt{2}\sin\left(x-\frac{\pi}{4}\right)$

$\Rightarrow$ $y = e^{\sqrt{2}\sin\left(x-\frac{\pi}{4}\right)}$

$\therefore$ Maximum value of $y = e^{\sqrt{2}}$.

84. (c)

Cauchy Mean Value theorem

$$\frac{p'(c)}{q'(c)} = \frac{p(a)-p(b)}{q(a)-q(b)}$$

$$\frac{e^{\frac{c}{2}}}{-e^{-\frac{c}{2}}} = \frac{e^{\frac{a}{2}}-e^{\frac{b}{2}}}{e^{-\frac{a}{2}}-e^{-\frac{b}{2}}}$$

$$-e^{c} = -e^{\frac{a}{2}}\cdot e^{\frac{b}{2}}$$

$$e^{c} = e^{\frac{a+b}{2}}$$

$$c = \frac{(a+b)}{2}$$

85. (b)

$$f(x) = \begin{cases} x-2 & x<2 \\ -x+2 & x>2 \\ 0 & x=2 \end{cases}$$

$f(2) = 0$

$\lim_{x\to 2^-} f(x) = x-2$, $\lim_{h\to 0} f(2-h) = 2-h-2$

$= -h$, decreasing function

$\lim_{x\to 2^+} f(x) = -x+2$, $\lim_{h\to 0} f(2+h) = -(2+h)+2$

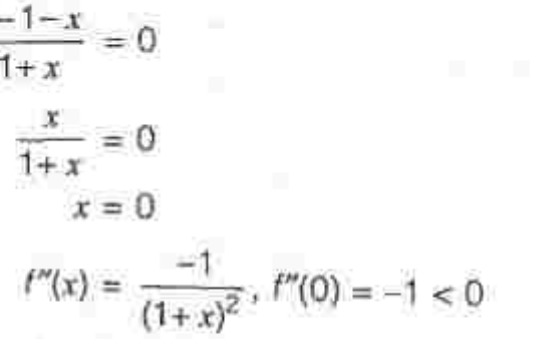
$= -h$, decreasing function

Hence at $x = 2$, function have maxima.

86. (c)

$$y = x^2-8x+16 = (x-4)^2$$

$y(2) = 4$, $y(5) = 1$, $y(4) = 0$



$\therefore$ Maximum value of y over the interval 2 to 5 will be at $x = 4$.

87. (6)

$f(x) = 2x^3-9x^2+12x-3$

$f'(x) = 6x^2-18x+12$

$f'(x) = 0$

$6x^2-18x+12 = 0$

$x^2-3x+2 = 0$

$x = 1, 2$

Hence critical points are {0, 1, 2, 3,}.

$f(x)$ attains its maximum value at one of these points.

$f(0) = -3$

$f(1) = 2$

$f(2) = 1$

$f(3) = 6$

88. (0)

$$f'(x) = \frac{1}{1+x}-1=0$$

$$\frac{1-1-x}{1+x} = 0$$

$$\frac{x}{1+x} = 0$$

$$x = 0$$

$$f''(x) = \frac{-1}{(1+x)^2},\ f''(0) = -1 < 0$$

$f(x)$ have maximum value at $x = 0$

$f(0) = hg(1+0)-0 = 0$

$f_{max} = 0$

89. (d)

$$f(x) = x^3 + 1$$

Put, $f'(x) = 0$

$\Rightarrow$ $3x^2 = 0$

$\Rightarrow$ $x = 0$ is the only critical point

At this critical point

$f''(x) = 6x$

$f''(0) = 6 \times 0 = 0$

Now, $f'''(x) = 6$ and

So, $f'''(0) = 6$ which is non zero

Since the first non zero derivative value occurs at the third derivative which is an odd derivative, this function has a point of inflection at $x = 0$.

90. (d)

$$f(t) = \frac{\sin t}{t}$$

$$f(t) = \frac{t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots}{t}$$

$$f(t) = 1 - \frac{t^2}{3!} + \frac{t^4}{5!} - \cdots$$

$$f'(t) = -\frac{2t}{3!} + \frac{4t^3}{5!} - \cdots$$

$$f''(t) = -\frac{2t}{3!} + \frac{4t^3}{5!} - \cdots$$

At $t = 0$, $f'(t) = 0$, $f''(t) < 0$

$\therefore$ $f(t)$ attains maxima.

91. (c)

$$f(x) = \frac{2x^2 + x + 9}{2x^2 + x + 1}$$

$$= \frac{2x^2 + x + 1 + 8}{2x^2 + x + 1}$$

$$f(x) = 1 + \frac{8}{2x^2 + x + 1}$$

$f(x)$, has maximum value when $g(x) = 2x^2 + x + 1$ has minimum value.

$$g(x) = 2x^2 + x + 1$$

Critical points,

$$g'(x) = 4x + 1 = 0$$

$$x = -\frac{1}{4}$$

$g''(x) = 4 > 0$, minima at $x = -\frac{1}{4}$

$$g\left(-\frac{1}{4}\right) = 2\left(-\frac{1}{4}\right)^2 - \frac{1}{4} + 1$$

$$= \frac{2}{16} - \frac{1}{4} + 1 = \frac{14}{16} = \frac{7}{8}$$

$$f(x) = 1 + \frac{8 \times 8}{7} = 1 + \frac{64}{7} = \frac{71}{7}$$

92. (b)

$f(x) = x^3 - 3x^2 - 24x + 100$ $x \in [-3,3$

$f'(x) = 3x^2 - 6x - 24$

$f'(x) = 0$ at $x = 4, -2$

Critical points are $\{-3, -2, 3\}$

$f(-3) = -27 - 27 + 72 + 100 = 118$

$f(-2) = -8 - 12 + 48 + 100 = 128$

$f(3) = 27 - 27 - 72 + 100 = 28$

Hence $f(x)$ has minimum value at $x = 3$ which is 28

93. (c)

$$f(x) = 2x^3 - 3x^2 - 36x + 20$$

Critical points,

$f'(x) = 0$

$6x^2 - 6x - 36 = 0$

$x^2 - x - 6 = 0$

$x^2 - 3x + 2x - 6 = 0$

$(x - 3)(x + 2) = 0$

$x = -2, 3$

Second derivative test

$f''(x) = 6(2x - 1)$

$f''(-2) = 6(-4 - 1)$

$= 6(-5)$

$= -30 < 0 \Rightarrow$ maxima

$f''(3) = 6(3(3) - 1)$

$= 6(8)$

$= 48 > 0 \Rightarrow$ minima

94. (12)

$f(x) = 3x^3 - 7x^2 + 5x + 6$ in[0,2

$f'(x) = 9x^2 - 14x + 5$

$f''(x) = 18x - 14$

$f'(x) = 0$

$x^2 - 14x + 5 = 0$

$x = 1, 0.55$

$x = 1$

$f''(1) = 18 - 14 = 4 > 0$ minima

$x = 0.55$

$f''(0.55) = -4.1 < 0$ maxima

Maximum $\{f(0), f(0.55), f(2)\}$

Maximum $\{6, 7.13, 12\} = 12$

95. **(-3.0)**

Given function, $y = x^3e^{-|x|}$

For $x < 0$, $y = x^3e^x$

$$\frac{dy}{dx} = 3x^2e^x + x^3e^x = 0$$

$$3x^2e^x = -x^3e^x$$

$$x = -3$$

$$\frac{d^2y}{dx^2} = 3x^2e^x + x^3e^x + 6xe^x + 3x^2e^x$$

$$= e^x(x^3 + 6x^2 + 6x)$$

$$= e^{-3}(-27 + 54 - 18)$$

$$= 9e^{-3} > 0$$

Hence, at $x = -3$ global minimum occurs.

96. **(c)**

Given function,

$$y = 3x^2 + 4x + 1$$

Mean value of function is given by, $\frac{1}{b-a}\int_a^b y\,dx$

$$= \frac{1}{2-(-1)}\int_{-1}^{2}(3x^2 + 4x + 1)\,dx$$

$$= \frac{1}{3}\left[x^3 + 2x^2 + x\right]_{-1}^{2}$$

$$= \frac{1}{3}(2^3 + 1 + 2)(2^2 - 1) + 3) = 6$$

97. **(c)**

We need absolute maximum of

$f(x) = x^3 - 9x^2 + 24x + 5$ in the interval [1, 6]

First find local maximum if any by putting $f'(x) = 0$.

i.e., $f'(x) = 3x^2 - 18x + 24 = 0$

i.e., $x^2 - 6x + 8 = 0$

$x = 2, 4$

Now $f''(x) = 6x - 18$

$f''(2) = 12 - 18 = -6 < 0$

(So $x = 2$ is a point of local maximum)

and $f''(4) = 24 - 18 = +6 > 0$

(So $x = 4$ is a point of local minimum)

Now tabulate the values of f at end point of interval and at local maximum point, to find absolute maximum in given range, as shown below:

x	$f(x)$
1	21
2	25
6	41

Clearly the absolute maxima is at $x = 6$ and absolute maximum value is 41.

98. **(a)**

$$f(x) = xe^{-x}$$

$$f'(x) = e^{-x} - xe^{-x} = 0$$

$$e^{-x}(1 - x) = \infty,$$

$\Rightarrow$ $x = 1$ (since $e^{-x} = 0$ only when $x = \infty$ which does not belong to the given interval)

Now, we need to check whether at $x = 1$, we have a maximum, minimum or saddle point.

$$f''(x) = -e^{-x} - (e^{-x} - xe^{-x})$$

$$= -2e^{-x} + xe^{-x} = e^{-x}(x - 2)$$

$f''(1) = -e^{-1}$ which is < 0

So at $x = 1$, we have a maximum.

The maximum value is

$$f(1) = 1e^{-1} = e^{-1}$$

99. **(a, d)**

$$f(x) = -(x^2 - 2)^4$$

Critical points,

$$f'(x) = 0$$

$$-4(x^2 - 2)(2x) = 0$$

$$x = 0, \pm\sqrt{2}$$

Second derivative test

$$f''(x) = -8(6x^2 - 2)$$

$$f''(0) = +16 > 0 \Rightarrow \text{minima}$$

$$f''(\sqrt{2}) = -8\left(6\sqrt{2}^2 - 2\right)$$

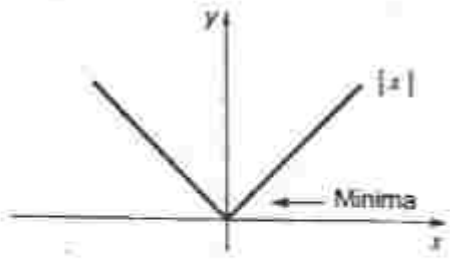
$$= -8(10)$$

$$= -80 < 0 \Rightarrow \text{maxima}$$

$$f''(-\sqrt{2}) = -80 < 0 \Rightarrow \text{maxima}$$

Hence $f(x)$ have two maxima and one minima.

100. **(a)**



101. **(a)**

$$\int_a^b f(x)dx = f(\xi)(b - a)$$

102. **(d)**

$$f(x) = x^3 - 27x + 3$$

Critical point,

$$f'(x) = 0$$

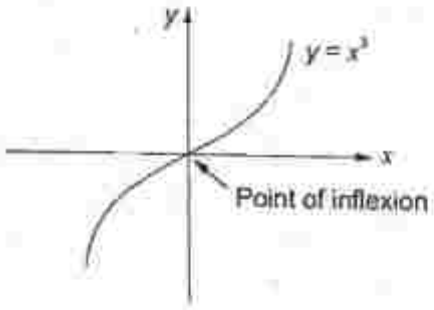
$$3x^2 - 27 = 0$$
$$x^2 = 9$$
$$x = \pm 3$$

Given interval (–3, 3)

Hence both critical points are not present in internal hence no maximum value between (–3, 3).

103. (d)

$f(x) = x^3$ at $x = 0$



At $x = 0$, the function $y = x^3$ has neither minima nor maxima.

104. (b)

$$f(x) = (x^2 - 4)^2$$
$$f'(x) = 2(x^2 - 4) \times 2x = 4x(x^2 - 4) = 0$$
$x = 0$, $x = 2$ and $x = -2$ are the stationary pts.
$$f''(x) = 4[x(2x) + (x^2 - 4) \times 1]$$
$$= 4[2x^2 + x^2 - 4] = 4[3x^2 - 4]$$
$$= 12x^2 - 16$$
$f''(0) = -16 <$ - maxima at $x = 0$
$f'(2) = (12)2^2 - 16$
$= 32 > 0$ minima at $x = 2$
$f'(-2) = 12(-2)^2 - 16$
$= 32 > 0$ minima at $x = -2$

∴ There is only one maxima and only two minima for this function.

105. (c)

$$f(x) = 2x - x^2 + 3$$
$$f'(x) = 2 - 2x = 0$$
$\Rightarrow$ $x = 1$ is the stationary point
$$f''(x) = -2$$
$\Rightarrow$ $f''(1) = -2 < 0$

So at $x = 1$ we have a relative maxima.

106. (a, b)

We need absolute maximum of

$f(x) = x^3 - 9x^2 + 24x + 5$ in the interval [1, 6]

First find local maximum if any by putting $f'(x) = 0$.

i.e., $f'(x) = 3x^2 - 18x + 24 = 0$

i.e., $x^2 - 6x + 8 = 0$

$x = 2, 4$

Now $f''(x) = 6x - 18$

$f''(2) = 12 - 18 = -6 < 0$

(So $x = 2$ is a pt. of local maximum)

and $f''(4) = 24 - 18 = +6 > 0$

(So $x = 4$ is a pt. of local minimum)

107. (c)

$$f(x) = (x - 1)^{2/3}$$
$$= \left(\sqrt[3]{x-1}\right)^2$$

As $f(x)$ is square of hence its minimum value be 0 where at $x = 1$.

108. (b)

$$f(x) = x^3 - 3x^2 - 24x + 100 \; x \in [-3, 3]$$
$$f'(x) = 3x^2 - 6x - 24$$
$f'(x) = 0$ at $x = 4, -2$

Critical points are {–3, –2, 3}

$f(-3) = -27 - 27 + 72 + 100 = 118$

$f(-2) = -8 - 12 + 48 + 100 = 128$

$f(3) = 27 - 27 - 72 + 100 = 28$

Hence $f(x)$ has minimum value at $x = 3$ which is 28.

109. (b)

$$\lim_{x \to 0} \frac{\sin(8x^2)}{\cos(2x)}$$

$$= \lim_{x \to 0} \frac{\frac{\sin(8x^2)}{8x^2} \times 8x^2}{\cos(0)} \quad \left\{\lim_{x \to 0} \frac{\sin(x)}{x} = 1\right\}$$

$$= \lim_{x \to 0} \frac{8x^2}{1} = 0$$

110. (a, c)

$$f(x) = x^{2P} \log_e x$$
$$f'(x) = \frac{x^{2P}}{x} + \log_e x \cdot 2P x^{2P-1}$$
$$f'(x) = x^{2P}\left(\frac{1}{x} + \frac{\log_e x}{x}\right)$$

By Rolle's theorem

$$f'(c) = 0$$

Case 1:

$$c^{2P} = 0$$

$$\therefore \quad P = -\infty \notin [0, 1]$$

Case 2:

$$\frac{1}{c} + \frac{\log_e c}{c} = 0$$

$$1 + 2P\log_e c = 0$$

$$2P\log_e c = -1$$

$$\log_e c = -\frac{1}{2P}$$

$$c = e^{-\frac{1}{2P}}$$

For $P = 3, \frac{1}{2}, c \in [0, 1]$

111. (1)

$$\int_0^{\pi/2} \frac{\cos x}{2\sqrt{1-\sin x}}dx$$

Put $\sin x = t$, $\cos x\, dx = dt$. Now t will vary from 0 to 1.

$$\int_0^1 \frac{dt}{2\sqrt{1-t}}$$

$$\int_0^1 \frac{1}{2}(1-t)^{-1/2} \qquad \text{Now, } \frac{1}{2}(1-t)^{1/2} * -2$$

$\Rightarrow$ $-(1-t)^{1/2}$ range from 0 to 1 you will get $(-1) = 1$

So, value is 1.

112. (7.28)

$$\int_0^{3\pi/2} |x\sin x|dx$$

$$= \int_0^{\pi}(x\sin x)dx + \int_{\pi}^{3\pi/2} -(x\sin x)dx$$

$$= [-x\cos x + \sin x]_0^{\pi} + \left(-[-x\cos x + \sin x]\right)_{\pi}^{\frac{3\pi}{2}}$$

$$= \pi + -(-1-\pi) = \pi + 1 + \pi$$

$$= 7.28$$

113. (0.5)

Let, $u = \sin x$

$du = \cos x\, dx$

So $\int_0^{\pi/2} \sin x\cos x\, dx = \int_0^1 u\, dx$

$$= \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2} = 0.5$$

114. (b)

$$\int_0^{\pi/6} \cos^4 3\theta \sin^3 6\theta\, d\theta$$

As we know ($\sin 2\theta = 2\sin\theta\cos\theta$)

$$= \int_0^{\pi/6} \cos^4 3\theta (2\sin^3 \cos 3\theta)^3$$

$$= 8\int_0^{\pi/6} \cos^4 3\theta \sin^3 3\theta \cos^3 3\theta$$

$$= 8\int_0^{\pi/6} \cos^7 3\theta \sin^3 3\theta$$

Let, $3\theta = t$

$$d\theta = \frac{dt}{3}$$

$$= \frac{8}{3}\int_0^{\pi/2} \cos^7 3\theta \sin^3 3\theta$$

As we know that,

$$\int_0^{\pi/2} \cos^m\theta \sin^n\theta$$

$$= \frac{(m-1)(m-3)(m-5)...(n-1)(n-3)(n-5)}{(m+n)(m+n-2)(m+n-4).....}$$

[* $\pi/2$, if m and n are even]

$$= \frac{8}{3} * \frac{6\times4\times2\times2}{10\times8\times6\times4\times2} = \frac{1}{15}$$

115. (1)

Let $\ln x = t$

Then, $dt = \frac{1}{x}dx$

Hence,

$$I = \int \sin t\, dt = -\cos t + c = -\cos(\ln x) + c$$

116. (b)

$$I = \int_0^{\pi} \log(1+\cos x) \qquad ...(1)$$

Using $f(x) = f(a-x)$

$$\Rightarrow \qquad I = \int_0^{\pi} \log(1-\cos x) \qquad ...(2)$$

Add eq. (1) + eq. (2)

$$\Rightarrow \quad 2I = \int_0^{\pi} [\log(1+\cos x) + \log(1-\cos x)]dx$$

$$\Rightarrow \quad 2I = \int_0^{\pi} \log(1-\cos^2 x)dx$$

$$\Rightarrow \quad 2I = \int_0^{\pi} \log\sin^2 x\, dx$$

$$\Rightarrow \quad 2I = 2\int_0^{\pi} \log\sin x\, dx$$

$$\Rightarrow \quad I = \int_0^{\pi} \log\sin x\, dx \qquad ...(3)$$

if $f(a-x) = f(x)$ then $2\int_0^{a/2} f(x)dx$

$I = 2\int_0^{\pi/2} \log\sin x\, dx$ ...(4)

$f(x) = f(a-x)$

$I = 2\int_0^{\pi/2} \log\cos x\, dx$ ...(5)

Add equation (4) and (5)

$\Rightarrow \quad 2I = 2\int_0^{\pi/2} \log\sin x\cos x\, dx$

$\Rightarrow \quad I = \int_0^{\pi/2} \log\frac{\sin 2x}{2} dx$

$I = \int_0^{\pi/2} \log\sin 2x - \int_0^{\pi/2} \log 2\, dx$

$I = \int_0^{\pi/2} \log\sin 2x - \frac{\pi}{2}\log 2$ ...(6)

Let $I' = \int_0^{\pi/2} \log\sin 2x\, dx$

$2x = t$

$2dx = dt$

$dx = \frac{dt}{2}$

$I' = \frac{1}{2}\int_0^{\pi} \log\sin t\, dt$

$I' = \frac{1}{2}I$ [from eq. (3)]

Putting in eq. (6)

$\Rightarrow \quad I = \frac{1}{2}I - \frac{\pi}{2}\log 2$

$\Rightarrow \quad \frac{I}{2} = \frac{-\pi}{2}\log 2$

$\Rightarrow \quad I = -\pi \log 2$

117. (d)

$$I = \int_0^{3\pi/2} \frac{\cos x - \sin x}{\cos x + \sin x + 2} dx$$

Let, $t = \cos x + \sin x + 2$

$dt = (-\sin x + \cos x)\, dx$

$\therefore$ Changing integral limit values

Lower limit

$t = \cos x + \sin x + 2 \; [x = 0] = 3$

Upper limit

$t = \cos x + \sin x + 2 \quad \left\{x = \frac{3\pi}{2}\right\}$

$t = 1$

$I = \int_3^1 \frac{dt}{(t)} = \ln(t)\big|_3^1$

$= \ln(1) - \ln(3) = -\ln(3)$

118. (b)

$$\int_0^1 \frac{x^{1/4}}{1+\sqrt{4}} dx$$

Put, $\sqrt{x} = t^2$

$x = t^4$

$dx = 4t^3 dt$

$= \int_0^2 \frac{t \cdot 4t^3 dt}{1+t^2} = 4\int_0^2 \frac{t^4 - 1 + 1}{1+t^2} dt$

$= 4\int_0^2 (t^2 - 1) + \frac{1}{1+t^2} dt$

$= 4\left[\frac{t^3}{3} - t + \tan^{-1} t\right]_0^2$

$= 4\left[\frac{8}{3} - 2 + \tan^{-1} 2\right]$

$= \frac{8}{3} + 4\tan^{-1} 2$

119. (c)

$$I = \int_{-\infty}^{0} e^{-\left(\frac{x^2}{2}\right)} dx$$

Comparing with area under the standard normal curve from $-\infty$ to 0.

We get

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0} e^{-\frac{x^2}{2}} dx = 0.5$$

So, the required integral

$$\int_{-\infty}^{0} e^{-\frac{x^2}{2}} dx = 0.5 \times \sqrt{2\pi} = \sqrt{\frac{\pi}{2}}$$

121. (2)

$$\int_0^1 \frac{1}{\sqrt{1-x}} dx = -2\int_0^1 \frac{1}{2\sqrt{1-x}} dx$$

$= -2\left(\sqrt{1-x}\right)\big|_0^1$

$= -2(0-1) = 2$

122. (a)

$$\int_0^{\pi} x^2 \cos x\, dx = x^2(\sin x) - 2x(-\cos x) + 2(-\sin x)\Big|_0^{\pi}$$

$= \pi^2 \cdot 0 + 2\pi(-1) - 0 = -2\pi$

123. (d)

$$\int_{-\infty}^{\infty}\frac{dx}{1+x^2} = \left[\tan^{-1}x\right]_{-\infty}^{\infty}$$
$$= \tan^{-1}(\infty) - \tan^{-1}(-\infty)$$
$$= \frac{\pi}{2} - \left[\frac{-\pi}{2}\right] = \pi$$

124. (b)

Let $$I = \int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}}dx \quad ...(i)$$

Since $$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$I = \int_0^a \frac{\sqrt{a-x}}{\sqrt{a-x}+\sqrt{x}}dx \quad ...(ii)$$

(i) + (ii) $\Rightarrow$ $$2I = \int_0^a \frac{\sqrt{x}+\sqrt{a-x}}{\sqrt{x}+\sqrt{a-x}}dx$$

$\Rightarrow$ $$2I = \int_0^a dx$$

$\Rightarrow$ $$2I = a$$

$\Rightarrow$ $$I = \frac{a}{2}$$

125. (d)

If $f(x)$ is even function then

$$\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$$

126. (a)

$$I = \int_0^{\pi/2}\frac{\cos x}{\sin x+\cos x}dx \quad ...(i)$$

$$I = \int_0^{\pi/2}\frac{\cos\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$I = \int_0^{\pi/2}\frac{\sin x}{\cos x+\sin x}dx \quad ...(ii)$$

Adding equation (i) and (ii)

$$2I = \int_0^{\pi/2}\frac{\sin+\cos x}{\cos x+\sin x}dx = \int_0^{\pi/2}dx$$

$$2I = x\Big|_0^{\pi/2} = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

127. (b)

$$P = \int_0^1 x\,e^x\,dx$$

Integrating by parts:

Let, $u = x$,
$dv = e^x\,dx$
$du = dx$,
$v = \int e^x\,dx = e^x$

Now, $\int u\,dv = uv - \int v\,du$

$\therefore$ $\int x\,e^x dx = x\,e^x - \int e^x\,dx$
$= x\,e^x - e^x + c$

$$\int_0^1 x\,e^x dx = [xe^x - e^x]_0^1$$
$$= (1\cdot e^1 - e^1) - (0\cdot e^0 - e^0)$$
$$= 0 - (-1) = 1$$

128. (b)

Taylor's series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \ldots\ldots\infty$$

If $f(x) = \sin x$

Then,

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots\infty$$

129. (a)

Method 1:

Applying by-parts integration,

$$\Rightarrow \int_{-\pi/2}^{\pi/2}(x\cos x)dx = x\sin x\Big|_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} 1\cdot\sin x\,dx$$

$$= \frac{\pi}{2}\cdot 1 - \left(\frac{-\pi}{2}(-1)\right) = 0$$

Method 2:

$f(x) = x\cos x$

$\Rightarrow$ $f(-x) = -x\cos(-x) = -x\cos x$

$\Rightarrow$ $f(x) \rightarrow$ odd function

$$\Rightarrow \int_{-\pi/2}^{\pi/2} x\cos x\,dx = 0$$

130. (c)

$$S = \int_1^{\infty} x^{-3}\,dx = \left[\frac{x^{-2}}{-2}\right]_1^{\infty} = -\left[\frac{1}{2x^2}\right]_1^{\infty}$$

$$= -\left[\frac{1}{\infty} - \frac{1}{2}\right] = \frac{1}{2}$$

131. (c)

$$I = \int_0^{\pi} \sin^3\theta\, d\theta$$

$$\int_0^{\pi}\left(1-\cos^2\theta\right)\sin\theta\, d\theta,$$

Let, $\cos\theta = t$

$-\sin\theta\, d\theta = dt,$

At $\theta = 0,\ t = \cos 0 = 1$

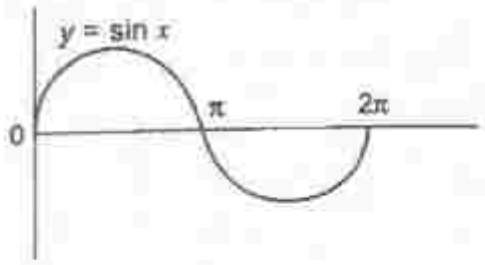
At $\theta = \pi,\ t = \cos\pi = -1$

So, $$I = -\int_{-1}^{+1}\left(1-t^2\right)dt$$

$$= \left|t - \frac{t^3}{3}\right|_{-1}^{1} = \left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right)$$

$$I = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$$

132. (4)



$$\Rightarrow \int_0^{2\pi} |x\sin x|\,dx = k\pi$$

$$\Rightarrow \int_0^{\pi} |x\sin x|\,dx + \int_{\pi}^{2\pi} |x\sin x|\,dx = k\pi$$

$$\Rightarrow \int_0^{\pi} x\sin x\,dx + \int_{\pi}^{2\pi} -(x\sin x)\,dx = k\pi$$

$$\Rightarrow (-x\cos x + \sin x)\Big|_0^{\pi} - (-x\cos x + \sin x)\Big|_{\pi}^{2\pi} = k\pi$$

$$\Rightarrow 4\pi = k\pi$$

$$\Rightarrow k = 4$$

133. (d)

$$I = \int_0^{2\pi} x\sin\left(\frac{x}{2}\right)dx = k\pi^2$$

$$= \left(-2x\cos\left(\frac{x}{2}\right) + 4\sin\left(\frac{x}{2}\right)\right)\Big|_0^{2\pi}$$

$$\Rightarrow \quad -4\pi = k\pi^2$$

$$k = -\frac{4}{\pi}$$

134. (b)

$$I = \int_0^2 \frac{(x-1)^2\sin(x-1)}{(x-1)^2 + \cos(x-1)}\,dx$$

Taking $x - 1 = z$

$\Rightarrow \quad dx = dz$

for $x = 0,\ z \to -1$ and $x = 2,\ z \to 1$

$\therefore$ $$I = \int_{-1}^{1} \frac{z^2\sin z}{z^2 + \cos z}\,dz$$

Let $$f(z) = \frac{z^2\sin z}{z^2 + \cos z}$$

$$f(-z) = -\frac{z^2\sin z}{z^2 + \cos z}$$

$f(z) = -f(z)$ function is ODD

$\therefore \quad I = 0$

135. (a)

$f(x) = \frac{1}{x^2}$ is not defined at $x = 0$

So, $$I = \int_{-1}^{0} \frac{1}{x^2}dx + \int_0^1 \frac{1}{x^2}dx$$

$$= \left(-\frac{1}{x}\right)_{-1}^{0} + \left(-\frac{1}{x}\right)_0^1 = \infty$$

$\frac{1}{x}$ is not defined at $x = 0$

Therefore, the integral does not converge

136. (d)

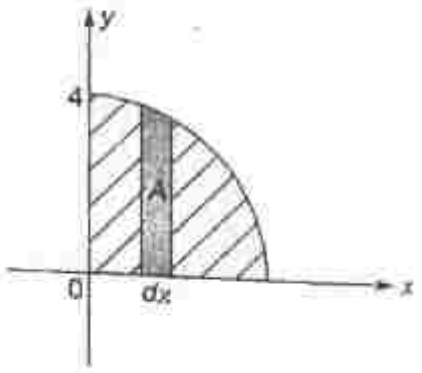
Comparing with $\int_a^b y\,dx$, for given integral

$$y = \sqrt{16 - x^2}$$

It can also be written as

$$x^2 + y = 16$$

This is equation of a circle of radius 4 centered origin as shown below.



Thus, $\int_0^4 \sqrt{16-x^2}\,dx$ = Area of circle in I quadrant

$$= \frac{1}{4}[\pi(16)] = 4\pi$$

137. (c)

$$\int_0^2 \frac{3x}{\sqrt{2x^2+1}}\,dx$$

Let, $2x^2 + 1 = t$

$\Rightarrow$ $4x\,dx = dt$

$$x\,dx = \frac{dt}{4}$$

$$\therefore \int_0^2 \frac{3x}{\sqrt{2x^2+1}}\,dx = \int_1^9 \frac{3}{\sqrt{t}}\left(\frac{dt}{4}\right) = \frac{3}{4}\int_1^9 \frac{1}{\sqrt{t}}\,dt$$

$$= \frac{3}{4}\left[\frac{\sqrt{t}}{\frac{1}{2}}\right]_1^9 = \frac{3}{2}\left(\sqrt{9}-\sqrt{1}\right) = 3$$

138. (a)

$$\int_0^{2\pi} \frac{3}{9+\sin^2\theta}\,d\theta = \int_0^{2\pi} \frac{3\sec^2\theta}{9\sec^2\theta+\tan^2\theta}\,d\theta$$

$$= 4\int_0^{\pi/2} \frac{3\sec^2\theta}{9\sec^2\theta+\tan^2\theta}\,d\theta = 4\int_0^{\pi/2} \frac{3\sec^2\theta}{9+10\tan^2\theta}\,d\theta$$

$$= \frac{12}{10}\int_0^{\pi/2} \frac{\sec^2\theta}{\frac{9}{10}+\tan^2\theta}\,d\theta$$

	Limits
Let, $\tan\theta = t$	$\theta = 0 \quad t = 0$
$\sec^2\theta\,d\theta = dt$	$\theta = \frac{\pi}{2} \quad t \to \infty$

$$= \frac{12}{10} \times \int_0^{\pi/2} \frac{dt}{\left(\sqrt{\frac{9}{10}}\right)^2 + t^2}$$

$$= \frac{12}{10}\cdot\left[\frac{1}{\frac{3}{\sqrt{10}}}\tan^{-1}\frac{t}{\frac{3}{\sqrt{10}}}\right]_0^{\infty}$$

$$= \frac{4}{\sqrt{10}}\left[\frac{\pi}{2}-0\right] = \frac{2\pi}{\sqrt{10}}$$

139. (a)

Let, $\sin^{-1} x = t$

$$\frac{dx}{\sqrt{1-x^2}} = dt$$

$$I = \int_0^{\pi/2} t^2\,dt = \left[\frac{t^3}{3}\right]_0^{\pi/2} = \frac{\pi^3}{24}$$

140. (c)

$$\int_0^{\pi/2} \frac{dx}{1+\sqrt{\frac{\cos x}{\sin x}}} = \int_0^{\pi/2} \frac{\sqrt{\sin x}\,dx}{\sqrt{\sin x}+\sqrt{\cos x}}$$

$$x \to \frac{\pi}{2} - x$$

Since, $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}\,dx}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos x}\,dx}{\sqrt{\sin x}+\sqrt{\cos x}}$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x}+\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx = \int_0^{\pi/2} dx$$

$$I = \frac{\pi}{4} = 0.78$$

141. (d)

$u = \sin x$

$du = \cos x\,dx$

$x = \frac{\pi}{2} \Rightarrow u = \sin\frac{\pi}{2} = 1$

$x = -\pi \Rightarrow u = \sin(-\pi) = 0$

$$\int_{-\pi}^{\pi/2} \cos(x)\cos(\sin(x))dx = \int_0^1 \cos u\,du$$

$$= |\sin u|_0^1 = (\sin 1) - \sin(0) = \sin 1$$

142. (a)

$$|x-2| = \begin{cases} -(x-2); & x<2 \\ (x-2); & x>2 \end{cases}$$

$$\int_1^3 \frac{|x-2|}{x}dx = \int_1^2 \frac{-(x-2)}{x}dx + \int_2^3 \frac{x-2}{x}dx$$

$$= \int_1^2 \left(-1+\frac{2}{x}\right)dx + \int_2^3 \left(1-\frac{2}{x}\right)dx$$

$$= [-x]_1^2 + [2\ln x]_1^2 + [x]_2^3 - 2[\ln x]_2^3$$

$$= -(2-1) + 2\ln 2 - 2\ln\frac{3}{2} + (3-2)$$

$$= 2\ln 2 - 2\ln\frac{3}{2} = 2\ln\frac{2}{\frac{3}{2}} = 2\ln\frac{4}{3}$$

143. (d)

Let $u = \sqrt{x}$

Then $du = \frac{1}{2\sqrt{x}}dx$

$\therefore dx = du \cdot 2\sqrt{x}$

$$\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx = \int \frac{\sin u}{\sqrt{x}} \cdot 2\sqrt{x}\,du$$

$$= 2\int \sin u\,du$$

$$= -2\cos\sqrt{x} + c$$

144. (d)

$$I = \int_0^{\pi/2} \log\left(\frac{\sin x}{\cos x}\right)dx$$

$$= \int_0^{\pi/2} [\log(\sin x)dx - \log(\cos x)dx]$$

$$= \int_0^{\pi/2} \log\sin\left(\frac{\pi}{2}-x\right)dx - \int_0^{\pi/2} \log(\cos x)dx$$

$I = 0$

145. (1)

Let, $\ln x = t$

Then, $dt = \frac{1}{x}dx$

Hence, $I = \int \sin t\,dt$

$= -\cos t + c = -\cos(\ln x) + c$

146. (d)

$u = \sin x$

$du = \cos x\,dx$

$x = \frac{\pi}{2} \Rightarrow u = \sin\frac{\pi}{2} = 1$

$x = -\pi \Rightarrow u = \sin(-\pi) = 0$

$$\int_{-\pi}^{\pi/2} \cos(x)\cos(\sin(x))dx = \int_0^1 \cos u\,du$$

$= |\sin u|_0^1 = (\sin 1) - \sin(0) = \sin 1$

147. (30)

$$\int_{-3}^{4} (|x+2| + |x-1|)dx$$

$$\Rightarrow \int_{-3}^{4} |x+2|dx + \int_{-3}^{4} |x-1|dx$$

$$\Rightarrow \int_{-3}^{-2} -(x+2)dx + \int_{-2}^{4} (x+2)dx + \int_{-3}^{+1} -(x+1)dx + \int_{+1}^{4} (x-1)dx$$

$$\Rightarrow -\left[\left(\frac{x^2}{2}+2x\right)\right]_{-3}^{-2} + \left[\left(\frac{x^2}{2}+2x\right)\right]_{-2}^{4} + \left[-\left(\frac{x^2}{2}-x\right)\right]_{-3}^{1} + \left[\left(\frac{x^2}{2}-x\right)\right]_{1}^{4}$$

$$= -\frac{1}{2} + 18 + 8 + \frac{9}{2} = 30$$

UNIT 3

Probability and Statistics

1. Counting Techniques

Q.1 It is required to divide the $2n$ members of a club into n disjoint teams of 2 members each. The teams are not labelled. The number of ways in which this can be done is:

(a) $\frac{(2n!)}{(2^n)}$ (b) $\frac{(2n!)}{n!}$

(c) $\frac{(2n!)}{(2^n \cdot n!)}$ (d) $\frac{(n!)}{2}$

Q.2 Let A be a sequence of 8 distinct integers sorted in ascending order. How many distinct pairs of sequences, B and C are there such that (i) each is sorted in ascending order, (ii) B has 5 and C has 3 elements, and (iii) the result of merging B and C gives A?

(a) 2 (b) 30

(c) 56 (d) 256

Q.3 In how many ways can 5 people be arranged in a row so, a particular person always in middle?

(a) 120 (b) 24

(c) 48 (d) 60

Q.4 Mala has a colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour pairs used to colour any two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

(a) 9 (b) 8

(c) 7 (d) 6

Q.5 Out of 7 consonants and 4 vowels, how many words of 8 consonants and 2 vowels can be formed?

(a) 25200

(b) 21300

(c) 24400

(d) 210

Q.6 There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that:

- The fastest computer gets the toughest job and the slowest computer gets the easiest job.
- Every computer gets at least one job.

The number of ways in which this can be done is _____.

Q.7 In a group of 6 boys and 4 girls, four children to be selected. In how many different ways can they can be selected such that atleast one boy be there?

(a) 212 (b) 209

(c) 159 (d) 201

Q.8 In how many ways can 11 men and 8 women sit in a row if all the men sit together and all the women sit together.

(a) 11! 8! (b) $\frac{11!\,8!}{2}$

(c) 2.11! 8! (d) None of these

Q.9 In how many ways can we distribute 5 distinct balls, $B_1, B_2, \ldots, B_5$ in 5 distinct cells, $C_1, C_2, \ldots, C_5$ such that Ball B_i is not in cell C_i, $\forall_i = 1, 2, \ldots, 5$ and each cell contains exactly one ball?

(a) 44 (b) 96

(c) 120 (d) 3125

Q.10 Suppose that a state's license plates consist of 3 letters followed by 3 digit then total number of different places can be formed (no repetitions allowed) _____?

Q.11 There are 15 teachers in a school in which 8 are male teachers and 7 are female teachers. The director selects a male teacher and a female teacher for Headmaster of the school. In how many ways the director can make this selection?

(a) 30 (b) 105

(c) 56 (d) 60

Q.12 In how many ways can be the letters of word ABACUS be rearranged such that the vowels appear together?

(a) $\frac{(6+3)!}{2!}$ (b) $\frac{6!}{2!}$

(c) $\frac{3!3!}{2!}$ (d) $\frac{4!3!}{2!}$

Q.13 In how many different ways can be the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

Q.14 Find the number of arrangement that can be made from the letters of the word MADEEASY, without changing the place of vowels in the word?

(a) 6 (b) 24

(c) 120 (d) None of these

Q.15 Find the number of possible 5 character passwords, if all characters must be lower case letters and distinct. Assume characters are only from *a* to *z*.

(a) $(26)^5$ (b) $\frac{26!}{16!}$

(c) $\frac{26!}{21!}$ (d) ${}^{26}C_5$

Q.16 There are 12 intermediate stations between two places A and B. Find the number of ways in which a train can be made to stop at 4 of these intermediate stations so that no two stopping stations are consecutive?

Q.17 How many 10 letter words can be formed using the letters of the word "MEGATE2026" such that all consonants come together? _____

Q.18 In how many ways can 4 girls and 5 boys be arranged in a row so that all the 4 girls are together?

(a) 1440

(b) 4320

(c) 17280

(d) 86400

Q.19 In how many ways can 30 identical apples be divided among 10 boys.

(a) $\frac{30!}{10!}$ (b) ${}^{40}C_{10}$

(c) ${}^{39}C_9$ (d) ${}^{30}C_{10}$

Q.20 The number of ways of distributing 52 cards to 4 people with a hand of 5 cards each is

(a) $\frac{52!}{(5!)^4\,4!}$ (b) $\frac{52!}{(5!)^4}$

(c) $\frac{52!}{(5!)^4\,(32)!}$ (d) $\frac{52!}{(5!)^4\,(32)!\,4!}$

Q.21 The number of permutation of {1, 2, 3, 4, 5} that keeps at least one integer fixed is

(a) 81 (b) 76

(c) 120 (d) 60

Q.22 In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada. 9 persons speak both English and Hindi, 11 person speak both Hindi and Kannada whereas 13 persons speak both Kannada and English. How many people speak all three languages?

(a) 9 (b) 8

(c) 7 (d) 6

Q.23 If the letters of word COMPUTER be arranged in random order, the number of arrangements in which the three vowels O, U and E occur together is

(a) 8! (b) 6!

(c) 3!6! (d) None of these

Q.24 Which of the following statement is/are true?

(a) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_{r+1}$

(b) ${}^nC_x = {}^nC_y \Rightarrow x = y$

(c) $n \cdot {}^{n-1}C_{r-1} = (n - r + 1)\,{}^nC_{r-1}$

(d) ${}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_r$

Q.25 Consider all possible words obtain by arranging all the letters of the word AGAIN. These words are now arranged in the alphabetical order, as in dictionary. The fiftieth word in this arrangement is

(a) IAANG (b) NAAGI

(c) NAAIG (d) IAAGN

Q.26 It is required to divide the $2n$ members of a club into 'n' disjoint teams of 2 members each. The teams are not labelled. The number of ways in which this can be done is:

(a) $\frac{(2n)!}{2^n}$ (b) $\frac{(2n)!}{n!}$

(c) $\frac{(2n)!}{2^n \cdot n!}$ (d) $\frac{n!}{2}$

Q.27 A new flag of gate club is to be designed with 5 vertices strips using some or all of the four vertices strips using some or all the four colours: green, maroon, red and yellow. In how many ways this can be done so that no two adjacent strips have the same colour?

(a) 120 (b) 424
(c) 324 (d) 526

Q.28 If $\binom{n}{r+1} = 36$, $\binom{n}{4} = 84$ and $\binom{n+1}{r+1} = 126$ then what is value of r? _____

Q.29 There are n kingdoms and $2n$ champions. Each kingdom gets 2 champion. The number of ways in which this can be done is:

(a) $\frac{(2n)!}{2^n}$ (b) $\frac{(2n)!}{n!}$
(c) $\frac{(2n)!}{n!2^n}$ (d) $\frac{n!}{2}$

Q.30 A three-persons committee has to formed from a group of 9 friends. _____ such distinct committees can be formed.

Q.31 Which of the following statement(s) is/are correct?

(a) A bag contains 25 balls such as 10 balls are red, 7 are white and 8 are blue. Then the minimum number of balls that must be picked up from the bag blindfolded (without replacing any of it) to be assured of picking at least one ball of each colour is 15.
(b) A bag contains 25 balls such as 10 balls are red, 7 are white and 8 are blue. Then the minimum number of balls that must be picked up from the bag blindfolded (without replacing any of it) to be assured of picking at least one ball of each colour is 18.
(c) In a get-together party, every person present shakes the hand of every other person. If there were 105 handshakes in all, Then number of persons were present at the party are 14.
(d) In a get-together party, every person present shakes the hand of every other person. If there were 105 handshakes in all, Then number of persons were present at the party are 18.

Q.32 The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

(a) n (b) n^2
(c) $\frac{n(n-1)}{2}$ (d) $\frac{n(n+1)}{2}$

Q.33 Suppose you can travel from a place P to a place Q by 2 auto, from place Q to place R by 3 buses, from place R to place S by 4 train and from place S to place T by 3 buses. In how many ways can you travel from P to T?

Q.34 How many ways the letters of the word "MADEEASY" can be arranged?

(a) 10080 (b) 20106
(c) 40320 (d) None of these

Q.35 Two girls have picked 10 roses, 15 sunflowers and 14 daffodils. What is the number of ways they can divide the flowers among themselves?

(a) 1638 (b) 2100
(c) 2640 (d) None of these

Q.36 The number of five digit numbers, all digits of which are even is (without repetition) _____.

(a) 120 (b) 60
(c) 96 (d) 192

Q.37 The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is _____.

Q.38 In a steamer there are stalls for 10 pets and there are cats, dogs and rabbits (not less than 10 each) ready to be shipped. In how many ways they can be loaded?

(a) 10^3 (b) 10!
(c) 3^{10} (d) None of these

Q.39 n couples are invited to a party with the condition that every husband should be accompanied by his wife. However, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is

(a) $\binom{2n}{n} * 2^n$ (b) 3^n
(c) $\frac{(2n)!}{2^n}$ (d) $\binom{2n}{n}$

Q.40 In how many different ways can the letters of the word "ANIRUDDHA" be arranged in such a way that the vowels occupy only the even positions?

(a) 720 (b) 1440
(c) 120 (d) 576

Q.41 The number of arrangements of six identical balls in three identical bins is _____.

Q.42 A bag contains 4 white balls, 2 black balls and 3 red balls. In how many ways can 4 balls be drawn from the bag if at least one red ball is to be included in the draw?

(a) 111 (b) 126
(c) 141 (d) None of these

Q.43 A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

(a) $\frac{1}{5}$ (b) $\frac{4}{25}$
(c) 1.4 (d) $\frac{2}{5}$

Q.44 There are two containers, with one containing 4 red and 3 green balls and the other containing 3 blue and 4 green balls. One ball is drawn at random from each container. The probability that one of the balls is red and the other is blue will be

(a) $\frac{1}{7}$ (b) $\frac{9}{49}$
(c) $\frac{12}{49}$ (d) $\frac{3}{7}$

Q.45 A product is an assemble of 5 different components. The product can be sequentially assembled in two different ways. If 5 components are placed in a box and these are drawn at random from the box, then probability of getting a correct sequence is

(a) $\frac{2}{5!}$ (b) $\frac{2}{5}$
(c) $\frac{2}{(5-2)!}$ (d) $\frac{2}{(5-3)!}$

2. Fundamentals of Probability

Q.46 A die is rolled thrice. What is the probability getting an odd number each time.

(a) $\frac{3}{3}$ (b) $\frac{1}{8}$
(c) $\frac{7}{8}$ (d) $\frac{3}{4}$

Q.47 There are 11 men and 9 female in a group. If five people are selected at random, what is the probability that 3 female and 2 male are selected

(a) 0.28 (b) 0.32
(c) 0.203 (d) 0.15

Q.48 Let A and B be any two arbitrary events, then which one of the following is true?

(a) $P(A \cap B) = P(A)\,P(B)$
(b) $P(A \cup B) = P(A) + P(B)$
(c) $P(A \mid B) = P(A \cap B)/P(B)$
(d) $P(A \cup B) < P(A) + P(B)$

Q.49 A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\left(\frac{1}{8}\right)$ and the probability of wife's selection is $\left(\frac{1}{6}\right)$. What is the probability that only one of them is selected?

(a) $\frac{2}{7}$ (b) $\frac{1}{7}$
(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

Q.50 A bag contains 10 white balls and 15 black balls. Two balls are drawn in succession. The probability that one of them is black and the other is white is

(a) $\frac{2}{3}$ (b) $\frac{4}{5}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Q.51 A speaks truth in 60% of cases and B in 70% of cases. In what percentage of cases are they likely to contradict each other, narrating the same incident?

(a) 70% (b) 60%
(c) 46% (d) 50%

Q.52 Two dice are thrown simultaneously. The probability that at least one of them will have 6 facing up is

(a) $\frac{1}{36}$ (b) $\frac{1}{3}$

(c) $\frac{25}{36}$ (d) $\frac{11}{36}$

Q.53 The probability that top and bottom cards of a randomly shuffled deck are both aces is

(a) $\frac{4}{52}\times\frac{4}{52}$ (b) $\frac{4}{52}\times\frac{3}{52}$

(c) $\frac{4}{52}\times\frac{3}{51}$ (d) $\frac{4}{52}\times\frac{4}{51}$

Q.54 The probability that it will rain today is 0.5. The probability that it will rain tomorrow is 0.6. The probability that it will rain either today or tomorrow is 0.7. What is the probability that it will rain today and tomorrow?

(a) 0.3 (b) 0.25

(c) 0.35 (d) 0.4

Q.55 In a class, 40% of the students offered English, 30% offered Hindi and 20% offered both. If a student is selected at random, what is the probability that he has offered English but not Hindi?

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$

(c) $\frac{1}{5}$ (d) $\frac{2}{5}$

Q.56 A die is rolled three times. The probability that exactly one odd number turns up among the three outcomes is

(a) $\frac{1}{6}$ (b) $\frac{3}{8}$

(c) $\frac{1}{8}$ (d) $\frac{1}{2}$

Q.57 Two dice are thrown simultaneously and the sum of the numbers obtained is found to be odd. What is the probability that the number 5 has appeared only on one dice?

(a) $\frac{1}{6}$ (b) $\frac{1}{4}$

(c) $\frac{2}{3}$ (d) $\frac{1}{4}$

Q.58 Suppose that the expectation of a random variable X is 5. Which of the following statements is true?

(a) There is a sample point at which X has the value 5.

(b) There is a sample point at which X has value greater than 5.

(c) There is a sample point at which X has a value greater than or equal to 5.

(d) None of the above

Q.59 Consider two events E_1 and E_2 such that probability of E_1, $\Pr[E_1] = \frac{1}{2}$, probability of E_2, $\Pr[E_2] = \frac{1}{3}$, and probability of E_1 and E_2, $\Pr[E_1 \text{ and } E_2] = \frac{1}{5}$.

Which of the following statements is/are true?

(a) $\Pr[E_1 \text{ or } E_2]$ is $\frac{2}{3}$

(b) Events E_1 and E_2 are independent

(c) Events E_1 and E_2 are not independent

(d) $\Pr\left[\frac{E_1}{E_2}\right] = \frac{4}{5}$

Q.60 E_1 and E_2 are events in a probability space satisfying the following constraints:

- $\Pr(E_1) = \Pr(E_2)$
- $\Pr(E_1 \cup E_2) = 1$
- E_1 and E_2 are independent

The value of $\Pr(E_1)$, the probability of the event E_1, is

(a) 0 (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) 1

Q.61 Seven (distinct) car accidents occurred in a week. What is the probability that they all occurred on the same day?

(a) $\frac{1}{7^7}$ (b) $\frac{1}{7^6}$

(c) $\frac{1}{2^7}$ (d) $\frac{7}{2^7}$

Q.62 Four fair coins are tossed simultaneously. The probability that at least one head and one tail turn up is

(a) $\frac{1}{16}$ (b) $\frac{1}{8}$

(c) $\frac{7}{8}$ (d) $\frac{15}{16}$

Q.63 Let P(E) denote the probability of the event E. Given P(A) = 1, P(B) = $\frac{1}{2}$, and A and B are independent, the values of P(A | B) and P(B | A) respectively are

(a) $\frac{1}{4}, \frac{1}{2}$ (b) $\frac{1}{2}, \frac{1}{4}$

(c) $\frac{1}{2}, 1$ (d) $1, \frac{1}{2}$

Q.64 Two dice are tossed. The probability of getting a sum of 4 is

(a) $\frac{1}{9}$ (b) $\frac{1}{12}$

(c) $\frac{1}{13}$ (d) $\frac{1}{6}$

Q.65 A sample space has two events A and B such that probabilities

$$P(A \cap B) = \frac{1}{2},\ P(\bar{A}) = \frac{1}{3},\ P(\bar{B}) = \frac{1}{3}$$

What is $P(A \cup B)$?

(a) $\frac{11}{12}$ (b) $\frac{10}{12}$

(c) $\frac{9}{12}$ (d) $\frac{8}{12}$

Q.66 Two card is randomly drawn from a pack of 52 cards. What is the probability that the cards drawn are the face cards (Jack, Queen or King) or Ace cards.

(a) $\frac{13}{221}$ (b) $\frac{11}{221}$

(c) $\frac{12}{221}$ (d) $\frac{1}{21}$

Q.67 Suppose we uniformly and randomly select a permutation from the 20! permutations of 1, 2, 3, ..., 20. What is the probability that 2 appears at an earlier position that any other even number in the selected permutation?

(a) $\frac{1}{2}$ (b) $\frac{1}{10}$

(c) $\frac{9!}{20!}$ (d) None of these

Q.68 A box contains 10 screws, 3 of which are defective. Two screws are drawn at random with replacement. The probability that none of the two screws is defective will be

(a) 100% (b) 50%

(c) 49% (d) None of these

Q.69 A box contains 5 black and 5 red balls. Two balls are randomly picked one after another from the box, without replacement. The probability for both balls being red is

(a) $\frac{1}{90}$ (b) $\frac{1}{2}$

(c) $\frac{19}{90}$ (d) $\frac{2}{9}$

Q.70 For a post three persons *X*, Y and *Z* appear in the interview. The probability of *X* being selected is half that of *Y* and the probability of *Y* being selected is thrice that of *Z*, what is the product of the individual probabilities of *X*, *Y*, *Z* being selected? _____

Q.71 If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

(a) $\frac{3}{8}$ (b) $\frac{1}{2}$

(c) $\frac{5}{8}$ (d) $\frac{3}{4}$

Q.72 Let *A*, *B* and *C* be three independent events each having the same probability p. Let $\bar{C}$ denote the complement of *C*. Then the conditional probability $P\left\{\frac{\bar{C}}{A \cap (B \cup C)}\right\}$ equals

(a) $\frac{1-p}{2-p}$ (b) $\frac{p}{1+p}$

(c) $1 - p$ (d) $\frac{1}{1+p}$

Q.73 A single die is thrown twice. What is the probability that the sum is neither 8 nor 9?

(a) $\frac{1}{9}$ (b) $\frac{5}{36}$

(c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.74 If P and Q are two random events, then the following is TRUE

(a) Independence of P and Q implies that probability $(P \cap Q) = 0$

(b) Probability $(P \cup Q) \geq$ Probability (P) + Probability (Q)

(c) If P and Q are mutually exclusive, then they must be independent

(d) Probability $(P \cap Q) \leq$ Probability (P)

Q.75 Let A and B be two events. Given that $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{8}$ and $P(A \cap B) = \frac{1}{7}$, then the value of $P\left(\frac{A}{B}\right) - P\left(\frac{B}{A}\right)$ is

(a) $\frac{1}{7}$ (b) $\frac{3}{7}$

(c) $\frac{2}{9}$ (d) $\frac{2}{7}$

Q.76 Two fair dice are rolled and the sum r of the numbers turned up is considered

(a) $\Pr(r > 6) = \left(\frac{1}{6}\right)$

(b) $\Pr(\frac{r}{3}$ is an integer$) = \left(\frac{5}{6}\right)$

(c) $\Pr(r = 8 \mid \frac{r}{4}$ is an integer$) = \left(\frac{5}{9}\right)$

(d) $\Pr(r = 6 \mid \frac{r}{5}$ is an integer$) = \left(\frac{1}{18}\right)$

Q.77 A box contains 20 defective items and 80 non-defective items. If two items are selected at random without replacement, what will be the probability that both items are defective?

(a) $\frac{1}{5}$ (b) $\frac{1}{25}$

(c) $\frac{20}{99}$ (d) $\frac{19}{495}$

Q.78 A fair dice is tossed two times. The probability that the second toss results in a value that is higher than the first toss is

(a) $\frac{2}{36}$ (b) $\frac{2}{6}$

(c) $\frac{5}{12}$ (d) $\frac{1}{2}$

Q.79 Suppose A and B are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$. Let $\bar{A}$ and $\bar{B}$ be their complements. Which one of the following statements is FALSE?

(a) $P(A \cap B) = P(A)\,P(B)$

(b) $P(A/B) = P(A)$

(c) $P(A \cup B) = P(A) + P(B)$

(d) $P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$

Q.80 If $P(X) = \frac{1}{4}$, $P(Y) = \frac{1}{3}$, and $P(X \cap Y) = \frac{1}{12}$, the value of $P\left(\frac{Y}{X}\right)$ is

(a) $\frac{1}{4}$ (b) $\frac{4}{25}$

(c) $\frac{1}{3}$ (d) $\frac{29}{50}$

Q.81 X and Y are two random independent events. It is known that $P(X) = 0.40$ and $P(X \cup Y^c) = 0.7$. Which one of the following is the value of $P(X \cup Y)$?

(a) 0.7 (b) 0.5

(c) 0.4 (d) 0.3

Q.82 If X and Y are independent events, then which of the following statement is/are true?

(a) $\bar{X}$ and $\bar{Y}$ are independent events.

(b) X and $\bar{Y}$ are independent events.

(c) $\bar{X}$ and Y are independent events.

(d) None of these

Q.83 A six-faced fair dice is rolled five times. The probability (in%) of obtaining "ONE" at least four times is

(a) 33.3 (b) 3.33

(c) 0.33 (d) 0.0033

Q.84 A bag contains 50 balls of which 10 are red and the remainder black. If two balls are drawn successively from the bag at random, what is the probability of selecting one red and one black ball?

(a) $\frac{9}{245}$ (b) $\frac{156}{245}$

(c) $\frac{16}{49}$ (d) $\frac{156}{49}$

3. Expectation and Variance

Q.85 If X and Y are independent random variables then which of the following statement is/are true?

(a) $E(2X + 3Y) = 2E(X) + 3E(Y) + 12E(XY)$

(b) $V(2X + 3Y) = 4V(X) + 3V(Y)$

(c) Covariance $(X, Y) \neq 0$

(d) $E(X/Y) = E(X)$

Q.86 Which one of the following statements is NOT true?

(a) The measure of skewness is dependent upon the amount of dispersion

(b) In a symmetric distribution, the values of mean, mode and median are the same

(c) In a positively skewed distribution: mean > median > mode

(d) In a negatively skewed distribution: mode > mean > median

Q.87 If 10 is added to each and every item of a data, then the arithmetic mean

(a) is increased by 10 times

(b) is not increased

(c) is greater by 10 (d) None of these

Q.88 Let X and Y be two independent random variables. Which one of the relations between expectation (E), variance (Var) and covariance (Cov) given below is FALSE?

(a) $E(XY) = E(X)\,E(Y)$

(b) Cov $(X, Y) = 0$

(c) Var $(X + Y)$ = Var (X) + Var (Y)

(d) $E(X^2\,Y^2) = (E(X))^2\,(E(Y))^2$

Q.89 Standard deviation of

27, 35, 40, 35, 36, 36, 29 is

(a) 17.44 (b) 4.14

(c) 34 (d) None of these

Q.90 The density function of a random variable X is given by $f(x) = \frac{x}{2}$ $0 < x < 2 = 0$ otherwise.

Then the mean and variance of X are

(a) $\frac{4}{3}, \frac{2}{9}$ (b) $\frac{2}{3}, \frac{4}{9}$

(c) $\frac{4}{3}, \frac{4}{9}$ (d) $\frac{2}{3}, \frac{2}{9}$

Q.91 If the variance for random variable X is 225, then X has a Standard Deviation equal to _____.

Q.92 The random variable X takes on the values 1, 2 (or) 3 with probabilities $\frac{2+5P}{5}, \frac{1+3P}{5}$ and $\frac{1.5+2P}{5}$ respectively the values of P and $E(X)$ are respectively.

(a) 0.05, 1.87 (b) 1.90, 5.87

(c) 0.05, 1.10 (d) 0.25, 1.40

Q.93 If mean = (3 Median – Mode) X, then the value of 'X' is

(a) 1 (b) 2

(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

Q.94 Consider a Gaussian distributed random variable with zero mean and standard deviation σ. The value of its cumulative distribution function at the origin will be

(a) 0 (b) 0.5

(c) 1 (d) 10 s

Q.95 Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean –1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$ the standard deviation of Y is

(a) 3 (b) 2

(c) $\sqrt{2}$ (d) 1

Q.96 The probability distribution for a discrete random variable X is given below:

X	1	2	3	4
$P(X)$	0.3	0.4	0.2	0.1

Find:

(i) The expected value of X

(ii) Variance of X

Q.97 If a random variable X satisfies the Poisson's distribution with a mean value of 2, then probability that $X \geq 2$ is

(a) $2e^{-2}$ (b) $1 - 2e^{-2}$

(c) $3e^{-2}$ (d) $1 - 3e^{-2}$

Q.98 If the difference between the expectation of the square of a random variable $(E[x^2])$ and the square of the expectation of the random variable $(E[x])^2$ is denoted by R, then

(a) $R = 0$ (b) $R < 0$

(c) $R \geq 0$ (d) $R > 0$

Q.99 Let X be a normal random variable with mean 1 and variance 4. The probability $P[X < 0]$ is
(a) 0.5
(b) greater than zero and less than 0.5
(c) greater than 0.5 and less than 1.0
(d) 1.0

Q.100 Consider the following dataset 3, 5, 7, 9, 11, 13. What is the standard deviation?
(a) 12 (b) 11
(c) 11.66 (d) 3.4

Q.101 Marks obtained by 100 students in an examination are given in the table:

Sr. No.	Marks obtained	No. of students
1	25	20
2	30	20
3	35	40
4	40	20

What would be mean, median and mode of marks obtained by the students?
(a) Mean 33; Median 35; Mode 40
(b) Mean 35; Median 32; Mode 40
(c) Mean 33; Median 35; Mode 35
(d) Mean 35; Median 32; Mode 35

Q.102 The mean of 30 values was calculated as 64.5, as it was found that a value 58 was miss read as 85. What is the correct mean?
(a) 68 (b) 72
(c) 68.5 (d) 63.6

Q.103 Arithmetic mean of the natural numbers from 1 to n is
(a) $\frac{n(n+1)}{2}$ (b) $\frac{(n+1)}{2}$
(c) $\frac{n}{2}$ (d) None of these

Q.104 In the following table, x is a discrete random variable and $p(x)$ is the probability density. The standard deviation of x is

x	1	2	3
$p(x)$	0.3	0.6	0.1

(a) 0.18 (b) 0.36
(c) 0.54 (d) 0.6

Q.105 A sample of 15 data is as follows:
17, 18, 17 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3
The mode of the data is
(a) 4 (b) 13
(c) 17 (d) 20

Q.106 If X is a normal variable with mean 30 and standard deviation 5, what is probability $(26 \le X \le 34)$, given $A(z = 0.8) = 0.2881$?
(a) 0.2881 (b) 0.5762
(c) 0.8181 (d) 0.1616

Q.107 A probability distribution with right skew is shown in the figure.

The correct statement for the probability distribution is
(a) Mean is equal to mode
(b) Mean is greater than median but less than mode
(c) Mean is greater than median and mode
(d) Mode is greater than median

Q.108 If X and Y are random variables such that $E[2X + Y] = 0$ and $E[X + 2Y] = 33$, then $E[X] + E[Y] =$ _____.

Q.109 In a sample of 100 students, the mean of the marks (only integers) obtained by them in a test is 14 with its standard deviation of 2.5 (marks obtained can be fitted with a normal distribution). The percentage of students scoring 16 marks is
(a) 36 (b) 23
(c) 12 (d) 10
(Area under standard normal curve between $z = 0$ and $z = 0.6$ is 0.2257; and between $z = 0$ and $z = 1.0$ is 0.3413)

Q.110 The variance of the first 'n' natural number is
(a) $\frac{n^2 - 1}{12}$ (b) $\frac{n^2 - 1}{6}$
(c) $\frac{n^2 + 1}{6}$ (d) $\frac{n^2 + 1}{12}$

Q.111 If x is the mean of data 3, x, 2 and 4, then the mode is _____.

Q.112 If x is a random variable with the expected value of 5 and the variance of 1, then the expected value of x^2 is
(a) 36 (b) 26
(c) 25 (d) 24

Q.113 Calculate the arithmetic mean for following data:
32, 30, 28, 21, 26, 17, 27, 31, 34, 18, 22
(a) 26 (b) 27
(c) 22 (d) 33

Q.114 Consider the following dataset:
55, 100, 75, 80, 90, 85, 95, 45, 70, 70, 55
Find the median.
(a) 75
(b) 85
(c) 90
(d) None of these

Q.115 The sum of two normally distributed random variables X and Y is
(a) normally distributed, only if X and Y have the same standard deviation
(b) always normally distributed
(c) normally distributed, only if X and Y have the same mean
(d) normally distributed, only if X and Y are independent

Q.116 The mean and variance, respectively, of a binomial distribution for n independent trials with the probability of success as p, are
(a) $\sqrt{np}, \sqrt{np(1-p)}$
(b) $np, np(1-p)$
(c) np, np
(d) $\sqrt{np}, np(1-2p)$

Q.117

Class	0 – 10	10 – 20	20 – 30
Frequency	5	14	6

mean is
(a) 8.55 (b) 8
(c) 9 (d) 15.4

Data for Q.118 to Q123 questions:

Let X be a random variable and its probability mass function is shown below: $(k > 0)$

x	0	1	2	3	5	6
$P(X = x)$	$\frac{k}{3}$	$\frac{k}{6}$	$\frac{k}{4}$	$\frac{3k}{16}$	$\frac{k}{32}$	$\frac{k}{32}$

Note:
1. $P(X = x)$ represents the probability at $X = x$.
2. $E(X)$ represents the expectation of X.
3. $V(X)$ represents the variance of X.
4. SD (X) represents the standard deviation of X

Q.118 What is the value of $P(X = 1.5)$? _____

Q.119 What is the value of $P(X \le 2)$? _____

Q.120 What is the value of $E(X)$?
(a) $\frac{115}{96}$ (b) $\frac{151}{96}$
(c) $\frac{145}{96}$ (d) None of these

Q.121 What is the value of $V(X)$?

Q.122 What is the value of $E(3X + 1)$? _____

Q.123 What is the value of $V(2X - 4)$? _____

Q.124 The spot speeds (expressed in km/hr) observed at a road section are 60, 56, 39, 73, 26, 45, 50, 54, 77, and 33. The median speed (expressed in km/hr) is _____.

Q.125 Arithmetic mean of
24, 28, 29, 34, 18, 22, 26, 30, 32, 24, 20 is
(a) 26.09 (b) 26
(c) 24 (d) 34

Q.126 Which of the following statements are true?
(a) The dispersion measure characterizes the extent to which data items differ from the central tendency value.
(b) The central tendency measure indicates the average value of data.
(c) Standard deviation is example of central tendency measure.
(d) Mean is example of central tendency measure.

Q.127 Suppose X is a continuous random variable with probability density function:

$$f(x) = \begin{cases} 1+x; & -1 \le x \le 0 \\ 1-x; & 0 < x < 1 \end{cases}$$

Then the mean standard deviation of X are

Q.128 The probability mass function of an random variable X is given as

$$P(X, a) = \begin{cases} a, & \text{if } X = 0 \\ 1-a, & \text{if } X = 1 \\ 0, & \text{otherwise} \end{cases}$$

If $a = 0.6$, then variance of X is _____
(a) 0 (b) 0.36
(c) 0.24 (d) 0.76

Q.129 A random variable X has probability density function:

$$f(x) = \begin{cases} a + bx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Given the expected value $E(X) = \frac{2}{3}$ the $\Pr[X < 0.5]$ is _____.

Q.130 Suppose the probability density function of a Poisson distribution is given by $f(x) = \frac{e^{-4}4^x}{x!}$ then its mean is _____.

Q.131 If mean and variance of a binomial distribution are $\frac{10}{8}$ and $\frac{30}{32}$ respectively, the probability of occurrence p is _____.

Q.132 In a binomial distribution, the mean is 9 and the standard deviation (σ) is $\sqrt{6}$. The value of n (total number of trials) and q (probability of failure of the event in each trial) respectively are:

(a) $27, \frac{1}{3}$ (b) $27, \frac{2}{3}$

(c) $36, \frac{3}{4}$ (d) $18, \frac{1}{2}$

Q.133 If x is a random variable which is uniformly distributed in the interval [0, 1], then the value $E[x^4]$ will be _____. (where $E[\ .\ ]$ signifies expectation operator)

Q.134 Let X be a random variable with distribution.

X	1	2	3
$P(X)$	0.4	0.5	0.1

The standard deviation of X is _____.

Q.135 If X is a variable having Poisson's distribution with mean λ, $\lambda \neq 0$, such that $P(2) = 9P(4) + 90P(6)$, then the value of mean λ is

(a) ± 1 (b) ± 2

(c) ± 3 (d) ± 4

Q.136 If 'x' is a zero mean unit variance Gaussian Random variable, then the value of $E[|5x|]$ is _____. [$E[\cdot]$ represents expectation operator]

Q.137 The mean of a set of number is '$\bar{X}$', if each number is increased by λ, the mean of new set is

(a) $\bar{X}$ (b) $\bar{X} + \lambda$

(c) $\lambda\bar{X}$ (d) λ

Q.138 A random variable X has $E(X^2) = \sqrt{5}$ and $E(X) = 1$. What is value of $V(2X - 1) + SD(X)$?
(Where $V(X)$ is variance of X and $SD(X)$ is Standard Deviation of X.

Q.139 Consider the following frequency distribution:

Class	Frequency
0 – 10	4
10 – 20	5
20 – 30	7
30 – 40	10
40 – 50	12
50 – 60	8
60 – 70	4

(i) What is the mean of the distribution?
(a) 37.2 (b) 38.1
(c) 39.2 (d) 40.2

(ii) What is the median of the distribution?
(a) 37 (b) 38
(c) 39 (d) 40

(iii) What is the mode of the distribution?
(a) 38.33
(b) 40.66
(c) 42.66
(d) 43.33

4. Random Variables and Probability Distributions

Q.140 Let a discrete random variable X is binomially distributed such that $X \sim \text{Bin}(30, 0.8)$. Then what will be the value of variance of X? _____

Q.141 Consider the continuous random variable with probability density function

$$f(t) = 1 + t \text{ for } -1 \le t \le 0$$
$$= 1 - t \text{ for } 0 \le t \le 1$$

The standard deviation of the random variable is

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{6}}$

(c) $\frac{1}{3}$ (d) $\frac{1}{6}$

Q.142 Let a discrete random variable X is binomially distributed such that $X \sim \text{Bin}(500, 0.6)$. Then what will be the value of standard deviation of X? (Nearest integer) _____

Q.143 Let X be a continuous random variable and its probability density function is defined as

$$f(x)=\begin{cases}\frac{x}{2}; & 0<x\le\frac{1}{2}\\ 1-x; & \frac{1}{2}\le x\le 1\\ 0; & \text{Otherwise}\end{cases}$$

If $F(X=x)$ is cumulative distribution function at $X=x$, then what is the value of $F(X=0.3)+F(X=0.8)$?

Q.144 Assume that the duration in minutes of a telephone conversion follows the exponential distribution $f(x)=\frac{1}{5}e^{-x/5}$, $x\ge 0$. The probability that the conversion will exceed five minutes is

(a) $\frac{1}{e}$ (b) $1-\frac{1}{e}$

(c) $\frac{1}{e^2}$ (d) $1-\frac{1}{e^2}$

Q.145 For random variable $x(-\infty<x<\infty)$ following normal distribution, the mean is $\mu=100$. If probability is $P=\alpha$ for $x\ge 110$. Then probability of x lying between 90 and 110 i.e., $P(90\le x\le 110)$ is equal to

(a) $1-2\alpha$ (b) $1-\alpha$

(c) $1-\frac{\alpha}{2}$ (d) 2α

Q.146 A random variable is uniformly distributed over the interval 2 to 10. Its variance will be

(a) $\frac{16}{3}$ (b) 6

(c) $\frac{256}{9}$ (d) 36

Data for Q.147 to Q.148 :

A continuous random variable X has probability density function

$$f(x)=\begin{cases}kx(2-x); & 0<x\le 2\\ 0; & \text{Otherwise}\end{cases}$$

Q.147 What is the value of k? _____

Q.148 If $F(X=x)$ is a cumulative distribution function at $X=x$, then what is the value of $F(X=2.5)+F(X=0.2)$? _____

Q.149 If probability density function of a ran variable x is

$f(x)=x^2$ for $-1\le x\le 1$, and
$=0$ for any other value of x

then, the percentage probability $P\left(-\frac{1}{3}\le x\le\right.$ is

(a) 0.247 (b) 2.47
(c) 24.7 (d) 247

Q.150 Consider a random variable X that takes valu +1 and −1 with probability 0.5 each. The valu of the cumulative distribution function $F(x)$ $x=-1$ and $+1$ are

(a) 0 and 0.5 (b) 0 and 1
(c) 0.5 and 1 (d) 0.25 and 0.75

Q.151 Find the value of λ such that function $f(x)$ is a probability density function

$$f(x)=\begin{cases}\lambda(x-1)(2-x) & 1\le x\le 2\\ 0 & \text{otherwise}\end{cases}$$

Q.152 A continuous random variable X has a probabil density function $f(x)=e^{-x}$, $0<x<\infty$. The $P(X>1)$ is

(a) 0.368 (b) 0.5
(c) 0.632 (d) 1.0

Q.153 If probability density function of a rando variable X is

$f(x)=x^3$ for $-1\le x\le 1$, and
$=0$ for any other value of x

then, the probability $P\left(-\frac{1}{2}\le x\le\frac{1}{2}\right)$ is ____

Q.154 The probability density function of evaporation on any day during a year in a watershed is giv by

$$f(E)=\begin{cases}\frac{1}{5} & 0\le E\le 5 \text{ mm/day}\\ 0 & \text{otherwise}\end{cases}$$

The probability that E lies in between 2 an 4 mm/day in a day in the watershed is (in decim

_____.

Q.155 Let a discrete random variable X is binomial distributed such that $X\sim \text{Bin}(12, 0.2)$. Then wha will be the value of mean of X? _____

Q.156 Let X be a random variable with probability density function:

$$f(x)=\begin{cases}0.2, & \text{for } |x|\le 1\\ 0.1, & \text{for } 1<|x|\le 4\\ 0, & \text{otherwise}\end{cases}$$

The probability $P(0.5 < X < 5)$ is _____.

Q.157 A random variable X has probability density function $f(x)$ as given below:

$$f(x)=\begin{cases}a+bx & \text{for } 0<x<1\\ 0 & \text{otherwise}\end{cases}$$

If the expected value $E[X]=\frac{2}{3}$, then $\Pr[X<0.5]$ is _____.

Q.158 Two random variables X and Y are distributed according to

$$f_{XY}(x,y)=\begin{cases}(x+y) & 0\le x\le 1 \quad 0\le y\le 1\\ 0 & \text{otherwise}\end{cases}$$

The probability $P(X+Y\le 1)$ is _____.

Q.159 Let x be a continuous variable defined over the interval $(-\infty, \infty)$, and $f(x)=e^{-x-e^{-x}}$. The integral $g(x)=\int f(x)dx$ is equal to

(a) $e^{e^{-x}}$ (b) $e^{-e^{-x}}$

(c) $e^{-e^{x}}$ (d) e^{-x}

Q.160 X is a random variable with uniform probability density function in the interval $[-2,10]$. For $Y=2X-6$, the conditional probability $P(Y\le 7 \mid X\ge 5)$ (rounded off to three decimal places) is _____.

Q.161 X is a continuous random variable with probability density function given by

$f(x) = kx \quad (0 \le x < 2)$

$= 2k \quad (2 \le x < 4)$

$= -kx + 6k \quad (4 \le x < 6)$

The value of k will be

(a) $\frac{2}{3}$ (b) $\frac{1}{8}$

(c) 1 (d) 8

Q.162 The first moment about origin of binomial distribution is

(a) np (b) npq

(c) $n(1-p)$ (d) $n(1-p)q$

Q.163 The shape of the cumulative distribution function of Gaussian distribution is

(a) Bell-shaped (b) S-shaped

(c) Horizontal line

(d) Straight line at 45 degree angle

Q.164 $R_x(x)=Pe^{-4|x|}+Qe^{-|x|}$ is the probability density function for the real random variable X over the entire x axis. P and Q are both positive real numbers. The equation relating P and Q is

(a) $2P+\frac{Q}{2}=1$ (b) $2P+\frac{Q}{3}=1$

(c) $\frac{P}{2}+2Q=1$ (d) $P+2Q=1$

Q.165 Let $f(x)$ be the continuous probability density function of a random variable x, the probability that $a < x \le b$, is

(a) $f(b-a)$ (b) $f(b)-f(a)$

(c) $\int_a^b f(x)dx$ (d) $\int_a^b xf(x)dx$

Q.166 If x is a random variable which is uniformly distributed in the interval [0, 1], then the value $E[x^4]$ will be _____. (where E [.] signifies expectation operator)

Q.167 If the pdf of a Poisson distribution is given by $f(x)=\frac{e^{-2}2^x}{x!}$ then its mean is

(a) 2^x (b) 2

(c) -2 (d) 1

Q.168 Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that max $[X, Y]$ is less than $\frac{1}{2}$ is

(a) $\frac{3}{4}$ (b) $\frac{9}{16}$

(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

Q.169 If $f(x)=\begin{cases}cx & \text{if } (0<x<4)\\ 0 & \text{otherwise}\end{cases}$

Compute $P(1\le X\le 2)$?

(a) $\frac{4}{16}$ (b) $\frac{3}{16}$

(c) $\frac{5}{8}$ (d) $\frac{6}{9}$

Q.170 Let X be a real valued random variable-taking value in the interval $[-1, 1]$ with a density function of the form $f(x) = x^2 + a$ for $-1 \le x \le 1$. Find the value of a?

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$

(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.171 Suppose the probability density function of a Poisson distribution is given by $f(x) = \frac{e^{-4}4^x}{x!}$ then its mean is _____.

Q.172 If mean and variance of a binomial distribution are $\frac{10}{8}$ and $\frac{30}{32}$ respectively, the probability of occurrence p is _____.

Q.173 Which of the following statements is/are correct?

(a) Binomial distribution and Poisson distribution are two discrete probability distribution.

(b) The Poisson distribution is often used as an approximation to the Binomial distribution for large n and small p, as the Poisson probabilities are easier to calculate.

(c) Binomial works only for finite trials.

(d) None of these

Q.174 Consider the following set X which is linearly dependent. If the value of λ is of the from $\frac{a}{b}$ find the value of $a + b$ _____.

$X = \{(1, -2, \lambda), (2, -1, 5), (3, -5, 7\lambda)\}$

Q.175 In a binomial distribution, the mean is 9 and the standard deviation (σ) is $\sqrt{6}$. The value of n (total number of trials) and q (probability of failure of the event in each trial) respectively are:

(a) $27, \frac{1}{3}$ (b) $27, \frac{2}{3}$

(c) $36, \frac{3}{4}$ (d) $18, \frac{1}{2}$

Data for Q.176 to Q.178 :

For some constant c, the random variable X has probability density function

$$f(x) = \begin{cases} cx^{n-1}; & 0 < x < 1 \\ 0; & \text{Otherwise} \end{cases}$$

Q.176 What is the value of expectation of X i.e. EX?

(a) $\frac{n+1}{n-1}$ (b) $1 - \frac{1}{n+1}$

(c) $1 - \frac{1}{n+2}$ (d) $1 - \frac{2}{n+1}$

Q.177 What is the value of $c^2 + 2c$?

(a) $n^2 + 1$

(b) $n(n+2)$

(c) $n^2 - 1$

(d) $(n-1)(n-1)$

Q.178 The probability $P(X > x)$ for $0 < x < 1$, is equal to

(a) x^{n+1}

(b) $1 - x^{n+1}$

(c) $x^{n+1} - 1$

(d) None of these

Q.179 The function $p(x)$ is given by $p(x) = \frac{A}{x^{\mu}}$ where A and μ are constants with $\mu > 1$ and $1 \le x < \infty$ and $p(x) = 0$ for $-\infty < x < 1$. For $p(x)$ to be a probability density function, the value of A should be equal to

(a) $\mu - 1$ (b) $\mu + 1$

(c) $\frac{1}{(\mu - 1)}$ (d) $\frac{1}{(\mu + 1)}$

Q.180 If 'x' is a zero mean unit variance Gaussian Random variable, then the value of $E[|5x|]$ is _____. [$E[\cdot]$ represents expectation operator]

Q.181 If X is a random variable with PDF given by

$$f(x) = \begin{cases} kx & 0 \le x \le 2 \\ 2k & 2 \le x \le 4 \\ -kx + 6k & 4 \le x \le 6 \end{cases}$$

The value k and mean value of X are respectively

(a) $\frac{1}{8}, \frac{11}{6}$ (b) $\frac{1}{8}, \frac{13}{6}$

(c) $\frac{11}{6}, \frac{13}{6}$ (d) $\frac{1}{6}, \frac{13}{6}$

Q.182 Probability density function of a random variable X is distributed uniformly between 0 and 10. The probability that X lies between 2.5 to 7.5 and the mean square value of X are respectively

(a) $\frac{1}{2}$ and $\frac{100}{3}$ (b) 5 and 100

(c) 5 and $\frac{100}{3}$ (d) $\frac{1}{2}$ and 100

Q.183 Let X be a random variable with probability density function

$$f(x) = \begin{cases} 0.3 & \text{for } |x| \le 1 \\ 0.2 & \text{for } 1 < |x| \le 4 \\ 0 & \text{Otherwise} \end{cases}$$

The probability $P(0.6 < x < 5)$ is _____.

Q.184 A continuous random variable X has a probability density $f(x) = e^{-x}$, $0 < x < \infty$. Then $P\{X > 1\}$ is

(a) 0.368 (b) 0.5
(c) 0.632 (d) 1.0

5. Statistical Theorem and Testing

Q.185 Consider the following statements:

S_1: The acceptance region for $\alpha = 0.05$ is larger than the acceptance region for $\alpha = 0.01$.

S_2: If α increases, then the rejection region decreases.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.186 A sample of size n is 10 has a mean of $\bar{x}$ is 8.2. The hypothesized population mean is $\mu = 7.5$ and the sample standard deviation is $s = 1.4$. Find the t-distribution.

Q.187 A sample of size is 9 has standard deviation is 2.7.

The population mean is μ is 5 and the computed t-statistic is 1.5.

What is the sample mean?

(a) 1.35 (b) 9.75
(c) 6.35 (d) 8.25

Q.188 A sample of size is 16 has sample mean overline $\bar{x}$ is 12.5.

The population mean is 11 and the computed t-statistic is 2.

Find the sample standard deviation(s).

(a) 2 (b) 3
(c) 4 (d) 5

Q.189 The following sample of 6 values is drawn from a population:

4, 5, 6, 7, 8, 10

Assume that the population mean is 6.

Find the t-statistic for this sample.

Q.190 A sample of 5 values: 2, 4, 5, 6, 9

The population mean is assumed to be 5 and the t-statistic is known to be 0.5. Using only this information, find the value of the sample mean.

Q.191 A sample of 100 fluorescent light bulbs has a mean life of 1570 hours and a standard deviation of 120 hours. The company claims that the average life of its bulbs is 1600 hours. Using a 5% level of significance, test whether the company's claim is acceptable.

(Assume population is normally distributed.)

(a) The claim is not acceptable because the calculated $t = 2.5$ is greater than the critical value $t_{0.025,99} = 1.984$, so we reject the null hypothesis.

(b) The claim is acceptable because the calculated $t = 2.5$ is less than the critical value $t_{0.025,99} = 1.984$.

(c) The claim is acceptable because the sample size is large and the sample mean is close to the claimed mean.

(d) The claim is not acceptable because the sample standard deviation is large.

Q.192 A machine produces bolts with a population standard deviation of 0.5 mm. A sample of 36 bolts has an average length of 10.3 mm. Test the hypothesis that the population mean length is 10 mm.

Find the z-test statistic.

Q.193 A population has a known standard deviation $\sigma = 4$. A sample of size 25 yields a z-test value of 2.5 for testing $H_0: \mu = 50$. Find the sample mean ($\bar{x}$).

(a) 54 (b) 50
(c) 52 (d) 56

Q.194 A sample of size 16 is taken from a population with mean 100. The sample mean is 104 and the z-test statistic is 2. Find the population standard deviation.

Q.195 If the expected frequency for a category is 20 and its contribution to the chi-square statistic is 1.25, find the observed frequency F_0.

(a) (15, 25) (b) (10, 20)
(c) (10, 25) (d) (15, 20)

Q.196 A random sample of 100 people was asked their favorite ice cream flavor among four flavors. The observed frequencies are 30, 40, 20, and 10 respectively. The expected proportions for these flavors are 25%, 35%, 25%, and 15%. Calculate the chi-square statistic to test if the observed distribution matches the expected proportions.

Q.197 Which of the following statements are correct?

(a) z-distribution is used when the population standard deviation is known.

(b) t-distribution is used when the sample size is small and population variance is unknown.

(c) z-distribution has heavier tails than t-distribution.

(d) Both z and t distributions are symmetric.

Q.198 Match the following distributions with their properties:

Distributions:

A. Z-distribution

B. T-distribution

C. Chi-squared distribution

Properties:

1. Symmetric, heavier tails
2. Skewed and non-negative
3. Symmetric, light tails

Codes:

	A	B	C
(a)	3	1	2
(b)	1	3	2
(c)	2	3	1
(d)	3	2	1

Q.199 Which of the following statements is/are true for the chi-squared distribution?

(a) It is symmetric when degrees of freedom are small.

(b) It is always positively skewed.

(c) It is used to test hypotheses about population variance

(d) It is used in the test of independence in contingency tables.

Q.200 Which of the following best defines a p-value?

(a) The probability that the null hypothesis is true.

(b) The probability that the alternative hypothesis is true.

(c) The probability of observing a result as extreme or more extreme than the one observed, assuming H_0 is true.

(d) The smallest probability that supports the alternative hypothesis.

Q.201 Consider the following statements:

S_1: p-value is the smallest level of significance at which the null hypothesis can be rejected.

S_2: p-value is the probability that the null hypothesis is false.

Which of the following is correct?

(a) Only S_1

(b) Only S_2

(c) Both S_1 and S_2

(d) None of these

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Answers Probability and Statistics

1.	(c)	**2.**	(c)	**3.**	(b)	**4.**	(d)	**5.**	(a)	**6.**	(65)		(b)		
8.	(c)	**9.**	(a)	**10.**	(11232000)	**11.**	(c)	**12.**	(d)	**13.**	(720)	**14.**	(b)		
15.	(c)	**16.**	(126)	**17.**	(120960)	**18.**	(c)	**19.**	(c)	**20.**	(c)	**21.**	(b)		
22.	(d)	**23.**	(c)	**24.**	(c)	**25.**	(c)	**26.**	(c)	**27.**	(c)	**28.**	(4)		
29.	(a)	**30.**	(84)	**31.**	(b, c)	**32.**	(d)	**33.**	(72)	**34.**	(a)	**35.**	(c)		
36.	(c)	**37.**	(12)	**38.**	(c)	**39.**	(b)	**40.**	(a)	**41.**	Sol.	**42.**	(a)		
43.	(a)	**44.**	(c)	**45.**	(a)	**46.**	(b)	**47.**	(a)	**48.**	(c)	**49.**	(d)		
50.	(c)	**51.**	(c)	**52.**	(d)	**53.**	(c)	**54.**	(d)	**55.**	(c)	**56.**	(b)		
57.	(a)	**58.**	(c)	**59.**	(c)	**60.**	(d)	**61.**	(b)	**62.**	(c)	**63.**	(d)		
64.	(b)	**65.**	(b)	**66.**	(c)	**67.**	(d)	**68.**	(c)	**69.**	(d)	**70.**	(0.027)		
71.	(a)	**72.**	(a)	**73.**	(d)	**74.**	(d)	**75.**	(d)	**76.**	(c)	**77.**	(d)		
78.	(c)	**79.**	(c)	**80.**	(c)	**81.**	(a)	**82.**	(b, c)	**83.**	(c)	**84.**	(c)		
85.	(d)	**86.**	(d)	**87.**	(c)	**88.**	(d)	**89.**	(b)	**90.**	(a)	**91.**	(15)		
92.	(a)	**93.**	(c)	**94.**	(b)	**95.**	(a)	**96.**	(2.1, 0.89)	**97.**	(d)	**98.**	(c)		
99.	(b)	**100.**	(d)	**101.**	(c)	**102.**	(d)	**103.**	(b)	**104.**	(d)	**105.**	(c)		
106.	(b)	**107.**	(c)	**108.**	(11)	**109.**	(c)	**110.**	(a)	**111.**	(3)	**112.**	(b)		
113.	(a)	**114.**	(a)	**115.**	(*)	**116.**	(b)	**117.**	(d)	**118.**	(0)	**119.**	(0.75)		
120.	(a)	**121.**	(2.2)	**122.**	(4.593)	**123.**	(8.8)	**124.**	(52)	**125.**	(a)	**126.**	(a, b, d)		
127.	(0, 0.4)	**128.**	(c)	**129.**	(0.25)	**130.**	(4)	**131.**	(0.25)	**132.**	(b)	**133.**	(0.20)		
134.	(0.64)	**135.**	(a)	**136.**	(3.99)	**137.**	(b)	**138.**	(18)	**139.**	(i) 37.2, (ii) 39, (iii) 43.33)				
140.	(4.8)	**141.**	(b)	**142.**	(11)	**143.**	(0.94)	**144.**	(a)	**145.**	(a)	**146.**	(a)		
147.	(0.75)	**148.**	(1.028)	**149.**	(b)	**150.**	(c)	**151.**	(6)	**152.**	(a)	**153.**	(0.031)		
154.	(0.4)	**155.**	(2.4)	**156.**	(0.4)	**157.**	(0.25)	**158.**	(0.33)	**159.**	(b)	**160.**	(0.3)		
161.	(b)	**162.**	(a)	**163.**	(b)	**164.**	(a)	**165.**	(c)	**166.**	(0.20)	**167.**	(b)		
168.	(b)	**169.**	(b)	**170.**	(b)	**171.**	(4)	**172.**	(0.25)	**173.**	(a, b, c)	**174.**	(19)		
175.	(b)	**176.**	(b)	**177.**	(b)	**178.**	(d)	**179.**	(a)	**180.**	(3.99)	**181.**	(a)		
182.	(a)	**183.**	(0.72)	**184.**	(a)	**185.**	(d)	**186.**	(1.583)	**187.**	(c)	**188.**	(b)		
189.	(0.756)	**190.**	(5.579)	**191.**	(a)	**192.**	(3.6)	**193.**	(c)	**194.**	(8)	**195.**	(a)		
196.	(4.381)	**197.**	(a, b, d)	**198.**	(a)	**199.**	(b, c, d)	**200.**	(c)	**201.**	(a)				

Explanations Probability and Statistics

1. (c)

In member to be n teams with 2 member each and teams are unordered so we can exchange n team member among them.

$$= \frac{(2n!)}{(2^* 2^* \ldots n \text{ times } {}^* n!)}$$

$$= \frac{(2n!)}{(2^n \cdot n!)}$$

So, option (c) is correct.

2. (c)

This corresponds to an ordered partition of 8 elements into two groups, the first with 5 elements and second with 3 elements. The number of ways of doing this is

$$P(8; 5, 3) = \frac{8!}{5!\,3!} = 56$$

3. (b)

A particular person have one choice

$$\underset{\smile}{\ } \overset{1}{-} \underset{\smile}{\ }$$

Rest four places, 4 person arrange in any order

$$= 4! = 24$$

4. (d)

The problem reduces to finding how many distinct ordered colour pairs (C_1, C_2) are possible with k colors.

Since the first color C_1 can be any one of the k colours and the second color C_2 also can be any one of the k colors (both prints of a letter can be colored with same color), the total no. of such order color pairs is equal to $k \times k = k^2$.

Since each pair of letters must be colored with different color pairs, at least 26 color pairs are required to do this.

Therefore the requirement is $k^2 \geq 26$.

The minimum value of k that satisfies this equation is $k = 6$.

5. (a)

Number of ways of selecting 3 consonants from $7 = {}^7C_3$

Number of ways of selecting 2 vowels from $4 = {}^4C_2$

Number of ways of selecting 3 consonants from 7 and 2 vowels from $4 = {}^7C_3 \times {}^4C_2$

$$= \left(\frac{7\times6\times5}{3\times2\times1}\right) \times \left(\frac{4\times3}{2\times1}\right) = 210$$

It means, we can have 210 groups where each group contains total 5 letters (3 consonants and 2 vowels)

Number of ways arranging 5 letters among themselves = 5! = 5 × 4 × 3 × 2 = 1 = 120

Hence, Required number of ways = 210 × 120

= 25200

Hence, option (a) is correct.

6. (65)

Let computers be A, B and C

Toughest job assigned to fastest computer (Say A) is 1 way.

Easiest job assigned to shortest computer (Say B) is 1 way.

Remaining 4 jobs to be assigned to 3 computers so that the computer C gets at least one job since A and B already assigned a job.

Number of ways 4 jobs assigned to 3 computers $= 3^4$.

Number of ways 4 jobs assigned to 3 computers so that computer C does not get any job $= 2^4$

Required number of ways = $3^4 - 2^4 = 81 - 16$

= 65 ways

7. (b)

In a group of 6 boys and 4 girls, four children are to be selected such that atleast one boy should be there.

Hence we have 4 options as given below

We can select 4 boys .... (option a)

Number of ways to this = ${}^6C_4$

We can select 2 boys and 2 girls .... (option b)

Number of ways to this = ${}^6C_2 \times {}^4C_2$

We can select 1 boy and 3 girls .... (option d)

Number of ways to this = ${}^6C_1 \times {}^4C_3$

Total number of ways

$= {}^6C_4 + {}^6C_3 \times {}^4C_1 + {}^6C_2 \times {}^4C_2 + {}^6C_1 \times {}^4C_3$

$= {}^6C_2 + {}^6C_3 \times {}^4C_1 + {}^6C_2 \times {}^4C_2 + {}^6C_1 \times {}^4C_3$

$= 15 + 80 + 90 + 24 = 209$

So, option (b) correct.

8. (c)

Men permutation $\Rightarrow$ 11 ! ways
Women permutation $\Rightarrow$ 8 ! ways
There are two choices

$$\Rightarrow \left.\begin{array}{c}\text{Men follow the women}\\ \text{or}\\ \text{Women follow the men}\end{array}\right\} \Rightarrow 2\times 11!\times 8!$$

9. (a)

We want every one of the 5 balls to be in the wrong box. This is nothing but the number of derangements of a set of 5 elements = D_5.
i.e. we need to compute D_5

$$D_n = \sum_{r=2}^{n}(-1)^r\frac{n!}{r!},\quad D_5 = \sum_{r=2}^{5}(-1)^r\frac{5!}{r!}$$

$$= \frac{5!}{2!}-\frac{5!}{3!}+\frac{5!}{4!}-\frac{5!}{5!}$$

$$= 60 - 20 + 5 - 1 = 44 \text{ ways}$$

10. (11232000)

$$^{26}P_3 \times {}^{10}P_3$$

$$= 26 \times 25 \times 24 \times 10 \times 8 \times 9 = 11232000$$

11. (c)

Select a male teacher = $^8C_1$
Select a female teacher = $^7C_1$

$$\text{Total ways} = {}^8C_1 \times {}^7C_1 = 56$$

12. (d)

Take 'AAU' together and treat it like 1 entity. Now arrange 'AAU' BCS in 4! ways. Then, the AAU can be arranged in $\frac{3!}{2!}$ ways because A has been repeated twice. So,

$$\text{Total arrangements} = \frac{4!3!}{2!}$$

So, option (d) is correct.

13. (720)

The word 'OPTIMAL' has 7 letters. It has the vowels 'O' 'I', 'A', in it and these 3 vowels should always come together. Hence these three vowels can be grouped and considered as a single letter. This is, PTCL(OIA).
Hence, we can assume total letter as 5 and all these letters are different.
Number of ways to arrange these letters = 5! = 120.
All the 3 vowels (OIA) are different.
Number of ways to arrange these vowels among themselves = 3! = 6.
Hence, required number of ways = 120 × 6 = 720.
So, answer is 720.

14. (b)

M A D EEA SY

4 positions are fixed.
Remaining 4 positions can be arranged in $^4P_4$
= 4! = 24 ways
So option (b) is correct.

15. (c)

$$\underset{26}{\square}\ \underset{25}{\square}\ \underset{24}{\square}\ \underset{23}{\square}\ \underset{22}{\square}$$

$$= 26\times 25\times 24\times 23\times 22 = \frac{26!}{21!}$$

16. (126)

Required number of ways = $9C_4$

$$= \frac{9\times 8\times 7\times 6}{4\times 3\times 2\times 1} = 126$$

17. (120960)

Total number of consonants = 3 (make a single group)

$$\frac{\text{Consonants arrange themselves in a group}}{\text{Total number of words}}$$

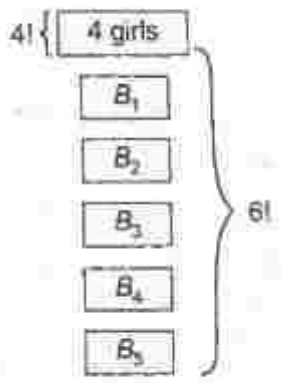
$$= \frac{3!}{\text{Consonant group + Remaining letters}}$$

$$= 7 + 1 = 8$$

Total number of times E repeated = 2

$$\text{Total ways} = \frac{8!}{2!}\times 3! = 120960$$

18. (c)



$\therefore$ Total ways = 6! × 4! = 17280

19. (c)

$n = 10, k = 30$

Required number of ways $= (k + n - 1)C(n - 1)$

$= 39C_9$

So, option (c) is correct.

20. (c)

The problem is ordered partitions of type

$$(52; 5, 5, 5, 5, 32) = \frac{52!}{(5!)^4 32!}$$

21. (b)

It is based on inclusion-exclusion principle.

Total arrangements – dearrangements $= 5! - [5! - 5C_1(4!) + 5C_2(3!) - 5C_3(2!) + 5C_4(1!) - 5C_5(0!)]$

$= 5C_1(4!) - 5C_2(3!) + 5C_3(2!) - 5C_4(1!) + 5C_5(0!)$

$= 120 - 60 + 20 - 5 + 1 = 76$

22. (d)

E : Persons speaks English

H : Persons speaks Hindi

K : Persons speaks Kannada

Assume that everyone in the room speaks at least one of the languages.

Given, $n(E \cup H \cup K) = 28$

$n(E) = 18$

$n(H) = 15$

$n(K) = 22$

$n(E \cap H) = 9$

$n(H \cap K) = 11$

$n(K \cap E) = 13$

$n(K \cap E \cap H) = ?$

By principle of inclusion and exclusion

$n(E \cup H \cup K) = n(E) + n(H) + n(K) - n(E \cap H) - n(H \cap E) - n(K \cap E) + n(K \cap E \cap H)$

$28 = 18 + 15 + 22 - 9 - 11 - 13 + n(K \cap E \cap H)$

$n(k \cap E \cap H) = 28 - 55 + 33 = 61 - 55 = 6$

Therefore, the number of people who speak all three languages are 6.

23. (c)

Group of vowels together and make it a single packet = OUE.

Rest of the words are : CMPTR.

Therefore, total words to be grouped including one packet = (OUE)CMPTR = 6!

But the packet itself can be arranged in 3! ways.

Hence answer 6!3!

25. (c)

Given word is : AGAIN

After arranging in alphabetic order : AAGIN

So, the number of arrangements are:

Starting with A : AAGIN = 4! = 24 ways

Starting with G : $\frac{4!}{2!}$ = 12 ways (since, A appears 2 times)

Starting with I: $\frac{4!}{2!}$ = 12 ways (since, a appears 2 times)

Now, Total ways = 24 + 12 + 12 = 48 ways

So, 49th word is : NAAGI

and 50th word is : NAAIG

So, option (c) is correct.

26. (c)

$2n$ member to be n teams with 2 member each and terms are unordered so we can exchange n team member among them.

$$= \frac{(2n)!}{\underbrace{(2!2!2!)\ldots}_{n \text{ times}} \times n!} = \frac{2n!}{2^n \times n!}$$

So, option (c) is correct.

27. (c)

Colour of the first strip can be selected from 4 colours in 4 ways.

Colour for next strip can be selected in 3 ways (since we can't allow adjacent strips to have same colour).

For similar reason,

Colour for each of the next 3 strips can be selected in 3 ways.

Hence, total number of ways of colouring the strips of the flag

$= 4 \times 3 \times 3 \times 3 \times 3 = 324$ ways

So, option (c) is correct.

28. (4)

$${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}, r = 4$$

29. (a)

We have n kingdoms as $k_1, k_2 \ldots k_n$

Firstly we can select 2 champions from $2n$ champions and assign to

$$k_1 = \binom{2n}{2} \text{ ways} \quad \ldots \text{eq.(1)}$$

Then we can select next 2 champions and assign to

$$k_2 = \binom{2n-2}{2} \text{ ways} \qquad \text{...eq. (2)}$$

and so on,

for last kingdom, we have 2 champion left

$$= \binom{2}{2} \text{ ways} \qquad \text{...eq. (n)}$$

Total ways for assigning $2n$ champions to n kingdoms

$$= w_1 \times w_2 \times .... \times w_n$$

$$= \binom{2n}{2} \times \binom{2n-2}{2} \times ..... \times \binom{2}{2} = \frac{(2n)!}{2^n}$$

So, answer is option (a).

30. (84)

Number of people in committee = 3

Number of people in group = 9

Out of 9 people, number of ways of form three-person committee = $\binom{n}{r} = \binom{9}{3} = 84$

Hence, answer is 84.

31. (b, c)

Consider three buckets red, white and blue and we want the total number of balls such that each bucket contain at least one ball. Now consider the state of picking up a ball without replacement: (normally you consider the worst case scenario in these cases) Starting 10 balls all are red and thus goes to bucket name Red. Now again picking up the ball gives 7 balls which are of same colour and put all of them in a bucket named White. The next pick will definitely be of different colour thus: we picked 10 + 7 + 1 = 18.

- Let the total number of persons present at the party be m, Then, $\left[\frac{[x \times (x-1)]}{2}\right] = 105$.

$$x = 14$$

32. (d)

For a string of length n :

The number of substrings of length 1 = n

The number of substrings of length 2 = $n - 1$

The number of substrings of length 3 = $n - 2$

and so on...

The number of substrings of length n is 1

So total number of substrings

$= n + (n-1) + ... + 1$

= Sum of n natural numbers = $\frac{n(n+1)}{2}$

33. (72)

$$P \xrightarrow[2]{} Q \xrightarrow[3]{} R \xrightarrow[4]{} S \xrightarrow[3]{} T$$

Total ways = $2 \times 3 \times 4 \times 3 = 72$

34. (a)

$$\text{Total way} = \frac{8!}{2!2!}$$

(E and A repeated twice)

35. (c)

Number of ways for distributing r similar things among n different things = ${}^{(n-1+r)}C_r$

The number of ways for distributing 10 roses among the two girls = ${}^{(2-1+10)}C_{10} = 11$.

Similarly number of ways for distributing 15 sunflowers among two girls = ${}^{(2-1+15)}C_{15}$

$= {}^{16}C_{15} = {}^{16}C_1 = 16$

Number of ways for distributing 14 daffodils among the two girls

$= {}^{(2-1+14)}C_{14} = {}^{15}C_{14} = {}^{15}C_1 = 15$

∴ Total number of ways = $11 \times 16 \times 15 = 2640$

36. (c)

Even digits = 0, 2, 4, 6, 8

LHS digit cannot be zero, otherwise it become four digit number.

Total ways = $4 \times 4 \times 3 \times 2 \times 1 = 96$

37. (12)

Since both L's are indistinguishable.

First L's can be arranged in 3 positions 2, 3, or 5 in ${}^3C_2 = 3$ ways as follows:

_ L _ L _

or _ L _ _ L

or _ _ _ L L

Now the letters I, A, C can be deranged in $2 \times 2!$ ways. Example in _ L _ L _ C cannot occupy 5th position, so only 2 ways.

Remaining I and A can be arranged in remaining 2 position in 2! ways = 2 ways.

So answer is $3 \times 2 \times 2! = 12$

38. (c)

Total stalls = 10

Each stalls have 3 choices.

Total ways = 3^{10}

39. (b)

For each of the n couples invited to the party one of three thing is possible:

1. Both husband and wife attend the party.
2. Wife only attends the party.
3. Neither husband nor wife attends the party.

Since there are n such couples. total number of possibilities = 3^n.

40. (a)

Total letters = 9

Number of even position = 4

Number of vowels = 4

'D', 'A' repeat 2 times.

$$\text{Total ways} = \frac{4!}{2!} \times \frac{5!}{2!} = 720$$

41. Sol.

Given that object are 6 identical balls this means that it does not matter. Which objects are grouped together, it only matters how many objects go into each bins. In addition to this the bins are identical, this means that ordering of bins does not matter. This problem can be modeled as partition of '6' into exactly '3' parts as follows.

(6, 0, 0) (5, 1, 0) (4, 2, 0) (4, 1, 1) (3, 3, 0) (3, 2, 1) (2, 2, 2)

42. (a)

Number of ways to draw 4 balls = $^9C_4 = 126$

Number of ways no red ball draw = $^6C_4 = 15$

Number of ways, atleast one ball draw

$$= 126 - 15 = 111$$

43. (a)

The five cards are {1, 2, 3, 4, 5}

Sample space = 5 × 4 ordered pairs.

[Since there is a Ist card and IInd card we have to take ordered pairs]

p(Ist card = IInd card + 1)

$$= p\{(2, 1), (3, 2), (4, 3), (5, 4)\}$$

$$= \frac{4}{5 \times 4} = \frac{1}{5}$$

44. (c)

p(one ball is Red and another is blue)

$$= p(\text{first is Red and second is Blue})$$

$$= \frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$$

45. (a)

$$P = \frac{\text{Number of correct ways}}{\text{Possible number of ways}}$$

Number of correct ways = 2

Possible number of ways

$$= 5c_1 \times 4c_1 \times 3c_1 \times 2c_1 \times 1c_1 = 5!$$

$$\Rightarrow \quad P = \frac{2}{5!}$$

46. (b)

Probability of getting an odd number on a die

$$= \frac{3}{6} = \frac{1}{2}$$

Probability of an odd number when a die rolled thrice

$$= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

47. (a)

Total ways to select five people = $^{20}C_5$

Total ways to select three female = $^9C_3$

Total ways to select two male = $^{11}C_2$

Probability of 3 female and 2 male

$$= \frac{^9C_3 \times {}^{11}C_2}{^{20}C_5} = 0.28$$

48. (c)

(a) $P(A \cap B) = P(A)\,P(B)$ is false since this is true if and only if A and B are independent events

(b) $P(A \cup B) = P(A) + P(B)$ is false since $P(A \cap B)$ is zero if and only if A and B are mutually exclusive.

(c) $P(A \mid B) = P(A \cap B)/P(B)$ is true.

(d) $P(A \cup B) < P(A) + P(B)$ is false.

Since $P(A \cup B) \leq P(A) + P(B)$

49. (d)

Probability Husband's select $P(H) = \frac{1}{8}$

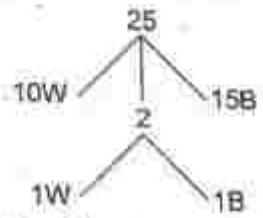
Probability Wife's select $P(W) = \frac{1}{6}$

Probability one of them select

$$= P(H)P(\bar{W}) + P(\bar{H})P(W)$$

$$= \frac{1}{8}\times\frac{5}{6}+\frac{7}{8}\times\frac{1}{6}=\frac{1}{4}$$

50. (c)



Bag contains 10 white balls and 15 black balls.
Required probability

$$= \frac{{}^{10}C_1\times{}^{15}C_1}{{}^{25}C_2} = \frac{10\times15}{\frac{25\times24}{2}}=\frac{1}{2}$$

51. (c)

$$P(A) = \frac{6}{10},\ P(B) = \frac{7}{10}$$

Probability of contradict each other

$$= P(A)P(\bar{B}) + P(\bar{A})P(B)$$

$$= \frac{6}{10}\times\frac{3}{10}+\frac{4}{10}\times\frac{7}{10}$$

$$= \frac{46}{100} = 46\%$$

52. (d)

P(atleast one of dice will have 6 facing

$$= 1 - P(\text{none of dice have 6 facing up})$$

$$= 1-\left[\frac{5}{6}\times\frac{5}{6}\right]=1-\frac{25}{36}=\frac{11}{36}$$

53. (c)

The probability that the bottom card of a randomly shuffled deck is ace $= \frac{4}{52}$ because there are 4 aces out of total 52 cards.
From the remaining 3 aces out of 51 cards the probability that the top card is also an ace $= \frac{3}{51}$.

So required probability $= \frac{4}{52}\times\frac{3}{51}$.

54. (d)

P(rain today) = 0.5
P(rain tomorrow) = 0.6
P(rain today ∪ rain tomorrow) = 0.7
P(rain today ∩ rain tomorrow) = ?
P(rain today ∩ rain tomorrow)
= P(rain today) + P(rain tomorrow) – P(rain today ∩ rain tomorrow)
So; 0.7 = 0.5 + 0.6 – P(rain today ∩ rain tomorrow)
P(rain today ∩ rain tomorrow)

$$= 0.5 + 0.6 - 0.7 = 0.4$$

So, the probability that it will rain today and tomorrow is 0.4.

55. (c)

$$P(E) = \frac{4}{10},\ P(H) = \frac{3}{10}$$

$$P(E\cup H) = \frac{4}{10}+\frac{3}{10}-\frac{2}{10}=\frac{5}{10}$$

$$P(E\cap\bar{H}) = P(E) - P(E\cap H)$$

$$= \frac{4}{10}-\frac{2}{10}=\frac{2}{10}=\frac{1}{5}$$

56. (b)

Probability of getting an odd number in rolling of a die $= \frac{3}{6}=\frac{1}{2}$.

Now using binomial distribution
P(Exactly one odd number among three outcomes)

$$= {}^3C_1\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^2 = 3\times\left(\frac{1}{2}\right)^3=\frac{3}{8}$$

57. (a)

Die one show 5, $P(D_1) = \frac{1}{6}$

Die second show 5, $P(D_2) = \frac{1}{6}$

Probability, die shows other than 5 and the sum is odd with other die $= \frac{3}{6} = P(DX_1) = P(DX_2)$

Probability 5 appear on one die

$$= P(D_1)P(DX_2) + P(DX_1)P(D_2)$$

$$= \frac{1}{6}\times\frac{3}{6}+\frac{3}{6}\times\frac{1}{6}=\frac{1}{6}$$

58. (c)

If all the points have $X < 5$, then expectation of a random variable X is surely less than 5. So according to this there should be atleast a sample point at which $X \geq 5$.

59. (c)

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{3}$$

and $$P(E_1 \cap E_2) = \frac{1}{5}$$

(a) $P(E_1 \text{ or } E_2)$

$$= P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{5} = \frac{19}{30}$$

However given in option (a) is $\frac{2}{3}$.

So option (a) is not true.

(b) For independent events

$P(E_1 \cap E_2) = P(E_1)\, P(E_2)$

Here, $P(E_1 \cap E_2) = \frac{2}{5}$

$$P(E_1)\, P(E_2) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

So, $(P(E_1 \cap E_2) \neq P(E_1)\, P(E_2)$

So event E_1 and E_2 are not independent.

Option (b) is not true.

(c) Since E_1 and E_2 are not independent

So option (c) is true.

(d) $P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{3}{5}$

So option (d) $P\left(\frac{E_1}{E_2}\right) = \frac{4}{5}$ is false.

60. (d)

Constraints are

(i) $P(E_1) = P(E_2) = x$

(ii) $P(E_1 \cup E_2) = 1$

(iii) E_1 and E_2 are independent so

$P(E_1 \cap E_2) = P(E_1)\, P(E_2)$

$$= x \times x = x^2$$

Now, $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$$1 = x + x - x^2 = 2x - x^2$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

So: $P(E_1) = P(E_2) = x = 1$

61. (b)

Sample space = 7^7

All accidents on the same day = 7 ways (all on Monday, all on Tuesday...)

So, required probability = $\frac{7}{7^7} = \frac{1}{7^6}$

62. (c)

p(atleast one head and one tail)

= 1 – p(no head or no tail)

= 1 – [p(no head) + p(no tail) – p(no head and no tail)]

= 1 – [p(all tails) + p(all heads) – 0]

$$= 1 - \left[\frac{1}{2^4} + \frac{1}{2^4} - 0\right]$$

$$= 1 - \frac{2}{16} = 1 - \frac{1}{8} = \frac{7}{8}$$

or

Alternate Method:

We need atleast one head (≥ 1 H) and atleast one tail (≥ 1 T). First we satisfy ≥ 1H as follows.

1 H, 3 T

2 H, 2 T

3 H, 1 T

and 4 H, 0 T

now to satisfy the second condition of ≥ 1 T, we have to remove 4 H, 0 T.

So, the favourable cases are only 1 H, 2 H and 3 H.

The probability of this by binomial distribution formula is

$$= \frac{{}^4C_1}{2^4} + \frac{{}^4C_2}{2^4} + \frac{{}^4C_3}{2^4} = \frac{7}{8}$$

63. (d)

Given, $P(A) = 1$

$$P(B) = \frac{1}{2}$$

Since both events are independent

$$P(A \mid B) = P(A) = 1$$

$$P(B \mid A) = P(B) = \frac{1}{2}$$

64. (b)

Total sample output = 6 × 6 = 36

Total ways, sum of dice is 4

= (1, 3), (2, 2), (3, 1)

= 3 ways

Probability of getting a sum of 4 = $\frac{3}{36} = \frac{1}{12}$

65. (b)

We know that:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\overline{P(A)} = 1 - P(A) \text{ and } \overline{P(B)} = 1 - P(B)$$

So, $P(A) = (1 - \overline{P(A)})$ and $P(B) = (1 - \overline{P(B)})$

$$P(A \cup B) = (1 - \overline{P(A)}) + (1 - \overline{P(B)}) - P(A \cap B).$$

$$= \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{3}\right) - \left(\frac{1}{2}\right)$$

$$= \left(\frac{2}{3}\right) + \left(\frac{2}{3}\right) - \left(\frac{1}{2}\right)$$

$$= \left(\frac{4}{3}\right) - \left(\frac{1}{2}\right) = \left(\frac{5}{6}\right) = \left(\frac{10}{12}\right)$$

66. (c)

Total ways to draw two cards $= {}^{52}C_2$

Total ways to draw Ace cards $= {}^{4}C_2$

Total ways to draw face cards $= {}^{12}C_2$

$$\text{Probability} = \frac{{}^{4}C_2 + {}^{12}C_2}{{}^{52}C_2} = \frac{12}{221}$$

67. (d)

Number of permutations with '2' in the first position $= 19!$

Number of permutations with '2' in the second position $= 10 \times 18!$

(Fill the first space with any of the 10 odd numbers and the 18 spaces after the 2 with 18 of the remaining numbers in 18! ways)

Number of permutations with '2' in 3rd position $= 10 \times 9 \times 17!$

(Fill the first 2 places with 2 of the 10 odd numbers and then the remaining 17 places with remaining 17 numbers) and so on until '2' is in 11th place.

After that it is not possible to satisfy the given condition, since there are only 10 odd numbers available to fill before the '2'.

So the desired number of permutations which satisfies the given condition is

$19! + 10 \times 18! + 10 \times 9 \times 17! + 10 \times 9 \times 8 \times 16! + \ldots + 10! \times 9!$

Now the probability of this happening is given by

$$\frac{19! + 10 \times 18! + 10 \times 9 \times 17! \ldots + 10! \times 9!}{20!}$$

Which is clearly not choices (a), (b) or (c)

∴ Answer is (d) none of these.

68. (c)

This problem is to be solved by binomial distribution, since although population is finite, sampling is done with replacement and so probability does not change from trial to trial.

Here, $n = 2$

$x = 0$ (no defective)

$$p = p(\text{defective}) = \frac{3}{10}$$

So, $p(x = 0) = 2C_0 \left(\frac{3}{10}\right)^0 \left(1 - \frac{3}{10}\right)^2 = 0.49 = 49\%$

69. (d)

Probability of drawing two red balls

= p(first is red) × p(second is red given that first is red)

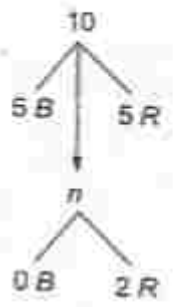
$$= \frac{5}{10} \times \frac{4}{9} = \frac{2}{9}$$

Alternatively this problem can be done as hypergeometric distribution, since it is sampling without replacement from finite population.

From above diagram,

$$p(X = 2) = \frac{{}^{5}C_2}{{}^{10}C_2}$$

$$= \frac{5 \times 4}{10 \times 9} = \frac{2}{9}$$



70. (0.027)

$$p(X) = \frac{Y}{2}, \; p(Y) = 3Z$$

$$p(X) + p(Y) + p(Z) = 1$$

$$\Rightarrow \quad p(X) = \frac{3}{11}, \; p(Y) = \frac{6}{11}, \; p(Z) = \frac{2}{11}$$

$$= p(X) \cdot p(Y) \cdot p(Z) = 0.027$$

71. (a)

The condition getting 2 heads and 2 tails is same as getting exactly 2 heads out of 4 tosses.

Given, $p = P(H) = \frac{1}{2}$

Applying the formula for binomial distribution, we get,

$$P(X = 2) = {}^{4}C_2 \left(\frac{1}{2}\right)^2 \left(1 - \frac{1}{2}\right)^{4-2}$$

$$= {}^{4}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{{}^{4}C_2}{2^4} = \frac{6}{16} = \frac{3}{8}$$

72. (a)

Given: A, B, C are independent with

$$P(A) = P(B) = P(C) = p$$

Now,

$$P\{\bar{C} \mid A \cap (B \cap C)\} = \frac{P\{\bar{C} \cap (A \cap (B \cup C))\}}{P\{A \cap (B \cup C)\}}$$

Using boolean algebra we can simplify

$$\bar{C} \cap \{A \cap (B \cup C)\} = C'\{A(B+C)\}$$

$$= C'(AB + AC) = C'AB + 0 = C'AB$$

$$= \bar{C} \cap A \cap B$$

So requires probability

$$= \frac{P(\bar{C} \cap A \cap B)}{P\{A \cap (B \cup C)\}}$$

$$= \frac{P(\bar{C}) \times P(A) \times P(B)}{P(A) \times \{P(B) + P(C) - P(B) \cap P(C)\}}$$

$$= \frac{(1-p)p^2}{p(p+p-p \times p)} = \frac{1-p}{2-p}$$

73. (d)

Sample space $= (6)^2 = 36$

Total ways in which sum is either 8 or 9 is (2, 6), (3, 5), (3, 6), (4, 4), (4, 5), (5, 3), (5, 4), (6, 2), (6, 3) = 9 ways

$\therefore$ Probability of coming sum 8 or 9 $= \frac{9}{36} = \frac{1}{4}$

So probability of not coming sum 8 or 9 $= 1 - \frac{1}{4} = \frac{3}{4}$

74. (d)

(a) is false since if P and Q are independent

$$pr(P \cap Q) = pr(P) * pr(Q)$$

which need not be zero.

(b) is false since

$$pr(P \cup Q) = pr(P) + pr(Q) - pr(P \cap Q)$$

$\therefore$ $pr(P \cup Q) \leq pr(P) + pr(Q)$

(c) is false since independence and mutually exclusive are unrelated properties.

(d) is true

Since $P \cap Q \subseteq P$

$\Rightarrow$ $n(P \cap Q) \leq n(P)$

÷ both sides by $n(S)$ we get,

$$\frac{n(P \cap Q)}{n(S)} \leq \frac{n(P)}{n(S)}$$

$\Rightarrow pr(P \cap Q) \leq pr(P)$

75. (d)

$$\Rightarrow \quad P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{8}{7}$$

$$\Rightarrow \quad P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{6}{7}$$

$$\Rightarrow \quad P(A|B) - P(B|A)$$

$$= \frac{2}{7}$$

76. (c)

If two fair dices are rolled the probability distribution of r where r is the sum of the numbers on each die is given by

Y	2	3	4	5	6	7	8	9	10	11	12
P(Y)	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The above table has been obtained by taking all different ways of obtaining a particular sum. For example, a sum of 5 can be obtained by (1, 4), (2, 3), (3, 2) and (4, 1).

$$\therefore \quad p(x = 5) = \frac{4}{36}$$

Now let us consider choice (a).

$$Pr(r > 6) = Pr(r \geq 7)$$

$$= \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{21}{36} = \frac{7}{12}$$

$\therefore$ Choice (a) $Pr(r > 6) = \frac{1}{6}$ is wrong.

Consider choice (b).

$$Pr\left(\frac{r}{3} \text{ is an integer}\right) = Pr(r = 3) + Pr(r = 6) + Pr(r = 9) + Pr(r = 12)$$

$$= \frac{2}{36} + \frac{5}{36} + \frac{4}{36} + \frac{1}{36} = \frac{12}{36} = \frac{1}{3}$$

$\therefore$ Choice (b) $Pr\ \frac{r}{3}$ is an integer $= \frac{5}{6}$ is wrong

Consider choice (c)

$$Pr\left(r = 8 \mid \frac{r}{4} \text{ is an integer}\right) = \frac{1}{36}$$

Now, $Pr\left(\frac{r}{4} \text{ is an integer}\right)$

$$= Pr(r = 4) + Pr(r = 8) + Pr(r = 12)$$

$$= \frac{3}{36} + \frac{5}{36} + \frac{1}{36} = \frac{9}{36} = \frac{1}{4}$$

$\Pr(r = 8 \text{ and } \frac{r}{4} \text{ is an integer}) = \Pr(r = 8) = \frac{5}{36}$

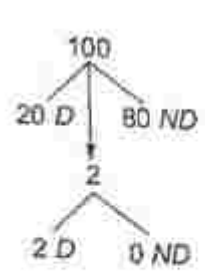
$\therefore \Pr(r = 8 \mid \frac{r}{4} \text{ is an integer}) = \frac{\frac{5}{36}}{\frac{1}{4}} = \frac{20}{36} = \frac{5}{9}$

$\therefore$ Choice (c) is correct.

77. (d)

Problem can be solved by hypergeometric distribution

$$p(X=2) = \frac{20C_2 \times 80C_0}{100C_2} = \frac{19}{495}$$



78. (c)

Total cases = 36 [(1, 1) (1, 2) (1, 3) and so on]
Favourable case = $(x_1 > x_2) = 15$

$$P[x_1 > x_2] = \frac{15}{36} = \frac{5}{12}$$

79. (c)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since,

$$P(A \cap B) = p(A)\,p(B)$$

(not necessarily equal to zero).

So, $P(A \cup B) = P(A) + P(B)$ is false.

80. (c)

$$P\left(\frac{Y}{X}\right) = \frac{P(Y \cap X)}{P(X)} = \frac{\frac{1}{12}}{\frac{1}{4}} = \frac{1}{3}$$

81. (a)

$$P(X \cup Y^c) = 0.7$$

$\Rightarrow P(X) + P(Y^c) - P(X)\,P(Y^c) = 0.7$

(Since X, Y are independent events)

$\Rightarrow P(X) + 1 - P(Y) - P(X)\,[1 - P(Y)] = 0$

$\Rightarrow P(Y) - P(X \cap Y) = 0.3 \quad \ldots(i)$

$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$

$= 0.4 + 0.3 = 0.7$

83. (c)

A dice is rolled 5 times

$n = 5$

P = (Probability of getting 1) $= \frac{1}{6}$

$q = 1 - \frac{1}{6} = \frac{5}{6}$

Probability of getting 1 at least 4 times is

$P(x \geq 4) = P(x = 4) + P(x = 5)$

$= {}^nC_4 p^4 q^{n-4} + {}^nC_5 p^5 q^{n-5}$

$= {}^5C_4 p^4 q^1 + {}^5C_5 p^5 q^0$

$= 5 \times \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 + 1 \times \left(\frac{1}{6}\right)^5 \times \left(\frac{5}{6}\right)^0$

$= \frac{25}{(6)^5} + \frac{1}{(6)^5} = \frac{26}{(6)^5} = 0.0033$

% Probability = 0.33%

84. (c)

$P(E) = P$ (1 red and 1 black)

$= \frac{{}^{10}C_1 \times {}^{40}C_1}{{}^{50}C_2} = \frac{400}{\frac{50 \times 49}{2}} = \frac{16}{49}$

86. (d)

a, b, c are true but (d) is not true since in a negatively skewed distribution, mode > median > mean.

87. (c)

If 10 is added to each and every item of a data then arithmetic mean is greater by 10

For example 1, 2, 3, 4

Arithmetic mean $\bar{x} = \frac{1+2+3+4}{4} = 2.5$

Now 10 is added to each term then we get 11, 12, 13, 14.

The arithmetic mean for these number is 12.5.

So arithmetic mean is greater by 10.

88. (d)

(a) is true, (b) is true, (c) is true.

(d) is false.

Since,

$E(X^2 Y^2) = E(X^2)\,E(Y^2)$

But since X is not independent of X,

$E(X^2) \neq [E(X)]^2$

$\therefore E(X^2 Y^2) = E(X^2)\,E(Y^2)$

$\neq [E(X)]^2\,[E(Y)]^2$

89. (b)

$$\text{Mean} = \bar{x} = \frac{27+35+40+35+36+36+29}{7} = 34$$

$$\text{Variance is } \sigma_x^2 = \frac{1}{n}\sum x_i^2 - \bar{x}^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$$

$$\frac{1}{7}\begin{Bmatrix}(27-34)^2 + (35-34)^2 + (40-34)^2 \\ +(35-34)^2 + (36-34)^2 + (36-34)^2 \\ +(29-34)^2\end{Bmatrix} = 16.8571$$

$$\text{Standard deviation} = \sqrt{16.8571} = 4.14$$

90. (a)

$$\text{Mean of } x = E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$$

$$= \int_0^2 x \cdot \left(\frac{x}{2}\right) dx = \frac{4}{3}$$

$$\text{Variance of } x = E(x^2) - (E(x))^2$$

$$E(x)^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^2 x^2 \left(\frac{x}{2}\right) dx = 2$$

$$\therefore \quad \text{Variance of } x = \frac{2}{9}$$

91. (15)

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

$$= \sqrt{225} = 15$$

92. (a)

For a discrete random variable, probability mass function,

$$\Sigma P(X) = 1$$

$$\Rightarrow \frac{2+5P}{5} + \frac{1+3P}{5} + \frac{1.5+2P}{5} = 1$$

$$\Rightarrow \quad 4.5 + 10P = 5$$

$$\Rightarrow \quad P = 0.05$$

Also, expectation,

$$E(X) = \Sigma X \cdot P(X)$$

$$= 1\left[\frac{2+5P}{5}\right] + 2\left[\frac{1+3P}{5}\right] + 3\left[\frac{1.5+2P}{5}\right] \quad \ldots(1)$$

Substituting $P = 0.05$ in (1)

$E(X) = 1.87$

93. (c)

We know that

$$\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$$

Given: $\text{mean} = (3\,\text{Median} - \text{Mode})\,X$

$\text{mean} = (3\,\text{median} - (3\,\text{median} - 2\,\text{mean}))X$

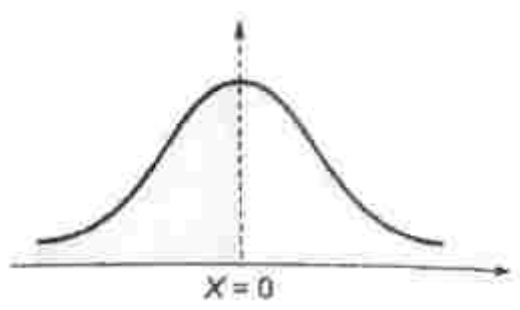
$$\Rightarrow \quad \text{mean} = 2\,(\text{mean})\,X$$

$$X = \frac{1}{2}$$

94. (b)

Cumulative distribution function

$$= \int_{-\infty}^{x} (\text{Probability density function})\,dx$$



Since,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \text{ [complete area of normal curve]} \quad \ldots(i)$$

Also, cumulative distribution function at origin is (shaded area),

$$\int_{-\infty}^{0} f(x)\,dx = \frac{1}{2} \text{ [using (i)]}$$

Alternate solution:

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{(x-\mu_x)^2}{2\sigma_x^2}}$$

If mean $\mu_x = 0$

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-\frac{x^2}{2\sigma_x^2}}$$

$$\int_{-\infty}^{\infty} f_x(x)\,dx = 1$$

Since mean is zero therefore,

$$\int_{-\infty}^{0} f_x(x)\,dx = \int_0^{\infty} f_x(x) = \frac{1}{2}$$

and $$F_x(x) = \int_{-\infty}^{x^+} f_x(x)\,dx$$

$$F_X(0) = \int_{-\infty}^{0^+} f_x(x)\,dx = \frac{1}{2} = 0.5$$

95. (a)

Given, $\mu_X = 1, \sigma_x^2 = 4 \Rightarrow \sigma_x = 2$

Also given, $\mu_Y = -1$ and σ_y is unknown

given,

$$p(X \le -1) = p(Y \ge 2)$$

Converting into standard normal variates,

$$p\left(z \le \frac{-1-\mu_x}{\sigma_x}\right) = p\left(z \ge \frac{2-\mu_y}{\sigma_y}\right)$$

$$p\left(z \le \frac{-1-1}{2}\right) = p\left(z \ge \frac{2-(-1)}{\sigma_y}\right)$$

$$p(z \le -1) = p\left(z \ge \frac{3}{\sigma_y}\right) \quad \text{... (i)}$$

Now since us know that in standard normal distribution,

$$p(z \le -1) = p(z \ge 1) \quad \text{... (ii)}$$

Comparing (i) and (ii) we can say that

$$\frac{3}{\sigma_y} = 1 \Rightarrow \sigma_y = 3$$

96. (2.1, 0.89)

(i) $E(X) = \sum_{i=1}^{4} x_i f(x_i)$

$= x_1 f(x_1) + x_2 f(x_2) + x_3 f(x_3) + x_4 f(x_4)$

$= (1 \times 0.3) + (2 \times 0.4) + (3 \times 0.2) + (4 \times 0.1) = 2.1$

(ii) $\text{Var}(x) = E(x^2) - (E(x))^2$

$E(X^2) = \sum_{i=1}^{4} x_i^2 f(x_i)$

$= (1 \times 0.3) + (4 \times 0.4) + (9 \times 0.2) + (16 \times 0.1)$

$= 0.3 + 1.6 + 1.8 + 1.6 = 5.5$

$\text{Var}(x) = E(x^2) - (E(x))^2 = 5.3 - (2.1)^2 = 0.89$

97. (d)

The probability mass function of Poisson's distribution is,

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!} \text{ where, } \lambda \to \text{mean}$$

so, $P(X \ge 2) = 1 - [P(X = 0) + P(X = 1)]$

$$= 1 - \left[\frac{e^{-2} \times 2^0}{0!} + \frac{e^{-2} \times 2^1}{1!}\right] = 1 - 3e^{-2}$$

98. (c)

$$V(x) = E(x^2) - [E(x)]^2 = R$$

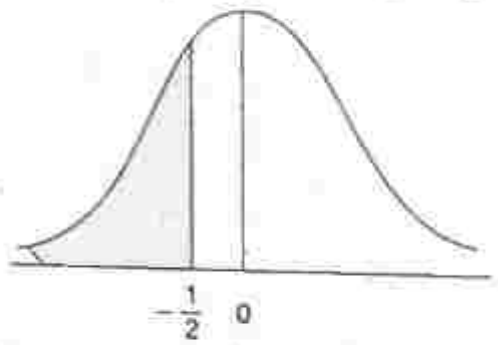
where $V(x)$ is the variance of x.

Since variance is σ_x^2 and hence never negative, $R \ge 0$.

99. (b)

Here, $\sigma^2 = 4 \Rightarrow \sigma = 2$

$$P(x < 0) = p\left(z < \frac{0-\mu}{\sigma}\right) = p\left(z < \frac{0-1}{2}\right) = p\left(z < -\frac{1}{2}\right)$$



Which is the shaded area in the picture and its value is clearly between 0 and 0.5

100. (d)

$$\text{Mean } \bar{x} = \frac{3+5+7+9+11+13}{6} = 8$$

$$\text{Variance is } \sigma_x^2 = \frac{1}{n}\sum_{i=1}^{6}(x_i - \bar{x})^2$$

$$= \frac{1}{6}\{(3-8)^2 + (5-8)^2 + (7-8)^2 + (9-8)^2 + (11-8)^2 + (13-8)^2\}$$

$$\sigma_x^2 = 11.6666$$

Standard deviation

$$\sigma_x = \sqrt{11.6666} = 3.41 \text{ (approx.)}$$

101. (c)

$$\text{Mean} = \frac{\text{Total marks}}{\text{Total students}}$$

$$= \frac{25 \times 20 + 30 \times 20 + 35 \times 40 + 40 \times 20}{20 + 20 + 40 + 20}$$

$\Rightarrow$ Mean = 33 [so answer is either A or C option]

Mode $\to$ Marks with highest frequency = 35

[40 students obtained 35 marks]

so answer is (c)

Median $\to$ Total number of observations [i.e., students] = 20 + 20 + 40 + 20 = 100

So median

$$= \frac{\left(\frac{100}{2}\right)^{th} \text{ student's marks} + \left[\frac{100}{2} + 1\right]^{th} \text{ students marks}}{2}$$

$$= \frac{35 + 35}{2} = 35$$

102. (d)

$$\frac{\sum_{i=1}^{30} x_i}{n} = 64.5$$

$$\Rightarrow \frac{\sum_{i=1}^{30} x_i}{30} = 64.5$$

$$\Rightarrow \sum_{i=1}^{30} x_i = 1935$$

$$\text{Correct } \sum_{i=1}^{30} x_i = 1935 - 85 + 58 = 1908$$

$$\text{Correct mean is } \frac{\sum_{i=1}^{30} x_i}{30} = \frac{1908}{30} = 63.6$$

103. (b)

Arithmetic mean of natural numbers 1, 2, ..., n is

$$\bar{x} = \frac{1+2+.....+n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

104. (d)

Mean,

$\bar{x} = \Sigma\, xp(x) = 1\times0.3 + 2\times0.8 + 3\times0.1 = 1.8$

Standard deviation,

$$\sigma = \left(\Sigma x^2 p(x) - (\Sigma x p(x))^2\right)^{1/2}$$

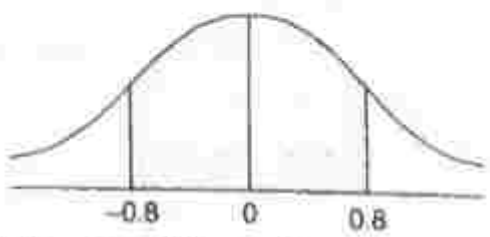
$$\therefore \quad \sigma = \left(0.3\times1^2 + 0.6\times2^2 + 0.1\times3^2 - 1.8^2\right)^{1/2}$$

$$= (3.6 - 1.8^2)^{1/2} = (0.36)^{1/2} = 0.6$$

105. (c)

Mode means highest number of observations or occurrence of data most of the time as data 17, occurs four times, i.e., highest time. So mode is 17.

106. (b)



Given that, $\mu = 30,\ \sigma = 5$

$$P\left(\frac{26-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{34-\mu}{\sigma}\right)$$

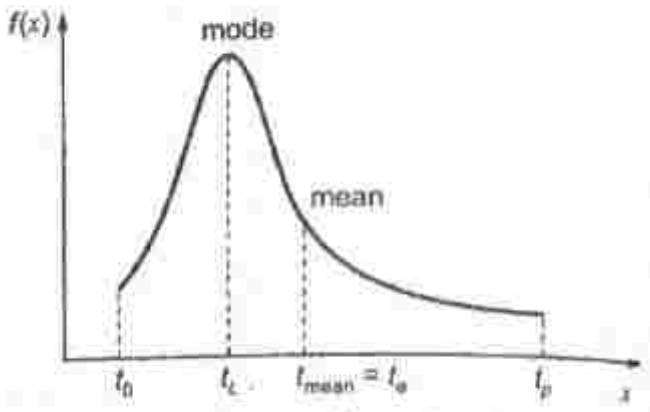
$$P\left(\frac{26-30}{5} < z < \frac{34-30}{5}\right)$$

$$P(-0.8 < z < 0.8) = 2\int_0^{0.8} \phi(z)\, dz$$

Given that, $z = 0.8$ is 0.2881

$= 2(0.2881) = 0.5762$

107. (c)



$t_L < t_{mean}$ = Curve is skew to right.

mode < mean

i.e., mean > median and mode

Mean is greater than the mode and the median. This is common for a distribution that is skewed to the right [i.e., bunched up toward the left and a 'tail' stretching toward the right].

108. (11)

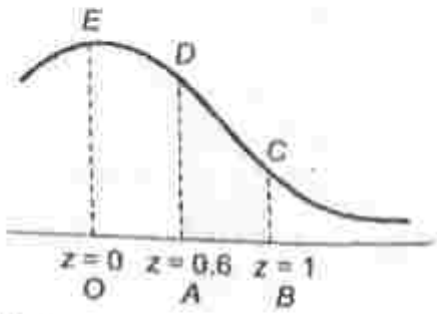
$E[2X + Y] = 0$ and $E[X + 2Y] = 33$

Then, $2E[X] + E[Y] = 0$ and $E[X] + 2E[Y] = 33$

$3E[X] + 3E[Y] = 0 + 33 = 33$

$E[X] + E[Y] = 11$

109. (c)



Given that, $\mu = 14,\ \sigma = 2.5$

$$P(X = 16) = P(15.5 < X < 16.5)$$

$$= P\left(\frac{15.5-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{16.5-\mu}{\sigma}\right)$$

$$= P\left(\frac{15.5-\mu}{2.5} < z < \frac{16.5-\mu}{2.5}\right)$$

$= P(0.6 < z < 1)$

= Area of $OABCDE$ – Area of $OADE$

$= 0.3413 - 0.2257 = 0.1156$

Number of students scoring 16 marks

$= 100 \times 0.1156 = 11.5,\ 12$

110. (a)

Mean of 1st 'n' natural numbers $= \frac{1+2+.....+n}{n}$

$$\bar{X} = \frac{n+1}{2}$$

Variance is $\sigma_x^2 = \frac{1}{n}\sum X_i^2 - \bar{X}^2$

$$= \frac{1}{n}\{1^2+2^2.....+n^2\} - \frac{(n+1)^2}{4}$$

$$= \frac{1}{n}n\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= \left(\frac{n+1}{2}\right)\left\{\frac{2n+1}{3} - \frac{n+1}{2}\right\}$$

$$= \left(\frac{n+1}{2}\right)\left\{\frac{4n+2-3n-3}{6}\right\}$$

$$= \left\{\frac{n+1}{2}\right\}\left\{\frac{n-1}{6}\right\} = \frac{n^2-1}{12}$$

111. (3)

Mean, $x = \frac{3+x+2+4}{4}$

$4x = 9 + x$

$3x = 9$

$x = 3$

So the given numbers are 3, 3, 2, 4

Mode = 3

Mode = Maximum times repeated value

112. (b)

$E(x) = 5$

$\text{Var}(x) = 1$

Then we know that

$\text{Var}(x) = E(x^2) - (E(x))^2$

$\Rightarrow E(x^2) = \text{Var}(x) + [E(x)]^2$

$= 1 + (5)^2 = 26$

113. (a)

Arithmetic mean of 32, 30, 28, 21, 26, 17, 27, 31, 34, 18, 22

$$\bar{x} = \frac{32+30+28+21+.....+22}{11} = 26$$

114. (a)

The median of 55, 100, 75, 80, 90, 85, 95, 45, 70, 70, 55

Arranging them in increasing order

45, 55, 55, 70, 70, 75, 80, 85, 90, 95, 100

Median = 75 (Middle most observation)

115. (*)

Let X be standard normal take:

(a) $Y = X$ if $|X| < 1$ and $Y = -X$ if $|X| >= 1$.

(b) take $Y = X$.

(c) $Y = 2X$

(d) $Y = X + 1$.

That may be reason to give marks to all.

116. (b)

Mean $= np$

Variance $= npq = np(1-p)$

117. (d)

$$\text{Mean} = \frac{\sum x_i f_i}{\sum f_i}$$

$$= \frac{5\times5+15\times4+25\times6}{5+14+6} = 15.4$$

118. (0)

$P(X = 1.5) = 0$

As we do not have any probability in PMF at $x = 1.5$.

119. (0.75)

$P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$

$$= \frac{k}{3} + \frac{k}{6} + \frac{k}{4} \quad ...(i)$$

$\sum P(X \le 6) = 1$

$$\frac{k}{3}+\frac{k}{6}+\frac{k}{4}+\frac{3k}{16}+\frac{k}{32}+\frac{k}{32} = 1$$

$\Rightarrow k = 1$

In equation (i) put $k = 1$

$$= \frac{1}{3}+\frac{1}{6}+\frac{1}{4}$$

$$\Rightarrow \frac{3}{4} = 0.75$$

120. (a)

$$E(X) = \sum_{i=0}^{6} x_i P(x_i)$$

$$= 0\times\frac{1}{3}+1\times\frac{1}{6}+2\times\frac{1}{4}+3\times\frac{3}{16}+5\times\frac{1}{32}+\frac{6}{32}$$

$$= \frac{115}{96}$$

121. (2.2)

$$V(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum_{i=0}^{i=6} x_i^2 P(x_i) = 3.635$$

$$E(X) = 1.197$$

$$V(X) = 3.635 - (1.197)^2 = 2.199 = 2.2$$

122. (4.593)

$$E(3X+1) = 3E(X) + 1$$

$$= 3\left(\frac{115}{96}\right) + 1 = 4.593$$

123. (8.8)

$$V(2X-4) = 2^2 V(X)$$

$$= 4V(X) = 4(2.2) = 8.8$$

124. (52)

Median speed is the speed at the middle value in series of spot speeds that are arranged in ascending order. 50% of speed values will be greater than the median 50% will be less than the median.

Ascending order of spot speed studies are

26, 33, 39, 45, 50, 54, 56, 60, 73, 77

$$\text{Median speed} = \frac{50+54}{2} = 52 \text{ km/hr}$$

125. (a)

Arithmetic mean of 24, 28, 29, 34, 18, 22, 26, 30, 32, 24, 20 is

$$\bar{x} = \frac{24+28+29+\ldots.+20}{11} = 26.09$$

126. (a, b, d)

- Standard deviation, variance and coefficient of variation are examples of dispersion measures.
- Mean, median and mode are some examples of central tendency measure.

127. (0 and 0.4)

Here $f(x)$ is an even function

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)dx = 0$$

($\because x f(x)$ is an odd function)

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx$$

$$= \int_{-1}^{0} x^2(1+x)dx + \int_{0}^{1} x^2(1-x)dx = \frac{1}{6}$$

$$\text{Variance of } X = E(X^2) - (E(X))^2 = \frac{1}{6}$$

$$\text{Standard deviation} = \frac{1}{\sqrt{6}}$$

128. (c)

Given, $a = 0.6$

X	0	1
$P(X)$	0.6	0.4

$$\text{Required value} = V(X) = E(X^2) - [E(X)]^2$$

$$E(X) = \sum_i X_i P_i$$

$$= 0 \times 0.6 + 1 \times 0.4 = 0.4$$

$$E(X^2) = \sum_i X_i^2 P_i$$

$$= 0^2 \times 0.6 + 1^2 \times 0.4 = 0.4$$

$$\therefore \quad V(X) = E(X^2) - [E(X)]^2$$

$$= 0.4 - 0.16 = 0.24$$

129. (0.25)

$$f(x) = \begin{cases} a+bx & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \frac{2}{3}$$

$$\Rightarrow \int_0^1 x\, f(x)dx = \frac{2}{3}$$

$$\Rightarrow \int_0^1 x(a+bx)dx = \frac{2}{3}$$

$$\Rightarrow a\left(\frac{x^2}{2}\right)_0^1 + b\left(\frac{x^3}{3}\right)_0^1 = \frac{2}{3}$$

$$a\left(\frac{1}{2}\right) + b\left(\frac{1}{3}\right) = \frac{2}{3}$$

$$\Rightarrow \quad 3a + 2b = 4$$

Now, $\int_0^1 f(x)dx = 1$

Total probability is always equal to 1.

$$\int_0^1 (a+bx)dx$$

$$\left(ax + \frac{bx^2}{2}\right)_0^1 = 1$$

$$a + \frac{b}{2} = 1$$

$$2a + b = 2 \quad \text{...(ii)}$$

On solving equation (i) and (ii)

$$a = 0, \quad b = 2$$

So, $f(x) = \begin{cases} 2x, & \text{for } 0 < x < 1 \\ 0, & \end{cases}$

Now, we need $\int_0^{\frac{1}{2}} 2x\,dx = \frac{1}{4} = 0.25$

130. (4)

Probability density of a function is given by

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \text{...(i)}$$

Given, $f(x) = \frac{e^{-4}4^x}{x!}$ ...(ii)

On comparing equation (i) and (ii) we get $\lambda = 4$.

131. (0.25)

$$\text{Mean} = np = \frac{10}{8}$$

$$\text{Variance} \quad npq = \frac{30}{32}$$

$$q = \frac{30}{32} \times \frac{8}{10} = \frac{3}{4}$$

$$p = 1 - \frac{3}{4} = \frac{1}{4}$$

132. (b)

For binomial distribution,

$$\text{Mean} = np = 9 \quad \text{...(i)}$$

$$\text{Variance} = npq = \sigma^2 = 6 \quad \text{...(ii)}$$

From (i) and (ii), $q = \frac{6}{9} = \frac{2}{3}$

$$\therefore \quad p = 1 - q = \frac{1}{3}$$

$$\therefore \quad np = n \times \frac{1}{3} = 9$$

$$n = 27$$

133. (0.20)

$\because$ x is a random variable which is uniformly distributed,

$$\therefore \quad f(x) = \begin{cases} \frac{1}{1-0} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore \quad E[x^4] = \int_{-\infty}^{\infty} x^4 f(x)\,dx$$

$$\Rightarrow \quad E[x^4] = \int_0^1 1 \times (x^4)\,dx$$

$$= \left[\frac{x^5}{5}\right]_0^1 = \frac{1}{5} = 0.2$$

134. (0.64)

Standard deviation $(\sigma) = \sqrt{E(X^2) - (E(X))^2}$

$$E(X) = \sum_{i=1}^{3} X_i \cdot P(X_i)$$

$$= 1(0.4) + 2(0.5) + 3(0.1) = 1.7$$

$$E(X^2) = \sum_{i=1}^{3} X_i^2 \cdot P(X_i)$$

$$= 1^2(0.4) + 2^2(0.5) + 3^2(0.1) = 4.1$$

$$\therefore \quad \sigma = \sqrt{3.3 - (1.7)^2} = 0.64$$

135. (a)

$$P(2) = 9P(4) + 90P(6)$$

For Poisson's distribution,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!},$$

where λ is the mean of Poisson's distribution

Hence, $\frac{e^{-\lambda}\lambda^2}{2!} = 9\frac{e^{-\lambda}\lambda^4}{4!} + 90\frac{e^{-\lambda}\lambda^6}{6!}$

$$\Rightarrow \quad \frac{e^{-\lambda}\lambda^2}{2} = 9\frac{e^{-\lambda}\lambda^4}{24} + 90\frac{e^{-\lambda}\lambda^6}{720}$$

$$\Rightarrow \quad \frac{e^{-\lambda}\lambda^6}{8} + \frac{3e^{-\lambda}\lambda^4}{8} - \frac{e^{-\lambda}\lambda^2}{2} = 0$$

$$\Rightarrow \quad \frac{e^{-\lambda}\lambda^2}{2}\left[\frac{\lambda^4}{4} + \frac{3\lambda^2}{4} - 1\right] = 0$$

Given: $\lambda \neq 0$

$$\therefore \quad \left[\frac{\lambda^4}{4} + \frac{3\lambda^2}{4} - 1\right] = 0$$

$$\Rightarrow \quad \lambda^4 + 3\lambda^2 - 4 = 0$$

$$\Rightarrow \quad \lambda^4 + 4\lambda^2 - \lambda^2 - 4 = 0$$

$$\Rightarrow \quad \lambda^2(\lambda^2 + 4) - 1(\lambda^2 + 4) = 0$$

$$\Rightarrow \quad (\lambda^2 - 1)(\lambda^2 + 4) = 0$$

$$\Rightarrow \quad \lambda^2 = 1, \lambda^2 + 4 \neq 0$$

$$\Rightarrow \quad \lambda = \pm 1$$

136. (3.99)

∵ 'x' is a Gaussian random variable.

Hence, $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$; $-\infty < x < \infty$

∵ Mean, $\mu = 0$, Variance $= \sigma^2 = 1$

So, $f(x) = \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-x^2}{2}\right)}$

$$E[|5x|] = \int_{-\infty}^{\infty} (5x)\frac{1}{\sqrt{2\pi}} e^{\left(\frac{-x^2}{2}\right)} dx$$

Put, $\frac{x^2}{2} = t$

$\therefore$ $x\,dx = dt$

$$\text{Expected value} = 2\int_0^{\infty} \frac{5}{\sqrt{2\pi}} e^{-t} dt$$

$$= \frac{10}{\sqrt{2\pi}}\int_0^{\infty} e^{-t} dt = \frac{10}{\sqrt{2\pi}}\left[\frac{e^{-t}}{-1}\right]_0^{\infty}$$

$$= \frac{10}{\sqrt{2\pi}}[-0+1] = \frac{10}{\sqrt{2\pi}} = 3.989 \approx 3.99$$

137. (b)

The mean of a set of numbers $= \bar{X}$

If each number is increased by λ, than mean of new set is $\bar{X} + \lambda$.

138. (18)

$$V(X) = E(X^2) - [E(X)]^2 = 5 - 1 = 4$$

$$V(2X-1) = 2^2 V(X) = 4V(X) \quad \text{...(i)}$$

$$SD(X) = \sqrt{V(X)} \quad \text{...(ii)}$$

Equation (i) + (ii)

$$= 4V(X) + \sqrt{V(X)}$$

$$= 4(4) + \sqrt{4} = 18$$

139. (i) 37.2, (ii) 39, (iii) 43.33)

Given:

Class	Frequency
0 – 10	4
10 – 20	5
20 – 30	7
30 – 40	10
40 – 50	12
50 – 60	8
60 – 70	4

(i) Mean is

$$\bar{x} = \frac{5\times4 + 15\times5 + 25\times7 + 35\times10 + 45\times12 + 55\times8 + 65\times4}{4+5+7+10+12+8+4} = 37.2$$

(ii) $\frac{N+1}{2} = \frac{50+1}{2} = 25.5$

Class	Frequency	Cumulative Frequency
0 – 10	4	4
10 – 20	5	9
20 – 30	7	16
30 – 40	10	26 ← Median class
40 – 50	12	38
50 – 60	8	46
60 – 70	4	50

where, $l = 30$, $f = 10$, $m = 16$, $N = 50$

$$\text{The median} = l + \left(\frac{\frac{N}{2} - m}{f}\right) C$$

$$= 30 + \left(\frac{\frac{50}{2} - 16}{10}\right) 10 = 33$$

(iii)

Class	Frequency	
0 – 10	4	
10 – 20	5	
20 – 30	7	
30 – 40	10	← modal class
40 – 50	12	
50 – 60	8	
60 – 70	4	

$l = 40$, $f = 12$, $f_{-1} = 10$, $f_1 = 8$

$\Delta_1 = f - f_{-1} = 2$

$\Delta_2 = f - f_1 = 4$

$$\text{Mode} = l + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) C$$

$$= 40 + \left(\frac{2}{2+4}\right) 10 = 43.33$$

140. (4.8)

$X \sim \text{Bin}(30, 0.8)$

$n = 20$

$p = 0.8$

$\text{var}(X) = npq \quad [q = 1 - p]$

$= 30 \times 0.8 \times 0.2 = 4.8$

141. (b)

$$\text{Mean } \mu_t = E(t) = \int_{-\infty}^{\infty} t \cdot f(t) \cdot dt = \int_{-1}^{1} t \cdot f(t) \cdot dt$$

$$= \int_{-1}^{0} t(1+t)dt + \int_{0}^{1} t(1-t)dt$$

$$= \left[\frac{t^2}{2} + \frac{t^3}{3}\right]_{-1}^{0} + \left[\frac{t^2}{2} - \frac{t^3}{3}\right]_{0}^{1}$$

$$= -\left[\frac{1}{2} - \frac{1}{3}\right] + \left[\frac{1}{2} - \frac{1}{3}\right] = -\left[\frac{1}{6} + \frac{1}{6}\right] = 0$$

$$\text{Variance} = E(t^2) - [E(t)]^2$$

$$= \int_{-\infty}^{\infty} t^2 f(t)dt - [E(t)]^2$$

$$= \int_{-\infty}^{\infty} t^2 f(t)dt - (0)^2 = \int_{-\infty}^{\infty} t^2 f(t)dt$$

$$= \int_{-1}^{0} t^2(1+t)dt + \int_{0}^{1} t^2(1-t)dt$$

$$= \int_{-1}^{0} (t^2 + t^3) \cdot dt + \int_{0}^{1} t^2(1-t)dt$$

$$= \left[\frac{t^3}{3} + \frac{t^4}{4}\right]_{-1}^{0} + \left[\frac{t^3}{3} - \frac{t^4}{4}\right]_{0}^{1}$$

$$= \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \frac{1}{\sqrt{6}}$$

142. (11)

$X \sim \text{Bin}(500, 0.6)$

$$n = 500$$

$$p = 0.6$$

$$\text{Standard deviation} = \sqrt{npq} \quad (q = 1-p)$$

$$= \sqrt{500 \times 0.6 \times 0.4}$$

$$= \sqrt{5 \times 6 \times 4} = \sqrt{120} = 11$$

143. (0.94)

$$F(X = 0.3) + F(X = 0.8)$$

$$F(X = 0.3) = \int_{0}^{0.3} \frac{x}{2} dx = 0.0225 \quad \ldots(i)$$

$$F(X = 0.8) = \int_{0}^{0.5} \frac{x}{2} dx + \int_{0.5}^{0.8} (1-x)dx$$

$$= 0.0625 + 0.855 = 0.9175 \quad \ldots(ii)$$

$$(i) + (ii) = 0.0225 + 0.9175 = 0.94$$

144. (a)

Given: $f(x) = \frac{1}{5}e^{-x/5}, x \geq 0$

Comparing the general function

$$f(x) = \begin{cases} \lambda e^{-\lambda x}; & x \geq 0 \\ 0; & x < 0 \end{cases} \Rightarrow \lambda = \frac{1}{5}$$

The cumulative distributive function

$$f(a) = 1 - e^{\lambda a}$$

$$\Rightarrow \quad P(E > 5) = 1 - f(5) = 1 - (1 - e^{\lambda 5})$$

$$= e^{-\frac{1}{5} \times 5} = e^{-1} = 1/e$$

Alternate solution:

$$P(x > 5) = \int_{5}^{\infty} f(x)\,dx = \int_{5}^{\infty} \frac{1}{5} e^{-x/5} dx$$

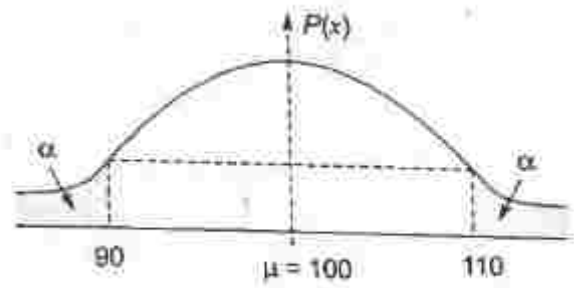
$$= \frac{1}{5} \int_{5}^{\infty} e^{-x/5} dx$$

$$= \frac{1}{5}\left[-5e^{-x/5}\Big|_{5}^{\infty}\right] = -1\left[e^{-\infty} - e^{-5/5}\right]$$

$$= e^{-1} = \frac{1}{e}$$

145. (a)

Since a follows normal distribution with mean = 100. Since normal curve is symmetrical about mean so $P(x \geq 110) = P(x \leq 90) = \alpha$



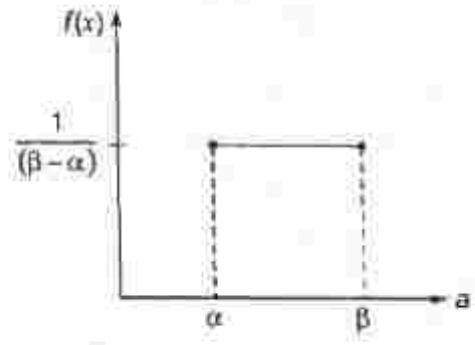
Also $P(-\infty < x < \infty) = 1$

$$\Rightarrow \quad P(90 \geq x \geq 110) = P(-\infty < x < \infty) - P(x \leq 90) - P(x \geq 110)$$

$$\Rightarrow P(90 \geq x \geq 110) = 1 - 2\alpha$$

146. (a)

For uniform distribution



Here, $\alpha = 2$ and $\beta = 10$

Variance;

$$V(x) = \left(\frac{\beta-\alpha}{12}\right)^2 = \frac{(10-2)^2}{12}$$

$$= \frac{64}{12} = \frac{16}{3}$$

147. (0.75)

$$P(x) = \int_0^2 f(x)dx = 1$$

$$= \int_0^2 k(2x-x^2)dx = 1$$

$$k = \frac{3}{4} = 0.75$$

148. (1.028)

$$\Rightarrow \quad F(X = 2.5) + F(X = 0.2)$$

$$= 1 + \int_0^{0.2} kx(2-x)dx$$

$$= 1 + 0.028 = 1.028$$

149. (b)

Given, $f(x) = x^2 \quad -1 \le x \le 1$

$= 0 \quad$ elsewhere

$$p\left(-\frac{1}{3} \le x \le \frac{1}{3}\right) = \int_{-\frac{1}{3}}^{\frac{1}{3}} f(x)\,dx$$

$$= \int_{-\frac{1}{3}}^{\frac{1}{3}} x^2\,dx = \left[\frac{x^3}{3}\right]_{-\frac{1}{3}}^{\frac{1}{3}} = \frac{2}{81}$$

The probability expressed in percentage,

$$p = \frac{2}{81} \times 100 = 2.469\% = 2.47\%$$

150. (c)

The p.d.f. of the random variable is

x	−1	+1
$P(x)$	0.5	0.5

The cumulative distribution function $F(x)$ is the probability upto x as given below:

x	−1	+1
$F(x)$	0.5	1.0

So correct option is (c).

151. (6)

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$f(x) = \begin{cases} \lambda(-x^2+3x-2) & 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore \quad \int_1^2 \lambda(-x^2+3x-2)dx = 1$$

$$\Rightarrow \quad \lambda\left[-\frac{x^3}{3}+3\frac{x^2}{2}-2x\right]_1^2 = 1$$

$$\Rightarrow \lambda\left[-\left(\frac{8}{3}-\frac{1}{3}\right)+\frac{3}{2}(4-1)-2(2-1)\right] = 1$$

$$\Rightarrow \quad \lambda\left[-\frac{7}{3}+\frac{9}{2}-2\right] = 1$$

$$\Rightarrow \quad \lambda\left[\frac{-14+27-12}{6}\right] = 1$$

$$\Rightarrow \quad \lambda = \frac{6}{1} = 6$$

$$\lambda = 6$$

152. (a)

$$P = \int_1^{\infty} f(x)dx = \int_1^{\infty} e^{-x}dx = -e^{-x}\Big|_1^{\infty}$$

$$= -(e^{-\infty} - e^{-1}) = e^{-1} = 0.368$$

153. (0.031)

Given, $f(x) = x^3 \quad -1 \le x \le 1$

$= 0 \quad$ elsewhere

$$P\left(-\frac{1}{2} \le x \le \frac{1}{2}\right) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x)\,dx$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} x^3\,dx = \left[\frac{x^4}{4}\right]_{-\frac{1}{2}}^{\frac{1}{2}}$$

$$= \frac{1}{64} + \frac{1}{64} = \frac{1}{32} = 0.031$$

154. (0.4)

$$f(E) = \begin{cases} \frac{1}{5} & 0 \le E \le \text{mm/day} \\ 0 & \text{Otherwise} \end{cases}$$

$$P(2< E < 4) = \int_2^4 f(E)dE = \int_2^4 \frac{1}{5}dE = \frac{1}{5}[E]_2^4$$

$$= \frac{1}{5}(4-2) = \frac{2}{5} = 0.4$$

155. (2.4)

$X \sim$ Bin (12, 0.2)

$n = 12, \quad p = 0.2$

$E(X) = np$

$= 12 \times 0.2 = 2.4$

156. (0.4)

Probability $(0.5 < X < 5)$

$$= \int_{0.5}^{5} f(x)dx$$

$$= \int_{0.5}^{1} 0.2dx + \int_1^4 0.1\,dx + \int_4^5 0dx$$

$$= 0.2[1-0.5] + 0.1[4-1] + 0[5-4]$$

$$= 0.2 \times 0.5 + 0.1 \times 3 = 0.1 + 0.3 = 0.4$$

157. (0.25)

$$f(x) = \begin{cases} a+bx & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

Now given $E(X) = \frac{2}{3}$

$$\Rightarrow \int_0^1 xf(x)dx = \frac{2}{3}$$

$$= \int_0^1 x(a+bx)\,dx = \frac{2}{3}$$

$$\Rightarrow a\left(\frac{x^2}{2}\right)_0^1 + b\left(\frac{x^3}{3}\right)_0^1 = \frac{2}{3}$$

$$a\left(\frac{1}{2}\right) + b\left(\frac{1}{3}\right) = \frac{2}{3}$$

$$\Rightarrow 3a + 2b = 4 \qquad ...(i)$$

Now, $\int_0^1 f(x)dx = 1$

(Total probability is always equal to 1)

$$\Rightarrow = \int_0^1 (a+bx)\,dx$$

$$= \left(ax + \frac{bx^2}{2}\right)_0^1 = 1$$

$$\Rightarrow a + \frac{b}{2} = 1$$

$$\Rightarrow 2a + b = 2 \qquad ...(ii)$$

Now solving (i) and (ii), we get

$a = 0, \quad b = 2$

So $f(x) = \begin{cases} 2x & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$

Now we need $\int_0^{1/2} 2x\,dx = \frac{1}{4}$

158. (0.33)

$$P(X+Y \le 1) = \int_{x=0}^{1}\int_{y=0}^{(1-x)} f_{xy}(x,y)dx\,dy$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} (x+y)dx\,dy$$

$$= \int_{x=0}^{1}\left(xy + \frac{y^2}{2}\right)_0^{1-x}$$

$$= \int_{x=0}^{1}\left(x(1-x) + \frac{(1-x)^2}{2}\right)dx$$

$$= \int_{x=0}^{1}\left(\frac{1}{2} - \frac{x^2}{2}\right)dx = \left(\frac{x}{2} - \frac{x^3}{6}\right)_0^1$$

$$= \frac{1}{2} - \frac{1}{6} = \frac{1}{3} = 0.33$$

159. (b)

$$f(x) = e^{-x-e^{-x}} = e^{-x} \cdot e^{-e^{-x}}$$

$$y(x) = \int f(x)dx = \int e^{-x} \cdot e^{-e^{-x}}dx$$

Let $e^{-x} = t$

$-e^{-x}\,dx = dt$

$$\int f(x)dx = \int e^{-t}\cdot(-dt)$$

$$= \frac{e^{-t}}{-1}\cdot(-d) = e^{-t}$$

$$= e^{-(e^{-x})} = e^{-e^{-x}}$$

160. (0.3)

x follows uniform distribution over [-2, 10]

$$\therefore \quad f(x) = \frac{1}{b-a} = \frac{1}{10-(-2)} = \frac{1}{12}$$

Given: $y = 2x - 6$

$$\Rightarrow \quad x = \frac{y+6}{2}$$

For $y = 7$

$$x = \frac{7+6}{2} = \frac{13}{2} = 6.5$$

$$P\left[\frac{y<7}{x>5}\right] = P\left[\frac{x<6.5}{x>5}\right] = \frac{P[5<x<6.5]}{P[x>5]}$$

$$= \frac{P[x>5 \text{ and } x<6.5]}{P[x>5]}$$

$$= \frac{\int_5^{6.5} f(x)dx}{\int_5^{10} f(x)dx} = \frac{\int_5^{6.5} \frac{1}{12}dx}{\int_5^{10} \frac{1}{12}dx}$$

$$= \frac{(x)_5^{6.5}}{(x)_5^{10}} = \frac{1.5}{5} = 0.3$$

161. (b)

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^2 kxdx + \int_2^4 2kdx + \int_4^6 (-kx+6k)dx = 1$$

$$k\left(\frac{x^2}{2}\right)_0^2 + 2k(x)|_2^4 + \left(-k\frac{x^2}{2} + 6kx\right)_4^6 = 1$$

$$k(2-0) + 2k(4-2) + (-18k+36k) - (-8k+24k) = 1$$

$$\Rightarrow \quad 2k + 4k + 18k - 16k = 1$$

$$\Rightarrow \quad 2k + 4k + 2k = 1$$

$$8k = 1$$

$$\therefore \quad k = \frac{1}{8}$$

162. (a)

The first moment about the origin of Binomial distribution is mean of Binomial distribution $E(x) = np$.

164. (a)

Given: A probability density function is,

$$R_x(x) = Pe^{-4|x|} + Qe^{-|x|}$$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} R_x(x)dx = 1$$

$$\int_{-\infty}^{\infty} \left(Pe^{-4|x|} + Qe^{-|x|}\right)dx = 1$$

$P_x(x)$ is even function as $|x|$ is even function.

So, by the property of even function,

$$2\int_0^{\infty} \left(Pe^{-4x} + Qe^{-x}\right)dx = 1$$

$$2P\left(\frac{e^{-4x}}{-4}\right)_0^{\infty} + 2Q\left(\frac{e^{-x}}{-1}\right)_0^{\infty} = 1$$

$$-\frac{P}{2}\left(e^{-\infty} - e^0\right) - 2Q\left(e^{-\infty} - e^0\right) = 1$$

$$-\frac{M}{2}(0-1) - 2Q(0-1) = 1$$

$$\frac{P}{2} + 2Q = 1$$

165. (c)

If $f(x)$ is the continuous probability density function of a random variable X then,

$$p(a<x\le b) = p(a\le x\le b)$$

$$= \int_a^b f(x)dx$$

166. (0.20)

$\because$ x is a random variable which is uniformly distributed,

$$\therefore \quad f(x) = \begin{cases} \frac{1}{1-0} & 0<x<1 \\ 0 & \text{Otherwise} \end{cases}$$

$$\therefore \quad E[x^4] = \int_{-\infty}^{\infty} x^4 f(x)dx$$

$$\Rightarrow \quad E[x^4] = \int_0^1 1\times(x^4)dx = \left[\frac{x^5}{5}\right]_0^1 = \frac{1}{5} = 0.2$$

167. (b)

Probability density function

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \ldots(i)$$

Given that

$$f(x) = \frac{e^{-2}2^x}{x!} \quad \ldots(ii)$$

Comparing both (i) and (ii)

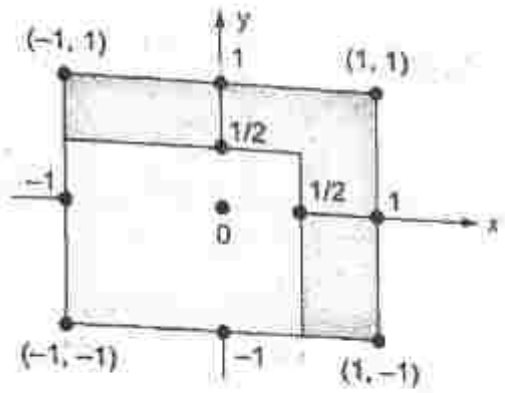
We get, $\lambda = 2$

In Poisson distribution mean $(\mu) = \lambda$

So, $\mu = 2$

168. (b)

$-1 \le x \le 1$ and $-1 \le y \le 1$ is the entire rectangle.
The region in which maximum of (x, y) is less than $\frac{1}{2}$ is shown below as shaded region inside this rectangle.



$$P\left(\max(x, y) < \frac{1}{2}\right) = \frac{\text{Area of shaded region}}{\text{Area of entire rectangle}}$$

$$= \frac{\frac{3}{2}\times\frac{3}{2}}{2\times 2} = \frac{9}{16}$$

169. (b)

$$\int_0^4 C_x\,dx = 1$$

$$\Rightarrow \quad c = \frac{1}{8}$$

$$P(1 \le x \le 2) = \int_1^2 C_x = C\int_1^2 x = \frac{1}{8}\int_1^2 x$$

$$= \frac{1}{8}\left[\frac{x^2}{2}\right]_1^2 = \frac{3}{16}$$

So, option (b) is correct.

170. (b)

Given,

$$f(x) = x^2 + a \text{ for } -1 \le x \le 1$$

Density function:

$$\int_{-1}^{1} f(x)\cdot dx = 1 \Rightarrow \int_{-1}^{1}(x^2 + a) = 1$$

$$\Rightarrow \left[\frac{x^3}{3} + ax\right]_{-1}^{1} = 1 \Rightarrow \frac{2}{3} + 2a = 1$$

$$\Rightarrow \quad a = \frac{1}{6}$$

So option (b) is correct.

171. (4)

Probability density of a function is given by

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \ldots(i)$$

Given, $$f(x) = \frac{e^{-4}4^x}{x!} \quad \ldots(ii)$$

On comparing equation (i) and (ii) we get $\lambda = 4$.

172. (0.25)

$$\text{Mean} = np = \frac{10}{8}$$

$$\text{Variance, } npq = \frac{30}{32}$$

$$q = \frac{30}{32}\times\frac{8}{10} = \frac{3}{4}$$

$$p = 1 - \frac{3}{4} = \frac{1}{4}$$

173. (a, b, c)

All three options (a), (b) and (c) are correct.

174. (19)

Given set is linearly dependent so its determinant will be zero.

$$\begin{vmatrix} 1 & -2 & \lambda \\ 2 & -1 & 5 \\ 3 & -5 & 7\lambda \end{vmatrix} = 0$$

$$1(-7\lambda + 25) + 2(14\lambda - 15) + \lambda(-10 + 3) = 0$$

$$-7\lambda + 25 + 28\lambda - 30 - 7\lambda = 0$$

$$14\lambda - 5 = 0$$

$$\lambda = \frac{5}{14}$$

So, $a = 5, \quad b = 14$

Then, $a + b = 19$

175. (b)

For binomial distribution,

$$\text{Mean} = np = 9 \quad \ldots(i)$$

$$\text{Variance} = npq = \sigma^2 = 6 \quad \ldots(ii)$$

From (i) and (ii), $q = \frac{6}{9} = \frac{2}{3}$

$$\therefore \quad p = 1 - q = \frac{1}{3}$$

$$\therefore \quad np = n\times\frac{1}{3} = 9$$

$$n = 27$$

176. (b)

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^1 cx^{n-1}dx = 1$$

$$c = n$$

$$E(X) = \int_0^1 x\,cx^{n-1}dx = \int_0^1 cx^n dx$$

$$= \frac{(n)}{(n+1)}$$

177. (b)

$$c = (n)$$

$$c^2 + 2c = n^2 + 2n = n(n+2)$$

178. (d)

$$P(X > x) = \int_x^{\infty} f(x)dx$$

$$= \int_x^1 c \cdot x^{n-1}dx = 1 - x^n$$

179. (a)

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_{-\infty}^{1} f(x)dx + \int_1^{\infty} f(x)dx = 1$$

$$0 + \int_1^{\infty} \frac{A}{x^{\mu}}dx = 1$$

$$A\left(\frac{x^{-\mu+1}}{-\mu+1}\right)\Bigg|_1^{\infty} = 1$$

$$\frac{A(0-1)}{1-\mu} = 1$$

$$A = \mu - 1$$

180. (3.99)

∵ 'x' is a Gaussian random variable,

Hence, $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$

∵ Mean, $\mu = 0$, Variance $= \sigma^2 = 1$

So, $f(x) = \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-x^2}{2}\right)}$

$$E[|5x|] = \int_{-\infty}^{\infty} (5x)\frac{1}{\sqrt{2\pi}} e^{\left(\frac{x^2}{2}\right)}dx$$

Put, $\frac{x^2}{2} = t$

$\therefore \quad xdx = dt$

Expected value $= 2\int_0^{\infty} \frac{5}{\sqrt{2\pi}} e^{-t}dt$

$$= \frac{10}{\sqrt{2\pi}}\int_0^{\infty} e^{-t}dt = \frac{10}{\sqrt{2\pi}}\left[\frac{e^{-t}}{-1}\right]_0^{\infty}$$

$$= \frac{10}{\sqrt{2\pi}}[-0+1] = \frac{10}{\sqrt{2\pi}} = 3.989 \approx 3.99$$

181. (a)

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^2 kxdx + \int_2^4 2kdx + \int_4^6 (-kx + 6k)dx = 1$$

$$\frac{kx^2}{2}\Bigg|_0^2 + 2kx\Big|_2^4 + \frac{-kx^2}{2} + 6kx\Bigg|_4^6 = 1$$

$$\frac{k}{2}(2^2 - 0) + 2k(4-2) - \frac{k}{2}(6^2 - 4^2) + 6k(6-4) = 1$$

$$2k + 4k - 10k + 12k = 1$$

$$8k = 1$$

$$k = \frac{1}{8}$$

$$\text{Mean} = \int_{-\infty}^{\infty} xf(x)dx$$

$$= \int_0^2 \frac{1}{8}x^2dx + \int_2^4 \frac{1}{4}xdx + \int_4^6 \left(-\frac{1}{8}x^2 + \frac{2}{3}x\right)dx$$

$$= \frac{1}{8}\frac{x^3}{3}\Bigg|_0^2 + \frac{1}{4}\frac{x^2}{2}\Bigg|_2^4 - \frac{1}{8}\frac{x^3}{3}\Bigg|_4^6 + \frac{2}{3}\frac{x^2}{2}\Bigg|_4^6$$

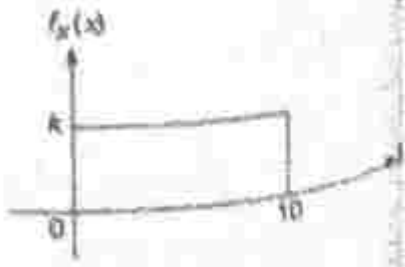
$$= \frac{1}{3} + \frac{3}{2} - \frac{19}{3} + \frac{20}{3} = \frac{13}{6}$$

182. (a)

$$\int_0^{10} kdx = 1$$

$$kx\Big|_0^{10} = 1$$

$$10k = 1$$



$$k = \frac{1}{10}$$

$$P(2.5 \le X \le 7.5) = \int_{2.5}^{7.5} \frac{1}{10} dx = \frac{1}{10} x \Big|_{2.5}^{7.5}$$

$$= \frac{1}{10}(7.5 - 2.5) = \frac{1}{2}$$

$$\text{Mean square value} = \int_0^{10} x^2 f(x) dx$$

$$\int_0^{10} \frac{1}{10} x^2 dx = \frac{1}{10} \frac{x^3}{3} \Big|_0^{10}$$

$$= \frac{10^3 - 0^3}{30} = \frac{1000}{30} = \frac{100}{3}$$

183. **(0.72)**

$$\text{Probability } (0.6 < n < 5) = \int_{0.6}^{5} f(x) dx$$

$$= \int_{0.6}^{1} 0.3 dx + \int_{1}^{4} 0.2\, dx + \int_{4}^{5} 0 dx$$

$$= 0.3[1 - 0.6] + 0.2[4 - 1] + 0[5 - 4]$$

$$= 0.3 \times 0.4 + 0.2 \times 3 = 0.72$$

184. **(a)**

$$P = \int_1^{\infty} f(x) dx = \int_1^{\infty} e^{-x} dx$$

$$= e^{-x} \Big|_{\infty}^{1} = e^{-1} = 0.368$$

185. **(d)**

- Larger α (0.05) means larger rejection region → smaller acceptance region. So acceptance region for 0.05 is smaller than for 0.01. So, S_1 is incorrect.
- Increasing a increases the rejection region, not decreases it. So, S_2 is incorrect.

186. **(1.583)**

Given, $n = 10$, $\bar{x} = 8.2$, $\mu = 7.5$, $s = 1.4$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{8.2 - 7.5}{\frac{1.4}{\sqrt{10}}} = \frac{0.7}{\frac{1.4}{3.162}} = \frac{0.7}{0.4423} = 1.583$$

187. **(c)**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\Rightarrow \bar{x} = t \cdot \frac{s}{\sqrt{n}} + \mu$$

$$\bar{x} = 1.5 \cdot \frac{2.7}{\sqrt{9}} + 5.0 = 1.5 \cdot \frac{2.7}{3} + 5.0$$

$$= 1.5 \cdot 0.9 + 5.0$$

$$= 1.35 + 5.0 = 6.35$$

188. **(b)**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\Rightarrow s = \frac{\bar{x} - \mu}{t} \cdot \sqrt{n}$$

$$s = \frac{12.5 - 11}{2} \cdot \sqrt{16}$$

$$= \frac{1.5}{2} \times 4 = 0.75 \times 4 = 3$$

189. **(0.756)**

Sample size $n = 6$

Sample mean $\bar{x}$:

$$\bar{x} = \frac{4 + 5 + 6 + 7 + 8 + 10}{6}$$

$$= \frac{40}{6} = 6.6667$$

Compute deviations and squared deviations:

$(4 - 6.6667)^2 = 7.1111$

$(5 - 6.6667)^2 = 2.7778$

$(6 - 6.6667)^2 = 0.4444$

$(7 - 6.6667)^2 = 0.1111$

$(8 - 6.6667)^2 = 1.7778$

$(10 - 6.6667)^2 = 11.1111$

Sum = 7.1111 + 2.7778 + 0.4444 + 0.1111 + 1.7778 + 11.1111 = 23.3333

$$s = \sqrt{\frac{23.333}{6 - 1}} = \sqrt{4.6667} \approx 2.1602$$

$$t = \frac{6.6667 - 6}{\frac{2.1602}{\sqrt{6}}} = \frac{0.6667}{\frac{2.1602}{2.4495}}$$

$$= \frac{0.6667}{0.8819} = 0.756$$

190. (5.579)

$$\bar{x}_{\text{rough}} = \frac{2+4+5+6+9}{5} = \frac{26}{5} = 5.2$$

Squared deviations from 5.2:

$(2-5.2)^2 = 1024,\ (4-5.2)^2 = 1.44,\ (5-5.2)^2 = 0.44$

$(6-5.2)^2 = 0.64,\ (9-5.2)^2 = 14.44$

Sum = 10.24 + 1.44 + 0.04 + 0.64 + 14.44 = 26.8

$$s = \sqrt{\frac{26.8}{5-1}} = \sqrt{6.7} = 2.588$$

Use t-formula to find $\bar{x}$:

$$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$$

$$\Rightarrow \bar{x} = t\cdot\frac{s}{\sqrt{n}} + \mu$$

$$\bar{x} = 0.5\cdot\frac{2.588}{\sqrt{5}} + 5 = 0.5\cdot\frac{2.588}{2.236} + 5$$

$$= 0.5 \cdot 1.158 + 5 = 0.579 + 5 = 5.579$$

191. (a)

Hypotheses:

$$H_0 : \mu = 1600,\ H_1 : \mu \neq 1600$$

Test statistic:

$$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{1570-1600}{\frac{120}{\sqrt{100}}} = \frac{-30}{12} = -2.5$$

Degree of freedom:

$$df = 100 - 1 = 99$$

Critical t-value: For two-tailed test at

$$\alpha = 0.05,\ t_{0.025,99} \neq 1.984$$

Since, $|t| = 2.5 > 1.984$ reject H_0

192. (3.6)

Given:

$$\sigma = 0.5,\ n = 36,\ \bar{x} = 10.3,\ \mu_0 = 10$$

Find: z

$$z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{10.3-10}{\frac{0.5}{\sqrt{36}}} = \frac{0.3}{\frac{0.5}{6}} = \frac{0.3}{0.0833} = 3.6$$

193. (c)

We know that,

Given: $\sigma = 4,\ n = 25,\ z = 2.5,\ \mu_0 = 50$

$$z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$\Rightarrow \bar{x} = \mu_0 + z\times\frac{\sigma}{\sqrt{n}}$$

$$\bar{x} = 50 + 2.5\times\frac{4}{\sqrt{25}} = 50 + 2.5\times\frac{4}{5}$$

$$= 50 + 2.5 \times 0.8 = 50 + 2 = 52$$

194. (8)

We know that,

Given:

$$n = 16,\ \mu_0 = 100,\ \bar{x} = 104,\ z = 2$$

$$z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$\Rightarrow \sigma = \frac{\bar{x}-\mu_0}{z}\times\sqrt{n}$$

$$\sigma = \frac{104-100}{2.0}\times\sqrt{16}$$

$$= \frac{4}{2}\times 4 = 2\times 4 = 8$$

195. (a)

$$\frac{(F_0-20)^2}{20} = 1.25$$

$$\Rightarrow (F_0-20)^2 = 25$$

$$F_0 - 20 = \pm 5$$

$$\Rightarrow F_0 = 25 \text{ or } 15$$

196. (4.381)

Calculate expected frequencies

F_e = Expected % × Total = {25, 35, 25, 15}

Calculate each term:

Flavor	F_o	F_e	$\frac{(F_o-F_e)^2}{F_e}$
Vanilla	30	25	$\frac{(30-25)^2}{25} = 1$
Chocolate	40	35	$\frac{(40-35)^2}{35} = 0.714$
Strawberry	20	25	$\frac{(20-25)^2}{25} = 1$
Mango	10	15	$\frac{(10-15)^2}{15} = 1.667$

$$\chi^2 = 1 + 0.714 + 1 + 1.667 = 4.381$$

197. (a, b, d)

- z-distribution is applied when population standard deviation (σ) is known and the sample is large (or population is normal). So, option (a) is correct.
- t-distribution is used when the sample size is small ($n < 30$) and population variance is unknown. So, option (b) is correct.
- The t-distribution has heavier tails than the z-distribution. This accounts for extra uncertainty in estimating s from a small sample. So, option (c) is incorrect.
- Both z and t distributions are symmetric about the mean (which is 0 in both cases). So, option (d) is correct.

199. (b, c, d)

- The chi-squared distribution is not symmetric, especially for small degrees of freedom (df). It is skewed to the right.
  As the df increases, the distribution becomes more symmetric, but not for small df. So, option (a) is incorrect.
- It is always positively skewed, particularly for small and moderate degrees of freedom. So, option (b) is correct.
- It is commonly used in testing hypotheses about population variance (e.g., testing if a sample variance matches a claimed population variance). So, option (c) is correct.
- Chi-squared test is also used in contingency tables to test the independence between categorical variables. So, option (d) is correct.

200. (c)

The p-value is the probability of obtaining the observed value of the test statistic (or something more extreme), under the assumption that H_0 is true.
It does not measure the probability that H_0 or H_1 is true or false.

201. (a)

- The p-value tells us the smallest significance level (a) at which we would reject the null hypothesis. This is a standard interpretation of the p-value in hypothesis testing. So, S_1 is correct.
- The p-value is not the probability that the null hypothesis is false. It is the probability of observing the given sample statistic (or more extreme), assuming that the null hypothesis is true. So, S_2 is incorrect.

■■■■

UNIT 4

Machine Learning

Q.1 Back propagation is a learning technique that adjusts weights in the neural network by propagating weight changes.

(a) Forward from source to sink
(b) Backward from sink to source
(c) Forward from source to hidden nodes
(d) Backward from sink to hidden nodes

Q.2 Linear Regression Model is

(a) Predicts Continuous value
(b) Predicts Binary outcomes
(c) Both (a) and (b)
(d) None of these

Q.3 Identify the following activation function:

$$\phi(V) = Z + \frac{1}{1+\exp(-x*V+Y)}$$

Z, X, Y are parameters

(a) Step function (b) Ramp function
(c) Sigmoid function (d) Gaussian function

Q.4 In a binary classification task using MLP, which of the following combinations is most appropriate for the output activation function and loss function?

(a) ReLU and Mean Squared Error
(b) Tanh and Categorical Cross-Entropy
(c) Sigmoid and Binary Cross-Entropy
(d) Softmax and Binary Cross-Entropy

Q.5 Match the following:

List-I	List-II
A. Supervised learning	1. The decisions system receives rewards for its action at the end of a sequence of steps.
B. Unsupervised learning	2. Manual labels of inputs are not used.
C. Reinforcement learning	3. Manual labels of inputs are used.
D. Inductive learning	4. System learns by example

Codes:

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	1	4
(c)	3	2	4	1
(d)	3	2	1	4

Q.6 Consider the following statements?

S_1: Linear Regression Model is Maximize the sum of squared errors.

S_2: Linear Regression Model is Minimize the sum of Squared errors.

Which of the following is true?

(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) None of these

Q.7 Which of the following activation functions is least likely to suffer from the vanishing gradient problem?

(a) Sigmoid (b) Tanh
(c) ReLU (d) Logistic

Q.8 Match the following:

List-I	List-II
A. Single linkage	1. Maximum distance
B. Decision Tree	2. Minimum distance
C. Complete linkage	3. Classification

Codes:

	A	B	C
(a)	2	3	1
(b)	2	1	3
(c)	3	2	1
(d)	1	2	3

Q.9 Which of the following can be used for clustering of data?

(a) Single layer perception
(b) Multilayer perception
(c) Self organizing map
(d) Radial basis function

Q.10 Consider a soft margin SVM with a linear Kernel and regularization parameter is 5. The total penalty term due to slack variables is 3.0. There are three slack variables $\xi_1 = 0.3$, $\xi_2 = 0.2$ and ξ_3 (unknown). Find the value of ξ_3.

Q.11 Which of the following statements are correct?

(a) SVM is a discriminative classifier, not a generative model.

(b) LDA is a generative classifier.

(c) LDA assumes normal distribution of features within each class.

(d) The Kernel trick avoids the need to compute high-dimensional feature vectors explicitly.

Q.12 An artificial neuron receives n inputs $x_1, x_2, \ldots, x_n$ with weights $w_1, w_2, \ldots, w_n$ attached to the input links. The weighted sum _____ is computed to be passed on to a non-linear filter ϕ called activation function to release the output.

(a) $\sum w_i$ (b) $\sum x_i$

(c) $\sum w_i + \sum x_i$ (d) $\sum w_i \cdot x_i$

Q.13 Consider the following data set:

x	y
1	2
2	3
3	5

A linear regression model predicts the output using the formula:

$$\hat{y} = 1 + 1x$$

Calculate the Sum of Squared Errors (SSE).

Q.14 In a multilayer perceptron, which of the following is true?

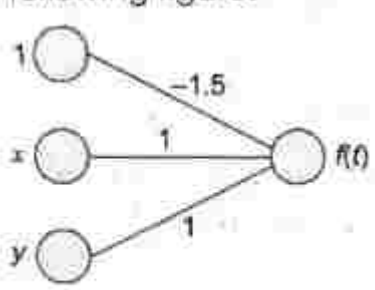
(a) All layers must have the same number of neurons

(b) Output of one layer becomes input to the next layer

(c) Learning rate is different for each layer

(d) Activation functions must be linear for convergence

Q.15 Consider a single perception with weights as given in the following figure:



and $f(t)$ defined as $f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \le 0 \end{cases}$

The above perception can solve

(a) OR problem (b) AND problem

(c) XOR problem (d) All of these

Q.16 The decision tree classifier is a widely used technique for _____.

(a) Classification (b) Association

(c) Partition (d) Clustering

Q.17 A simple linear regression model is given by:

$$\hat{y} = 2 + 0.5x$$

Given the following actual values:

x	y
2	3
4	5
6	7

Find the Sum of Squared Errors (SSE).

Q.18 In a single perceptron, the updation rule of weight vector is given by

(a) $w(n+1) = w(n) + \eta[d(n) - y(n)]$

(b) $w(n+1) = w(n) - \eta[d(n) - y(n)]$

(c) $w(n+1) = w(n) + \eta[d(n) - y(n)]^* \times (n)$

(d) $w(n+1) = w(n) - \eta[d(n) - y(n)]^* \times (n)$

Q.19 Consider the following statements about decision trees:

S_1: Entropy is zero when all examples belong to the same class.

S_2: A split with highest information gain always has the smallest Gini index.

Which is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.20 Match the following learning modes w.r.t. characteristics of available information for learning:

List-I

A. Supervised

B. Recording

C. Reinforcement

D. Unsupervised

List-II

1. Instructive information on desired responses, explicitly specified by a teacher.

2. A priori design information for memory storing.

3. Partial information about desired responses, or only "right" or "wrong" evaluative information.
4. No information about desired responses.

Codes:

	A	B	C	D
(a)	1	2	3	4
(b)	1	3	2	4
(c)	2	4	3	1
(d)	2	3	4	1

Q.21 A dataset contains 40 samples belonging to two classes: Class A (25 samples) and Class B (15 samples). What is the Gini index of this dataset (node)?

Q.22 Which of the following is correct regarding Support Vector Machine (SVM) and its capacity to handle non-linearly separable data?

(a) SVM cannot handle non-linear decision boundaries under any circumstance.
(b) SVM uses Kernel functions to implicitly map data into higher-dimensional spaces.
(c) SVM always requires feature scaling to function correctly.
(d) SVM's soft margin parameter increases the bias and decreases the variance.

Q.23 Consider a single-layer perceptron with input vector $x = [3, -2]$ weights $w = [0.5, 1.5]$ and bias $b = -1$. Calculate the output using the step activation function.

(a) 1 (b) 0
(c) –1 (d) None of these

Q.24 Consider the following statements about a perception:

I. Feature detector can be any function of the input parameters.
II. Learning procedure only adjusts the connection weights to the output layer.

Identify the correct statement out of the following:

(a) I is false and II is false
(b) I is true and II is false
(c) I is false and II is true
(d) I is true and II is true

Q.25 The actual values of the response variable are:

$$y = [4, 6, 8, 10]$$

And the predicted values from a linear model are:

$$\hat{y} = [3.5, 6.5, 7, 9]$$

Calculate the Sum of Squared Errors (SSE)

Q.26 Consider the following statements:

S_1: Decision trees can handle both categor and numerical attributes.

S_2: Decision trees always produce the m compact tree.

Which of the following is correct?

(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) None of these

Q.27 What is the role of backpropagation in train MLPs?

(a) To generate random initial weights
(b) To compute gradients for weight update
(c) To evaluate the output layer
(d) To normalize data inputs

Q.28 Match the following:

List-I

A. Classification
B. Clustering
C. Feature Extraction
D. Feature Selection

List-II

1. Principal component analysis
2. Branch and Bound
3. K-nearest neighbour
4. K-means

Codes:

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	1	2
(c)	3	4	1	2
(d)	4	3	2	1

Q.29 Consider the two class classification task th consists of the following points:

Class C_1: $[-1, -1]$, $[-1, 1]$, $[1, -1]$

Class C_2: $[1, 1]$

The decision boundary between the two classe C_1 and C_2 using single perceptron is given by

(a) $x_1 - x_2 - 0.5 = 0$
(b) $-x_1 + x_2 - 0.5 = 0$
(c) $0.5(x_1 + x_2) - 1.5 = 0$
(d) $x_1 + x_2 - 0.5 = 0$

Q.30 Which of the following is a true statement about ReLU activation?
(a) It is bounded in the range [0, 1]
(b) Its derivative is always 1
(c) It is non-differentiable
(d) It can cause neurons to die if they output 0 constantly

Q.31 Consider a standard additive model consisting of rules of the form of
If x is A_i AND y is B_i THEN z is C_i.
Given crisp inputs $x = x_0$, $y = y_0$, the output of the model is :
(a) $z = \sum_i \mu_{A_i}(x_0)\mu_{B_i}(y_0)\mu_{C_i}(z)$
(b) $z = \sum_i \mu_{A_i}(x_0)\mu_{B_i}(y_0)$
(c) $z = \text{centroid}\left(\sum_i \mu_{A_i}(x_0)\mu_{B_i}(y_0)\mu_{C_i}(z)\right)$
(d) $z = \text{centroid}\left(\sum_i \mu_{A_i}(x_0)\mu_{B_i}(y_0)\right)$

Q.32 Which of the following statements is/are correct about decision tree pruning?
(a) Pruning helps reduce overfitting
(b) Pruning is performed after the tree is fully grown
(c) Pruning increases training error
(d) None of these

Q.33 Consider the two class classification task that consists of the following points :
Class C_1 : [1 1.5] [1 −1.5]
Class C_2: [−2 2.5] [−2 −2.5]
The decision boundary between the two classes using single perceptron is given by:
(a) $x_1 + x_2 + 1.5 = 0$ (b) $x_1 + x_2 - 1.5 = 0$
(c) $x_1 + 1.5 = 0$ (d) $x_1 - 1.5 = 0$

Q.34 A perceptron has input weights $W_1 = -3.9$ and $W_2 = 1.1$ with threshold value $T = 0.3$. What output does it give for the input $x_1 = 1.3$ and $x_2 = 2.2$?
(a) −2.65 (b) −2.30
(c) 0 (d) 1

Q.35 Any decision tree that sorts n elements has height _____.
(a) $\Omega(\log n)$ (b) $\Omega(n)$
(c) $\Omega(n \log n)$ (d) $\Omega(n^2)$

Q.36 Consider the following statements:
S_1 : In Leave-One-Out Cross-Validation, the training set in each fold has only one example left out.
S_2 : LOOCV often has lower variance in performance estimates compared to 10-fold CV.
Which of the following is correct?
(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.37 A neuron with 3 inputs has the weight vector $[0.2 \; -0.1 \; 0.1]^T$ and a bias $\theta = 0$. If the input vector is $X = [0.2 \; 0.4 \; 0.2]^T$ then the total input to the neuron is :
(a) 0.20 (b) 1.0
(c) 0.02 (d) −1.0

Q.38 Which of the following neural networks uses supervised learning?
(i) Multilayer perceptron
(ii) Self organizing feature map
(iii) Hopfield network
(a) (i) only
(b) (ii) only
(c) (i) and (ii) only
(d) (i) and (iii) only

Q.39 Which of the following is not a Clustering method?
(a) K - Mean method
(b) Self Organizing feature map method
(c) K - nearest neighbor method
(d) Agglomerative method

Q.40 Match **List-I** with **List-II**:

List-I
A. Silhouette Coefficient
B. Gini Index
C. One-vs-Rest (OVR)

List-II
1. Cluster Evaluation
2. Impurity Measure
3. Multiclass Classification

Codes:

	A	B	C
(a)	1	2	3
(b)	2	1	3
(c)	1	3	2
(d)	3	2	1

Q.41 Which of the following is correct about K-Fold and Leave-One-Out Cross-Validation?

(a) LOOCV has lower computational cost than 10-Fold CV.

(b) K-Fold CV always results in higher variance than LOOCV.

(c) LOOCV uses n models for nnn samples, while 10-Fold CV uses 10 models.

(d) K-Fold CV with $k = nk = nk = n$ is equivalent to training on the full dataset without validation.

Q.42 Consider the following statements with respect to Single Linkage Clustering:

S_1 : Single linkage clustering is known to produce the chaining effect.

S_2 : In single linkage clustering, the distance between two clusters is the maximum of all pairwise distances between their members.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.43 Suppose 15000 water sample tested for pollutants. Out of them, 10000 are actually clean and 5000 are actually polluted. For the polluted sample, the test was positive for 3800 and negative for 1200. For the clean samples, the test was positive for 600 and negative for 9400. Calculate the F'-score of the model.

Q.44 Which of the following is true for the Gini Index used in decision trees?

(a) Gini Index can handle multiclass classification

(b) Gini Index is always greater than 1

(c) Gini Index measures purity using log terms

(d) Gini Index favors splits with maximum entropy

Q.45 Consider a single perceptron with sign activation function. The perceptron is represented by weight vector $[0.4 - 0.3\ 0.1]'$ and a bias $\theta = 0$. If the input vector to the perceptron is $X = [0.2\ 0.6\ 0.5]$ then the output of the perceptron is:

(a) 1 (b) 0

(c) -0.05 (d) -1

Q.46 Which of the following is an examples of unsupervised neural network?

(a) Back-propagation network

(b) Hebb network

(c) Associative memory network

(d) Self-organizing map

Q.47 K-mean clustering algorithm has clustered the given 8 observations into 3 clusters after 1st iteration as follows:

C1 : {(3, 3), (5, 5), (7, 7)}

C2 : {(0, 6), (6, 0), (3, 0)}

C3 : {(8, 8), (4, 4)}

What will be the Manhattan distance for observation (4, 4) from cluster centroid C1 in second iteration?

(a) 2 (b) $\sqrt{2}$

(c) 0 (d) 18

Q.48 Which of the following statements is/are correct?

(a) Support Vector Machines can be extended to multi-class classification using one-vs-one or one-vs-rest strategies.

(b) LDA is not applicable when the classes have unequal covariance matrices.

(c) The bias-variance trade-off shows that both high bias and high variance can lead to high test error.

(d) In K-Fold Cross-Validation, higher values of k tend to reduce the variance in the estimate.

Q.49 A soft margin SVM has slack variables $\xi = [0.4, 0.6, 0.2, 0.8]$. The total penalty term in the objective function due to slack variables is 3.0. Find the regularization parameter c.

Q.50 Consider the following learning algorithms:

I. Logistic regression

II. Multi layer perceptron

III. Linear regression

Which of the following option represents classification algorithms?

(a) I and II only (b) I and III only

(c) II and III only (d) I, II and III

Q.51 Given a dataset with these actual and predicted values:

y (Actual)	$\hat{y}$ (Predicted)
10	9
8	7
6	6
4	5

Calculate the Mean Squared Error (MSE).

Q.52 In a feed forward neural network with the following specifications: Input layer has 4 neurons, hidden layer has 3 neurons and output layer has 2 neurons using the sigmoid activation function for given input values [0.5, 0.8, 0.2, 0.6] as well as the initial weights for the connections.

W1 : [0.1, 0.3, 0.5, 0.2]
W2 : [0.2, 0.4, 0.6, 0.2]
W3 : [0.3, 0.5, 0.7, 0.2]
} Input layer to hidden layer weights

W4 : [0.4, 0.1, 0.3]
W5 : [0.5, 0.2, 0.4]
} Hidden layer to output layer weights

What is the output of the output layer when the given input values are passed through neural network? Round the answer to two decimal places.

(a) [0.62, 0.68] (b) [0.72, 0.78]
(c) [0.82, 0.88] (d) [0.92, 0.98]

Q.53 Arrange the following steps in the correct sequence for applying an unsupervised learning technique such as K-means clustering is to a data set:

(A) Randomly initialize cluster centroids.
(B) Assign each data point to nearest cluster centroid.
(C) Update the cluster centroids based on the mean of data points assigned to each cluster.
(D) Specify the number of clusters (K) to partition the data into.
(E) Repeat steps B and C until convergence criteria are met.

Choose the **correct** answer from the options given below:

(a) (D), (A), (B), (C), (E)
(b) (A), (B), (C), (D), (E)
(c) (C), (B), (A), (D), (E)
(d) (D), (C), (A), (B), (E)

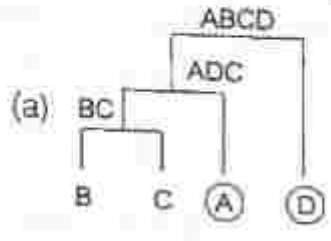
Q.54 Which of the following is correct regarding the comparison between LDA and SVM in classification problems?

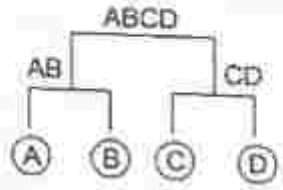
(a) LDA always outperforms SVM in non-linear classification tasks.
(b) SVM works well even when class covariance matrices are unequal, unlike LDA.
(c) LDA is preferred over SVM when there is no need for interpretability.
(d) SVM cannot be used when the number of features exceeds the number of samples.

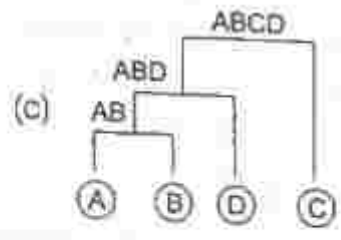
Q.55 Consider the following distance matrix of 4 points A, B, C, D:

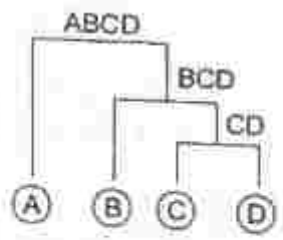
	A	B	C	D
A	0	2	5	7
B	2	0	4	3
C	5	4	0	6
D	7	3	6	0

Use Agglomerative Hierarchical Clustering with Single Linkage to form clusters step-by-step. Show the order of merges.

(a) 

(b) 

(c) 

(d) 

Q.56 Which of the following statements are correct?

(a) LDA produces linear boundaries even when classes are not linearly separable.
(b) SVM with RBF kernel can handle non-linearly separable data.
(c) Bias-variance trade-off is not relevant in high-dimensional data.
(d) LOOCV tends to have high variance in error estimation.

Q.57 Let $x_i = [1, 2]$ and $x_j = [3, 0]$.

Consider a polynomial Kernel of degree 2 defined as:

$$K(x_i, x_j) = (x_i \cdot x_j + 1)^2$$

Compute $K(x_i, x_j)$.

Q.58 Which of the following statements is/are true about Single Linkage Clustering?

(a) It merges clusters based on the minimum pairwise distance.
(b) It tends to produce compact spherical clusters.
(c) It is less sensitive to outliers compared to complete linkage.
(d) It may result in the chaining effect.

Q.59 Arrange the following steps in a proper sequence for the process of training a neural network.

(A) Weight initialization
(B) Feed forward
(C) Back Propagation
(D) Loss Calculation
(E) Weight Update

Choose the **correct** answer from the options given below:

(a) (A), (B), (D), (C), (E)
(b) (D), (B), (A), (C), (E)
(c) (A), (C), (D), (B), (E)
(d) (E), (C), (B), (D), (A)

Q.60 For a 2-layer MLP, input vector $x = [1, 0]$ hidden layer weights matrix:

$$W_1 = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, b_1 = [0, 0]$$

The activation function in the hidden layer is sigmoid. The output layer weights are $W_2 = [1, 1]$ bias $b_2 = (0,$ and output activation is identity. Find the final output.

Q.61 Consider the following statements:

S_1 : LDA can be used for both binary and multi-class classification.

S_2 : SVM cannot find a solution if the data is not linearly separable.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.62 Which of the following is/are true regarding support vector machines (SVM)?

(a) SVM seeks the hyperplane that maximizes the margin between classes.
(b) Only support vectors determine the decision boundary.
(c) The decision boundary of SVM is probabilistic in nature.
(d) Kernels allow SVM to handle non-linear classification.

Q.63 Match **List-I** with **List-II**:

List-I

(A) The activation function
(B) The learning method of perceptron
(C) Areas of application of artificial neural network include
(D) The output of the perceptron

List-II

(I) is called the delta rule.
(II) is one of the key components of the perceptron as in the most common neural network architecture.
(III) is always boolean like a switch.
(IV) system identification and control.

Choose the **correct** answer from the options given below:

(a) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
(b) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(c) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(d) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q.64 Which of the following components introduces non-linearity into an MLP?

(a) Input layer
(b) Weight matrix
(c) Activation function
(d) Bias term

Q.65 A neuron uses identity activation. Input $x = 1.0$, weight $w = 0.5$ bias $b = 0$ output $y = 0.5$ and target $t = 1.0$. Loss function is Mean Squared Error (MSE). Find the gradient of loss with respect to w, i.e., $\frac{dL}{dw}$.

(a) –0.5 (b) –0.25
(c) 0.5 (d) 0.25

Q.66 Consider the following data:

x	y
2	4
3	6
5	9
6	12

A regression model predicts the values using

$$\hat{y} = 2x$$

Calculate the Root Mean Squared Error (RMSE)

Q.67 Which of the following is/are true about backpropagation in MLP?

(a) It uses gradient descent to minimize loss
(b) It updates weights from output layer to input layer
(c) It requires differentiable activation functions
(d) All of the above

Q.68 Consider the following statements:

S_1 : ReLU activation suffers from vanishing gradient problem.

S_2 : Sigmoid activation saturates and leads to slow learning.

Which of the following is correct?

(a) Only S_1

(b) Only S_2

(c) Both S_1 and S_2

(d) None of these

Q.69 Match List-I with List-II:

List-I

(A) Natural language processing

(B) Reinforcement learning

(C) Support vector machine

(D) Expert system

List-II

(I) A method of training algorithm by rewarding desired behaviour and/or punishing undesired one.

(II) System designed to emulate the decision making abilities of a human expert.

(III) A branch of AI focused on understanding and generating human language.

(IV) A machine learning technique that finds the hyper plane that best separates different class in a feature space.

Choose the **correct** answer from the options given below:

(a) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(b) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(c) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(d) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q.70 Let $x_i = [2, 2]$, $x_j = [2, 2]$.

Consider the Kernel $K(x_i, x_j) = (x_i \cdot x_j + 1)^2$.

Which of the following is correct?

(a) $K(x_i, x_j) = 100$ (b) $K(x_i, x_j) = 81$

(c) $K(x_i, x_j) = 49$ (d) $K(x_i, x_j) = 64$

Q.71 A dataset of 60 samples, with 30 samples of class X and 30 samples of class Y, is split by feature F in two ways:

Split 1:

- Left has 25 samples, which include 20 samples of class X and the remaining from class Y.
- Right has 35 samples, which include 10 samples of class X and the remaining from class Y.

Split 2:

- Left has 30 samples, which include 15 samples of class X and the remaining from class Y.
- Right has 30 samples, which include 15 samples of class X and the remaining from class Y.

Which of the following is correct?

(a) Gini index left is 0.32 in split 1.

(b) Weighted Gini index of split 1 is 0.36.

(c) Weighted Gini index of split 2 is 0.36.

(d) Gini index right and left of split 2 both are same.

Q.72 Which of the following is correct in the context of bias-variance trade-off in supervised learning?

(a) A high-bias model tends to overfit the training data.

(b) High-variance models tend to perform poorly on training data but well on test data.

(c) Reducing variance often comes at the cost of increased bias.

(d) Bias and variance are independent and can be minimized simultaneously.

Q.73 Let $x_i, x_j \in \mathbb{R}^n$ and define

$$K(x_i, x_j) = (x_i \cdot x_j + 1)^2$$

Which of the following statements is correct?

(a) The Kernel computes the square of the Euclidean distance.

(b) The Kernel maps data to a higher-dimensional space implicitly.

(c) The Kernel is not suitable for classification tasks.

(d) The Kernel ignores dot product information between x_i and x_j.

Q.74 Given the following data points:

(3, 4), (4, 6), (6, 9), (7, 11), (9, 15)

Find the line of best fit using the Least Squares method.

Determine the slope m and intercept c.

(a) $y = 2.1x + 1.4$

(b) $y = 2.3x + 0.9$

(c) $y = 2.5x + 1.2$

(d) $y = 1.75x - 1.38$

Q.75 Consider the following statements:

S_1 : SMA* gives the optimal solution if reachable; otherwise, it returns the best reachable optimal solution.

S_2 : Greedy Best-First Search is incomplete even if the search space is finite.

Which of the above statements is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.76 An MLP has 4 input neurons, 5 hidden neurons, and 2 output neurons. All neurons are fully connected, and biases are included. How many total weights (including biases) are there?

(a) 30 (b) 32

(c) 35 (d) 37

Q.77 Consider the following statements:

S_1 : Decision trees use only Mean Squared Error (MSE).

S_2 : The Naive Bayes algorithm assumes all features are dependent given the class label.

Which of the above statements is/are incorrect?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.78 The following data points are given:

(2,5), (4, 9), (5, 13), (7, 16), (9, 20)

Using the Least Squares method, determine the line of best fit.

What is the slope m?

(a) 2.15 (b) 2.8

(c) 3.2 (d) 3.0

Q.79 Consider the following statements:

S_1 : Gradient Descent can converge to a local minimum for non-convex functions, which may not be global.

S_2 : Stochastic Gradient Descent (SGD) updates weights individually for each training instance.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.80 Consider an SVM model with weight vector $W = [3, -1]$ and bias $b = 5$. Given an input vector $x = [2, 4]$, find the value of the decision function $f(x)$.

(a) 7 (b) 5

(c) 3 (d) 9

Q.81 Consider the following statements:

S_1 : Linear Discriminant Analysis (LDA) maximizes between-class variance while minimizing within-class variance.

S_2 : In LDA, the projection is chosen such that best separates class means while ignoring covariance.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.82 Match the items in **Column-I** with the items in **Column-II** in the following table

Column-I

A. Decision tree

B. Bias-Variance Trade-Off

C. Naive Bayes classifier

D. Cross-Validation

Column-II

1. Model Complexity
2. Discriminative model
3. Generative model
4. Model Validation Technique

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	2	1	4	3
(c)	1	2	4	3
(d)	1	4	2	3

Q.83 Which of the following statements is/are correct?

(a) Single linkage clustering is also known as the minimum distance or nearest neighbor method.

(b) In single linkage clustering, the distance between two clusters is defined as the maximum distance between any two points in the clusters.

(c) Single linkage clustering is sensitive to noise and outliers, as they can cause chaining where clusters are forced together.

(d) Single linkage clustering can result in clusters with a higher degree of overlap compared to complete linkage clustering.

Q.84 For an SVM with weight vector $W = [5, -2]$ and input point $x = [1, 3]$ the decision function value $f(x)$ is 4. Find the bias term b.

(a) 3 (b) 4

(c) 5 (d) 6

Q.85 Given a dataset with the following points and a regularization parameter $C = 5$.

x_1	x_2	Class
1	1	+1
2	2	+1
3	3	−1
4	4	−1

The soft margin SVM has the following stack variables:

$$\xi_1 = 0.4, \xi_2 = 0.3, \xi_3 = 0.5, \xi_4 = 0.6$$

Compute the total value of the slack variable penalty term in the objective function.

Q.86 Which of the following statements is/are true?

S_1: In linear regression, the R-squared value indicates the proportion of variance explained by the model.

S_2: Decision trees use only mean squared error for regression problems.

S_3: Naive Bayes assumes that features are dependent given the class label.

S_4: Support Vector Machines (SVM) can handle non-linear decision boundaries using the Kernel trick.

Which of the above is true?

(a) S_1 and S_4
(b) S_1, S_2, S_4
(c) S_1, S_3, S_4
(d) S_1, S_2, S_3, S_4

Q.87 Consider the following statements:

S_1: Decision criterion in a decision tree is used to determine the best feature to split the data.

S_2: Decision criterion compares feature values against a purity measure make splitting decisions.

Which of the following statements is/are correct?

(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) None of these

Q.88 An SVM model has $W = [1, 2]$, $b = -5$. Predict the class label of input $x = [3, 2]$ assuming class +1 if $f(x) > 0$, else −1.

(a) +1 (b) −1
(c) 0 (d) Cannot be determined

Q.89 Match List-I with List-II:

List-I

A. Principal Component Analysis
B. Support Vector Machine (SVM)
C. Logistic Regression

List-II

1. Dimensionality Reduction
2. Margin Maximization
3. Classification

Codes:

	A	B	C
(a)	1	2	3
(b)	3	2	1
(c)	1	3	2
(d)	2	1	3

Q.90 Consider the following statements:

S_1: Leave-One-Out (LOO) cross-validation uses each data point as a test set once.

S_2: The dual problem of SVM involves minimizing the Lagrangian dual function.

S_3: K-means clustering assigns data points to the nearest cluster centroid based on distance.

Which of the following is correct?

(a) Only S_1 and S_2
(b) Only S_1 and S_3
(c) Only S_2 and S_3
(d) None of these

Q.91 A dataset of 14 samples has 9 positive and 5 negative samples. It is split into two subsets: Subset 1 has 4 positive, and 0 negative and Subset 2 has 5 positive, and 5 negative Calculate the information gain after this split.

(a) 0.22 (b) 0.42
(c) 0.34 (d) 0.62

Q.92 Which of the following statements is/are true about K-Fold Cross-Validation?

(a) In Stratified K-Fold, each fold maintains the same class proportion as the full dataset.
(b) The test error estimate from K-Fold CV is always unbiased.
(c) Lower values of k can lead to high bias in error estimation.
(d) K-Fold CV with $k = nk = nk = n$ is equivalent to Leave-One-Out Cross-Validation.

Q.93 Which of the following statements about the Support Vector Machine (SVM) is/are correct?

S_1: The decision function for an SVM is given by $f(x) = w^T x + b$.

S_2: In the context of a soft-margin SVM, the regularization parameter C controls the trade-off between maximizing the margin and minimizing the classification error.

S_3: The margin of an SVM is defined as $\frac{2}{\|w\|}$.

S_4: The Kernel function in an SVM allows the transformation of the input data into a higher-dimensional space to make it linearly separable.

(a) Only S_1
(b) Only S_2 and S_3
(c) All of the above
(d) None of these

Q.94 Consider an SVM with a polynomial Kernel of degree 2:

$$K(x_i, x_j) = (x_i \cdot x_j + 1)^2$$

For $x_i = [2, 1]$ and $x_j = [1, 3]$, compute $K(x_i, x_j)$.

Q.95 Which of the following statements about entropy is true in decision tree learning?

(a) Entropy is zero when classes are perfectly mixed.
(b) Entropy is maximum when classes are equally mixed.
(c) Entropy cannot measure impurity.
(d) Entropy decreases as dataset size increases.

Q.96 Match List-I with List-II:

List-I
A. Naive Bayes Classifier
B. Decision Tree
C. Bagging (Bootstrap AGGreg.)

List-II
1. Generative Model
2. Greedy Splitting
3. Ensemble Technique

Codes:

	A	B	C
(a)	1	2	3
(b)	2	3	1
(c)	3	1	2
(d)	2	1	3

Q.97 Consider a soft margin SVM with regularization parameter is 3. The slack variables for three support vectors are $\xi_1 = 0.7$, $\xi_2 = 0.5$, and $\xi_3 = 0.8$.

What is the penalty term (regularization term by slack variables) included in the SVM objective function?

(a) 4 (b) 3
(c) 2 (d) 6

Q.98 Given an SVM with weight vector $W = [4, -2]$ and $b = -6$ what is the equation of the decision boundary?

(a) $4x_1 - 2x_2 = 6$
(b) $4x_1 - 2x_2 = -6$
(c) $2x_1 + 4x_2 = 6$
(d) $2x_1 - 4x_2 = 6$

Q.99 Consider an MLP model with two hidden layers and one output layer. The first hidden layer has 8 neurons, the second hidden layer has 5 neurons, and the output layer has 3 neurons. The input to the MLP is a 4-dimensional vector. Each neuron is fully connected to every neuron in the previous layer, and a bias term is included for each neuron.

Calculate the total number of trainable parameters in this MLP model.

Q.100 A dataset has an entropy of 0.9 before splitting on a feature. After splitting, the weighted average entropy of the subsets is 0.4. What is the information gain obtained by this split?

(a) 1.8
(b) 0.5
(c) 1.3
(d) 0.4

Q.101 A dataset contains 50 instances, divided into two classes:

30 instances belong to the approved class and 20 to the rejected class

A particular feature splits the dataset into two subsets:

- Left subset: 20 instances → 15 approved, 5 rejected
- Right subset: 30 instances → 15 approved, 15 rejected

Find the difference between weighted Gini index and left subset Gini index.

Q.102 Which of the following statements about Support Vector Machines (SVMs) is true?

S_1: SVMs can be used for both classification and regression tasks.

S_2: SVMs always require the use of Kernel functions to handle non-linearly separable data.

S_3: The hyperplane in SVM is chosen to maximize the margin between the classes.

S_4: The complexity of an SVM model is always linear in terms of the number of training samples.

Which of the following is/are correct?

(a) Only S_1 and S_2 (b) Only S_2 and S_3
(c) Only S_1 and S_3 (d) Only S_2 and S_4

Q.103 A node contains 100 samples distributed among three classes:

Class 1: 40 samples
Class 2: 35 samples
Class 3: 25 samples

Calculate the Gini index of the node.

Q.104 Why are non-linear activation functions used in the hidden layers of a Multilayer Perceptron (MLP)?

(a) To improve weight initialization
(b) To ensure faster convergence
(c) To introduce non-linearity into the model
(d) To normalize the output

Q.105 Match **List-I** with **List-II** and select the correct answer using the codes given below the lists:

List-I

A. Decision tree
B. Bias-Variance Trade-Off
C. Multi-Layer Perceptron (MLP)
D. Cross-Validation

List-II

1. Model Complexity
2. Discriminative Model
3. Back propagation
4. Model Validation Technique

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	1	2	3	4
(c)	1	3	2	4
(d)	2	3	4	1

Q.106 A fully connected MLP has 3 input neurons, 1 hidden layer with 4 neurons, and 1 output neuron. How many total weights (excluding biases) are there?

(a) 12 (b) 15
(c) 16 (d) 13

Q.107 For an SVM with weight vector $W = [3, 4]$, what is the width of the margin?

(a) 5 (b) 0.4
(c) 0.5 (d) 2.5

Q.108 Consider a K-class dataset containing V points, where each sample is described by D continuous features. You decide to use a Gaussian Naive Bayes classifier. How many parameters need to be estimated for this model?

(a) 2DK + K (b) DK
(c) 3DK (d) 2D + K

Q.109 Given the dataset : (0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7) and assuming the initial centroids for K = 3 means clustering to be $C_1 = (0, 0)$, $C_2 = (3, 3)$ and $C_3 = (6, 6)$. one iteration of the Expectation - Maximization for K-means clustering will update C_2 to (____, ____).

(a) (2.5, 2.5) (b) (7.5, 7.5)
(c) (1.5, 1.5) (d) (3, 3)

Q.110 A k-nearest neighbors (k-NN) classifier is trained on a small dataset and achieves 100% accuracy by setting $k = 1$. Which of the following models trained on the same dataset is most likely to also achieve 100% accuracy?

(i) A decision tree with maximum depth equal to the number of training samples.
(ii) A linear support vector machine (SVM).
(iii) A logistic regression model.
(iv) A polynomial kernel SVM with degree 2.

(a) (i) (b) (i) and (iv)
(c) All of the above (d) None of the above

Q.111 Consider the following statements:

S_1: Slack variables $\xi_i \geq 0$ represent the amount of violation of margin constraints in soft margin SVM.

S_2: If c is set very high, the SVM behaves like a hard margin SVM.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.112 Consider a polynomial Kernel of the form

$$K(x_i, x_j) = (x_i \cdot x_j + c)^d$$

Let $x_i = [1, 1] = [2, 0]$, $c = 2$, $d = 3$

Compute $K(x_i, x_j)$.

Q.113 Which of the following is typically used as the activation function in the output layer of an MLP for multi-class classification?

(a) ReLU (b) Tanh
(c) Sigmoid (d) Softmax

Q.114 In a dataset with three class, class DA has 35 instances class CS has 25 instances and class ME has 45 instance. What is the Gini index for this dataset?

Q.115 For an SVM with a Racial Basis Function (RBF) Kernel compute the value of the Kernel function $K(x_i, x_j)$ for $\gamma = 0.8$, $x_i = [3, 3]$ and $x_j = [5, 7]$.

(a) 1.125×10^{-6} (b) 1.125×10^{-13}
(c) 1.125×10^{-9} (d) 1.125×10^{-7}

Q.116 An SVM uses the following weight vector and bias term:

$$w = [4, 5],\ b = -2$$

Calculate the margin of the separating hyperplane, if the following support vector are given

$$x_1 = [0, 1],\ x_2 = [2, -1]$$

(a) $\frac{1}{41}$ (b) $\frac{1}{\sqrt{5}}$
(c) $\frac{1}{4}$ (d) $\frac{2}{\sqrt{41}}$

Q.117 Consider the following confusion matrix of a data model which classifies the given flower as Rose.

	Actual Rose	Actual Lilly
Predicted Rose	8	6
Predicted Lilly	10	14

F'-score of the model is _____?

Q.118 Match List-I with List-II:

List-I

A. L1 Regularization
B. K-Means Clustering
C. Random Forest

List-II

1. Sparsity in Coefficients
2. Unsupervised Learning
3. Ensemble of Decision Trees

Codes:

	A	B	C
(a)	1	2	3
(b)	3	1	2
(c)	1	3	2
(d)	2	1	3

Q.119 Given class probabilities $p_1 = 0.5$, $p_2 = 0.3$, $p_3 = 0.2$ compute the entropy of the dataset.

(a) 1.485 (b) 1.3
(c) 1.2 (d) 1.1

Q.120 Consider the following statements:

S_1: The discriminant function in LDA is linear in the input variables.

S_2: The decision boundary produced by a kernel SVM is always linear in the original input space.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.121 Consider the following predictions of a loan approval classifier model that predicts where a loan application will be "approved" or "denied". Calculate the F'-score and error rate of model in predicting "approved" application.

Sample	Actual o/p (y)	Predicted o/p (ŷ)
1	Approved	Approved
2	Denied	Approved
3	Approved	Denied
4	Approved	Approved
5	Denied	Approved
6	Denied	Denied
7	Approved	Denied
8	Approved	Approved
9	Denied	Denied
10	Approved	Approved

(a) (0.667, 0.6) (b) (0.982, 0.6)
(c) (0.667, 0.4) (d) (0.982, 0.4)

Q.122 Which of the following statements is correct?

(a) LDA and SVM both require the assumption of Gaussian class distributions.
(b) LDA finds a projection that minimizes the margin between classes.
(c) SVM maximizes the margin between classes and does not require probabilistic assumptions.

(d) SVM and LDA always produce the same decision boundary.

Q.123 Which of the following statements about MLPs is true?

(a) MLP can only solve linearly separable problems.
(b) MLP requires weight sharing across layers.
(c) MLP with one hidden layer and sufficient neurons can approximate any continuous function.
(d) MLPs do not use activation functions.

Q.124 A parent node in a decision tree contains 100 samples, out of which 60 belong to class *A* and 40 belong to class *B*. This node is split into two child nodes: the left child and the right child. The left child contains 70 samples, of which 50 belong to class *A* and the remaining 20 to class *B*. The right child contains the remaining 30 samples, with 20 belonging to class *B* and the remaining 10 to class *A*.

Calculate the reduction in the Gini index (also known as Gini gain) due to this split.

Q.125 Consider a dataset with 10 samples: 6 belong to class A and 4 belong to class B.

Calculate the entropy of the dataset

(a) 0.84 (b) 0.88
(c) 0.46 (d) 0.97

Q.126 Which of the following measures is used to select the best attribute at each node in a decision tree (ID3 algorithm)?

(a) Gini Index (b) Information Gain
(c) Chi-Square (d) Mean Squared Error

Q.127 Consider the following statements:

S_1 : MLP with one hidden layer and non-linear activation is a universal approximation.

S_2 : Without activation functions, MLP becomes equivalent to a single-layer perceptron.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.128 Match List-I with List-II:

List-I

A. Gradient Descent
B. Multilayer Perceptron
C. ROC Curve

List-II

1. Optimization
2. Universal Function Approximator
3. Classifier Evaluation

Codes:

	A	B	C
(a)	1	2	3
(b)	3	1	2
(c)	2	1	3
(d)	1	3	2

Q.129 Which of the following is/are true about the bias-variance trade-off?

(a) Increasing model complexity tends to reduce bias and increase variance.
(b) A model with zero training error must have low bias and low variance.
(c) Simpler models usually exhibit higher bias and lower variance.
(d) Reducing variance always leads to lower test error.

Q.130 Let W_{ij} represented weight between node i at layer k and node j at layer $(k-1)$ of a given multilayer perceptron. The weight updation using gradient descent method is given by

(a) $W_{ij}(t+1) = W_{ij}(t) + \alpha \frac{\partial E}{\partial W_{ij}}, 0 \le \alpha \le 1$

(b) $W_{ij}(t+1) = W_{ij}(t) - \alpha \frac{\partial E}{\partial W_{ij}}, 0 \le \alpha \le 1$

(c) $W_{ij}(t+1) = \alpha \frac{\partial E}{\partial W_{ij}}, 0 \le \alpha \le 1$

(d) $W_{ij}(t+1) = -\alpha \frac{\partial E}{\partial W_{ij}}, 0 \le \alpha \le 1$

■■■■

Answers Machine Learning

1.	(b)	**2.**	(a)	**3.**	(c)	**4.**	(c)	**5.**	(d)	**6.**	(b)	**7.**	(c)
8.	(a)	**9.**	(c)	**10.**	(0.1)	**11.**	(a, b, c, d)	**12.**	(d)	**13.**	(1)	**14.**	(b)
15.	(b)	**16.**	(a)	**17.**	(5)	**18.**	(c)	**19.**	(a)	**20.**	(a)	**21.**	(0.4688)
22.	(b)	**23.**	(b)	**24.**	(d)	**25.**	(2.5)	**26.**	(a)	**27.**	(b)	**28.**	(c)
29.	(d)	**30.**	(d)	**31.**	(c)	**32.**	(a, b, c)	**33.**	(c)	**34.**	(c)	**35.**	(c)
36.	(a)	**37.**	(c)	**38.**	(a)	**39.**	(c)	**40.**	(a)	**41.**	(c)	**42.**	(a)
43.	(0.806)	**44.**	(a)	**45.**	(d)	**46.**	(d)	**47.**	(a)	**48.**	(a, b, c)	**49.**	(1.5)
50.	(a)	**51.**	(0.75)	**52.**	(a)	**53.**	(a)	**54.**	(b)	**55.**	(c)	**56.**	(a, b, d)
57.	(16)	**58.**	(a, c, d)	**59.**	(a)	**60.**	(1.684)	**61.**	(a)	**62.**	(a, b, d)	**63.**	(c)
64.	(c)	**65.**	(a)	**66.**	(0.5)	**67.**	(d)	**68.**	(b)	**69.**	(c)	**70.**	(b)
71.	(a, b, d)	**72.**	(c)	**73.**	(b)	**74.**	(d)	**75.**	(c)	**76.**	(d)	**77.**	(c)
78.	(a)	**79.**	(c)	**80.**	(a)	**81.**	(a)	**82.**	(a)	**83.**	(a, c, d)	**84.**	(c)
85.	(9)	**86.**	(a)	**87.**	(c)	**88.**	(a)	**89.**	(a)	**90.**	(b)	**91.**	(a)
92.	(a, c, d)	**93.**	(c)	**94.**	(36)	**95.**	(b)	**96.**	(a)	**97.**	(d)	**98.**	(a)
99.	(103)	**100.**	(b)	**101.**	(0.45)	**102.**	(c)	**103.**	(0.655)	**104.**	(c)	**105.**	(a)
106.	(b)	**107.**	(b)	**108.**	(a)	**109.**	(d)	**110.**	(a)	**111.**	(b)	**112.**	(64)
113.	(d)	**114.**	(0.661)	**115.**	(d)	**116.**	(d)	**117.**	(0.496)	**118.**	(a)	**119.**	(a)
120.	(a)	**121.**	(c)	**122.**	(c)	**123.**	(c)	**124.**	(0.060)	**125.**	(d)	**126.**	(b)
127.	(c)	**128.**	(a)	**129.**	(a, c)	**130.**	(b)						

Explanations Machine Learning

1. (b)

Back propagation algorithm is a learning technique that adjusts weights in neural network by **propagating** weight changes backward from the sink to the source nodes.

2. (a)

Linear Regression Model is used to predict continuous values, such as prices, temperatures, or sales. It models the relationship between the dependent variable and one or more independent variables by fitting a linear equation to observed data.

3. (c)

A **Sigmoid function** is a mathematical function having an '*S*' shape (Sigmoid curve). A sigmoid function is a bounded differentiable real function that is defined for all real input values and has a positive derivative everywhere.

4. (c)

- Sigmoid squashes output into (0,1), which represents probabilities.
- Binary Cross-Entropy is the appropriate loss for comparing predicted probability vs actual class (0 or 1).

5. (d)

- **Supervised learning** is the machine learning task of inferring a function from labeled training data. There may **manual labels of inputs are used.**
- **Unsupervised learning** is a type of machine learning algorithm used to draw inferences from data sets consisting of **input data without labelled responses.**

- **Reinforcement learning** is an area of machine learning inspired by behaviorist psychology, concerned with how software agents ought to take actions in an environment so as to maximize some nation of **cumulative reward.**
- **Inductive learning** uses **lean by example system.**

6. (b)

Linear Regression Model aims to minimize the sum of squared errors (SSE) between the actual values and the predicted values. The model chooses coefficients that minimize:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

7. (c)

ReLU outputs 0 for negative inputs and a linear identity for positive inputs. For positive values, its gradient is constant (1), which avoids the vanishing gradient problem that is common in sigmoid and tanh due to their flat saturation zones.

9. (c)

A **Self-Organizing Map** (SOM) is a type of Artificial Neural Network (ANN) that is trained using unsupervised learning to produce a low dimensional maps. In maps consisting of thousands of nodes, it is possible to perform **cluster** operations on the map itself.

10. (0.1)

Given, $c = 5$

$$\text{Penalty} = 3$$

$$\text{Penalty} = c\times\sum_{i=1}\xi_i$$

$$3 = 5 \times [0.3 + 0.2 + \xi_3]$$

$$\Rightarrow \quad 3 = 5 \times [0.5 + \xi_3] = 2.5 + 5\xi_3$$

$$5\xi_3 = 0.5$$

$$\xi_3 = 0.1$$

11. (a, b, c, d)

- SVM models the decision boundary between classes directly. It does not estimate the underlying class distributions, making it a discriminative model. So, option (a) is correct.
- LDA models the distribution of each class and then uses Baye's theorem to classify — this makes it a generative model. So, option (b) is correct.
- LDA assumes that features within each class follow a multivariate normal distribution with a common covariance matrix. So, option (c) is correct.
- The Kernel trick allows computing inner products in high-dimensional space without ever explicitly mapping the data — it uses a Kernel function like RBF, polynomial, etc., for this. So, option (d) is correct.

12. (d)

A step of algorithm iterate through the input patterns x_i of the training set using the weight set; i.e., compute the weighted sum of inputs.

$$\text{net } j = \sum_{i=1}^{n} x_i w_i$$

For each input pattern j.

13. (1)

Predicted values using the regression formula:

x	y(actual)	$\hat{y}$(predicted)	$y-\hat{y}$	$(y-\hat{y})^2$
1	2	1+ 1(1) = 2	0	0
2	3	1+ 1(2) = 3	0	0
3	5	1+1(3) = 4	1	1

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{SSE} = 0 + 0 + 1 = 1$$

14. (b)

MLPs are feedforward networks. Each layer processes input and passes the result to the next layer. There is no requirement for identical neuron count or linear activations.

15. (b)

Perception after multiplying weights with input will give:

$$\Rightarrow \quad 1(-1.5) + x\cdot 1 + y\cdot 1 = f(t)$$

$$\Rightarrow \quad f(t) = x + y - 1.5$$

According to given definition of $f(t)$,

If $f(t) = 1$, if $t > 0$ else $f(t) = 0$

So, truth table is:

x	y	$f(t)$
0	0	0
0	1	0
1	0	0
1	1	1

e.g. if $x = 0, y = 0$

$\Rightarrow$ $f(t) = -1.5 < 0,$

So, $f(t) = 0$

So, table is truth table of AND.

16. (a)

Decision tree classifier is a simple and widely used classification technique. It applies a straightforward idea to solve the classification problem. Decision tree classifier poses a series of carefully crafted questions about the attributes of the test record. Each time it receive an answer a follow-up question is asked until a conclusion about the class label of the record is reached.

17. (5)

Compute predicted values:

For $x = 2$: $\hat{y} = 2 + 0.5(2) = 3$

For $x = 4$: $\hat{y} = 2 + 0.5(4) = 4$

For $x = 6$: $\hat{y} = 2 + 0.5(6) = 5$

Compute squared errors:

$$(3-3)^2 = 0$$
$$(5-4)^2 = 1$$
$$(7-5)^2 = 4$$

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{SSE} = 0 + 1 + 4 = 5$$

18. (c)

The **perception learning algorithm**, which can be put into the following form:

$$w(n+1) = w(n) + \eta[d(n) - y(n)] \times x(n)$$

where, η is the step size, y is the network output and d is the desired response.

19. (a)

- If all examples belong to one class, there is no uncertainty (entropy = 0). So, S_1 is correct.
- Information Gain and Gini Index are different measures and may select different splits. So, S_2 is incorrect.

20. (a)

- **Recording** is a priori design information for memory storing.
- **Supervised** is instructive information on desired responses, explicitly specified by a teacher.
- **Unsupervised** has no information about desired responses.
- **Reinforcement** is partial information about desired responses, or only "right" or "wrong" evaluative informations.

21. (0.4688)

Total sample = 40

Class A = 25, Class B = 15

$$p_A = \frac{25}{40} = 0.625, \quad p_B = \frac{15}{40} = 0.375$$

$$\text{Gini index} = 1 - \sum_i p_i^2 = 1 - \left(p_A^2 + p_B^2\right)$$

$$\text{Gini index} = 1 - (0.625)^2 + (0.375)^2$$
$$= 1 - (0.3906 + 0.1406)$$
$$= 1 - 0.5312 = 0.4688$$

22. (b)

- SVM can handle non-linearly separable data using kernels.
- This is correct - kernels (like RBF, polynomial) transform data into higher dimensions where a linear separator can be applied.
- While feature scaling helps SVM performance, it is not a strict requirement for all kernels.
- The soft margin parameter CCC controls the trade-off between bias and variance - higher CCC means lower bias and higher variance.

23. (b)

x_1	3	w_1	0.5
x_2	-2	w_2	1.5

$b = -1$

$$z(\text{net input}) = \sum_{i=1}^{n} x_i w_i + b$$

$$z = (3 \times 0.5) + (-2 \times 1.5) + (-1)$$
$$= 1.5 - 3 - 1 = -2.5$$

By using step activation function:

$$\text{Output} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

Since $z = -2.5 < 0$ the output is 0.

24. (d)

- In general neural network models, a feature detector can be any function of the input parameters, such as linear combinations, non-linear activations, or even complex transformations depending on the architecture. I is true.

- In the context of a single-layer perceptron, where learning involves updating only the connection weights leading to the output layer. Hence, both statements are valid under typical interpretations of the perceptron model. II is true.

25. (2.5)

We know that,

$$\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$(y_1 - \hat{y}_1)^2 = (4 - 3.5)^2 = (0.5)^2 = 0.25$$

$$(y_2 - \hat{y}_2)^2 = (6 - 6.5)^2 = (-0.5)^2 = 0.25$$

$$(y_3 - \hat{y}_3)^2 = (8 - 7)^2 = (1)^2 = 1$$

$$(y_4 - \hat{y}_4)^2 = (10 - 9)^2 = (1)^2 = 1$$

$$\text{SSE} = 0.25 + 0.25 + 1 + 1 = 2.5$$

26. (a)

- Decision tree algorithms (like CART and C4.5) can handle both types of data. So, S_1 is correct.
- Decision trees are prone to overfitting, and pruning is needed to reduce complexity. So, S_2 is correct.

27. (b)

Backpropagation uses the chain rule to compute the derivative of the loss function with respect to each weight in the network. These gradients are then used by an optimizer (e.g., gradient descent) to update the weights to minimize loss.

28. (c)

- **Classification:** K-nearest neighbour.
- **Clustering:** K-means.
- **Feature extraction:** Principal component analysis (PCA).
- **Feature selection:** Branch and Bound.

30. (d)

If the input to a ReLU neuron is negative for all data, the neuron always outputs 0 — effectively making it inactive or "dead".

31. (c)

$$Z = \text{Centroied}\left(\sum_i \mu_{A_i}(x_0)\,\mu_{B_i}(y_0)\,\mu_{C_i}(z)\right)$$

32. (a, b, c)

Pruning is a post-processing step done after the tree is built. It reduces overfitting by simplifying the tree structure. It may slightly increase training error but improves generalization.

33. (c)

The way the perceptron predicts the output in each iteration is by following the equation.

$$y = f\left[w^T x\right] = f\left[\vec{w}.\vec{x}\right]$$

$$= f[w_0 + w_1 x_1 + w_2 x_2 + \ldots + x_n w_n]$$

Therefore, $x_1 + 1.5 = 0$

34. (c)

$$W_1 \times x_1 + W_2 \times x_2$$

$$= -3.9 \times 1.3 + 1.1 \times 2.2$$

$$= -5.07 + 2.42 = -2.65$$

$$-2.65 < T = 0.3$$

Thus, output = 0

35. (c)

For any **comparison based algorithm** with n elements, there are $n!$ possible ordering of elements, so corresponding decision tree needs to have atleast $n!$ leaf nodes to produce such permutation. Since tree must have $n!$ leaves atleast, with height 'h'

$$2^h \geq l \geq n!$$

$$\Rightarrow \quad h = \log(n!) = \Omega(n \log n)$$

36. (a)

- In LOOCV, each fold leaves out exactly one observation as test data, and uses all the rest for training. This results in n training cycles (where n is the number of samples). So, S_1 is correct.
- While LOOCV gives a nearly unbiased estimate of test error, it often has higher variance, because each test set is just one point - small changes in data can cause large swings in error. 10-fold CV tends to have lower variance in practice. So, S_2 is incorrect.

37. (c)

In **Neural** networks:

$$f(x) = W^T \times X + \Theta$$

$$= 0.2 \times 0.2 - 0.1 \times 0.4 + 0.1 \times 0.2$$

$$= 0.04 - 0.04 + 0.02 = 0.02$$

38. (a)

Multilayer perceptron network and support vector machines are examples of **supervised learning.** while self organizing feature maps and hopfield networks are examples of **unsupervised learning.**

39. (c)

Clustering is the process of making a group of abstract objects into classes of similar objects.

Types of Clustering method:

1. Partitioning method
2. Hierarchical method
3. Density-based method
4. Grid based method
5. Model based method
6. Constraint based method

- K-means belongs to partitioning method.
- Agglomerative and divisive belongs to hierarchical method.

40. (a)

- Silhouette Coefficient → Measures clustering quality
- Gini Index → Impurity measure used in decision trees
- OVR → Strategy for multiclass classification with binary classifiers

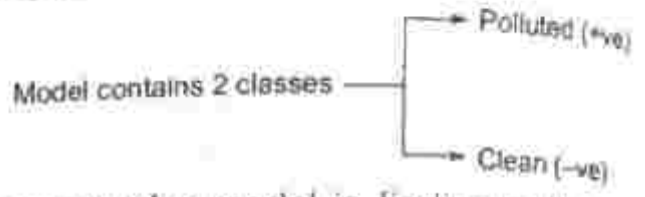
41. (c)

- LOOCV is more computationally expensive, as it trains nnn models.
- LOOCV has higher variance because each model is trained on nearly the same data.
- Correct: LOOCV requires n training iterations (one per data point left out), whereas K-Fold needs k training iterations.
- K-Fold with $k = nk = nk = n$ is equivalent to LOOCV, not to training on the entire dataset.

42. (a)

- Single linkage often creates long, chain-like clusters by linking points based on the nearest neighbors, leading to the chaining effect. So, S_1 is correct.
- This describes complete linkage, where the distance between clusters is based on the maximum pairwise distance. Single linkage uses the minimum distance. So, S_2 is incorrect.

43. (0.806)



Objective of a model is finding polluted so, polluted is +ve & clean is –ve. Actual data

Number of water samples = 15000

Number of clean samples = 10000

Number of polluted samples = 5000

	Actual label polluted	Actual label clean	
Predicted label polluted	Polluted (TP) 3800 Polluted	Clean (FP) 600 Polluted	Σ = 4400 4400 samples predicted as polluted
Predicted label clean	Polluted (FN) 1200 Clean	Clean (TN) 9400 Clean	Σ = 10600 10600 samples predicted as clean
	Σ : 5000 (5000 samples Actual polluted samples)	Σ : 10000 (10000 samps Clean polluted samples)	

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{3800}{3800+600} = \frac{3800}{4400} = 0.86$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{3800}{3800+1200} = \frac{3800}{5000} = 0.76$$

$$F^1\text{-score} = \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$$

$$= \frac{2\times0.86\times0.76}{(0.86+0.76)} = \frac{1.3072}{1.62} = 0.8069$$

44. (a)

The Gini Index is a measure of impurity and is suitable for both binary and multiclass classification problems. It is computed as:

$$\text{Gini(D)} = 1 - \sum_{i=2}^{k} p_i^2$$

where p_i is the probability of class i.

47. (a)

Centroid for cluster C1 after 1st iteration

$$\frac{1}{3}(3+5+7),\ \frac{1}{3}(3+5+7) = (5, 5)$$

Manhattan distance between (x_1, y_1) and (x_2, y_2)

$$= |x_1 - y_1| + |x_1 - y_1|$$
$$= |5-4| + |5-4| = 1 + 1 = 2$$

48. (a, b, c)

- SVM is inherently a binary classifier, but it can be extended to multi-class classification using strategies like One-vs-One and One-vs-Rest. These are standard extensions widely used in practical machine learning systems.
  So, option (a) is correct.
- LDA assumes equal covariance for all classes. If the covariance matrices are unequal, the assumption is violated, and LDA may perform poorly. In such cases, Quadratic Discriminant Analysis (QDA) is more appropriate.
  So, option (b) is correct.
- This is a foundational concept in learning theory.
  High bias → underfitting → poor performance
  High variance → overfitting → poor generalization
  The total test error is minimized by balancing the two. So, option (c) is correct.
- As k increases (e.g., from 5 to 10 to LOOCV), the variance of the validation estimate increases because the training sets are more similar and test sets smaller. It's the bias that decreases, not variance. So, option (d) is incorrect.

49. (1.5)

$$\text{Penalty} = 3$$

$$\text{Penalty term} = c \times \sum_i \xi_i$$

$$3 = c \times [0.4 + 0.6 + 0.2 + 0.8]$$

$$3 = c \times 2$$

$$c = \frac{3}{2} = 1.5$$

50. (a)

Logistic Regression and Multi-Layer Perceptron (MLP) are classification algorithms because they are used to predict discrete class labels. Logistic regression predicts the probability of classes using a sigmoid function, while MLP uses layers of neurons with activation functions to classify data into categories. On the other hand, Linear Regression is a regression algorithm used for predicting continuous numeric values, not class labels.

51. (0.75)

Compute squared errors for each data point.

$$(y_1 - \hat{y}_1)^2 = (10 - 9)^2 = (1)^2 = 1$$

$$(y_2 - \hat{y}_2)^2 = (8 - 7)^2 = (1)^2 = 1$$

$$(y_3 - \hat{y}_3)^2 = (6 - 6)^2 = (0)^2 = 0$$

$$(y_4 - \hat{y}_4)^2 = (4 - 5)^2 = (-1)^2 = 1$$

Sum of squared errors (SSE).

$$\text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\text{SSE} = 1 + 1 + 0 + 1 = 3$$

So, Mean Squared Error (MSE).

$$\text{MSE} = \frac{\text{SSE}}{n} = \frac{3}{4} = 0.75$$

52. (a)

Given, input value [0.5, 0.8, 0.2, 0.6]
Weight of input layer to hidden layer
W_1 : [0.1, 0.3, 0.5, 0.2]
W_2 : [0.2, 0.4, 0.6, 0.2]
W_3 : [0.3, 0.5, 0.7, 0.2]
Weight of hidden layer to output layer
W_4 : [0.4, 0.1, 0.3]
W_5 : [0.5, 0.2, 0.4]
By using the sigmoid function formula:

$$z = \sum_{i=1}^{m} w_i x_i$$

Input layer to hidden layer

$$z_1 = w_i \times x_i = \sum_{i=1}^{m} w_i x_i$$

$$= [0.1 \times 0.5 + 0.3 \times 0.8 + 0.5 \times 0.2 + 0.2 \times 0.6]$$

$$= [0.05 + 0.24 + 0.1 + 0.12] = 0.512$$

$$z_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-0.51}} = 0.6248$$

$$z_2 = \sum_{i=1}^{m} w_i x_i$$

$$= [0.2 \times 0.5 + 0.4 \times 0.8 + 0.6 \times 0.2 + 0.2 \times 0.6]$$

$$= [0.1 + 0.32 + 0.12 + 0.12] = 0.66$$

$$z_2 = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-0.64}} = 0.6592$$

$$z_3 = [0.3 \times 0.5 + 0.5 \times 0.8 + 0.7 \times 0.2 + 0.2 \times 0.6]$$
$$= [0.15 + 0.4 + 0.14 + 0.12] = 0.81$$
$$z_3 = \frac{1}{1+e^{-z_3}} = \frac{1}{1+e^{-0.81}} = 0.6921$$

Hidden layer to output layer

Here, new value of x : 0.6248, 0.6547, 0.6921

For,

$$z_4 = (0.4 \times 0.6248 + 0.1 \times 0.6547 + 0.3 \times 0.6921)$$
$$z_4 = (0.2499 + 0.06547 + 0.2076) = 0.52297$$
$$z_4 = \frac{1}{1+e^{-z_4}} = \frac{1}{1+e^{-52297}} = 0.6278$$

For,

$$z_5 = (0.5 \times 0.6248 + 0.2 \times 0.6547 + 0.4 \times 0.6921)$$
$$= 0.3124 + 0.13094 + 0.27684 = 0.72048$$
$$z_5 = \frac{1}{1+e^{-z_5}} = \frac{1}{1+e^{-72018}} = 0.68$$

So, output of output layer is [0.62, 0.68]

54. (b)

- LDA performs poorly in non-linear classification tasks without transformation. SVM with kernels works better in such cases.
- LDA assumes equal covariance matrices for the classes. If this assumption fails, LDA struggles. SVM has no such assumption, making it more flexible.
- SVM is less interpretable than LDA; if interpretability is not a concern, SVM may be used, but that's not a defining reason.
- SVM can handle high-dimensional data (e.g., text classification, gene expression), even when features > samples.

55. (c)

	A	B	C	D
A	0			
B	2	0		
C	5	4	0	
D	7	3	6	0

Select the small value in the matrix and form the cluster.

Small value = 2

So, A and B points are more similar.

From the cluster.

i.e., (AB)

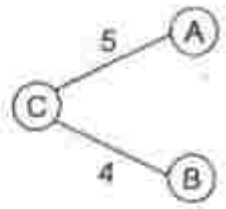
Re-form the distance matrix based on the single linkage

	AB	C	D
AB	0		
C	4	0	
D	3	6	0

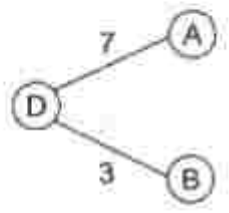
minimum

Single linkage focussed on minimum distance

$d_{min}((AB), C)$ and $d_{min}((AB), D)$



$$d_{min}(5, 4) = 4$$



$$d_{min}(7, 3) = 3$$

Select the small value and form the new cluste i.e.,

Small value = 3

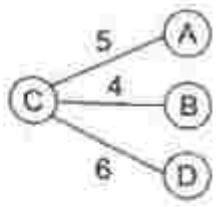
Means, (AB) and D are more similar.

∴ Form the cluster.

i.e., (ABD)

	ABD	C
ABD	0	
C	4	0

$d_{min}((ABD), C)$

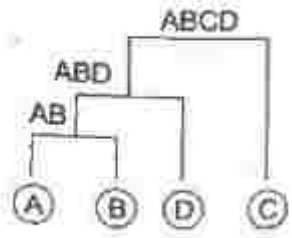


$d_{min}(5, 4, 6)$

$$d_{min} = 4$$

Merge the cluster ABCD then distance matri

	ABCD
ABCD	0



56. (a, b, d)

- LDA always produces linear decision boundaries, even if the classes are not perfectly linearly separable. It does the best it can based on Gaussian assumptions and common covariance. So, option (a) is correct.
- The RBF Kernel allows SVM to map data into a higher-dimensional space, enabling separation of non-linear patterns in the original space. So, option (b) is correct.
- The bias-variance trade-off remains important regardless of dimensionality. In high dimensions, variance can become more problematic due to overfitting. So, option (c) is incorrect.
- LOOCV gives different models for small changes in training data and uses only one point for testing, leading to high variance in test error estimates. So, option (d) is correct.

57. (16)

$$K(x_i, x_j) = 1 \cdot 3 + 2 \cdot 0 = 3$$
$$K(x_i, x_j) = (3 + 1)^2 = 16$$

58. (a, c, d)

- Single linkage merges clusters based on the minimum distance between any pair of points from two clusters.
- It does not form compact spherical clusters; it often results in elongated shapes.
- Since it uses only the closest points, it is less sensitive to outliers than complete linkage, which considers farthest points.
- It can lead to a chaining effect, where distant points are linked through intermediate close neighbors.

59. (a)

- **Weight Initialization:** Assign initial values to weights and biases, typically using random values or specific initialization techniques.
- **Feed Forward:** Process the input through the network layers to generate an output.
- **Loss Calculation:** Measure the difference between the predicted output and the actual output using a suitable loss function.
- **Back Propagation:** Compute gradients of the loss function concerning network parameters to adjust them effectively.
- **Weight Update:** Modify the weights using an optimization algorithm, such as gradient descent, to minimize the loss.

60. (1.684)

Net input to hidden layer neuron before applying the activation function

$$(z_1) = x \times W_1 + b_1$$
$$= [1, 0] \times \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + [0, 0]$$

$[(1 \times 1) + (0 \times 2)\ (1 \times 3) + (0 \times 4)] = [1, 3]$

Hidden layer output h by using sigmoid activation function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\sigma(1) = \frac{1}{1+e^{-1}} = 0.731$$

$$\sigma(3) = \frac{1}{1+e^{-3}} = 0.952$$

So, the hidden layer output vector (h)

$$h = [0.731, 0.953]$$

The net input to the output layer:

$$z_2 = h \cdot W_2^T + b_2$$
$$= (0.731 \times 1) + (0.953 \times 1) + 0 = 1.684$$

Since output activation is identity, output

$$= z_2 = 1.684$$

61. (a)

- LDA naturally extends to multi-class classification. It projects the data to a lower-dimensional space where the class separation is maximized. So, S_1 is correct.
- Even for non-linearly separable data, SVM with soft margin (using slack variables) or Kernel trick allows a solution to be found. This is a key advantage of SVM. So, S_2 is incorrect.

62. (a, b, d)

- SVM tries to find a hyperplane with maximum margin — the largest distance between the closest data points (support vectors) of each class. So, option (a) is correct.
- Only the support vectors contribute to defining the optimal hyperplane. The remaining training points do not affect it. So, option (b) is correct.

- SVM is a deterministic classifier. Unlike probabilistic models (e.g., logistic regression), it does not output class probabilities by default. So, option (c) is incorrect.
- Kernels map data to higher-dimensional spaces, allowing SVM to create non-linear decision boundaries. So, option (d) is correct.

64. (c)

Without non-linear activation functions (like ReLU, sigmoid), the entire network behaves like a linear model. The activation function allows the MLP to learn complex, non-linear mappings.

65. (a)

We know that,

$$L = \frac{1}{2}(y-t)^2$$

$$\frac{dL}{dy} = \frac{d}{dy}\left[\frac{1}{2}(y-t)^2\right]$$

$$= (y-t) = (0.5-1) = -0.5$$

Since activation is identity:

$$y = z = wx + b$$

$$\Rightarrow \quad \frac{dy}{dw} = x = 1.0$$

By using Chain rule:

$$\frac{dL}{dw} = \frac{dL}{dy} \times \frac{dy}{dw}$$

$$= -0.5 \times 1.0 = -0.5$$

66. (0.5)

Compute the predicted values ($\hat{y}$) using the model.

x	y (actual)	$\hat{y}$ (predicted)	$y-\hat{y}$	$(y-\hat{y})^2$
2	4	2×2 = 4	0	0
3	6	2×3 = 6	0	0
5	9	2×5 = 10	–1	1
6	12	2×6 = 12	0	0

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$SSE = 0 + 0 + 1 + 0 = 1$$

Mean Squared Error (MSE).

$$MSE = \frac{SSE}{n} = \frac{1}{4} = 0.25$$

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{0.25} = 0.5$$

67. (d)

Backpropagation computes gradients using the chain rule, requires differentiable activations, and updates weights by propagating errors backward from output to input layer.

68. (b)

- ReLU helps to reduce the vanishing gradient problem (though it may cause dead neurons). So, S_1 is incorrect.
- Sigmoid saturates for large positive or negative inputs, causing gradients to vanish and learning to slow down. So, S_2 is correct.

69. (c)

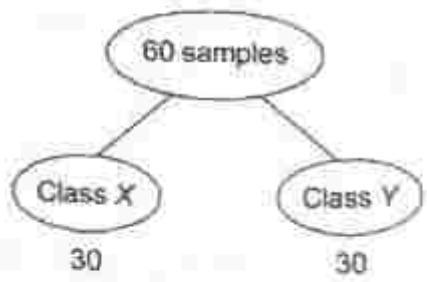
- **Natural Language Processing:** NLP is a branch of AI focused on understanding and generating human language.
- **Reinforcement Learning:** RL trains models by rewarding desired behavior and punishing undesired behavior.
- **Support Vector Machine:** SVM is a machine learning technique that finds the best hyperplane to separate different classes.
- **Expert System:** Expert systems are designed to emulate human decision-making using rule-based AI.

70. (b)

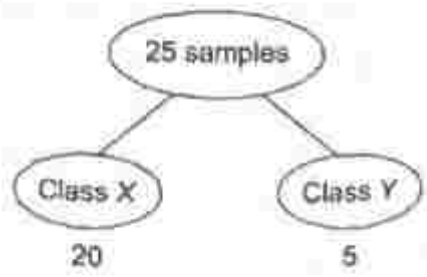
$$K(x_i, x_j) = 2 \cdot 2 + 2 \cdot 2 = 4 + 4 = 8$$

$$K(x_i, x_j) = (8+1)^2 = 9^2 = 81$$

71. (a, b, d)



Split 1: Left subset



$$p_X = \frac{20}{25} = 0.8, \quad p_Y = \frac{5}{25} = 0.2$$

$$\text{Gini Index}_{left} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - (0.8)^2 - (0.2)^2$$

$$= 1 - 0.64 - 0.04 = 1 - 0.68 = 0.32$$

Right subset

$$p_X = \frac{10}{35} = 0.285, \quad p_Y = \frac{25}{35} = 0.714$$

$$\text{Gini Index}_{right} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - (0.285)^2 - (0.714)^2$$

$$= 1 - 0.081 - 0.509$$

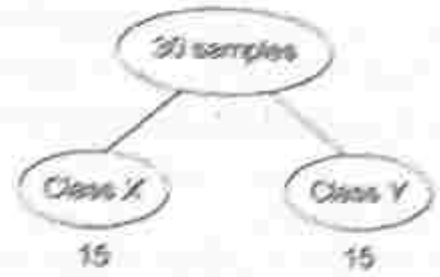
$$= 1 - 0.59 = 0.41$$

$$\text{Weight Gini}_{split1} = \frac{25}{60} \times 0.32 + \frac{35}{60} \times 0.4082$$

$$= 0.41 \times 32 + 0.58 \times 0.41$$

$$= 0.13 + 0.23 = 0.36$$

Split 2: Left subset



$$p_X = \frac{15}{30} = 0.5, \quad p_Y = \frac{15}{30} = 0.5$$

$$\text{Gini Index}_{left} = 1 - \sum_{i=1}^{c} p_i^2$$

$$= 1 - (0.5)^2 - (0.5)^2 = 1 - 0.25 - 0.25$$

$$= 1 - 0.5 = 0.5$$

Same distribution as left in right.

So, $\text{Gini index}_{right} = 0.5$

$$\text{Weighted Gini index}_{split2} = \frac{30}{60} \times 0.5 + \frac{30}{60} \times 0.5$$

$$= 0.5 \times 0.5 + 0.5 \times 0.5$$

$$= 0.25 + 0.25 = 0.5$$

Difference between both Weighted Gini index

$$= \text{Gini index}_{split1} - \text{Gini index}_{split2}$$

$$= 0.5 - 0.36 = 0.14$$

72. (c)

- High-bias models underfit training data by oversimplifying the function.
- High-variance models typically fit training data well but perform poorly on unseen data (test set).
- This is the essence of the trade-off: reducing variance (e.g., by regularization or simpler models) usually increases bias.
- Bias and variance are interrelated — reducing one can increase the other.

73. (b)

Polynomial Kernels implicitly map inputs to a higher-dimensional feature space without explicitly computing that mapping — enabling non-linear decision boundaries.

74. (d)

Calculate the required sums.

x	y	xy	x^2
3	4	12	9
4	6	24	16
6	9	54	36
7	11	77	49
9	15	135	81

$$\Sigma x = 3 + 4 + 6 + 7 + 9 = 29$$

$$\Sigma y = 4 + 6 + 9 + 11 + 15 = 45$$

$$\Sigma xy = 12 + 24 + 54 + 77 + 135 = 302$$

$$\Sigma x^2 = 9 + 16 + 26 + 49 + 81 = 191$$

$$n = 5$$

We know that,

$$m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n(\Sigma x^2 - (\Sigma x)^2}$$

$$m = \frac{5(302) - (29)(45)}{5(191) - (29)^2}$$

$$= \frac{1510 - 1305}{955 - 841} = \frac{205}{114} = 1.79$$

$$c = \frac{\Sigma y - m\Sigma x}{n}$$

$$c = \frac{45 - 1.8(29)}{5} = \frac{45 - 51.81}{5} = \frac{-1.91}{5} = -1.382$$

Thus, the equation of the line of best fit is:

$$y = 1.75x - 1.382$$

75. (c)

- SMA* is designed to find an optimal solution if one exists; if the optimal solution is not reachable, it returns the best reachable solution. So, S_1 is correct.
- Greedy Best-First Search can get stuck in local optima and may fail to find a solution even in finite search spaces, making it incomplete. So, S_2 is correct.

76. (d)

Weights between input and hidden layer:

$$4 \times 5 = 20$$

Add biases for hidden layer neurons: 5

Calculate weights between hidden and output layer:

$$5 \times 2 = 10$$

Add biases for output layer neurons: 2

Total weights including biases:

$$20 + 5 + 10 + 2 = 37$$

77. (c)

- Decision trees typically use the Mean Squared Error (MSE) or other criteria like Mean Absolute Error (MAE) for regression tasks, but the common practice is to use MSE for splitting. So, S_1 is incorrect.
- The Naive Bayes algorithm indeed assumes that all features are independent given the class label. So, S_2 is incorrect.

78. (a)

x	y	xy	x^2
2	5	10	4
4	9	36	16
5	13	65	25
7	16	112	49
9	20	180	81

$$\Sigma x = 2 + 4 + 5 + 7 + 9 = 27$$
$$\Sigma y = 5 + 9 + 13 + 16 + 20 = 63$$
$$\Sigma xy = 10 + 36 + 65 + 112 + 180 = 403$$
$$\Sigma x^2 = 4 + 16 + 25 + 49 + 81 = 175$$
$$n = 5$$

Calculate the slope m.

$$m = \frac{5(403) - (27)(63)}{5(175) - (27)^2} = \frac{2015 - 1701}{875 - 729} = \frac{314}{146} = 2.15$$

Thus, the slope m is approximately 2.15, which matches option (a).

79. (c)

- Gradient Descent can converge to a local minimum for non-convex functions, which may not be the global minimum. So, S_1 is correct.
- Stochastic Gradient Descent (SGD) updates the weights for each training instance individually, using only one sample (or a small batch) at a time to compute the gradient. So, S_2 is correct.

80. (a)

$$f(x) = w^T x + b$$
$$= (3)(2) + (-1)(4) + 5 = 6 - 4 + 5 = 7$$

81. (a)

- LDA works by projecting the data in a direction that maximizes the ratio of between-class variance to within-class variance. This is the core objective of LDA and ensures maximum separability. So, S_1 is correct.
- LDA does not ignore covariance. It uses the pooled within-class covariance matrix in its optimization to ensure that the direction of projection accounts for class spread (intra-class scatter). So, S_2 is incorrect.

82. (a)

- Decision tree → Discriminative Model
- Bias-Variance Trade-Off → Model Complexity
- Cross-Validation → Model Validation Technique

83. (a, c, d)

- Single linkage clustering is indeed called the "nearest neighbor" or "minimum distance" method. So option (a) is correct.
- The definition given applies to complete linkage clustering, not single linkage. In single linkage, the distance between two clusters is the minimum distance between any points in the two clusters. So, option (b) is incorrect.
- Single linkage clustering is known for being sensitive to noise and outliers, leading to chaining. So, option (c) is correct.

- Single linkage clustering tends to form clusters with more overlap compared to complete linkage clustering. So, option (d) is correct.

84. (c)

$$f(x) = w^T x + b$$
$$= 5 \times 1 + (-2) \times 3 + b$$
$$= 5 - 6 + b = -1 + b$$
Given, $f(x) = 4$
$$= -1 + b = 4 \Rightarrow b = 5$$

85. (9)

Slack variable penalty $= C\sum_{i=1}^{n}\xi_i$

Penalty $= 5 \times (0.4 + 0.3 + 0.5 + 0.6) = 5 \times 1.8 = 9$

86. (a)

- The R-squared value indicates the proportion of variance in the dependent variable explained by the independent variables in Linear Regression. So, option (a) is correct.
- Decision Trees can use various criteria like mean squared error for regression, but they are not limited to just that. So, option (b) is incorrect.
- Naive Bayes assumes that features are independent given the class label, not dependent. So, option (c) is incorrect.
- SVM uses the kernel trick to handle non-linear decision boundaries, which is correct, but per your requirements, only one statement should be correct. So, option (d) is incorrect.

87. (c)

- The decision criterion helps choose the best feature for splitting the data at each node of the decision tree. So, S_1 is correct.
- the decision criterion often compares feature values to a threshold (such as in Gini impurity, entropy, or other splitting metrics) to determine how to split the data. So, S_2 is correct.

88. (a)

$$f(x) = w^T x + b$$
$$= 1 \times 3 + 2 \times 2 - 5$$
$$= 3 + 4 - 5 = 2 > 0$$

So, class is +1.

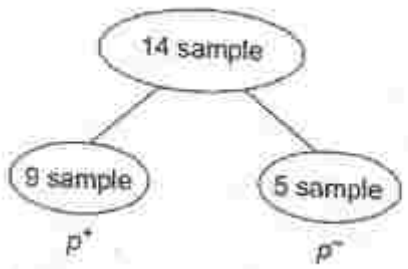
89. (a)

- PCA → Dimensionality Reduction
- SVM → Maximizes the margin between classes
- Logistic Regression → Classification algorithm

90. (b)

- LOO cross-validation, where each individual data point is used as a test set exactly once, while the rest of the data points serve as the training set. So, S_1 is correct.
- In SVM, the primal problem aims to minimize a cost function, while the dual problem focuses on maximizing the dual function. So, S_2 is incorrect.
- K-means clustering operates, assigning data points to the nearest cluster centroid. So, S_3 is correct.

91. (a)



Initial entropy:

$$p_+ = \frac{9}{14} = 0.6428$$

$$p_- = \frac{5}{14} = 0.3571$$

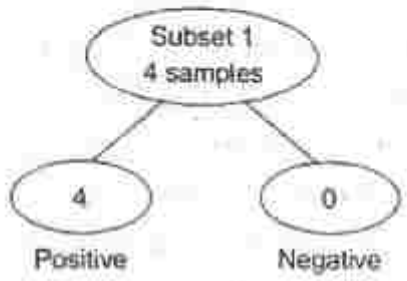
Entropy $(E) = -\sum p_i \log_2 p_i$

$= -[(0.6428 \log_2 0.6428 + 0.3571 \log_2 0.3571)]$

$= -[0.6428 \times (-0.6375) + 0.3571 \times (-1.4856)]$

$= -[-0.4098 - 0.5305] = 0.9403$

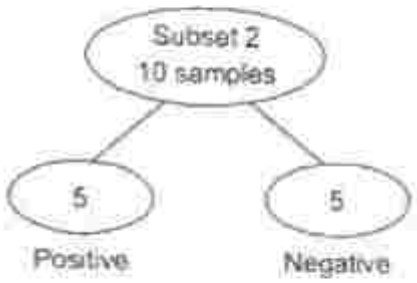
Now, entropy of subset 1:



$$p_+ = \frac{4}{4} = 1$$

It is pure subset so, entropy is 20.

Entropy of subset 2:



$$p_+ = \frac{5}{10} = 0.5, \quad p_- = \frac{5}{10} = 0.5$$

$$\text{Entropy } (E) = -\sum p_i \log_2 p_i$$
$$= -[0.5 \log_2 0.5 + 0.5 \log_2 0.5]$$
$$= -[0.5 \times (-1) + 0.5 \times (-1)]$$
$$= -(-1) = +1$$

Weight entropy after split

$$\text{Entropy}_{\text{split}} = \frac{4}{14} \times 0 + \frac{10}{14} \times 1$$
$$= 0 + 0.7142 = 0.7142$$

$$\text{Information gain} = \text{Entropy}_{\text{initial}} - \text{Entropy}_{\text{split}}$$
$$= 0.9403 - 0.7142 = 0.226\%$$

92. (a, c, d)

- Stratified K-Fold is used in classification problems to ensure that each fold has the same distribution of classes as the original dataset. So, option (a) is correct.
- While K-Fold CV provides a reasonable estimate, it's not guaranteed to be unbiased. Bias depends on the data distribution and model type. So, option (b) is incorrect.
- With small k (e.g., $k = 2$), each fold has a lot of data held out for testing. This leads to high bias in the estimated test error because the training set is small. So, option (c) is correct.
- LOOCV is a special case of K-Fold where $k = nk = nk = n$, i.e., one sample is left out each time. So, option (d) is correct.

93. (c)

- The decision function for an SVM is indeed $f(x) = w^T x + b$. So, S_1 is true.
- In a soft-margin SVM, C controls the trade-off between maximizing the margin and minimizing the slack variables (classification error). So, S_2 is true.
- The margin of an SVM is $\frac{2}{\|w\|}$, where $|w|$ is the norm of the weight vector. So, S_3 is true.
- The Kernel function enables the transformation of data into a higher-dimensional space to facilitate linear separation. So, S_4 is true

94. (36)

Kernel function calculation:

$$K(x_i, x_j) = (x_i \cdot x_j + 1)^2$$

Compute $x_i \cdot x_j$:

$$x_i \cdot x_j = [2, 1] \cdot [1, 3]$$
$$= 2 \times 1 + 1 \times 3$$
$$= 2 + 3 = 5$$

Kernel value

$$K(x_i, x_j) = (5 + 1)^2 = 6^2 = 36$$

The Kernel function value $K(x_i, x_j)$ is 36.

95. (b)

Entropy measures impurity or uncertainty.
When all samples belong to one class (perfect purity), entropy = 0.
When classes are equally split (maximum uncertainty), entropy is maximum.

96. (a)

- Naive Bayes → Generative Model
- Decision Tree → Uses Greedy Splitting (Information Gain/Gini)
- Bagging → Ensemble technique (bootstrap aggregation)

97. (d)

Given, $c = 3$

$$\text{Penalty} = c \times \sum_{i=1}^{3} \xi_i$$
$$= 3 \times (0.7 + 0.5 + 0.8)$$
$$= 3 \times 2.0 = 6$$

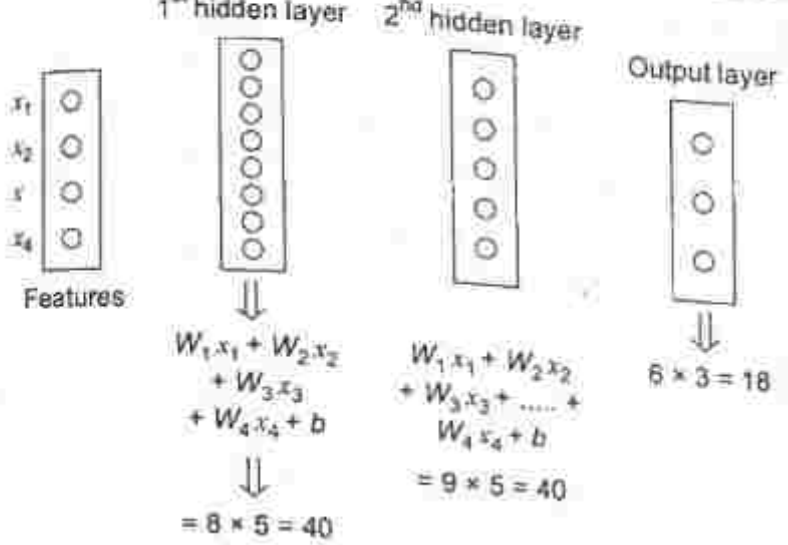
98. (a)

Decision boundary satisfies:

$$w^T x + b = 0$$
$$\Rightarrow \quad 4x_1 - 2x_2 - 6 = 0$$
$$\text{or} \quad 4x_1 - 2x_2 = 6$$

99. (103)

Given,
2 hidden layer, 1 output layer
1st hidden layer takes 8 neurons
2nd hidden layer takes 5 neurons



Total number of trainable parameters = 40 + 45 + 18 = 103

100. (b)

Entropy before split = 0.9
Weighted entropy after split = 0.4
Information gain = Entropy before split – Weighted entropy after split
Information gain = 0.9 – 0.4 = 0.5

101. (0.45)

Left subset:

Total instances = 20
Approved instances = 15
Rejected instance = 5

$$P_{approved} = \frac{15}{20} = 0.75,\ P_{rejected} = \frac{5}{20} = 0.25$$

$$\text{Gini Index}_{left} = 1 - \sum_{i=1}^{c} P_i^2$$

$$= 1 - (0.75)^2 - (0.25)^2$$
$$= 1 - 0.5625 - 0.0625$$
$$= 1 - 0.625 = 0.375$$

Right subset:

Total instances = 30
Approved instances = 15
Rejected instance = 15

$$P_{approved} = \frac{15}{30} = 0.5,\quad P_{rejected} = \frac{15}{30} = 0.5$$

$$\text{Gini Index}_{right} = 1 - \sum_{i=1}^{c} P_i^2$$

$$= 1 - (0.5)^2 - (0.5)^2 = 1 - 0.25 - 0.25)$$
$$= 1 - 0.5 = 0.5$$

Weighted Gini index :

$$= \frac{\text{Instances}_{left}}{\text{Total instance}} \cdot \text{Gini}_{left} + \frac{\text{Instances}_{right}}{\text{Total instance}} \cdot \text{Gini}_{right}$$

$$= \frac{20}{50} \times 0.375 + \frac{30}{50} \times 0.5$$
$$= 0.4 \times 0.375 + 0.6 \times 0.5$$
$$= 0.15 + 0.3 = 0.45$$

102. (c)

- SVMs can be used for classification (SVC) and regression (SVR) tasks. So, S_1 is correct.
- Kernel functions are used when the data is not linearly separable, but SVMs can work without Kernels if the data is linearly separable. So, S_2 is incorrect.
- The SVM algorithm aims to find a hyperplane that maximizes the margin between the classes. So, S_3 is correct.
- The complexity of SVMs is quadratic with respect to the number of training samples, not linear. So, S_4 is incorrect.

103. (0.655)

$$P_1 = \frac{40}{100} = 0.4$$

$$P_2 = \frac{35}{100} = 0.35$$

$$P_3 = \frac{25}{100} = 0.25$$

$$\text{Gini Index} = 1 - \sum_{i=1}^{n} P_i^2$$

$$= 1 - [(0.4)^2 + (0.35)^2 + (0.25)^2]$$
$$= 1 - [0.16 + 0.1225 + 0.0625]$$
$$= 1 - 0.345 = 0.655$$

104. (c)

Linear functions, when stacked (i.e., multiple layers), still produce a linear output. Without non-linear activation functions like ReLU or Sigmoid, an MLP would collapse to a linear model, regardless of its depth. The non-linear activation enables the MLP to model complex decision boundaries and approximate non-linear functions.

105. (a)

- Decision Tree → Discriminative Model
- Bias-Variance Trade-Off → Model Complexity
- Multi-Layer Perceptron (MLP) → Back propagation
- Cross-Validation → Model Validation Technique

106. (b)

Input to Hidden Layer:

3 inputs × 4 hidden = 12 weights

Hidden to Output:

4 hidden × 1 output = 4 weights

Total = 12 + 4

= 16 weights

107. (b)

Margin width is given by:

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{3^2+4^2}} = \frac{2}{5} = 0.4$$

108. (a)

- Class prior probability estimation = K class labels given
  = K parameters
- Mean → Each feature within class mean value calculation i.e., D * K
- Variance → Each feature within class variance calculation i.e., D * K

Total number of parameters = 2DK + K

109. (d)

I[st] iteration :

C_1 : (0,0)	C_2 : (3,3)	C_3 : (6,6)
C_1	C_2	C_3
(0,0)	(2,2)	(5,5)
(1,1)	(3,3)	(6,6)
	(4,4)	(7,7)

$$\text{New mean of } C_2 = \left[\left(\frac{2+3+4}{3}\right), \frac{(2+3+4)}{3}\right]$$

$$= (3, 3)$$

110. (a)

- A decision tree with a maximum depth equal to the number of training samples can perfectly fit the training data, achieving 100% accuracy.
- A linear SVM and logistic regression may not achieve 100% accuracy if the data is not linearly separable.
- A polynomial kernel SVM with degree 2 could achieve 100% accuracy if the data is polynomially separable, but it's not guaranteed like the decision tree in this case.

111. (b)

- Slack variables quantify margin violations and are non-negative. So, S_1 is correct.
- Very high c penalizes slack variables heavily, forcing no margin violation, effectively a hard margin SVM. So, S_2 is correct.

112. (64)

$$K(x_i, x_j) = 1 \cdot 2 + 1 \cdot 0 = 2$$

$$K(x_i, x_j) = (2+2)^3 = 4^3 = 64$$

113. (d)

For multi-class classification, we require class probabilities that sum to 1. The softmax activation converts raw scores (logits) into a probability distribution, which is essential for computing categorical cross-entropy loss.

114. (0.661)

Proportion of each class

$$P_{DA} = \frac{35}{35+25+45} = \frac{35}{105} = \frac{1}{3} = 0.33$$

$$P_{CS} = \frac{25}{35+25+45} = \frac{25}{105} = 0.23$$

$$P_{ME} = \frac{45}{35+25+45} = \frac{45}{105} = 0.42$$

$$\text{Gini index} = 1 - \sum_{i=1}^{n} (P_i)^2$$

$$= 1 - \left[(P_{DA})^2 + (P_{CS})^2 + (P_{ME})^2\right]$$

$$= 1 - \left[(0.35)^2 + (0.23)^2 + (0.42)^2\right]$$

$$= 1 - [0.1089 + 0.0529 + 0.1764]$$

$$= 1 - 0.3382 = 0.6618$$

115. (d)

We know that,

$$\text{RBF Kernel } K(x_i, x_j) = \bar{e}\left[\gamma \|x_i - x_j\|^2\right] \quad \ldots(i)$$

$$\|x_i - x_j\|^2 = (3-5)^2 + (3-7)^2$$

$$= (-2)^2 + (-4)^2$$

$$= 4 + 16 = 20$$

Put the value in equation (i)

$$= \bar{e}(0.8 \times 20) = \bar{e}(16)$$

$$= 01.125 \times 10^{-7}$$

116. (d)

Euclidean norm ($\|w\|$) of the weight vector $w = [4, 5]$

$$\|w\| = \sqrt{w_1^2 + w_2^2} = \sqrt{4^2 + 5^2} = \sqrt{41}$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{41}}$$

117. (0.496)

$$\text{Precision (P)} = \frac{TP}{TP + FP} = \frac{8}{8+6} = 0.57$$

$$\text{Recall (R)} = \frac{TP}{TP + FN} = \frac{8}{8+10} = 0.44$$

$$\text{F'-score} = \frac{2 \times P \times R}{P + R}$$

$$= \frac{2 \times 0.57 \times 0.44}{0.57 + 0.44}$$

$$= \frac{0.5016}{1.01} = 0.496$$

118. (a)

- L1 Regularization (Lasso) → Promotes sparsity (sets some weights to 0)
- K-Means → Unsupervised clustering technique
- Random Forest → Ensemble of decision trees with bootstrapped samples

119. (a)

$$\text{Entropy}(E) = -\sum p_i \log_2 p_i$$

$$= -[(0.5 \log_2 0.5) + (0.3 \log_2 0.3) + (0.2 \log_2 0.2)]$$

$$= -[0.5 \times (-1) + 0.3 \times (-1.737) + 0.2 \times (-2.322)]$$

$$\text{Entropy}(E) = -(-0.5 - 0.5211 - 0.4644)$$

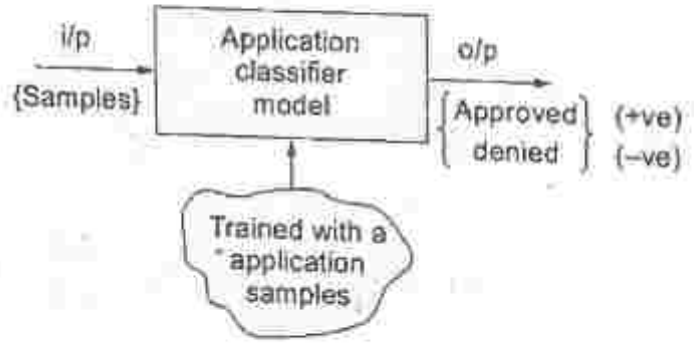
$$= 1.4855 = 1.485$$

120. (a)

- In LDA, the discriminant function is of the form $w^T x + b$, which is clearly a linear function of the input vector x. That's why it is called Linear Discriminant Analysis. So, S_1 is correct.
- In Kernel SVMs, we transform data to a higher-dimensional space using a Kernel function. The decision boundary becomes nonlinear in the original input space, although it's linear in the transformed feature space. So, S_2 is incorrect.

121. (c)

	Actual class label (+ ve) (Approved)	Actual class label (–ve) (Denied)	
Predicted class label (+ ve) (Approved)	Approved 4 (TP) Approved	Denied 2 (FP) Approved	Σ = 6 Application number of sample predicted as +ve
Predicted class label (–ve) (Denied)	Approved 2 (FN) Denied	Denied 2 (TN) Denied	Σ = 4 Application number of sample predicted as –ve
	Σ : 6 Application sample actually +ve	Σ : 4 Application sample actually (–ve)	



$$\text{Accuracy} = \frac{\text{Number of samples correctly predicted}}{\text{Total number of samples in the training set}}$$

$$= \frac{6}{10} = 0.6$$

$$\text{Error} = 1 - 0.6 = 0.4$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{4}{4+2} = \frac{4}{6} = 0.667$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{4}{4+2} = \frac{4}{6} = 0.667$$

$$\text{F'-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} = \frac{0.889}{1.334} = 0.667$$

122. (c)

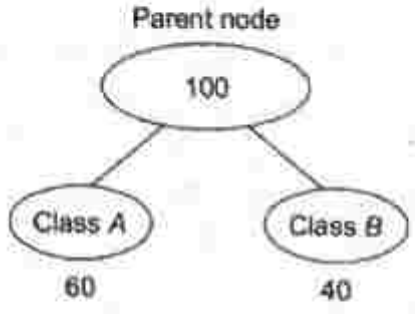
- Only LDA assumes Gaussian class conditional distributions with equal covariance; SVM does not rely on distributional assumptions.
- LDA maximizes class separability, not minimizes margin.

- SVM maximizes the margin between classes and does not assume any probabilistic model — it's purely geometric.
- SVM and LDA often produce different decision boundaries, especially when the assumptions of LDA do not hold.

123. (c)

This is the Universal Approximation Theorem — a single hidden layer MLP with a non-linear activation function (like sigmoid or ReLU) and enough neurons can approximate any continuous function on compact subsets.

124. (0.060)



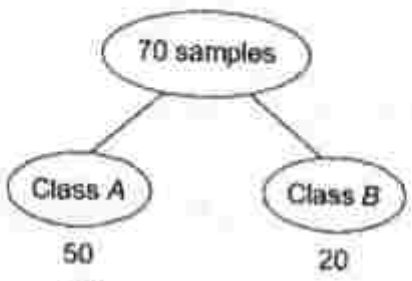
$$P_A = \frac{60}{100} = 0.6, \quad P_B = \frac{40}{100} = 0.4$$

$$\text{Gini Index}_{\text{parent}} = 1 - \sum_{i=1}^{n} P_i^2$$

$$= 1 - (0.6)^2 - (0.4)^2$$

$$= 1 - 0.36 - 0.16) = 1 - 0.52 = 0.48$$

Left child:



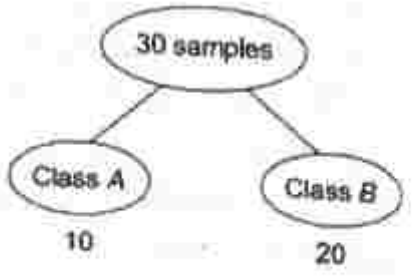
$$P_A = \frac{50}{70} = 0.714, P_B = \frac{20}{70} = 0.285$$

$$\text{Gini Index}_{\text{left}} = 1 - \sum_{i=1}^{n} P_i^2$$

$$= 1 - (0.714)^2 - (0.285)^2$$

$$= 1 - 0.509 - 0.081 = 0.41$$

Right child:



$$P_A = \frac{10}{30} = 0.33, \quad P_B = \frac{20}{30} = 0.66$$

$$\text{Gini Index}_{\text{right}} = 1 - \sum_{i=1}^{n} P_i^2$$

$$= 1 - (0.33)^2 - (0.66)^2$$

$$= 1 - 0.1089 - 0.4356$$

$$= 1 - 0.5445 = 0.45$$

Weighted Gini index after splitting:

$$\text{Gini Indx}_{\text{split}} = \frac{70}{100} \times 0.41 + \frac{30}{100} \times 0.45$$

$$= 0.7 \times 0.41 + 0.3 \times 0.45$$

$$= 0.287 + 0.135 = 0.42$$

Reduction in Gini index:

$$= \text{Gini Index}_{\text{parent}} - \text{Gini Index}_{\text{split}}$$

$$= 0.48 - 0.42 = 0.060$$

125. (d)

Class probabilities:

$$p_A = \frac{6}{10} = 0.6, \quad p_B = \frac{4}{10} = 0.4$$

$$\text{Entropy } (E) = -\sum p_i \log_2 p_i$$

$$= -(p_A \log_2 p_A + p_B \log_2 p_B)$$

$$= -[(0.6 \log_2 0.6) + (0.4 \log_2 0.4)]$$

$$= -[0.6 \times (-0.73697) + 0.4 \times (-1.32193)]$$

$$\text{Entropy } (E) = -(-0.44218 - 0.52877)$$

$$= 0.97095 = 0.97$$

126. (b)

The ID3 algorithm selects the attribute that maximizes Information Gain, which is based on the concept of entropy. It helps in identifying the attribute that most effectively splits the dataset.

127. (c)

- According to the Universal Approximation Theorem, an MLP with just one hidden layer and non-linear activation (like ReLU or sigmoid) can approximate any continuous function on a compact subset of $R^{(n)}$, given enough neurons. So, S_1 is correct.
  If no activation function is used, each layer performs a linear transformation.
- Multiple linear transformations without non-linearity collapse into a single linear transformation.
  So the MLP becomes equivalent to a single-layer perceptron (i.e., linear model). So, S_2 is correct.

128. (a)

- Gradient Descent → Optimization algorithm
- MLP → Universal approximator in theory (Hornik's theorem)
- ROC Curve → Used to evaluate classifier performance

129. (a, c)

- As complexity increases (e.g., using deeper trees or higher-degree polynomials), the model fits training data better (lower bias) but may become too sensitive to fluctuations (higher variance). So, option (a) is correct.
- Zero training error could mean the model is overfitting, i.e., it may have low bias but very high variance, which leads to poor generalization. So, option (b) is incorrect.
- Simpler models, like linear regression for nonlinear data, are less flexible, so they underfit (high bias) but do not change much with different training sets (low variance). So, option (c) is correct.
- Reducing variance might increase bias. It's not always beneficial - the goal is to minimize total test error, which depends on the right balance between bias and variance. So, option (d) is incorrect.

■■■■

UNIT 5

Artificial Intelligence

Q.1 A* algorithm is guaranteed to find an optimal solution if
(a) h' is always 0.
(b) g is always 1.
(c) h' never overestimates h.
(d) h' never underestimates h.

Q.2 A search algorithm uses depth-limited search with a maximum depth limit f. The only goal is at depth d. Which of the following is true?
(a) if $f = d$, the algorithm will find the goal node and the path will be optimal.
(b) If $f < d$, the algorithm will still find the goal node but not necessarily the shortest path.
(c) If $f > d$, the algorithm will fail to find the goal.
(d) If $f = d$ the algorithm may fail to find the goal.

Q.3 What is the goal of exact inference in probabilistic reasoning?
(a) To estimate probabilities using random sampling.
(b) To compute the exact posterior probabilities given evidence.
(c) To simplify computations by ignoring evidence.
(d) To use heuristics for probability estimation.

Q.4 $\neg (X \wedge Y) \rightarrow (\neg X \vee (\neg X \vee Y))$ is logically equivalent to
(a) $\sim(X \vee Y)$ (b) $\sim X \vee Y$
(c) $\sim X \wedge Y$ (d) $\sim(X \wedge Y)$

Q.5 A^* algorithm uses $f' = g + h'$ to estimate the cost of getting from the initial state to the goal state, where g is a measure of the cost of getting from initial state to the current node and the function $h¢$ is an estimate of the cost of getting from the current node to the goal state. To find a path involving the fewest number of steps, we should set
(a) $g = 1$ (b) $g = 0$
(c) $h' = 0$ (d) $h' = 1$

Q.6 The strategy used to reduce the number of the branches and the number of static evaluations applied in case of a game tree is
(a) Minmax strategy
(b) Alpha-beta pruning strategy
(c) Constraint satisfaction strategy
(d) Static max strategy

Q.7 A spam filtering system correctly classifies spam emails with a probability of 0.95 and has a false positive rate of 0.10 (i.e., the probability of classifying a non-spam email as spam). If 0.5% of the emails received are actually spam, what is the probability that an email classified as spam is indeed spam?
(a) 0.0455
(b) 0.095
(c) 0.10425
(d) 0.95

Q.8 Match **List-I** with **List-II**:

List-I	List-II
(A) Hill climbing	(I) $O(b^a)$
(B) Best first search	(II) $O(bd)$
(C) A* Search	(III) $O(1)$
(D) Depth first search	(IV) $O(b^m)$

Choose the **correct** answer from the options given below :
(a) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(b) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(c) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(d) (A)-(I), (B)-(III), (C)-(II), (D)-(I)

Q.9 If h^* represents an estimate of the cost of getting from the current node N to the goal node and h represents actual cost of getting from the current node to the goal node, then A^* algorithm gives an optimal solution if
(a) h^* is equal to h
(b) h^* overestimates h
(c) h^* underestimates h
(d) None of these

Q.10 Which one of the following Boolean expresses is NOT a tautology?

(a) $((a \to b) \wedge (b \to c)) \to (a \to c)$

(b) $(a \leftrightarrow b) \to (\neg b \to (a \wedge c))$

(c) $(a \wedge b \wedge c) \to (c \vee a)$

(d) $a \to (b \to a)$

Q.11 Consider the following statements:

S_1: Uniform-cost search will never expand more nodes than depth-first search.

S_2: Depth-first search will always expand more nodes than breadth-first search.

S_3: A* search will always expand fewer nodes than uniform-cost search.

Which of the following is incorrect?

(a) Only S_1 and S_2 (b) Only S_2 and S_3

(c) Only S_1 and S_3 (d) All of these

Q.12 The mean-end analysis process centers around the detection of differences between the current state and goal state. Once such a difference is isolated, an operator that can reduce the difference must be found. But perhaps that operator can not be applied to the current state. So a sub-problem of getting to a state in which it can be applied is set up. The kind of backward chaining in which operators are selected and then sub goals are set up to establish the precondition of operators is called

(a) backward planning

(b) goal stack planning

(c) operator subgoaling

(d) operator overloading

Q.13 Which of the following is true?

(a) DFS expands all nodes at a given depth before moving deeper

(b) DFS may go deep into the search tree without finding the solution

(c) DFS is not memory efficient

(d) DFS is guaranteed to find the optimal solution

Q.14 In alpha-beta pruning, _____ is used to cut off the search at maximizing level only and _____ is used to cut off the search at minimizing level only.

(a) alpha, beta (b) beta, alpha

(c) alpha, alpha (d) beta, beta

Q.15 Which one of the following is not an informed search technique?

(a) Hill climbing search

(b) Best first search

(c) A* search

(d) Depth first search

Q.16 The map colouring problem can be solved using which of the following technique?

(a) Means-end analysis

(b) Constraint satisfaction

(c) AO* search

(d) Breadth first search

Q.17 Consider a depth-limited search with depth cutoff f. The unique goal is at depth d. What happens if $f < d$?

(a) The algorithm reaches the goal node but may return a longer path.

(b) The algorithm cannot find the goal node because the depth limit is too low.

(c) The algorithm is guaranteed to find the goal optimally.

(d) The algorithm finds multiple goals at different depths.

Q.18 Which of the following is true?

(i) $\exists x\{P(x) \wedge Q(x)\} \equiv \exists x P(x) \wedge \exists x Q(x)$

(ii) $\exists x\{P(x) \wedge Q(x)\} \Rightarrow \exists x P(x) \wedge \exists x Q(x)$

(iii) $\exists x P(x) \wedge \exists x Q(x) \equiv \exists x P(x) \wedge \exists y Q(y)$

(a) (i) only

(b) (ii) and (iii) only

(c) (ii) only

(d) None of these

Q.19 An A^* algorithm is a heuristic search technique which

(a) is like a depth-first search where most promising child is selected for expansion

(b) generates all successor nodes and computes an estimate of distance (cost) from start node to a goal node through each of the successors. It then chooses the successor with shortest cost.

(c) saves all path lengths (costs) from start node to all generated nodes and chooses shortest path for further expansion.

(d) None of the above

Q.20 Consider $f(N) = g(N) + h(N)$
Where function g is a measure of the cost of getting from the start node to the current node N and h is an estimate of additional cost of getting from the current node N to the goal node. Then $f(N) = h(N)$ is used in which one of the following algorithms?
(a) A^* algorithm (b) AO^* algorithm
(c) Greedy best first search algorithm
(d) Iterative A^* algorithm

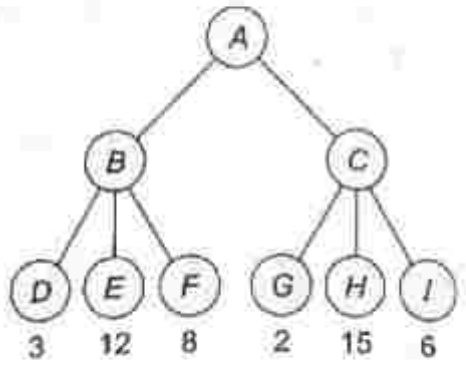
Q.21 What is the space complexity of BFS assuming branching factor b and solution depth d?
(a) $O(b)$ (b) $O(d)$
(c) $O(b^d)$ (d) $O(\log b)$

Q.22 Consider the following statements:
S_1 : In systematic sampling, every element in the population has an equal probability of being selected.
S_2 : Systematic sampling is often used when the population is arranged in some specific order.
Which of the above is true?
(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.23 _____ is used in game trees to reduce the number of branches of the search tree to be traversed without affecting the solution.
(a) Best first search (b) Goal stack planning
(c) Alpha-beta pruning procedure
(d) Min-max search

Q.24 How does randomized hill-climbing choose the next move each time?
(a) It generates a random move from the moveset, and accepts this move.
(b) It generates a random move from the whole state space, and accepts this move.
(c) It generates a random move from the moveset, and accepts this move only if this move improves the evaluation function.
(d) It generates a random move from the whole state space, and accepts this move only if this move improves the evaluation function.

Q.25 Consider the following game tree in which root is a maximizing node and children are visited left to right. What nodes will be pruned by the alpha-beta pruning?



(a) I (b) HI
(c) CHI (d) GHI

Q.26 Either everything is material or there are some things that are not material.
(a) $(\forall x)M(x) \vee (\exists x) \sim M(x)$
(b) $\forall x(M(x) \vee \sim M(x))$
(c) $(\forall x)M(x) \wedge (\exists x) \sim M(x)$
(d) $(\forall x)M(x) \vee (\exists y) \sim M(y)$

Q.27 Match **List-I** with **List-II** and choose the correct answer from the code given below:
List-I
A. Greedy Best-First Search
B. A* Search
C. Recursive Best-First Search
D. Iterative-deepening A* Search
List-II
1. Selects a node for expansion if optimal path to that node has been found.
2. Avoids substantial overhead associated with keeping the sorted queue of nodes.
3. Suffers from excessive node generation.
4. Time complexity depends on the quality of heuristic.

Codes:

	A	B	C	D
(a)	1	4	3	2
(b)	4	3	2	1
(c)	1	2	3	4
(d)	4	1	2	3

Q.28 Consider following two rules R1 and R2 in logical reasoning in Artificial Intelligence (AI):
R1: From $\alpha \supset \beta$
$\frac{\text{and } \alpha}{\text{Inter } \beta}$ is known as Modus Tollens (MT)
R2: From $\alpha \supset \beta$
$\frac{\text{and } \neg\beta}{\text{Inter } \neg\alpha}$ is known as Modus Ponens (MP)

(a) Only R1 is correct.
(b) Only R2 is correct.
(c) Both R1 and R2 are correct.
(d) Neither R1 nor R2 is correct.

Q.29 Which of the following statements are true?
(a) Minimax search is breadth-first; it processes all the nodes at a level before moving to a node in next level.
(b) The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined.
(c) The alpha-beta search algorithm computes the same optimal moves as minimax algorithm.
(d) Optimal play in games of imperfect information does not require reasoning about the current and future belief states of each player.

Q.30 In Artificial Intelligence (AI), what is present in the planning graph?
(a) Sequence of levels
(b) Literals (c) Variables
(d) Heuristic estimates

Q.31 What is the best method to go for the game playing problem?
(a) Optimal Search (b) Random Search
(c) Heuristic Search (d) Stratified Search

Q.32 If b is the branching factor and m is the maximum depth of the search tree, what is the space complexity of greedy search?
(a) $O(b + m)$ (b) $O(bm)$
(c) $O(b^m)$ (d) $O(m^b)$

Q.33 In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics $h_1 \ldots\ldots h_m$ is available for a problem and none of them dominates any of the others, which should we choose?
(a) $h(n) = \max\{h_1(n), \ldots, h_m(n)\}$
(b) $h(n) = \min\{h_1(n), \ldots, h_m(n)\}$
(c) $h(n) = \text{avg}\{h_1(n), \ldots, h_m(n)\}$
(d) $h(n) = \text{sum}\{h_1(n), \ldots, h_m(n)\}$

Q.34 Consider following sentences regarding A*, an informed search strategy in Artificial Intelligence (AI).
A. A* expands all nodes with $f(n) < C^*$.
B. A* expands no nodes with $f(n) \geq C^*$.
C. Pruning is integral to A*.
Here, C^* is the cost of the optimal solution path.
Which of the following is correct with respect to the above statements?
(a) Both statement (A) and statement (B) are true.
(b) Both statement (A) and statement (C) are true.
(c) Both statement (B) and statement (C) are true.
(d) All the statements (A), (B) and (C) are true.

Q.35 Consider the following statements:
S_1: A heuristic is admissible if it never over-estimates the cost to reach the goal.
S_2: A heuristic is monotonous if it follows triangle inequality property.
Which one of the following is true referencing the above statements?
(a) Statement S_1 is true but statement S_2 is false.
(b) Neither of the statements S_1 and S_2 are true.
(c) Both the statements S_1 and S_2 are true.
(d) Statement S_1 is false but statement S_2 is true.

Q.36 Which of the above two are equivalent?
1. $\neg\forall x(P(x))$
2. $\neg\exists x(P(x))$
3. $\neg\exists x(\neg P(x))$
4. $\exists x(\neg P(x))$

Select the correct option:
(a) 1 and 3 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

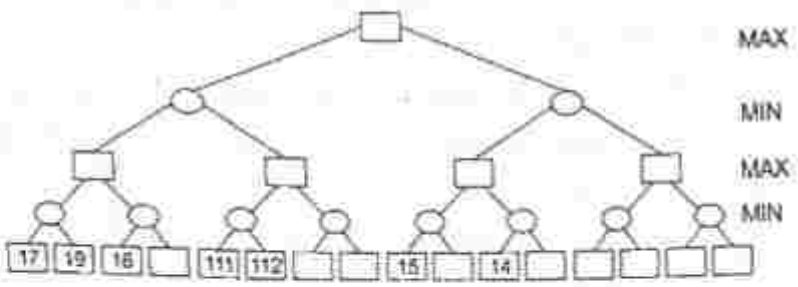
Q.37 Consider the following terminology and match **List-I** with **List-II** and choose the correct answer from the code given below:
b = Branching factor
d = Depth of the shallowest solution
m = Maximum depth of the search tree
l = Depth limit

List-I	List-II
A. BFS search	1. $O(bd)$
B. DFS search	2. $O(b^d)$
C. Depth limited search	3. $O(bm)$
D. Iterative deepening search	4. $O(bl)$

Codes:

	A	B	C	D
(a)	2	3	4	1
(b)	1	3	4	2
(c)	1	2	4	3
(d)	3	2	4	1

Q.38 Consider the game tree given below:



Here ○ and □ represent MIN and MAX nodes respectively. The value of the root node of the game tree is:

(a) 14 (b) 17
(c) 111 (d) 112

Q.39 Which of the following is NOT true in problem solving in artificial intelligence?

(a) Implements heuristic search techniques
(b) Solution steps are not explicit
(c) Knowledge is imprecise
(d) It works on or implements repetition mechanism

Q.40 Consider the following statements about Variational Inference:

S_1: It provides exact inference for tree-structured graphs.

S_2: It approximates the posterior by minimizing KL-divergence.

S_3: It is useful when scalability is a concern in large probabilistic models.

Which of the following options is correct?

(a) Only S_1 and S_2
(b) Only S_2 and S_3
(c) Only S_1 and S_3
(d) All of the above

Q.41 In a game of Go, the branching factor is exceptionally high, making traditional search algorithms inefficient. How might you adapt traditional adversarial search techniques to address the complexities of Go, and what additional strategies or algorithms can be employed to enhance the search process?

(a) Reduce the search depth and employ alpha-beta pruning.
(b) Use Monte Carlo Tree Search (MCTS) with random rollouts and UCT for move selection.
(c) Increase the search depth and utilize iterative deepening.
(d) Implement a large opening book with pre-determined moves.

Q.42 What is the worst-case time complexity of DFS in a tree of depth d and branching factor b?

(a) $O(b + d)$ (b) $O(b * d)$
(c) $O(b^d)$ (d) $O(\log b)$

Q.43 Given below are two statements:

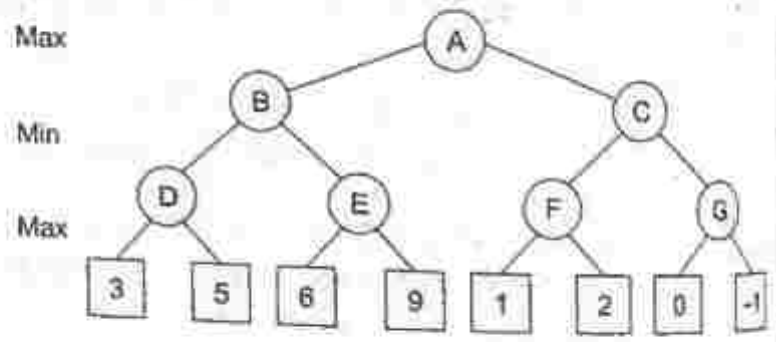
Statement-I: A genetic algorithm is a stochastic hill-climbing search in which a large population of states is maintained.

Statement-II: In non-deterministic environments, agents can apply AND-OR search to generate contingent plans that reach the goal regardless of which outcomes occur during execution.

In the light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is correct but Statement II is false
(d) Statement I is incorrect but Statement II is true

Q.44 Consider the given tree below. Calculate the value at the root of the tree using alpha-beta pruning algorithm.



(a) 3 (b) 5
(c) 6 (d) 9

Q.45 Consider the following statements:

S_1: Rejection sampling can be inefficient if the proposal distribution has a much wider range than the target distribution.

S_2: Gibbs sampling involves iterating over all variables in a multivariate distribution in a fixed sequence.

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.46 $(\exists!\, x)\, P(x)$ is equivalent to _____.

(a) $\forall x\, \exists y(P(x) \wedge [P(y) \rightarrow (x! = y)])$
(b) $\exists y\, (\forall x P(x) \wedge [P(y) \rightarrow (x = y)])$
(c) $\exists x\, (P(x) \wedge \forall y [P(y) \rightarrow (x = y)])$
(d) None of these

Q.47 In the content of Alpha Beta pruning in game trees which of the following statements are correct regarding cut off procedures?

(A) Alpha Beta pruning can eliminate subtrees with certainly when the value of a node exceeds both the alpha and beta bonds.

(B) The primarily purpose of Alpha-Beta pruning is to save computation time by searching fewer nodes in the same tree.

(C) Alpha Beta pruning guarantees the optimal solution in all cases by exploring the entire game tree.

(D) Alpha and Beta bonds are initialized to negative and positive infinity respectively at the root note.

Choose the **correct** answer from the options given below:

(a) (A), (C), (D) Only (b) (B), (C), (D) Only
(c) (A), (B), (D) Only (d) (C), (B) Only

Q.48 Match **List-I** with **List-II**:

List-I

A. Greedy Best-First Search
B. A*
C. Recursive Best-First Search
D. SMA*

List-II

I. Space complexity is $O(d)$ where d = depth of the deepest optimal solution.
II. Incomplete even if the search space is finite.
III. Optimal if optimal solution is reachable: otherwise returns the best reachable optimal solution.
IV. Computation and space complexity is too high.

Choose the correct answer from the options given below:

(a) A-II, B-IV, C-I, D-III (b) A-II, B-III, C-I, D-IV
(c) A-III, B-II, C-IV, D-I (d) A-III, B-IV, C-II, D-I

Q.49 Which among the following statement(s) is(are) FALSE?

(a) Greedy best-first search is not optimal but is often efficient.

(b) A* is complete and optimal provided $h(n)$ is admissible or consistent.

(c) Recursive best-first search is efficient in terms of time complexity but poor in terms of space complexity.

(d) $h(n) = 0$ is an admissible heuristic for the 8-puzzle.

Q.50 Match **List-I** with **List-II**:

List-I

(A) Greedy Best first search
(B) A*
(C) Recursive best first search
(D) SMA*

List-II

(I) The space complexity as $O(d)$ where d = depth of the deepest optimal solution
(II) Incomplete even if the search space is finite
(III) Optimal if optimal solution is reachable otherwise return the best reachable optimal solution
(IV) Computation and space complexity is two light

Choose the **correct** answer from the options given below:

(a) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(b) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(c) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
(d) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q.51 In a genetic algorithm optimization problem the fitness function is defined as $f(x) = x^2 - 4x + 4$. Given a population of four individuals with values of x : [1.5, 2.0, 3.0, 4.5].

What is the fitness value of the individual that will be selected as the parent for reproduction in one generation?

(a) 2.25 (b) 6.0
(c) 0.0 (d) 6.25

Q.52 What is correct translation of the following statement into mathematical logic?

"Some real number are rational"

(a) $\exists x$ (real $(x) \vee$ rational (x))
(b) $\forall x$ (real $(x) \rightarrow$ rational (x))
(c) $\exists x$ (real $(x) \wedge$ rational (x))
(d) $\exists x$ (rational $(x) \rightarrow$ real (x))

Q.53 Consider the following statements:

S_1: Mean-field approximation assumes independence among variables.

S_2: Mean-field approximation is more computationally expensive than structured methods.

S_3: Structured variational methods consider dependencies between variables.

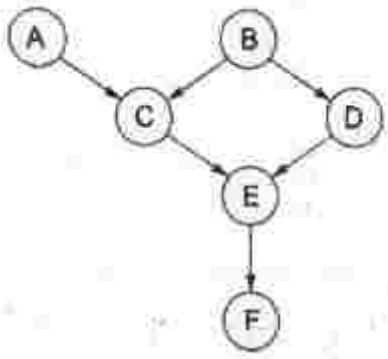
Which of the following options is correct?

(a) Only S_1 and S_2 (b) Only S_2 and S_3
(c) Only S_1 and S_3 (d) All of the above

Q.54 What is role of the importance function in importance sampling?

(a) Assign higher weights to more relevant samples
(b) Ignore less probable variables
(c) Compute probabilities deterministically
(d) Reduce iterations for exact inference

Q.55 Assume the random variables in the graph shown below are Boolean. How many parameters are needed in total to fully specify this Bayesian network?



(a) 6 (b) 8
(c) 10 (d) 5

Q.56 Which of the following is true for space complexity in A search using a consistent heuristic?

(a) A* requires linear space.
(b) A* has space complexity $O(b^d)$.
(c) A* discards nodes to save memory.
(d) A* only keeps the current path in memory.

Q.57 In a neighbourhood, 90% children were falling sick due to flu 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu.

(a) 0.83 (b) 0.43
(c) 0.74 (d) 0.95

Q.58 Exact inference guarantees correct results in:

(a) All Bayesian networks
(b) Only tree-structured Bayesian networks
(c) Networks with high-dimensional variables
(d) Loopy Bayesian networks

Q.59 The statement formula:

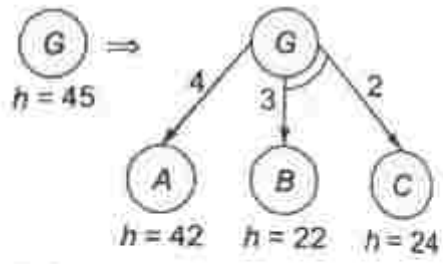
$((a \vee b) \wedge (\sim a \vee c) \wedge \sim (b \vee c))$ is _____.

(a) a tautology (b) a contradiction
(c) a contingency (d) None of these

Q.60 Which informed search algorithm is known for its ability to guarantee both completeness and optimality when provided with an admissible heuristic?

(a) A* Search
(b) Greedy Best-First Search
(c) Depth-First Search
(d) Hill Climbing

Q.61 Consider the following AO graph:



Which is the best node to expand next by AO* algorithm?

(a) A (b) B
(c) C (d) B and C

Q.62 What distinguishes Informed Search Algorithms from uninformed ones?

I. They consider all possible states.
II. They use a predefined set of actions.
III. They use a heuristic function to estimate the cost to the goal.
IV. They are always faster.

(a) I and III (b) II only
(c) III only (d) I, II and IV

Q.63 In a game of chess, if the branching factor (average number of legal moves) at a given state is 30, and the search depth is limited to 3 levels, how many terminal nodes will be evaluated using the minimax algorithm?

(a) 90 (b) 810
(c) 27,000 (d) 81,000

Q.64 Unlike A*, which guarantees optimality under certain conditions, what can be said about the optimality of Greedy Best-First Search?

(a) Greedy Best-First Search is guaranteed to find the optimal solution.
(b) Greedy Best-First Search is guaranteed to find a solution, but it may not be optimal.
(c) Greedy Best-First Search is not guaranteed to find any solution.
(d) Greedy Best-First Search is always faster than A*.

Q.65 In the context of sampling-based approximate inference, what is the sampling variance, and how does it impact the quality of the inference?

(a) Sampling variance is the spread of random samples, and high sampling variance indicates more accurate results.

(b) It is the expected value of a random sample, and it does not affect the quality of inference.

(c) Sampling variance is the square of the standard deviation of the estimates, and high sampling variance indicates less reliable results.

(d) It is a measure of the correlation between variables in a probabilistic model.

Q.66 Which one of the following is the most appropriate logical formula to represent the statement:

"Gold and silver ornaments are precious"

The following notations are used:

$G(x)$: x is a gold ornament

$S(x)$: x is a silver ornament

$P(x)$: x is precious

(a) $\forall x(P(x) \rightarrow (G(x) \wedge S(x)))$

(b) $\forall x(G(x) \wedge (S(x) \rightarrow P(x)))$

(c) $\exists x((G(x) \wedge S(x)) \rightarrow P(x))$

(d) $\forall x((G(x) \vee S(x)) \rightarrow P(x))$

Q.67 Select the concepts or methods used in approximate inference through sampling in AI.

(a) Markov Chain Monte Carlo (MCMC)

(b) Bayesian Networks

(c) Rejection Sampling

(d) Exact Inference

Q.68 In the grid-based pathfinding problem described, with the heuristic function $h(n)$ representing the Manhattan distance, consider the following scenario:

Start State (S) → Intermediate State (I) → Goal State (G)

The heuristic function values for each state are:

$$h(S) = 6,\ h(I) = 4,\ h(G) = 0$$

The actual path cost from the start state to the intermediate state is: $g(S, I) = 3$

What is the total estimated cost (f-value) for the intermediate state (I) when applying the A* search algorithm in this scenario?

(a) 6 (b) 7

(c) 8 (d) 9

Q.69 A disadvantage of approximate inference is:

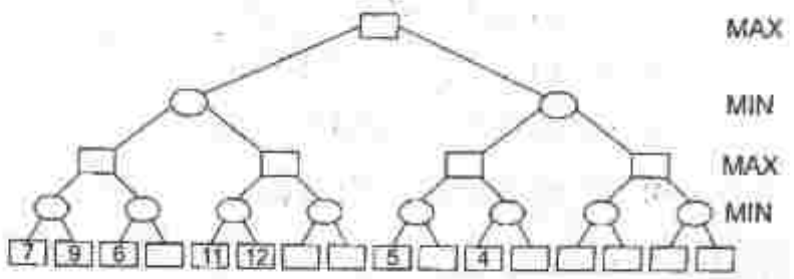
(a) It is computationally infeasible

(b) It does not guarantee accuracy in the results

(c) It ignores evidence in computations

(d) It only applies to specific network types

Q.70 Consider the game tree given below:



Here ○ and □ represents MIN and MAX nodes respectively. The value of the root node of the game tree is

(a) 4 (b) 7

(c) 11 (d) 12

Q.71 Exact inference in Bayesian networks is computationally expensive because:

(a) It ignores conditional independence properties.

(b) It requires summing over all possible variable combinations.

(c) It uses random sampling for probabilities.

(d) It only applies to tree-structured networks.

Q.72 Approximate inference is used when:

(a) Exact inference is computationally infeasible

(b) The model is deterministic

(c) Evidence is ignored to reduce complexity

(d) Exact probabilities are required

Q.73 Which of the following is a type of approximate inference?

(a) Enumeration

(b) Belief propagation in tree networks

(c) Variable elimination

(d) Gibbs sampling

Q.74 In which scenario is approximate inference is critical

(a) Computation resources are unlimited

(b) Exact results are mandatory

(c) Real-time decisions are required under uncertainty

(d) Bayesian networks have no cycles

Q.75 Consider the following predicates:
$S(x)$: "x is student"
GATE (x, y) : "x has written gate in every stream".
Find the equivalent predicate logic for the following statement:
"There does not exist a student who has written a GATE in every stream."
(a) $\exists y \exists x [S(x) \wedge \sim GATE(x, y)]$
(b) $\forall y \exists x [\sim S(x) \vee \sim GATE(x, y)]$
(c) $\exists y \forall x [\sim S(x) \vee \sim GATE(x, y)]$
(d) $\exists y \exists x [\sim S(x) \wedge \sim GATE(x, y)]$

Q.76 The primary advantage of exact inference is:
(a) It reduces the dimensionality of variables
(b) It works for cyclic networks
(c) It uses sampling techniques for faster computation
(d) It provides accurate and reliable results

Q.77 Which is the logical transaction of the following statement?
"None of my friends are perfect"
(a) $\exists x(F(x) \wedge P(x))$
(b) $\exists x(\neg F(x) \wedge P(x))$
(c) $\exists x(\neg F(x) \wedge \neg P(x))$
(d) $\neg \exists x(F(x) \wedge P(x))$

Q.78 Which of the following is an example of Best-First Search algorithm?
(a) Depth-First Search
(b) A* Search (c) Breadth-First Search
(d) Uniform Cost Search

Q.79 In Depth-First Search, which data structure is primarily used for storing nodes to be expanded?
(a) Queue (b) Stack
(c) Priority Queue (d) Set

Q.80 Under which of the following conditions does Depth First Search (DFS) guarantee completeness?
(a) DFS explores all possible paths before backtracking
(b) The solution path always exists
(c) The search tree is finite
(d) The branching factor of the tree is infinite

Q.81 A* search is an informed search algorithm because it uses:
(a) A heuristic function that estimates the cost to the goal
(b) A uniform cost function
(c) A random exploration function
(d) Depth-first search to explore the nodes

Q.82 Which one of the first order predicate calculus statements given below correctly expresses the following English statement?
Tigers and lions attack if they are hungry or threatened.
(a) $\forall x$ [(tiger $(x) \wedge$ lion $(x)) \rightarrow$ [(hungry (x) $\vee$ threatened $(x)) \rightarrow$ attacks (x)]]
(b) $\forall x$ [(tiger $(x) \vee$ lion $(x)) \rightarrow$ [(hungry (x) $\vee$ threatened $(x)) \wedge$ attacks (x)]]
(c) $\forall x$ [(tiger $(x) \vee$ lion $(x)) \rightarrow$ [(attacks (x) $\rightarrow$ (hungry $(x) \vee$ Threatened $(x))$]]
(d) $\forall x$ [(tiger $(x) \vee$ lion $(x)) \rightarrow$ [(hungry (x) $\vee$ threatened $(x)) \rightarrow$ attacks (x)]]

Q.83 Which condition is required for A Search* to guarantee an optimal solution?
(a) The tree must be finite
(b) The heuristic function must be admissible
(c) The heuristic function must be consistent
(d) The solution must be reachable from the start node

Q.84 Beam Search can be viewed as a compromise between:
(a) Depth-First Search and Breadth-First Search
(b) Depth-First Search and A* Search
(c) A* Search and Best-First Search
(d) Best-First Search and Depth-First Search

Q.85 **Statement:** Last year, the only Book I read were adventure stories.
Logical representation of the above statement is?
(a) Adventure story $\rightarrow$ book that I read last year
(b) Book that I read last year $\rightarrow$ adventure story
(c) Not a book I read last year $\rightarrow$ not adventure story
(d) None of these

Q.86 "If X, then Y unless Z" is represented by which of the following formulae in propositional logic?
(a) $(X \wedge Y) \rightarrow \neg Z$
(b) $(X \wedge \neg Z) \rightarrow Y$
(c) $(X \rightarrow (Y \wedge \neg Z)$
(d) $Y \rightarrow (X \wedge \neg Z)$

Q.87 Bidirectional Search can only be used when:
(a) The search space is large
(b) The goal is known
(c) The search space is small
(d) The graph is unidirectional

Q.88 The Minimax algorithm assumes that the opponent plays:
(a) Suboptimally
(b) Optimally
(c) Randomly
(d) With limited knowledge

Q.89 In the best-case scenario, what is the time complexity of the Mini-Max algorithm with alpha-beta pruning, assuming optimal move ordering?
(a) $O(b^d)$ (b) $O(b^{(d/2)})$
(c) $O(d^b)$ (d) $O(\log b^d)$

Q.90 In Bidirectional Search, the goal is to:
(a) Search from the goal towards the start node
(b) Perform a deep search in both directions simultaneously
(c) Use heuristics to find the optimal solution
(d) Search from the start and goal node simultaneously and meet in the middle

Q.91 The Greedy Best-First Search selects the node based on:
(a) The path cost
(b) The sum of the path cost and heuristic value
(c) The heuristic function alone
(d) The lowest depth

Q.92 In Depth-First Search, the algorithm gets stuck when:
(a) The node with the best heuristic is selected
(b) The shallowest node is expanded first
(c) The highest-cost node is expanded
(d) A loop occurs due to unhandled cycles

Q.93 Consider the following statements about the operation of Loopy Belief Propagation (LBP):
S_1: Loopy Belief Propagation uses iterative message passing between neighboring nodes.
S_2: Observed evidence cannot be handled in Loopy Belief Propagation.
S_3: The messages in Loopy Belief Propagation depend on local potentials and messages from neighboring nodes.
Which of the following options is correct?
(a) Only S_1 and S_2
(b) Only S_2 and S_3
(c) Only S_1 and S_3
(d) All of the above

Q.94 The following propositional statement is
$(P \rightarrow (Q \vee R)) \rightarrow ((P \wedge Q) \rightarrow R)$
(a) Satisfiable but not valid
(b) Valid
(c) A contradiction
(d) None of the above

Q.95 $(p \wedge q) \wedge \sim (p \vee q)$ is a negation of
(a) Tautology
(b) Fallacy
(c) Both (a) and (b)
(d) None of these

Q.96 Arrange the following encoding strategies used in Genetic Algorithms (GAs) in the correct sequence starting from the initial step and ending with the final representation of solutions:
(A) Binary encoding
(B) Real valued encoding
(C) Permutation encoding
(D) Gray coding
Choose the **correct** answer from the options given below:
(a) (D), (B), (A), (C)
(b) (B), (D), (A), (C)
(c) (C), (D), (A), (B)
(d) (B), (C), (A), (D)

Q.97 Consider the following statements regarding Loopy Belief Propagation (LBP):
S_1: Loopy Belief Propagation is guaranteed to converge on any loopy graph.
S_2: Loopy Belief Propagation gives exact marginals only when the graph is a tree.
S_3: On graphs with cycles, Loopy Belief Propagation is an approximate inference method.
Which of the following options is correct?
(a) Only S_1 and S_2
(b) Only S_2 and S_3
(c) Only S_1 and S_3
(d) All of the above

Q.98 Which of the following formulas is a formalization of the sentence:
"There is a computer which is not used by any student".

(a) $\exists x(\text{computer}(x) \wedge \forall y(\neg \text{student}(y) \rightarrow \text{uses}(y, x)))$

(b) $\exists x(\text{computer}(x) \rightarrow \forall y(\text{student}(y) \rightarrow \neg \text{uses}(y, x)))$

(c) $\exists x(\text{computer}(x) \wedge \forall y(\text{student}(y) \rightarrow \neg \text{uses}(y, x)))$

(d) $\exists x(\text{computer}(x) \vee \forall y(\text{student}(y) \rightarrow \neg \text{uses}(y, x)))$

Q.99 Which of the following statements is correct?

(a) Breadth-First Search is incomplete when the branching factor is finite.

(b) Depth-First Search is always complete for infinite-depth spaces.

(c) Depth-Limited Search is complete if depth limit depth of the shallowest goal.

(d) Uniform Cost Search is incomplete for graphs with positive edge costs.

Q.100 Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$

(a) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$

(b) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$

(c) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$

(d) $\forall x(\exists y(\neg\alpha) \vee \exists z(\neg\beta))$

Q.101 A search time complexity depends on which two factors when using an admissible heuristic?

(a) Branching factor and path cost

(b) Branching factor and depth

(c) Heuristic accuracy and branching factor

(d) Cost function and depth limit

Q.102 Given that

$A(x)$ means "x is an alligator".

$H(x)$ means "x is an Human", and

$E(x, y)$ means "x eats y".

Which of the given choices is the best English translation for the following first order logic statement?

$$\forall x\left(H(x) \rightarrow \forall y\left[E(y,x) \rightarrow A(y)\right]\right)$$

(a) All humans eat alligators.

(b) Alligators eat only humans.

(c) Every Alligators eats humans.

(d) Only alligators eat humans.

Q.103 A logical binary relation $\odot$, is defined as follows

A	B	$A \odot B$
True	True	True
True	False	True
False	True	False
False	False	True

Let ~ be the unary negation (NOT) operator, with higher precedence, than $\odot$. Which one of the following is equivalent to $A \wedge B$?

(a) $(\sim A \odot B)$ (b) $\sim (A \odot \sim B)$

(c) $\sim (\sim A \odot \sim B)$ (d) $\sim (\sim A \odot B)$

Q.104 "If GATE rank is needed, i will not write the GATE exam, if i do not join MADEEASY. GATE rank is needed and i will join MADEEASY. Therefore i will write the GATE exam."
Identify the above inference?

(a) Valid

(b) Invalid

(c) One argument is missed

(d) More than one argument is missed.

Q.105 Which of the following correctly represents the time complexity of Breadth-First Search (BFS)?

(a) $O(b^d)$ (b) $O(d^b)$

(c) $O(b * d)$ (d) $O(d^2)$

Q.106 Consider the following two statements:

S_1: If a candidate is known to be corrupt, then he will not be elected.

S_2: If a candidate is kind, he will be elected

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

(a) If a person is known to be corrupt, he is kind

(b) If a person is not known to be corrupt, he is not kind

(c) If a person is kind, he is not known to be corrupt

(d) If a person is not kind, he is not known to be corrupt

Q.107 Consider the statement, "If you will give me a cow, then I will give you magic beans." Then which of the following statement does not represent the contrapositive statement?

(a) If you will give me a cow, then I will not give you magic beans.

(b) If I will not give you magic beans, then you will not give me a cow.

(c) If I will give you magic beans, then you will give me a cow.
(d) If you will not give me a cow, then I will not give you magic beans.

Q.108 Which of the following is tautology?
(a) $(\neg P \wedge (P \rightarrow q)) \rightarrow \neg q$
(b) $\neg(P \rightarrow q) \rightarrow \neg q$
(c) $[(\neg P \wedge q) \wedge [q \rightarrow (P \rightarrow q)]] \rightarrow \neg r$
(d) Both (b) and (c)

Q.109 Which of the following is correct?
(a) BFS can use backtracking
(b) DFS can use backtracking
(c) Both (a) and (b)
(d) None of these

Q.110 What is the predicate logic for the following statement?
There are atmost two males in the class.
(a) $\forall x \forall y((\text{Male}(x) \wedge \text{Male}(y)) \rightarrow (x = y \vee y = x))$
(b) $\exists x \exists y\ (\text{Male}(x) \wedge \text{Male}(y) \wedge x \neq y \wedge \forall z(\text{Male}(z) \rightarrow (z = x \vee z = y)))$
(c) $\forall x \forall y \forall z\ ((\text{Male}(x) \wedge \text{Male}(y) \wedge \text{Male}(z)) \rightarrow (x = y \vee x = z \vee y = z))$
(d) None of these

Q.111 Which of the following algorithm(s) is/are complete?
(a) A* algorithm
(b) Breadth-First Search (BFS)
(c) Uniform Cost Search (UCS)
(d) Greedy Best-First Search

Q.112 Which of the following are valid?
(a) $\exists x\ \exists y\ P(x, y) \rightarrow \exists y\ \exists x\ P(x, y)$
(b) $\forall x\ \exists y\ Q(x, y) \rightarrow \exists y\ \forall x\ Q(x, y)$
(c) $\exists x\ \forall y\ P(x, y) \rightarrow \forall y\ \exists x\ R(x, y)$
(d) Not $[\exists x\ S(x)]$ IFF $\forall x$ NOT $[S(x)]$

Q.113 Consider the following statements:
S_1: DFS can use less memory than BFS.
S_2: BFS can use less memory than DFS.
S_3: DFS maintains all nodes at the current level in memory.
Which of the following options is incorrect?
(a) Only S_1 and S_2
(b) Only S_1 and S_3
(c) Only S_2 and S_3
(d) All of the above

Q.114 Which of the following is a type of Approximate inference?
(a) Junction method
(b) Loopy Belief Propagation
(c) Monte Carlo Methods
(d) None of these

Q.115 Let p and q be two propositions. Consider the following two formulae in prepositional logic.
$S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$
$S_2 : q \rightarrow (\neg p \wedge (p \vee q))$
Which one of the following choices is correct?
(a) Neither S_1 nor S_2 is a tautology.
(b) Both S_1 and S_2 are tautologies.
(c) S_1 is a tautology but S_2 is not a tautology.
(d) S_1 is not a tautology but S_2 is a tautology.

Q.116 Which of the following is correct?
(a) Greedy Best-First Search uses a stack to hold nodes to explore, sorted by heuristic value
(b) Greedy Best-First Search uses a priority queue to hold nodes to explore, sorted by heuristic value
(c) Greedy Best-First Search uses a FIFO queue to hold nodes to explore, sorted by heuristic value
(d) None of these

Q.117 Consider the following expressions:
(i) False (ii) Q
(iii) True (iv) $P \vee Q$
(v) $\neg Q \vee P$
The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____.

Q.118 Which data structure is used to implement BFS?
(a) Stack
(b) Queue
(c) Priority queue
(d) Hash Table

Q.119 Which statement is true about the heuristic used in Greedy Best-First Search?
(a) It must be admissible and consistent to guarantee optimality.
(b) It only considers the cost already incurred.
(c) It is solely based on the estimated cost to reach the goal from the current node.
(d) It is the sum of path cost and heuristic estimate.

Q.120 Consider the following statements:

S_1: Loopy Belief Propagation is widely used because it provides good approximate results in practice.

S_2: The message update rules in Loopy Belief Propagation are different from those in standard belief propagation.

S_3: Loopy Belief Propagation is useful when exact inference methods are computationally infeasible.

Which of the following options is correct?

(a) Only S_1 and S_2 (b) Only S_2 and S_3

(c) Only S_1 and S_3 (d) All of the above

Q.121 Consider the following logical inferences:

I_1 : If it rains then the cricket match will not be played.

The cricket match was played.

Inference: There was no rain.

I_2 : If it rains then the cricket match will not be played.

It did not rain.

Inference: The cricket match was played.

Which of the following is TRUE?

(a) Both I_1 and I_2 are correct inferences

(b) I_1 is correct but I_2 is not a correct inference

(c) I_1 is not correct but I_2 is a correct inference

(d) Both I_1 and I_2 are not correct inferences

Q.122 The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$ II. $q \Rightarrow p$

III. $(\neg q) \vee p$ IV. $(\neg p) \vee q$

(a) I only (b) I and IV only

(c) II only (d) II and III only

Q.123 Consider the following statements:

S_1: The minimax algorithm explores the complete game tree using depth-first search

S_2: The minimax algorithm explores the complete game tree using best-first search.

Which of the following is true?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.124 Which of the following is a disadvantage of the Junction Tree Algorithm?

(a) It cannot handle observed evidence.

(b) It becomes computationally expensive on large or dense graphs.

(c) It is based on sampling, so results are not reliable.

(d) It gives only approximate results.

Q.125 Which one of the following is not equivalent to $p \leftrightarrow q$?

(a) $(\neg p \vee q) \wedge (p \vee \neg q)$

(b) $(\neg p \vee q) \wedge (q \rightarrow p)$

(c) $(\neg p \wedge q) \vee (p \wedge \neg q)$

(d) $(\neg p \wedge \neg q) \vee (p \wedge q)$

Q.126 In Greedy Best-First Search, the node to be expanded next is selected based on:

(a) The lowest cumulative path cost from the start node.

(b) The lowest heuristic estimate of distance to the goal.

(c) The highest path cost from the start node

(d) Random selection among all nodes.

Q.127 Which of the following is/are true?

(a) $\forall x\,(P(x) \vee y)$ is equivalent to $(\forall x\,P(x) \vee y)$

(b) $(\forall x\,(P(x) \vee \exists y\,P(y)))$ is equivalent to $\exists x\,P(x)$

(c) $\forall x\,(P(x) \vee y)$ is not equivalent to $(\forall x\,P(x) \wedge y)$

(d) $\forall x\,(P(x) \wedge y)$ is not equivalent to $(\forall x\,P(x) \wedge y)$

Q.128 Which of the following statements about Uniform Cost Search (UCS) are true?

(a) UCS always finds the least cost path to the goal if all step costs are strictly positive.

(b) UCS guarantees to find a solution if one exists, regardless of step costs.

(c) UCS may fail to find a solution if the step costs include zero or negative values.

(d) UCS is guaranteed to find the shortest path in terms of number of steps rather than cost.

Q.129 Which of the following is correct?

(a) Uniform Cost Search explores nodes by expanding the node with the highest cumulative cost first.

(b) Uniform Cost Search explores nodes randomly to avoid bias in the path.

(c) Uniform Cost Search always expands the least costly node first, progressing towards the goal.

(d) Uniform Cost Search may not find the optimal path even if the goal is reached.

Q.130 The space complexity of DFS in the worst case is:

(a) $O(b^d)$ (b) $O(b * d)$
(c) $O(d)$ (d) $O(b)$

Q.131 Which of the following is correct about the heuristic function in Greedy Best-First Search?

(a) It estimates the cost from the start node to the current node.
(b) It estimates the cost from the current node to the goal node.
(c) It guarantees finding the optimal path.
(d) It ignores the goal node distance completely.

Q.132 Consider the following statement over the domain of natural number.

"No prime number except 7 is divisible by 7"

Find the equivalent predicate logic for the above statement.

(a) $\forall x \in N\ [x \neq 7 \wedge \text{Prime}\ (x) \rightarrow (\neg\ \text{divisible by}\ 7\ (x))]$
(b) $\neg\ \exists x \in N\ [x \neq 7 \wedge \text{Prime}\ (x) \wedge \text{divisible by}\ 7\ (x)]$
(c) Both (a) and (b) (d) None of these

Q.133 Which search algorithm has the best space efficiency among the options below?

(a) BFS (b) UCS
(c) DFS (d) A*

Q.134 Consider the following statements:

S_1: The minimax algorithm without alpha-beta pruning always selects the optimal move for the current player.
S_2: When using alpha-beta pruning, ordering the moves from best to worst helps maximize the number of pruned branches.

Which of the following is correct?

(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are true
(d) Neither S_1 nor S_2 is true

Q.135 The CORRECT formula for the sentence, "not all rainy days are cold" is

(a) $\forall d(\text{Rainy}\ (d) \wedge \sim \text{Cold}\ (d))$
(b) $\forall d(\sim \text{Rainy}\ (d) \rightarrow \text{Cold}\ (d))$
(c) $\exists d(\sim \text{Rainy}\ (d) \rightarrow \text{Cold}\ (d))$
(d) $\exists d(\text{Rainy}\ (d) \wedge \sim \text{Cold}\ (d))$

Q.136 What is the time complexity of Depth-Limited Search (DLS) with depth limit l and branching factor b?

(a) $O(b^l)$ (b) $O(b^d)$
(c) $O(l^b)$ (d) $O(b * l)$

Q.137 Which of the following algorithms is guaranteed to find the optimal solution but may require exponential space?

(a) Depth-First Search
(b) Depth-Limited Search
(c) Uniform Cost Search
(d) Random Search

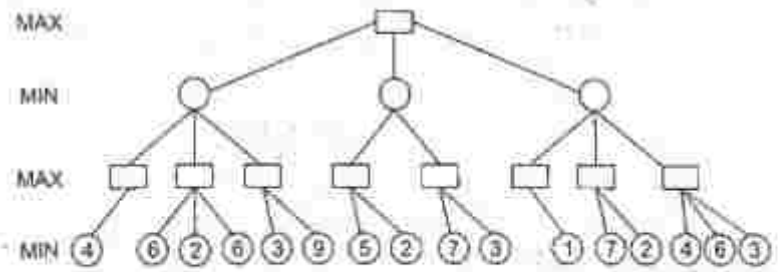
Q.138 Consider the following statements:

S_1: Rejection sampling involves accepting samples from a proposal distribution based on their likelihood of matching the target distribution.
S_2: Gibbs sampling is an MCMC method that samples from conditional distributions iteratively to approximate a multivariate distribution.

Which of the following statements is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.139 Consider the following minimax game tree search:



What will be the value propagated at the root?

(a) 3 (b) 6
(c) 4 (d) 5

Q.140 Which of the following statements is/are true?

(a) BFS is more memory-efficient than UCS.
(b) UCS finds the shortest path with non-negative edge costs.
(c) BFS uses heuristics to guide its search.
(d) DFS is complete only if the search space is finite and the goal is reachable.

Q.141 Which of the following is the negation of

$$[(\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, \gamma))]$$

(a) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \gamma))]$
(b) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg\gamma))]$

(c) $[\forall x, \neg\alpha \rightarrow (\exists y, \neg\beta \rightarrow (\forall u, \exists v, \neg\gamma))]$

(d) $[\forall x, \alpha \wedge (\exists y, \beta \wedge (\exists u, \forall v, \neg\gamma))]$

Q.142 Consider the following statements is/are correct?

(a) UCS expands nodes with the lowest cumulative cost first.

(b) The branching factor in IDA* determines the number of successors generated at each level of the search tree.

(c) Pruning (alpha-beta) reduces nodes by eliminating irrelevant branches in Min-Max search.

(d) A* uses less space than IDDFS due to its efficient node storage in open and closed lists.

Q.143 Consider a 3-puzzle where, like in the usual 8-puzzle game, a tile can only move to an adjacent empty space. Given the initial state

1	2
	3

, which of the following state cannot be reached?

(a)
3	1
	2

(b)
	3
2	1

(c)
1	3
	2

(d)
	2
1	3

Q.144 What is the time complexity of Uniform Cost Search (UCS) when the optimal solution is at depth d and the cost of every action is at least ε?

(a) $O(b^d)$ (b) $O\left(b^{\left(\frac{C}{\varepsilon}\right)}\right)$

(c) $O(d * b)$ (d) $O(C^b)$

Q.145 Match List-I with List-II:

List-I

A. Greedy best-first

B. Lowest cost-first

C. A* Algorithm

List-II

1. Minimal cost $(p) + h(p)$
2. Minimal $h(p)$
3. Minimal cost (p)

Choose the correct option from those given below:

Codes:

	A	B	C
(a)	1	2	3
(b)	3	2	1
(c)	1	3	2
(d)	2	3	1

Q.146 Which of the following is/are tautology.

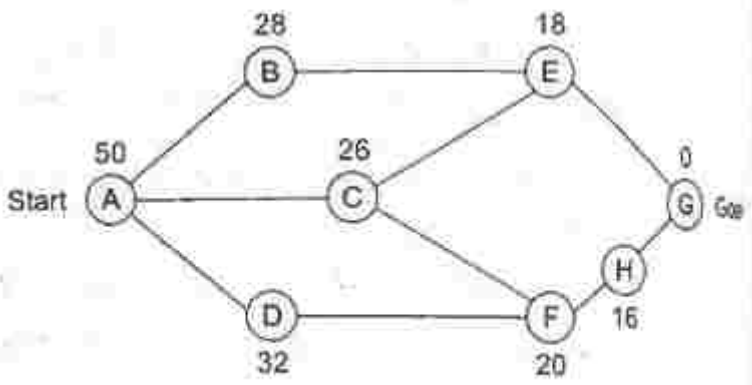
(a) $(a \vee b) \rightarrow (b \wedge c)$

(b) $(a \wedge b) \rightarrow (b \vee c)$

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

Q.147 Consider the following graph use the best first search algorithm to find the path from node A to node G (Assume node A is start node and node G is goal node)



Which of the following path sequence is correct?

(a) A → D → C → G

(b) A → C → E → G

(c) A → C → F → H → G

(d) A → D → F → G

Q.148 Consider the following statements:

S_1: The space complexity of min-max algorithm $O(bm)$.

S_2: The space complexity of min-max algorithm $O(\log m)$.

Which of the following is true?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.149 Consider a Bayesian network with three binary variables A, B and C. The join probability distribution $P(A, B, C)$ is given in the table below

A	B	C	P(A, B, C)
0	0	0	0.4
0	0	1	0.6
0	1	0	0.4
0	1	1	0.4
1	0	0	0.4
1	0	1	0.6
1	1	0	0.4
1	1	1	0.4

Using variable elimination, calculate the conditional probability $P(A = 1 \mid B = 1)$.

Q.150 Consider a Bayesian network representing climates patterns, there are two variables: *W*(weather) and *R*(rain). The conditional probability tables are follows:

W (weather)	Probability *P*(*W*)
Sunny	0.6
Cloudy	0.4

Rain (*R*)	*W* (weather)	Probability (*R* = 1|*W*)
1	Sunny	0.3
1	Cloudy	0.7

Compute the marginal probability *P*(*R* = 1) _____.

Q.151 Consider the following statements:

S_1 : If the heuristic $h(n)$ overestimates the actual cost to the goal, then A* algorithm will still be optimal.

S_2 : IDDFS guarantees the optimal solution for all types of problems.

S_3 : IDDFS performs better than DFS in the term of space complexity.

Which of the following is false?

(a) Only S_1 (b) Only S_1 and S_2

(c) Only S_1 and S_2 (d) Only S_2 and S_3

■■■■

Answers Artificial Intelligence

1.	(c)	2.	(a)	3.	(b)	4.	(b)	5.	(a)	6.	(b)	7.	(a)
8.	(a)	9.	(c)	10.	(b)	11.	(d)	12.	(c)	13.	(b)	14.	(b)
15.	(d)	16.	(b)	17.	(b)	18.	(b)	19.	(b)	20.	(c)	21.	(c)
22.	(b)	23.	(c)	24.	(c)	25.	(b)	26.	(a)	27.	(a)	28.	(c)
29.	(b, c)	30.	(a)	31.	(c)	32.	(c)	33.	(a)	34.	(b)	35.	(c)
36.	(b)	37.	(a)	38.	(b)	39.	(d)	40.	(b)	41.	(b)	42.	(c)
43.	(a)	44.	(b)	45.	(c)	46.	(c)	47.	(c)	48.	(a)	49.	(c)
50.	(a)	51.	(d)	52.	(c)	53.	(c)	54.	(a)	55.	(c)	56.	(b)
57.	(b)	58.	(b)	59.	(b)	60.	(a)	61.	(a)	62.	(c)	63.	(c)
64.	(b)	65.	(c)	66.	(d)	67.	(a, c)	68.	(b)	69.	(b)	70.	(d)
71.	(b)	72.	(a)	73.	(d)	74.	(c)	75.	(c)	76.	(d)	77.	(d)
78.	(b)	79.	(b)	80.	(c)	81.	(a)	82.	(d)	83.	(b)	84.	(d)
85.	(b)	86.	(b)	87.	(b)	88.	(b)	89.	(b)	90.	(d)	91.	(c)
92.	(d)	93.	(c)	94.	(a)	95.	(a)	96.	(c)	97.	(b)	98.	(c)
99.	(c)	100.	(a)	101.	(c)	102.	(d)	103.	(d)	104.	(b)	105.	(a)
106.	(c)	107.	(a, c, d)	108.	(b)	109.	(b)	110.	(c)	111.	(a, b, c)	112.	(a, c, d)
113.	(c)	114.	(b, c)	115.	(c)	116.	(b)	117.	(4)	118.	(b)	119.	(c)
120.	(c)	121.	(b)	122.	(d)	123.	(a)	124.	(b)	125.	(c)	126.	(b)
127.	(a, c)	128.	(a, b)	129.	(c)	130.	(c)	131.	(b)	132.	(c)	133.	(c)
134.	(c)	135.	(d)	136.	(a)	137.	(c)	138.	(b)	139.	(d)	140.	(b, d)
141.	(d)	142.	(a, b, c)	143.	(c)	144.	(b)	145.	(d)	146.	(b)	147.	(b)
148.	(a)	149.	(0.5)	150.	(0.46)	151.	(c)						

Explanations Artificial Intelligence

1. (c)

- If the **heuristic function** h is admissible, meaning that it **never overestimates** the actual minimal cost of reaching the goal, then A^* is itself admissible — (or optimal) if we do not use a closed set. If a closed set is used, then h must also be monotonic (or consistent) for A^* to be optimal.
- If we can guarantee that h' **never overestimates** h. In that case the A^* algorithms is guaranteed to find an optimal path to a goal, if one exists.

2. (a)

Depth-limited search explores all nodes up to depth l.

- If $l = d$, the search reaches the goal depth exactly.
- Since the goal is unique and at depth d, it is found and the path is shortest - optimal.
- If $l < d$, the goal is beyond search depth not found.
- If $l > d$, the search explores beyond the goal, so it will still find the goal (the option saying failure is incorrect).

Hence, option (a) is correct.

3. (b)

Exact inference in probabilistic reasoning refers to the process of computing the exact posterior probabilities of variables given some observed evidence.

It relies on methods like enumeration, variable elimination, or belief propagation.

This approach ensures mathematically accurate results by fully considering all relevant dependencies and evidence.

However, it can be computationally expensive for large networks.

4. (b)

$$\neg(X \wedge Y) \rightarrow (\neg X \vee (\neg X \vee Y))$$
$$= \neg(X \wedge Y) \rightarrow (\neg X \vee Y)$$
$$= (X \wedge Y) \vee (\neg X \vee Y)$$
$$= \underline{(X \wedge Y) \vee \neg X} \vee Y$$
$$= (X \vee \neg X) \wedge (Y \vee \neg X) \vee Y$$
$$= T \wedge (Y \vee \neg X) \vee Y$$
$$= Y \vee \neg X = \neg X \vee Y$$

So option (b) is correct.

5. (a)

If we want to find a path involving the fewest number of steps then we set the cost of going from a node to its successor (i.e. g) as a constant usually 1.

6. (b)

Alpha-Beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the minimum algorithm in its search tree.

7. (a)

Given, $P(S) = 0.005$

$P(\text{Not } S) = 1 - 0.005 = 0.995$

$P(C \mid S) = 0.95$

$P(C \mid \text{Not } S) = 0.10$

By using Baye's theorem

$$P(S \mid C) = \frac{P(C \mid S) * P(S)}{P(C)} \quad \ldots(i)$$

By using total probability theorem,

$$P(C) = P(C \mid S) \times P(S) + P(C \mid \text{Not } S) \times P(\text{Not } S)$$
$$P(C) = 0.95 \times 0.005 + 0.10 \times 0.995$$
$$P(C) = 0.00475 + 0.0995 = 0.10425$$

Put the value of P(C) in equation (i)

$$P(S \mid C) = \frac{0.95 \times 0.005}{0.10425} = \frac{0.00475}{0.10425} = 0.0455$$

9. (c)

Technically, the A^* algorithm should be called simply A if the heuristic is an underestimate of the actual cost. However it will continued to be called A^* because the implementation is the same and the game programming community does not distinguish A from A^*.

10. (b)

$(a \leftrightarrow c) \rightarrow (\neg b \rightarrow (a \wedge c))$

$(a'c' + ac) \rightarrow (b + (ac))$

$(a'c + ac') + (b + ac)$

$a(c + c') + a'c + b$

$a + a'c + b$

$a + c + b$

So not tautology but contingency.

11. (d)

- Uniform-cost search may expand more nodes than depth-first search, which can quickly go deep. So, S_1 is incorrect.
- Depth-first search does not always expand more nodes than breadth-first search; it depends on the problem. So, S_2 is incorrect.
- A* may expand the same number of nodes as uniform-cost search if the heuristic is zero, so it does not always expand fewer nodes. So, S_3 is incorrect.

12. (c)

- The **means-ends analysis** process centres around the detection of differences between current state and the goal state. Once such a difference is isolated, an operator that can reduce the difference must be found. But perhaps that operator can not be applied to the current state. So, we set up a subproblem of getting to a state in which it can be applied. The kind of backward chaining in which operators are selected and subgoals are setup to establish the preconditions of the operator is called operator subgoaling.
- **Operator subgoaling** is a computer problem solving method in which the inability of the computer to take the desired next step at any point in the problem solving process leads to a subgoal of making that step feasible.

13. (b)

Depth-First Search explores a path as deep as possible before backtracking.
It may miss shallow solutions and is not guaranteed to find the optimal one.
However, it is memory-efficient, using space proportional to the depth of the tree.

14. (b)

Two variables are used during alpha-beta pruning. Alpha is the lower bound encountered so far, and beta is the upper bound. At maximizing levels, only beta is used to cut off the search and at minimizing levels only alpha is used.

15. (d)

Hill climbing is a mathematical optimization technique which belongs to the family of local search. It is an informed search.

A* search is an informed search algorithm used for path-finding and graph traversal.
Informed methods add domain – specific information to select the best path to continue searching along.
Depth first search is not informed search algorithm.

16. (b)

Constraint satisfaction problems are mathematical problems defined as a set of objects whose state must satisfy a number of constraints or limitations. Examples of simple problems that can be modeled as a constant satisfaction - problem include :

- Eight queens puzzle
- Map coloring problem
- Crosswords, Sudoku, other logic puzzles.

17. (b)

Depth-Limited Search (DLS) only explores paths up to a fixed depth limit *f*.
If the goal node is at depth d and $f < d$, DLS does not expand nodes beyond level *f*.
Therefore, it cannot reach or find the goal node that lies deeper than the limit.

18. (b)

III is true since once a variable is bound to a qualifier it's name does not matter.
So $\exists x Q(x)$ is same $\exists y Q(y)$ and so II is true since LHS and RHS is same.
I is false, since
LHS: some value of x satisfies both P and Q
RHS: some values satisfies P and some value satisfies Q, but these 2 values need not be same.
II is true, since
If the same value satisfies both P and Q surely some value satisfies P and some values satisfies Q.
In other words LHS of implication is stronger than RHS and hence implication will be true.

19. (b)

At each node along a path to the goal, the **A* algorithm** generates all successor nodes and computes an estimate of the distance (cost) from the start node to a goal node through each of the successors. It then chooses the successor with the shortest estimated distance for expansion. The

successors for this node are then generated, their distances estimated and the process continues until goal is found or the search ends in failure.
Option (c) defines **branch and bound search**.
Option (a) defines **Hill climbing** search.

20. (c)

- **Greedy BFS:** Using a greedy algorithm, expand the first successor of the parent. After a successor is generated. If the successor's heuristic is better than its parent, the successor is set at the front of the queue (with the parent reinserted directly behind it), and the loop restarts.
- **Best first search:** Use an evaluation function $f(n)$ for each node. $f(n)$ provides an estimates for the total cost. Expand the node n with smallest $f(n)$. Implementation is order the nodes in fringe increasing order of cost. Using A^* search.
- Evaluation function $f(n) = g(n) + h(n)$
- Uniform cost search has $f(n) = g(n)$
- Best first search has $f(n) = h(n)$

21. (c)

BFS stores all nodes at the current frontier, leading to $O(b^d)$ space complexity.

22. (b)

- Systematic sampling does not give each element an equal chance of selection due to the fixed periodic interval. So, S_1 is incorrect.
- Systematic sampling is often used when the population has a specific order or structure. So, S_2 is correct.

23. (c)

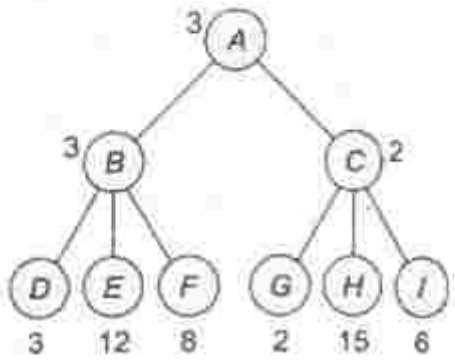
- **Alpha-Beta pruning** is a search algorithm that seeks to decrease the number of nodes that are evaluated by **minmax algorithm** in its search tree. It is an adversarial search algorithm used commonly for machine playing of two-player games.
- Avoiding searching a part of a tree is called pruning is an example of **Alpha-Beta** pruning.

24. (c)

It generates a random move from the moveset, and accepts this move only if this move improves the evaluation function.

25. (b)

Alpha-beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the **minimax algorithm** in its search tree.



Since, here we have to maximize A and minimize B and C (as per min-max tree) min of B is 3 and min of C is 2 for sure. So there is no need of further expanding H and I as whatever value. So, H and I are pruned.

26. (a)

"Either everything is material or there are somethings that are not material"
Either-OR means XOR.
We will define P as $\forall x$ (material(x))
q as $\exists x$ ($\neg$ material (x))
Either P or $q \equiv (\neg P \wedge q) \vee (P \wedge \neg q)$
$\Rightarrow$ $(\neg \forall x$ (material (x)) $\wedge \exists x$ ($\neg$ material (x)))$\vee$
$(\forall x$(material (x)) $\wedge \neg \exists x(\neg$ material (x)))
$\Rightarrow$ $(\exists x(\neg$ material (x)) $\wedge \exists x(\neg$ material (x)))
$(\forall x$(material (x)) $\wedge \forall x$(material (x)))
$\Rightarrow$ $(\exists x(\neg$ material (x)) $\vee (\forall x$ (material (x)))
Hence, option (a) is correct.

28. (c)

According to **Modus Tollens**:
if $\alpha \supset \beta$ and α then Inter β.
According to **Modus Ponens**:
if $\alpha \supset \beta$ and $\neg\beta$ then Inter $\neg\alpha$.

29. (b, c)

- Minimax is a depth-first search, not breadth-first. It recursively goes down to terminal states to evaluate utility and then backs up the values. So, option (a) is incorrect.
  Correct node ordering (e.g., best moves first) can lead to maximum pruning.

- In the best case, alpha-beta pruning reduces time complexity from $O(b^d)$ to $O(b^{d/2})$. So, option (b) is correct.
  Alpha-beta is an optimization of minimax.
- It prunes branches that cannot affect the final decision, but the result optimal move) remains the same as minimax. So, option (c) is correct.
- In imperfect information games (like poker), players must reason about belief states i.e., what they believe the opponent knows or might do.
  Optimal strategies often include randomization and probabilistic reasoning based on partial knowledge. So, option (d) is incorrect.

30. (a)

A planning graph consists of sequence of levels corresponding to time steps.

31. (c)

Heuristic approach is used in game playing problem as it will find out brute force computation, looking at hundreds of thousands of position.

32. (c)

If b is branching factor and m is depth of the search tree. Space complexity will be $O(b^m)$ for greedy search.

33. (a)

In heuristic search algorithm in Artificial Intelligence (AI), if a collection of admissible heuristic $h_1, h_2 \ldots\ldots h_m$ is available for a problem and none of them dominates any of other then $h(n) = \max\{h_1(n), h_2(n), h_m(n)\}$.

35. (c)

- A heuristic function also called heuristic is a function that ranks alternative in search algorithm at each branching step based on available information to decide which branch to follows.
- A heuristic is said to be admissible if it never overestimates the cost of reaching the goal. In other words the cost if estimates to reach goal is not higher than the lowest possible cost from the around point in path.
- The triangle inequality states that for any triangle the sum of lengths of two sides must be greater or equal to third side.

So both are correct.

36. (b)

According to negation property of universal qualifier and existential quantifier.

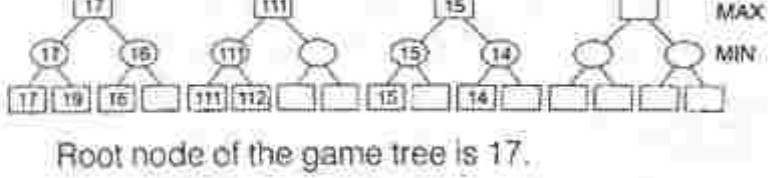
$$\neg\forall x \in \chi P(x) \equiv \exists x \in \chi \neg P(x)$$

So, option (b) is correct.

37. (a)

BFS search $\rightarrow O(b^d)$ worst case complexity
DFS search $\rightarrow O(bm)$ worst case space complexity
Depth-limited search $\rightarrow O(bl)$
Iterative deepening search $\rightarrow O(bd)$

38. (b)



Root node of the game tree is 17.

40. (b)

- Variational Inference does not produce exact results, even on trees. So, S_1 is incorrect.
- VI formulates inference as KL-divergence minimization. So, S_2 is correct.
- It is often used in large-scale models like topic models or Bayesian neural networks due to efficiency. So, S_3 is correct.

41. (b)

Go is a complex board game with a high branching factor, making traditional search algorithms inefficient. Monte Carlo Tree Search (MCTS) is a more suitable approach for Go. It uses random rollouts to explore the game tree and the UCT (Upper Confidence Bound for Trees) algorithm for move selection. MCTS is well-suited for games like Go due to its ability to handle high branching factors and the absence of a clear evaluation function.

Other options:

(a) Reducing the search depth and employing alpha-beta pruning: This option is not specific to addressing Go's complexities.

(c) Increasing the search depth and utilizing iterative deepening: Increasing search depth may not be a practical solution due to Go's high branching factor.

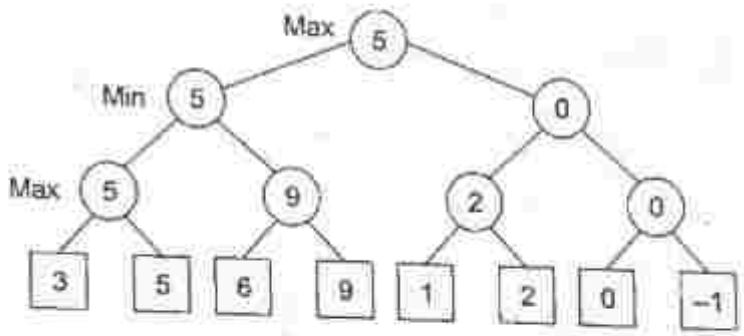
(d) Implementing a large opening book with pre-determined moves: While an opening book can be useful, it's not the primary technique for addressing Go's complexities.

42. (c)

In the worst case, DFS visits all nodes down to depth d, resulting in $O(b^d)$ time complexity.

44. (b)

Use min-max algorithm, instead of alpha-beta pruning in small graph, to save time.



45. (c)

- Rejection sampling can be inefficient if the proposal distribution is too broad compared to the target distribution, leading to many rejections. So, S_1 is correct.
- Gibbs sampling involves iterating over variables in a fixed sequence, sampling from their conditional distributions to approximate the joint distribution. So, S_2 is correct.

47. (c)

Alpha-Beta pruning does not find the optimal solution but achieves this by avoiding the need to explore the entire tree, minimizing the number of nodes evaluated and speeding up the search process.

48. (a)

A. Greedy Best-First Search: Incomplete even if the search space is finite.

B. A* : Computation and space complexity is too high.

C. Recursive Best-First Search: Space complexity is $O(d)$ where d = depth of the deepest optimal solution

D. SMA* : Optimal if optimal solution is reachable otherwise returns the best reachable optimal solution.

49. (c)

- Greedy best-first search selects nodes based on $h(n)$ (heuristic only). It is not guaranteed to be optimal, but can be fast in practice. So, option (a) is correct.
- If $h(n)$ is admissible (never overestimates) then A* is optimal.
  If $h(n)$ is consistent, A* is also complete and optimal. So, option (b) is correct.
- Recursive Best-First Search (RBFS) is Space-efficient (uses linear space like DFS), not as time-efficient (may revisit states) So, option (c) is incorrect.
- An admissible heuristic never overestimates the cost to reach the goal.
  $h(n) = 0$ is trivially admissible, though not informative. So, option (d) is correct.

51. (d)

Given, fitness function $f(x) = x^2 - 4x + 4$

Population with values x : {1.5, 2.0, 3.0, 4.5}

Fitness value for individual value of x

$$f(x) = x^2 - 4x + 4$$
$$f(1.5) = (1.5)^2 - 4 \times (1.5) + 4$$
$$= 2.25 - 6 + 4$$
$$= 6.25 - 6 = 0.25$$
$$f(2.0) = (2)^2 - 4 \times (2) + 4$$
$$= 4 - 8 + 4 = 0$$
$$f(3.0) = (3)^2 - 4(3) + 4 = 9 - 12 + 4 = 1$$
$$f(4.5) = (4.5)^2 - 4(4.5) + 4$$
$$= 24.25 - 18 = 6.25$$

Fitness value {0.25, 0, 1, 6.25}

Highest fitness value for individual at $x = 4.5$ with fitness value of 6.25.

52. (c)

Option (c) is correct, other are incorrect because reason is

(a) "There exist some numbers which are either real or rational".

(b) "All real numbers are rational".

(c) "There exist some numbers which are both real and rational".

(d) "There exist some numbers for which rational implies real".

53. (c)

- Mean-field approximation assumes all variables are independent, simplifying calculations by factoring the joint distribution. This reduces complexity but may lower accuracy when dependencies exist. So, S_1 is correct.
- Mean-field is computationally simpler and faster than structured methods because it ignores dependencies, whereas structured methods involve more complex calculations. So, S_2 is incorrect.
- Structured variational methods model dependencies between variables, providing better accuracy but requiring more computation than mean-field. So, S_3 is correct.

54. (a)

In importance sampling, the importance function helps focus sampling on the more relevant parts of the probability distribution by assigning higher weights to important samples. This improves the efficiency and accuracy of the estimation compared to uniform sampling. It does not ignore variables or compute probabilities deterministically.

55. (c)

- Node A has no parents, thus we need 1 parameter (P(A)).
- Node B is also a root node with no parents, so it requires 1 parameter (P(B)).
- Node C has two parents (A and B). Since both are Boolean, there are 4 possible combinations of the parents' values (True/True, True/False, False/True, False/False), but we only need to specify 3 parameters (P(C | A = True, B = True), P(C | A = True, B = False), P(C | A = False, B = True)), as the fourth can be calculated knowing that the probabilities must sum up to 1.
- Node D has one parent (B). With one Boolean parent, we need 1 parameter (P(D | B = True)) since the probability P(D | B = False) can be inferred.
- Node E has two parents (C and D). Like with node C, we need 3 parameters for the same reason.
- Node F has one parent (E), so again we need 1 parameter (P(F | E = True)).

Adding these up:

- 1 parameter for A
- 1 parameter for B
- 3 parameters for C
- 1 parameter for D
- 3 parameters for E
- 1 parameter for F

In total, we need 10 parameters to fully define the conditional probability distributions for the entire network.

56. (b)

A* keeps all generated nodes in memory to ensure optimality, leading to $O(b^d)$ space usage.

57. (b)

Let, F = Children with flu

M = Children with measles

R = Children showing the symptom of rash

$P(F) = 90\% = 0.9,\ P(M) = 10\% = 0.1$

$$P\left(\frac{R}{F}\right) = 0.08,\ P\left(\frac{R}{M}\right) = 0.95$$

By using Baye's theorem

$$P\left(\frac{F}{R}\right) = \frac{P\left(\frac{R}{F}\right)P(F)}{P\left(\frac{R}{M}\right)P(M)+P\left(\frac{R}{F}\right)P(F)}$$

$$= \frac{0.08\times0.9}{0.95\times0.1+0.08\times0.9}$$

$$= \frac{0.072}{(0.095)+0.072} = \frac{0.072}{0.167} \cong 0.43$$

So, $P\left(\frac{F}{R}\right) = 0.43$

58. (b)

Exact inference guarantees correct and efficient results only in tree-structured Bayesian networks (polytrees). In such networks, algorithms like belief propagation compute exact posteriors efficiently. For general networks, exact inference is correct but computationally expensive.

59. (b)

It can be written as $(a + b)(a' + c)(b + c)$

$= (a+b)(a'+c)(b'c')$ by Demorgan's law

$= (ab'c')(a'+c)$ by commutativity $= 0$

Hence a contradiction.

So, option (b) is correct.

60. (a)

A* Search uses both the actual cost from the start ($g(n)$) and the estimated cost to the goal ($h(n)$).
When the heuristic $h(n)$ is admissible (never overestimates), A* is complete and optimal.
It finds the least-cost path and always terminates if a solution exists.
This makes A* suitable for informed, optimal path finding problems.

62. (c)

Informed search algorithms use a heuristic function to estimate the cost from a state to the goal, guiding the search intelligently.
This is what sets them apart from uninformed (blind) search methods, which do not use such estimates.
They do not necessarily explore all states, are not always faster, and both informed and uninformed may use predefined actions.

63. (c)

Given, $b = 30,\ d = 3$
We know that,

$$O(b^d) = 30^3 = 27000$$

64. (b)

Greedy Best-First Search uses only the heuristic function $h(n)$ to estimate the cost to the goal and selects the node that appears closest to the goal.
It does not consider the cost so far $g(n)$, which means it may choose a suboptimal path.
While it is often faster and memory-efficient, it is not guaranteed to find the optimal solution.
However, if a solution exists, Greedy Best-First Search will eventually find one.
Its performance heavily depends on the quality of the heuristic.
This distinguishes it from A*, which guarantees optimality when the heuristic is admissible.

65. (c)

In sampling-based approximate inference, sampling variance measures how much the estimates vary across different samples.
It is mathematically the variance (square of standard deviation) of the estimate distribution.
High sampling variance means that the estimate is unstable and less reliable, reducing the quality of inference.
Lower variance implies that repeated sampling would give consistent results, improving accuracy.
Controlling sampling variance is key to obtaining trustworthy approximate inference.

66. (d)

The correct translation of "Gold and silver ornaments are precious" is choice (d)

$$\forall x((G(x) \vee S(x)) \rightarrow P(x))$$

which is read as "if an ornament is gold or silver then it is precious".
Now since a given ornament cannot be both gold and silver at the same time.
Choice (b) $\forall x((G(x) \wedge S(x)) \rightarrow P(x))$ is incorrect.

67. (a, c)

Correct options:

(a) Markov Chain Monte Carlo (MCMC): MCMC is a method used in approximate inference through sampling to estimate complex probabilistic distributions.

(c) Rejection Sampling: Rejection sampling is another method used in approximate inference to sample from a target distribution when the exact distribution is difficult to sample from directly.

Incorrect options:

(b) Bayesian Networks: Bayesian networks are a graphical model used to represent and infer probabilistic relationships but are not specifically a method for approximate inference.

(d) Exact Inference: Exact inference refers to the exact computation of probabilities in probabilistic models and is not a method used in approximate inference.

68. (b)

Given, $h(S) = 6,\ h(I) = 4$
$h(G) = 0$
$g(S, I) = 3$
For node I

$$g(I) = 3,\ h(I) = 4$$

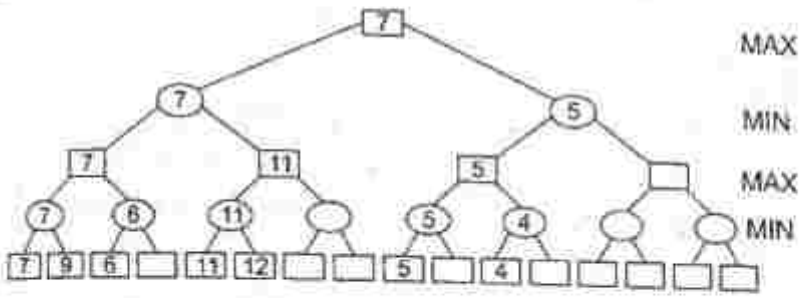
We know that,
A* search

$$f(n) = g(n) + h(n)$$
$$f(I) = g(I) + h(I)$$
$$= 7$$

69. (b)

A disadvantage of approximate inference is that it does not guarantee exactness or accuracy of the results. These methods provide estimated posterior probabilities and may introduce approximation errors or bias. While computationally efficient, the trade-off is that the results can be less precise compared to exact inference.

70. (d)



71. (b)

Exact inference in Bayesian networks is computationally expensive because it involves marginalizing over multiple hidden variables.
This requires summing over all possible assignments of non-evidence and non-query variables.
The number of such combinations grows exponentially with the number of variables.
Hence, exact inference becomes intractable in large or densely connected networks.

72. (a)

Approximate inference is employed when exact inference is intractable due to the high computational complexity of summing over exponentially many variable configurations. It uses methods like sampling or variational approximations to estimate posterior distributions efficiently.

74. (c)

Approximate inference is critical when quick decisions are needed despite uncertainty. It offers faster probability estimates compared to exact inference, which can be too slow or computationally expensive in real-time applications.

75. (c)

$$\sim\left[\exists x\, S(x) \wedge \forall_y \text{GATE}(x, y)\right]$$
$$= \sim \forall y \exists x \left[S(x) \wedge \text{GATE}(x, y)\right]$$
$$= \exists y \forall x \left[\sim S(x) \vee - \text{GATE}(x, y)\right]$$

76. (d)

Exact inference computes true posterior probabilities without approximation, ensuring accuracy and reliability. Unlike approximate methods, it does not rely on sampling or heuristics, making it dependable for precise probabilistic reasoning.

77. (d)

$F(x) = x$ is my friend. $P(x) = x$ is perfect.

(a) There exist some friends which are not perfect.
(b) There are some people who are not my friend and are perfect.
(c) There exist some people who are not my friend and are not perfect.
(d) There does not exist any person who is my friend and perfect.

78. (b)

A* Search is a best-first search algorithm because it uses a heuristic to prioritize nodes that appear closer to the goal. Depth-First and Breadth-First Search do not use heuristics, while Uniform Cost Search prioritizes by path cost only.

80. (c)

DFS is complete only when the search space is finite.
In infinite trees or graphs, DFS can get stuck exploring an endless path.
A finite tree guarantees that DFS will terminate and find a solution if one exists.

81. (a)

A* search is informed because it uses a heuristic $h(n)$ to estimate the cost from a node to the goal. This helps guide the search efficiently toward the goal.
It combines actual cost $g(n)$ and estimated cost $h(n)$ to make decisions.

82. (d)

The given statement should be read as
"If an animal is a tiger or a lion, then (if the animal is hungry or threatened, then it will attack). Therefore the correct translation is
$\forall x$ [(tiger (x) $\vee$ lion (x)) $\rightarrow$ ((hungry (x) $\vee$ threatened (x)) $\rightarrow$ attacks (x))]
which is choice (d).

83. (b)

In A* search, admissibility of the heuristic function means it never overestimates the true cost to reach the goal from any node.

This property ensures that A* will always find the least-cost (optimal) path if one exists.

Although consistency (option c) is stronger and useful for graph search, only admissibility is required to guarantee optimality in tree search.

Finite and reachability are basic assumptions but not sufficient alone to guarantee optimality.

84. (d)

Beam Search is a heuristic search that limits memory by exploring only a fixed number of best nodes (beam width) at each level.

It uses the node ranking of Best-First Search but restricts expansion like Breadth-First Search.

This makes it more memory-efficient but not guaranteed to find the optimal solution.

85. (b)

This states the original accurately. If something is a book that I read last year, then it is guaranteed to be adventure story. The equivalent rule is that if a book is not an adventure story, then i definitely did not read it last year.

86. (b)

If X, then Y unless Z is equivalent to:

$X \rightarrow (Z' \rightarrow Y)$

Now, $X \rightarrow (Z' \rightarrow Y)$

$= X \rightarrow (Z + Y)$

$= X' + Z + Y$

So, this match is with option (b),

So, option (b) is correct.

87. (b)

Bidirectional search requires the goal node to be known so that a reverse search can be started from it. Without a known goal, the backward search cannot proceed, making the algorithm inapplicable.

88. (b)

Minimax assumes the opponent always makes the best possible moves to minimize your advantage. It models an opponent who plays optimally, so you plan for the worst-case scenario.

89. (b)

The standard Mini-Max algorithm has a time complexity of $O(b^d)$, where b is the branching factor and d is the depth. However, if alpha-beta pruning is applied with optimal move ordering, it can reduce the effective branching factor, improving the time complexity to $O(b^{(d/2)})$. This makes alpha-beta pruning significantly more efficient in the best case.

Therefore, the correct answer is (b).

90. (d)

Bidirectional Search is a search algorithm used to find the shortest path from a start node to a goal node. Instead of searching forward from the start node until the goal is found, it simultaneously searches forward from the start and backward from the goal, stopping when the two searches meet.

91. (c)

Greedy Best-First Search selects nodes based only on the heuristic function, which estimates the closeness to the goal. It does not consider the path cost traveled so far. This makes it fast but not always optimal.

92. (d)

Depth-First Search explores nodes deeply along a branch before backtracking.

If the graph contains cycles and no mechanism to track visited nodes is used, it can enter an infinite loop.

This happens because it keeps revisiting the same nodes without reaching a goal or terminating.

93. (c)

- LBP involves iterative message-passing, where each node sends messages to its neighbors based on the messages it receives from others. So, S_1 is correct.
- LBP can incorporate observed evidence by clamping the observed variable to its value and modifying the local potentials accordingly. So, S_2 is incorrect.
- Messages are calculated based on the local potential functions and the incoming messages from neighboring nodes. So, S_3 is correct.

Thus, only statements S_1 and S_3 are correct

94. (a)

$(P \to (Q \vee R)) \to ((P \wedge Q) \to R)$

$\equiv (P \to Q + R) \to (PQ \to R)$

$\equiv [P' + Q + R] \to [(PQ)' + R]$

$\equiv [P' + Q + R] \to [P' + Q' + R]$

$\equiv (P' + Q + R)' + P' + Q' + R$

$\equiv P\,Q'\,R' + P' + Q' + R$

$\equiv Q' + Q'\,P\,R' + P' + R$

$\equiv Q' + P' + R$ (by absorption law)

Which is a contingency (i.e. satisfiable but not valid).

95. (a)

After drawing truth table of the statement it is found that it is fallacy hence negation of fallacy = tautology.

97. (b)

- Loopy Belief Propagation is not guaranteed to converge when applied to graphs with cycles. In some cases, it may diverge or oscillate. So, S_1 is incorrect.
- The belief propagation algorithm produces exact marginal probabilities only for tree-structured graphs (i.e., graphs without loops). So, S_2 is correct.
- In graphs with loops, the algorithm becomes approximate and is known as Loopy Belief Propagation. So, S_3 is correct.

Hence, only statements S_2 and S_3 are correct.

99. (c)

- Breadth-First Search is complete if the branching factor is finite and the goal exists at some finite depth. This statement wrongly implies it's not complete. So, option (a) is incorrect.
- Depth-First Search is not complete in infinite-depth spaces or with loops. It can get stuck going down an infinite path and never find the goal. So, option (b) is incorrect.
- Depth-Limited Search is complete only if the depth limit is equal to or greater than the depth of the shallowest goal node. Otherwise, it may miss the goal. So, option (c) is correct.
- Uniform Cost Search is complete for graphs with positive edge costs, even if the graph is infinite. It guarantees finding the least-cost path to a goal. So, option (d) is incorrect.

100. (a)

Let, $\forall y(\alpha) = P$, $\forall z(\beta) = Q$

Then, $\exists y(\neg\alpha) = \neg P$ and $\exists z(\neg\beta) = \neg Q$

Given, $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$

$= \neg\exists x(P \wedge Q) = \neg\forall x(\neg P \vee \neg Q)$

$= \forall x(P \to \neg Q)$

(a) $\forall x(\exists z(\neg\beta) \to \forall y(\alpha)) = \forall x(\neg Q \to P)$

(b) $\forall x(\forall z(\beta) \to \exists y(\neg\alpha)) = \forall x(Q \to \neg P)$

$= \forall x(P \to \neg Q)$

(c) $\forall x(\forall y(\alpha) \to \exists z(\neg\beta)) = \forall x(P \to \neg Q)$

(d) $\forall x(\exists y(\neg\alpha) \vee \exists z(\neg\beta)) = \forall x(\neg P \vee \neg Q)$

$= \forall x(P \to \neg Q)$

$\therefore$ Only (a) is not logically equivalent to $\forall x(P \to \neg Q)$.

101. (c)

The better the heuristic and the lower the branching factor, the fewer nodes A* will expand. Time complexity depends heavily on heuristic quality and b.

102. (d)

Option (d) is most appropriate translation for above predicate logic statement.

103. (d)

By using min terms we can define

$A \odot B = AB + AB' + A'B' = A + A'B'$

$= (A + A') \cdot (A + B') = A + B'$

(a) $\sim A \odot B = A' \odot B = A' + B'$

(b) $\sim(A \odot \sim B) = (A \odot B')' = (A + (B')')'$

$= (A + B)' = A'B'$

(c) $\sim(\sim A \odot \sim B) = (A' \odot B')' = (A' + (B')')'$

$= (A' + B)' = AB'$

(d) $\sim(\sim A \odot B) = (A' \odot B)' = (A' + B')'$

$= A \cdot B = A \wedge B$

$\therefore$ Only, choice (d) $\equiv A \wedge B$

[**Note:** This problem can also be done by constructing truth table for each choice and comparing with truth table for $A \wedge B$]

104. (b)

Let, p : GATE rank is needed

q : I will write the GATE exam

r : I will join in MADEEASY.

Given arguments:

1. If GATE rank is needed, I will not write GATE exam, if i do not join MADEEASY.

2. GATE rank is needed : p
3. I will join MADEEASY : r
4. I will write the GATE exam : q

Inference is: $(p \wedge \neg r) \rightarrow \neg q$

$$\frac{p \quad r}{q}$$

We can also write the above inference as following: $(p \wedge \neg r)$

$$\big(\big[(p \wedge \neg r) \rightarrow \neg q\big] \wedge p \wedge r\big) \rightarrow q$$

If above proposition is tautology then given inference is valid.

Above given arguments are independent of q, because p and r are true.

$\therefore$ Given inference is invalid and we cannot judge about missing arguments.

105. (a)

BFS explores all nodes level by level up to depth d with b branching factor, so time complexity is $O(b^d)$.

106. (c)

C : Person is corrupt, K : Person is kind

E : Person is elected

$S_1 : C \rightarrow \bar{E}$, $S_2 : K \rightarrow E$, $S_2 \equiv \bar{E} \rightarrow \bar{K}$

So from S_1 and S_2 : $(C \rightarrow \bar{E}) \wedge (\bar{E} \rightarrow \bar{K}) \equiv C \rightarrow \bar{K}$

We can conclude $C \rightarrow \bar{K}$ which is same as $K \rightarrow \bar{C}$, which is same as option (c).

107. (a, c, d)

The converse is "If I will give you magic beans, then you will give me a cow." The contrapositive is "If I will not give you magic beans, then you will not give me a cow." All the other statements are neither the converse nor contrapositive.

108. (b)

Let take each options

(a) $(\neg P \wedge (P \rightarrow q)) \rightarrow \neg q == (P'(P' + Q'))' + Q'$

$== (P' + P'Q)' + Q' == P(P + Q') + Q'$

$== P + PQ' + Q'$

$== P + Q'$ which is contingency.

(b) $\neg(P \rightarrow q) \rightarrow \neg q == ((P' + q)')' + q'$

$== P' + q + q'$

$== 1$ which is tautology.

(c) $\big[(\neg P \wedge q) \wedge [q \rightarrow (P \rightarrow q)]\big] \rightarrow r == ((P'q)$

$(q' + P' + q))' + r'$

$== (P'q)' + r' == P + q' + r'$

which is contingency.

109. (b)

- Breadth-First Search (BFS) explores nodes level by level and maintains all frontier nodes in a queue. It does not rely on backtracking; once a node is explored, BFS moves forward to the next level. So, option (a) is incorrect.
- Depth-First Search (DFS) explores deep into the tree or graph. If a path leads to a dead-end, it backtracks to explore other branches. This is a key characteristic of DFS. So, option (b) is correct.

111. (a, b, c)

A*, BFS, and UCS are complete if the branching factor is finite and step costs are positive. Greedy Best-First Search is not complete, as it can get stuck in loops or overlook better paths due to relying only on the heuristic.

112. (a, c, d)

Quantifiers of the same type (existential or universal) can be reordered without altering the meaning of the statement.

For example, let $Q(x, y)$ be $(y > x)$ and suppose the domain is nonnegative integers. Then the left side asserts that for every nonnegative integer, there is a larger nonnegative integer, which is true. The right side asserts that there exists a nonnegative integer greater than every other nonnegative integer, which is false. Therefore, the implication as a whole is false, and the statement is not valid.

Suppose that the left side is true. Then there exists an x^0 such that for all y, $R(x^0, y)$ is true. Thus, for all y, there exists an x(namely x^0) such that $R(x, y)$ is true. Therefore, the right side is also true, and the statement is valid.

If it's not true that there is an element in the domain with property S, then every element in the domain must not have property S. Conversely, if every element in the domain does not have property S, it can't be true that some element in the domain has property S.

113. (c)

- DFS typically uses less memory than BFS because it stores only the current path from the root to a leaf (depth), which is $O(b \times m)$ where b is branching factor and m is maximum depth. So, S_1 is correct.
- BFS generally uses more memory than DFS because it must keep all nodes at a given level before proceeding. It requires $O(b^d)$ memory, where d is the depth of the shallowest solution. So, S_2 is incorrect.
- It is BFS, not DFS, that maintains all nodes at the current level in memory. DFS does not maintain nodes level-wise; it goes deep along a branch. So, S_3 is incorrect.

115. (c)

$S_1 : (\neg p \wedge (p \vee q)) \rightarrow p$

$S_2 : q \rightarrow (\neg p \wedge (p \vee q))$

$S_1 : [p' (p + q)] \rightarrow q$

$\equiv (p'p + p'q) \rightarrow q$

$\equiv p'q \rightarrow q$

$\equiv (p'q)' + q \equiv p + q' + q$

$\equiv p + 1 \equiv 1$ (tautology)

$S_2 : q \rightarrow (p' (p + q))$

$\equiv q \rightarrow (p'p + p'q)$

$\equiv q \rightarrow p'q \equiv q' + p'q$

$\equiv (q' + p') (q' + q)$

$\equiv q' + p'$ (contigency) (not a tautology)

So, option (c) S_1 is a tautology and S_2 is not a tautology is correct.

116. (b)

Greedy Best-First Search selects the node with the lowest heuristic value ($h(n)$), so it uses a priority queue to efficiently manage nodes based on their heuristic. Stacks and FIFO queues are used in DFS and BFS respectively, not suitable for this purpose.

117. (4)

$p \wedge (p \Rightarrow q) = p(p' + q) \equiv pq$

Take (i) false

$pq \Rightarrow$ false $\equiv pq \Rightarrow 0$

$\equiv (pq)' + 0 \equiv p' + q' + 0$

$\equiv$ not valid

Take (ii)

$pq \Rightarrow q \equiv (pq)' + q \equiv (pq)' + q$

$\equiv p' + 1 \equiv 1 \equiv$ valid

Take (iii)

$pq \Rightarrow$ true $\equiv pq \Rightarrow 1$

$\equiv (pq)' + 1 \equiv 1 \equiv$ valid

Take (iv)

$pq \Rightarrow p + q \equiv (pq)' + p + q$

$\equiv p' + q' + p + q$

$\equiv 1 + 1 \equiv 1 \equiv$ valid

Take (v)

$pq \Rightarrow q' + p \equiv (pq)' + q' + p$

$\equiv p' + q' + q' + p \equiv 1 \equiv$ valid

So the number of expressions that are logically implies by $p \wedge (p \Rightarrow q)$ is 4.

118. (b)

BFS uses a queue to explore nodes level-by-level, maintaining the order in which nodes are discovered.

119. (c)

- Admissibility and consistency are required for A* to guarantee optimality.
  Greedy Best-First Search can work even with inadmissible heuristics, though it may give non-optimal solutions. So, option (a) is incorrect.
- This describes the $g(n)$ (path cost so far), which Greedy Best-First Search ignores completely. So, option (b) is incorrect.
- This matches the behavior of Greedy Best-First Search: it expands nodes only based on $h(n)$ the estimated cost to the goal. So, option (c) is correct.
- This describes $f(n) = g(n) + h(n)$, used in A* Search not in Greedy Best-First Search. So, option (d) is incorrect.

120. (c)

- Despite lacking guarantees of convergence or correctness, LBP is widely used due to its empirical effectiveness in many real-world problems. So, S_1 is correct.
- The message update rules remain the same as in standard belief propagation; the difference lies in the graph structure (presence of loops). So, S_2 is incorrect.
- Loopy Belief Propagation is often used as an approximate method when exact algorithms (e.g., junction tree) are computationally prohibitive. So, S_3 is correct.

Thus, only statements S_1 and S_3 are correct.

121. (b)

Let p : It rains

q : cricket match will not be played.

I_1 : $p \Rightarrow q$

$\sim q$

$\therefore$ $\sim p$

Clearly I_1 is correct since it is in the form of Modus Tollens (rule of contrapositive)

I_2 : $p \Rightarrow q$

$\sim p$

$\therefore$ $\sim q$

which corresponds $[p \Rightarrow q \wedge \sim p] \Rightarrow \sim q$

$\equiv [(p' + q)\, p'] \Rightarrow q'$

$\equiv [p' + qp'] \Rightarrow q' \equiv p' \Rightarrow q'$

$\equiv (p')' + q' \equiv p + q'$

which is not a tautology.

So I_2 is incorrect inference.

122. (d)

Statement $p' \Rightarrow q' \equiv p + q'$

I. $p \rightarrow q \equiv p' + q$

II. $q \rightarrow p \equiv q' + p$

III. $(\neg q \vee p) \equiv q' + p$

IV. $(\neg p \vee q) \equiv p' + q$

So, clearly the given statement is same as II and III only.

123. (a)

- The minimax algorithm uses depth-first search (DFS) to explore the complete game tree, aiming to evaluate the best possible moves for both players by going down each branch of the game tree to the terminal nodes. So, S_1 is correct.
- The minimax algorithm does not use best-first search. Best-first search is commonly used in algorithms like A* but is not a characteristic of the minimax algorithm, which systematically explores all moves to find the minimax value. So, S_2 is incorrect.

124. (b)

- The Junction Tree Algorithm can efficiently incorporate observed evidence by adjusting the clique potentials before message passing, ensuring accurate inference. So, option (a) is incorrect.
- The algorithm's complexity depends on the size of the largest clique in the junction tree. For large or densely connected graphs, the cliques become very large, causing exponential growth in computation time and memory, making it impractical for such cases. So, option (b) is correct.
- The Junction Tree Algorithm is a deterministic exact inference method using message passing, not a sampling-based approach like Monte Carlo methods. So, option (c) is incorrect.
- The Junction Tree Algorithm provides exact inference results, not approximations. So, option (d) is incorrect.

125. (c)

Here, option (a) and (b) can be reduced to $(p \rightarrow q) \wedge (q \rightarrow p)$ and hence $\equiv p \leftrightarrow q$

Option (d) is $p'q' + pq \equiv p \leftrightarrow q$.

Option (c) is $p'q + pq' = p \oplus q$ which is not equivalent to $p \leftrightarrow q$.

126. (b)

- This describes Uniform Cost Search, which expands based on cumulative cost from the start ($g(n)$), not heuristic. So, option (a) is incorrect.
- Greedy Best-First Search selects the node with the lowest $h(n)$ the estimated cost from the current node to the goal.
  It ignores the path cost so far. So, option (b) is correct.
- Selecting the node with the highest path cost is never a valid strategy in informed search algorithms. So, option (c) is incorrect.
- Greedy Best-First Search is not random. It is guided entirely by the heuristic function. So, option (d) is incorrect.

128. (a, b)

- UCS expands the node with the lowest path cost first, so if all step costs are strictly positive, it is guaranteed to find the path with the least cumulative cost. So, option (a) is true.
- UCS is complete, meaning it will eventually find a solution if one exists (assuming finite branching factor and step costs not zero or negative that break termination). So, option (b) is true.

- UCS can fail or behave incorrectly if zero or negative costs are present because it relies on increasing path costs to avoid loops and ensure progress. So, option (c) is false.
- UCS optimizes total cost, not the number of steps; the shortest path by number of steps is not guaranteed. So, option (d) is false.

129. (c)

- UCS does not expand the node with the highest cost. It uses a priority queue to always expand the node with the lowest cumulative path cost, ensuring optimal exploration. So, option (a) is incorrect.
- UCS does not explore randomly. Its search is deterministic and guided entirely by path cost. There is no randomness involved in the node expansion strategy. So, option (b) is incorrect.
- This is a true statement. UCS always expands the node with the least cumulative cost first, moving progressively toward the goal. This is the core exploration principle of UCS. So, option (c) is correct.
- UCS does guarantee the optimal path to the goal. The first time it expands the goal node, it is assured that this path is the least costly path. So, option (d) is incorrect.

130. (c)

DFS stores only the current path from root to leaf, so space complexity is $O(d)$, where d is the depth of the tree.

131. (b)

- This describes the path cost ($g(n)$), which is used in algorithms like UCS and A*.
  Greedy Best-First Search does not use this; it uses only the heuristic function ($h(n)$). So, option (a) is incorrect.
- The heuristic function in Greedy Best-First Search estimates the cost from the current node to the goal node.
  This is what guides the search toward the goal quickly, hence the name "greedy". So, option (b) is correct.
- Greedy Best-First Search does not guarantee optimality. It may find a suboptimal path because it ignores actual path cost. So, option (c) is incorrect.
- The only thing Greedy Best-First Search uses is the estimated cost to the goal, i.e., goal node distance is central to the heuristic. So, option (d) is incorrect.

133. (c)

DFS only needs to store one path from root to leaf (and its siblings), hence space complexity is $O(d)$ much lower than others.

134. (c)

- The minimax algorithm (without pruning) explores the entire game tree up to a fixed depth and always selects the optimal move assuming perfect play by both players. So, S_1 is correct.
- In alpha-beta pruning, if you order moves from best to worst, pruning becomes more effective because bad branches get eliminated early, saving computation. So, S_2 is correct.

135. (d)

Not (all rainy days are cold):

$$\sim (\forall d\ (\text{Rainy}\ (d) \rightarrow \text{Cold}\ (d)))$$

$$\equiv - (\forall d\ (\sim \text{Rainy}\ (d) \vee \text{Cold}\ (d)))$$

$$\equiv \exists d\ (\text{Rainy}\ (d) \wedge \sim \text{Cold}\ (d))$$

Alternate Method:

Not all rainy days are cold is same as some rainy days are not cold which is same as

$$\exists d\ (\text{Rainy}\ (d) \wedge - \text{Cold}\ (d))$$

136. (a)

DLS performs DFS up to a limit l. It explores $O(b^l)$ nodes in the worst case.

137. (c)

UCS guarantees optimality but stores all generated nodes, hence may require exponential space:

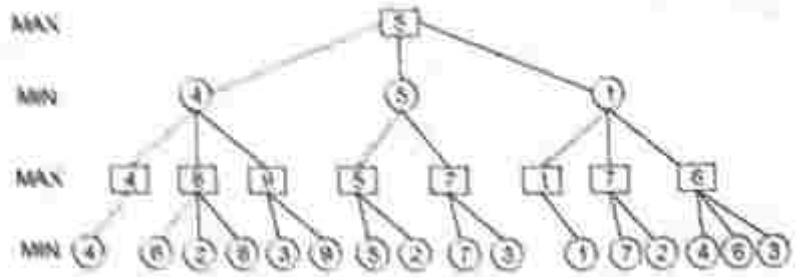
$$O\left(b^{\left(\frac{C}{\varepsilon}\right)}\right)$$

138. (b)

- Rejection sampling uses a likelihood ratio to decide whether to accept or reject samples, not just accepting based on likelihood alone. So, S_1 is incorrect.

- Gibbs sampling is indeed an MCMC method for generating samples by iterating through conditional distributions. So, S_2 is correct.

139. (d)



140. (b, d)

- UCS generally requires more memory than BFS because UCS stores nodes with their path costs in a priority queue, while BFS stores nodes level-by-level. So, option (a) is incorrect.
- UCS finds the shortest path when all edge costs are non-negative. So, option (b) is correct.
- BFS does not use heuristics; it expands nodes level-by-level. So, option (c) is incorrect.
- DFS is complete in a finite search space if the goal is reachable, but it may fail in infinite spaces or if the goal is not reachable. So, option (d) is correct.

141. (d)

The given predicate is

$[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, \gamma))]$

The negation is of this predicate is

$\neg[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow \forall u, \exists v, \gamma)]$

$\neg[\forall x, \alpha \rightarrow (\forall y, \neg\beta \vee \forall u, \exists v, \gamma)]$

$\neg[\exists x, \neg\alpha \vee (\forall y, \neg\beta \vee \forall u, \exists v, \gamma)]$

$[\forall x, \alpha \wedge (\exists y, \beta \wedge (\exists u, \forall v, \neg\gamma))]$

Which is option (d).

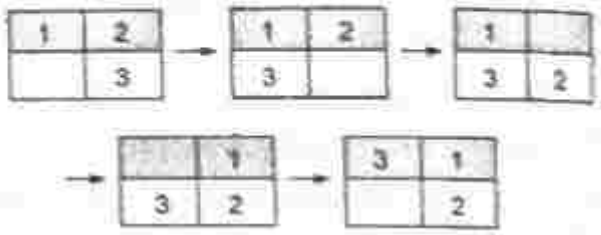
142. (a, b, c)

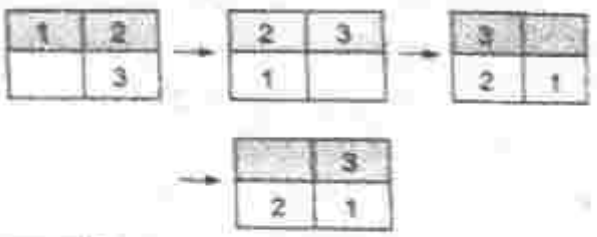
- Uniform Cost Search (UCS) expands the node with the lowest total cost from the start node to the current node. So, option (a) is correct.
- The branching factor in Iterative Deepening A* (IDA*) is the average number of successors each node has, which influences the search tree's width at each level. So, option (b) is correct
- Alpha-beta pruning skips evaluating branches of the search tree that cannot influence the final decision, improving the Min-Max search's efficiency. So, option (c) is correct
- A* typically requires more space than IDDFS because it stores both the open and closed lists, while IDDFS only needs to store nodes at the current depth. So, option (d) is incorrect

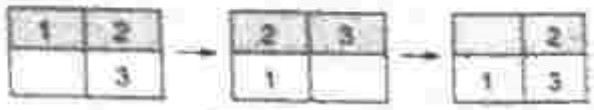
143. (c)

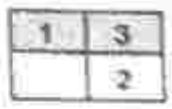
Odd vs even permutation:

Given:

(a) 

(b) 

(d) 

But in option (c)  can not be achieved

3 is always left of 1, it can not be right.

144. (b)

UCS expands nodes in order of path cost. If C is the cost of the optimal solution, the number of nodes with cost $\leq C$ is bounded by $O\left(b^{\left(\frac{C}{\epsilon}\right)}\right)$.

146. (b)

(a) $(a \vee b) \rightarrow (b \wedge c)$

$\equiv (a + b)' + bc$

$\equiv a'b' + bc$

Therefore, $((a \vee b) \rightarrow (b \wedge c))$ is contingency and not tautology.

(b) $(a \wedge b) \rightarrow (b \vee c)$

$\equiv ab \rightarrow b + c$

$\equiv (ab)' + b + c$

$\equiv a' + b' + b + c$

$\equiv a' + 1 + c \equiv 1$

So $((a \wedge b) \rightarrow (b \vee c))$ is tautology.

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

$\equiv (a + b) \rightarrow (b' + c)$

$\equiv (a + b)' + b' + c$

$\equiv a'b' + b' + c$

$\equiv b' + c$

So $((a \vee b) \rightarrow (b \rightarrow c))$ is contingency but not tautology.

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

$\equiv (a' + b) \rightarrow (b' + c)$

$\equiv (a' + b)' + b' + c$

$\equiv ab' + b' + c$

$\equiv b' + c$

Therefore, $((a \rightarrow b) \rightarrow (b \rightarrow c))$ is contingency but not tautology.

147. (b)

We are starting from A

Direct pathA → B [with heuristics value is 28]

A → C [with heuristics value is 26]

A → D [with heuristics value is 32]

So, best first search algorithm and choose the path with lowest heuristics value.

So, A → C is lowest.

Now, from C

C have direct path C → E [18 heuristics value]

C → F [20 heuristics value]

So, best first search algorithm and choose the path with lowest heuristics value.

So, C → E

Now, from E

E have direct path to go to the goal node G (with heuristics value of 0)

So we will go from E to G

E → G

So, now the goal node G has been reached and path will follow is A → C → E → G.

148. (a)

Space complexity of min-max algorithm = $O(bm)$.

149. (0.5)

To find $P(A = 1 \mid B = 1)$

$$P(A = 1 \mid B = 1) = \frac{P(A = 1 \mid B = 1)}{P(B = 1)}$$

First calculate

$P(A = 1 \mid B = 1)$ by summing out C

$$P(A = 1 \mid B = 1) = \sum_{C} P(A = 1, B = 1, C)$$

$P(A = 1, B = 1, C = 0) = 0.4$

$P(A = 1, B = 1, C = 1) = 0.4$

Adding these probabilities

$P(A = 1, B = 1) = 0.4 + 0.4 = 0.8$

Calculate $P(B = 1)$ by summing out A and C

$$P(B = 1) = \sum_{AC} P(A, B = 1, C)$$

From the relevant probability

$P(B = 1) = 0.4 + 0.4 + 0.4 + 0.4 = 1.6$

$$P(A = 1 \mid B = 1) = \frac{P(A = 1 \mid B = 1)}{P(B = 1)} = \frac{0.8}{1.6} = 0.5$$

Conditional probability $P(A = 1 \mid B = 1)$ is 0.5.

150. (0.46)

To compute the marginal probability $P(R = 1)$

We know that the law of total probability

$$P(R = 1) = \sum_{w \in \{\text{sunny, cloudy}\}} P(R = 1 \mid W = w) \cdot P(W = w)$$

$P(R = 1 \mid W = \text{sunny}) = 0.3$

$P(W = \text{sunny}) = 0.6$

$P(R = 1 \mid W = \text{sunny}) \cdot P(W = \text{sunny}) = 0.3 \times 0.6 = 0.18$

$P(R = 1 \mid W = \text{cloudy})$

$P(R = 1 \mid W = \text{cloudy}) = 0.7$

$P(W = \text{cloudy}) = 0.4$

$P(R = 1 \mid W = \text{cloudy}) \cdot P(W = \text{cloudy}) = 0.7 \times 0.4 = 0.28$

Total marginal probability $P(R = 1)$

$P(R = 1) = 0.18 + 0.28 = 0.46$

■■■■

UNIT 6

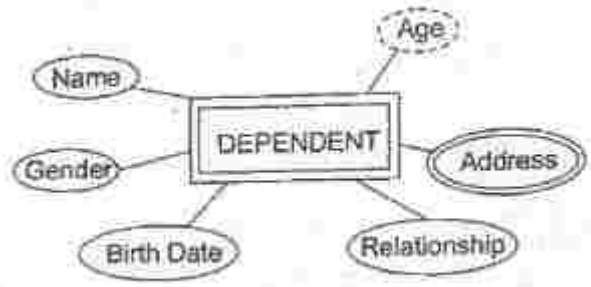
Databases

1. The Relational Model

Q.1 In the Relational Model, the number of attributes and number of tuples in a relation are termed as _____ and _____ respectively.

(a) Cardinality, domain
(b) Degree, cardinality
(c) Domain, degree
(d) Cardinality, degree

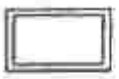
Q.2 The following diagram represents the dependent entity from an Entity Relationship diagram.



Select the characteristics which are not represented by the above diagram.

(a) Age is a derived attribute
(b) Gender is an atomic attribute
(c) Address is a multi-valued attribute
(d) Name is a key attribute

Q.3 Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A. 	1. Identifying relationship
B.	2. Weak entity
C.	3. Derived attribute
D.	4. Multi-valued attribute

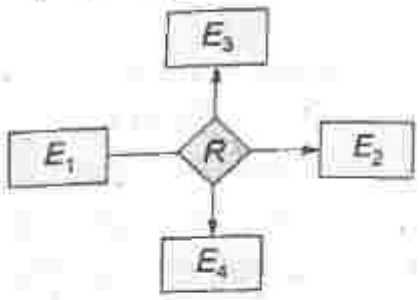
Codes:

	A	B	C	D
(a)	1	3	4	2
(b)	2	4	3	1
(c)	2	3	4	1
(d)	1	4	3	2

Q.4 Which one of the following is used to represent the supporting many-one relationships of a weak entity set in the entity relationship diagram?

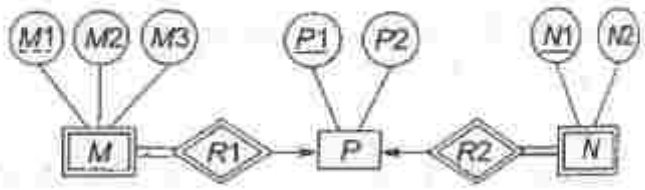
(a) Diamonds with double/bold border
(b) Rectangles with double/bold border
(c) Ovals with double/bold border
(d) Ovals that contain underline identifies

Q.5 Consider the following ER diagram with three entity sets E_1, E_2, E_3 and relationship set R.



If E_1, E_2 and E_3 has 50, 30, 100 and 400 records respectively. What is the maximum number of records of entities that could be in the relationship set R?

Q.6 Consider the following ER diagram:



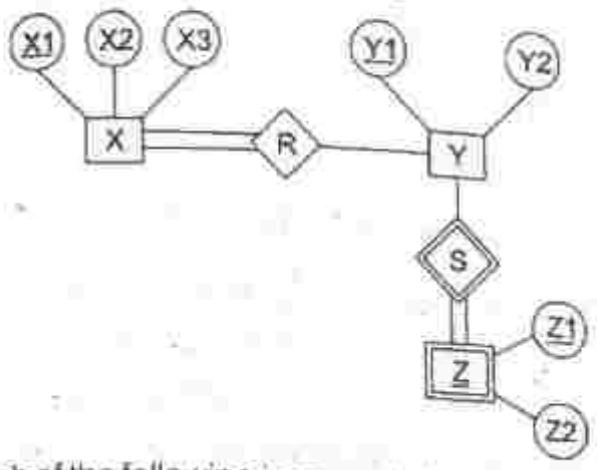
The minimum number of table needed to represent M, N, P, $R1$, $R2$ is

(a) 2 (b) 3
(c) 4 (d) 5

Q.7 Which of the following is/are correct?

(a) An Entity set is a set of entities of the same type that share the same properties, or attributes.
(b) The attribute AGE is calculated from DATE_OF_BIRTH. The attribute AGE is Composite.
(c) Not applicable condition can be represented in relation entry as NULL.
(d) Data integrity constraints are used to improve the quality of data entered for a specific property

Q.8 Consider the following ER diagram:



Which of the following is correct when ER-diagram is converted into minimum tables?

(a) Minimum 4 tables are required to represent X, Y, Z, R and S
(b) The one of the table will be {X1, X2, X3, Y1} where X1 is primary key and Y1 references Y
(c) The one of the table will be {Y1, Z1, Z2} where Y1 is primary key
(d) {Y1, Y2} is one of the table where Y1 is primary key.

Q.9 Which scenario best illustrates uncontrolled data redundancy in a database?

(a) Storing the same student details across multiple normalized tables.
(b) Having a single table where each employee's manager name is repeated for every record.
(c) Using primary key-foreign key relationship between tables.
(d) Using views for accessing data.

Q.10 Which of the following ensures that only valid data is entered into a relational database?

(a) Concurrency control
(b) Checkpoints (c) Integrity constraints
(d) Triggers

Q.11 What is a major drawback of file-based systems when compared to a DBMS?

(a) Ease of implementation
(b) Absence of metadata
(c) Poor data sharing and isolation control
(d) Availability of indexing

Q.12 Which feature is inherent in DBMS but not typically found in traditional file systems?

(a) Storage of large volumes of data
(b) Query processing
(c) Data redundancy
(d) Manual backup control

Q.13 Which component of a DBMS handles crash recovery and concurrency control?

(a) Query optimizer
(b) Transaction manager
(c) Storage manager
(d) Data definition language compiler

Q.14 What does the external level in 3-schema architecture represent?

(a) Actual data stored on disk
(b) Logical organization of data
(c) Specific user views
(d) Index structure for optimization

Q.15 Which of the following mappings occurs between conceptual schema and internal schema?

(a) Logical mapping
(b) Physical mapping
(c) External mapping
(d) View mapping

Q.16 What does logical data independence allow you to do?

(a) Modify internal schema without changing external views
(b) Change user interface without affecting database
(c) Change conceptual schema without altering application programs
(d) Update metadata without affecting data

Q.17 In which of the following cases would physical data independence be violated?

(a) Moving from magnetic disk to SSD storage changes access logic
(b) Adding a new table to the database
(c) Renaming an attribute in a view
(d) Changing SQL syntax from SELECT to PROJECT

Q.18 A database administrator splits a table into two normalized tables without affecting existing user applications. Which type of data independence is demonstrated?

(a) Physical Data Independence
(b) Schema Independence
(c) Logical Data Independence
(d) Storage Independence

Q.19 Which of the following is always a superkey but not necessarily a candidate key?
(a) Primary key
(b) Foreign key
(c) Any attribute set containing the candidate key
(d) Minimal attribute set that uniquely identifies tuples

Q.20 If a relation has only one candidate key, how many superkeys does it have at minimum?
(a) 1
(b) Equal to number of attributes
(c) Infinite
(d) Depends on number of non-key attributes

Q.21 Which of the following must be minimal for uniqueness but not necessarily used as the primary key?
(a) Superkey (b) Candidate key
(c) Foreign key (d) Alternate key

Q.22 Which of the following always refers to a candidate key in another relation?
(a) Primary key
(b) Superkey
(c) Foreign key
(d) Alternate key

Q.23 A relation has two candidate keys: (A, B) and (C). Which of the following can be a minimal composite key?
(a) (A, B) (b) (C)
(c) (A, B, C) (d) (B, C)

Q.24 If a foreign key references a primary key, what happens if the referenced primary key is deleted?
(a) The referencing row is automatically deleted
(b) An error occurs if no ON DELETE action is defined
(c) The referencing key becomes primary
(d) It becomes a candidate key

Q.25 Which is true about weak entities in an ER diagram?
(a) They do not have attributes
(b) They do not participate in any relationship
(c) They depend on strong entities for identification
(d) They act as primary keys

Q.26 What is the minimum number of keys a relation must have?
(a) 0
(b) 1
(c) 2
(d) Depends on attributes

Q.27 Which of the following is true about total participation in ER diagrams?
(a) Entity participates in all relationships
(b) Each entity instance must be in the relationship
(c) It implies disjoint specialization
(d) Every relationship has multiple entities

Q.28 In ER diagrams, double rectangles are used for
(a) Composite attributes
(b) Weak entities
(c) Strong entities
(d) Derived attributes

Q.29 Which statement is incorrect about candidate keys?
(a) They must uniquely identify each tuple
(b) They can have null values
(c) There can be more than one candidate key
(d) All candidate keys are superkeys

Q.30 Which condition makes a relationship ternary?
(a) It connects 3 entity sets
(b) It uses 3 attributes
(c) It connects 2 entity sets with 3 attributes
(d) It connects 3 weak entities

Q.31 An attribute derived from other stored values in the database is called:
(a) Primary attribute (b) Multi-valued attribute
(c) Derived attribute (d) Key attribute

Q.32 Which one best describes specialization in an ER diagram?
(a) Child entity inherits from parent
(b) Parent entity contains all attributes of child
(c) Strong entity depends on weak entity
(d) Relationship with weak participation

Q.33 In an ER model, which of the following is used to identify a weak entity uniquely?
(a) Foreign key
(b) Strong entity's key + Partial key
(c) Derived attribute
(d) Candidate key

Q.34 In an ER model, when a relationship itself needs to be treated as an entity for another relationship, what is used?
(a) Specialization
(b) Aggregation
(c) Ternary Relationship
(d) Weak Relationship

Q.35 Which of the following correctly distinguishes generalization from specialization in ER modeling?
(a) Generalization moves from specific to general; specialization from general to specific
(b) Both mean the same
(c) Generalization applies only to weak entities
(d) Specialization creates parent entities

Q.36 Which scenario is best modeled using a ternary relationship instead of three binary relationships?
(a) Student enrolls in course
(b) Doctor prescribes drug to patient
(c) Customer buys product
(d) Employee manages department

Q.37 In an ER diagram, a total specialization of entity "person" into "Student" and "Employee" implies:
(a) Some persons may be neither student nor employee
(b) A person must be either a student or an employee or both
(c) Person cannot be a student
(d) Student and Employee are the same entity

Q.38 Which of the following is not true about aggregation in ER modeling?
(a) It allows a relationship to participate in another relationship
(b) It is represented using a double a rectangle covering entities
(c) It helps in representing higher-level abstractions
(d) It is different from ternary relationships

Q.39 Consider entities Doctor, Patient, and Time. Which relationship should be modeled as ternary rather than three binary relationships?
(a) Doctor meets patient at time
(b) Doctor treats patient
(c) Patient visits doctor
(d) Doctor is available at time

Q.40 In specialization, when each entity belongs to only one subclass, the specialization is:
(a) Disjoint
(b) Overlapping
(c) Total
(d) Partial

Q.41 What is the main reason we prefer aggregation over ternary relationships in some ER diagrams?
(a) Aggregation reduces memory
(b) Aggregation allows attribute sharing
(c) Aggregation emphasizes conceptual clarity when a relationship needs to be involved in another relationship
(d) Aggregation increases redundancy

Q.42 Which is a valid representation in an ER diagram to model manager manages a project through an employee (manager assigns employee to project)?
(a) Binary relationship between Manager and Project
(b) Aggregation of (Manager-Employee) related to Project
(c) Ternary relationship among Manager, Employee, and Project
(d) Binary relationships between Manager and Employee, and Employee and Project

Q.43 In an ER diagram for a hotel booking system, a "Room" entity cannot exist without a "Hotel" entity. Which notation correctly enforces this dependency?
(a) Double diamond + Dashed underline
(b) Double rectangle + Identifying relationship
(c) Single rectangle + Partial participation
(d) Thick line + Total participation

Q.44 In a project management system, "Employees" use "Equipment" to complete "Tasks". How would you model a scenario where the assignment of equipment to tasks is tracked per project?
(a) Ternary relationship between Employee, Equipment, and Task
(b) Aggregation of (Employee, Equipment, Task) into a higher-level entity linked to Project
(c) Two binary relationships: Employee → Equipment and Equipment → Task
(d) Specialization of Equipment into ProjectEquipment

Q.45 A university system records which instructor teaches a specific course to a student. Why is a ternary relationship preferred over multiple binary relationships?
(a) To avoid redundancy in instructor assignments
(b) Open force total participation of all entities
(c) To capture the simultaneous association of Student, course, and Instructor
(d) To reduce the number of tables in the relational model

Q.46 In a hospital ER model, "Patient" is specialized into "Outpatient" and "Inpatient". If a patient cannot be both outpatient and inpatient, which constraint is applied?
(a) Overlapping, partial
(b) Overlapping, total
(c) Disjoint, partial
(d) Disjoint, total

Q.47 In an ER diagram for employee management, how would you distinguish between an employee acting as a "Supervisor" and a "Subordinate" in the same relationship?
(a) Role indicator labels on the relationship lines
(b) Separate entities for Supervisor and Subordinate
(c) Aggregation of the "Supervises" relationship
(d) Specialization of Employee into Supervisor

Q.48 In an ER diagram, how is a "Address" attribute (composed of Street, City, ZIP) represented?
(a) Multi-valued attribute
(b) Derived attribute
(c) Composite attribute
(d) Weak attribute

Q.49 A student can enroll in many courses, and each course can have many students. What is the cardinality ratio of the "Enrolls" relationship?
(a) 1 : 1 (b) 1 : N
(c) M : N (d) N : 1

Q.50 In an ER diagram, how is total participation of "Employee" in the "Works_for" relationship with "Department" depicted?
(a) Thick line on the Employee side
(b) Double diamond on the relationship
(c) Dashed line on the Department side
(d) Arrow pointing to Department

Q.51 In a university, "Staff" can be specialized into "Faculty" and "Administrator", with some staff members being both. Which constraint applies?
(a) Disjoint, total
(b) Overlapping, partial
(c) Disjoint, partial
(d) Overlapping, total

Q.52 In converting an ER diagram to a relational schema, if a total participation of a weak entity is not enforced, what could be the effect in the resulting schema?
(a) Redundant tuples
(b) Insertion anomalies
(c) Loss of dependency
(d) Weak entity rows without foreign key reference

Q.53 If a many-to-many relationship in an ER diagram is converted into a relation, which is true about the resulting table?
(a) It must have a single primary key
(b) It contains foreign keys of both entities
(c) It cannot be in 2NF
(d) It contains only one attribute

Q.54 When an ER diagram with a multivalued attribute is converted to a relational schema, how is that attribute represented?
(a) As a single column with multiple values
(b) As a separate relation
(c) As a derived attribute
(d) It is ignored in the schema

Q.55 While converting an ER diagram with a generalization hierarchy, if we use "table per subclass" strategy, what is a likely drawback?
(a) Redundant tuples
(b) Nulls in tables
(c) Join required to get full data
(d) Cannot preserve keys

Q.56 Which method of mapping generalization avoids nulls but may increase redundancy?
(a) Table per subclass
(b) Table per hierarchy (single table for all classes)
(c) Table per relationship
(d) Table per attribute

Q.57 Which of the following is not true about converting ER diagram into relational schema in 3NF?
(a) Transitive dependencies are removed
(b) All non-prime attributes depend on the key
(c) It preserves all functional dependencies
(d) It allows partial dependencies

Q.58 In an ER diagram, entity B is a weak entity dependent on entity A with total participation. How is this converted to a relational schema?
(a) Table B with foreign key A_id (nullable)
(b) Table B with composite primary key (A_id, B_weak_key) and foreign key A_id (NOT NULL)
(c) Merge A and B into a single table
(d) Table B with primary key B_weak_key

Q.59 A ternary relationship R connects entities C, D, and E, where C has total participation. How is R mapped to a relational schema while avoiding NULL values?
(a) Add foreign keys for C, D, E in the table for C
(b) Create a separate table for R with foreign keys C_id, D_id, E_id
(c) Merge R into the table for D
(d) Use a composite key in C's table for D and E

Q.60 An entity "Vehicle" is specialized into "Car" and "Bike" with total participation and disjoint constraint. How is this mapped to 3NF-compliant tables?
(a) Single table with attributes Car_specific, Bike_specific (nullable)
(b) Three tables: Vehicle, Car (Vehicle_id), Bike (Vehicle_id)
(c) Two tables: Car and Bike with all Vehicle attributes
(d) Single table with a discriminator attribute "Type"

Q.61 A multivalued attribute "Phone_numbers" in entity Student is converted to a relational schema. Which design ensures 3NF?
(a) Column Phone_numbers (array type) in Student table
(b) Separate table Phone (Student_id, Number) with Student_id as foreign key
(c) Merge Phone_numbers into another entity
(d) Store Phone_numbers as a derived attribute

Q.62 A 1 : 1 relationship between "Manager" and "Department" with total participation on Manager is mapped to a schema. Which design minimizes NULLs?
(a) Separate tables Manager and Department with foreign keys
(b) Merge Manager and Department into one table
(c) Add Department_id (nullable) in Manager table
(d) Add Manager_id (nullable) in Department table

Q.63 In a relation derived from an ER diagram, attribute X is functionally dependent on attribute Y, which is not a candidate key. Which step ensures 3NF?
(a) Keep the table as-is
(b) Create a new table with Y and X, and remove X from the original table
(c) Make Y the primary key
(d) Store X as a derived attribute

Q.64 A composite attribute "Name" (First, Middle, Last) in entity Employee is converted to a schema. Which design is correct?
(a) Single column Name (varchar)
(b) Columns First, Middle, Last in Employee table
(c) Separate table for Name linked to Employee
(d) Store Name as a multivalued attribute

Q.65 Overlapping specialization of "Person" into "Student" and "Employee" allows NULLs. How is this mapped to 3NF tables?
(a) Single table Person with boolean flags is_student, is_empioyee
(b) Three tables: Person, Student (Person_id), Employee (Person_id)
(c) Two tables: Student and Employee with all Person attributes
(d) Merge Student and Employee into one table

Q.66 Consider a relation Student(CourseID, StudentID, Name) where the primary key is the combination (CourseID, StudentID). Can this composite key be replaced by a single surrogate key SID without losing any normalization benefit?
(a) Yes, always
(b) No, it causes redundancy
(c) Yes, but only if CourseID functionally determines StudentID
(d) No, because it violates 1NF

Q.67 Which of the following may cause an insertion anomaly in a relation Orders(OrderID PK, CustomerID FK)?
(a) Inserting an order with a non-existent CustomerID
(b) Inserting a duplicate OrderID
(c) Inserting an order with NULL in CustomerID
(d) All of the above

Q.68 In a table Employee(EID PK, ManagerID FK) where ManagerID references EID, what happens if an employee is promoted and their ManagerID is set to their own EID?
(a) Allowed, creates a self-loop
(b) Not allowed due to FK constraint
(c) Allowed only if ON UPDATE CASCADE is set
(d) Allowed only if PK is composite

Q.69 If T_1(A PK, B), T_2(B FK references T_1(B), (C PK) - Is this schema valid?
(a) Yes, foreign key can point to non-primary attribute
(b) No, foreign key must reference a primary key
(c) Yes, only if T_1(B) is declared UNIQUE
(d) Invalid unless B is composite

Q.70 In a database with Course(CourseID PK, Name) and Enrollment(SID, CourseID FK), what happens when a course is deleted and ON DELETE SET NULL is set on the foreign key?
(a) All enrollments for that course are deleted
(b) SID is set to NULL
(c) CourseID in Enrollment becomes NULL
(d) Delete is blocked

Q.71 Consider two tables:
Employees (EmpID, Name, DeptID)
Departments (DeptID, ManagerID, Budget)
If a foreign key constraint is added to Employees.DeptID referencing Departments. DeptID with ON DELETE CASCADE, what happens when a department is deleted?
(a) Employees in that department are set to NULL in DeptI(d)
(b) Employees in that department are automatically deleted.
(c) The deletion is blocked if employees exist in the department.
(d) The DeptID in Employees is set to a default value.

Q.72 A table Orders (OrderID, ProductID, Quantity) has a composite primary key (OrderID, ProductID). Can ProductID alone be referenced as a foreign key in another table?
(a) Yes, because any subset of a composite key can be a foreign key.
(b) No, because a foreign key must reference the entire primary key.
(c) Yes, if ProductID is declared as a unique key in Orders.
(d) No, because composite keys cannot be referenced in foreign keys.

Q.73 In a table Students (RollNo, Name, CourseID) with RollNo as the primary key, suppose CourseID is a foreign key referencing Courses(CourseID). What happens if you try to insert a student with a CourseID not present in Courses?
(a) The insertion succeeds with a warning.
(b) The insertion fails with a foreign key constraint violation.
(c) The CourseID is set to NULL automatically.
(d) The Courses table is updated to include the new Coursel(d)

Q.74 Table A (X, Y) has a composite primary key (X, Y). Table B (P, Q) references A via a foreign key (P) pointing to A.X. What is the constraint here?
(a) Valid, as X is part of the composite key in A.
(b) Invalid, because the foreign key must include all columns of A's primary key.
(c) Valid, only if X is declared as unique in A.
(d) Invalid, because foreign keys cannot reference composite keys partially.

Q.75 If a table Book (ISBN, Title, AuthorID) has AuthorID as a foreign key referencing Author(AuthorID), and the foreign key is defined with ON UPDATE SET NULL, what happens when the AuthorID in Author is updated?
(a) All corresponding AuthorID in Book are set to NULL.
(b) The update in Author is blocked.
(c) The AuthorID in Book is updated to match the new value.
(d) Only the Author table is updated, and Book remains unchanged.

Q.76 Consider a table Student with columns (RollNo, Email, PhoneNo). Which of the following ensures that no two students can have the same combination of Email and PhoneNo, even if their RollNo is different?
(a) Declaring (Email, PhoneNo) as PRIMARY KEY
(b) Declaring (Email, PhoneNo) as UNIQUE
(c) Declaring Email as UNIQUE and PhoneNo as UNIQUE separately
(d) Adding NOT NULL constraint on Email and PhoneNo

Q.77 Which of the following is invalid for a PRIMARY KEY constraint in standard SQL?
(a) It can be a composite of two attributes
(b) It implicitly applies NOT NULL on all participating attributes
(c) A table can have multiple primary keys
(d) It enforces uniqueness on the key attributes

Q.78 Which of the following violates referential integrity?
(a) Inserting a row with a foreign key referencing a non-existent primary key
(b) Deleting a primary key row that is referenced in another table
(c) Updating a foreign key to a value not in the parent table
(d) All of the above

Q.79 What happens if a composite primary key has one attribute set as NULL during insertion?
(a) Insertion is allowed
(b) NULL is ignored during constraint checking
(c) It causes constraint violation
(d) Only warning is raised

Q.80 You want to ensure that every employee's ManagerID exists as an EmpID in the same table. Which constraint is suitable?
(a) Self foreign key
b) Check constraint
(c) Unique constraint
(d) Recursive key

Q.81 Which of the following is possible in SQL but violates normalization principles?
(a) Making a column both UNIQUE and FOREIGN KEY
(b) Making a table with composite key and non-atomic attribute
(c) Using CHECK for range constraints
(d) Using NULLs in non-key attributes

Q.82 Consider a table with a UNIQUE constraint on column A. Which statement is TRUE?
(a) A can have multiple NULL values.
(b) A can have exactly one NULL value.
(c) A cannot have any NULL values.
(d) The UNIQUE constraint is equivalent to a PRIMARY KEY.

Q.83 Table R has a composite primary key (A, B). Can a foreign key in table S reference only column A of R?
(a) Yes, if A is declared as UNIQUE in R.
(b) Yes, because A is part of the primary key.
(c) No, a foreign key must reference the entire primary key.
(d) No, unless A is a candidate key.

Q.84 Why can't a CHECK constraint enforce referential integrity across tables?
(a) CHECK constraints cannot use subqueries.
(b) CHECK constraints are limited to row-level conditions.
(c) Both (a) and (b)
(d) CHECK constraints can enforce cross-table integrity.

Q.85 What is the difference between a PRIMARY KEY and a UNIQUE KEY constraint?
(a) A table can have multiple PRIMARY KEYs.
(b) A UNIQUE constraint allows NULLs.
(c) PRIMARY KEY allow NULL value.
(d) There is no difference.

Q.86 When will ON DELETE SET NULL fail for a foreign key?
(a) If the foreign key column is NOT NULL.
(b) If the referenced table has no matching rows.
(c) If the foreign key is part of a composite key.
(d) All of the above

Q.87 Exclusion constraints (e.g., in PostgreSQL) differ from UNIQUE constraints because they:
(a) Prevent overlapping ranges.
(b) Allow conditional uniqueness.
(c) Use operator-based comparisons.
(d) All of the above

2. Database Design and Normalization

Q.88 A relation R with two attribute is always in
(a) 1 NF (b) 2 NF
(c) 3 NF (d) BCNF

Q.89 Consider the relation schema $S = \{A, B, C, D\}$ and the following functional dependencies on S.

$A \rightarrow BCD$

$B \rightarrow C$

$CD \rightarrow A$

Which of the following is true?
(a) S is in 3 NF and also in BCNF
(b) S is in 3 NF but not in BCNF
(c) S is in 2 NF but not in 3 NF
(d) S is in BCNF but not in 3 NF

Q.90 Consider a relation $R(A\ B\ C\ D\ E)$ with sets of FD's $F\{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$ Given R is in which highest normal forms?
(a) 1 NF (b) 2 NF
(c) 3 NF (d) BCNF

Q.91 Given FD = $\{AB \rightarrow C, CD \rightarrow E, E \rightarrow FG, G \rightarrow H\}$. Find which of the following can be a possible deduction from given functional dependencies.
(a) $DFG \rightarrow E$ (b) $ACD \rightarrow B$
(c) $ABD \rightarrow E$ (d) None of these

Q.92 Consider the following relation:
Vehicle (vehicle No, sold_date, salesmanNo, commission%, discount_amt)
Assume that a vehicle may be sold by multiple salesman and vehicle (vehicle No, salesmanNo) is Primary key. Additional dependencies are Sold_date → discount_amt and salesmanNo → commission%.
Given relation is in which normal form?
(a) 1 NF (b) 2 NF
(c) 3 NF (d) None of these

Q.93 Find the number of FD's in the minimal cover of following FD set FD = $\{A \rightarrow B, CD \rightarrow A, CB \rightarrow D, CE \rightarrow D, AE \rightarrow F, AC \rightarrow D\}$.
(a) 2 (b) 3
(c) 4 (d) 5

Q.94 Given the relational schema $R(A, B, C, D, E)$ with the following functional dependencies.

$A \rightarrow B, BD \rightarrow E, C \rightarrow AD, E \rightarrow C$

Find which of the following is not a candidate key?
(a) E (b) C
(c) BD (d) A

Q.95 Consider the relation $R(A, B, C, D, E)$ and the set

$F = \{AB \rightarrow CE, E \rightarrow AB, C \rightarrow D\}$

What is the highest normal form of this relation?
(a) 1NF (b) 2NF
(c) 3NF (d) BCNF

Q.96 Find the number of possibilities to choose a primary key of the relation $R(A, B, C, D, E)$.
(a) 5 (b) 25
(c) 31 (d) 120

Q.97 Let R(ABCDEFGH) be a relation with the following functional dependencies

$FD = \{AB \rightarrow CDEF$
$A \rightarrow C$
$B \rightarrow D$
$C \rightarrow FG$
$G \rightarrow H$
$H \rightarrow F$
$G \rightarrow F\}$

Find the maximum normal form applicable to R
(a) 1NF (b) 2NF
(c) 3NF (d) BCNF

Q.98 A relation (from the relational database model) consists of a set of tuples, which implies that
(a) all tuples in a relation must be distinct.
(b) relational model supports multi-valued attributes whose values can be represented in sets.
(c) for any two tuples, the values associated with all of their attributes may be the same.
(d) all tuples in a particular relation may have different attributes.

Q.99 Suppose that we decompose the schema $R(A, B, C, D, E)$ into $R_1(A, B, C)$ and $R_2(A, D, E)$ if the set F are functional dependencies hold on R.

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

The decomposition is a
(a) Lossless decomposition
(b) Lossy decomposition
(c) May or may not be lossless decomposition
(d) None of these

Q.100 Consider the relation given below and find the maximum normal form applicable to them

1. $R(A, B)$ with productions $\{A \rightarrow B\}$
2. $R(A, B)$ with productions $\{B \rightarrow A\}$
3. $R(A, B)$ with productions $\{A \rightarrow B, B \rightarrow A\}$
4. $R(A, B, C)$ with productions $\{A \rightarrow B, B \rightarrow A, AB \rightarrow C\}$

(a) 1, 2 and 3 are in 3NF and 4 is in BCNF
(b) 1 and 2 are in BCNF and 3 and 4 are in 3NF
(c) All are in 3NF
(d) All are in BCNF

Q.101 Consider the relation $R(S, T, U, V)$ and the set $F = \{S \rightarrow T, T \rightarrow U, U \rightarrow V$ and $V \rightarrow S\}$. Let R is decomposed into R_1 and R_2 such that $R_1 \cap R_2 = \phi$, the decomposition is

(a) Lossless decomposition
(b) Lossy decomposition
(c) Dependency preservation
(d) Cannot say

Q.102 $f = \{A \rightarrow BC, B \rightarrow A, C \rightarrow A\}$
$g = \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

(a) f covers g
(b) g covers f
(c) f and g are equivalent
(d) None of these

Q.103 Consider the following set of functional dependencies F on the schema (A, B, C)

$F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$

The canonical cover of this set is

(a) $A \rightarrow BC$ and $B \rightarrow C$
(b) $A \rightarrow BC$ and $AB \rightarrow C$
(c) $A \rightarrow BC$ and $A \rightarrow B$
(d) $A \rightarrow B$ and $B \rightarrow C$

Q.104 Consider the following relation and its sample data. (Consider that these are the only tuples for the given relation)?

Emp No	Dept No	Proj No
1001	01	12
1001	01	13
1002	01	12
1003	01	14

1. The functional dependency (Emp No, Dept No)→ ProjNo holds over R.
2. The functional dependency DeptNo → ProjNo holds over R.
3. The functional dependency EmpNo → DeptNo holds over R.
4. The functional dependency Proj No → DeptNo holds over R.
5. The functional dependency (EmpNo, ProjNo) → DeptNo holds over R,

(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 3, 4 and 5 (d) All of these

Q.105 Choose the correct statement:

(a) An alternative key is a candidate key that is not a primary key.
(b) An alternate key is a primary key that is not a candidate key.
(c) An alternate key is a candidate key that is also a primary key.
(d) None of these

Q.106 Relation produced from an ER model will always be in:

(a) 1 NF (b) 2 NF
(c) 3 NF (d) 4 NF

Q.107 Consider the following FD set:

$\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, EC \rightarrow FA, CF \rightarrow BD, D \rightarrow E\}$

Which of the following FD set is minimal cover of given FD set?

(a) $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, BE \rightarrow C, EC \rightarrow F, CF \rightarrow D, D \rightarrow E\}$
(b) $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, BE \rightarrow C, EC \rightarrow F, CF \rightarrow B, D \rightarrow E\}$
(c) $\{AB \rightarrow C, C \rightarrow A, CD \rightarrow B, EC \rightarrow F, CF \rightarrow B, D \rightarrow E\}$
(d) $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, BE \rightarrow C, EC \rightarrow A, CF \rightarrow B, D \rightarrow E\}$

Q.108 A functional dependency of the form $X \rightarrow Y$ is trivial if:

(a) $Y \subseteq X$
(b) $Y \subset X$
(c) $X \subseteq Y$
(d) $X \subset Y$ and $Y \subset X$

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Q.109 Given the function dependencies:

$X \rightarrow W$; $X \rightarrow Y$; $Y \rightarrow Z$ and $Z \rightarrow PQ$

Which of the following does not hold good?

(a) $X \rightarrow Z$ (b) $W \rightarrow Z$
(c) $X \rightarrow WY$ (d) None of these

Q.110 Which-one of the following statements about normal forms is TRUE?

(a) BCNF is stricter than 3NF.

(b) Lossless decomposition and dependency preserving in 3NF is always possible.

(c) Lossless decomposition dependency preserving in BCNF is always possible.

(d) Any relation with two attributes is BCNF.

Q.111 Consider the following relational schema $R(ABCDE)$. The number of super keys in relation R if every two attributes of relation R is candidate keys are _____.

Q.112 Consider $A(P, Q, R, S, T, V, W)$ and the following FD's:

$$W \rightarrow VS,\ T \rightarrow S,\ WS \rightarrow RT,\ QS \rightarrow P$$

Which of the following is minimal cover of the given FD's?

(a) $\{W \rightarrow V,\ T \rightarrow S,\ W \rightarrow R,\ WS \rightarrow T,\ QS \rightarrow P\}$

(b) $\{W \rightarrow V,\ W \rightarrow S,\ T \rightarrow S,\ W \rightarrow R,\ QS \rightarrow P\}$

(c) $\{W \rightarrow V,\ T \rightarrow S,\ W \rightarrow R,\ WS \rightarrow R,\ QS \rightarrow P\}$

(d) $\{W \rightarrow V,\ T \rightarrow S,\ W \rightarrow R,\ W \rightarrow T,\ QS \rightarrow P\}$

Q.113 Consider the following relational schema:

$R(ABCGHI)$ with FD's $\{A \rightarrow B,\ A \rightarrow C,\ CG \rightarrow A,\ CG \rightarrow I,\ B \rightarrow H\}$

The number of candidate keys for relation R are _____.

Q.114 Let $R(a, b, c)$ and $S(d, e, f)$ be two relations in which d is the foreign key of S that refers to the primary key of R. Consider the following four operations R and S

1. Insert into R
2. Insert into S
3. Delete from R
4. Delete from S

Which of the following is true about the referential integrity constraint above?

(a) None of 1, 2, 3 or 4 can cause its violation

(b) All of 1, 2, 3 and 4 can cause its violation

(c) Both 1 and 4 can cause its violation

(d) Both 2 and 3 can cause its violation

Q.115 Consider the following relation:

$R(A_1, A_2, \ldots, A_n)$ and Every $(n-2)$ attributes of R forms Candidate key. Which of the following represents the number of super keys are there in R?

(a) ${}^{n}C_{n-2} * 2^2$ (b) ${}^{n}C_{n-2} * n$

(c) ${}^{n}C_{n-2} * [3]$ (d) ${}^{n}C_{n-2}$

Q.116 The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade.

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

What is the number of tuples that must be additionally deleted to preserve referential integrity when the tuple (4, 3) is deleted?

(a) 4 (b) 5

(c) 6 (d) 7

Q.117 Consider the relation schema $S(A, B, C, D)$ and the following functional dependencies on

$$S\{A \rightarrow BCD,\ B \rightarrow C,\ CD \rightarrow A\}$$

If we decompose S into $S_1 = ABD$ and $S_2 = BC$ then fill in the following blanks for A, B and C

1. The decomposition is _____.
2. The decomposition is _____.
3. Valid decomposition into _____.

(a) lossless, dependency preserving, BCNF

(b) lossless, not dependency preserving, BCNF

(c) lossy, not dependency preserving, 3NF

(d) lossy, dependency preserving, BCNF

Q.118 Let R = (A, B, C, D, E, F) be a relation scheme with the following dependencies $C \rightarrow F$, $E \rightarrow A$, $EC \rightarrow D$, $A \rightarrow B$. Which of the following is a key for R?

(a) CD (b) EC

(c) AE (d) AC

Q.119 From the following instance of a relation scheme $R(A, B, C)$, we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

(a) A functionally determines B and B functionally determines C
(b) A functionally determines B and B does not functionally determines C
(c) B does not functionally determines C
(d) A does not functionally determines B and B does not functionally determines C

Q.120 Consider a relation scheme R = (A, B, C, D, E, H) on which the following functional dependencies hold:

$\{A \rightarrow B,\ BC \rightarrow D,\ E \rightarrow C,\ D \rightarrow A\}$

What are the candidate keys of R?
(a) AE, BE
(b) AE, BE, DE
(c) AEH, BEH, BCH
(d) AEH, BEH, DEH

Q.121 In a schema with attributes A, B, C, D and E following set of functional dependencies are given

$A \rightarrow B,\ A \rightarrow C,\ CD \rightarrow E,\ B \rightarrow D,\ E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set
(a) $CD \rightarrow AC$ (b) $BD \rightarrow CD$
(c) $BC \rightarrow CD$ (d) $AC \rightarrow BC$

Direction for Question 122 :

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values.

$F = \{CH \rightarrow G,\ A \rightarrow BC,\ B \rightarrow CFH,\ E \rightarrow A,\ F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F is exactly the set of FDs that hold for R.

Q.122 How many candidate keys does the relation R have?
(a) 3 (b) 4
(c) 5 (d) 6

Q.123 Which of the following is NOT a superkey in a relational schema with attributes *V, W, X, Y, Z* and primary key *V Y*?
(a) *V X Y Z* (b) *V W X Z*
(c) *V W X Y* (d) *V W X Y Z*

Q.124 A prime attribute of a relation scheme *R* is an attribute that appears
(a) in all candidate keys of *R*.
(b) in some candidate key of *R*.
(c) in a foreign key of R.
(d) only in the primary key of *R*.

Q.125 BCNF is not used for cases where a relation has
(a) Two (or more) candidate keys
(b) Two candidate keys and composite
(c) The candidate key overlap
(d) Two mutually exclusive foreign keys

Q.126 Purpose of 'Foreign Key' in a table is to ensure
(a) Null Integrity
(b) Referential Integrity
(c) Domain Integrity
(d) Null and Domain Integrity

Q.127 Consider the following dependencies and the BOOK table in a relational database design. Determine the normal form of the given relation.

ISBN → Title
ISBN → Publisher
Publisher → Address

(a) First Normal Form
(b) Second Normal Form
(c) Third normal Form
(d) BCNF

Q.128 Every time the attribute *A* appears, it is matched with the same value of attribute *B* but not the same value of attribute *C*. Which of the following is true?
(a) $A \rightarrow (B, C)$
(b) $A \rightarrow B,\ A \twoheadrightarrow C$
(c) $A \rightarrow B,\ C \twoheadrightarrow A$
(d) $A \twoheadrightarrow B,\ B \rightarrow C$

Q.129 Consider the schema $R(A, B, C, D)$ and the functional dependencies $A \rightarrow B$ and $C \rightarrow D$. If the decomposition is made as $R_1(A, B)$ and $R_2(C, D)$, then which of the following is TRUE?
(a) Preserves dependency but cannot performs lossless join
(b) Preserves dependency and performs lossless join
(c) Does not preserve dependency and cannot performs lossless join
(d) Does not preserve dependency but performs lossless join

Q.130 If every non-key attribute is functionally dependent on the primary key, then the relation will be in
(a) First normal form
(b) Second normal form
(c) Third normal form
(d) Fourth normal form

Q.131 Consider a relation $R(A, B, C, D)$ with FD's $\{A \rightarrow D, B \rightarrow D, D \rightarrow BC\}$. What is the minimum number of decomposition required to make above relation BCNF?

(a) 1 (b) 2
(c) 3 (d) 4

Q.132 Relation R has attributes $A, B, C, D, E, F, G, H, I, J$ and satisfies FD's

$$ABD \rightarrow E,\ C \rightarrow J,\ AB \rightarrow G,$$
$$CI \rightarrow I,\ B \rightarrow F,\ G \rightarrow HI$$

Find candidate key

(a) A, B, C, I (b) A, B, C, D, G
(c) A, B, C, D, E (d) A, B, C, D

Q.133 A set of FDs is given for a relation $R(A, B, C, D, E, F, G)$

$$\{A \rightarrow B,\ BC \rightarrow DE,\ AEF \rightarrow G\}$$

What is the closure of $\{A, C\}^+$ under this set?

(a) A, B, C (b) A, B, C, D
(c) A, B, C, D, E, F (d) A, B, C, D, E

Q.134 Consider a relation schema $R_1 = (N, R, C, T, Z)$ on which of the following functional dependencies hold $\{N \rightarrow RCT,\ RCT \rightarrow Z,\ Z \rightarrow CT\}$ then what is/are the candidate keys of R_1?

(a) N (b) R
(c) Z, N (d) C, T

Q.135 For relation $R(A, B, C, D, E, F)$ the set of FD's is $\{A \rightarrow C,\ B \rightarrow D,\ C \rightarrow E,\ D \rightarrow E,\ E \rightarrow A,\ F \rightarrow B\}$ What is the candidate key for R?

(a) A (b) C
(c) D (d) F

Q.136 Below is an instance of R(A1, A2, A3, A4). Choose the FD which may hold on R

A1	A2	A3	A4
1	2	3	4
1	2	3	5
6	7	8	2
2	1	3	4

(a) A4 → A1 (b) A2 A3 → A4
(c) A2 A3 → A1 (d) None of these

Q.137 Let R(P Q R S T U V) satisfy the following FDs:
P → Q, RU → P, Q → T, QV → R, TU → S, QT → V
Which of the following FDs is also guaranteed to be satisfied by R?

(a) PQT → RS (b) QTU → SV
(c) RV → PQ (d) All of these

Q.138 Consider the relation R(A, B, C, D, E, F, G) with functional dependencies:
{AB → C, CD → E, EF → G, FG → E, DE → F, BC → A}
Which of the following is not a candidate key for R?

(a) BDF (b) BDFG
(c) ABDF (d) BDEF

Q.139 Given the following set of functional dependencies for a relation R(P, Q, R, S, T, U):
PQ → RS, ST → U, R → T, U → S, PT → R, T → P
Which of the following statements about attribute closures is correct?

(a) The closure of {P, T} is exactly {P, R, S, T}
(b) The closure of {S, T} includes all attributes in R
(c) The closure of {Q, T} is equal to {P, Q, R, S, T, U}
(d) The closure of {P, Q} is {P, Q, R, T, U}

Q.140 You are given a relation R(P, Q, R, S, T, U) with the following functional dependencies:
{PQ → RS, R → P, Q → T, S → Q, T → U}
How many candidate keys does this relation have?

Q.141 Suppose, a relational schema R(v, w, x, y, z) and set of functional dependencies F and G are as follow:
F : {w → x, wx → y, z → wy, z → v}
G : {w → xy, z → wx}
Check the equivalency of functional dependences F and G.

(a) F ⊆ G (b) G ⊆ F
(c) F ⊄ G (d) None of these

Q.142 Suppose a relational schema R(A B C D E F G H I) and set of functional dependencies.
F : {AB → C, AD → GH, BD → EF, A → I, H → I}

(a) 3NF not in BCNF
(b) 2NF not in 3NF
(c) 1NF not in 2NF
(d) None of these

Q.143 Consider the relation R(P, Q, S, T, X, Y, Z, W) with the following functional dependencies.

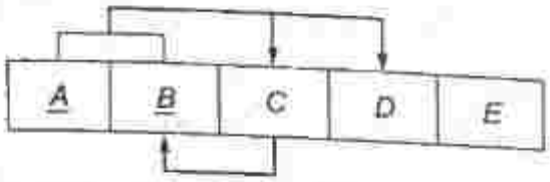
$$PQ \rightarrow X;\ PY \rightarrow X;\ Q \rightarrow Y;\ Y \rightarrow ZW$$

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

D_1: R = [(P, Q, S, T, X); (P, T, X); (Q, Y); (Y, Z, W)]
D_2: R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]
Which one of the following options is correct?
(a) D_1 is a lossy decomposition, but D_2 is a lossless decomposition.
(b) Both D_1 and D_2 are lossless decompositions.
(c) Both D_1 and D_2 are lossy decompositions.
(d) D_1 is a lossless decomposition, but D_2 is a lossy decomposition.

Q.144 A relation in which every non-key attribute is fully functionally dependent on the primary key and which has no transitive dependencies is said to be in:
(a) BCNF (b) 2NF
(c) 3NF (d) 4NF

Q.145 Consider the following table :



The table is in which normal form?
(a) First Normal Form
(b) Second Normal Form
(c) Third Normal Form but not BCNF
(d) Third Normal Form and BCNF

Q.146 Which of the following is true statement:
S_1: 2NF does not allow partial dependency in relation.
S_2: 3NF doesnot allow transitive dependency in relation.
S_3: Every 3NF decomposition is lossless and dependency preserving
(a) S_1, S_2, S_3 is true
(b) S_1, S_2 is true S_2 is false
(c) S_1, S_2, S_3 are false
(d) S_1, S_2 are false S_3 is true

Q.147 Which of the following terms refers to the method used for structuring database tables so that data duplication is reduced, consistency is maintained, and updates become easier?
(a) Design rationalization
(b) Class diagram
(c) Data normalization
(d) Database design

Q.148 Consider the following statements about a relation R(A,B) with only two attributes:
S_1: R is automatically in BCNF regardless of its functional dependencies.
S_2: If R has no functional dependencies, it satisfies all normal forms up to BCNF.
Which of the following is CORRECT?
(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are true
(d) Neither S_1 nor S_2 is true

Q.149 Given FDs {A → B, BC → D, D → E, E → C}, which FD violates 3NF if the relation has attributes A – E?
(a) BC → DA (b) A → B
(c) D → E (d) E → C

Q.150 Which of the following is TRUE about a relation that is in 3NF?
(a) It can have transitive dependencies
(b) It cannot have multiple candidate keys
(c) Every functional dependency has a superkey on the LHS
(d) It cannot have a composite key

Q.151 What distinguishes Boyce-Codd Normal Form (BCNF) from Third Normal Form (3NF)?
(a) BCNF allows transitive dependencies if the determinant is a superkey.
(b) 3NF permits non-trivial FDs where the determinant is not a superkey.
(c) BCNF requires all candidate keys to be single-attribute.
(d) 3NF is stricter than BCNF.

Q.152 Which one of the following does not allow a dependency-preserving decomposition into BCNF for the given relation R(P, Q, R, S)?
(a) P → Q, Q → R S
(b) P → Q, Q → R, R → S
(c) PQ → R, R → P S
(d) P → Q R S

Q.153 Consider a relational schema T(P, Q, R, S, T, U, V) with the following set of functional dependencies:
F = PR → Q, R → TUV, U → R, T → R
The relation is decomposed into two sub-relations:
- T_1(P, Q, R, S, T) with dependencies: PR → Q, R → T, T → R
- T_2(P, R, V) with dependencies: R → V

Which of the following statements is correct regarding the above decomposition?

(a) The decomposition is lossless and dependency preserving.
(b) The decomposition is lossy but dependency preserving.
(c) The decomposition is lossless but not dependency preserving.
(d) The decomposition is lossy and not dependency preserving.

Q.154 Why is 3NF preferred over BCNF in practical database design?

(a) 3NF always removes redundancy
(b) BCNF is never dependency preserving
(c) 3NF ensures dependency preservation with lossless join
(d) BCNF always increases number of relations

Q.155 Which of the following is true about a decomposition that is dependency preserving but lossy?

(a) It ensures redundancy removal
(b) It may result in extra tuples after join
(c) It guarantees candidate keys remain the same
(d) It is better than lossless non-preserving decomposition

3. Relational Algebra

Q.156 Which of the following sets of operations represent a complete set of relational algebra operations?

(a) $\{\sigma, \pi, \cup, -, X\}$ (b) $\{\sigma, \pi, \cup, \div, X\}$
(c) $\{\sigma, \pi, \cap, -, X\}$ (d) $\{\pi, \cup, \sigma, -, \div\}$

Q.157 Select the relational expression which could possibly return the following result.

a	c
1	2
2	3

(a) $\Pi_{a,c}(\sigma_{a=c}R)$ (b) $\Pi_{a<c}(\Pi_{a,c}R)$
(c) $\Pi_{a<2}R$ (d) $\sigma_{a<c}(\Pi_{a,c}R)$

Q.158 Consider the following two tables:

R_1

A	B	C
1	2	3
1	2	4
2	1	3
3	1	3

R_2

A	B	D
1	2	1
2	1	5
4	2	1
3	2	1

The number of rows where null entries are present in the table $R_1 \ltimes R_2$ (R_1 natural full outer join R_2) is _____.

Q.159 Let $R_1(A, B, C)$ and $R_2(D, E)$ be two relation schema, where the primary keys are shown underlined, and let C be a foreign key in R_1 referring R_2. Suppose there is no violation of the above referential integrity constraint in the corresponding relational instances r_1 and r_2. Which one of the following relational algebra expressions would necessarily produce an empty relation?

(a) $\pi_D(r_2) - \pi_C(r_2)$ (b) $\pi_C(r_1) - \pi_D(r_2)$
(c) $\pi_D(r_1 \triangleright\!\triangleleft_{C=D} r_2)$ (d) $\pi_C(r_1 \triangleright\!\triangleleft_{C=D} r_2)$

Q.160 Find the number of tuples returned by the following query? [ρ is used to rename]

$$\pi_{AD}(R \times S) - \rho_{A \leftarrow B}(\pi_{BD}(R \triangleright\!\triangleleft_{B=C} S))$$

(a) 4 (b) 5
(c) 6 (d) 7

Q.161 Assume table R has two attributes A and B. Similarly table S has two attributes A and B. Which of the following relational algebra expressions are not equivalent to $R \cap S$?

(a) $R \bowtie S$
(b) $(R \cup S) - ((R - S) \cup (S - R))$
(c) $S - (S - R)$
(d) $R - (S - R)$

Q.162 Consider the following relations R and S:

R

A	B	C
1	2	3
4	5	6
6	7	8
7	8	9

S

B	C	D
2	3	10
2	3	11
6	7	12
7	8	10

The number of tuples in $R \triangleright\!\triangleleft S$ [right outer join] are _____.

Q.163 Consider two relations $R(A_1, A_2, A_3, \ldots, A_m)$ and $S(A_1, A_2, A_3, \ldots, A_n)$ where $m > n$. Find the number of attributes that appears in R/S.

(a) $\left\lfloor \frac{n}{m} \right\rfloor$ (b) $\left\lfloor \frac{m}{n} \right\rfloor$
(c) $m - n$ (d) $m + n$

Q.164 Consider two relations enrolled and course as shown below:

Enrolled		
Sid	Cid	Fees
S1	C1	10
S1	C2	20
S2	C3	30
S3	C4	40

Course		
Cid	Cname	Dept
C1	ALGO	CS
C2	DS	CS
C3	TOC	CS
C4	THERMO	ME

$\pi_{Sid,Cid}(\text{Enrolled}) / \pi_{Cid}(\sigma_{Dept = EE}(\text{Course}))$.

If above relational algebra query executes over above data base table, then how many tuples are there in the result of query?

Q.165 Assume $A(P, Q)$ and $B(R, S)$ relations have the following instances

A

P	Q
1	2
2	1
3	3

B

R	S
1	2
2	4
2	8

The number of tuples returned by the following query (ρ is used to rename the attribute) is

$$\pi_{PS}(A \times B) - \rho_{P \leftarrow Q}\left(\pi_{QS}\left(A \bowtie_{Q=R} B\right)\right)$$

Note: Q is renamed as P.

Q.166 With respect to relational algebra, which of the following options is correct?

1. $\sigma_{c_1 \text{AND } c_2 \text{AND} \ldots c_n}(R) \equiv \sigma_{c_1}(\sigma_{c_2}(\ldots(\ldots(\sigma_{c_n}(R)\ldots))$
2. $\pi_{List_1}(\pi_{List_2}(\ldots(\pi_{List_N}(R))\ldots)) \equiv \pi_{List_1}(R)$, where $List_1 \subseteq List_2 \subseteq List_3 \ldots List_n$
3. $\pi_{A_1 A_2 \ldots A_n}(\sigma_c(R) \equiv \sigma_c(\pi_{A_1, A_2, \cdots A_n}(R))$

(a) Only 1 and 3 are correct
(b) Only 2 and 3 are correct
(c) Only 1 and 2 are correct
(d) All 1, 2 and 3 are correct

Q.167 Consider the following relation schema:

Student (Sid, Sname)
Course (Cid, Cname, Dname)
Takes (Sid, Cid, Dname)

"Find the name of all the students who have taken all the course of the department '*CS*".

Which of the following will represents the above query?

(a) $\{t \mid \exists r \in \text{Student } (r[Sid] = t[Sid]) \wedge (\forall u \in \text{Course } (u[Dname] = \text{"}CS\text{"} \Rightarrow \exists s \in \text{Takes } (t[Sid] = s[Sid] \wedge s[cid] = u[cid]))\}$

(b) $\{t \mid \exists r \in \text{Student } (r[Sid] = t[Sid]) \wedge (\forall u \in \text{Course } (u[Dname] \neq \text{"}CS\text{"} \vee \exists s \in \text{Takes } (t[Sid] = s[Sid] \wedge s[cid] = u[cid]))\}$

(c) Both (a) and (b)
(d) None of the above

Q.168 If P and Q are predicates and e is the relational algebra expression, then which of the following equivalence is valid?

(a) $\sigma_P(\sigma_Q(e)) = \sigma_Q(\sigma_P(e))$
(b) $\sigma_P(\sigma_Q(e)) = \sigma_{P \wedge Q}(e)$
(c) $\sigma_Q(\sigma_P(e)) = \sigma_{P \wedge Q}(e)$
(d) All of the above

Q.169 Given the relations:

Employee (name, salary, deptno) and
Department (deptno, deptname, address)

Which of the following queries cannot be expressed using the basic relational algebra operation $(\sigma, \pi, \times, \bowtie, \cup, \cap, -)$?

(a) Department address of every employee
(b) Employees whose name is the same as their department name
(c) The sum of all employees salaries
(d) All the employees of a given department

Q.170 What does the following query do

{*S*. rollNo, *S*. name | student (*S*) ∧ *S*.sex = '*F*' ∧ (∃*d*) (department (*d*) ∧ *d*. name = 'Maths' ∧ *d*. deptId = *S*. deptNo)}

(a) Name of all students in Maths dept.
(b) Obtain the rollNo, name of all girl students in the Maths Dept.
(c) Obtain the name of all male students in the Maths Dept.
(d) None of the above

Q.171 The following query in TRC selects:

{*S*. name | student (*S*) ∧ (∃*e*) (∃*t*) (enrollment (*e*) ∧ teaching (*t*) ∧ *e*. courseId = *t*. courseId ∧ *e*. rollNo = *S*. rollNo ∧ *t*. empId = *S*. advisor}

(a) Students who have taken all courses taught by their advisor.

(b) Students who have taken courses, not taught by their advisor.
(c) Students who have taken at least one course taught by their advisor.
(d) None of the above

Q.172 Consider the following relations:

$R_1(A\ B\ C)$ and $R_2(A\ D\ E)$

R_1 has 1000 records and R_2 has 2000 records. The non-Null attribute 'A' in R_2 is referencing attribute 'A' in R_1. Let X be minimum number of records in $R_1 \rhd\lhd R_2$ and Y be the maximum number of records in $R_1 \bowtie R_2$. The sum of $(X + Y)$ is _____.

Q.173 Consider two relations **Enrolment** and **Course** as shown below :

Enrolment

Sid	Cid	Fee
S_1	C_1	1000
S_1	C_2	2000
S_1	C_3	3000
S_2	C_2	2000

Course

Cid	Cname	Instructor
C_1	DB	Srinivas
C_2	CN	Siddharth
C_3	CO	Aditya
C_4	OS	Aashish

If following relation algebra query executes over above database table then the number of record(s) that present in result of the query are _____

$\pi_{Sid, Cid}(Enrolment) / \pi_{Cid}(\sigma_{Instructor = 'Shiva'}(Course))$

Q.174 Let $R_1, R_2, \ldots, R_n$ be a decomposition of schema U. Let $u(U)$ be a relation, and the $r_i = \Pi r_i(U)$. Which of the following holds true?

(a) $u \subseteq r_1 \rhd\lhd r_2 \rhd\lhd r_3, \ldots, \rhd\lhd r_n$
(b) $u = r_1 \rhd\lhd r_2 \rhd\lhd r_3, \ldots, \rhd\lhd r_n$
(c) $u \supseteq r_1 \rhd\lhd r_2 \rhd\lhd r_3, \ldots, \rhd\lhd r_n$
(d) None of these

Q.175 Consider the following relations:

$R(\underline{A}\ B)$ A primary key

$S(\underline{C}\ B)$ C primary key and attribute "B" of relation "R" is foreign key references to S and no NULL values in both relations and both R and S consists more than two records. Which of the following statement is false?

(a) $\pi_B(R) - \pi_C(S)$ always empty
(b) $\pi_C(S) - \pi_B(R)$ may not empty
(c) $\pi_B\left(R \underset{R.B=S.C}{\rhd\lhd} S\right) - \pi_C(S)$ always empty
(d) $\pi_B\left(R \underset{R.B \neq S.C}{\rhd\lhd} S\right) - \pi_C(S)$ may not empty

Q.176 One of the following four expressions of Relational Algebra is not equivalent to the other three. They are based on the relation $R(A, B)$ and $S(B, C)$

(a) $\pi_{AB}(R \rhd\lhd S)$
(b) $R \rhd\lhd \pi_B(S)$
(c) $R \cap (\pi_A(R) \times \pi_B(S))$
(d) $\pi_{A, R.B}(R \times S)$

Q.177 Assume $R(A, B)$ and $S(C, D)$ relations have the following instances

R

A	B
1	2
2	1
3	3

S

C	D
1	2
3	4
3	5

Find number of tuples returned by the following query (r is used to rename the attribute)

$\pi_{AD}(R \times S) - \rho_{A \leftarrow B}\left(\pi_{BD}\left(R \rhd\lhd_{B=C} S\right)\right)$

Q.178 Consider the relation $P(A, B, C)$, $Q(C, D, E)$ and $R(E, F)$ having tuples 200, 300 and 100 respectively. Estimate the number of tuples in relation $P \rhd\lhd Q \rhd\lhd R$.

Q.179 Consider the join of a relation R with a relation S. If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

(a) $m + n$ and 0
(b) mn and 0
(c) $m + n$ and $|m - n|$
(d) mn and $m + n$

Q.180 Given the relations:

- employee (name, salary, dept_no), and
- department (dept_no, dept_name, address)

Which of the following queries can be expressed using the basic relational algebra operations ($\pi, \times, \rhd\lhd, \cup, \cap, -$)?

(a) Department address of every employee
(b) Employees whose name is the same as their department name
(c) The sum of all employees' salaries
(d) All employees of a given department

Q.181 The relational algebra expression equivalent to the following tuple calculus expression:
$\{t \mid t \in r \wedge (t[A] = 10 \wedge t[B] = 20)\}$ is
(a) $\sigma_{(A=10 \vee B=20)}(r)$
(b) $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
(c) $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
(d) $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

Q.182 A library relational database system uses the following schema
USERS (User #, UserName, HomeTown)
BOOK (Book #, BookTitle, AuthorName)
ISSUED (Book #, User #, Date)
Explain in one English sentence, what each of the following relational algebra queries is designed to determine.
(a) $\sigma_{User\# = 6}(\pi_{User\#, BookTitle}((USERS \bowtie ISSUED) \bowtie BOOK))$
(b) $\sigma_{AuthorName}(BOOK \bowtie (\sigma_{HomeTown=Delhi}(USERS \bowtie ISSUED)))$

Q.183 Given the relations:
employee (name, salary, deptno), and
department (deptno, deptname, address)
Which of the following queries cannot be expressed using the basic relational algebra operations ($\sigma, \pi, \times, \bowtie, \cup, \cap, -$)?
(a) Department address of every employee
(b) Employees whose name is the same as their department name
(c) The sum of all employees salaries
(d) All employees of a given department

Q.184 Let R and S be relational schemes such that R = {a, b, c} and S = {c}. Now consider the following queries on the database :
I. $\pi_{R-S}(r) - \pi_{R-S}(\pi_{R-S}(r) \times S - \pi_{R-S,S}(r))$
II. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s(\exists v \in r(u = v[s] \wedge t = v[R - S]))\}$
III. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r(\exists u \in s(u = v[s] \wedge t = v[R - S]))\}$
IV. Select R.a, R.b
from R, S
where R.c = S.c
Which of the above queries are equivalent?
(a) I and II
(b) I and III
(c) II and IV
(d) III and IV

Q.185 Let R and S be two relations with the following schema:

R(P, Q, R1, R2, R3)

S(P, Q, S1, S2)

where (P, Q) is the key for both schemas. Which of the following queries are equivalent?
I. $\Pi_P(R \bowtie S)$
II. $\Pi_P(R) \bowtie \Pi_P(S)$
III. $P_P(P_{P,Q}(R) \cap P_{P,Q}(S))$
IV. $P_P(P_{P,Q}(R) - (P_{P,Q}(R) - P_{P,Q}(S)))$
(a) Only I and II
(b) Only I and III
(c) Only I, II and III
(d) Only I, III and IV

Q.186 Consider the following relations P(X, Y, Z), Q(X, Y, T) and R(Y, V).

P		
X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Q		
X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R	
Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?
$\Pi_x(\sigma_{(PY = RY \wedge RV = V2)}(P \times R)) - \Pi_x(\sigma_{(QY = RY \wedge QT > 2)}(Q \times R))$

Q.187 Relational schema $R(A, B, C)$ and $S(C, D, E)$ with $\{C \to A, A \to B, C \to D, D \to E\}$ functional dependencies and R has n tuples S has m tuples.
How many maximum tuples resulted by $R \rhd\lhd S$?
(a) $n \times m$
(b) $n + m$
(c) $\min(n, m)$
(d) $\max(n, m)$

Q.188 Let r and s be two relations over the relation schemas R and S respectively, and let A be an attribute in R. Then the relational algebra expression
$\sigma_{A=a}(r \rhd\lhd s)$ is always equal to
(a) $\sigma_{A=a}(r)$
(b) r
(c) $\sigma_{A=a}(r) \rhd\lhd s$
(d) None of these

Q.189 Which of the following is wrong?

(a) $\pi_{L_1 \cup L_2}(E_1 \bowtie_\theta E_2) = (\pi_{L_1}(E_1)) \bowtie_\theta (\pi_{L_1}(E_2))$

(b) $\sigma_p(E_1 - E_2) = \sigma_p(E_1) - E_2$

(c) $\sigma_{\theta_1 \wedge \theta_2}(E) = \sigma_{\theta_1}(\sigma_{\theta_2}(E))$

(d) $E_1 \bowtie_\theta E_2 = E_2 \bowtie_\theta E_1$

Q.190 Select the relational expression which could possibly return the following result:

a	c
1	2
2	3

(a) $\Pi_{a,c}(\sigma_{a<c} R)$ (b) $\Pi_{a<c}(\Pi_{a,c} R)$

(c) $\Pi_{a<2} R$ (d) $\sigma_{a<c}(\Pi_{a,c} R)$

Q.191 Given relations $R(w, x)$ and $S(y, z)$, the result of

SELECT DISTINCT w, x

FROM R, S

is guaranteed to be same as R, if

(a) R has no duplicates and S is non-empty

(b) R and S have no duplicates

(c) S has no duplicates and R is non-empty

(d) R and S have the same number of tuples

Common Data Questions 192 and 193:

Branch-scheme = (Branch - name, assets, branch-city)

Customer-scheme = (Customer-name, street, customer-city)

Deposit-scheme = (Branch-name, account-number, customer-name, balance)

Borrow-scheme = (Branch-name, loan-number, customer-name, amount)

Client-scheme = (Customer-name, banker-name)

Q.192 Using relational algebra, the query that finds customers who have a balance of over 1000 is

(a) $\pi_{\text{customer_name}}(\sigma_{\text{balance} > 1000}(\text{Deposit}))$

(b) $\sigma_{\text{customer_name}}(\pi_{\text{balance} > 1000}(\text{Deposit}))$

(c) $\pi_{\text{customer_name}}(\sigma_{\text{balance} > 1000}(\text{borrow}))$

(d) $\sigma_{\text{customer_name}}(\pi_{\text{balance} > 1000}(\text{borrow}))$

Q.193 Which of the following Domain relational calculus finds all customers who have a loan amount of over 1200?

(a) $\{<C> \mid \exists\, b, 1, a (< b, 1, c, a > \in \text{borrow} \vee a > 1200)\}$

(b) $\{<C> \mid \exists\, b, 1, a (< b, 1, c, a > \in \text{borrow} \wedge a > 1200)\}$

(c) $\{<C> \mid \exists <b, 1, c, a> \in \text{borrow} \wedge a > 1200\}$

(d) $\{<C> \mid <b, 1, c, a> \in \text{borrow} \wedge a > 1200\}$

Q.194 If P and Q are predicates and P is the relational algebra expression, then which of the following equivalence are valid?

(a) $\sigma_P(\sigma_Q(e)) = \sigma_Q(\sigma_P(e))$

(b) $\sigma_P(\sigma_Q(e)) = \sigma_{P \wedge Q}(e)$

(c) $\sigma_Q(\sigma_P(e)) = \sigma_{P \wedge Q}(e)$

(d) All of these

Q.195 Which of the following query transformation (i.e. replacing the LHS expression by the RHS expression) is incorrect? R_1 and R_2 are relations, C_1, C_2 are selection conditions and A_1, A_2 are attributes of R_1.

(a) $\pi_{A_2}(\sigma_{C_1}(R_1)) \rightarrow \sigma_{C_1}(\pi_{A_2}(R_1))$

(b) $\sigma_{C_1}(\pi_{A_1}(R_1)) \rightarrow \pi_{A_1}(\sigma_{C_1}(R_1))$

(c) $\sigma_{C_1}(R_1 \cup R_2) \rightarrow \sigma_{C_1}(R_1) \cup \sigma_{C_1}(R_2)$

(d) $\sigma_{C_1}(\sigma_{C_1}(R_1)) \rightarrow \sigma_{C_2}(\sigma_{C_2}(R_1))$

Q.196 Consider the following schemas:

Branch_Schema = (branch_name, assets, city)

Customer_Schema = (cutstomer_name, street, city)

Deposit_Schema = (branch_name, account_number, customer_name, balance)

Borrow_Schema = (branch_name, loan_number, customer_name, amount)

Which of the following tuple relational calculus finds all customers who have loan amount more than ₹ 12,000?

(a) $\{t(\text{customer_name}) \mid t \in \text{borrow} \wedge t[\text{amount}] > 12000\}$

(b) $\{t \mid t(\text{customer_name}) \mid t \in \text{borrow} \wedge t[\text{amount}] > 12000\}$

(c) $\{t \mid s \in \text{borrow} \wedge (t(\text{customer_name} = s(\text{customer_name})) s[\text{amount}] > 12000\}$

(d) $\{t \mid s \in \text{borrow} \wedge (t(\text{customer_name}) \wedge s[\text{amount}] > 12000\}$

Q.197 Consider the following three tables R, S and T. In this question, all the join operations are natural joins ($\bowtie$). (π) is the projection operation of a relation :

R	
A	B
1	2
3	2
5	6
7	8
9	8

S	
B	C
6	2
2	4
8	1
8	3
2	5

T	
A	C
7	1
1	2
9	3
5	4
3	5

Possible answer tables for this question are also given as below :

(i)

A	B	C
1	2	4
1	2	5
3	2	4
3	2	5
5	6	2
7	8	1
7	8	3
9	8	1
9	8	3

(ii)

A	B	C
1	2	2
3	2	5
5	6	4
7	8	1
9	8	3

(iii)

A	B	C
1	6	2
3	2	5
5	2	4
7	8	1
9	8	3

(iv)

A	B	C
3	2	5
7	8	1
9	8	3

What is the resulting table of

$\pi_{A,B}(R \bowtie T) \bowtie \pi_{B,C}(S \bowtie T)$?

(a) (i) (b) (ii)
(c) (iii) (d) (iv)

Q.198 Consider the following schemas :
Branch = (Branch-name, Assets, Branch-city)
Customer = (Customer-name, Bankname, Customer-city)
Borrow = (Branch-name, loan number, customer account-number)
Deposit = (Branch-name, Accountnumber, Customer-name, Balance) Using relational Algebra, the Query that finds customers who have balance more than 10,000 is _____.

(a) $\pi_{\text{customer-name}}(\sigma_{\text{balance} > 10000}(\text{Deposit}))$
(b) $\sigma_{\text{customer-name}}(\sigma_{\text{balance} > 10000}(\text{Deposit}))$
(c) $\pi_{\text{customer-name}}(\sigma_{\text{balance} > 10000}(\text{Borrow}))$
(d) $\sigma_{\text{customer-name}}(\pi_{\text{balance} > 10000}(\text{Borrow}))$

Q.199 Which of the following operation results same output table as given input table?

P	Q
3	4
6	8
9	11
10	14

(a) $\Pi_{P,Q}(\sigma_{P=Q}R)$ (b) $\Pi_{P<Q}(\Pi_{P,Q}R)$
(c) $\Pi_{P<2}R$ (d) $\sigma_{P<Q}(\Pi_{P,Q}R)$

Q.200 Consider two relations Enrolment and Course as shown below :

Enrolment

Sid	Cid	Fee
S_1	C_1	1000
S_1	C_2	2000
S_1	C_3	3000
S_2	C_2	2000

Course

Cid	Cname	Instructor
C_1	DB	Srinivas
C_2	CN	Siddharth
C_3	CO	Aditya
C_4	OS	Aashish

If following relation algebra query executes over above database table then the number of record(s) that present in result of the query are _____.

$\pi_{\text{Sid, Cid}}(\text{Enrolment}) / \pi_{\text{Cid}}(\sigma_{\text{Instructor = Shiva}}(\text{Course}))$

Q.201 Consider the following relations :

$R_1(P\ Q\ R)$ and $R_2(P, S, T)$

R_1 has 1000 records and R_2 has 2000 records. The non-Null attribute 'P' in R_2 is referencing attribute 'P' in R_1. Let X be minimum number of records in $R_1 \bowtie R_2$ and Y be the maximum number of records in $R_1 \bowtie R_2$. $X + 2Y$ is _____.

Q.202 Consider drinker data base with the following relations:
Drinker (drinkerName, age, city)
Bar (barName, owner, city)
Frequent (drinkerName, barName)
Which of the following query finds all drinkers who frequents only bars on same city where he lives?

(a) $\pi_{\text{drinkerName}}\left(\text{Frequent} \rhd \left(\text{Drinker} \rhd_{\text{Drinker.city = Bar.city}} \text{Bar}\right)\right)$

(b) $\pi_{\text{drinkerName}}(\text{Frequent}) - \pi_{\text{drinkerName}}\left(\text{Frequent} \rhd \left(\text{Drinker} \rhd_{\text{Drinker.city} \neq \text{Bar.city}} \text{Bar}\right)\right)$

(c) Both (a) and (b)
(d) None of these

4. Structured Query Language (SQL)

Q.203 Consider the set of relations given below and the SQL query that follows:
Students: (Roll_number, Name, Date_of_birth)
Courses: (Course_number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)

```
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE Students.Roll_number =
Grades.Roll_number
        AND Courses.Instructor = Sriram
        AND Courses.Course_number =
        Grades.Course_number
        AND Grades.Grade = A
```

Which of the following sets is computed by the above query?

(a) Names of Students who have got an A grade in all courses taught by Sriram
(b) Names of Students who have got an A grade in all courses
(c) Names of Students who have got an A grade in at least one of the courses taught by Sriram
(d) None of the above

Q.204 Consider the following schema:
Sailors(sid, sname, rating, age)
Boats(bid, bname, colour)
Reserves(sid, bid, day)
Two boats can have the same name but the colour differentiates them.
The two relations
ρ(Tempsids, ($\pi_{sid, bid}$ Reserves)/
(π_{bid} ($\sigma_{bname = \text{'Ganga'}}$ Boats))),
π_{sname} (Tempsids $\bowtie$ Sailors)
If / is division operation, the above set of relations represents the query

(a) Names of sailors who have reserved all boats called Ganga,
(b) Names of sailors who have not reserved any Ganga boat
(c) Names of sailors who have reserved at least one Ganga boat
(d) Names of sailors who have reserved at most one Ganga boat

Q.205 Consider the following schema:
Employee (name, sex, salary, deptName)
Consider the following SQL query

```
SELECT deptName
FROM Employee
WHERE sex = 'M'
GROUP by DeptName
Having avg (Salary) > {select avg (Salary) from
Emp}
```

The output will be

(a) Average salary of male employee is the average salary of the organization
(b) Average salary of male employee is less than the average salary of the organization
(c) Average salary of male employee is equal to the average salary of the organization
(d) Average salary of male employee is more than the average salary of the organization

Q.206 Consider the following relational query on the above database:

```
SELECT S.name
FROM Suppliers S
Where S.sid NOT IN (SELECT C.sid
    FROM Catalog C
    WHERE C.pid NOT IN (SELECT P.pid
        FROM Parts P
        WHERE P.color < > "blue"))
```

Assume that relations corresponding to the above schema are not empty. Which of the following is the correct interpretation of the above query?

(a) Find the names of all suppliers who have supplied non-blue part.
(b) Find the names of all suppliers who have not supplied non-blue part.
(c) Find the names of all suppliers who have supplied only non-blue parts
(d) Find the names of all suppliers who have not supplied only non-blue part.

Q.207 The SQL expression:

```
Select distinct T.branch_name from
branch T, branch S
where T.assets > S.assets and S.branch_city =
"Mumbai" finds the names of
```

(a) All branches that have greater assets than some branch located in Mumbai.
(b) All branches that have greater assets than all branches in Mumbai.
(c) The branch that has greatest asset in Mumbai.
(d) Any branch that has greater assets than any branch in Mumbai.

Q.208 The employee information of an Organization is stored in the relation:

Employee (name, sex, salary, deptname)

Consider the following SQL query

```
Select deptname from Employee
    Where sex = 'M'
    group by deptname
    having avg (salary) >
    (select avg (salary) from Employee)
```

Output of the given query corresponds to

(a) Average salary of employee more than average salary of the organization.

(b) Average salary less than average salary of the organization.

(c) Average salary of employee equal to average salary of the organization.

(d) Average salary of male employees in a department is more than average salary of the organization.

Q.209 Consider a database table *R* with attributes *A* and *B*. Which of the following SQL queries is illegal?

(a) SELECT A FROM R;

(b) SELECT A, COUNT(*) FROM R;

(c) SELECT A, COUNT(*) FROM R GROUP BY A;

(d) SELECT A, B, COUNT(*) FROM R GROUP BY A, B;

Q.210 Consider the following ORACLE relations:

$R(A, B, C) = \{<1, 2, 3>, <1, 2, 0>, <1, 3, 1>, <6, 2, 3>, <1, 4, 2>, <3, 1\ 4>\}$

$S(B, C, D) = \{<2, 3, 7>, <1, 4, 5>, <1, 2, 3>, <2, 3, 4>, <3, 1, 4>\}$.

Consider the following two SQL queries SQ_1 and SQ_2:

SQ_1 :
```
SELECT RB, AVG (SB)
FROM R, S
WHERE R·A = S·C AND S·D < 7
GROUP BY R·B;
```

SQ_2 :
```
SELECT DISTINCT S·B, MIN (S·C)
FROM S GROUP BY S·B
HAVING COUNT (DISTINCT S·D) > 1;
```

If *M* is the number of tuples returned by SQ_1 and *N* is the number of tuples returned by SQ_2 then

(a) $M = 4, N = 2$ (b) $M = 5, N = 3$

(c) $M = 2, N = 2$ (d) $M = 3, N = 3$

Q.211 Consider the following relation :

Works (emp_name, company_name, salary)

Here, emp_name is primary key.

Consider the following SQL query

```
Select emp_name
From works T
where salary > (select avg (salary)
                from works S
                where T.company _ name =
                      S.company _ name)
```

The above query is for following :

(a) Find the highest paid employee who earns more than the average salary of all employees of his company.

(b) Find the highest paid employee who earns more than the average salary of all the employees of all the companies.

(c) Find all employees who earn more than the average salary of all employees of all the companies.

(d) Find all employees who earn more than the average salary of all employees of their company.

Q.212 What is SQL primarily used for in the context of relational databases?

(a) To design user interfaces

(b) To create and manipulate databases

(c) To display data on web pages

(d) To format printed reports

Q.213 Given below are two statements to find the sum of salaries of all employees of the English department as well as the maximum, minimum and average salary in English department.

STATEMENT-I:

```
SELECT SUM (SALARY), MAX (SALARY), MIN
(SALARY),
AVG (SALARY) FROM EMPLOYEE,
DEPARTMENT
WHERE DEPTNO = DID
AND DNAME = 'ENGLISH';
```

STATEMENT-II:

```
SELECT SUM (SALARY), MAX (SALARY), MIN
(SALARY),
AVG (SALARY) FROM EMPLOYEE,
DEPARTMENT
WHERE DNAME = 'ENGLISH';
```

In the light of the above statements, choose the correct answer from the options given below:

(a) Both Statement-I and Statement-II are true
(b) Both Statement-I and Statement-II are false.
(c) Statement-I is correct but Statement-II is false.
(d) Statement-I is incorrect but Statement-II is true.

Q.214 Which of the following query/queries return the employee ID and name of employees whose salary is greater than the salary of all employees in department number 20 of university. Order result by employee ID (refer table structures given above).

A. SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY>(SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20) ORDER BY EID;

B. SELECT EID, NAME
FROM EMPLOYEE WHERE SALARY > (SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20);

C. SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY > ALL(SELECT SALARY FROM EMPLOYEE WHERE DEPTNO = 20) ORDER BY EID

Choose the correct answer from the options given below:

(a) A and B only (b) A and C only
(c) B only (d) C only

Q.215 Which of the following is/are correct?

(a) An SQL query automatically eliminates duplicates
(b) An SQL query will not work if there are no indexes on the relations
(c) SQL permits attribute names to be repeated in the same relation
(d) None of the above

Common Data for Questions 216 :

Consider the following relations A, B and C:

A.

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B.

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C.

Id	Phone	Age
10	2200	02
99	2100	01

Q.216 How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > All (SELECT B.Age
                   FROM B
                   WHERE B.Name = 'Arun')
```

(a) 4 (b) 3
(c) 0 (d) 1

Q.217 The relation book (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL:

```
select title
from book as B
    where (select count (*)
        from book as T
        where T. price > B. Price) < 5
```

(a) Titles of the four most expensive books
(b) Title of the fifth most inexpensive book
(c) Title of the fifth most expensive book
(d) Titles of the five most expensive books

Q.218 Consider the set of relations given below and the SQL query that follows:

Students: (Roll_number, Name, date_of_birth)
Courses: (Course_number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)

```
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE  Students. Roll_number = Grades. Roll_number
    AND Course. Instructor = Korth
    AND Courses.Course_number = Grades. Course_number AND Grades. Grade = A
```

Which of the following sets is computed by the above query?

(a) Names of students who have got an A grade in all courses taught by Korth.
(b) Names of students who have got an A grade in all courses.
(c) Names of students who have got an A grade in at least one of the courses taught by Korth.
(d) None of the above

Q.219 Consider the following relational schema pertaining to a students database.
Student: (rollno, name, address)
Enroll: (rollno, courseno, coursename)
Where the primary keys are shown underlined. The number of tuples in the Student and the Enroll tables are 120 and 8 respectively. What are the maximum and the minimum number of tuples that can be present in (Student * Enroll), where * denotes natural join?
(a) 8, 8
(b) 120,8
(c) 960, 8
(d) 960, 120

Q.220 The employee information in a company is stored in the relation. Assume name is the primary key:
Employee: (name, sex, salary, deptName)
Consider the SQL query;

```
SELECT deptName
FROM Employee
WHERE sex = M
GROUP By deptName
HAVING avg(salary) > (SELECT avg(salary)
                      FROM Employee)
```

It returns the names of the departments in which the average salary
(a) Is more than the average salary in the company.
(b) Of the male employees is more than the average salary of all the male employees in the company.
(c) Of the male employees is more than the average salary of the employees in the same department.
(d) Of the male employees is more than the average salary in the company.

Q.221 There are two relations X and Y. Relation X has arity 1 and cardinality 2, relation Y has arity 3 and cardinality 4. Indicate the result of the SQL statement SELECT COUNT (*) FROM X, Y.
(a) 4
(b) 6
(c) 8
(d) Can not be calculated from given information

Common Data for Questions 222 and 223 :
Consider the relation S (S#, Sname, Status, City), P (P#, Pname, color, weight, city) and SP (S#, P#, Qty) where S supplier and P is the parts database.

Q.222 Consider the following SQL query :

```
SELECT SP.P#
FROM SP
GROUP By SP.P#
HAVING Count (SP.S#) > 1;
```

The above query represent which of the following?
(a) Get part number for part supplied by one supplier.
(b) Get part number for part supplied by more than one supplier.
(c) Get part number of supplier who supply more than on parts.
(d) None of the above

Q.223 Consider the following SQL query :

```
SELECT Distinct S.name
FROM S
WHERE Exists SP.P#
(SELECT * From SP
WHERE SP.S# = S.S# AND SP.P# = 'P2');
```

The above query represents which of the following?
(a) Get supplier names for suppliers who supply part P_2.
(b) Get supplier names who doesn't supply part P_2.
(c) Get supplier names whose supply part P_2 is null.
(d) None of the above

Q.224 Which of the following is correct?
(a) An SQL query automatically eliminates duplicates.
(b) AN SQL query will not work if there are no indexes on the relations.
(c) SQL permits attribute names to be repeated in the same relation.
(d) None of the above

Q.225 In SQL, relations can contain null values, and comparisons with null values are treated as unknown. Suppose all comparisons with a null value are treated as false. Which of the following pairs is not equivalent?

(a) $x = 5$ not (not $(x = 5)$)

(b) $x = 5, x > 4$ and $x < 6$, where x is an integer

(c) $x \neq 5$ not $(x = 5)$

(d) None of the above

Q.226 Which of the following is correct statement about SQL?

(a) Data Control Language provides facilities to retrieve data.

(b) Data Manipulation Language can delete all records and tables in a database.

(c) Data Definition Language provides facilities to add more data into a database while providing integrity of data.

(d) Data Definition Language provides facilities to modify structures of tables.

Q.227 Consider two tables in a relational database with columns and rows as follows:

Table : Student

Roll_no	Name	Dept_id
1	ABC	1
2	DEF	2
3	GHI	3
4	JKL	4

Table : Department

Dept_id	Dept_name
1	A
2	B
3	C

Roll_no is the primary key of the student table, Dept_id is the primary key of the department table and student. Dept_id is a foreign key from Department. Dept_id

What will happen if we try to execute the following two SQL statements?

1. Update student set Dept_id = Null where Roll_no = 1
2. Update Department set Dept_id = Null where Dept_id = 1

(a) Both 1 and 2 will fail

(b) 1 will fail 2 will succeed

(c) 1 will succeed but 2 will fail

(d) Both 1 and 2 will succeed

Q.228 Let $R(A, B, C)$ and $S(B, C, D)$ be the relations as given below:

R

A	B	C
a1	b1	c2
a2	b4	c1
a3	b3	c3
a2	b3	c5
a1	b1	c2

S

B	C	D
b1	c1	d2
b2	c4	d1
b3	c3	d4

What is the result of following query?

SELECT R.A FROM R WHERE R.B = NOT ALL (SELECT S.B FROM S WHERE R.C. = S.C)

(a)

A
a1
a3
a2
a1

(b)

A
a1
a3
a2

(c)

A
a3
a2

(d)

A
a1
a3
a2
a3

Q.229 A company has a storeroom, which consists of few employees who order products from different shops. Following is the snapshot of the "product-order" table maintained at company's database

Product-Order

Product-no	Employee-no	Shop-no
P1	E6	S2
P2	E2	S1
P3	E7	S4
P4	E5	S3
P5	E2	S3
P6	E7	S2

Consider the following query:

SELECT A.product_no, A.employee_no
FROM product_order A, product_order B
WHERE A.employee_no = B.employee_no
and A.shop_no < > B.shop_no;

The number of tuples contained in the output will be

(a) 3 (b) 4

(c) 5 (d) 2

Q.230 In SQL the statement Select * from R, S is equivalent to
(a) Select * From R natural join S
(b) Select * From R cross join S
(c) Select * From R union join S
(d) Select * From R inner join S

Q.231 Consider the following schema :
$R(\underline{C}, D)$, $Q(\underline{B}, C)$, $P(\underline{A}, B)$
C is foreign key in Q referencing $R(C)$ on delete cascade, B is foreign key in P referencing $Q(B)$ on delete set null. Suppose current content of P, Q, R as follows.

P

A	B
a	a
b	b

Q

B	C
a	a
b	a

R

C	D
a	a
b	a

After executing the following query, what are the tuples in P?

delete from R

(a) (a, NULL) and (b, b)
(b) (a, NULL) and (b, NULL)
(c) (b, b) only
(d) P will not be changed

Q.232 Consider the relations :
Supplier (S_no, S_name, city) and
Item (Item_no, Item_name, Item_Brand, S_no)
Which of the following SQL statement gives suppliers names who have submitted at least one 'Nestle' brand item?
(a) SELECT S_name FROM Supplier S, Item I WHERE I. Item_Brand = 'Nestle'
(b) SELECT S_name FROM Supplier S WHERE S_no IN (SELECT S_no From Item WHERE Item_Brand = 'Nestle')
(c) SELECT S_name FROM Supplier S WHERE Item_Brand = 'Nestle'
(d) SELECT S_name FROM Supplier WHERE S_no EXISTS (SELECT S_no FROM Supplier, Item WHERE Item_Brand = 'Nestle')

Q.233 The standard SQL "ORDER BY" clause
(a) can be used to order the columns used in select.
(b) may only be used when "GROUP BY" is used.
(c) can be used any place of query to order the tuples.
(d) can only be used as the last clause of a query to order the tuples.

Q.234 Which of the following SQL query is illegal
(a) Select max (count(*)) from R group by A;
(b) Select count(*), max (B) from R group by A;
(c) Select count(*) from R group by A;
(d) Select count(*) from R;

Q.235 There are two relations X and Y. Relation X has arity 1 and cardinality 2, relation Y has arity 3 and cardinality 4. Indicate the result of the SQL statement SELECT COUNT (*) FROM X, Y.
(a) 4 (b) 6
(c) 8 (d) 3

Q.236 Consider the following relations:
Student (snum, sname, major, level, age)
Class (name, meets_at, room, fid)
Enrolled (snum, cname)
Faculty (fid, fname, deptid)
What does the following query find?
SELECT C.name
FROM Class C
WHERE C.room = 'R128'
or C.name in (SELECT E.Cname) FROM Enrolled E GROUP BY E.Cname HAVING count(*) >= 5)
(a) Finds the names of all classes that either meet in room R128 and have five or more students enrolled
(b) Finds the names of all classes that either meet in room R128 or have five or more students enrolled
(c) Finds the name of atmost one class that either meet in room R128 or have five or more students enrolled
(d) Finds the names of all classes that meet in room R128 that has five or more students enrolled in it

Q.237 Consider the following relation SQL query:

R

A	B	C
4	Null	5
5	6	Null
6	Null	6
10	8	Null
8	8	5

Select *
From R
Where $A \geq 5$ and ($B \geq C$ or $C \geq 5$)

How many records resulted by above query on given instant of relation R?
(a) 1 (b) 2
(c) 3 (d) 4

Q.238 Consider the following statements:

S_1: Select operation in SQL is equivalent to the projection operation in relational algebra, except that select SQL contain duplicates.

S_2: A SQL query can have a 'having' clause without containing 'group by' clause.

S_3: Operator 'between' can be used between main query and the sub query.

S_4: If two relations R and S are joined, then the non-matching tuples of both R and S are ignored in full outer join.

Which of the above statements are correct?

(a) Only S_1
(b) Only S_1, S_3 and S_4
(c) Only S_1 and S_3
(d) All of these

Q.239 Consider the following relational schema:

R	A	B	C
	4	5	6
	3	4	5
	7	2	9
	8	5	6

The number of records resulted by below SQL query are _____.

```
Select *
From R
Where C = ALL (select B
               From R
               Where A > 10)
```

Q.240 SQL allows duplicate tuples in relations, and correspondingly defines the multiplicity of tuples in result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

select * from R where a in (select S.a from S)

(a) select R.* from R, S where R.a = S.a
(b) select distinct R.* from R,S where R.a = S.a
(c) select R.* from R, (select distinct a from S) as S1 where R.a = S1.a
(d) select R.* from R, S where R.a = S.a and is unique R

Q.241 Consider the following relations :

Bank(bname, city)
Travel(pname, city)

Query: SELECT T1.pname
FROM Travel T1
WHERE NOT EXISTS (SELECT B.city
From Bank B
WHERE B.bname = 'SBI'
EXCEPT
SELECT T2.city
FROM Travel T2
WHERE T1.pname = T2.pname)

The above query finds name of the persons

(a) Who have not travelled in any city where SBI is located
(b) Who have not travelled in all city where SBI is located
(c) Who have travelled in all city where SBI is located
(d) Who have travelled in any city where SBI is located

Q.242 An athletics meeting involves several competitors who participate in a number of events. The database is intended to record who is to take part in which event and to record the outcome of each event. As results become available the winner attribute will be updated with the cid of the appropriate competitor.

Competitor (cid, name, nationality)
Even (eid, description)
Competes (cid, eid)

Competitor		
cid	name	nationality
01	Pat	British
02	Hilary	British
03	Sven	Swedish
04	Pierre	French

Event	
eid	description
01	running
02	jumping
03	throwing

Competes	
cid	eid
01	01
02	01
03	02
04	02
04	03

What would be the result of the following SQL statement.

```
SELECT eid FROM Competes, Competitor
WHERE Competes. cid = Competitor. cid
AND nationality = 'Swedish';
```

(a) 01 (b) 02
(c) 04 (d) 03

Q.243 Which is the subset of SQL commands used to manipulate Database Structures, including tables?

(a) Data Definition Language
(b) Data Manipulation Language
(c) Data Described Language
(d) Data Retrieval Language

Q.244 Consider the following relational schema :
Teacher (Tid, Tname, Dept) is Teacher relation
Course (Cid, Tid, Grade, Dept) is course enroll relation
What is the output of following SQL query?

```
Select *
from Teacher T
Where not Exists (Select *
        from course C
        where C.T id = T.Tid and C.Grade < 9)
```

(a) Displays Teachers who never got a grade above 9 in madeeasy.
(b) Displays Teachers who never got a grade less than or equal to 9 in some course in madeeasy.
(c) Displays Teachers who got a grade greater than equals to 9 in all course in madeeasy.
(d) Displays Teachers who got a grade less than 9 in all course in madeeasy.

Q.245 Consider the following database table :
Emp (Eid, Ename, age)
Project (Pid, Pname, budget)
Works for (Eid Pid)

```
Select Eid
From Emp E
where age > 30 and not Exists (select Pid
        From project P
        where Pname = 'database' and not
        exist (select Pid
                from works W
                where W.Eid = E.Eid
                and W.Pid = P.Pid))
```

Which of the fo'llowing sets is computed by the above query retrieves employees whose
(a) age more than 30 and works for every project with project name database.
(b) age more than 30 and works for some project with project name database.
(c) age more than 30 and not works for every project with project name database.
(d) age more than 30 and not works for any project with project name database.

Q.246 Consider the following relational schema $R(\underline{AB})$ and $S(\underline{CD})$:

```
Query 1: Select A
         From R
         Where EXISTS (Select *
                       From S
                       Where R. A > S.C
```

Query 2: $\pi_A(R \bowtie_{R.A > S.C} S)$
Query 3:

```
Select A
From R
Where A > ALL (Select C
               From S)
```

Which of the following statement is true?
(a) Query 1 and Query 2 results same but not Query 3.
(b) Query 1 and Query 3 results same but not Query 2.
(c) Query 2 and Query 3 results same but not Query 1.
(d) All Query 1, Query 2, Query 3 results same.

Q.247 Edges of the directed graph stored in database relation Adj(X, Y) [i.e. if there exists edge from vertex A to B then (A, B) is one record in Adj relation]. Which of the query correct representation to retrieve vertices with only degree two?

I.
```
SELECT X
FROM Adj
Group by X
having count (*) = 2
```

II. $\pi_X(Adj \bowtie_{X = X1 \wedge Y \neq Y1} \rho_{X1, Y1}(Adj))$

(a) Only I correct (b) Only II correct
(c) Both I, II correct (d) Both I, II incorrect

Q.248 Consider the following relational schema :

R	A	B	C
	4	5	6
	3	4	5
	7	2	9
	8	5	6

The number of records resulted by below SQL query are _____.

```
Select *
From R
Where C = ALL (select B
               From R
               Where A > 10)
```

Q.249 Consider a relation Employee (E-id, Ename, Salary) and the given query is executed on the relation

```
SELECT Ename, SUM (Salary)
from Employee
group by Ename
Having sum (Salary) < 10000;
```

Which of the following will represents the output of above query?

(a) Name of all the Employee with their respective Salary is less than 10000.
(b) Name of all the Employee having same name with their individual Salary is less than 10000.
(c) Name of the Employee having same name with their total Salary less than 10000.
(d) None of these

Q.250 There is a database of the dealers of a Products_deal. Following is the snapshot.

Product_deal

Dealer-no	Part-no	Color-id
D_6	P_2	C_1
D_2	P_1	C_2
D_7	P_4	C_3
D_5	P_3	C_4
D_2	P_3	C_5
D_7	P_2	C_6

Consider the following query:
SELECT A.Color-id, A.Dealer-no
FROM Product_deal *A*, Product_deal *B*
WHERE A.Dealer-no = *B*.Dealer-no
and *A*.Part-no < > *B*.Part-no;
The number of tuples contained in the output will be _____.

Q.251 Consider the following SQL queries on relation Student (Name, Marks).

S_1 : SELECT S.name
FROM Student S
WHERE S.marks = ANY (SELECT marks
FROM Student
WHERE S.name = 'Deepti');

S_2 : SELECT S.name
FROM Student S
WHERE S.marks IN (SELECT marks
FROM Student
WHERE S.name = 'Deepti');

S_3 : SELECT S.name
FROM Student S
WHERE S.marks = ALL (SELECT marks
FROM Student
WHERE S.name = 'Deepti');

Which one of the following is true?

(a) S_1, S_2 are same
(b) S_2, S_3 are same
(c) S_1, S_2, S_3 are same
(d) None of these

Q.252 Consider Employee Table has 1000 records, it has a non-null 'Salary' column which is also unique. Consider the following SQL query.

SELECT COUNT(*)
FROM Employee
WHERE Salary > ANY (SELECT Salary FROM Employee);

What will be the value printed by above query?

(a) 1000 (b) 999
(c) 500 (d) 100

5. File Organization and Indexing

Q.253 Given a block can hold either 3 records or 10 key pointers. A database contains *n* records, then how many blocks do we need to hold the data file and the dense index

(a) $\frac{13n}{30}$ (b) $\frac{n}{3}$
(c) $\frac{n}{10}$ (d) $\frac{n}{30}$

Q.254 Calculate the order of leaf (P_{leaf}) and non leaf (P) nodes of a B^+ tree based on the information given below:

Search key field = 12 field
Record pointer = 10 bytes
Block pointer = 8 bytes
Block size = 1 KB

(a) $P_{leaf} = 51$ and $p = 46$
(b) $P_{leaf} = 47$ and $p = 52$
(c) $P_{leaf} = 46$ and $p = 51$
(d) $P_{leaf} = 52$ and $p = 47$

Q.255 A B-tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

(a) 5 (b) 4
(c) 1 (d) 2

Q.256 Which of the following is dense index?

(a) Primary index
(b) Clusters index
(c) Secondary index
(d) Secondary non key index

Q.257 B^+ tree are preferred to binary tree in Database because

(a) Disk capacity are greater than memory capacities

(b) Disk access is much slower than memory access

(c) Disk data transfer rates are much less than memory data transfer rate

(d) Disks are more reliable than memory

Q.258 A/B^+ tree index is to be built on the name attribute of the relation STUDENT. Assume that all students names are of length 8 bytes, disk block are of size 512 bytes and index pointers are of size 4 bytes. Given this scenario what would be the best choice of the degree (i.e. the number of pointers per node) of the B^+ tree?

(a) 16 (b) 42

(c) 43 (d) 44

Q.259 For any B-tree of minimum degree $t \geq 2$, every node other than the root must have atleast _____ keys and every node can have at most _____ keys.

(a) $t-1, 2t+1$ (b) $t+1, 2t+1$

(c) $t-1, 2t-1$ (d) $t+1, 2t-1$

Q.260 The upper bound and lower bound for the number of leaves in a B-tree of degree K with height h is given by :

(a) K^h and $2\left\lceil \frac{K}{2} \right\rceil^{h-1}$

(b) $K*h$ and $2\left\lfloor \frac{K}{2} \right\rfloor^{h-1}$

(c) K^h and $2\left\lfloor \frac{K}{2} \right\rfloor^{h-1}$

(d) $K*h$ and $2\left\lceil \frac{K}{2} \right\rceil^{h-1}$

Q.261 A clustering index is created when _____.

(a) primary key is declared and ordered,

(b) no key ordered

(c) foreign key ordered

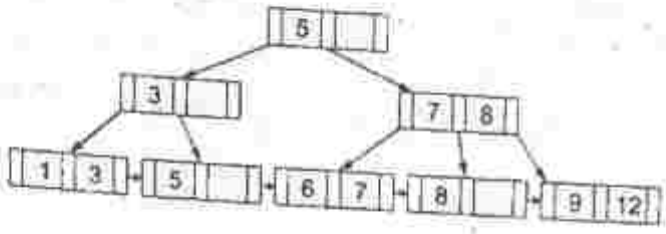
(d) there is no key and no order

Q.262 If following sequence of keys are inserted in a B^+ tree with $K(=3)$ pointers :

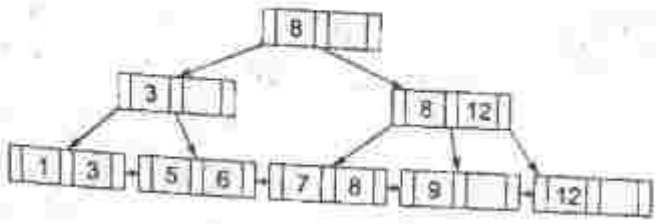
8, 5, 1, 7, 3, 12, 9, 6

Which of the following shall be correct B^+ tree?

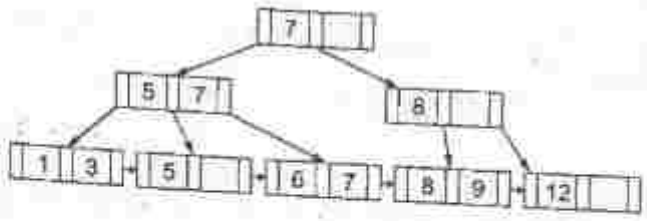
(a)



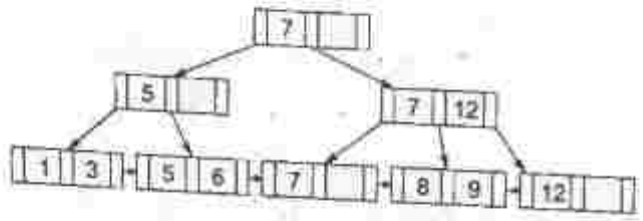
(b)



(c)



(d)



Q.263 Which of the following is correct?

(a) B-trees are for storing data on disk and B^+ trees are for main memory.

(b) Range queries are faster on B^+ trees.

(c) B-trees are for primary indexes and B^+ trees are for secondary indexes.

(d) The height of a B^+ tree is independent of the number of records.

Q.264 The order of an internal node in a B^+ tree index is the maximum number of children it can have. Suppose that a child pointer takes 6 bytes, the search field value takes 14 bytes, and the block size is 512 bytes. What is the order of the internal node?

(a) 24 (b) 25

(c) 26 (d) 27

Q.265 Which one of the following is a key factor for preferring B^+ trees to binary search trees for indexing database relations?

(a) Database relations have a large number of record

(b) Database relations are sorted on the primary key

(c) B^+ trees require less memory than binary search trees

(d) Data transfer from disks is in blocks

Q.266 Consider a B^+ tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

(a) 1 (b) 2
(c) 3 (d) 4

Q.267 A file is organized so that the ordering of data records is not the same as or close to the ordering of data entries in some index. Then that index is called

(a) Dense (b) Sparse
(c) Clustered (d) Secondary

Q.268 Which of the following is not a file operation?

(a) FindNext (b) Find
(c) Reset (d) None of these

Q.269 Index sequential file is made of all of these expect

(a) Primary data storage area
(b) Overflow area
(c) Hierarchy of indices
(d) Address of prime data track

Q.270 B^+ trees are considered BALANCED because

(a) The lengths of the paths from the root to all leaf nodes are all equal.
(b) The lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
(c) The number of children of any two non-leaf sibling nodes differ by at most 1.
(d) The number of records in any two leaf nodes differ by at most 1.

Q.271 A clustering index is defined on the fields which are of type

(a) Non-key and ordering
(b) Non-key and non-ordering
(c) Key and ordering
(d) Key and non-ordering

Q.272 Which allocation scheme would work best for a file system implemented on a device that can only be accessed sequentially, a tape drive, for instance?

(a) Contiguous allocation
(b) Non Contiguous allocation
(c) Indexed allocation
(d) None of the above

Q.273 Consider the following statements:

S_1 : Anchor block contains first records of chain.
S_2 : Anchor record is first record of a block.

Which of the above statements is/are true?

(a) S_1 only
(b) S_2 only
(c) Both S_1 and S_2 are true
(d) Both S_1 and S_2 are false

Q.274 Given a block can hold either 3 records or 30 key pointers. A database contains n records. How many blocks do we need to hold the data file and the dense index.

(a) $\frac{13n}{30}$ (b) $\frac{n}{3}$
(c) $\frac{n}{10}$ (d) $\frac{n}{30}$

Q.275 A B tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

(a) 5 (b) 4
(c) 3 (d) 2

Q.276 Which is the best suitable for sequential access of data?

(a) B tree (b) B^+ tree
(c) Both (a) and (b) (d) None of these

Q.277 In a database file structure, the search key field is 9 bytes long, the block size is 512 bytes, a record pointer is 7 bytes and a block pointer is 6 bytes. The largest possible order of a leaf node in a B^+ tree implementing this file structure is

(a) 23 (b) 24
(c) 34 (d) 44

Q.278 Which of the following statement is true?

(a) A secondary index should be sparse.
(b) A second level index should be dense.
(c) For queries of the form $\sigma_{A=a}(R)$ hashing is usually better than a B^+ tree.
(d) For queries of the form $\sigma_{A>a}(R)$ hashing is usually better than a B^+ tree.

Q.279 In a database file structure, the search key field is 9 bytes long, the block size is 512 bytes, a record pointer is 7 bytes and a block pointer is 6 bytes. The largest possible order of a non-leaf node is a B^+ tree implementing this file structure is

(a) 23 (b) 24
(c) 34 (d) 44

Q.280 Each block of *B* tree of order '*n*' can have
(a) $n-1$ key values, and n pointers
(b) $n+1$ key values, and n pointers
(c) n key values, and $n+1$ pointers
(d) Cannot be determined

Q.281 What will be the order (p) of a B^+ tree with a database of 5,00,000 records of 200 bytes each and the search key is 15 bytes? Assume the tree and data pointers are 5 bytes each and the index node (data block size) is 1024 bytes.
(a) 50 (b) 51
(c) 1024 (d) 15

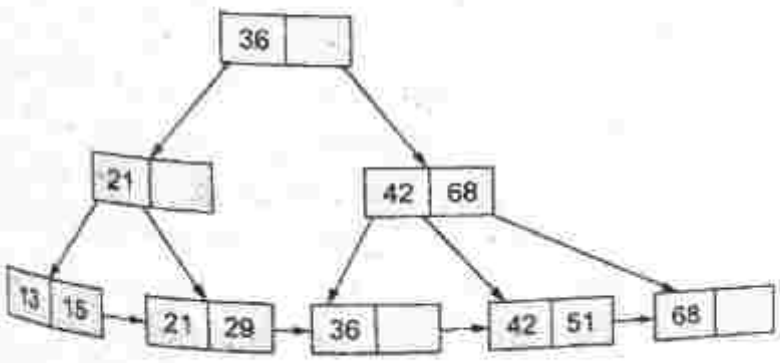
Q.282 A database relation has 5000 records block can hold either 10 records or 15 keys and pointer pairs. If sparse index is used at 1st level and multilevel indexing is used in system, then the number of disk block required to store relation and index is _____.

Q.283 Which of the following file organization provides very fast access to any arbitrary record of a file?
(a) Ordered file
(b) Unordered file
(c) Hashed file
(d) B^+ tree

Q.284 Assume a system has the following parameters to construct a B^+ tree:
- System block size of 232 bytes.
- Block pointers are of 12 bytes each.
- Record pointers are of 12 bytes each.
- Index keys are of 8 bytes each.

Find the maximum order of the B^+ tree, if the block can contain at most one B^+ node in the system?
(a) 9 (b) 10
(c) 11 (d) 12

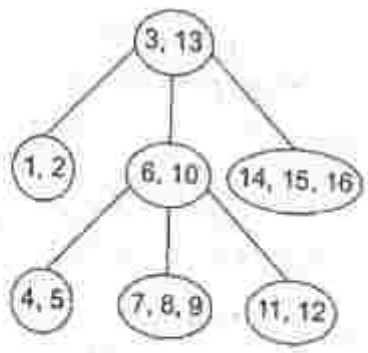
Q.285 Consider the following B^+ tree with the order of internal and leaf nodes as 3 and 2 respectively:

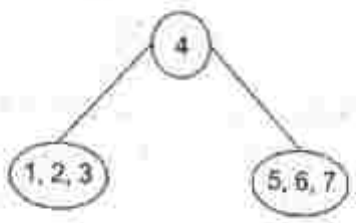


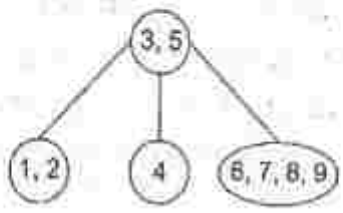
The minimum number of key insertions that causes a new level to be introduced in the above B^+ tree _____. (Assume key redistribution is not allowed)

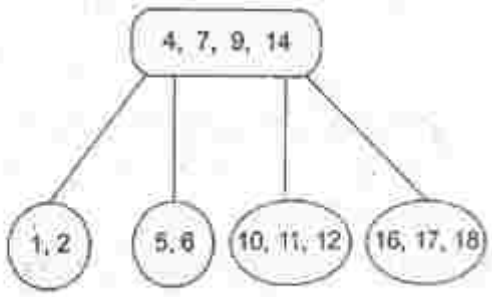
Q.286 In a *B*-Tree, the block size is 512 bytes, search key is 2 bytes, block pointer size is 6 bytes and record pointer size is 8 bytes. Every node contains $n-1$ record pointers, $n-1$ search keys, n block pointers. Find out the average number of search keys that can be accommodated if the fill factor of B-tree is 67%?
(a) 20 (b) 21
(c) 31 (d) None of these

Q.287 Which of the following is a legal *B*-tree? Assume minimum branching factor is 3.

(a)


(b)


(c)


(d)


Q.288 The following key values are inserted into an empty B^+ tree where each node have two key values. Find sum of keys present at height one (assume root is present at height 0) and left biasing.

8, 5, 1, 7, 3, 12, 9, 6

Q.289 A file is organized so that the ordering of data record is same as or close to the ordering of data block in some index. Then that index is called

(a) Dense (b) Sparse
(c) Clustered (d) Unclustered

Q.290 The maximum number of records that can be indexed for a B^+ tree of 4 levels with order 10 and root at level 1 are _____.

Q.291 Consider the following statement :

(i) Primary index is always sparse.
(ii) Secondary index may or may not be dense.

Which of the above statement is/are false?

(a) Only (i) (b) Only (ii)
(c) Both (i) and (ii) (d) None of these

Q.292 Suppose size of disk block 1000 bytes and search key of 12 bytes, pointer size 8 bytes. How many minimum number of records in data file which leads 3 level dense B^+ tree index? (Assume minimum node fill factor $\lceil P/2 \rceil$ pointers where P is the maximum pointer per node).

(a) 1200 (b) 1250
(c) 15000 (d) 15625

Q.293 Database file consist 50000 records with record size 100 bytes, block size 512 bytes. If sparse B^+ tree index build over given database file with search key size 20 bytes both block pointer and record pointer sizes 12 bytes each. How many maximum index blocks required if node order P is defined as between $\lceil P/2 \rceil$ to P pointers per node?

(a) 1635 (b) 1631
(c) 716 (d) 713

Q.294 Database file consist 1250 records. Block can hold either 3 records or (10 keys, 11 pointers). The maximum number of levels of index required for sparse B^+ tree index for database file are _____.

Q.295 Consider a B-tree with degree d and the number of children C of any internal node except the root is such that $d \le C \le 2d-1$. Which of the following are correct?

(a) The minimum key possible at height h is $2 * d^h * (d-1)$.
(b) The minimum key possible at height h is $2 * d^{h-1} * (d-1)$.
(c) The maximum key possible at height h is $(2d-1)^h * (2d-2)$.
(d) The maximum key possible at height h is $2d^h * (2d-2)$.

Q.296 A file consists of records of size 64 bytes each including key field of size 14 bytes. An address of a disk block takes 2 bytes. Assume the disk block size is 512 bytes and there are 16 K records then which of the following is true?

(a) Blocking factor of record file is 8.
(b) Blocking factor of index file is 24.
(c) Number of blocks in the first level is 64.
(d) Number of blocks in the second level index is 10.

Q.297 Which of the following is/are correct?

(a) In ordered indices the file containing the records is sequentially ordered, a Clustered index is an index whose search key also defines the sequential order of the file.
(b) In ordered indices the file containing the records is sequentially ordered, a Non clustered is an index whose search key also defines the sequential order of the file.
(c) Indices whose search key specifies an order different from the sequential order of the file are called non-clustered indices.
(d) Indices whose search key specifies an order different from the sequential order of the file are called secondary indices.

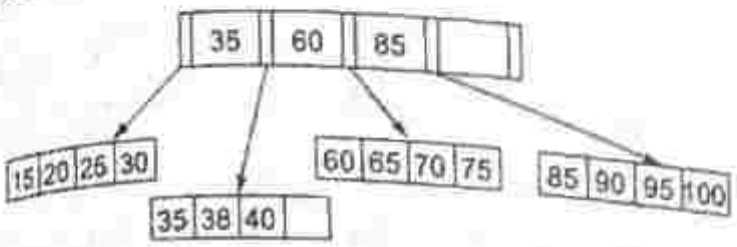
Q.298 Consider the following structure of an student record:

```
struct student
{   int sno;
    char sname[22];
    float marks;
    char subj[10];
};
```

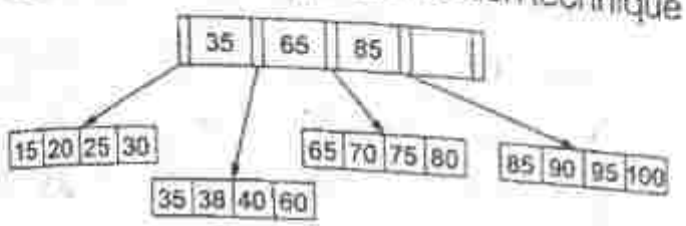
Assume integer, character and float data of 2 bytes, 1 byte and 4 bytes respectively. Block size of 512 bytes is used in unspanned organization and each file has 500 records. Find the unused space in the last block of a file?

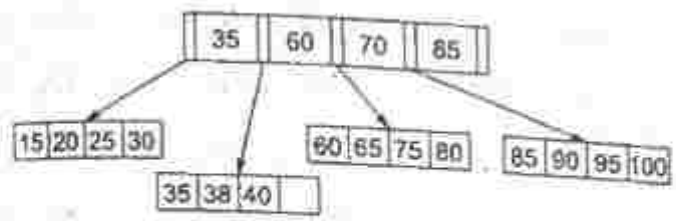
(a) 18 bytes (b) 156 bytes
(c) 266 bytes (d) 284 bytes

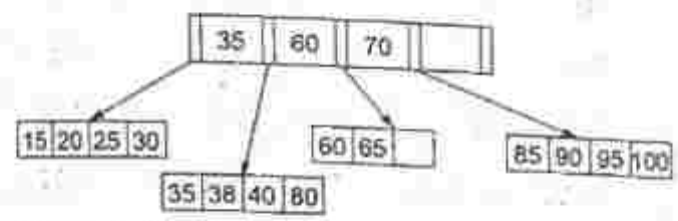
Q.299 Consider the following B^+ tree:



What will be the resulting B^+ tree after inserting 80. Assume B^+ tree incorporate rotation technique

(a)


(b)


(c)


(d) None of the above

Q.300 Suppose that we store records on a disk device having the following characteristics:
Average access motion time : 0.02 sec
Disk rotation speed: 3,000 revolutions per minute
Data transfer rate: 250,000 bytes per sec
What is the expected data transfer time (in sec.) for a randomly accessed physical record that is 500 bytes in length?

(a) 0.0212 (b) 0.0230
(c) 0.0240 (d) 0.0320

Q.301 Consider B tree of degree 'n' with height 'h'. Calculate maximum number of records that can be indexed by the above tree (assume root is at level '0')

(a) $n^h - 1$ (b) n^h
(c) $n^{h+1} - 1$ (d) $n - 1$

Q.302 The record pointer, key field and block pointer of a 'B' tree are 8 B, 10 B and 6 B respectively. Calculate the order of the tree if the block size is 1 kB

(a) 44 (b) 43
(c) 24 (d) None of these

■■■■

Answers Databases

1. (b) 2. (d) 3. (c) 4. (a) 5. (50) 6. (a) 7. (a, c, d)
8. (b, d) 9. (b) 10. (c) 11. (c) 12. (b) 13. (b) 14. (c)
15. (b) 16. (c) 17. (a) 18. (c) 19. (c) 20. (a) 21. (b)
22. (c) 23. (a) 24. (b) 25. (c) 26. (b) 27. (b) 28. (b)
29. (b) 30. (a) 31. (c) 32. (a) 33. (b) 34. (b) 35. (a)
36. (b) 37. (b) 38. (b) 39. (a) 40. (a) 41. (c) 42. (c)
43. (b) 44. (b) 45. (c) 46. (d) 47. (a) 48. (c) 49. (c)
50. (a) 51. (b) 52. (d) 53. (b) 54. (b) 55. (c) 56. (b)
57. (d) 58. (b) 59. (b) 60. (b) 61. (b) 62. (b) 63. (b)
64. (b) 65. (b) 66. (b) 67. (d) 68. (a) 69. (c) 70. (c)
71. (b) 72. (c) 73. (b) 74. (d) 75. (a) 76. (b) 77. (c)
78. (d) 79. (c) 80. (a) 81. (b) 82. (a) 83. (c) 84. (c)
85. (b) 86. (a) 87. (d) 88. (d) 89. (b) 90. (c) 91. (c)
92. (d) 93. (d) 94. (a) 95. (b) 96. (c) 97. (a) 98. (a)
99. (a) 100. (d) 101. (b) 102. (c) 103. (d) 104. (c) 105. (a)
106. (a) 107. (b) 108. (a) 109. (b) 110. (a, b, d) 111. (26) 112. (d)
113. (2) 114. (d) 115. (b) 116. (c) 117. (b) 118. (b) 119. (c)
120. (d) 121. (b) 122. (b) 123. (b) 124. (b) 125. (b) 126. (d)
127. (b) 128. (b) 129. (a) 130. (c) 131. (b) 132. (d) 133. (d)
134. (a) 135. (d) 136. (c) 137. (b) 138. (c) 139. (c) 140. (4)
141. (a, b, c) 142. (c) 143. (d) 144. (c) 145. (a) 146. (c) 147. (c)
148. (c) 149. (b) 150. (a) 151. (b) 152. (c) 153. (c) 154. (c)
155. (b) 156. (a) 157. (d) 158. (3) 159. (b) 160. (c) 161. (d)
162. (4) 163. (c) 164. (3) 165. (6) 166. (c) 167. (c) 168. (d)
169. (c) 170. (b) 171. (c) 172. (4000) 173. (2) 174. (a) 175. (d)
176. (d) 177. (6) 178. (100) 179. (b) 180. (a, b, d) 181. (c) 182. (Sol.)
183. (c) 184. (c) 185. (d) 186. (1) 187. (c) 188. (c) 189. (d)
190. (d) 191. (a) 192. (a) 193. (b) 194. (d) 195. (d) 196. (c)
197. (a) 198. (a) 199. (d) 200. (2) 201. (6000) 202. (b) 203. (c)
204. (a) 205. (d) 206. (a, c) 207. (a) 208. (d) 209. (b) 210. (a)
211. (d) 212. (b) 213. (c) 214. (d) 215. (d) 216. (b) 217. (d)
218. (c) 219. (d) 220. (d) 221. (c) 222. (b) 223. (a) 224. (c)
225. (c) 226. (d) 227. (c) 228. (b) 229. (b) 230. (b) 231. (b)
232. (b) 233. (d) 234. (a) 235. (c) 236. (b) 237. (b) 238. (a)
239. (4) 240. (c) 241. (c) 242. (b) 243. (a) 244. (c) 245. (a)
246. (a) 247. (a) 248. (4) 249. (c) 250. (4) 251. (a) 252. (b)
253. (a) 254. (c) 255. (a) 256. (c) 257. (b) 258. (c) 259. (c)

260. (a) **261.** (a) **262.** (a) **263.** (b) **264.** (c) **265.** (d) **266.** (b)
267. (c) **268.** (c) **269.** (d) **270.** (a) **271.** (a) **272.** (a) **273.** (c)
274. (a) **275.** (a) **276.** (b) **277.** (c) **278.** (c) **279.** (c) **280.** (a)
281. (a) **282.** (538) **283.** (c) **284.** (c) **285.** (3) **286.** (a) **287.** (b)
288. (18) **289.** (b) **290.** (9000) **291.** (c) **292.** (a) **293.** (b) **294.** (4)
295. (b, c) **296.** (a, c) **297.** (a, c, d) **298.** (d) **299.** (a) **300.** (d) **301.** (c)
302. (b)

Explanations Databases

1. (b)

Number of attributes are called as degree while number of tuples are called as cardinality.

2. (d)

Dependent is a weak entity.
Age is a derived attribute since, inside dotted oval.
Gender is an atomic since under solid oval.
Address is multivalued attribute because it is double oval.
Name is not a key attribute since no underline.

3. (c)

= Weak entity

= Derived attribute

= Multivalued attribute

= Identifying relationship

4. (a)

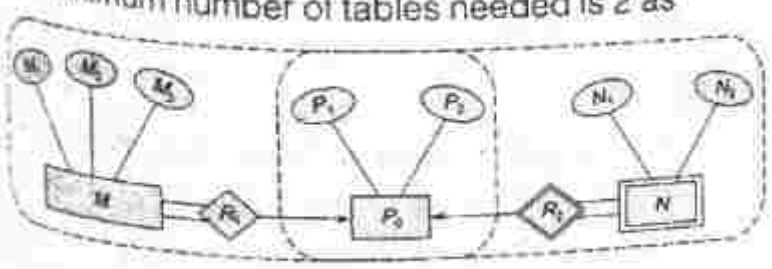
Diamonds with double/bold border are used to represent a relationship for a weak entity set.

5. (50)

For relationship set R candidate and E_1 candidate key is same because between E_1 to E_3, E_1 to E_2 and E_1 to E_4 there is one to many relationship.

6. (a)

Minimum number of tables needed is 2 as



1st table have the relation between M and P and 2nd table as P and N.
Hence option (a) is correct.

8. (b, d)

Tables:

XR(X1, X2, X3, Y1)

SZ(Y1, Z1, Z2)

Y(Y1, Y2)

Total 3 tables are required.

9. (b)

Repeating manager name in every employee record leads to data duplication.
Option (a) is avoided through normalization.
Option (c) prevents redundancy.
Option (d) is a logical layer.

10. (c)

Integrity constraints (like NOT NULL, UNIQUE, CHECK) maintain data consistency by restricting invalid entries.

11. (c)

File systems lack proper isolation, atomicity, and concurrent access handling, which is a strong point of DBMS.

12. (b)

Only DBMS provides high-level query language (like SQL) for efficient data retrieval.

13. (b)

Transaction manager ensures ACID properties - especially recovery and isolation.

14. (c)

External level is tailored to individual user views, hiding implementation and logical details.

15. (b)

Physical mapping translates how logical data structures are stored on disk.

16. (c)

Logical data independence allows you to change table definitions, add new fields, etc., without rewriting app-level queries.

17. (a)

Physical data independence means changing how data is physically stored should not affect the logical layer or applications.

18. (c)

Logical data independence allows changes to the conceptual schema (e.g., normalization) without altering external schemas or applications

19. (c)

A superkey may contain extra attributes; only candidate keys are minimal superkeys.

20. (a)

The candidate key itself is a superkey. Other superkeys include it with extra attributes.

21. (b)

Candidate keys are minimal superkeys. One is chosen as primary; others are alternate.

22. (c)

Foreign key refers to a candidate key (usually primary) in another relation.

23. (a)

A composite key contains more than one attribute, so (A, B) qualifies.

24. (b)

Without ON DELETE CASCADE or similar, deletion of referenced key violates referential integrity.

25. (c)

Weak entities rely on owner (strong) entity + partial key to be uniquely identified.

26. (b)

Every relation must have at least one candidate key to ensure tuple uniqueness.

27. (b)

Total participation means every instance of the entity must be in the relationship.

28. (b)

Weak entities are represented using double rectangles.

29. (b)

Candidate keys cannot have null values because they must identify every row uniquely.

30. (a)

Ternary relationships involve 3 entity sets, not just 3 attributes.

31. (c)

Derived attributes are not stored directly (like age = current_date - DOB).

32. (a)

In specialization, child entities inherit from a common parent entity.

33. (b)

Weak entities have no key of their own. They use strong entity's primary key + their partial key.

34. (b)

Aggregation is used when a relationship becomes an entity to participate in another relationship.

35. (a)

In generalization, multiple sub-entities merge into a generalized entity. In specialization, a parent entity is divided into child entities.

36. (b)

A ternary relationship is essential when all three entities together define the association uniquely (e.g., doctor, drug, and patient together define a prescription).

37. (b)

Total specialization (double line) means every instance of the parent must appear in at least one child entity.

38. (b)

Aggregation is represented using a box around a relationship diamond, not a double rectangle.

39. (a)

In (A), Doctor, Patient, and Time together define one interaction - hence ternary is required to preserve semantics.

40. (a)

Disjoint specialization ensures no entity belongs to more than one subclass at the same time.

41. (c)

Aggregation is chosen when we need to relate a relationship to another entity, unlike ternary which just relates 3 entities.

42. (c)

A ternary relationship is needed to reflect the dependency of all three entities in the context of assigning work.

43. (b)

Weak entities (e.g., Room) are represented with a double rectangle and depend on an identifying relationship (double diamond) with the owner entity (Hotel). Total participation is shown with a thick line, but the dependency requires an identifying relationship.

44. (b)

Aggregation groups the relationship "uses" (Employee, Equipment, Task) into a single entity (Assignment) linked to Project, preserving the context of assignments per project. Ternary relationships lose this higher-level grouping.

45. (c)

A ternary relationship (Student, Course, Instructor) preserves the unique combination where all three entities interact. Splitting into binary relationships (e.g., Student-Course and Course-Instructor) loses this specific triplet dependency.

46. (d)

"Disjoint" ensures a patient belongs to only one subclass. "Total" participation means all patients must be either outpatient or inpatient.

47. (a)

Role indicators (e.g., labels like "Supervisor" and "Subordinate") clarify the different roles an entity (Employee) plays in a recursive relationship.

48. (c)

Composite attributes are hierarchical (e.g., Address → Street + City + ZIP). Multivalued attributes (e.g., Phone Numbers) use double ellipses.

49. (c)

M : N (many-to-many) cardinality is used when both entities participate multiple times in the relationship.

50. (a)

Total participation (all employees must belong to a department) is shown with a thick line connecting the Employee entity to the relationship.

51. (b)

Overlapping allows entities to belong to multiple subclasses (e.g., a staff member can be both faculty and administrator). Partial participation means not all staff need to be in subclasses.

52. (d)

Total participation ensures every weak entity is linked to its owning entity. Without it, nulls can appear in foreign keys, violating design rules.

53. (b)

Many-to-many relationships are converted into a separate table with foreign keys referencing both participating entities.

54. (b)

Multivalued attributes are always converted into a separate relation with a foreign key referencing the original entity.

55. (c)

In "table-per-subclass", full entity details require joins across multiple tables to reconstruct original entity.

56. (b)

Single-table inheritance (table per hierarchy) puts all entities in one table - avoids nulls but may increase redundancy or unused columns.

57. (d)

In 3NF, partial dependencies are not allowed. They are removed to meet the second normal form requirement.

58. (b)

Weak entities require a composite primary key (owner's PK + partial key) and total participation enforces NOT NULL on the foreign key (A_id).

59. (b)

Ternary relationships require a separate table with all three foreign keys. Total participation of C ensures all C entries exist in R, but storing foreign keys in C's table would introduce NULLs if D/E are optional.

60. (b)

Disjoint specialization with total participation maps to separate tables for subclasses linked to the superclass via primary keys. This avoids NULLs and ensures 3NF.

61. (b)

Multivalued attributes violate 1NF if stored as arrays. A separate table with Student_id as FK avoids redundancy and satisfies 3NF.

62. (b)

Merging into one table avoids NULLs because every Manager must map to a Department, and vice-versa (total participation).

63. (b)

3NF requires removing transitive dependencies. A new table (Y, X) eliminates dependency of X on non-key Y.

64. (b)

Composite attributes map to individual atomic columns (First, Middle, Last) in the same table for 1NF compliance.

65. (b)

Overlapping specialization uses separate tables for subclasses linked to the superclass. This avoids NULLs (e.g., a Person can be in both Student and Employee).

66. (b)

Replacing composite key with a surrogate key may cause redundancy and loss of dependencies unless proper decomposition is applied. Composite keys are essential when each component carries unique semantics.

67. (d)

(a) : Violates referential integrity
(b) : Violates primary key constraint
(c) : Foreign key cannot be NULL if not allowed by schema
Hence, all three can lead to anomalies.

68. (a)

Self-referencing foreign keys are allowed if the value exists in the same table. It creates a loop, which may be logically weird, but not a violation.

69. (c)

A foreign key must reference a candidate key in another table. If T_1(B) is not UNIQUE, then it cannot be referenced safely.

70. (c)

ON DELETE SET NULL sets the foreign key (CourseID) to NULL in the referencing table (Enrollment), not the primary key or unrelated columns.

71. (b)

ON DELETE CASCADE ensures that deleting a department also deletes all employees in that department.

72. (c)

A foreign key can reference any candidate key (e.g., a unique ProductID), not just the primary key. If ProductID is unique in Orders, option C is valid.

73. (b)

Foreign key constraints enforce referential integrity. Insertion fails if the referenced CourseID does not exist.

74. (d)

A foreign key referencing a composite primary key must include all columns of the composite key. Partial references are invalid unless the referenced column is a standalone candidate key.

75. (a)

ON UPDATE SET NULL modifies the foreign key value in the child table (Book) to NULL when the referenced primary key (AuthorID) is updated.

76. (b)

A UNIQUE constraint on a combination of fields ensures no duplicate pairs. Separate UNIQUEs allow duplicate combinations.

77. (c)

A table can have only one primary key, though it can span multiple columns.

78. (d)

All three violate referential integrity, unless ON DELETE/UPDATE actions are defined

79. (c)

All attributes in a composite primary key must be NOT NULL - NULL violates primary key rules.

80. (a)

A self-referencing foreign key ensures that values like ManagerID refer to another value in the same table.

81. (b)

Having non-atomic values (like a list in a column) violates 1NF - though SQL allows it.

82. (a)

In SQL, a UNIQUE constraint allows multiple NULL values (as NULL is considered distinct from other NULLs in this context).
A PRIMARY KEY, however, implicitly enforces NOT NULL and uniqueness.

83. (c)

A foreign key must reference a full candidate key (primary key). Referencing a subset of a composite primary key is invalid unless the subset itself is a candidate key and if subset is candidate key, it no longer a composite key.

84. (c)

CHECK constraints cannot reference other tables or use subqueries (SQL standard restriction). They validate only row-level conditions.

85. (b)

A PRIMARY KEY implicitly enforces NOT NULL and uniqueness. A UNIQUE constraint allows NULL values (unless explicitly restricted).

86. (a)

ON DELETE SET NULL sets the foreign key column to NULL when the referenced row is deleted. If the column is NOT NULL, this action violates the constraint.

87. (d)

Exclusion constraints use operators (e.g., && for overlapping ranges) to enforce that no two rows satisfy a specified condition.

88. (d)

Let the relation is $R(A, B)$
If A is key then $A \rightarrow B$ which is in BCNF
If B is key then $B \rightarrow A$ which is also in BCNF
If AB is key then $AB \rightarrow AB$ which is in BCNF
So $R(A, B)$ will always in BCNF

89. (b)

$$F = \{A \rightarrow BCD, B \rightarrow C, CD \rightarrow A\}$$
$$A^+ = A\,B\,C\,D$$
$$CD^+ = A\,B\,C\,D$$

So candidate keys are $\{A, CD\}$
Prime attributes are $\{A, C, D\}$
Check for BCNF :
$A \rightarrow BCD$ ✓ ($\because$ A is candidate key)
$B \rightarrow C$ X ($\because$ B is not candidate key)
$CD \rightarrow A$ ✓ (CD is candidate key)
Check for 3NF :
$A \rightarrow BCD$ ✓ ($\because$ A is candidate key)
$B \rightarrow C$ ✓ ($\because$ C is prime attribute)
$CD \rightarrow A$ ✓ ($\because$ CD is candidate key)

90. (c)

$$F = \{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$$
$$A^+ = A, B,$$
$$(BC)^+ = B, C, E$$
$$(ED)^+ = E, D, A, B$$

So candidate keys may be $\{EDC, BCD, ACD\}$ and prime attributes are $\{A, B, C, D, E\}$
BCNF: If $x \rightarrow y$ then x should be candidate key then the relation will be in BCNF. In the given FDs set $\{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$, x may be A or BC

or *ED* which are not candidate key so given relation is not in BCNF.

3NF: If $x \to y$ then (i) x should be candidate key, or (ii) y should be prime attribute in our given relation y may be *B*, *E*, *A*, which are prime attributes. So given relation is in 3 NF.

91. (c)

$AB \to C$

Augment '*D*'

$ABD \to CD$ and we have $CD \to E$.

∴ Transitively : $ABD \to E$ is a possible deduction.

OR

$(ABD)^+ = ABCDEFGH$

∴ $ABD \to E$ is possible

92. (d)

In the given relation salesman No. has multiple attribute so it is not in 1NF.

93. (d)

$AC \to D$ can be eliminated, it can be derived from $A \to B$ and $CB \to D$ using augmentation and transitive rule.

$A \to B \Rightarrow AC \to BC$

$\Rightarrow AC \to D$

and remaining FDs are not possible to eliminate.

∴ 5 FDs are there in the minimal cover.

94. (a)

$(E)^+ = ECADB = ABCDE$

$(C)^+ = ABCDE$

$(BD)^+ = ABCDE$

$(A)^+$ can not determine all attributes.

∴ *A* is not a candidate key.

95. (b)

Two possible keys of the relation are *AB* and *E*. Therefore prime attributes are, *A*, *B* and *E*. Non prime attributes are *C* and *D*. The relation is in 2NF because there are no partial dependency on any of the key.

The relation is not in 3NF because of $C \to D$ functional dependency.

96. (c)

Number of possible primary keys are :

$2^5 - 1 = 31$ (with zero attributes not possible)

97. (a)

Given relation: R(A, B, C, D, E, F, G, H)

$AB \to CDEF$

$A \to C$

$B \to D$

$C \to FG$

$G \to H$

$H \to F$

$G \to F$

Here, AB+ = ABCDEFGH

Thus, AB is a candidate key.

1. $A \to C$: it a partial dependency. Thus, above relation is not in 2NF. Hence, only 1NF possible

98. (a)

In a relational database model, all types in a relation must be distinct.

99. (a)

Given relation R(A, B, C, D, E) decomposed into R_1(A, B, C) R_2(A, D, E)

R_1 | A | B | C | R_2 | A | D | E |

1. For lossless join:

   There must be 3 conditions that needs to be followed in order to satisfy lossless join property.

   - $R_1 \cup R_2 = R$

     {ABC} ∪ {ADE} = {ABCDE} True
   - $R_1 \cap R_2 \neq \phi$ and must be a superkey of atleast one of the relation.

     {ABC} ∩ {ADE} = A

R_1 | A | B | C | R_2 | A | D | E |

$A \to BC$ $E \to A$

$BC \to A$

We can clearly see, A is the superkey of relation R_1. Thus, decomposition is lossless.

100. (d)

Consider the relations 1, 2 and 3. All are binary relations. A binary relation always be in BCNF. Coming to 4th relation it is also in BCNF because *A*, *B* and *AB* all are super keys.

101. (b)

Whenever $R_1 \cap R_2 = \phi$ then it is the condition of lossy decomposition and hence, option (b) is true

102. (c)

$f = \{A \to BC, B \to A, C \to A\}$

$g = \{A \to B, B \to C, C \to A\}$

To check if f covers g, every FD in g logically implies in f.

FD in g : $A \to B$, $B \to C$, $C \to A$

Check for $A \to B$,

$A^+ \to ABC$ $\{A \to BC$ in $f\}$

Check for $B \to C$,

$B^+ \to BAC$, $\{B \to A, A \to BC$ in $f\}$

Check for $C \to A$,

$C^+ \to CAB$ $\{C \to A, A \to BC$ in $f\}$

Therefore, f covers g.

To check if g covers f:

FD in f: $A \to BC$, $B \to A$, $C \to A$

Check for $A \to BC$,

$A^+ \to ABC$ $\{A \to B, B \to C$ in $g\}$

Check for $B \to A$,

$B^+ \to BCA$ $\{B \to C, C \to A$ in $g\}$

Check for $C \to A$,

$C^+ \to A$ $\{C \to A$ in $g\}$

Therefore, g covers f.

Both f and g covers each other, so they are equivalent $f \equiv g$.

103. (d)

$$\text{Old set} \begin{cases} A \to BC \\ B \to C \\ A \to B \\ AB \to C \end{cases}$$

Now check extraneous in L.H.S. Take B as extraneous attribute

$$\text{New set} \begin{cases} A \to BC \\ B \to C \\ A \to B \\ A \to C \end{cases}$$

Now old FD's imply the new FD's

So B is an extraneous attribute.

Similarly if we check for C, it is also an extraneous attribute. But we will remove only one attribute to preserve the dependency in the set.

Now the reduced FD's will be

$$\left.\begin{matrix} A \to BC \\ B \to C \end{matrix}\right\} \text{old}$$

But the last two dependencies can be removed because they can be derived from $A \to BC$. Now, the FD set will be

$$\left.\begin{matrix} A \to BC \\ B \to C \end{matrix}\right\} \text{old}$$

Now we check extraneous attribute in R.H.S. Take C as extraneous

$$\left.\begin{matrix} A \to B \\ B \to C \end{matrix}\right\} \text{new}$$

Here new set of FD's implies old set of FD's. So C is an extraneous attribute.

Similarly B is not an extraneous attribute.

104. (c)

The FD (EmpNo, DeptNo.) → ProjNo does not hold because for EmpNo 1001 and DeptNo of there are two ProjNo (12, 13), which violates the FD.

Similarly, FD DeptNo → ProjNo violates the FD.

105. (a)

An alternative key is a candidate key that is not a primary key.

106. (a)

Relation produced from an E-R will always be in 1NF.

107. (b)

$[AB \to C$
$C \to A$
$BC \to D$
$ACD \to B$
$BE \to C$
$EC \to F$
$EC \to A$
$CF \to B$
$CF \to D$
$D \to E]$

After ⟹ removal of extraneous attributes

$[AB \to C$
$C \to A$
$BC \to D$
$CD \to B$
$BE \to C$
$EC \to F$
$EC \to A$
$CF \to B$
$CF \to D$
$D \to E]$

After ⟹ removal of redundant FD's

$[AB \to C$
$C \to A$
$BC \to D$
$BE \to C$
$EC \to F$
$CF \to B$
$D \to E]$
Minimal cover

108. (a)

Functional dependency $X \to Y$ is trivial when Y is subset of X i.e., $Y \subseteq X$

Hence option (a) is true.

109. (b)

Given functional dependency:

$X \to W$

$X \to Y$

$Y \to Z$

$Z \to PQ$

$X^+ = XYZWPQ$

Thus, option (a) and (c) are false. $W^+ = W$.

Thus, option (b) is true.

111. (26)

$^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 26$

112. (d)

Checking $QS \to P$, $Q^+ = Q$, $S^+ = S$.

Hence $QS \to P$ is essential.

Checking $WS \to R$, $WS \to T$

$W^+ \to WVSRT$, Hence it can be decomposed to

$W \to R$, $W \to T$

So, the dependencies remained are

$W \to V$, $W \to S$, $T \to S$, $W \to R$, $W \to T$, $QS \to P$

Now, $\{W \to T, T \to S\}$ by transitive rule $W \to S$ can be obtained.

Hence minimal cover is: $W \to V$, $T \to S$, $W \to R$, $W \to T$, $QS \to P$.

113. (2)

$(AG)^+ = ABCGHI$

$\vdots$

$(CG)^+ = ABCGHI$

$\{\underline{AG}, \underline{CG}\}$ candidate keys.

114. (d)

Referential integrity constraint:

In relational model, two relation are related to each other over the basis of attributes. Every value of referencing attribute must be null or be available in the referenced attribute.

$R(\underline{a}, b, c)$ and $S(d, \underline{e}, f)$

Here d is the foreign key of S that refers to the primary key of R.

1. Insert into R will not cause any violation.
2. Insert into S may cause violation because for each entry in 'S' it must be in 'R'.
3. Delete from R may cause violation because for the deleted entry in R there may be referenced entry in the relation S.
4. Delete from S will not cause any violation.

Hence (d) is the correct option.

115. (b)

$$^nC_{n-2} \Rightarrow \frac{n(n-1)(n-2)!}{(n-2)! \times 2} = \frac{n(n-1)}{2}$$

Candidate keys.

If every $(n-2)$ attributes are Candidate keys then every $(n-1)$ and n attributes are also Superkeys

$$\text{Total Superkeys} = {}^nC_{n-2} + {}^nC_{n-1} + {}^nC_n$$
$$= {}^nC_{n-2} + n + 1$$
$$= \frac{n(n-1)}{2} + n + 1$$

116. (c)

If (4, 3) is deleted then 4 is the primary key but in (2, 4), (6, 4) and (3, 4), 4 is the foreign key so these must be deleted.

The primary key for (2, 4), (3, 4) and (6, 4) is 2, 3 and 6 respectively must be deleted but in (5, 2) and (7, 2) with primary key 5 and 7 is the foreign key also deleted and (9, 5) is also deleted because 5 is primary key which is already deleted.

117. (b)

Given relation: $S(A\ B\ C\ D)$

$S\{A \to BCD,$

$B \to C,$

$CD \to A\}$

$S_1(A, B, D)$ $\quad$ $S_2(B, C)$

$A \to BD$ $\quad$ $B \to C$

$BD \to A$

1. For lossless join there are two conditions that must be satisfied:

   (i) $S_1 \cup S_2 = S$

   $\{ABD\} \cup \{BC\} = ABCD$ True

   (ii) $S_1 \cap S_2 \neq \phi$ and must be superkey of one of the two relation.

   i.e., $\{ABD\} \cap \{BC\} = B$

   and B is super key of S_2.

   Thus, Lossless
2. Dependency is not preserved
3. Both S_1 and S_2 is in BCNF. Hence, option (b) is true.

118. (b)

We can choose the correct option by finding the closure of each option

(a) CD

$(CD)^+ = \{C, D, F\}$

$\therefore$ CD is not a key for R.

(b) $(EC)^+ = \{A, B, C, D, E, F\}$

$\therefore$ EC is a key for R.

(c) $(AE)^+ = \{A, B, E\}$

$\therefore$ AE is not a key for R

(d) $(AC)^+ = \{A, B, C, F\}$

$\therefore$ AC is not a key for R

Therefore, EC is a key for R.

119. (c)

Since given snapshot is only an instance of the database. Just by checking the snapshot, we can't determine whether something is functional determined or not but on the contrary to this, we can correctly determine something which is not functionally determined.

Hence B does not functionally determine C is the correct answer.

120. (d)

R = (A, B, C, D, E, H)

FD = {A → B, BC → D, E → C, D → A,}

The candidate keys of R is AEH, BEH, DEH because both of these generates the {A, B, C, D, E, H}

121. (b)

Apply membership test for all the given Functional Dependencies.

1. CD → AC

   CD^+ = CDEAB

2. BD → CD

   BD^+ = BD

i.e. BD cannot derive CD and hence is not implied.

Similarly for rest two can be done.

122. (b)

Relation contain 8 attributes ABCDEFGH

F = {CH → G, A → BC, B → CFH, E → A, F → EG}

Candidate key : AD, ED, BD, FD

123. (b)

In *V W X Z*, no complete candidate key exists.

So, it is not super key.

124. (b)

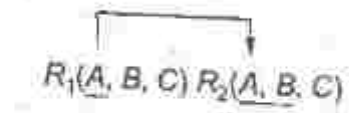
A prime attribute is an attribute that appears in some candidate key of given relation *R*.

125. (b)

BCNF is not used for cases where a relation has two candidate keys and composite.

126. (d)

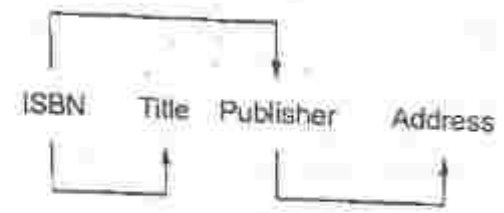
The main purpose of 'foreign key' in table is to ensure referential integrity i.e., the foreign key of one table can have only the values that are in some primary key of referenced table.



So, *A* in R_2 is foreign key.

127. (b)

Finding candidate key:



$(ISBN)^+ \rightarrow$ ISBN, Title, Publisher, Address

Hence, ISBN is the candidate key. Considering the third functional dependency, the attribute on both LHS and RHS are non-prime attributes which results in transitive dependency. Hence, the higher normal form is second normal form.

128. (b)

Every time the attribute *A* appears, it is matched with the same value of attribute *B* means functional dependency $A \rightarrow B$ where *b* is candidate key.

Every time the attribute *A* appears, is not matched the same value of attribute *C* i.e. A is repeating for many value of *C* and *C* also repeating so $A \rightarrow C$.

129. (a)

R(*A*, *B*, *C*, *D*) with functional dependency $A \rightarrow B$ and $C \rightarrow D$.

Decomposition into $R_1(A, B)$ with functional dependency $A \rightarrow B$ and $R_2(C, D)$ functional dependency $C \rightarrow D$. Since dependency is preserved but decomposition relation are not lossless since R_1 intersection R_2 = Phi.

130. (c)

Consider, $X \to A$

Since, X is a non key attribute and A is the prime attribute which in turn is consisting of prime attributes the given relation is free from transitive dependency and hence the given relation will be in 3NF.

131. (b)

$R(A, B, C, D) : \{A \to D, B \to D, D \to BC\}$

Candidate keys: A

So above relation is in 2NF.

Since $D \to B$ is creating transitive dependency.

So, $R_1(A, D)$ $A \to D$ $R_2(B, C, D)$ $B \to D$, $D \to BC$

Both R_1 and R_2 are dependency preserving with loss less decomposition and in BCNF.

So 2 tables are required.

132. (d)

$R(A, B, C, D, E, F, G, H, I, J)$

$A^+ = \{A\}$

$(A\,B\,C\,D)^+ = \{A, B, C, D, G, E, F, H, I\}$

$B^+ = \{B, F\}$

$C^+ = \{C, J\}$

Hence $(A\,B\,C\,D)$ is the candidate key

$D^+ = \{D\}$

$E^+ = \{E\}$

$F^+ = \{F\}$

$G^+ = \{G, H, I\}$

Hence (d) is correct option.

133. (d)

$R(A, B, C, D, E, F, G)$

As $A \to B$

$BC \to DE$

$\therefore$ $(AC)^+ = \{A, B, C, D, E\}$

Hence (d) is the correct option.

134. (a)

$R_1 = (N, R, C, T, Z)$

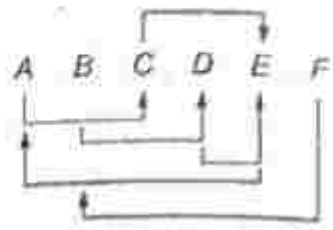
FD $= \{N \to RCT, RCT \to Z, Z \to CT\}$

Here, $N^+ = (N, R, C, T, Z)$

[using $N \to NCT, RCT \to Z$]

135. (d)

According to the given set of functional dependencies



$A^+ = ACE$ (hence not a key)

$C^+ = CEA$ (hence not a key)

$D^+ = DEAC$ (hence not a key)

$F^+ = ABDEFC$ (key)

136. (c)

For FD : $X \to Y$

$\therefore$ For single value of X, Y value remain same.

For different value of X, Y value may be different or same.

So,

(a) A4 $\to$ A1:

FD : $\{4 \to 1, 4 \to 2\}$ is not possible.

(b) A2 A3 $\to$ A4:

FD : $\{2, 3 \to 4, 2, 3 \to 5\}$ is not possible.

(c) A2 A3 $\to$ A1:

FD : may not hold in R.

137. (b)

(a) PQT $\not\to$ S, (b) QTU $\not\to$ SV, (c) RV $\not\to$ PQ

138. (c)

ABDF is not a candidate key because it contains AB, which is not necessary since BDF alone is sufficient.

139. (c)

(a) $\{PT\}^+ = PT$, not generating attributes {R, S}

(b) $\{ST\}^+ = STUP$, not generating attributes {Q, R}

(c) $\{QT\}^+ = QTPRSU$

(d) $\{PQ\}^+ = PQRSTU \to$ generating extra attributes

140. (4)

$\{PQ\}^+ = \{P, Q, T, U, R, S\}$

$\{PD\}^+ = \{P, Q, R, S, T, U\}$

$\{QR\}^+ = \{Q, R, P, S, T, U\}$

$\{RS\}^+ = \{R, S, P, Q, T, U\}$

Therefore, there are 4 candidate keys.

141. (a, b, c)

Closure of left side attribute in F using FDs in G

$w^+ = wxy$

$wx^+ = wxy$

$z^+ = wxyz$

$F \not\subset G$ because in FD set F :

$w \to x$ and w^+ also have x,
$wx \to y$ and wx^+ also have y,
$z \to wy$ and z^+ also have wy
$z \to v$ but z^+ does not have v
Therefore F ⊄ G but G ⊄ F
Closure of left side attribute in G using FDs in F :

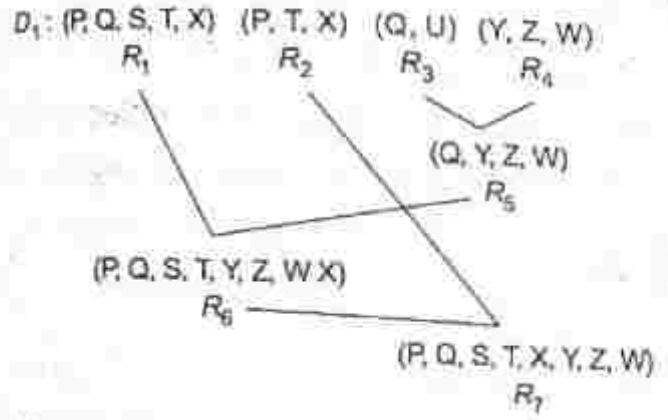
$$w^+ = wxy$$
$$z^+ = wyvzx$$

G⊆F because in FD set G :
$w \to xy$ and w^+ also have xy
$z \to wx$ and z^+ also have wx

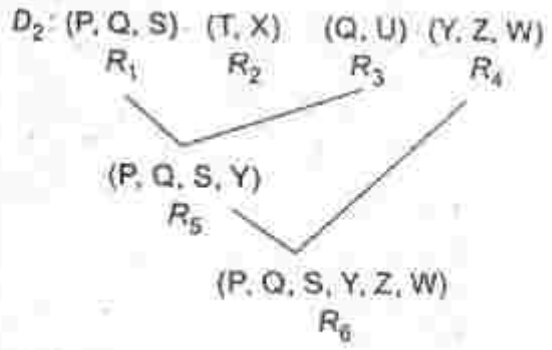
142. (c)

ABD is candidate key A → I is partial dependency not allowed in 2NF every given relation always in 1NF

143. (d)



Hence, D_1 is lossless.



Number of way to join R_2 and R_5.
Hence lossy decomposition.

145. (a)

Candidate key = {ABE} having partial dependency B → C so not in 2NF.

146. (c)

2NF have partial dependency if RHS is Prime attribute
3NF have transitive dependency is RHS is prime attribute
Not every decomposition for 3NF is lossless and dependency preserving.

148. (c)

- With only two attributes, any FD must have a superkey as determinant (BCNF requirement).
- Without any FDs, there are no normalization violations by default.

149. (b)

A is not a superkey neither B is prime attribute
FD: B → D is not valid

150. (a)

3NF can have trnasitive dependency if we RHS is prime attribute.

151. (b)

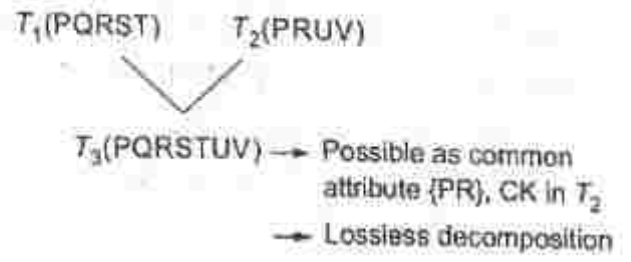
- BCNF is stricter: every determinant must be a superkey. 3NF allows dependencies where the determinant is a prirne attribute.
- Option (a) is incorrect (BCNF bans all non-superkey determinants). Option (d) reverses the hierarchy.

152. (c)

In relation (PQ → R, R → PS) PQ and RQ are the candidates key We cannot break tables such that it is BCNF and dependency preserving.

153. (c)

Checking lossless:



Dependency R → U and U → R not preserved.

154. (c)

- 3NF allows some redundancy but guarantees both lossless join and dependency preservation
- BCNF can result in loss of some dependencies, hence is not always dependency preserving
- This is why 3NF is often preferred practically

156. (a)

{σ, π, ∪, −, X} these contribute a complete set of relational algebra operation.

157. (d)

a	c
1	2
2	3

As we can see clearly both entries of $(a < c)$. Also for selecting the entries we have to project on R. Hence (d) $\sigma_{a<c}(\Pi_{a,c}R)$ is the correct option.

158. (3)

R_1

A	B	C
1	2	3
1	2	4
2	1	3
3	1	3

R_2

A	B	D
1	2	1
2	1	5
4	2	1
3	2	1

$R_1 \bowtie R_2$ tuples where both $R_1 \cdot A = R_2 \cdot A$ and $R_1 \cdot B = R_2 \cdot B$

$R_1 \bowtie R_2 \equiv$

A	B	C	D
1	2	3	1
1	2	4	1
2	1	3	5

$R_1 \overset{\text{L}}{\bowtie} R_2 \equiv$

A	B	C	D	
1	2	3	1	$R \bowtie S$
1	2	4	1	
2	1	3	5	
3	1	3	Null	Left outer join
4	2	Null	1	Right outer join
3	2	Null	1	

Hence, 3 tuples contains NULL entries.

159. (b)

As C is referring to R_2 and D is primary key of R_2, $\pi_C(r_1) - \pi_D(r_2)$ will give empty relation or empty table as number of values in C column of table r_1 will always refer to of respective values in D column of r_2.

160. (c)

$\pi_{AD}(R \times S) - \rho_{A \leftarrow B}(\pi_{BD}(R \bowtie_{B=C} S))$

A	D
a	b
a	d
a	e
b	b
b	d
b	e
c	b
c	d
c	e

$-\rho_{A \leftarrow B}$

B	D
a	b
c	d
c	e

$=$

A	D
a	d
a	e
b	b
b	d
b	e
c	b

$\therefore$ 6 tuples are returned by the given query.

161. (d)

$R - (S - R)$ is equivalent to R.

Except (d), remaining all relational algebra expressions are equivalent.

162. (4)

$R \overset{\text{R}}{\bowtie} S$ [right outer join of R and S]

A	B	C	D
1	2	3	10
1	2	3	11
6	7	8	10
NULL	6	7	12

$\therefore$ Number of tuples = 4

163. (c)

Number of attributes in R/S is $m - n$.

164. (3)

$|\pi_{Cid}(\sigma_{dept = 'EE'}(\text{Course}))| = 0$

$\pi_{Sid,Cid}(\text{Enrolled}) / \pi_{Cid}(\text{empty tuples in Dept})$

All the distinct Sid will be in the result.

165. (6)

$\pi_{PS}(A \times B)$

P	S	
1	2	✓
1	4	
1	8	
2	2	
2	4	
2	8	✓
3	2	
3	4	✓
3	8	

$-\rho_{P \leftarrow Q}$

Q	S
1	2
2	4
2	8

$=$

P	S
1	4
1	8
2	2
3	2
3	4
3	8

6 tuples are returned by the query.

166. (c)

1. A consecutive selection condition can be broken up into a cascade / sequence of individual σ operations.
2. In a cascade (sequence) of π operations all but the last one can be ignored.
3. If the selection condition c involves only those attributes $A_1, A_2, \ldots, A_n$ in the projection list the two operations can be commuted.

167. (c)

"Find the name of all the students who have taken all the course of the department '*CS*'"

$\{t \mid \exists r \in$ Student $(r[\text{Sid}]=t[\text{Sid}]) \wedge (\forall u \in$ Course $(u[\text{Dname}]=\text{"}CS\text{"} \Rightarrow \exists s \in$ Takes $(t[\text{Sid}] = s[\text{Sid}] \wedge s[\text{cid}] = u[\text{cid}]))\}$

Since $P \Rightarrow Q \equiv \sim P \vee Q$ therefore $\{t \mid \exists r \in$ Student $(r[\text{Sid}]=t[\text{Sid}]) \wedge (\forall u \in$ Course $(u[\text{Dname}] \neq \text{"}CS\text{"} \vee \exists s \in$ Takes $(t[\text{Sid}] = s[\text{Sid}] \wedge s[\text{cid}] = u[\text{cid}]))\}$.

168. (d)

Since selection operation is commutative thus all of the above are valid conditions basically they are evaluating $P \wedge Q$.

169. (c)

The sum operation can't be performed with the basic algebra operation.

170. (b)

Query gives the roll no. and name of those students who are female (sex=f) and who are in maths department (d.name= 'math').

171. (c)

From the options, option (c) is giving the best explanation of the query that it gives names of all students who have taken at least 1 course under their advisor.

172. (4000)

Since 'A' in R_2 is not key, hence all value of A may or may not be unique. Hence every entry under A in R_2 will match with A in R_1.

Hence maximum is 2000. But A in R_2 is foreign key referencing A in R_1.

Therefore minimum also 2000.

$\therefore$ Sum is 2000 + 2000 = 4000

173. (2)

There is no instructor with name 'Shiva' in Course Table so query results in all distinct Sid's from Enrolment table.

174. (a)

$r_i = \Pi r_i(U)$ implies that $t[R_i] \in r_i$, $1 \le i \le n$

Thus, $t[R_1] \bowtie t[R_2] \bowtie t[R_3] \ldots \bowtie t[R] \in$

$r_1 \bowtie r_2 \ldots \bowtie r_n$

By the definition of natural Join,

$t[R_1] \bowtie t[R_2] \bowtie \ldots \bowtie t[R_n] = \pi_a[\sigma_\beta(t[R_1] \times t[R_2] \times \ldots \times t[R_n])]$

where condition β is satisfied if values of attributes one tuple. The Cartesian product of single tuples generates one tuple. The selection process is satisfied because all attributes with the same name must have the same value since they are projection from the same tuple. Finally, the projection clause removes duplicate attribute name.

By the definition of decomposition, $U = R_1 \cup R_2 \cup \ldots \cup R_n$, what mean that all attributes of t are in $t[R_1] \bowtie t[R_2] \bowtie \ldots \bowtie t[R_n]$.

That is, t is equal to result of this join. Since t is any arbitrary tuple in u.

$u \subseteq r_1 \bowtie r_2 \bowtie \ldots \bowtie r_n$

175. (d)

$\pi_B\left(R \underset{RB \neq SC}{\bowtie} S\right)$ which selects all foreign key values.

$\pi_C\left(R \underset{RB \neq SC}{\bowtie} S\right) - \pi_C(S) \equiv$ Always empty

All foreign keys values – All primary key values ≡ Always empty

176. (d)

$\pi_{AB}(R \bowtie S)$:

R	A	B
	1	a
	2	b
	1	c

R	B	C
	a	Ram
	a	Shyam
	c	Radha

=

R	A	B
	1	a
	1	c

$R \bowtie \pi_B(S)$:

R	A	B
	1	a
	2	b
	1	c

R	B	C
	a	Ram
	a	Shyam
	c	Radha

=

R	A	B
	1	a
	1	c

$R \cap (\pi_A(R) \times \pi_B(S))$:

R	A	B
	1	a
	2	b
	1	c

R	B	C
	a	Ram
	a	Shyam
	c	Radha

=

R	A	B
	1	a
	1	c

$\pi_{A\,RB}(R \times S)$:

R

A	B
1	a
2	b
1	c

R

B	C
a	Ram
a	Shyam
c	Radha

=

R

A	B
1	a
2	b
1	c

177. **(6)**

$\pi_{AD}(R \times S)$

A	D	
1	2	✓
1	4	
1	5	
2	2	
2	4	
2	5	
3	2	
3	4	✓
3	5	✓

$- \rho_{A \leftarrow B}$

B	D
1	2
3	4
3	5

=

A	D
1	4
1	5
2	2
2	4
2	5
3	2

6 tuples are returned by the query.

178. **(100)**

$P \bowtie Q \Rightarrow$ Common attribute = C which is key for both P and

$R_1 = P \bowtie Q \Rightarrow \min(200, 300) = 200$ tuples

$R_1 \bowtie R \Rightarrow$ Common attribute = E which is key for both R_1 and R.

$R_1 \bowtie R \Rightarrow \min(200, 100) = 100$ tuples

So, $|P \bowtie Q \bowtie R| = 100$ tuples

179. **(b)**

If R has m tuples and S has n tuple.

Maximum size of join = mn

Minimum size of join = 0 (when nothing in common)

180. **(a, b, d)**

To calculate sum of all employee's salaries, we have to use aggregate function. So it cannot be expressed using basic relational algebra operation.

181. **(c)**

The given tuple calculus expression give tuples where A = 10 & B = 20 of relation r.

Therefore, relational algebra expression

$\sigma_{A=10}(r) \cap \sigma_{B=20}(r)$ is equivalent to given tuple calculus expression.

182. **Sol.**

(a) Determines titles of the books issued to user with user # 6

(b) Determines name of authors of books that are issued to users who have home town 'Delhi'.

183. **(c)**

Aggregate operations like sum, average, count can't be expressed in terms of basic relational algebra operations. Aggregate function require extended relational algebra. (Min, Max can be expressed using basic RA operations)

184. **(c)**

R(A, B, C) S(C)

I. $\pi_{R-S}(R) - \pi_{R-S}(\pi_{R-S}(R) \times S - \pi_{R-S,S}(R)) \equiv \pi_{AB}(R) - \pi_{AB}(\pi_{AB}(R) \times \pi_C(S) - \pi_{ABC}(R) \equiv R/S$

II. $\underbrace{\{r | t \in \pi(R)}_{R-S} \wedge \forall U \in S(\exists V \in R(U = V[S] \wedge t = V[R-S])\}$

"AB" of R and for every "C" of S there must be some C of R which equal to A, B and C.

Which also equal to R/S.

III. TRC query which equal to S/R.

IV. SQL query equal to R ⋈ S.

So query I and II are equal.

185. **(d)**

In I, Ps from natural join of R & S are selected. In III, all Ps from intersection of (P, Q) pairs present in R and S. IV is also equivalent to III because $(R - (R - S)) = R \bowtie S$.

II is not equivalent as it may also include Ps where Qs are not same in R and S.

186. **(1)**

P

X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

$\Pi_x(\sigma_{(P.Y = R.Y \wedge R.V = V2)}(P \times R)) \Rightarrow$

X
X2

Q

X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

$\Pi_x(\sigma_{(Q.Y = R.Y \wedge Q.T > 2)}(Q \times R)) \Rightarrow$

X
X1

I − II ⇒

X
X2

one record in result.

187. (c)

'C' common attribute which is key for both R and S. So one record of R maps with one record of S and vice-versa.

So, max matches are min (n, m).

188. (c)

The relational algebra expression:

"$\sigma_{A=a}(r \bowtie s)$" is interpreted as

Select from $(r \bowtie s)$ where "$A = a$"

(a) $\sigma_{A=a}(r)$ is not equivalent

(b) r is not equivalent

(c) $\sigma_{A=a}(r) \bowtie s$ does the same work

Hence (c) is the required option.

189. (d)

$E_1 \bowtie E_2 = E \bowtie_{\theta_1} E_1$ is not equivalent because θ join is not cumulative hence the entries in $E_1 \bowtie_{\theta} E_2$ and $E_2 \bowtie_{\theta_1} E_1$ may not be same.

Hence option (d) is correct.

190. (d)

a	c
1	2
2	3

As we can see clearly both entries of $(a < c)$. Also for selecting the entries we have to project on R.

Hence (d) $\sigma_{a<c}(\Pi_{a,c}R)$ is the correct option.

191. (a)

The given query

SELECT DISTINCT W, X

FROM R, S

is guaranteed to be same as R, if R has no duplicates and 'S' is non-empty.

Since, if R is having a duplicates, then the tuples selected by SELECT operation of the R and the given query will not be same also if 'S' is empty then the given query outputs null.

192. (a)

From the relation DEPOSIT first we have to select the those entries in which balance is over 1000. This can be obtained by $\sigma_{\text{balance} > 1000}$ (Deposit).

Now, we have to select those customers name who have balance over 1000 which can be obtained by simply projecting customer name in above sub-query.

193. (b)

All customers who have a lone amount of over 1200

DRC: $\{<C> \mid \exists\ b, 1, a\ (< b, 1, c, a > \in \text{borrow} \wedge <b, 1, a> \in \text{loan} \wedge a > 1200)\}$

194. (d)

All of the given option valid:

(a) $\sigma_P(\sigma_Q(e)) = \sigma_Q(\sigma_P(e))$

(b) $\sigma_P(\sigma_Q(e)) = \sigma_{P \wedge Q}(e)$

(c) $\sigma_Q(\sigma_P(e)) = \sigma_{P \wedge Q}(e)$

They simply find and selects the tuples which are common to both P and Q.

195. (d)

$\sigma_{c_1}(\sigma_{c_1}(R_1)) \longrightarrow \sigma_{c_2}(\sigma_{c_2}(R_1))$

The conditions c_1 and c_2 may not be same. (b)

All customers who have a lone amount of over 1200

DRC: $\{<C> \mid \exists\ b, 1, a\ (< b, 1, c, a > \in \text{borrow} \wedge <b, 1, a> \in \text{loan} \wedge a > 1200)\}$

196. (c)

Query: Find all customers who have loan amount more than ₹ 12000.

TRC: $\{t \mid s \in \text{borrow} \wedge (t(\text{customer_name} = s(\text{customer_name})) s[\text{amount}] > 12000\}$.

197. (a)

Given

R

A	B
1	2
3	2
5	6
7	8
9	8

S

B	C
6	2
2	4
8	1
8	3
2	5

T

A	C
7	1
1	2
9	3
5	4
3	5

$\pi_{A,B}(R \bowtie T) \bowtie \pi_{B,C}(S \bowtie T)$

A	B	C
1	2	4
1	2	5
3	2	4
3	2	5
5	6	2
7	8	1
7	8	3
9	8	1
9	8	3

198. **(a)**

Using relational algebra, the query that finds customers who have balance more than 10,000 is:

(*i*) From deposit

(*ii*) Apply condition balance > 10,000

(*iii*) Project customer name

Therefore, $\pi_{\text{customer-name}}(\sigma_{\text{balance}>10,000}(\text{Deposit}))$

199. **(d)**

P	Q
3	4
6	8
9	11
10	14

As we can see clearly both entries of $(P < Q)$. Also for selecting the entries we have to project on R.

Hence (d) $\sigma_{P<Q}(\Pi_{P,Q}R)$ is the correct option.

200. **(2)**

There is no instructor with name 'Shiva' in Course Table so query results in all distinct Sid's from Enrolment table.

201. **(6000)**

Since 'P' in R_2 is not key, hence all value of 'P' may or may not be unique. Hence every entry under 'P' in R_2 will match with 'P' in R_1.

Hence maximum is 2000. But 'P' in R_2 is foreign key referencing 'P' in R_1.

Therefore minimum is also 2000.

$$\therefore \quad X = 2000,$$
$$Y = 2000;$$
$$X + 2Y = 2000 + 4000$$
$$= 6000$$

202. **(b)**

(a) finds those drinkers who are frequents atleast one bar on same city where he lives.

203. **(c)**

In the SQL query it will find the DISTINCT Name of student who have got an A grade in at least one of the course which is taught by Sriram.

204. **(a)**

Given queries will result Names of sailors who have reserved all boats called Ganga.

205. **(d)**

```
Select deptName
from employee
where sex = 'M'
grouply deptName
having avg (salary)  >  (select avg (salary)
                         from employees)
```

[Avg salary of male employees in each department] > [Avg salary in the company]

It returns the names of departments in which the average salary of male employees is more than the average salary of the organization.

206. **(a, c)**

The inner subquery give pid of parts which are not blue. The outer subquery give sids of all those suppliers who have supplied blue parts. The outer query gives the name of all suppliers who have supplied a non-blue part.

207. **(a)**

Find all branches that have greater assets than some branch located in Mumbai:

```
Select branch_name
From branch
Where assets > Some (Select assets
                     From branch
                     Where_city = "Mumbai")
```

or

```
Select distinct T.branch_name
From branch as T, branch as S
Where T.assets > S.assets and
    S.branch_city = "Mumbai"
```

208. **(d)**

```
Select deptname
  From employee
  Where sex = 'M'
group by dept name
having avg (salary) > (select avg (salary) from employee)
```

We are selecting male employee by grouping all the male employee of a department whose average salary is greater than the average salary of organization.

209. **(b)**

When **aggregation functions** (sum, avg, count, max and min) are used in select clause then we need to use **group by clause** for grouping the attributes

210. (a)

SQ_1:

R			S		
A	B	C	B	C	D
3	1	4	2	3	7
3	1	4	2	3	4
1	2	3	3	1	4
1	2	0	3	1	4
1	3	1	3	1	4
1	4	2	3	1	4

$\Rightarrow$

R.B	Avg. (S.B)
1	2
2	3
3	1.5
4	1.5

Four tuple selected $m = 4$

SQ_2:

B	C	D
1	2	3
1	4	5
2	3	4
2	3	7
3	1	4

$\Rightarrow$

B	min C
1	2
2	3

Only two tuple selected $n = 2$

211. (d)

The given query will give all employees who each more than the average salary of all employees of their company.

212. (b)

SQL is primarily used to create and manipulate databases, including tasks like querying data, updating records, and defining database structures.

213. (c)

Statement-II is false because DEPTNO = DID is not in the condition which not give the correct solution.

214. (d)

```
SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY > ALL(SELECT SALARY
FROM EMPLOYEE WHERE DEPTNO = 20)
```

- Here ALL is required to test with all the salary of subquery and order by required to order then as asked in the question.

215. (d)

To eliminate duplicate rows, we need to use "DISTNICT" keyword with SELECT in SQL query. SQL not permits attribute names to be repeated in the same relation. Therefore option (d) is correct.

216. (b)

In this
```
(select B. Age
  From B
where B. Name = Arun)
        ⇓
        ϕ
```
So all A. Age will be selected, so answer 3.

217. (d)

```
Select title
FROM book as B
Where (Select count (*)
      FROM book as T
      Where T.price > B.price) < 5
```
Inner query counts number of books more price than B.price of count < 5 then B.title in result if count 0 means no other book more price ⇒ most expensive

If count 1, 2, 3, 4 more one, two three, four other books more price ⇒ 2nd, 3rd, 4th 5th most expensive books.

Query retrieves titles of fire most expensive books.

218. (c)

The WHERE condition checks for the Grade = 'A' and matching of course-number in the courses and Grades, where the instructor must be 'Korth'. Overall it checks for students to whom 'Korth' has instructed and they have received Grade = 'A' in atleast one of the courses taught by 'Korth'.

219. (d)

The maximum number of tuples results when each of the 120 students enrolls for each of the 8. Courses, giving $120 \times 8 = 960$ tuples. The minimum number of tuples results when all the 120 students enroll for the same course, giving $120 \times 1 = 120$ tuples.

220. (d)

The given query first calculates the average salary from the relation Employee, then it selects the department name where employees who have average salary more than the average salary of the company.

221. (c)

For relation X cardinality = 2
For relation Y cardinality = 4
Hence the SQL statement
Select COUNT (*) from X, Y
gives the number of rows for the relation obtained from cross product of X, Y. Hence, $2 \times 4 = 8$.

222. (b)

Query will return part number for part supplied by more than one supplier.

223. (a)

Query will return supplier names for suppliers who supply part P_2.

225. (c)

Considering each of the given pairs:

- $x = 5$; false not (not ($x = 5$)) = not(not(false)); false
- $x = 5$; false $x > 4$ and $x < 6$ = false and false = false
- $x \neq 5$; false not ($x = 5$) = not(false); true

227. (c)

Query-1 will succeed as dept_id of student table is foreign key. But in query-2, dept_id is of department table which is primary key and which can't be set to NULL.

228. (b)

Select $b.B$ from S where $R.C = S.C$

It will return $\begin{array}{|c|}\hline b_1 \\ \hline b_3 \\ \hline \end{array}$ as c_1 and c_3 matches in both R and S.

The total query will give, $\begin{array}{|c|}\hline a_1 \\ \hline a_3 \\ \hline a_2 \\ \hline \end{array}$

229. (b)

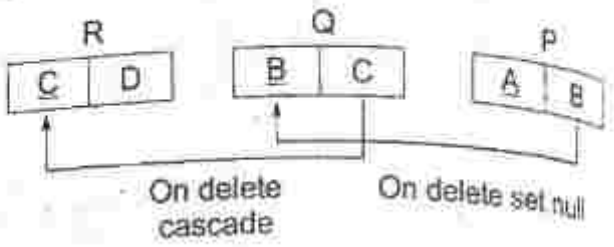
The output Table will be

Product_no.	employ_no.
P_2	E_2
P_3	E_7
P_5	E_2
P_6	E_7

230. (b)

Even through the R and S may have common attributes it would give output as cross-product unless we give explicit condition for tuple matching.
To make it equivalent to natural join we have to use WHERE condition to match the common attributes.

231. (b)



On deleting table R, the contents of table Q having attribute values of C as a or b will also gets deleted consequently, the contents of table P having attribute value of B as a or b will change their values to null.
Hence table P will looks like,

A	B
a	Null
b	Null

232. (b)

Option (b) is correct as 'IN' condition will give the respective S.No. and will equate both S_No. But EXISTS will check if any tuple exists and it will give wrong answer.

233. (d)

"Order by" clause is used to order the tuples and it can only be used as the last clause of a query

234. (a)

Option (a) query is illegal due to max (count(*))

235. (c)

Arity → Number of Columns
Cardinality → Number of Rows
For relation X cardinality = 2
For relation Y cardinality = 4
Hence the SQL statement
Select COUNT (*) from X, Y
gives the number of rows for the relation obtained from cross product of X, Y.
Hence, $2 \times 4 = 8$ is the correct option.

236. (b)

The query gives the names of those classes whose room number is 'R128' or have 5 or more students enrolled.

237. (b)

Given query will result tuple where $A \geq 5$ (i.e. A = 5, 6, 10, 8) and $B \geq C$ (i.e. $B = 8$ because we can not compare null values) or $C \geq 5$ (i.e. $C = 5, 6, 5$ because we can not compare null values) so the resulting tuple are 2 i.e. $A = 6$, $A = 8$.

238. (a)

- Select operation in SQL is equivalent to the projection operation in relation algebra, except that select in SQL contain duplicates.
- A SQL query can have a 'having' clause only if it has a 'group by' clause.
- Operator 'between' can't be used between main query and the subquery.
- If two relations R and S are joined, then the non-matching tuples of both R and S are ignored in natural join.

239. (4)

Because inner query result empty ALL clause condition true for all records of outer Query.

240. (c)

Select * from R where a in (select s.a from S)

≡

Select $R.*$ from R, (select distinct a from S) as $S1$ where $R.a = S1.a$.

241. (c)

Query computes name of persons who have travelled in every city where SBI is located.

242. (b)

where competes.cid = competitor.cid

It gives

Cid	01	02	03	04

and nationality = 'Swedish'

It gives

Cid
03

So corresponding eid = 02.

244. (c)

Given SQL query produces:
Displays Teachers who got a grade greater than equals to 9 in all course in madeeasy.

245. (a)

Select Eid ⇐ employee age max than 30
From Emp E
where age > 30 and
not Exists (select Pid ⇐ all the project id whose project name is database
From project P
where Pname = 'database' and
not exist (select Pid
⇐ the P.id where Eid is not in work relation
from works W
where W.Eid = E.Eid
and W.Pid = P.Pid))

246. (a)

Q_1 : Retrieves A which are more than some C.
Q_2 : Retrieves A which are more than some C.
Q_3 : Retrieves A which are more than every C.

247. (a)

Query I returns vertices whose out degree is only 2.
But Query II returns whose out degree is atleast 2.

248. (4)

Because inner query result empty ALL clause condition true for all records of outer Query.

249. (c)

Consider the following instance of the relation Employee

Employee		
E-id	Ename	Salary
1	Akbar	10000
2	Birbal	30000
3	Anand	3000
4	Anand	2000
5	Suresh	15000
6	Ramesh	25000

When the given query is executed then E-id 3, 4 are having same name and their total salary is less than 10000 will be return. The output will be

Ename	Salary
Anand	5000

250. (4)

The output table will be

Dealer-No.	Color-id
D_2	C_2
D_7	C_3
D_2	C_5
D_7	C_8

251. (a)

S_1 and S_2 are same as 'ANY' and 'IN' conditions will give the same result.

252. **(b)**

This query counts the number of tuples with salary > minimum salary. All the tuples will have unique salary. Hence only one tuple with minimum salary. Therefore 999 will be the result of the query (999 records will have salary greater than minimum salary).

253. **(a)**

Blocks required by data file = $\frac{n}{3}$

Number of records needed by pointer = $\frac{n}{10}$

Total blocks needed = $\frac{n}{3} + \frac{n}{10} = \frac{13n}{30}$.

254. **(c)**

For a leaf node, $n(K + R_p) + p_{next} \leq$ Block size

$\Rightarrow n(12 + 10) + 8 \leq 1024$ B

$\Rightarrow n \leq 46.18 = 46$

For a non-leaf node, $n(p) + n - 1(k) \leq$ Block size

$\Rightarrow n(8) + n - 1(12) \leq 1024$ B

$\Rightarrow 8n + 12n - 12 \leq 1024$

$\Rightarrow n \leq 51.8 = 51$

255. **(a)**

Suppose all nodes are completely full that means every node has $n - 1$ keys.

Tree has 4 levels. If a new key is inserted then at every level new node will be created.

In worst case root node will also be broken into two parts and we have 4 levels.

So answer should be 5 because tree will be increased with one more level.

256. **(c)**

In a dense index, an index entry appears for every search-key value in the file.

Primary index contain index entry for every block not for every search-key value.

Secondary index contain index entry for every search-key value.

So option (c) is correct.

257. **(b)**

The major **advantage of B+ tree** is in reducing the number of last level access which would be from disk in case of large data size. So, we need not to access disk every time.

258. **(c)**

In a B^+ tree we want an entire node content to be in a disk block. A node can contain up to p pointers to child nodes and up to $(p-1)$ key values for a B^+ tree of order P. Here, key size is 8 bytes and pointer size is 4 bytes. Thus,

$$8(p-1) + 4p \leq 512$$
$$12p \leq 520$$
$$p = 43$$

259. **(c)**

In B-tree every internal node other than root is at least half as full.

There are upper and lower bounds on the number of keys on a node. To specify these bounds we use a fined integer $t \geq 2$, the minimum degree of the B-tree:

- **Lower bound:** Every node other than root must have at least $(t-1)$ keys means at least t children.
- **Upper bound:** Every node can contain at most $(2t-1)$ keys means every internal node has at most $(2t)$ children.

260. **(a)**

The maximum number of leaves in a B-tree is when B-tree is complete. There will be k^h total number of leaves.

The minimum number of leaves in a B-tree when B-tree is satisfies just a minimum keep in internal node.

There will be $2\left\lceil \frac{k}{2} \right\rceil^{h-1}$ total number of leaves.

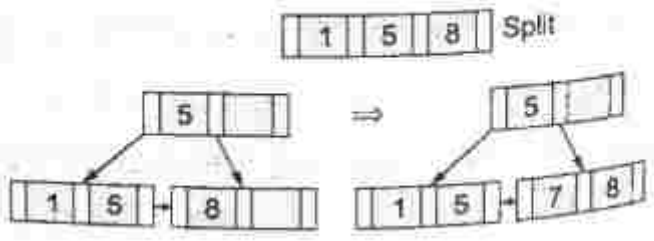
261. **(a)**

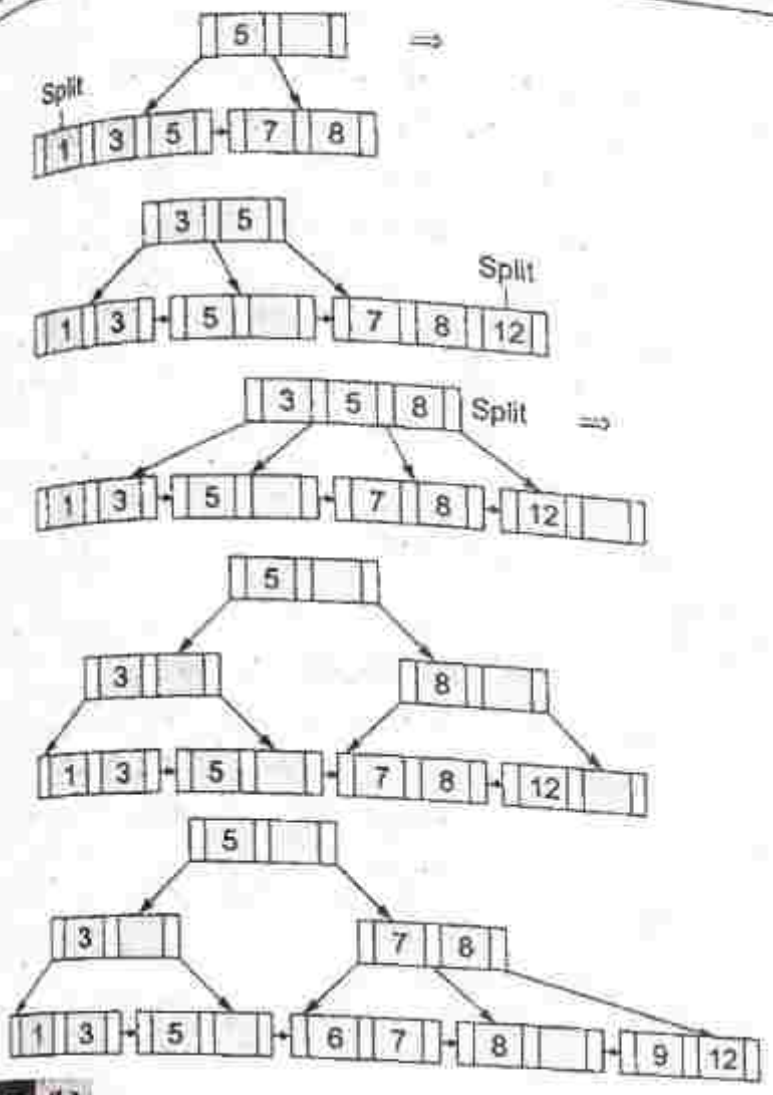
A **clustered index** is a special type of index that reorders the way records in the table are physically stored.

With a clustered index the rows are stored physically on the disk in the same orders as the index. Therefore, there can be only one clustered index.

262. **(a)**

B+ tree with maximum 3 pointers:





263. (b)

Most database systems use indexes built on some form of a B⁺-tree due to its many advantages, in particular its support for range queries. Leaf nodes are linked together in B⁺ trees hence range queries are faster.

264. (c)

Size of child pointer = 6 byte
Size of search field value takes 14 bytes
Block size = 512
The order of internal node = P
$\Rightarrow (P-1)14 + P * 6 \leq 512$
$P = 26$

265. (d)

The transfer of data from disk to primary memory is in the form of data blocks, if a data block is larger than indexing is easy due to this B⁺ tree is better than binary search tree data structure if large amount of data can be access.

266. (b)

In a B⁺ tree the root node have minimum two block pointer and maximum P block pointer where P = order, key = order – 1 and in the non-root node the minimum number of keys = $\left\lceil \frac{P}{2} \right\rceil - 1$

So in the question key = 5, order = 6
So minimum number of keys in non-root node = $\left\lceil \frac{6}{2} \right\rceil - 1 = 2$

267. (c)

File is organized so that the ordering of data records is not the same as or close to the ordering of data entries in some index. Then that index is called secondary index.

270. (a)

B^+ tree balanced because the length of the paths from the root to all leaf nodes are all equal and every internal node must filled by.

271. (a)

If records of a file are physically ordered on a non-key field which doesn't have a distinct value for each record that field is called the clustering field.

272. (a)

Contiguous allocation scheme would work best for a file system implementation on a device that can only be accessed sequentially, a tape drive.

273. (c)

- Anchor block contains first records of chain, while overflow block contains records other than those that are 1st records of a chain.
- Anchor record is first record of a block.

274. (a)

For storing the records number of blocks required $= \frac{n}{3}$ and
For storing the keys in dense index number of blocks required = $\frac{n}{10}$

So total blocks required = $\frac{n}{3} + \frac{n}{10} = \frac{13n}{30}$

275. (a)

Suppose all nodes are completely full that means every node has $n - 1$ keys.
Tree has 4 levels. If a new key is inserted then at every level new node will be created.
In worst case root node will also be broken into two parts and we have 4 levels.
So answer should be 5 because tree will be increased with one more level.

276. **(b)**

B^+ tree is best suitable for sequential access of data.

277. **(c)**

From the structure of B^+ tree we can get the following equation:

$n * p + (n-1) * (k + q) \le B$

where, n = Order, p = Tree/block/index pointer,
q = Record/data pointer,
B = Size of block

In B^+ tree, non leaf node has no record pointer:
(by putting $q = 0$)

$$n * p + (n-1)k \le B$$
$$n * 6 + (n-1) * 9 \le 512$$
$$n \le 34.77$$

Largest possible order of a non-leaf node = 34.

278. **(c)**

- A secondary index has to be dense since the file is not organized according to a secondary index. Otherwise, the index structure does not allow to find all values directly.
- If it were dense, we would have the same number of entries in the second level as in the first level index which defeats the purpose of the second level index.
- Yes, since we can directly access the record.
- Hashing does not support range queries since a hash function needs a specific key value (whereas we don't know the values of A given just $A > a$).

279. **(c)**

$p \times$ size of tree pointer + $(p-1) \times$ (size of key field) $\le$ Block size

Since it is B^+ tree. In order for each B^+ tree node to fit in a single disk block the maximum value of p is

$$p \times P + (p-1)\, V \le B$$
$$p \times 6 + (p-1) \times 9 \le 512$$
$$15p \le 521$$
$$p \le 34$$

280. **(a)**

Order of a B-tree refers to the number of block pointers it can accommodate and hence, $(n-1)$ key values and n-pointers are there in any block of B tree with order n.

281. **(a)**

Number of records = 5,00,000
Size of each record = 200B
Search key = 15 bytes
Tree/data pointers are 5B, Data block size = 1[illegible]
Order in B^+ tree must be defined specifically for leaf and non leaf nodes. But here, both tree and data pointer size is same so we can ignore it.
Thus calculating order for leaf node.

$$P(\text{Key} + Rp) + B_p \le 1024$$
$$P(15 + 5) + 5 \le 1024$$
$$P \le \frac{1019}{20},\ P \le 50.98 \text{ or } 50$$

Hence, option (a) is true.

282. **(538)**

Disk block size = 5000 records
Block size = 10 records or 15 (keys + Pointers)
Sparse index at 1st level. So number of disk block at 1st level is number of block in database.

$$\text{Data base} = \frac{5000}{10} = 500 \text{ blocks}$$

$$1^{st} \text{ level} = \left\lceil \frac{500}{15} \right\rceil = \lceil 33.33 \rceil = 34 \text{ blocks}$$

$$2^{nd} \text{ level} = \left\lceil \frac{34}{15} \right\rceil = \lceil 2.26 \rceil = 3 \text{ blocks}$$

3^{rd} level = 1 block
Total = [500 + 34 + 3 +1] blocks = 538 blocks

283. **(c)**

Hashed file can be accessed with O(1) complexity

284. **(c)**

Number of keys (Index key size + block pointer size) + Record pointer size $\le$ Block size

$$\Rightarrow\ n(8 + 12) + 12 \le 232$$
$$\Rightarrow\ n \le \frac{232 - 12}{20}$$
$$\Rightarrow\ n \le 11$$
$$\therefore\ \text{Maximum order} = 11$$

285. **(3)**

If we insert keys 45,48,55 in same order, then on insertion of key 55, root will be overflow and new level will be created.

286. **(a)**

If fill factor is 67% so $32 \times \frac{67}{100} = 21.44 = 21$

So, 21 block pointers
Number of keys = Number of block pointers -1
i.e., $(21 - 1) = 20$ keys

287. (b)

(a) All leafs are not at same level (not balanced). It is not *B*-tree.

(b) Root can have 1 key and remaining contains minimum 2 keys.
Hence it is *B*-tree.

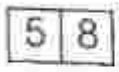
(c) "4" key has less than allowable size. (Minimum 2 keys)

(d) Right to 7, there is no key.

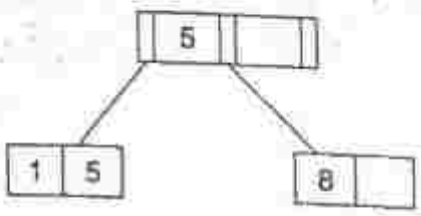
288. (18)

Construction of B^+ tree:

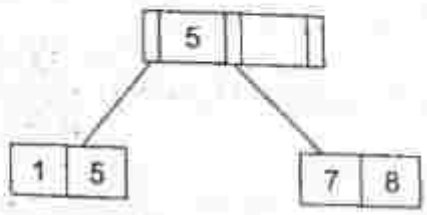
- Insert 8, 5



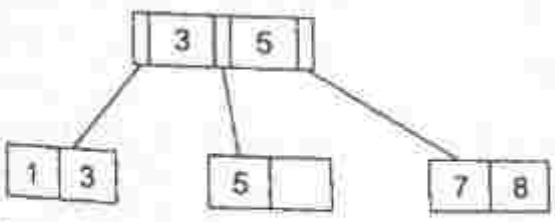
- Insert 1 (splitting with left biasing)



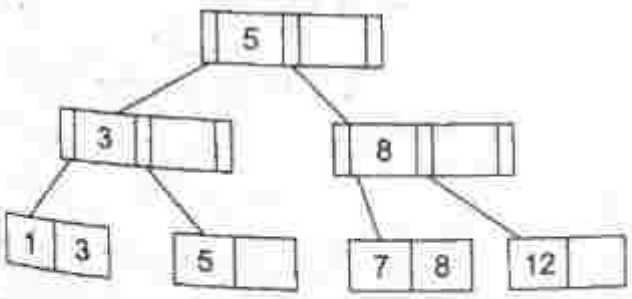
- Insert 7



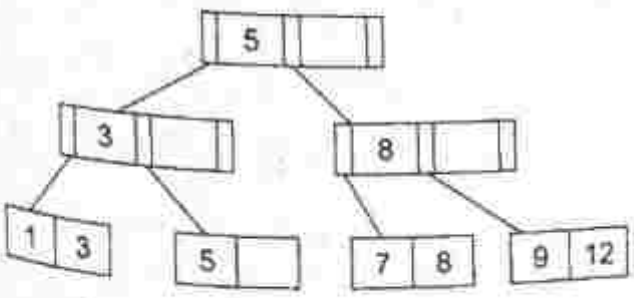
- Insert 3



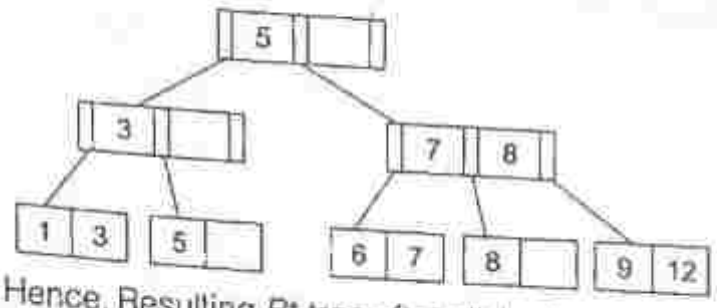
- Insert 12



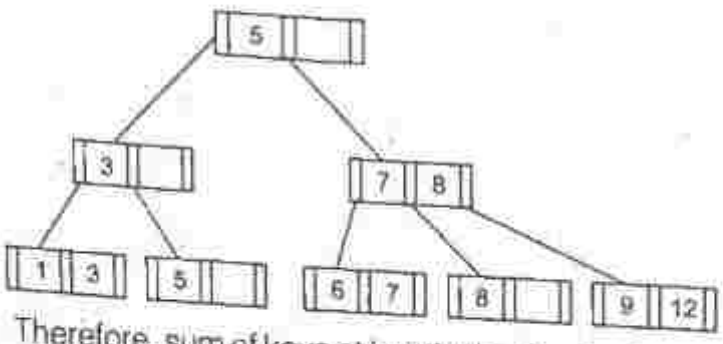
- Insert 9



- Insert 6



Hence, Resulting B^+ tree after all the insertions is shown below.



Therefore, sum of keys at height 1 is 3 + 7 + 8 = 18

289. (b)

- In sparse index number of entries depends on number of data block in previous level.
- In dense index number of entries depends on number of data record in previous level.

290. (9000)

In level 2 there will be 10 nodes as order is 10.
In level 3 there will be $10 \times 10 = 100$ nodes
In level 4 there will be $10 \times 100 = 1000$ nodes
At level 4 each node will have 9 record pointers.
So, therefore the maximum number of records that be indexed are $1000 \times 9 = 9000$.

291. (c)

Primary index may or may not be sparse.
Secondary index is always dense.

292. (a)

Order P : $P \times$ Pointer + $(P-1) \times$ Key $\leq$ Block

$$P \times 8 + (P-1)\,12 \leq 1000$$

$$20P \leq 1012$$

$$P = \left\lfloor \frac{1012}{20} \right\rfloor = 50$$

Level	Min nodes	Min B_p	Min keys
1	1	2	1
2	2	2 × 25	24
3	50	–	50 × 24
			1200

293. **(b)**

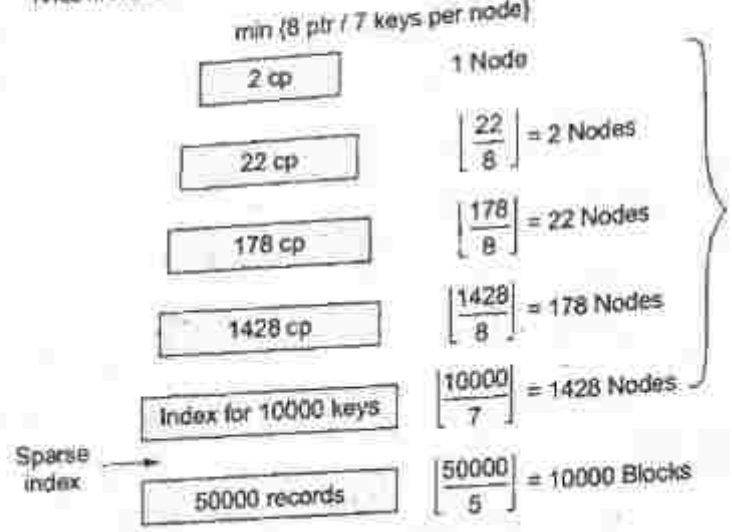
Order $P \Rightarrow P \times B_p + (P-1)\,\text{Key} \le \text{Block}$

$$P \times 12 + (P-1)\,20 \le 512$$

$$32P \le 532$$

$$P = \left\lfloor \frac{532}{32} \right\rfloor = 16$$

Maximum index nodes in index mean min fill factor



294. **(4)**

Sparse index and maximum (minimum filling in nodes)

Number of blocks in file $= \dfrac{1250}{3} = 417$ blocks

Minimum number of index blocks at 1st level

$$= \frac{417}{5} = \lfloor 83.4 \rfloor = 83$$

Minimum number of index blocks at 2nd level

$$= \left\lfloor \frac{83}{6} \right\rfloor = 13$$

Minimum number of index blocks at 3rd level

$$= \left\lfloor \frac{13}{6} \right\rfloor = 2$$

Minimum number of index blocks at 4th level = 1

So, maximum number of level are 4.

298. **(d)**

Number of blocks required for a file

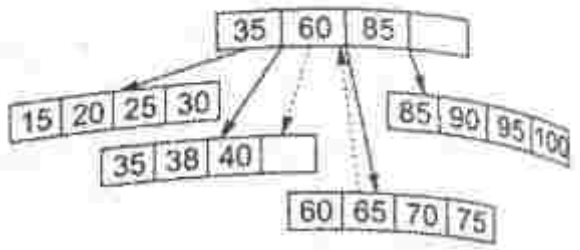
$$= \left\lceil \frac{500 \text{ records}}{13 \text{ records/block}} \right\rceil = 39$$

39 blocks required for 500 records and each block occupied by 13 records.

$39 \times 13 - 500 = 7$ records space is not used for file. The last block holds only 6 records for file and remaining space is wasted.

$\therefore 512 - 6 \times 38 = 284$ bytes of space unused.

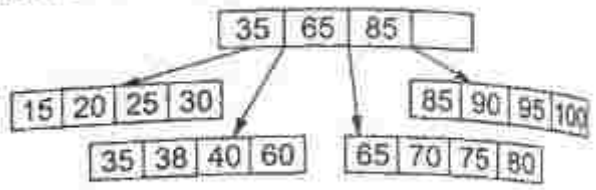
299. **(a)**



B^+ tree incorporate rotation technique.

Insertion of 80 would be done between 60 and 85. But in the node between 60 and 85, there is no space and also the child node of 85 and above has no space left. So due to rotation technique, 60 will be inserted into the child node of 35 and 60. 65 will be shifted upwards and thus 80 will be inserted.

So the tree will look like;



300. **(d)**

Average Access Time = Seek Time + Average Rotational Delay + Data Transfer Time

Seek Time = 0.02 (given)

$$\text{Average Rotational Delay} = \frac{60}{3000} \times \frac{1}{2} = 0.01$$

$$\text{Data Transfer Time} = \frac{500}{250000} = 0.002$$

So, Access Time = 0.02 + 0.01 + 0.002 = 0.32

301. **(c)**

At level '0', the number of records = $n-1$

At level '1', the number of records = $n(n-1)$

At level 'h' the number of records = $n^h(n-1)$

$$\text{Total} = (n-1)(1 + n + \ldots n^{h-1} + n^h)$$

$$= (n-1)\frac{(n^{h+1}-1)}{(n-1)} = n^{h+1}-1$$

302. **(b)**

$nBp + (n-1)\,\text{key} + (n-1)\,\text{record pointer} \le 1024$

$$n \times 6 + (n-1) \times 10 + (n-1) \times 8 \le 1024$$

$$6n + 10n + 8n - 18 \le 1024$$

$$24n \le 1024 + 18$$

$$n \le \frac{1042}{24}$$

$$n \le 43.41 \Rightarrow n = 43$$

UNIT 7

Data Warehouse

Q1 In binning, which method replaces each value in a bin with the median of the bin?

(a) Mean binning (b) Median binning
(c) Mode binning (d) Frequency binning

Q2 Which of the following statements is/are correct?

S_1: Roll-up operation in OLAP summarizes data by moving up a hierarchy, leading to higher-level aggregates.

S_2: Drill-down in OLAP increases data detail by navigating down a hierarchy, revealing finer granularity.

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q3 A dataset contains the values are

$25000, $30000, $35000, $40000,
$45000, $50000, $55000, $60000

If the mean income is $42500 and the Standard deviation is $10000, what is the Z-score of $50000?

Q4 Which of the following is true about Fact Constellation Schema?

(a) It allows data modeling for complex business processes.
(b) It uses separate dimension tables for every fact table.
(c) It supports only one subject area.
(d) It is less flexible than Star Schema.

Q5 Which of the following binning methods is primarily concerned with dividing the data into bins of equal width, regardless of how many data points fall into each bin?

(a) Mean binning
(b) Median binning
(c) Equal-width binning
(d) Equal-frequency binning

Q6 Consider the following statements regarding OLAP (Online Analytical Processing) and OLTP (Online Transaction Processing) in data warehousing:

S_1: OLAP is used for decision support and data analysis based on historical data.

S_2: OLTP ensures data integrity and consistency across operational systems.

S_3: OLAP databases are optimized for fast transaction processing and real-time updates.

S_4: OLTP databases are typically denormalized to improve query performance.

Which of the following is/are true?

(a) S_1 and S_2 (b) S_1 and S_4
(c) S_2 and S_3 (d) S_1, S_2, and S_3

Q7 During ETL, a standardized value of +1.25 was produced. If mean latency = 350 ms and Standard deviation = 80 ms, find the original latency.

Q8 A factory monitors the average number of defects per batch over the first 50 batches, which is recorded as 8. If the 51st batch has 12 defects, what is the updated average number of defects per batch after including the 51st batch? _____.

Q9 Suppose a group of 12 sales price records has been sorted methods as follows:

16, 28, 40, 64, 78, 82, 108,
118, 138, 208, 214, 236

Partition into four bins using equal-width partitioning. Which of the following statements is/are correct?

(a) Difference of sum of B_2 and B_4 is 270
(b) Difference of sum of B_2 and B_4 is 272
(c) Difference of sum of B_1 and B_3 is 10
(d) Difference of sum of B_1 and B_3 is 15

Q10 Which smoothing technique replaces the data in a bin with the closest boundary value (either minimum or maximum in the bin)?

(a) Mean binning
(b) Smoothing by bin boundaries
(c) Median binning
(d) Equal-frequency binning

Q.11 Consider the following statements:

S_1: Fact Constellation Schema is also called Galaxy Schema.

S_2: Each fact table in a constellation has its own separate dimension tables.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.12 For a product catalog, average product price is \$300. Standard deviation is \$40.
Which of the following prices is an outlier ($Z > 3$)?

(a) \$420 (b) \$410
(c) \$430 (d) \$405

Q.13 From a retail dataset:
A customer purchased 18 items in one visit.
Mean = 12, Standard deviation = 3.
What is the Z-score of this purchase?

Q.14 Consider the following statements:

S_1: Equal-frequency binning partitions data so that each bin has roughly the same number of data points.

S_2: Histogram-based discretization always uses bins of equal width.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.15 A salary of \$84000 has been normalized to a Z-score of 2. What must be the mean of the data if the standard deviation is \$4000?

(a) 76000 (b) 84000
(c) 72000 (d) 86000

Q.16 Which of the following are true about the data storage in OLTP systems?

(a) OLTP systems use normalized tables to reduce redundancy.
(b) OLTP systems store historical data for long-term analysis.
(c) OLTP systems require efficient indexing to ensure fast data retrieval.
(d) OLTP systems are typically used for strategic decision-making.

Q.17 Consider a dataset containing income values as follows:
\$45,000, \$52,000, \$60,000, \$78,000, \$85,000
If the income attribute is to be normalized to a new scale ranging from 2 to 6 using min-max normalization, calculate the normalized value corresponding to \$60,000. Additionally, determine the normalized value for \$52,000 within the same scale.

(a) Normalized value of \$60000 is 3.5
(b) Normalized value of \$60000 is 2.7
(c) Normalized value of \$52000 is 2.7
(d) Normalized value of \$52000 is 3.5

Q.18 Consider the following statements:

S_1: Snowflake Schema reduces data redundancy.

S_2: Snowflake Schema increases schema complexity.

Which of the following is correct?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q.19 Which of the following binning techniques is focused on equal number of data points per bin, not equal interval width?

(a) Equal-width binning
(b) Frequency binning
(c) Mean binning
(d) Smoothing by bin boundaries

Q.20 A customer demographic dataset contains age values: 20, 35, 50, 65, and 80 years.
When normalized to the range [0, 10], find the normalized age value corresponding to 50 years. Additionally, determine the original age that corresponds to a normalized value of 7.

Q.21 Which binning method divides the range of data into intervals of equal size, regardless of the number of data points in each?

(a) Equal-width binning
(b) Equal-frequency binning
(c) Mean binning
(d) Median binning

Q.22 Which of the following statements is/are correct?

S_1: Roll-up is an OLAP operation that reduces the dimensionality of data.

S_2: Drill-down operation can be used to decrease the number of dimensions in an OLAP cube.

(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 or S_2

Q.23 Consider a data set with "income" values as an attribute \$60000, \$70000, \$80000, \$90000, \$98000, and \$108000. To find the min-max normalized value of the "income" attribute set to \$108000. If the new range is [0, 4].
Calculate the normalized value using the min-max?

Q.24 In data preprocessing, which binning method replaces all values in a bin with the median of that bin?
(a) Mean binning
(b) Median binning
(c) Equal-frequency binning
(d) Equal-width binning

Q.25 After dividing a dataset into bins, each value in a bin is replaced with the nearest boundary value (either the lower or upper bound of the bin). Which binning method does this describe?
(a) Mean binning (b) Median binning
(c) Boundary binning
(d) Equal-width binning

Q.26 A student has taken 6 tests with an average score of 75. After a 7th test, their average rises to 77. What score did they get in the 7th test?
(a) 75 (b) 77
(c) 66 (d) 89

Q.27 Which of the following is a characteristic of OLAP (Online Analytical Processing) systems in data warehousing?
(a) OLAP is designed for high throughput and low-latency transaction processing.
(b) OLAP databases use denormalized data models to improve query performance.
(c) OLAP is used primarily for maintaining current operational data.
(d) OLAP supports complex querying and multidimensional analysis for business intelligence.

Q.28 Consider the following statements:
S_1: Min-max normalization scale the feature values to fixed range, typically [0, 1].
S_2: Min-max normalization is sensitive to outlier as outlier, can influence the scaling.
(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) Neither S_1 nor S_2

Q.29 The average marks of 30 students in section A is 70. The average marks of 20 students in section B is 80. What is the overall average?

Q.30 The heights (in cm) of a group are:
150, 160, 170, 180, 190.
If the normalized height corresponding to 170 cm is 0.5 on the scale [0, 1], find the normalized value of 180 cm.

Q.31 The average sales for 4 months is \$25000. To raise the average to \$26000 after 5 months, what should be the sales in the 5th month?

Q.32 Consider the following statement about hierarchies in data warehouse:
(a) A roll-up operation moves from a higher level of to a lower level.
(b) Hierarchies facilitate efficient data navigation and analysis at different aggregation levels.
(c) Star schema do not support hierarchical data structure.
(d) Hierarchical structures can improve query performance in data warehouse.
Which of the above is correct?

Q.33 Which of the following is a data smoothing technique that helps reduce noise in data?
(a) Binning (b) One-hot encoding
(c) Standardization (d) PCA

Q.34 A factory had a defect average of 6 per batch for 10 batches. After 5 more batches, the average drops to 5. What is the total number of defects in the last 5 batches?

Q.35 Which of the following is correct regarding the Star Schema?
(a) It normalizes all dimension tables to reduce redundancy.
(b) It creates a complex relationship between fact and dimension tables.
(c) It contains a central fact table and surrounding denormalized dimension tables.
(d) It uses snowflaked normalized structure for dimensions.

Q.36 A search engine query had a Z-score of 1.8. If the average query time is 0.4 scores and Standard deviation is 0.1 seconds, what was the actual time?

Q.37 Consider the following statements:

S_1: In Star Schema, dimension tables are typically normalized.

S_2: Star Schema enhances query performance by simplifying joins.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.38 Which of the following is/are true for Star Schema?

(a) It supports fast aggregations.

(b) Fact table stores dimensional descriptions.

(c) Dimension tables contain hierarchies in denormalized form.

(d) None of these

Q.39 Consider the following statements:

S_1 : Star Schema is suitable for large OLTP systems.

S_2 : Star Schema simplifies complex queries in OLAP.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.40 Which of the following is correct about Star Schema?

(a) All dimension tables are directly connected to the fact table.

(b) Dimension tables connect through intermediary tables.

(c) Dimension tables are normalized to reduce joins.

(d) Star Schema contains only numeric data.

Q.41 A customer demographic dataset contains age values that are normalized using min-max normalization to the range [0, 10]. The normalized value corresponding to age 50 years is 5.0, and the minimum age in the dataset is 20 years. Determine the maximum age in the original dataset.

Q.42 Which of the following is correct in the context of Star Schema?

(a) Star schema has more complex joins than Snowflake Schema

(b) Dimension tables are normalized to 3NF

(c) Star schema is ideal for reducing query response time

(d) All of the above

Q.43 Which of the following is correct regarding Snowflake Schema?

(a) Dimension tables are fully denormalized

(b) It avoids redundancy by normalizing dimension tables.

(c) It reduces the number of joins in queries

(d) Fact tables are normalized.

Q.44 Consider the following statements:

S_1: Snowflake Schema reduces storage requirements by normalization.

S_2: Snowflake Schema has faster query performance than Star Schema.

Which of the following is correct?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.45 If the Z-score of a value is greater than 3, it is considered a potential outlier.

In a dataset with mean \$100000 and Standard deviation \$5000, which of the following is an outlier?

(a) \$112000 (b) \$108000

(c) \$117000 (d) \$116000

Q.46 Which of the following is/are true for Snowflake Schema?

(a) All dimensions are denormalized.

(b) Suitable for complex queries with normalized structure.

(c) Provides faster query speed than Star Schema.

(d) Uses fewer tables than Star Schema.

Q.47 Which of the following is true about Snowflake Schema?

(a) Dimension tables are normalized into multiple related tables.

(b) Fact table stores textual data.

(c) It is faster than Star Schema in query processing.

(d) All dimension data is stored in a single table

Q.48 Which of the following is correct about Fact Constellation Schema?

(a) It has only one fact table shared by all dimensions.

(b) It consists of multiple fact tables that share dimension tables.

(c) It stores only summary-level data.

(d) It avoids reusability of dimensions

Q.49 A set of page load times (in seconds) is recorded as: 0.8, 1.2, 1.5, 2.0, and 3.2.
Given that the maximum value is 3.2 seconds and the normalized value of 1.5 seconds under min-max normalization to the [0, 1] range is 0.4, find the minimum page load time in the dataset.

Q.50 Consider the following statements:
S_1 : OLAP uses star and snowflake schemas for organizing data, enhancing query performance.
S_2 : OLAP systems primarily focus on individual transaction processing and daily operations.
Which of the following is correct?
(a) Only S_1 (b) Only S_2
(c) Both S_2 and S_2 (d) None of these

Q.51 Consider the following statement about hierarchies in data warehouse:
(a) A roll-up operation moves from a higher level of to a lower level.
(b) Hierarchies facilitate efficient data navigation and analysis at different aggregation levels.
(c) Star schema do not support hierarchical data structure.
(d) Hierarchical structures can improve query performance in data warehouse.
Which of the above is correct?

Q.52 Which of the following is true for Fact Constellation Schema?
(a) It avoids redundancy completely.
(b) It cannot support shared dimensions.
(c) It can represent many-to-many relationships using bridge tables.
(d) It is limited to transactional data.

Q.53 Which of the following describes Fact Constellation Schema?
(a) Contains a single fact table per schema.
(b) Dimension tables cannot be shared.
(c) Useful for representing complex, interrelated business processes.
(d) Based on denormalized dimension structures.

Q.54 Which of the following is/are true?
(a) Star Schema is easier for end-users to query.
(b) Snowflake Schema avoids redundancy.
(c) Fact Constellation Schema supports Single fact tables.
(d) None of these

Q.55 Which of the following is a drawback of Snowflake Schema compared to Star Schema?
(a) Redundancy in dimension tables
(b) Complex joins due to normalized structure
(c) Inefficient storage
(d) Inability to model hierarchies

Q.56 In which schema are dimension tables typically normalized into multiple related tables?
(a) Star Schema
(b) Snowflake Schema
(c) Fact Constellation Schema
(d) Flat File Schema

Q.57 A set of 15 temperature records has been sorted as follows:
14, 18, 22, 25, 29, 35, 38, 45,
50, 52, 60, 64, 70, 73, 80
Partition the data into three bins using equal-width partitioning. What will be the difference least values in Bin 3 and Bin 2?

Q.58 Consider the following statements:
S_1 : In a Snowflake schema, all dimension tables are normalized.
S_2 : In a Snowflake schema, some dimension tables are normalized.
Which of the following is correct?
(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 nor S_2

Q.59 Which of the following statements is/are correct?
(a) OLAP is used for decision support and data analysis based on historical data.
(b) OLTP ensures data integrity and consistency across operational systems.
(c) OLAP databases are optimized for fast transaction processing and real-time updates.
(d) OLTP databases are typically denormalized to improve query performance.

Q.60 Which of the following statements is/are correct?
(a) Binning without dependence on class labels or predefined threshold is classified as unsupervised discretization.
(b) Discretization by Binning consist of partitioning a continuous attribute into bins of equal width or frequency.

(c) Discretization by histogram analysis uses histogram bins to discretize continuous data based on frequency distribution.
(d) Discretization by histogram analysis always result in bins of equal width.

Q.61 Monthly sales quantities are recorded as follows: 250, 300, 350, 400, 450, and 500 units. If min-max
Normalization is applied to rescale the data into the range [100, 200], then what is the difference between the normalized value of a sales quantity of 375 unit and that of 300 units.

Q.62 In binning, which of the following methods replaces each value in a bin with the bin's mean value?
(a) Mean binning
(b) Median binning
(c) Equal-width binning
(d) Frequency binning

Q.63 Which of the following statements is/are correct?
S_1: Roll-up is an OLAP operation that reduces the dimensionality of data.
S_2: Drill-down operation can be used to decrease the number of dimensions in an OLAP cube.
(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 or S_2

Q.64 Which of the following statements is/are correct?
(a) Binning without dependence on class labels or predefined threshold is classified as unsupervised discretization.
(b) Discretization by Binning consist of partitioning a continuous attribute into bins of equal width or frequency.
(c) Discretization by histogram analysis uses histogram bins to discretize continuous data based on frequency distribution.
(d) Discretization by histogram analysis always result in bins of equal width.

Q.65 Suppose a group of 12 sales price records has been sorted methods as follows:
5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215
Partition into three bins using equal-width partitioning what will be the value of Bin 2

Q.66 Which of the following is an accurate description of the "equal-width binning" method in discretization?
(a) The range of data is divided into intervals of equal size, and each data point is assigned to the corresponding interval.
(b) Data is grouped based on frequency, ensuring each bin contains an equal number of data points.
(c) Data points are assigned to bins based on their proximity to the bin's mean value.
(d) The data range is split into bins based on predefined thresholds that vary in width.

Q.67 A company tracks the average monthly sales of its product and finds that the average sales over 12 months is \$ 1200. If the sales for the 13th months are \$ 1500. What is the updated average sales?

Q.68 Consider the following statements:
S_1: OLAP uses star and snowflake schemas for organizing data, enhancing query performance.
S_2: OLAP systems primarily focus on individual transaction processing and daily operations.
Which of the following is correct?
(a) Only S_1 (b) Only S_2
(c) Both S_2 and S_2 (d) None of these

Q.69 In a product rating system where ratings range from 1 to 10, the ratings are normalized to a scale from –5 to +5.
Find the min-max normalized value corresponding to a rating of 7. Furthermore, determine the original rating that corresponds to a normalized value of 0.
(a) (2.67, 5.5) (b) (2.03, 5.5)
(c) (2.5, 5.5) (d) (1.67, 5.5)

Q.70 Which of the following is an accurate description of the equal-width binning method in discretization?
(a) The range of data is divided into intervals of equal size, and each data point is assigned to the corresponding interval.
(b) Data is grouped based on frequency, ensuring each bin contains an equal number of data points.

(c) Data points are assigned to bins based on their proximity to the bin's mean value.
(d) The data range is split into bins based on predefined thresholds that vary in width.

Q.71 Consider the following sorted dataset of exam scores for a group of 15 students:
Score: 45, 55, 60, 62, 70, 73, 75, 80, 82, 85, 90, 92, 95, 100, 120
Using the equal frequency method, partition the dataset into 3 bins. What will be the difference between the sum of Bin 2 and Bin 3?

Q.72 Consider the following statements:
S_1: Binning reduces dimensionality by aggregating data points into fewer categories.
S_2: OLTP focuses on complex queries and aggregations typical in data warehouses.
Which of the following is correct?
(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 nor S_2

Q.73 Which binning technique divides the data range into bins having the same number of data points but does not necessarily replace values by the bin's mean or median?
(a) Mean binning
(b) Median binning
(c) Equal-frequency binning
(d) Equal-width binning

Q.74 Consider the following statements:
S_1: In Star Schema, dimensions are often normalized.
S_2: Snowflake Schema is good for minimizing disk space usage.
S_3: Fact Constellation Schema simplifies business analysis by merging all data into one table.
Which of the following is correct?
(a) Only S_1 and S_2 (b) Only S_2 and S_3
(c) Only S_1 and S_3 (d) None of these

Q.75 The mean of 200 data points is 150. If a new data point with a value of 180 is added, What will be the new mean?

■■■■

Answers Data Warehouse

1.	(b)	2.	(c)	3.	(0.75)	4.	(a)	5.	(c)	6.	(a)	7.	(450)		
8.	(8.07)	9.	(b, c)	10.	(b)	11.	(a)	12.	(a, c)	13.	(2)	14.	(a)		
15.	(c)	16.	(a, c)	17.	(a, c)	18.	(c)	19.	(b)	20.	(62)	21.	(a)		
22.	(d)	23.	(4)	24.	(b)	25.	(c)	26.	(d)	27.	(d)	28.	(c)		
29.	(74)	30.	(0.75)	31.	(30000)	32.	(b, d)	33.	(a)	34.	(15)	35.	(c)		
36.	(0.58)	37.	(b)	38.	(a, c)	39.	(b)	40.	(a)	41.	(80)	42.	(c)		
43.	(b)	44.	(a)	45.	(c, d)	46.	(b)	47.	(a)	48.	(b)	49.	(0.367)		
50.	(a)	51.	(b, d)	52	(c)	53.	(c)	54.	(a, b)	55.	(b)	56.	(b)		
57.	(22)	58.	(a)	59.	(a, b)	60.	(a, b, c)	61.	(30)	62.	(a)	63.	(d)		
64.	(a, b, c)	65.	(92)	66.	(a)	67.	(1223.07)	68.	(a)	69.	(d)	70.	(a)		
71.	(102)	72.	(a)	73.	(c)	74.	(a)	75.	(150.14)						

Explanations Data Warehouse

1. (b)

In median binning, every value in a bin is replaced with the median (middle value when sorted) of that bin. This method is good at reducing the effect of outliers.

2. (c)

- The roll-up operation in OLAP (Online Analytical Processing) aggregates data by moving up the hierarchy levels (e.g., from individual sales to regional totals). So, S_1 is correct.
- The drill-down operation in OLAP allows users to view more detailed data by moving down the hierarchy (e.g., from regional totals to individual sales). So, S_2 is correct.

3. (0.75)

Given,

Data point (X) = \$50000
Mean (μ) = \$42500
Standard deviation (σ) = \$10000

We know that,

$$\text{Z-score}(Z) = \frac{X-\mu}{\sigma}$$

$$Z = \frac{50000-42500}{10000}$$

$$= \frac{7500}{10000} = 0.75$$

4. (a)

Fact Constellation Schema supports data modeling for complex business processes by using multiple fact tables. This allows it to represent different subject areas within the same database.

5. (c)

Equal-width binning splits the entire range of the dataset into bins of equal numerical width. The focus here is purely on the size of the intervals, regardless of how many data points fall into each bin. This method can result in bins having unequal numbers of data points but provides a simple and consistent way to segment data based on value ranges.

6. (a)

- OLAP is indeed used for decision support and analysis based on historical data. So, S_1 is true
- OLTP ensures data integrity and consistency across operational systems. So, S_2 is true
- OLAP databases are not optimized for fast transaction processing and real-time updates; that is the role of OLTP. So, S_3 is false
- OLTP databases are typically normalized, not denormalized, to maintain data integrity and reduce redundancy. S_4 is false

7. (450)

Given, $Z = +1.25$
Mean latency (μ) = 350 ms
Standard deviation (σ) = 80 ms

We know that, $Z = \frac{X-\mu}{\sigma}$

$X = \mu + Z \cdot \sigma$
$X = 350 + 1.25 \times 80$
$= 350 + 100 = 450$

8. (8.07)

Calculate the total defects in the first 50 batches
Average defects per batch = 8
Total defects in 50 batches = $8 \times 50 = 400$
Add the defects from the 51st batch:
Defects in the 51st batch = 12
Total defects after 51 batches = 400 + 12 = 412
Calculate the updated average defects per batch:
Total number of batches = 51
Updated average defects per batch

$$= \frac{\text{Total defects}}{\text{Total number of batches}}$$

$$\text{Updated average} = \frac{412}{51} = 8.07$$

So, the updated average number of defects per batch after including the 51st batch is approximately 8.08.

9. (b, c)

Sales price records:
16, 28, 40, 64, 78, 82, 108, 118, 138, 208, 214, 236

Range of Data = Max – Min = 236 – 16 = 220
Number of Bins = 4

$$\text{Width of each Bin} = \frac{\text{Range of Data}}{\text{Number of Bins}} = \frac{220}{4} = 55$$

Calculate the Bin Intervals:

Bin 1 = From 16 to 16 + 55 = 71
Bin 2 = From 71 to 71 + 55 = 126
Bin 3 = From 126 to 126 + 55 = 181
Bin 4 = From 181 to 181 + 55 = 237

Allocated Bin:

Bin 1 = 16, 28, 40, 64, sum = 148
Bin 2 = 78, 82, 108, 118, sum = 386
Bin 3 = 138
So, Bin 4 = 208, 214, 236, sum = 658
Bin 4 – Bin 2 = 658 – 386 = 272
Bin 1 – Bin 3 = 148 – 138 = 10

10. (b)

In smoothing by bin boundaries, each data point is replaced by the closest extreme (boundary) value of the bin — either the minimum or the maximum value in the bin.

11. (a)

- Fact Constellation Schema is also known as Galaxy Schema, reflecting its complex structure with multiple fact tables. So, S_1 is correct.
- In this schema, dimension tables are often shared among fact tables rather than being separate for each. This sharing of dimensions is a key characteristic of the schema. So, S_2 is incorrect.

12. (a, c)

Given, Mean (μ) = 300 ms
Standard deviation (σ) = 40
An outlier as a value with $Z > 3$
We know that,

$$Z = \frac{X - \mu}{\sigma}$$

(a) \$420 $\quad Z = \frac{420 - 300}{40} = \frac{120}{40} = 3.0$

(b) \$410 $\quad Z = \frac{410 - 300}{40} = \frac{121}{40} = 2.75$

(c) \$430 $\quad Z = \frac{430 - 300}{40} = \frac{130}{40} = 3.25$

(d) \$405 $\quad Z = \frac{420 - 300}{40} = \frac{105}{40} = 2.625$

13. (2)

We are given

Purchase value (X) = 18, Mean (μ) = 12
Standard deviation (σ) = 3
We know that,

$$\text{Z-score}(Z) = \frac{X - \mu}{\sigma}$$

$$Z = \frac{18 - 12}{3} = \frac{6}{3} = 2$$

14. (a)

- Equal-frequency binning is a discretization technique where the continuous data is divided into bins such that each bin contains approximately the same number of data points. This helps in balancing the distribution across bins. So, S_2 is correct.
- Histogram-based discretization does not always use bins of equal width. While many histograms use equal-width bins, there are variants like equal-frequency histograms where bin widths vary to maintain an equal number of data points per bin. Thus, the term "always". So, S_2 is incorrect.

15. (c)

Given, Salary (X) = 84000
Z-score (Z) = 2
Standard deviation (σ) = 4000
Mean (μ) = ?

We know that, $Z = \frac{X - \mu}{\sigma}$

$$2 = \frac{84000 - \mu}{4000}$$

$\Rightarrow \quad 84000 - \mu = 8000$
$\Rightarrow \quad \mu = 84000 - 8000 = 76000$

16. (a, c)

(a) OLTP systems use normalized tables to reduce data redundancy and ensure data integrity. So, option (a) is correct.
(b) OLTP systems focus on current transaction data, not on storing historical data for long-term analysis, which is more typical of OLAP systems. So, option (b) is incorrect.

(c) Efficient indexing is crucial in OLTP systems to allow for quick retrieval of transaction data. So, option (c) is correct.

(d) OLTP systems are used for day-to-day transaction processing, not for strategic decision-making, which is the role of OLAP systems. So, option (d) is incorrect.

17. (a, c)

Min value, X_{min} = 45000

Max value, X_{max} = 85000

New range $[a, b]$ = [2, 6]

Values to normalize: $X = 60000$ and $X = 52000$

Min-max normalization formula:

For, $X = 60000$

$$X' = a + \frac{(x - x_{min})(b - a)}{x_{max} - x_{min}}$$

$$X' = 2 + \frac{(60000 - 45000)(6 - 2)}{85000 - 45000}$$

$$= 2 + \frac{15000 \times 4}{40000} = 2 + 1.5 = 3.5$$

Min-max normalization formula:

$$X' = a + \frac{(x - x_{min})(b - a)}{x_{max} - x_{min}}$$

For, $X = 52000$

$$X' = 2 + \frac{(52000 - 45000)(4)}{40000}$$

$$= 2 + \frac{7000 \times 4}{40000} = 2 + 0.7 = 2.7$$

Normalized value for \$60,000 is 3.5

Normalized value for \$52,000 is 2.7

18. (c)

- Snowflake Schema reduces data redundancy by normalizing dimension tables, which helps eliminate duplicate data and saves storage. However, this normalization increases the number of tables and relationships, making the schema more complex. As a result, while data consistency improves, query complexity and maintenance efforts also increase.

19. (b)

Frequency (or equal-frequency) binning divides the data into bins such that each bin has approximately the same number of elements, regardless of the range of values.

20. (62)

Given,

$$x_{min} = 20,\ x_{max} = 80$$

New range [0, 10]

x = 50, normalized x' to find

Normalized value x' = 7, original x to find

Normalize 50:

$$x' = 0 + \frac{(50 - 20)(10 - 0)}{80 - 20} = \frac{30 \times 10}{60} = 5$$

Inverse for $x' = 7$

$$x = x_{min} + \frac{(x' - a)(x_{max} - x_{min})}{b - a}$$

$$= 20 + \frac{7 \times 60}{10} = 20 + 42 = 62$$

Normalized age of 50 years = 5

Original age for normalized value 7 = 62 years

21. (a)

In equal-width binning, the entire range of values is divided into k intervals of the same width. Data points are grouped based on which interval they fall into.

22. (d)

S_1: Roll-up does not reduce dimensionality; it aggregates data within existing dimensions. So, S_1 is incorrect.

S_2: Drill-down is an OLAP operation that increases data details by moving down a concept hierarchy, providing more specific, granular data. So, S_2 is incorrect.

23. (4)

We know that the formula of min-max normalization

$$\text{Normalized value} = \frac{X - \min_A}{\max_A - \min_A} (\text{New max} - \text{New min}) + \text{New min}$$

Min value = \$60000

Max value = \$108000

Value to normalized (V) : \$ 98000

New range of normalization [0, 4]

$$\text{Normalized value} = \frac{108000 - 60000}{108000 - 60000} [4 - 0] + 0$$

$$= \frac{48000}{48000} \times 4 + 0$$

Normalized value = $1 \times 4 = 4$

24. (b)

Median binning replaces each value in a bin with the median value of that bin. The median is the middle value when the data in the bin is sorted, which makes this method less sensitive to outliers compared to mean binning. This approach helps smooth the data by reducing noise while preserving the central tendency in each bin.

25. (c)

Boundary binning replaces every value in a bin with the nearest boundary value, either the lower or upper limit of the bin. This method smooths the data by mapping all values in a bin to a fixed boundary point; which is typically the start or end of the bin's interval, rather than calculating a statistical measure like the mean or median. This helps preserve edge values and reduces the impact of noise, especially when values are skewed toward one side of the bin.

26. (d)

Given, Average of 6^{th} tests = 75

Total after 6 tests = $6 \times 75 = 450$

Average of after 7^{th} tests = 77

Total after 7 tests = $7 \times 77 = 539$

Now, 7^{th} test score = $539 - 450 = 89$

27. (d)

- OLAP is not designed for transaction processing; OLTP systems handle transactional data with high throughput and low latency.
- OLAP databases may use denormalized data models for performance reasons, but this is not a defining characteristic.
- OLAP does not focus on maintaining current operational data; OLTP, systems are designed for operational data management.
- OLAP systems provide capabilities for complex querying and multidimensional analysis, which are essential for business intelligence and decision support.

28. (c)

- Fixed range [0, 1].
- Extreme value can skew the minimum and maximum values, thus affecting the scaling of other values.

29. (74)

Given,

Section A :

Total number of student = 30

Average marks = 70

Total marks = $30 \times 70 = 2100$

Section B :

Total number of student = 20

Average marks = 80

Total marks = $20 \times 80 = 1660$

Both Section A and Section B :

Total marks = $2100 + 1600 = 3700$

$$\text{Overall average} = \frac{3700}{50} = 74$$

30. (0.75)

Given,

$X_{min} = 150$, $X_{max} = 190$

$X = 170$

Normalized $X' = 0.5$

Range = [0, 1]

Normalize at 180:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$X' = \frac{180 - 150}{40} = \frac{30}{40} = 0.75$$

Original height for $X' = 0.8$

$X = 150 + 0.8 \times 40 = 150 + 32 = 182$

Normalized value of 180 cm = 0.75

31. (30000)

Average sales for 4 months $25000

Total sales for 4 months = $4 \times 25000 = 100000$

Desired average after 5 months = $26000

Required total for 5 months = $5 \times 26000 = 130000$

5^{th} month sales = $130000 - 100000 = 30000$

32. (b, d)

(a) A roll-up operation moves from a lower level of granularity to a higher level, aggregating the data. So, option (a) is incorrect.

(b) Hierarchies facilitate efficient data navigation and analysis by organized data into level or tiers allow user to drill down or drill up to different levels of details as needed. So, option (b) is correct.

(c) Star schema do support hierarchical data structure although they are less normalized compared to snowflake schema. So, option (c) is incorrect.

(d) Hierarchical structures can improve query performance in data warehouse by organized data efficient for analysis. So, option (d) is correct.

33. (a)

Binning is a smoothing technique where continuous data is grouped into bins. It helps reduce the effects of minor observation errors or outliers by smoothing data values within bins.

34. (15)

First 10 batches $= 10 \times 6 = 60$

New average = 5 over 15 batches

$\Rightarrow$ Total $= 15 \times 5 = 75$

Last 5 batches $= 75 - 60 = 15$ defects

35. (c)

A Star Schema consists of a central fact table that holds measurable data, surrounded by denormalized dimension tables that provide descriptive context. This design simplifies queries and improves performance by reducing joins. It is commonly used in data warehousing for analytical processing.

36. (0.58)

Given, Z-score $= 1.8$

Mean $(\mu) = 0.4$ seconds

Standard deviation $(\sigma) = 0.1$ seconds

We know that, $Z = \dfrac{X-\mu}{\sigma}$

$$Z \cdot \sigma = X - \mu$$

$$X = \mu + Z \cdot \sigma$$

$$X = 0.4 + 1.8 \times 0.1$$

$$= 0.4 + 0.18 = 0.58$$

37. (b)

- In Star Schema, dimension tables are generally denormalized, not normalized, to make data access faster. So, S_1 is incorrect.
- this structure simplifies joins, making queries more efficient. Star Schema is designed to improve performance in data warehousing by reducing complexity. So, S_2 is correct.

38. (a, c)

- In a Star Schema, fast aggregations are supported because the fact table holds numeric measures optimized for summarization. So, option (a) is true.
- The fact table stores keys and measures, not descriptive information. So, option (b) is false.
- Dimension tables are denormalized and often include hierarchical attributes (such as year → quarter → month). So option (c) is also true.

39. (b)

- Star Schema is not designed for OLTP systems, which require highly normalized databases for transaction efficiency. So S_1 is incorrect.
- However, it is widely used in OLAP systems because its denormalized design simplifies complex analytical queries. So, S_2 correct.

40. (a)

In a Star Schema, all dimension tables are directly connected to the central fact table, creating a straightforward structure that simplifies querying and improves performance.

41. (80)

Normalized range $= [0, 10]$

$a = 0,\ b = 10$

Original age $(X) = 50$

Normalized value $(X') = 5$

Minimum age $(X_{min}) = 20$

$X_{max} = ?$

We know that,

$$X' = a + \frac{(X - X_{min})(b-a)}{X_{max} - X_{min}}$$

$$\Rightarrow \quad 5 = 0 + \frac{(50-20)10}{X_{max} - 20}$$

$$5 = \frac{300}{X_{max} - 20}$$

$$\therefore \quad 5(X_{max} - 20) = 300$$

$$5X_{max} - 100 = 400$$

$$X_{max} = \frac{400}{5} = 80$$

42. (c)

Star Schema is designed to improve query response time by using denormalized dimension tables and a simple structure with direct links to the fact table, which reduces the number and complexity of joins. Options (a) and (b) are incorrect because Star Schema has simpler joins than Snowflake Schema and its dimension tables are denormalized, not normalized to 3NF.

43. (b)

- In Snowflake Schema, dimension tables are normalized, not fully denormalized, to reduce redundancy. So option (a) is incorrect.
- Snowflake Schema normalizes dimension tables to avoid data duplication and improve consistency. So option (b) is correct.
- Normalization increases the number of tables, so Snowflake Schema usually increases joins, not reduces them. So option (c) is incorrect.
- Fact tables generally store numeric measures and keys and are not normalized in Snowflake Schema. So option (d) is incorrect.

44. (a)

- Snowflake Schema normalizes dimension tables, which reduces data redundancy and thus saves storage space. So, S_1 is correct.
- Snowflake Schema involves more joins due to normalization, which generally slows down query performance compared to the simpler Star Schema. So, S_1 is correct.

45. (c, d)

Given, Mean (μ) = 100000

Standard deviation (σ) = 5000

Outlier if Z-score > 3

We know that,

$$\text{Z-score} = \frac{X-\mu}{\sigma}$$

Option (a) : X = \$112000

$$Z = \frac{112000-100000}{5000} = \frac{12000}{5000} = 2.4$$

Option (b) : \$108000

$$Z = \frac{108000-100000}{5000} = \frac{8000}{5000} = 1.6$$

Option (c) : \$117000

$$Z = \frac{117000-100000}{5000} = \frac{17000}{5000} = 3.4$$

Option (d) : \$116000

$$Z = \frac{116000-100000}{5000} = \frac{16000}{5000} = 3.24$$

46. (b)

Snowflake Schema uses a normalized structure for dimension tables, which helps organize data efficiently and supports complex queries. Options (b), (c), and (d) are incorrect because dimensions are normalized (not denormalized), it generally has slower query speed due to more joins, and it uses more tables compared to Star Schema.

47. (a)

In Snowflake Schema, dimension tables are normalized into multiple related tables to reduce redundancy and improve data integrity. This results in a more complex structure with several layers of sub-tables representing different levels of detail.

48. (b)

Fact Constellation Schema has multiple fact tables sharing common dimension tables, allowing dimension reuse. This design supports complex data models and multiple business processes efficiently. It involves more than one fact table and stores both detailed and summary data. The schema promotes dimension reusability, improving consistency and reducing redundancy.

49. (0.367)

Given, $X = 1.5$

$X_{max} = 3.2$

Normalized value $X' = 0.4$

New range: [0, 1] (so $a = 0$, $b = 1$)

We know that, $X' = \frac{X - X_{min}}{X_{max} - X_{min}} = 0.4$

$$0.4 = \frac{1.5 - X_{min}}{3.2 - X_{min}}$$

$$\Rightarrow \quad 0.4(3.2 - X_{min}) = 1.5 - X_{min}$$

$$1.28 - 0.4\,X_{min} = 1.5 - X_{min}$$

$$1.28 - 0.4\,X_{min} - 1.5 + X_{min} = 0$$

$$-0.22 + 0.6\,X_{min} = 0$$

$$\Rightarrow \quad 0.6\,X_{min} = 0.22$$

$$\Rightarrow \quad X_{min} = \frac{0.22}{0.6} = 0.3667$$

Minimum page load time $X_{min} = 0.367$ seconds.

50. (a)

- OLAP systems utilize star and snowflake schemas to structure data for efficient retrieval and analysis. The star schema simplifies data organization, while the snowflake schema reduces redundancy by normalizing data. So, S_1 is correct.
- OLAP systems are designed for complex data analysis and reporting, not for processing individual transactions like OLTP systems. So, S_2 is incorrect.

51. (b, d)

(a) A roll-up operation moves from a lower level of granularity to a higher level, aggregating the data. So, option (a) is incorrect.

(b) Hierarchies facilitate efficient data navigation and analysis by organized data into level or tiers allow user to drill down or drill up to different levels of details as needed. So, option (b) is correct.

(c) Star schema do support hierarchical data structure although they are less normalized compared to snowflake schema. So, option (c) is incorrect.

(d) Hierarchical structures can improve query performance in data warehouse by organized data efficient for analysis. So, option (d) is correct.

52. (c)

Fact Constellation Schema can represent many-to-many relationships using bridge tables, allowing flexible linking between fact and dimension tables. It supports shared dimensions across multiple facts. The schema is not limited to transactional data and also handles analytical data. Additionally, it allows controlled redundancy to support complex business needs.

53. (c)

Fact Constellation Schema is useful for modeling complex and interrelated business processes by using multiple fact tables. It allows shared dimensions to reduce redundancy and maintain consistency. Unlike star schemas, it provides greater flexibility for enterprise-level data modeling.

54. (a, b)

- Star Schema is user-friendly due to its simple structure, making it easier to query.
- Snowflake Schema reduces redundancy by normalizing dimension tables.
- Fact Constellation Schema supports multiple fact tables, enabling modeling of complex business processes.

55. (b)

- In a Snowflake Schema, dimension tables are normalized to reduce redundancy and save storage, but this results in complex joins across multiple tables, slowing down queries.
- While it supports hierarchy modeling well, the increased complexity makes it harder to query compared to the Star Schema, which offers simpler joins and faster performance due to denormalized dimensions.

56. (b)

In a Snowflake Schema, dimension tables are normalized into multiple related sub-tables to eliminate redundancy and improve storage efficiency. This contrasts with the Star Schema, where dimensions are denormalized for faster query performance.

57. (22)

Find the range of the data:

Minimum value = 14, Maximum value = 80

Range = 80 – 14 = 66

Determine the bin width: Since we need to divide the data into three equal-width bins, the bin width is:

$$\text{Bin width} = \frac{\text{Range}}{\text{Number of bins}} = \frac{66}{3} = 22$$

Define the bin intervals:

Bin 1: [14, 14 + 22) = [14, 36)

Bin 2: [36, 36 + 22) = [36, 58)

Bin 3: [58, 80]

Assign the values to the bins:

Bin 1 (14 to 36): 14, 18, 22, 25, 29, 35

Bin 2 (36 to 58): 38, 45, 50, 52

Bin 3 (58 to 80): 60, 64, 70, 73, 80

Find the least values in Bin 2 and Bin 3:
Least value in Bin 2: 38
Least value in Bin 3: 60
Calculate the difference between the least values:

$$\text{Difference} = 60 - 38 = 22$$

58. (a)

- In a Snowflake schema, some dimension tables are normalized, but not all of them necessarily. The structure involves normalizing dimension tables into multiple related tables to reduce redundancy.

59. (a, b)

- OLAP is indeed used for decision support and analysis based on historical data. So, option (a) is correct.
- OLTP ensures data integrity and consistency across operational systems. So, option (b) is correct.
- OLAP databases are not optimized for fast transaction processing and real-time updates; that is the role of OLTP. So, option (c) is incorrect.
- OLTP databases are typically normalized, not denormalized, to maintain data integrity and reduce redundancy. So, option (d) is incorrect.

60. (a, b, c)

(a) Binning without dependence on class labels or predefined thresholds is indeed classified as unsupervised discretization. This technique focuses on dividing data based on statistical measures rather than predefined categories. So, option (a) is correct.

(b) Discretization by Binning involves partitioning a continuous attribute into bins, which can be of equal width or frequency. This allows continuous data to be transformed into categorical or ordinal data. So, option (b) is correct.

(c) Discretization by Histogram analysis does use histogram bins to discretize continuous data based on its frequency distribution. Histogram bins are intervals that group data points based on their occurrence within specific value ranges. So, option (c) is correct.

(d) Discretization by Histogram analysis does not always result in bins of equal width. The width of histogram bins varies based on the data distribution and how the bins are defined to accurately represent the data's frequency distribution. So, option (d) is incorrect.

61. (30)

Given, $X_{min} = 250,\ X_{max} = 500$

New range $[a, b] = [100, 200]$

Value normalize at $X = 375$

$$X' = a + \frac{(X - X_{min})(b - a)}{X_{max} - X_{min}}$$

$$X' = 100 + \frac{(375 - 250)(100)}{250}$$

$$= 100 + \frac{50 \times 100}{250} = 100 + 50 = 150$$

Value normalize at $X = 350$

$$X' = a + \frac{(X - X_{min})(b - a)}{X_{max} - X_{min}}$$

$$X' = 100 + \frac{(300 - 250)(100)}{250}$$

$$X' = 100 + \frac{50 \times 100}{250} = 100 + 20 = 120$$

$$\text{Difference} = 150 - 20 = 30$$

62. (a)

(a) Mean binning replaces each value in a bin with the mean (average) value of all the data points within that bin. So, option (a) is correct.

(b) Median binning would replace each value with the median, not the mean. So, option (b) is incorrect.

(c) Equal-width binning refers to dividing the data range into equal-sized bins, not replacing values with the mean. So, option (c) is incorrect.

(d) Frequency binning refers to bins containing equal numbers of data points, not mean replacement. So, option (d) is incorrect.

63. (d)

S_1: Roll-up does not reduce dimensionality; it aggregates data within existing dimensions. So, S_1 is incorrect.

S_2: Drill-down is an OLAP operation that increase data details by moving down a concept hierarchy, providing more specific, granular data. So, S_2 is incorrect.

64. **(a, b, c)**

(a) Binning without dependence on class labels or predefined thresholds is indeed classified as unsupervised discretization. This technique focuses on dividing data based on statistical measures rather than predefined categories. So, option (a) is correct.

(b) Discretization by Binning involves partitioning a continuous attribute into bins, which can be of equal width or frequency. This allows continuous data to be transformed into categorical or ordinal data. So, option (b) is correct.

(c) Discretization by Histogram analysis does use histogram bins to discretize continuous data based on its frequency distribution. Histogram bins are intervals that group data points based on their occurrence within specific value ranges. So, option (c) is correct.

(d) Discretization by Histogram analysis does not always result in bins of equal width. The width of histogram bins varies based on the data distribution and how the bins are defined to accurately represent the data's frequency distribution. So, option (d) is incorrect.

65. **(92)**

Sales Price Records: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

Find the Range of Data = Max – Min
= 215 – 5 = 210

Number of Bins = 3

$$\text{Width of Each Bin} = \frac{210}{3} = 70$$

Calculate the Bin Intervals:

Bin 1 = From 5 to 5 + 70 = 75
Bin 2 = From 75 to 75 + 70 = 145
Bin 3 = From 145 to 215

Allocate Records to Bins:

Bin 1 = 5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2 = 92
Bin 3 = 204, 215

So, Bin 2 = 92

66. **(a)**

- The range of data is divided into intervals of equal size, and each data point is assigned to the corresponding interval.

67. **(1223.07)**

Average monthly sales over 12 months $ 1200
Sales for 13th month S_{new} = $ 1500

Total sales = $1200 \times 12 = 14400$
Update total sales = Total sales + S_{new}
= $14400 + 1500 = 15900$
New sample size = $n + 1 = 12 + 1 = 13$

$$\text{Updated average sales} = \frac{\text{Updated total sales}}{\text{New sample size}}$$

$$= \frac{15900}{13} = 1223.07$$

68. **(a)**

- OLAP systems utilize star and snowflake schemas to structure data for efficient retrieval and analysis. The star schema simplifies data organization, while the snowflake schema reduces redundancy by normalizing data. So, S_1 is correct.
- OLAP systems are designed for complex data analysis and reporting, not for processing individual transactions like OLTP systems. So, S_2 is incorrect.

69. **(d)**

$x_{min} = 1, \quad x_{max} = 10$
New range $[a, b] = [-5, 5]$
Normalize $x = 7$

$$x' = a + \frac{(x - x_{min})(b - a)}{x_{max} - x_{min}}$$

$$= -5 + \frac{(7-1)(10)}{9} = -5 + \frac{60}{9}$$

$$= -5 + 6.67 = 1.67$$

To find original rating for $x' = 0$

$$x' = a + \frac{(x - x_{min})(b - a)}{x_{max} - x_{min}}$$

$$\Rightarrow \quad 0 = -5 + \frac{(x-1)(10)}{9}$$

$$\frac{(x-1)(10)}{9} = 5$$

$$\Rightarrow \quad (x-1) = \frac{5 \times 9}{10} = 4.5$$

$$x = 5.5$$

Normalized rating for 7 is 1.67
Original rating for normalized value 0 is 5.5

70. (a)

(a) Equal-width binning divides the entire data range into intervals (bins) of equal size, with each data point assigned to the appropriate bin based on its value. So, option (a) is correct.

(b) This describes equal-frequency binning, not equal-width binning. So, option (b) is incorrect.

(c) Bins in equal-width binning are not based on proximity to a mean but are of uniform width. So, option (c) is incorrect.

(d) In equal-width binning, the intervals have the same width, not varying widths. So, option (d) is incorrect.

71. (102)

Total records = 15

$$\text{Number of records/Bin} = \frac{\text{Total records}}{\text{Number of bins}}$$

$$= \frac{15}{3} = 5$$

Now, create bins:

Bin 1 = (45, 55, 60, 62, 70)

Bin 2 = (73, 75, 80, 82, 85)

Bin 3 = (90, 92, 95, 100, 120)

Sum of Bin 2 = 73 + 75 + 80 + 82 + 85 = 395

Sum of Bin 3 = 90 + 92 + 95 + 100 + 120 = 497

Difference = 497 – 395 = 102

72. (a)

- Binning does reduce dimensionality by grouping data points into fewer categories, often simplifying analysis. So, S_1 is correct.
- OLTP (Online Transaction Processing) focuses on handling short, transactional queries in real-time, such as inserting, updating, and deleting data, rather than complex queries or aggregations typically found in data warehouses, which are more associated with OLAP (Online Analytical Processing). So, S_2 is incorrect.

73. (c)

Equal-frequency binning divides the dataset into bins such that each bin contains approximately the same number of data points. This method focuses on how the bins are formed based on data frequency and does not inherently specify how the values within each bin are replaced, unlike mean or median binning.

74. (a)

- Star Schema may use slightly normalized dimensions, though it's generally denormalized. So, S_1 is correct.
- Snowflake Schema reduces redundancy and saves disk space via normalization. So, S_2 is correct.
- Fact Constellation Schema uses multiple fact tables-it doesn't merge all data into one table. So, S_3 is incorrect.

75. (150.14)

Mean of 200 data points = 150

A new data point with value of 180 is added,

Sum of original 200 data points = Mean × Number of data points

$$= 150 \times 200 = 30000$$

New sum of data points = 30000 + 180 = 30180

$$\text{So, New mean} = \frac{\text{New sum}}{\text{Number of data points}}$$

$$= \frac{30180}{201} = 150.14$$

■■■■

UNIT 8

Algorithms

1. Asymptotic Analysis of Algorithms

Q.1 The concept of order (Big O) is important because
(a) It can be used to decide the best algorithm that solves a given problem
(b) It determines the maximum size of a problem that can be solved in a given amount of time
(c) It is the lower bound of the growth rate of algorithm
(d) Both (a) and (b) above

Q.2 Each of the function $2^{\sqrt{n}}$ and $n^{\log n}$ has a growth rate .... than that of any polynomial.
(a) Greater (b) Less
(c) Equal to (d) Uncertain

Q.3 A machine needs a minimum of 100 sec to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately
(a) 50.2 sec (b) 6.7 sec
(c) 72.7 sec (d) 11.2 sec

Q.4 Suppose $f, g, h, k : N \to N$.
If $f = O(h)$ and $g = O(k)$, then
(a) $f + g = O(h + k)$
(b) $fg = (hk)$
(c) Both (a) and (b) above
(d) None of the above

Q.5 Consider the following two functions:

$$g_1(n) = \begin{cases} n^3 \text{ for } 0 \le n \le 10{,}000 \\ n^2 \text{ for } n \ge 10{,}000 \end{cases}$$

$$g_2(n) = \begin{cases} n \text{ for } 0 \le n \le 100 \\ n^3 \text{ for } n > 100 \end{cases}$$

Which of the following is true?
(a) $g_1(n)$ is $O(g_2(n))$ (b) $g_1(n)$ is $O(n^3)$
(c) $g_2(n)$ is $O(g_1(n))$ (d) $g_2(n)$ is $O(n)$

Q.6 Consider the following functions:

$f(n) = 3n^{\sqrt{n}}$

$g(n) = 2^{\sqrt{n}\log_2 n}$

$h(n) = n!$

Which of the following is true?
(a) $h(n)$ is $O(F(n))$
(b) $h(n)$ is $O(F(n))$
(c) $g(n)$ is not $O(F(n))$
(d) $F(n)$ is $\theta(g(n))$

Q.7 1. $\frac{1}{2}n^2 = \omega(n)$

2. $\frac{1}{2}n^2 = \omega(n^2)$

Which of the following is true?
(a) 1 is correct
(b) 2 is correct
(c) 1 and 2 both are correct
(d) None of these

Q.8 $f(n) = O(g(n))$ If and only if
(a) $g(n) = O(f(n))$ (b) $g(n) = \omega(f(n))$
(c) $g(n) = \Omega(f(n))$ (d) None of these

Q.9 Which one of the following is true?
1. $an = O(n^2)$ (small oh) $a \ge 0$
2. $an^2 = O(n^2)$ (big Oh) $a > 0$
3. $an^2 \ne O(n^2)$ (small oh) $a > 0$
(a) Only 1 and 2 are correct
(b) Only 1 is correct
(c) 1 and 3 are correct only
(d) All are correct

Q.10 Consider the following two functions:

$f(n) = n^3$, if $0 \le n < 10{,}000$
n^2, otherwise

$g(n) = n$, if $0 \le n < 100$
$n^2 + 5n$, otherwise

Which of the following is/are true?
1. $f(n)$ is $O(n^3)$
2. $g(n)$ is $O(n^3)$
3. $O(f(n))$ is same as $O(g(n))$
4. $g(n)$ is $O(n^2)$
(a) 3 and 4 (b) 1, 2 and 3
(c) 1, 2 and 4 (d) None of these

Q.11 $f(n) = O(g(n))$ if and only if
(a) $g(n) = \Omega(f(n))$
(b) $g(n) = \omega(f(n))$
(c) $g(n) = \Omega(f(n))$ and $g(n) = \omega(f(n))$
(d) None of these

Q.12 Using the standard algorithm, what is the time required to determine that a number n is prime?
(a) Constant time (b) Quadratic time
(c) Logarithmic time (d) Linear time

Q.13 Consider the following functions:
$f(n) = 2^n$
$g(n) = n!$
$h(n) = n^{\log n}$

Which of the following statements about the asymptotic behaviour of $f(n)$, $g(n)$ and $h(n)$ is true?
(a) $f(n) = O(g(n))$; $g(n) = O(h(n))$
(b) $f(n) = \Omega(g(n))$; $g(n) = O(h(n))$
(c) $g(n) = O(f(n))$; $h(n) = O(f(n))$
(d) $h(n) = O(f(n))$; $g(n) = \Omega(f(n))$

Q.14 Let $f(n) = n^2 \log n$ and $g(n) = n(\log n)^{10}$ be two positive functions of n.
Which of the following statements is correct?
(a) $f(n) = O(g(n))$ and $g(n) \neq O(f(n))$
(b) $g(n) = O(f(n))$ and $f(n) \neq O(g(n))$
(c) $f(n) \neq O(g(n))$ and $g(n) \neq O(f(n))$
(d) $f(n) = O(g(n))$ and $g(n) = O(f(n))$

Q.15 Match the following groups:

Group-I ($n > 0$)	Group-II
A. $3n + 4n^2 + 5n \log n$	1. $O(1)$
B. $n + \log n + \log \log n$	2. $O(\log n)$
C. $10 + n + n \log n + \log n$	3. $O(n)$
D. $10 + 10000 + 100000$	4. $O(n \log n)$
	5. $O(n^2)$

Codes:

	A	B	C	D
(a)	5	3	4	2
(b)	5	4	3	1
(c)	5	3	4	1
(d)	5	4	3	2

Q.16 Unrestricted use of Goto is harmful, because it
(a) increase running time of programs.
(b) makes debugging difficult.
(c) results in the compiler generating longer machine code.
(d) increase memory requirement of programs.

Q.17 Consider the following function:

```
void f(int n)
{
    int i, j, k, m;
    for (i = 0; i < 100; i ++)
    {
        for (j = 0; j < n; j++)
        {
            for (k = 0; k < j; k++)
                printf("%d", k);
        }
    }
}
```

What is the worst case running time of the function f for any positive value of n?
(a) $O(1)$ (b) $O(n)$
(c) $O(n^2)$ (d) $O(n^3)$

Q.18 Consider the following segment of c code:

```
int j, n;
j = 1;
while (j < = n)
```

The number of comparisons mode in the execution of the loop for any $n > 0$ is
(a) $\lceil \log_2 n \rceil + 1$ (b) n
(c) $\lceil \log_2 n \rceil$ (d) $\lfloor \log_2 n \rfloor + 1$

Q.19 What is time complexity of following code:

```
int a = 0;
for (i = 0; i < N; i++)
{
    for (j = N; j > i; j--)
    {
        a = a + i + j;
    }
}
```

(a) $O(N)$
(b) $O(N * (\log N)$
(c) $O(N * Sqrt(N))$
(d) $O(n^2)$

Q.20 Consider an array of n element with sorted order, if any element i appear more than half the number of element, what is the time complexity to count the number of occurrences of i?
(a) $O(\log n)$ (b) $O(1)$
(c) $O(n)$ (d) $O(\log \log n)$

Q.21 Find the time complexity of the following summation. Assume that k is a constant, $k > 0$

$$\sum_{i=1}^{n}\sum_{j=i+1}^{n}\frac{1}{k}$$

(a) $O(n)$ (b) $O(n^2)$
(c) $O(n^3)$ (d) None of these

Q.22 Consider the following functions:

1. $n!$
2. a^n, a is constant, $a > 0$
3. n^n
4. n^k, k is a constant, $k > 0$
5. e^n

Choose the correct statement which ranks all functions by order of growth.

(a) $2 < 4 < 5 < 1 < 3$
(b) $4 < 2 < 3 < 5 < 1$
(c) $4 < 5 < 2 < 1 < 3$
(d) $4 < 2 < 5 < 1 < 3$

Q.23 What is time complexity of following code:

```
int a = 0, i = N;
while (i > 0) {
    a = a + i;
    i /= 2;
}
```

(a) $O(N)$ (b) $O(Sqrt(N))$
(c) $O(N/2)$ (d) $O(\log N)$

Q.24 Consider the following recursive function:

```
int F (int array [ ], int n)
{
    int S = 0;
    if (n = = 0)
        return 0;
    S = F (array, n – 1)
    if (array [n – 1] < 0)
        S = S + 100;
    return S;
}
```

What is the worst case time complexity of the above function?

(a) $O(n)$ (b) $O(n \log n)$
(c) $O(n^2)$ (d) $O(\log n)$

Q.25 What will be the worst case time complexity for the following code segment?

```
int count = 0, N;
for (i = 0; i < N * 2; i++)
{
    for (j = 0; j < i/3; i++)
    {
        for (k = 0; k < j * j; k++)
        {
            count ++;
        }
    }
}
```

(a) $O(N^4)$ (b) $O(N^3)$
(c) $O(N^2)$ (d) $O(N)$

Q.26 Let $f(n) = n$ and $g(n) = n^{(1+\sin n)}$, where n is a positive integer. Which of the following statements is/are correct?

I. $f(n) = O(g(n))$
II. $f(n) = \Omega(g(n))$

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

Q.27 Let's look about the algorithm:

```
int temp, j, i;
for (i = 1; i < n; i++)
{
    temp = A[i];
    for (j = i – 1; j ≥ && (A[j] > temp); j--)
    A[j + 1] = A[j];
    A[j] = temp;
    }
}
```

If the array is sorted the complexity is

(a) $O(n)$ (b) $O(n^2)$
(c) $O(n \log n_2 n)$ (d) $O(\log_2 n)$

Q.28 Consider the following function:

```
Void func (int n) {
    int k = n;
    int i = 0;
    for(; i < n; i++){
    while (k > 1)q
    k >> = 1;
    }
}
```

What is worst case time complexity of the functior

(a) $O(n)$ (b) $O(\log n)$
(c) $O(n^2)$ (d) $O(n \log n)$

Q.29 You are given two sorting algorithms A and B that work in time $O(n \log n)$ and $O(n^2)$ respectively. Consider the following statements:

1. Algorithm A will sort any array faster than algorithm B.
2. On an average, algorithm A will sort a given array faster than algorithm B.
3. If we need to implement one of the two as the default sorting algorithm in a system, algorithm A will always be preferable to algorithm B.

Which of the statements above are true?

(a) 1, 2 and 3
(b) 1 and 3
(c) 2 and 3
(d) None of them

Q.30 If $f(n) = O(g(n))$ for f, g are non-decreasing function, then $f(n) * \log_2(f(n)^c) = O(g(n) * \log_2 g(n))$, where C = power of $f(n)$ is

(a) Always true
(b) Never
(c) Sometimes true, sometimes not depending on f, g.
(d) Sometimes true, sometimes false depending on C.

Q.31 Among the following asymptotic expression which of these functions grows slowest (as a function of n) asymptotically?

(a) $\left(\sqrt{\log n}\right)^{\log^2 n}$ (b) $(\log n)^{\sqrt{\log n}}$

(c) n^{10} (d) $2^{2^{\sqrt{\log \log n}}}$

Q.32 In above question if the array is in reverse sorted order then time complexities will be

(a) $O(n)$ (b) $O(n \log_2 n)$
(c) $O(n^2)$ (d) $O(\log_2 n)$

Q.33 How many times the following loop runs for given value of n?

```
for (int i = 2; i <= n; i = pow(i, 2))
{
   printf("%", i);
}
```

(a) $O(n)$
(b) $O(\log n)$
(c) $O(n^2)$
(d) $O(\log \log n)$

Q.34 Consider the following C function:

```
find (int n)
{   if (n < 2) then return;
    else
{   sum = 0;
    for (i = 1; i ≤ 4; i++)
        find (n/2);
    for (i = 1; i ≤ n * n; i ++)
    sum = sum +1;
    }
}
```

Assume that the division operation takes constant time and "sum" is global variable. What is the time complexity of "find (n)"?

(a) $\Theta(n^2)$ (b) $\Theta(n^2 \log n)$
(c) $\Theta(n^3)$ (d) None of these

Q.35 What is time complexity of this code:

```
int A(int)
{
    if (n < = 2) return 1;
    else
    return (A(√n) + n);
}
```

(a) $O(\log \log(n))$
(b) $O(\log n)$
(c) $O(n)$
(d) $O(\log n \log n)$

Q.36 What is relation between the asymptotic time complexities of $f(n)$ and $g(n)$?

$f(n) = n * 2^n$

$g(n) = e^n$

(a) $e^n = O(n * 2^n)$ (b) $e^n = \theta(n * 2^n)$
(c) $e^n = \omega(n * 2^n)$ (d) $e^n = o(n * 2^n)$

Q.37 Arrange the following functions is increasing order of growth rate (with $g(n)$ following $f(n)$ in your list if and only if $f(n) = O(g(n))$.

$$\sqrt{n}, 10^n, n^{1.5}, 2^{\sqrt{\log_2 n}}, n^{5/3}$$

(a) $\sqrt{n}, 10^n, n^{1.5}, 2^{\sqrt{\log_2 n}}, n^{5/3}$

(b) $n^{1.5}, n^{5/3}, \sqrt{n}, 2^{\sqrt{\log_2 n}}, 10^n$

(c) $\sqrt{n}, 2^{\sqrt{\log_2 n}}, n^{1.5}, n^{5/3}, 10^n$

(d) None of these

Q.38 Consider the following code segment. Assume n to be a positive integer

```
for (i = 1; i < = n; i++) {
for (j = 1; j < = n; j + i) {
printf("Hi");
}}
```

Let $T(n)$ represent the total running time of the above case segment. Specify which of the following statements are correct.

(a) $T(n) = O(n^2)$
(b) $T(n) = O(n \log n)$
(c) $T(n) = O(n^2 \log n)$
(d) $T(n) = O(\log n)$

Q.39 Consider the equality $\sum_{i=0}^{n} i^3 = X$ and the following choices for X.

1. $\theta(n^4)$
2. $O(n^5)$
3. $\theta(n^5)$
4. $\Omega(n^3)$

The equality above remains correct of X is replaced by:

(a) Only 1
(b) Only 2
(c) 1 or 3 or 4 but not 2
(d) 2 or 3 or 4 but not 1

Q.40 In the following C function, let $n \geq m$.

```
int gcd (n, m)
{   if (n% m = = 0) return m;
    n = n% m;
    return gcd (m, n);
}
```

How many recursive calls are made by this function?

(a) $\Theta(\log_2 n)$
(b) $\Omega(n)$
(c) $\Theta(\log_2 \log_2 n)$
(d) $\Theta(\sqrt{n})$

Q.41 Consider the following code segment of C-code:

```
{
    for(int i = 1; i < = n; i++)
    {
        int p = pow(i, k)
        for (int j = 1; j < = p; j ++)
        {
            // some O(1) work
        }
    }
}
```

Calculate time complexity of above code. The time complexity will be

(a) $O(n^k)$
(b) $O(n^{k+1})$
(c) $O(n^2)$
(d) $O(n^3)$

Q.42 Consider the following two functions:

$$f(n) = \begin{cases} n^3, & \text{if } 0 \leq n < 10{,}000 \\ n^2, & \text{otherwise} \end{cases}$$

$$g(n) = \begin{cases} n, & \text{if } 0 \leq n < 100 \\ n^2 + 5n, & \text{otherwise} \end{cases}$$

Which of the following is/are true?

1. $f(n)$ is $O(n^3)$
2. $g(n)$ is $O(n^3)$
3. $O(f(n))$ is same as $O(g(n))$
4. $g(n)$ is $O(n^2)$

(a) 3 and 4
(b) 1, 2 and 3
(c) 1, 2 and 4
(d) None of these

Q.43 Consider the three functions $f(n) = n^2 + n + 234$, $g(n) = 12n^2 + 34$ and $h(n) = 123n + 4$.

Which of the following statements is/are true?

(a) $f(n) = O(g(n))$
(b) $g(n) = O(f(n))$
(c) $f(n) = O(h(n))$
(d) $h(n) = O(f(n))$

Q.44 An algorithm is made up of two independent modules M_1 and M_2. If order of M_1 is $f(n)$ and M_2 is $g(n)$ then the order of algorithm is/are

(a) $\max(f(n), g(n))$
(b) $\min(f(n), g(n))$
(c) $f(n) + g(n)$
(d) $f(n) \times g(n)$

Q.45 Consider the following pseudo code:

```
sum = 0;
for (i = 0, i < n; i++) {
    for (j = 0, j < i; j++) {
        for (k = 0, k < j; k++) {
            sum = sum + 1;
        }
    }
}
```

What of the following options is/are correct with respect to the time complexity of the pseudo code given above?

(a) $\theta(n^3(\log n))$
(b) $\Omega(n^2)$
(c) $O(n^3(\log n))$
(d) $O(n^3)$

Q.46 Let $g(n) = \Omega(n)$, $f(n) = O(n)$ and $h(n) = \theta(n)$ then what is the time complexity of $[g(n)\ f(n) + h(n)]$

(a) $O(n)$
(b) $\theta(n)$
(c) $\Omega(n)$
(d) $\theta(n^2)$

Q.47 Consider the following functions:

$$f_1 = n^4,\ f_2 = 4^n,\ f_3 = n^{110/37},\ f_4 = \left(\frac{119}{37}\right)^n$$

Which of the following is correct order of increasing growth rate?

(a) f_1, f_3, f_2, f_4 (b) f_3, f_1, f_4, f_2
(c) f_3, f_1, f_2, f_4 (d) f_1, f_3, f_4, f_2

Q.48 Consider the following recursive function find:

```
int find (int A[ ], int n)
{
    int sum = 0;
    if (n = = 0) return 0;
    sum = find (A, n – 1)
    if (A [n – 1] < 0) sum = sum + 1;
    return sum;
}
```

What is the worst case running time of above function find (A [ ], n) when array A has 0 to $n-1$ elements?

(a) O(1) (b) O(log n)
(c) O(n) (d) O(n^2)

Q.49 Consider the following program segments:

```
main( )
{
    int i = 0, m = 0;
for (i = 1; i < n; i++) {
        for (j = 1; j < i * i; j++) {
            if ((j % i) = = 0)
        for (k = 1; k < j; k++) {
            m = m + 1; }
        }
    }
}
```

What is the time complexity of above program?

(a) O($n^2 \log n$) (b) O(n^3)
(c) O(n^4) (d) O($n \log n$)

Q.50 Choose the correct arrangement of the following functions in decreasing order [Highest function to lowest function].

I. $n^2 \log(n)^4$ II. $n^3 + n^2 + n + 1$
III. $1+\frac{1}{n}$ IV. 2^{1000}
V. $n^{11/5}$

(a) II, V, I, IV, III (b) II, I, V, IV, III
(c) II, V, I, III, IV (d) IV, II, I, V, III

2. Recurrence Relations

Q.51 Consider the following is true?

$$T(n) = 2T(\lfloor\sqrt{n}\rfloor) + 1,\ T(1) = 1$$

Which one of the following is true?

(a) $T(n) = \Theta(\log \log n)$ (b) $T(n) = \Theta(\log n)$
(c) $T(n) = \Theta(\sqrt{n})$ (d) $T(n) = \Theta(n)$

Q.52 The recurrence equation:

$T(1) = 1$
$T(n) = 2T(n-1) + n,\ n \geq 2$

Evaluates to

(a) $2^{n+1} - n - 2$ (b) $2^n - n$
(c) $2^{n+1} - 2n - 2$ (d) $2^n + n$

Q.53 When $n = 2^{2^k}$ for some $k \geq 0$, the recurrence relation $T(n) = \sqrt{2}T\left(\frac{n}{2}\right) + \sqrt{n}$, $T(1) = 1$ evaluates to:

(a) $\sqrt{n}(\log n + 1)$ (b) $\sqrt{n}\log n$
(c) $\sqrt{n}\log\sqrt{n}$ (d) $n\log\sqrt{n}$

Q.54 The recurrence relation that arises in relation with the complexity of binary search is

(a) $T(n) = T\left(\frac{n}{2}\right) + k$, where k is a constant
(b) $T(n) = 2T\left(\frac{n}{2}\right) + k$, where k is a constant
(c) $T(n) = T\left(\frac{n}{2}\right) + \log(n)$
(d) $T(n) = T\left(\frac{n}{2}\right) + n$

Q.55 Time complexity of an algorithm $T(n)$, where n is the input size is given by

$$T(n) = T(n-1) + \frac{1}{n},\ \text{if } n > 1$$
$$= 1, \text{otherwise}$$

The order of this algorithm is

(a) $\log n$ (b) n
(c) n^2 (d) n^n

Q.56 Let $T(n)$ be the function defined by $T(1) = 1$,

$$T(n) = T\left(\left\lfloor\frac{n}{2}\right\rfloor\right)\sqrt{n} \text{ for } n \geq 2$$

Which of the following statements is true?

(a) $T(n) = O(\sqrt{n})$ (b) $T(n) = O(n)$
(c) $T(n) = O(\log n)$ (d) None of these

Q.57 Solve the recurrence relation to find:

$$T(n),\ T(n) = 4T\left(\frac{n}{2}\right) + n$$

(a) $\Theta(\log_2 n)$ (b) $\Theta(n^2)$
(c) $\Theta(n^2 \log_2 n)$ (d) None of these

Q.58 Solve $T(n) = 4T\left(\frac{n}{2}\right) + n^2$

(a) $\Theta(n^2)$ (b) $\Theta(n^2 \log_2 n)$
(c) $\Theta(n \log_2 n)$ (d) None of these

Q.59 Let $T(n)$ be a function defined by the recurrence

$$T(n) = 2T\left(\frac{n}{2}\right) + \sqrt{n} \text{ for } n \geq 2 \text{ and } T(1) = 1$$

Which of the following statements is TRUE?

(a) $T(n) = \Theta(\log n)$
(b) $T(n) = \Theta(\sqrt{n})$
(c) $T(n) = \Theta(n)$
(d) $T(n) = \Theta(n \log n)$

Q.60 The recurrence relation $T(n) = mT\left(\frac{n}{2}\right) + an^2$ is satisfied by

(a) $T(n) = O(n^m)$ (b) $T(n) = O(\log m)$
(c) $T(n) = O(n^{\log m})$ (d) $T(n) = O(\log n)$

Q.61 The time complexity of an algorithm $T(n)$, where n is the input size is given by

$$T(n) = T(n-1) + \frac{1}{n}, \text{ if } n > 1$$

$$1, \quad \text{otherwise}$$

The order of this algorithm is

(a) $O(\log n)$ (b) $O(n)$
(c) $O(n^2)$ (d) $O(n^n)$

Q.62 The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with n discs is

(a) $T(n) = 2T(n-2) + 2$
(b) $T(n) = 2T(n-1) + n$
(c) $T(n) = 2T\left(\frac{n}{2}\right) + 1$
(d) $T(n) = 2T(n-1) + 1$

Q.63 What is time complexity of recurrence equation

$$T(n) = T\left(\sqrt{n}\right) + 1$$

(a) $O(\log n)$ (b) $O(n^2)$
(c) $O(n \log \log n)$ (d) $O(\log \log n)$

Q.64 Solve the recurrence:

$$T(n) = T\left(\frac{n}{2} - 1\right) + T\left(n - \frac{n}{2}\right) + \theta(n)$$

(a) $\theta(n \log n)$ (b) $\theta(n)$
(c) $\theta(n^2 \log n)$ (d) $\theta(n^2)$

Q.65 Consider the following recurrence relation

$$T(n) = 7T\left(\frac{n}{2}\right) + an^2$$

What will be the solution of above recurrence?

(a) $O(n^2)$ (b) $O(n^3)$
(c) $O(n^2 \log n)$ (d) $O(n^{2.81})$

Q.66 Suppose $T(n) = 2T\left(\frac{n}{2}\right) + n,\ T(0) = T(1) = 1$

Which one of the following is FALSE?

(a) $T(n) = O(n^2)$ (b) $T(n) = \Theta(n \log n)$
(c) $T(n) = \Omega(n^2)$ (d) $T(n) = O(n \log n)$

Q.67 The time complexity of the following C function is (assume $n > 0$)

```
int recursive (int n)
{  if (n == 1)
   return (1);
   else
   return (recursive (n - 1) + recursive (n - 1));
}
```

(a) $O(n)$ (b) $O(n \log n)$
(c) $O(n^2)$ (d) $O(2^n)$

Q.68 What is time complexity of the following recurrence relation and step to derive the same

$$T(n) = T(\sqrt{n}) + \log(\log n)$$

(a) $\log \log n$ (b) n
(c) $(\log \log n)^2$ (d) n^2

Q.69 What is the value of $T(n)$ for given recurrence relation

$$T(n) = T\left(\frac{n}{2}\right) + 2,\ T(1) = 1$$

when n is power of 2

(a) $2(\log n + 1)$ (b) $2 \log n$
(c) $\log n + 1$ (d) $2 \log n + 1$

Q.70 What is complexity of following recurrence relation

$$T(n) = \sqrt{n}T(\sqrt{n}) + 100n$$

(a) $\theta(n \log \log n)$ (b) $\theta(n \log n \log n)$
(c) $\theta(n)$ (d) $\theta(n \log n)$

Q.71 Consider the following recurrence relation:

$T(n) = 8T\left(\frac{n}{2}\right) + n^2 \log n,\ n > 1$ and

$T(1) = 1$

Find the time complexity of $T(n)$

(a) $O(n^2)$ (b) $O(n^2 \log n)$
(c) $O(n^3)$ (d) $O(n^3 \log n)$

Q.72 What is the time complexity of the following recurrence relation $T(k) = T\left(\frac{k}{2}\right) + T\left(\frac{k}{2}\right) + k^2$?

(a) $O(k)$ (b) $O(k^2)$
(c) $O(k \log k)$ (d) $O(k^2 \log k)$

Q.73 Which of the following can be solved using Master theorem?

(a) $T(n) = 2T\left(\frac{n}{2}\right) + \frac{n}{\log n}$

(b) $T(n) = 2T\left(\frac{n}{2}\right) + \log n$

(c) $T(n) = T\left(\frac{n}{2}\right) + \log n$

(d) None of these

Q.74 We solve TOH problem recursively breaking the task in three sections, which of the following recurrence will accord will with the approach, that is shows correct order of work done on each recursive step

(a) $T(n) = T(n-1) + 1 + T(n-1)$
(b) $T(n) = T(n-1) + 1T(n-1) + 1$
(c) $T(n) = 1 + T(n-1) + T(n-1)$
(d) $T(n) = T(n-1) + T(n-1) + 2$

Q.75 The running time of an algorithm $T(n)$, where n is the input size, is given by following:

$$T(n) = \begin{cases} 8T\left(\frac{n}{2}\right) + qn & \text{if } n > 1 \\ p & \text{if } n = 1 \end{cases}$$

where p and q are constants, the order of algorithm is

(a) n^2 (b) n^3
(c) n (d) n^n

Q.76 What is complexity of recurrence relation

$$T(n) = 3T\left(\frac{n}{2} + 47\right) + 2n^2 + 10n - \frac{1}{2}$$

(a) $O(n^2)$ (b) $O(n^{3/2})$
(c) $O(n \log n)$ (d) None of these

Q.77 Solve the recurrence relation and find complexity of it

$$T(n) = T\left(\frac{n}{2}\right) + \frac{n^2}{2} + n$$

(a) $O(n^3)$ (b) $O(n)$
(c) $O(n^2)$ (d) $O(n^2 \log n)$

Q.78 Let $T(n) = T(n-1) + \frac{1}{n}$. Then $T(n)$ is

(a) $\Theta(1)$ (b) $\Theta(\log n)$
(c) $\Theta(\log \log n)$ (d) $\Theta(n)$

Q.79 Which of the following function given by their recurrences grows the fastest asymptotically?

(a) $T(n) = 8T\left(\frac{n}{4}\right) + 100n^2$

(b) $T(n) = 81T\left(\frac{n}{9}\right) + 10n^2$

(c) $T(n) = 16T\left(\frac{n}{4}\right) + 100(n \log n)^{1.99}$

(d) $T(n) = 100T\left(\frac{n}{100}\right) + n \log^2 n$

Q.80 Let $T(n) = T(n-1) + \frac{1}{n}$; $T(1) = 1$; Then $T(n) = ?$

(a) $O(n^2)$ (b) $O(n \log n)$
(c) $O(\log n)$ (d) $O(n^2 \log n)$

Q.81 Which recurrence relation satisfy the sequence: 2, 3, 4, ..., for $n \geq 1$.

(a) $T(N) = 2T(N-1) - T(N-2)$
(b) $T(N) = 2T(N-1) + T(N-2)$
(c) $T(N) = N + 1$
(d) None of these

Q.82 $T(n) = T\left(\frac{2n}{3}\right) + 1$ then $T(n)$ is equal to

(a) $\Theta(\log_2 n)$ (b) $\Theta(n \log_2 n)$
(c) $\Theta(n^2)$ (d) $\Theta(n)$

Q.83 What is complexity of recurrence (n is a natural number)

$$T(n) = \begin{cases} 7T\left(\frac{n}{3}\right) + n^3; n > 2 \\ 1 \quad ; n \leq 2 \end{cases}$$

(a) $O(n)$ (b) $O(n^2)$
(c) $O(n^3)$ (d) $O(n \log n)$

Q.84 Let m, n be positive integers. Define $Q(m, n)$ as

$Q(m, n) = 0$, if $m < n$

$Q(m - n, n) + p$, if $m \geq n$

Then $Q(m, 3)$ is (a div b, gives the quotient when a is divided by b)

(a) a constant (b) $p \times (m \bmod 3)$

(c) $p \times (m \text{ div } 3)$ (d) $3 \times p$

Q.85 The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with n discs is

(a) $T(n) = 2T(n-2) + 2$

(b) $T(n) = 2T(n-1) + n$

(c) $T(n) = 2T\left(\frac{n}{2}\right) + 1$

(d) $T(n) = 2T(n-1) + 1$

Q.86 What is complexity of recurrence relation

$$T(n) = 3T\left(\frac{n}{2} + 47\right) + 2n^2 + 10n - \frac{1}{2}$$

(a) $O(n^2)$ (b) $O(n^{3/2})$

(c) $O(n \log n)$ (d) None of these

Q.87 What is complexity of recurrence relation

$$T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + n$$

(a) $\theta(n \log_3 n)$ (b) $\theta(n \log\log_{4/3} n)$

(c) $\theta(n^2 \log_{4/3} n)$ (d) $\theta(n \log_{4/3} n)$

Q.88 What is complexity of recurrence equation

$$T(n) = \text{sqrt}(2) + \left(\frac{n}{2}\right) + \text{sqrt}(n)$$

(a) $\theta(n(\log n + 1))$ (b) $\theta(n^2(\log n + 1))$

(c) $\theta(\sqrt{n}(\log n + 1))$ (d) $\theta(n^3(\log n + 1))$

Q.89 $T(n) = T(\sqrt{n}) + T(\log n) + 1; \quad n \geq 2$

When you solve the above recurrence relation using recursive tree method, what is the maximum height of the tree obtained?

(a) $\theta(\log \log n)$ (b) $\theta(\log n)$

(c) $\theta(\sqrt{n})$ (d) $\theta(n)$

Q.90 Find the worst case time complexity of the following recurrence relations:

$$T(n) = T(\sqrt{n}) + T(\log_2 n) + c$$

where c is a constant and $T(2) = 1$

(a) $\Omega(\log_2 \log_2 n)$ (b) $\Omega(\log_2 n)$

(c) $\Omega(n)$ (d) $\Omega(n \log_2 n)$

Q.91 Consider the following recurrence relation:

$$T(n) = 4T\left(\frac{n}{2}\right) + n^2 \log n; \quad T(1) = 1$$

Select the correct statements from the following

(a) A n-sized problem is divided into 4 sub-problems of $\frac{n}{2}$ size each.

(b) A n-sized problem is divided into 2 sub-problems of $\frac{n}{4}$ size each.

(c) $T(n) = O(n^2 (\log n)^2)$ (d) $T(n) = O(n^2 \log n)$

3. Divide and Conquer

Q.92 Which of the following sorting methods sorts a given set of items that is already in sorted order or in reverse sorted order with equal speed?

(a) Heap sort (b) Quick sort

(c) Insertion sort (d) Selection sort

Q.93 Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using randomized quicksort?

(a) $O(n)$ (b) $(n \log n)$

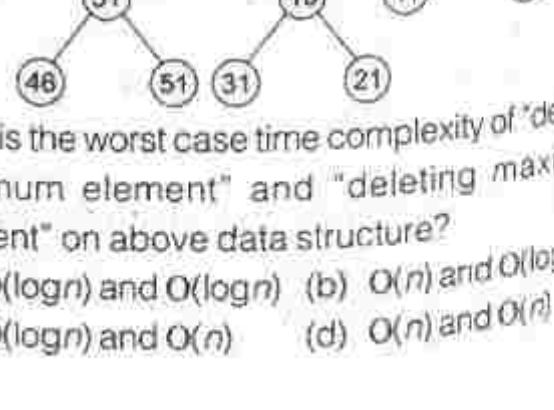
(c) $O(n^2)$ (d) $O(n!)$

Q.94 In quick sort for sorting of n numbers, the 75% greatest element is selected as pivot using $O(n)$ time complexity algorithm than what is the worst case time complexity of quick sort.

(a) $O(n^2)$ (b) $O(n \log n)$

(c) $O(n^3)$ (d) $O(n)$

Q.95 A **min-max heap** has the following property: "A almost complete binary tree where each node at an even level in the tree is less than all of its descendants and each of the node at an odd level in the tree is greater than all of its descendants"



What is the worst case time complexity of "deleting minimum element" and "deleting maximum element" on above data structure?

(a) $O(\log n)$ and $O(\log n)$ (b) $O(n)$ and $O(\log n)$

(c) $O(\log n)$ and $O(n)$ (d) $O(n)$ and $O(n)$

Q.96 Consider the following cases for quick sort to sort an array of n element $a[0 \ldots n-1]$.

1. Choosing the pivot element randomly from the given array.
2. Choosing median element as pivot.
3. Choosing middle element as pivot.

For which of the above cases quick sort always gives $O(n \log n)$ time complexity?

(a) 1 and 3
(b) 2 and 3
(c) 1 and 2
(d) None of these

Q.97 Consider the following sorting algorithms:

A. Merge sort
B. Quick sort
C. Heap sort
D. Insertion sort

Find the correct pair such that their best, average and worst case time complexities are same. (Assume that all elements are distinct)

(a) A, B (b) A, C
(c) A, D (d) None of these

Q.98 For merging two sorted lists of sizes m and n into a sorted list of size $m + n$. Find out the time complexity of this merging process.

(a) $O(m)$ (b) $O(n)$
(c) $O(m + n)$ (d) $O(\log(m) + \log(n))$

Q.99 Given n integers in the range of 0 to k we want to preprocess the input in such away that to any query about how many of the n-integers fall in the range $a \ldots b$ in $O(1)$. Then what will be the processes time?

(a) $O(n + k)$ (b) $O(n)$
(c) $O(\log n)$ (d) $O(k)$

Q.100 Quick sort gives $O(n \log n)$ worst case performance if the pivot is selected as:

(a) First element of array
(b) Median of first, last and middle elements
(c) Arithmetic mean of the elements
(d) None of above

Q.101 A sorting technique is called stable if

(a) It takes $O(n \log n)$ time.
(b) It maintain the relative order of occurrence of non-distinct elements.
(c) It uses divide and conquer paradigm.
(d) It takes $O(n)$ space.

Q.102 Suppose there are 4 sorted lists of $\frac{n}{2}$ elements each. If we merge these lists into a single sorted list of n elements, how many key comparisons are needed in the worst case using an efficient algorithm?

(a) $\frac{6}{4}n-3$ (b) $\frac{7}{4}n-3$
(c) $\frac{8}{4}n-3$ (d) $\frac{9}{4}n-3$

Q.103 A list of n strings, each of length n, is sorted into lexicographic order using the merge sort algorithm. The worst case running time of this computation is

(a) $O(n \log n)$ (b) $O(n^2 \log n)$
(c) $O(n^2 + \log n)$ (d) $O(n^2)$

Q.104 Consider an array containing 'n' elements. The elements present in an array are in arithmetic progression, but one element is missing in that order. What is the time complexing to find the position of the missing element using divide and conquer (by binary search).

(a) $O(n)$ (b) $O(\log n)$
(c) $O(n^2)$ (d) $O(n \log n)$

Q.105 How many term will be computed to determine the value of ${}^{10}C_8$ using a divide and conquer algorithms?

Q.106 What is time complexity to find the k^{th} largest element in a min heap?

(a) $O(n)$ (b) $O(\log n)$
(c) $O(n \log n)$ (d) $O(n^2)$

Q.107 Given two sorted list of size 'm' and 'n' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be

(a) $m \times n$ (b) maximum of m, n
(c) minimum of m, n (d) $m + n - 1$

Q.108 Consider an online sorting algorithm which is based on insertion sort. At some point of time, we have n elements which are already in sorted order. Now knew elements arrive and they need to be added to the list using the above mentioned online sorting algorithm. The worst-case time complexity to get the new sorted list with $(n + k)$ elements is:

Note: Initial elements are already sorted.

(a) $O(n(n+k))$ (b) $O((n+k)^2)$
(c) $O(k(n+k))$ (d) None of these

Q.109 Given "log n" sorted list each of $\sin\left(\frac{n}{\log n}\right)$, what is time required to merge them into one single list?

(a) $O(n \log n)$
(b) $O(n \log \log n)$
(c) $O(n \log n \log n)$
(d) $O(n)$

Q.110 "n" strings each of length "n" are given them, what is time taken to sort them, 2-way merge?

(a) $O(n)$ (b) $O(n \log n)$
(c) $O(n^2 \log n)$ (d) $O(\log n)$

Q.111 Given an array A of elements:

80, 70, 55, 90, 71, 72, 73, 77, 52, 50

What will be the output after 1 pass of 3 way merge sort?

(a) 70, 80, 55, 90, 71, 72, 73, 77, 50, 52
(b) 55, 70, 80, 71, 72, 90, 52, 73, 50, 77
(c) 50, 52, 55, 70, 71, 72, 73, 77, 80, 90
(d) 55, 70, 80, 71, 72, 90, 52, 73, 77, 50

Q.112 What is the worst case time complexity of Bubble Sort?

(a) $O(\log n)$ (b) $O(n)$
(c) $O(n \log n)$ (d) $O(n^2)$

Q.113 Consider an array having the following elements:

20, 50, 12, 45, 19, 29, 74, 37

Then, total number of comparisons required to sort the array using merge sort?

Q.114 Which of the following represent correct recurrence for worst case of Binary search with n element.

(a) $T(n) = T\left(\frac{n}{2}\right) + O(n)$

(b) $T(n) = 2T\left(\frac{n}{2}\right) + O(1)$

(c) $T(n) = 2T\left(\frac{n}{2}\right) + O(n)$

(d) $T(n) = T\left(\frac{n}{2}\right) + O(1)$

Q.115 Which of the following sorting algorithms has the lowest worst-case complexity?

(a) Merge sort (b) Bubble sort
(c) Quick sort (d) Selection sort

Q.116 Which of the following algorithms is not a comparison-based algorithm?

(a) Heap sort (b) Quick sort
(c) Radix sort (d) Topological sort

Q.117 Given a binary-max heap. The elements are stored in an array as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations?

(a) 14,13,8,12,10 (b) 14,12,13,10,8
(c) 14,13,12,8,10 (d) 14,13,12,10,8

Q.118 A list has n-strings where each string is of length n. Using divide and conquer approach, what is the worst case running time (tightest upper bound) to find the first string?

Note: First string is a string which appears first when all strings are sorted into the lexicographical order.

(a) $O(n^2)$ (b) $O(n)$
(c) $O(n \log n)$ (d) $O(n^2 \log n)$

Q.119 Given a sequence of n integers in the order 9, 8, 7, 6, 5, 4, 3, 2, 1, 0, after which iteration, the resulting sequence would be 1, 2, ,3 4, 5, 6, 7, 8, 9, 0 if you apply insertion sort? _____ th iteration

Q.120 In the standard merge sort algorithm on a list of size n, what is the maximum number of times an item can be compared?

(a) 2 (b) $\log n$
(c) $n-1$ (d) $n \log n$

Q.121 Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are TRUE?

I. Quick sort runs in $\Theta(n^2)$ time
II. Bubble sort runs in $\Theta(n^2)$ time
III. Merge sort runs in $\Theta(n)$ time
IV. Insertion sort runs in $\Theta(n)$ time

(a) I and II only
(b) I and III only
(c) II and IV only
(d) I and IV only

Q.122 Assume an array of n integers and a function Max( ) given below:

Max( ) = {find an element which is higher than its left and right element if any}

What is the worst case time complexity for Max( ) on an input array?

(a) $O(\log n)$ (b) $O(n)$
(c) $O(n \log n)$ (d) $O(n^2)$

Q.123 Consider an array consisting of the following elements in unsorted order (placed randomly), but 60 as first element.

60, 80, 15, 95, 7, 12, 35, 90, 55

Quick sort partition algorithm is applied by choosing first element as pivot element. How many total number of arrangements of array integers is possible preserving the effect of first pass of partition algorithm.

Q.124 Which one of the following does not represent the array if bubble sort to sort the elements in ascending order is run after one iteration

(a) 1, 2, 5, 7, 8, 6, 4, 3 9
(b) 3, 2, 5, 9, 8, 6, 4, 1, 7
(c) 1, 2, 5, 6, 7, 8, 4, 3, 9
(d) None of the above

Q.125 A machine needs a minimum of 100 sec to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately

(a) 50.2 sec (b) 6.7 sec
(c) 72.7 sec (d) 11.2 sec

Q.126 Which of the following can make for an improved version of bubble sort?

(a) Traverse the array left to right during old passes and right to left during even passes, in bubble sort.
(b) Divide the array into smaller arrays apply quick sort on each of the small arrays and apply the bubble sort on the entire array.
(c) Store the addresses to be swapped in a separate array and make all the swaps at one go.
(d) None of these

Q.127 In quick sort, for sorting n elements, the $\left(\frac{n}{4}\right)^{th}$ smallest element is selected as pivot using an $O(n)$ time algorithm. What is the worst case time complexity of the quick sort?

(a) $\Theta(n)$ (b) $\Theta(n \log n)$
(c) $\Theta(n^2)$ (d) $\Theta(n^2 \log n)$

Q.128 The usual $\Theta(n^2)$ implementation of Insertion Sort to sort an array uses linear search to identify the position where an element is to be inserted into the already sorted part of the array. If instead, we use binary search to identify the position, the worst case running time will

(a) remain $\Theta(n^2)$
(b) become $\Theta(n (\log n)^2)$
(c) become $\Theta(n \log n)$
(d) become $\Theta(n)$

Q.129 Match List-I (Divide Conquer) with List-II (Recurrence Relation (Worst-Case) and select the correct answer using the codes given below the lists:

List-I	List-II
A. Binary search	1. $T(n) = 2T\left(\frac{n}{2}\right) + C$
B. Merge sort	2. $T(n) = T\left(\frac{n}{2}\right) + 1$
C. Quick sort	3. $T(n) = T(n-1) + n - 1$
D. Max Min	4. $T(n) = 2T\left(\frac{n}{2}\right) + kn$

Codes:

	A	B	C	D
(a)	1	3	4	2
(b)	2	3	4	1
(c)	1	4	3	2
(d)	2	4	3	1

Q.130 A modified version of merge sort is devised with a modified merge method is used where the merge takes $O(n^2)$ time to merge lists that sums up to size n, the time complexity for such a version of merge sort is

(a) $O(n \log n)$
(b) $O(n)$
(c) $O(n^2 \log n)$
(d) $O(\sqrt{n})$

Q.131 What is worst case time complexity to delete an middle element from min heap of n distinct element?

(a) $O(\log n)$
(b) $O(n \log n)$
(c) $O(n)$
(d) $O(n^2)$

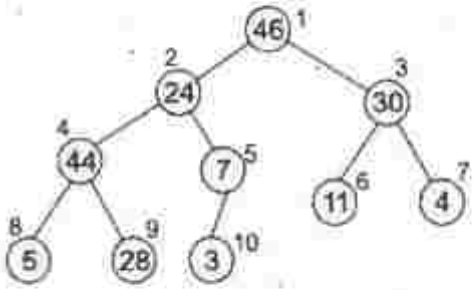
Q.132 An e-commerce site on search for a particular key word returns the search results in order to popularity by default also the user is allowed to sort by price (low to high) or (high to low) however if two or more products are of the same price then among the same price products should be presented to the user in the same order of popularity. Which of the following sorting algorithm cannot be used for such a scenario?

(a) Insertion sort (b) Bubble sort
(c) Quick sort (d) None of above

Q.133 Which of one the following is false?

(a) Heap sort is an inplace algorithm.
(b) Heap sort has $O(n \log n)$ average case time complexity.
(c) Heap sort is a stable sort.
(d) Heap sort is a comparison-based sorting algorithm.

Q.134 The essential part of heap sort is construction of max-heap. Consider the tree shown below, the node 24 violates the max-heap property. Once heapify procedure is applied to it, which position will it be in?



(a) 4 (b) 5
(c) 8 (d) 9

Q.135 The asymptotic running time of Randomized QuickSort on arrays of length n.

(a) In expectation is $\theta(n \log n)$
(b) In expectation is $\theta(n^2)$.
(c) In the worst case is $\theta(n \log n)$
(d) In the worst case $\theta(n^2)$.

Q.136 Which of the following statement(s) is/are correct?

(a) Consider an array A in which upto some index I, integers are stored and after that NULL values are stored. Let the size of array be n, Then time taken to find the value I would be $O(\log n)$.
(b) Consider an array A in which upto some index I, Real numbers are stored and after that NULL values are stored. Let the size of array be n, Then time taken to find the value I would be $O(\log n)$.
(c) Consider an array A in which upto some index I, integers are stored and after that NULL values are stored. Let the size of array be n, Then time taken to find the value I would be $O(n)$.
(d) Consider an array A in which upto some index I, Real numbers are stored and after that NULL values are stored. Let the size of array be n, Then time taken to find the value I would be $O(n)$.

Q.137 Which of the following sorting algorithm(s) does not have a worst case running time of $O(n^2)$?

(a) Insertion sort (b) Merge sort
(c) Quick sort (d) Bubble sort

Q.138 A sort method is said to be stable if the relative order of keys is the same after the sort as it was before the sort. In which of the following pairs both sorting algorithms are stable?

(a) Quick sort and Insertion sort
(b) Insertion sort and Bubble sort
(c) Quick sort and Heap sort
(d) Quick sort and Bubble sort

Q.139 A binary search tree contains the values -1, 2, 3, 4, 5, 6, 7 and 8. The tree is traversed in preorder and the values are printed out. Which of the following sequences is a valid output?

(a) 5 3 1 2 4 7 8 6 (b) 5 3 1 2 6 4 9 7
(c) 5 3 2 4 1 6 7 8 (d) 5 3 1 2 4 7 6 8

Q.140 Given two sorted list of size 'm' and 'n' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be

(a) $m \times n$
(b) maximum of m, n
(c) minimum of m, n
(d) $m + n - 1$

Q.141 For merging two sorted lists of sizes m and n into a sorted list of size $m + n$, we require comparisons of

(a) $O(m)$ (b) $O(n)$
(c) $O(m + n)$ (d) $O(\log m + \log n)$

Q.142 The average successful search time taken by binary search on a sorted array of 10 items is

(a) 2.6 (b) 2.7
(c) 2.8 (d) 2.9

Q.143 The number of swapping needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order, using bubble sort is

(a) 11 (b) 12
(c) 13 (d) 10

Q.144 A sorting technique is called stable if

(a) It takes $O(n \log n)$ time
(b) It maintains the relative order of occurrence of non-distinct elements
(c) It uses divide and conquer paradigm
(d) It takes $O(n)$ space

Q.145 Consider the following scenario during insertion sort when the array looks like the following:

{25, 75, 95, 125, 80, 5, 10}

The number of comparisons that it will further take for the array to be completely sorted are _____.

Q.146 The average number of comparisons performed by the merge sort algorithm, in merging two sorted lists of length 2 is

(a) $\frac{8}{3}$ (b) $\frac{8}{5}$
(c) $\frac{11}{7}$ (d) $\frac{11}{6}$

Q.147 A sorting technique that guarantees, that records with the same primary key occurs in the same order in the sorted list as in the original unsorted list is said to be

(a) Stable (b) Consistent
(c) External (d) Linear

Q.148 A list of n strings, each of length n, is sorted into lexicographic order using the merge sort algorithm. The worst case running time of this computation is

(a) $O(n \log n)$ (b) $O(n^2 \log n)$
(c) $O(n^2 + \log n)$ (d) $O(n^2)$

Q.149 The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are:

(a) $\Theta(n \log n)$, $\Theta(n \log n)$, and $\Theta(n^2)$
(b) $\Theta(n^2)$, $\Theta(n^2)$, and $\Theta(n \log n)$
(c) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n \log n)$
(d) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$

Q.150 The complexity of comparison based sorting algorithms is

(a) $\theta(n \log n)$ (b) $\theta(n)$
(c) $\theta(n^2)$ (d) $\theta(n A n)$

Q.151 An algorithm is made up of 2 modules M_1 and M_2. If time complexity of modules M_1 and M_2 are $h(n)$ and $g(n)$ respectively, the time complexity of the algorithm is

(a) $\min(h(n), g(n))$
(b) $\max(h(n), g(n))$
(c) $h(n) + g(n)$
(d) $h(n) * g(n)$

Q.152 Suppose that the splits at every level of quicksort are in the proportion $(1 - \alpha)$ to α, where $0 < \alpha \leq \frac{1}{2}$ is a constant. The minimum depth of a leaf in the recursion tree is approximately given by

(a) $-\frac{\lg n}{\lg(1-\alpha)}$ (b) $-\frac{\lg(1-\alpha)}{\lg n}$
(c) $-\frac{\lg n}{\lg \alpha}$ (d) $-\frac{\lg \alpha}{\lg n}$

Q.153 If there are n integers to sort, each integer has d digits and each digit is in the set $\{1, 2, \ldots, k\}$, radix sort can sort the numbers in :

(a) $O(d\,n\,k)$ (b) $O(d\,n^k)$
(c) $O((d+n)k)$ (d) $O(d(n+k))$

Q.154 Which of the following algorithms sort n integers, having the range 0 to $(n^2 - 1)$, in ascending order in $O(n)$ time?

(a) Selection sort (b) Bubble sort
(c) Radix sort (d) Insertion sort

Q.155 A priority queue is implemented as a max-heap. Initially, it has five elements. The level order traversal of the heap is as follows:

20, 18, 15, 13, 12

Two new elements '10' and '17' are inserted in the heap in that order. The level-order traversal of the heap after the insertion of the element is:

(a) 20, 16, 17, 15, 13, 12, 10
(b) 20, 18, 17, 12, 13, 10, 15
(c) 20, 18, 17, 10, 12, 13, 15
(d) 20, 18, 17, 13, 12, 10, 15

Q.156 If the array A contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array A after third pass of insertion sort?

(a) 67, 12, 10, 5, 4, 7, 23
(b) 4, 7, 10, 23, 67, 12, 5
(c) 4, 5, 7, 67, 10, 12, 23
(d) 10, 7, 4, 67, 23, 12, 5

4. Greedy Techniques

Q.157 Single shortest path problem can be implemented by Greedy algorithms using
(a) Singly linked list (b) Min Heap
(c) AVL tree (d) All of the above

Q.158 Kruskal's algorithm for finding a minimum spanning tree of a weighted graph G with n vertices and m edges has the time complexity of
(a) $O(n^2)$ (b) $O(mn)$
(c) $O(m+n)$ (d) $O(m \log n)$

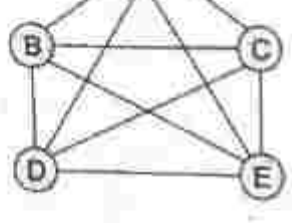
Q.159 To implement Dijkstra's shortest path algorithm on unweighted graphs so that it runs in linear time, then data structure to be used is
(a) Queue (b) Stack
(c) Heap (d) B-Tree

Q.160 Find the running time of Dijkstra's algorithm on complete graph of n-vertices.
(a) $O(n)$ (b) $O(n^2)$
(c) $O(n^2 \log n)$ (d) $O(n^3)$

Q.161 Consider the following steps:
S_1: Characterize the structure of an optimal solution.
S_2: Compute the value of an optimal solution in bottom-up fashion.
Which of the following step(s) is/are common to both dynamic programming and Greedy algorithm?
(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) Neither S_1 or S_2

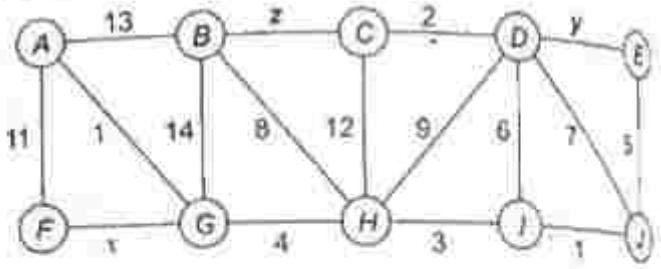
Q.162 The number of spanning trees for a complete graph with seven vertices is
(a) 2^5 (b) 7^5
(c) 3^5 (d) $2^{2\times5}$

Q.163 The number of spanning tree possible for the following graph is _____.



Q.164 Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w. Which of the following is FALSE?
(a) There is a minimum spanning tree containing e.
(b) If e is not in a minimum spanning tree T, then in the cycle formed by adding e to T, all edges have the same weight.
(c) Every minimum spanning tree has an edge of weight w.
(d) e is present in every minimum spanning tree.

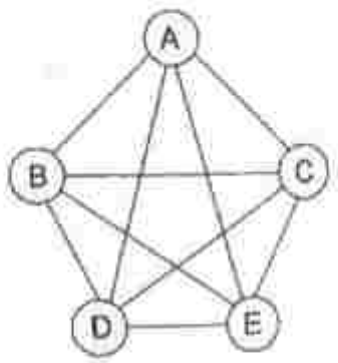
Q.165 Suppose that minimum spanning tree of the following edge weighted graph contains the edges with weights x, y and z



What is the maximum value of $x + y + z$?

Q.166 If Kruskal's algorithm is used for finding a minimum spanning tree of a weighted graph G with n vertices and m edges and edge weights are already given in a sorted list, then, what will be the time complexity to compute the minimum cost spanning tree given that union and find operations amortized O(1)?
(a) $O(m \log n)$ (b) $O(n)$
(c) $O(m)$ (d) $O(n \log m)$

Q.167 Consider a graph 'G' given below:



If edges are having weight 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10. Then maximum possible weight that minimum weight spanning tree of G can have

_____.

Q.168 Consider a weighted complete graph G on the vertex set $\{v_1, v_2, \ldots, v_n\}$ such that the weight of the edge (v_i, v_j) is $2|i-j|$. The weight of a minimum spanning tree of G is
(a) $n-1$ (b) $2n-2$
(c) $\left(\frac{n}{2}\right)$ (d) n^2

Q.169 Consider the following statements:

I. For every weighted graph and any two vertices s and t, Bellman-Ford algorithm starting at s will always return a shortest path to t.

II. At the termination of the Bellman-Ford algorithm, even if graph has negative weight cycle, a correct shortest path is found for a vertex for which shortest path is well-defined.

Which of the above statements are true?

(a) Only I
(b) Only II
(c) Both I and II
(d) None of these

Q.170 Let k_n be the complete graph on "n" vertices labelled $\{1, 2, \ldots n\}$ with $m = \frac{n(n-1)}{2}$ edges. What is the number of spanning trees of k_n?

(a) $\frac{m}{n-1}$
(b) m^{n-1}
(c) n^{n-2}
(d) n^{n-1}

Q.171 Consider a weight complete graph G on the vertex set $\{V_1, V_2, \ldots V_n\}$ such that the weight of edge (V_i, V_j) is $4|i-j|$. The weight of minimum cost spanning tree of G is:

(a) $4n^2$
(b) n
(c) $4n-4$
(d) $2n-2$

Q.172 Consider the following statements:

S_1: In a connected undirected graph $G = (V, E)$ with distinct edge costs, the cheapest edge belongs to every minimum spanning tree.

S_2: In a connected undirected graph $G = (V, E)$ with distinct edge costs, the most expensive edge is excluded from every minimum spanning tree.

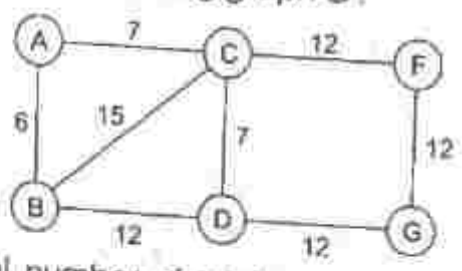
Which of the following is true?

(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 nor S_2

Q.173 Assume a graph G has negative weight edges. Which of the following statement is correct when the Dijkstra's algorithm is applied on G?

(a) Always produces correct results
(b) Always produces incorrect results
(c) It always terminates but may produce incorrect results
(d) None of the above

Q.174 Consider the following graph G:

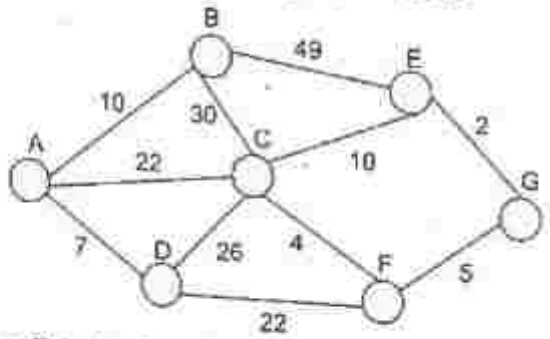


The total number of minimum spanning trees using Prim's or Kruskal's algorithm are _____

Q.175 Let $G(V, E)$ an undirected graph with positive edge weights. Dijkstra's single source-shortest path algorithm can be implemented using the binary heap data structure with time complexity?

(a) $O(|V|^2)$
(b) $O(|E|+|V|\log|V|)$
(c) $O(|V|\log|V|)$
(d) $O((|E|+|V|)\log|V|)$

Q.176 Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

(a) (E, G), (C, F), (F, G), (A, D), (A, B), (A, C)
(b) (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)
(c) (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)
(d) (A, D), (A, B), (D, F), (F, C), (F, G), (G, E)

Q.177 Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm?

P. Always finds a negative weighted cycle, if one exists.

Q. Finds whether any negative weighted cycle is reachable from the source

(a) P only
(b) Q only
(c) Both P and Q
(d) Neither P nor Q

Q.178 Let the weight of minimum spanning tree of given graph G as per Kruskal's algorithm is w_K and as per Prim's algorithm is w_P then

(a) $w_K \le w_P$
(b) $w_K \ge w_P$
(c) $w_K = w_P$
(d) Any of the above is possible depends on graph structure

Q.179 Which of the following statements is invalid?

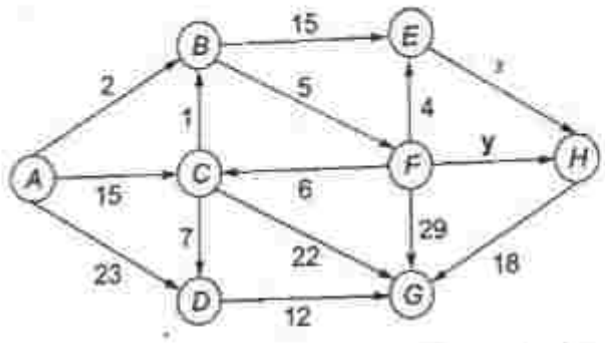
(a) A graph where all edge weights are distinct can have more than one shortest path between two vertices.

(b) Multiplying all edge weights by a positive number might change the cost of minimum spanning tree.

(c) Both (a) and (b)

(d) Neither (a) nor (b)

Q.180 Suppose that you are running Dijkstra's algorithm on the edge-weighted digraph below, starting from vertex *A*.



The table gives 'Distance' and 'Parent' entry of each vertex after vertex *E* has been deleted from the priority queue and relaxed.

Vertex	Distance	Parent
A	0	NULL
B	2	*A*
C	13	*F*
D	23	*A*
E	11	*F*
F	7	*B*
G	36	*F*
H	19	*E*

What could be the possible values of x and y?

(a) $x = 11, y = 7$

(b) $x = 8, y = 12$

(c) $x = 6, y = 10$

(d) $x = 10, y = 14$

Q.181 Given a set $A = \{A_1, A_2, \ldots, A_n\}$ of n activities with start and finish time (Si, fi), $1 \le i \le n$, select maximal set S of "non-overlapping" activities. If the given problem is solved using greedy algorithm, then what will be the time complexity?

(a) $O(n^3)$ (b) $O(n^2)$

(c) $O(n \log n)$ (d) $O(n^n)$

Q.182 The total number of spanning trees that can be drawn using five labelled vertices is _____?

Q.183 Consider the following statements:

1. Dijkstra's algo will terminate even if there is a –ve edge or –ve cycle.
2. At the termination of Bellman Ford, even if graph has –ve cycle, a correct shortest path is found for a vertex for which shortest path is well-defined.

Choose correct option:

(a) 1-True; 2-True (b) 1-True; 2-False

(c) 1-False; 2-True (d) 1-False; 2-False

Q.184 Which of the following statements is true?

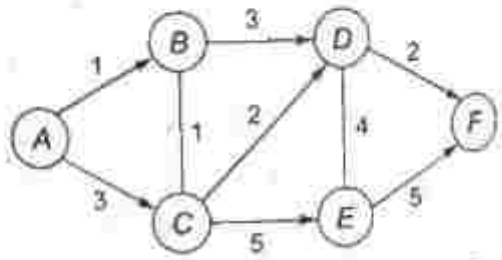
(a) Adding a constant to every edge weight in a directed graph can change the set of edges that belongs to shortest path tree. Assume unique weights.

(b) Adding a constant to every edge weight in an undirected graph can change the set of edges that belongs to the minimum spanning tree.

(c) Rerunning Dijkstra's algorithm on a graph V times will result in the correct shortest paths tree, even if there are negative edges (but no negative cycles).

(d) None of these

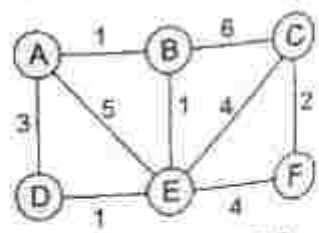
Q.185 Suppose we run Dijkstra's single source shortest path algorithm on the following edge weighted directed graph with vertex *A* as the source. What is the order of the vertices in which the shortest path is found by the algorithm?



(a) *A, B, C, D, F, E* (b) *A, B, C, D, E, F*

(c) *A, B, D, C, E, F* (d) *A, B, C, E, D, F*

Q.186 Consider the weighted, undirected, connected graph below. What is the minimum cost spanning tree and maximum cost spanning tree respectively?



(a) 9, 22 (b) 10, 22

(c) 9, 20 (d) 10, 20

Q.187 Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I

A. All pairs shortest path
B. Quick sort
C. Minimum weight spanning tree
D. Connected Components

List-II

1. Greedy
2. Depth-first search
3. Dynamic programming
4. Divide and conquer

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	3	4	1	2
(c)	3	4	2	1
(d)	4	1	2	3

Q.188 The number of spanning trees for a complete graph with seven vertices is

(a) 2^6 (b) 7^5
(c) 3^5 (d) $2^{2\times5}$

Q.189 The minimum number of edges in a connected cyclic graph on n vertices is

(a) $n-1$ (b) n
(c) $n+1$ (d) None of these

Q.190 In the following table, the left column contains the names of standard graph algorithms and the right column contains the time complexities of the algorithms. Match each algorithm with its time complexity.

List-I	List-II
1. Bellman-Ford algorithm	A. $O(m \log n)$
2. Kruskal's algorithm	B. $O(n^3)$
3. Floyd-Warshall algorithm	C. $O(nm)$
4. Topological sorting	D. $O(n+m)$

(a) 1-C, 2-A, 3-B, 4-D
(b) 1-B, 2-D, 3-C, 4-A
(c) 1-C, 2-D, 3-A, 4-B
(d) 1-B, 2-A, 3-C, 4-D

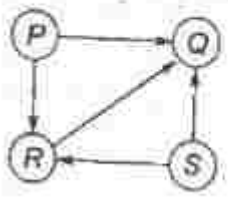
Q.191 Which of the following algorithms solves the all pair shortest path problem?

(a) Dijkstra's algorithm
(b) Floyd's algorithm
(c) Prim's algorithm
(d) Warshall's algorithm

Q.192 The maximum degree of any vertex in a simple graph with n vertices is

(a) n (b) $n-1$
(c) $n+1$ (d) $2n-1$

Q.193 Consider the directed graph given below:



Which one of the following is TRUE?

(a) The graph does not have any topological ordering.
(b) Both PQRS and SRQP are topological orderings.
(c) Both PSRQ and SPRQ are topological orderings.
(d) PSRQ is the only topological ordering.

Q.194 Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm?

P : Always finds a negative weighted cycle, if one exists.
Q : Finds whether any negative weighted cycle is reachable from the source.

(a) P only
(b) Q only
(c) Both P and Q
(d) Neither P nor Q

Q.195 Let $G = (V, E)$ be any connected undirected edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

I. Minimum Spanning Tree of G is always unique.
II. Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

Q.196 Complexity of Kruskal's algorithm for finding the minimum spanning tree of an undirected graph containing n vertices and m edges if the edges are sorted is

(a) $O(m)$ (b) $O(n)$
(c) $O(m+n)$ (d) None of these

Q.197 The time complexities of some standard graph algorithms are given. Match each algorithm with its time complexity? (n and m are number of nodes and edges respectively)

List-I

A. Bellman Ford algorithm
B. Kruskal algorithm
C. Floyd Warshall algorithm
D. Topological sorting

List-II

1. $O(m \log n)$
2. $O(n^3)$
3. $O(mn)$
4. $O(m+n)$

Codes :

	A	B	C	D
(a)	3	1	2	4
(b)	2	4	3	1
(c)	3	4	1	2
(d)	2	1	3	4

Q.198 Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I

A. 'V' invocations of Dijkstra's algorithm using min heap tree data structure.
B. 'V' invocations of Bellman-Ford algorithm.
C. Single invocation of Floyd-Warshall algorithm.

List-II

1. $O(V^3)$
2. $O(V^3 \log V)$
3. $O(V^4)$

[Given edges = $O(V^2)$, where V is vertices]

Codes :

	A	B	C
(a)	2	3	1
(b)	3	2	1
(c)	1	2	3

(d) None of these

Q.199 Select all the correct options:

(a) Dijkstra's shortest path algorithm always gives the correct output for a directed weighted graph.
(b) Shortest path between two vertices may change if every edge is increased by the same quantity.
(c) Bellman Ford algorithm is a dynamic programming based technique.
(d) Kruskal's algorithm may create disconnected intermediate spanning tree when it is used on an undirected graph.

Q.200 Consider a complete graph G with V vertices and E edges. Now, assume two nodes u and v. The shortest distance between them is 8. Assuming that all edge weights are positive, how many such path are possible if atmost 4 edges can be used in the shortest path?

(a) 180 (b) 121
(c) 150 (d) 165

5. Graph Based Algorithms

Q.201 G is an undirected graph with vertex set {v1, v2, v3, v4, v5, v6, v7} and edge set {v1v2, v1v3, v1v4, v2v4, v2v5, v3v4, v4v5, v4v6, v5v6, v6v7}. A breadth first search of the graph is performed with v1 as the root node. Which of the following is a tree edge?

(a) v2v4
(b) v1v4
(c) v4v5
(d) v3v4

Q.202 Which of the following is application of Breadth First Search on the graph?

(a) Finding diameter of the graph
(b) Finding bipartite graph
(c) Both (a) and (b)
(d) None of the above

Q.203 The number of edges in a regular graph of degree d and n vertices is

(a) maximum of n and d
(b) $n + d$
(c) nd
(d) $\frac{nd}{2}$

Q.204 Maximum number of BFS traversal possible of BST of height 3 is? (Root node is at height 0)

_____?

Q.205 Preorder is same as

(a) Depth-first order
(b) Breadth-first order
(c) Topological order
(d) Linear order

Q.206 Consider the following statements:

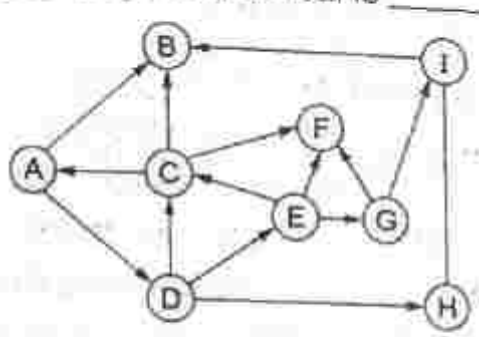
S_1: DFS of directed graph always produces the same number of edges in the traversal irrespective of starting vertex.

S_2: If we remove all of back edges found while DFS traversal on directed graph, the resultant graph is acyclic.

Find which of the following is true.

(a) Both S_1 and S_2 are valid
(b) Only S_1 valid
(c) Only S_2 Valid
(d) Neither S_1 nor S_2 is valid

Q.207 If DFS algorithm is applied to the given graph starting from vertex A the maximum height of the DFS tree for DFS traversal is _____?



Q.208 Which of the following is the correct match between the algorithm and their time complexity?

Algorithm

P. Dijkstra (Priority queue and adjacency cost)
Q. Prim's algorithm (Binary heap)
R. BFS
S. Floyd Warshall
T. Kruskal Algorithm
U. Bellman Ford algorithm

Time Complexity

1. $O(V + E)$
2. $O(E \log V)$
3. $O(V^3)$
4. $O((E + V) \log V)$
5. $O(VE)$

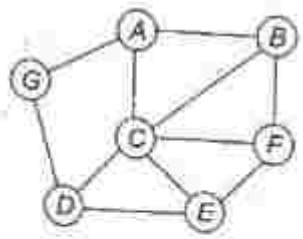
(a) P-5, Q-2, R-3, S-4, T-5, U-3
(b) P-4, Q-2, R-1, S-3, T-2, U-5
(c) P-5, Q-2, R-2, S-3, T-2, U-5
(d) P-2, Q-4, R-3, S-3, T-5, U-5

Q.209 Which of the following is application of depth-first search?

(a) Only topological sort.
(b) Only strongly connected components.
(c) Both topological sort and strongly connected components.
(d) Neither topological sort nor strongly connected components.

Q.210 Number of undirected graphs (not necessarily connected) can be constructed given set $V = \{1, 2, 3, 4\}$ of 4 vertices, are _____.

Q.211 Which of the following are the articulation points in the following graph?



(a) Only C
(b) D, C, F
(c) B, C, F
(d) None of these

Q.212 Let G be a graph with 'n' vertices and 'm' edges.

S_1: All its DFS forest (for traversals starting at different values) will have the same number of trees.

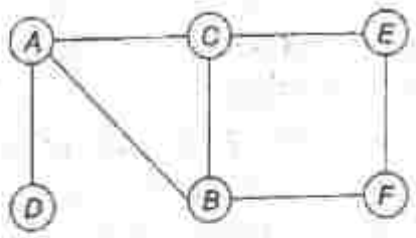
S_2: All its DFS forests will have the same number of tree edges and the number of back edges.

(a) S_1 - True; S_2 - True
(b) S_1 - True; S_2 - False
(c) S_1 - False; S_2 - True
(d) S_1 - False; S_2 - False

Q.213 A mother vertex in a graph $G = (V, E)$ is a vertex V such that all other vertices in G can be reached by a path from V. What is time complexity of an efficient algorithm to find the mother vertex in a directed connected graph?

(a) $O(V(V + E))$
(b) $O(V^2)$
(c) $O(V \log V)$
(d) $O(V + E)$

Q.214 The depth first search algorithm has been implemented using the stack data structure. One possible order of visiting the node of the following graph is

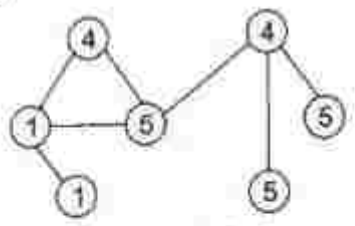


(a) $A\,D\,B\,C\,F\,E$
(b) $A\,C\,E\,B\,F\,D$
(c) $B\,A\,C\,D\,F\,E$
(d) $B\,C\,E\,F\,A\,D$

Q.215 Which of the following problem cannot be computed by DFS?

(a) Finding connected components in directed graph.
(b) Finding shortest path in unweighted graph.
(c) Finding directed cycle in directed graph.
(d) Finding path from a source vertex to all other reachable vertices in directed graph.

Q.216 The number of articulation points of the following graph is:

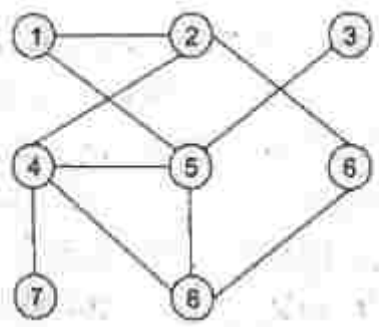


(a) 0 (b) 1
(c) 2 (d) 3

Q.217 Which of the following condition is sufficient to detect cycle in a directed graph?

(a) There is an edge from currently being visited node to an already visited node.
(b) There is an edge from currently being visited node to an ancestor of currently visited node in DFS forest.
(c) Every node is seen twice in DFS.
(d) None of the above

Q.218 Consider the following graph:



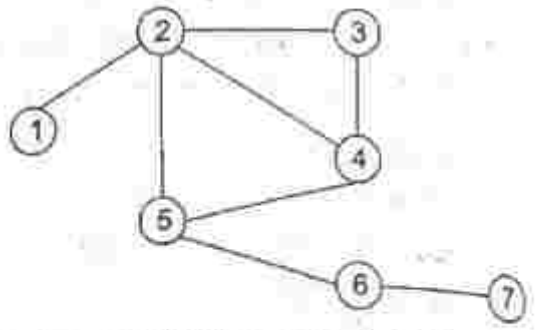
Which does not represent Depth First Search (DFS) sequence for the above graph?

(a) 12475386 (b) 12685437
(c) 12453867 (d) 12458637

Q.219 Which one of the following statements is false?

(a) Optimal binary search tree construction can be performed efficiently using dynamic programming.
(b) Breadth-first search cannot be used to find connected components of a graph.
(c) Given the prefix and postfix walks over a binary tree, the binary tree cannot be uniquely constructed.
(d) Depth-first search can be used to find connected components of a graph.

Q.220 Consider the following graph:



What is the depth first search (DFS) sequence for the above graph?

(a) 4, 5, 2, 1, 3, 6, 7
(b) 2, 3, 4, 5, 6, 1, 7
(c) 2, 1, 5, 3, 4, 6, 7
(d) 4, 2, 1, 5, 6, 3, 7

Q.221 Which of the following statements are true?

S_1: Suppose we do a DFS on a directed graph G. If we remove all of the back edges found, the resulting graph is now acyclic.
S_2: Both BFS and DFS requires Big-omega(V) storage for their operation. (That is, for working storage, above and beyond the storage needed to represent the input).
S_3: A depth-first search of a directed graph always produces the same number of tree edges (i.e., independent of the order in which the vertices are provided and independent of the order the adjacency lists).

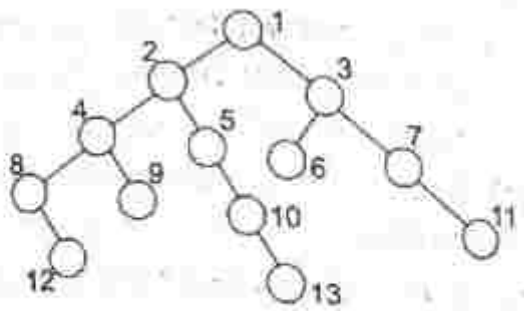
(a) S_1 and S_3 (b) S_1 only
(c) S_1 and S_2 (d) S_2 only

Q.222 Which of the following statements are true?

S_1: For all weighted graph and all vertices s and t, Bellman-Ford starting at s will always return a shortest path to t.
S_2: If all edges in a graph have distinct weights then the shortest path between two vertices is unique.
S_3: In an unweighted graph where the distance between any two vertices is atmost T, any BFS tree has depth atmost T but a DFS tree might have larger depth.
S_4: There is no edge in an undirected graph that jumps more than one level of any BFS tree of the graph.

(a) S_1 and S_3 (b) S_3 only
(c) S_3 and S_4 (d) S_1, S_2 and S_4

Q.223 In the following graphs, assume that if there is ever a choice amongst multiple nodes, both the DFS and BFS algorithms will choose the left-most node first.



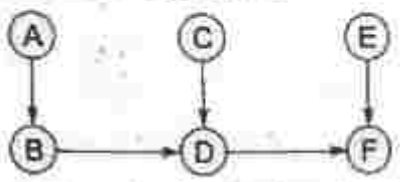
Starting from the root 1 at top, which algorithm will visit the least number of nodes before visiting the 6th node?

(a) BFS
(b) DFS
(c) Neither DFS or BFS will ever encounter the goal node in his graph.
(d) BFS and DFS encounter same number of nodes before encounter the goal node.

Q.224 Consider an undirected unweighted graph G. Let a breadth-first traversal of G be done starting from a node r. Let $d(r, u)$ and $d(r, v)$ be the lengths of the shortest paths from r to u and v respectively in G. If u is visited before v during the breadth-first traversal, which of the following statement is correct?

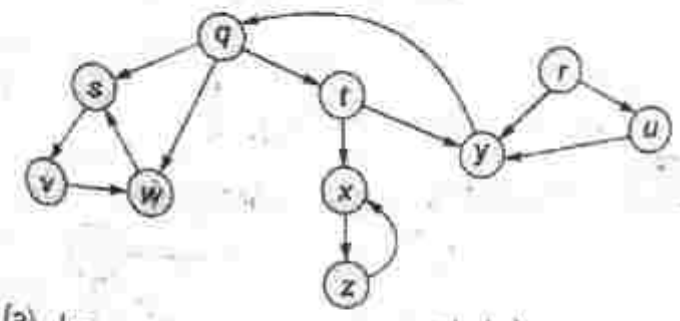
(a) $d(r, u) < d(r, v)$ (b) $d(r, u) > d(r, v)$
(c) $d(r, u) \leq d(r, v)$ (d) None of these

Q.225 Consider the following graph:



The number of topological orders for given graph are _____.

Q.226 What are the strongly connected components in the below figure?



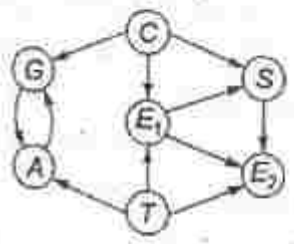
(a) {s, v, w}, {x, z}, {q, t, y}, {r}, {v}
(b) {q, s, v}, {s, v, w}, {t, w, z}, {q, y}, {r, v}
(c) {r}, {u}, {s, v, w}, {t, w, z}, {r, v, y}
(d) {s, v, w}, {x, z}, {q, t, y}, {r, v}

Q.227 Which of the following statements are true?

1. In a depth-first search of an undirected graph G, every edge of G is either a tree edge or a back edge.
2. Forward and cross edges never occur in a depth-first search of an undirected graph.
3. A directed graph is Acyclic if and only if a depth-first search yields no back edges.

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) All of these

Q.228 Find the number of strong components in the following graph. Strong component is the maximum subgraph which is strongly connected. Union of all strong components should contain all vertices of given graph.



(a) 1 (b) 2
(c) 3 (d) None of these

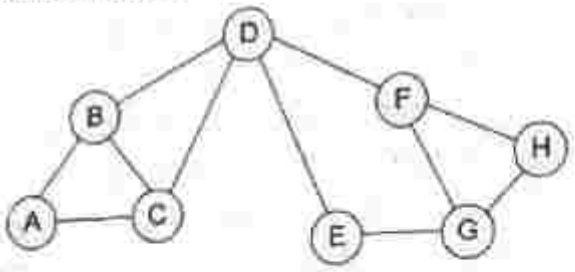
Q.229 Which of the following statement(s) is/are true?

(a) The different minimum cuts are there in a tree with n nodes is $n - 1$.
(b) The different minimum cuts are there in a tree with n nodes is n.
(c) On adding one extra edge to a directed graph G, the number of strongly connected components increases.
(d) On adding one extra edge to a directed graph G, the number of strongly connected components decreases.

Q.230 Consider a directed graph with distinct and nonnegative edge lengths and a source vertex s. Fix a destination vertex t, and assume that the graph contains at least one $s - t$ path. Which of the following statements is/are true?

(a) There is a shortest $s - t$ path with no repeated vertices (i.e. a "simple" or "loopless" such path).
(b) The shortest (i.e., minimum-length) $s - t$ path might have as many as $n - 1$ edges, where n is the number of vertices.
(c) The shortest $s - t$ path must include the minimum-length edge of G.
(d) The shortest $s - t$ path must exclude the maximum-length edge of G.

Q.231 What will be the path from A-H if BFS is used in the following graph?



(a) ABCDEFGH
(b) ACBDEFHG
(c) ACBDFEGH
(d) ACDBEFHG

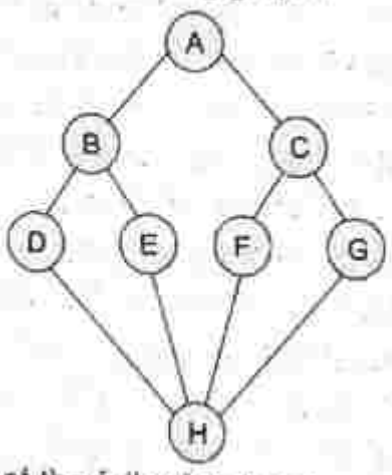
Q.232 Consider the following statement regarding Kruskal and Prim's Algorithm for generating minimum spanning tree:

(a) While generating MST, it is possible to obtain forest during execution of the algorithm in Kruskal's algorithm.
(b) The time complexity for generating MST is O(E log E) for Kruskal's algorithm.
(c) The time complexity for generating MST is O(E) for Prim's algorithm.
(d) While generating MST, it is possible to obtain forest during execution of the algorithm in Prim's algorithm.

Q.233 Which of the following problem(s) can be solved through BFS?

(a) Verifying given graph connected or not.
(b) Finding number of connected components in a graph.
(c) Checking given graph contain cycle or not.
(d) Single Source Shortest Path in given unweighted graph.

Q.234 Consider the following graph:



Which of the following is/are can be the depth first search traversal of the above graph?

(a) ACFHEBDG (b) ACFHGEBD
(c) ABDHFCGE (d) ABDEHFGH

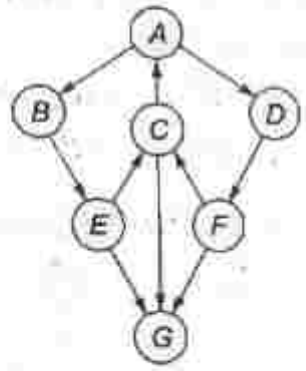
Q.235 Which of the following statements about Dijkstra's shortest-path algorithm are true for input graphs that might have some negative edge lengths?

(a) It is guaranteed to terminate.
(b) It may or may not correctly compute shortest-path distances (from a given source vertex to all other vertices), depending on the graph.
(c) It may or may not terminate (depending on the graph).
(d) It is guaranteed to correctly compute shortest-path distances (from a given source vertex to all other vertices).

Q.236 Suppose we run Prim's algorithm and Kruskal's algorithm on a graph G and the two algorithms produce minimum cost spanning trees T_p and T_k respectively. Then, which of the following is/are true?

(a) T_p must be identical to T_k.
(b) If e is a minimum cost edge in G, e belongs to both T_p and T_k.
(c) If T_p is different from T_k, some pair of edges in G have the same weight.
(d) If e is a maximum cost edge in G, e belongs to neither T_p nor T_k.

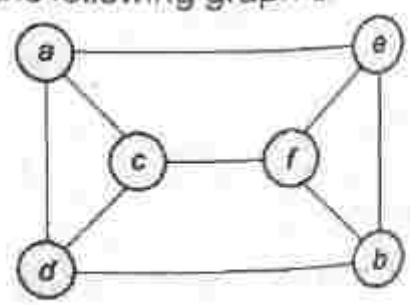
Q.237 Consider the following graph G:



Perform a DFS on G starting at vertex A and selection of adjacent vertex in DFS decided by the lexicographical order. In graph G, B and D are adjacent to A. First it selects B, because B comes first in lexicographical order. Find number of cross edges when the given DFS is performed on G?

(a) 1 (b) 2
(c) 3 (d) 4

Q.238 Consider the following graph G:



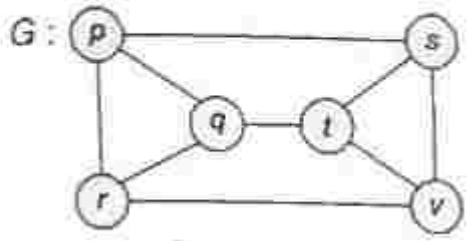
Modified BFS traversal on G applied as follows:

- Assume starting vertex is 'a'.
- At any level (breadth), vertices are visited in alphabetic order.
- In above graph G, after visiting vertex 'a', the order of vertices visited in G are *c*, *d* and *e*.
- It works same as BFS except the restriction on order of vertices visited.

Which of the following is not a cross edge during given BFS traversal on G?

(a) (d, c) (b) (d, b)
(c) (e, b) (d) (e, f)

Q.239 Consider the following graph *G*:



Modified DFS on *G* applied as follows:

- Starting vertex '*P*'.
- Vertex is visited based on alphabetic order.
- Vertices are visited in order *p*, *q*, *r*, *s*, *t*, *v*.
- It works same as DFS except the visiting order restriction.

What is number of back edges during the above DFS traversal on *G* _____?

Q.240 Consider the following statements:
Which of the following statement(s) is/are true?

(a) Kruskal's algorithm will never produce non-minimal spanning tree.
(b) Kruskal's algorithm can efficiently implemented using the disjoint-set data structure.
(c) Kruskal's algorithm is best suited for the dense graphs than the Prim's algorithm.
(d) Prim's algorithm is best suited for the dense graphs than the Kruskal's algorithm.

Q.241 Which of the following is/are true?

(a) Prim's algorithm can also be used for disconnected graphs.
(b) Kruskal's algorithm can also run on the disconnected graphs.
(c) Prim's algorithm is simpler than Kruskal's algorithm.
(d) In Kruskal's algorithm sort edges are added to MST in decreasing order of their weights.

Q.242 Given pseudo code of Dijkstra's Algorithm:

```
1.  //Initialise single source(G, s)
2.  S = 0
3.  Q = V[G]
4.  While Q! = 0
5.      Do u = extract-min(Q)
6.              S = S union {u}
7.              For each vertex v in adj[u]
8.              Do relax(u, v, w)
```

Then which of the following statement(s) is/are true when while loop in line 4 is changed to while Q > 1?

(a) While loop gets executed for V – 1.
(b) While loop gets executed for V times.
(c) Above algorithm will not give shortest path to every vertex in the graph.
(d) Above algorithm still work because that was just a more stronger condition.

Q.243 Consider the following statements:
Which of the following statement(s) is/are true?

(a) Kruskal's algorithm might produce a non-minimal spanning tree.
(b) Kruskal's algorithm can efficiently implemented using the disjoint-set data structure.
(c) Min priority queue is the most commonly used data structure for implementing Dijkstra's Algorithm.
(d) The maximum number of times the decrease key operation performed in Dijkstra's algorithm will be equal to total number of edges.

Q.244 Let $G = (V, E)$ be a directed graph. Each edge of *G* is represented as (i, j) with length $l[i, j]$. If there is no edge from *i* to *j* then $l[i, j] = \infty$. Assume *n* vertices in *V* and $d^k_{i,j}$ is the length of shortest path from *i* to *j* that does not pass through any vertex in $\{k + 1, k + 2, \ldots, n\}$.

$$d^k_{i,j} = \begin{cases} l[i, j] & \text{if } k = 0 \\ \min\{A, B\} & \text{if } 1 \le k \le n \end{cases}$$

If the above $d^k_{i,j}$ computed recursively to find all pairs shortest path, identify *A* and *B* respectively?

(a) $d^{k-1}_{i,j}$ and $d^{j-1}_{i,j} + d^{j-1}_{k,j}$
(b) $d^{k-1}_{i,j}$ and $d^{k-1}_{i,k} + d^{k-1}_{k,j}$
(c) $d^k_{i,j}$ and $d^k_{i,k} + d^k_{k,j}$
(d) $d^k_{i,j}$ and $d^k_{i,k} + d^k_{j,k}$

Q.245 Consider the following adjacency matrix 'D' for represented the directed graph with distances (costs) between every pair of vertices.

$$D = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 2 \\ 1 & 0 & 2 \\ 4 & 1 & 0 \end{bmatrix} \end{array}$$

If Warshall's algorithm is applied on 'D' to compute the shortest distances then following resultant matrix is obtained.

$$D^* = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 2 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} \end{array}$$

If the edge from B to C in the given directed graph is removed then how many shortest distances will be changed in D^*?

Q.246 In the Floyd-Warshall Algorithm for all pair shortest path. The matrix D represents

(a) The shortest path of length i in between a the vertices.

(b) The shortest path of the first i vertices amon themselves.

(c) Intermediate paths of length i in the shorte path between all pairs of vertices.

(d) None of the above

Answers Algorithms

1. (d) 2. (a) 3. (b) 4. (c) 5. (a) 6. (d) 7. (a)
8. (c) 9. (c) 10. (a) 11. (b) 12. (d) 13. (d) 14. (b)
15. (c) 16. (b) 17. (c) 18. (d) 19. (d) 20. (a) 21. (b)
22. (d) 23. (d) 24. (a) 25. (a) 26. (d) 27. (a) 28. (a)
29. (c) 30. (a) 31. (d) 32. (c) 33. (d) 34. (b) 35. (a)
36. (c) 37. (d) 38. (b) 39. (c) 40. (a) 41. (b) 42. (a)
43. (a, b, d) 44. (a, c) 45. (c, d) 46. (c) 47. (b) 48. (c) 49. (c)
50. (a) 51. (b) 52. (a) 53. (a) 54. (a) 55. (a) 56. (a)
57. (d) 58. (b) 59. (c) 60. (c) 61. (a) 62. (d) 63. (d)
64. (a) 65. (d) 66. (c) 67. (d) 68. (b) 69. (d) 70. (a)
71. (c) 72. (b) 73. (b) 74. (a) 75. (b) 76. (a) 77. (c)
78. (b) 79. (a) 80. (b) 81. (a) 82. (a) 83. (b) 84. (c)
85. (d) 86. (a) 87. (d) 88. (c) 89. (a) 90. (a) 91. (a, c)
92. (b) 93. (c) 94. (c) 95. (a) 96. (d) 97. (b) 98. (c)
99. (a) 100. (b) 101. (b) 102. (c) 103. (b) 104. (b) 105. (89)
106. (a) 107. (d) 108. (b) 109. (b) 110. (b) 111. (d) 112. (d)
113. (16) 114. (d) 115. (a) 116. (c) 117. (c) 118. (c) 119. (8)
120. (c) 121. (d) 122. (b) 123. (720) 124. (b) 125. (b) 126. (d)
127. (b) 128. (a) 129. (d) 130. (b) 131. (a) 132. (c) 133. (c)
134. (d) 135. (a, d) 136. (a, b) 137. (b) 138. (a) 139. (d) 140. (d)
141. (c) 142. (d) 143. (d) 144. (b) 145. (17) 146. (a) 147. (a)
148. (b) 149. (d) 150. (a) 151. (b) 152. (c) 153. (d) 154. (c)
155. (d) 156. (b) 157. (d) 158. (d) 159. (c) 160. (b) 161. (a)
162. (b) 163. (125) 164. (d) 165. (25) 166. (c) 167. (14) 168. (b)
169. (b) 170. (c) 171. (c) 172. (a) 173. (c) 174. (3) 175. (b)
176. (c) 177. (b) 178. (c) 179. (b) 180. (b) 181. (c) 182. (125)
183. (a) 184. (a) 185. (a) 186. (a) 187. (b) 188. (b) 189. (b)
190. (a) 191. (b) 192. (b) 193. (c) 194. (b) 195. (a) 196. (c)
197. (a) 198. (a) 199. (b, c d) 200. (d) 201. (b) 202. (c) 203. (d)
204. (128) 205. (a) 206. (c) 207. (6) 208. (b) 209. (c) 210. (64)
211. (d) 212. (d) 213. (d) 214. (d) 215. (b) 216. (d) 217. (b)
218. (b) 219. (b) 220. (a) 221. (c) 222. (c) 223. (a) 224. (c)
225. (15) 226. (a) 227. (d) 228. (c) 229. (a) 230. (a, b, c, d) 231. (a, b, c)
232. (a, b) 233. (a, b, c, d) 234. (a, b, c) 235. (a, b) 236. (c) 237. (b) 238. (b)
239. (4) 240. (a, b, d) 241. (b) 242. (a, c) 243. (b, c, d) 244. (b) 245. (1)
246. (d)

Explanations Algorithms

1. (d)

Big O notation gives worst case limit for a given problem. Also find out the least upper bound of problem.

2. (a)

$2^{\sqrt{n}}$ and $n^{\log n}$ grows exponentially which have growth rate greater than any polynomial.

3. (b)

In the best case **quick sort** algorithm makes $O(n \log (n))$ comparisons.

So $1000 \times \log (1000) = 9000$ comparisons, which takes 100 sec.

To sort 100 names a minimum of $100 (\log 100) = 600$ comparisons are needed.

This takes $\frac{100 \times 600}{9000} = 6.7$ sec.

4. (c)

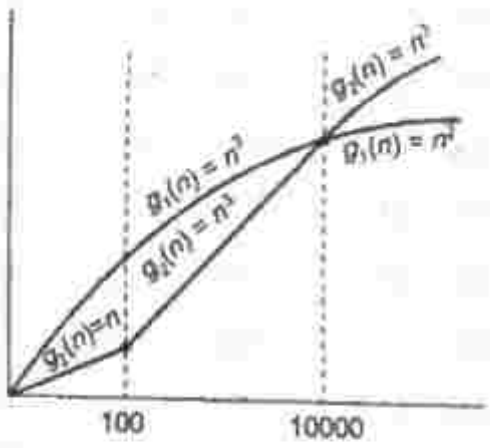
If $f = O(h)$

and $g = O(k)$

Then, $(f + g) = O(n + k)$

and $(f \times g) = O(nk)$

5. (a)



Therefore;

$n^2 \leq n^3$ for $N \geq 10000$

$g_1(n) = O(g_2(n))$

6. (d)

$$f(n) = 3n^{\sqrt{n}}$$

$$g(n) = 2^{\sqrt{n}\log_2 n}$$

$$h(n) = n!$$

$$f(n) = 3n^{\sqrt{n}}$$

$$g(n) = n^{\sqrt{n}\log_2 n}$$

$$h(n) = n(n-1)(n-2)\ldots$$

$$\equiv n^n$$

Here $f(n)$ is $O(g(n))$

$f(n)$ is $O(h(n))$

If $f(n)$ is $O(g(n))$

$f(n)$ is $O(g(n))$ and $g(n)$ is $0 (f(n))$

7. (a)

1. $\frac{1}{2}(n^2) = \omega(n)$

$$\frac{1}{2}(n^2) \geq C_1(n)$$

So true always

2. $\frac{1}{2}(n^2) = \omega(n^2)$

$$\frac{1}{2}(n^2) \geq C_1(n^2)$$

So false, since $\Omega(n^2)$ is true.

8. (c)

$$f(n) = O(g(n))$$

$$\Rightarrow \quad f(n) \leq c_1 \cdot g(n)$$

$$\Rightarrow \quad g(n) \geq \frac{1}{c_1} \cdot f(n)$$

$$\Rightarrow \quad g(n) \geq c_2 \cdot f(n)$$

$$\Rightarrow \quad g(n) = \Omega(f(n))$$

9. (c)

1. $an = O(n^2)$ for $a \geq 0$
   True
2. $an^2 = O(n^2)$ for $a > 0$
   False, big-Oh needed
3. $an^2 \neq O(n^2)$ for $a > 0$
   True, big-Oh needed

So, 1 and 3 are correct.

10. (a)

In this question total possibility is four but only two possibility will give solution and other two possibilities will not give solutions.

$$g(n) = O(f(n))$$

In case of $0 \le n < 100$ & & $0 \le n < 10{,}000$ (it is also not possible)

$g(n) = \Theta(f(n))$ in case of otherwise

From the above two equations the conclusion is $O(f(n))$ is same as $O(g(n))$ and $g(n)$ and $f(n)$ will be $O(n^2)$.

11. (b)

$f(n) = O(g(n))$

Then, $g(n) = \omega(f(n))$

12. (d)

It takes not more than $\frac{n}{2}$ comparison to check whether the given number n is prime or not.

13. (d)

$f(n) = 2^n \Rightarrow f(n) = O(2^n)$

$g(n) = n! \Rightarrow g(n) = O(n!)$

$h(n) = n^{\log n} \Rightarrow h(n) = O(n^{\log n})$

[$n < n^2$ means n grows more slowly than n^2]

The asymptotic order of function is as follows:

$1 < \log\log n < \log n < n^{\varepsilon} < n^c < n^{\log n} < c^n < n^n < c^{c^n} < n!$

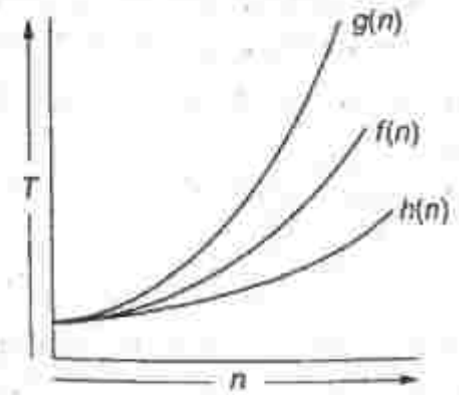
Where $0 < \varepsilon < 1 < c < n$

$\Rightarrow n^{\log n} < c^n < n!$

$\Rightarrow n^{\log n} < 2^n < n!$

Assume $c = 2$

$h(n) < f(n) < g(n)$



$h(n) = O(f(n)) \Rightarrow h(n) \le k.f(n)$ is true

$g(n) = \Omega(f(n)) \Rightarrow g(n) \ge k.f(n)$ is also true

14. (b)

$f(n) = n^2 \log n$

$g(n) = n(\log n)^{10}$

$n(\log n)^{10} \le n^2 \log n$

$\therefore \quad g(n) = O(f(n))$

whereas

$f(n) \ne O(g(n))$ because

$n^2 \log n \not\le n(\log n)^{10}$

15. (c)

$3n + 4n^2 + 5n \log n = O(n^2)$

$n + \log n + \log\log n = O(n)$

$10 + n + n \log n + \log n = O(n \log n)$

$10 + 10000 + 100000 = O(1)$

16. (b)

Unrestricted we of goto statement is harmful because it makes more difficult to verifying programs i.e., use of goto can results in unstructured code and there can be blocks with multiple entry and exit which can cause difficulty which debugging of program.

17. (c)

$$f(n) = \sum_{i=0}^{99} \sum_{j=0}^{n-1} \left(\sum_{k=0}^{j-1} 1 \right) = O(n^2)$$

18. (d)

Let the increment of j is $2^0, 2^1, \ldots, 2^i$ for some value of i so, according to the question for while loop; $2^i \le n$ or $i \le \log_2 n$.

One extra comparison required for the termination of while loop.

So, total number of comparisons

$= i + 1 = \lfloor \log_2 n \rfloor + 1$

19. (d)

The above code runs total number of times

$= N + (N-1) + (N-2) + \ldots + 1 + 0$

$= \frac{N * (N+1)}{2}$

$= \frac{1}{2} * N^2 + \frac{1}{2} * N = O(N^2)$ times

20. (a)

Time complexity is $O(\log n)$ to count number of occurrences of i.

21. (b)

$$\sum_{i=1}^{n} \sum_{j=i+1}^{n} \left(\frac{1}{k} \right) = \frac{1}{k} \sum_{i=1}^{n} \sum_{j=i+1}^{n} (1)$$

$$= \frac{1}{k} \sum_{i=1}^{n} [1 + 1 + 1 + \ldots + n - (i+1) + 1 \text{ times}]$$

$$= \frac{1}{k} \sum_{i=1}^{n} [n - i]$$

$$= \frac{1}{k}\left[n\sum_{i=1}^{n}(1) - \sum_{i=1}^{n}(i)\right] = \frac{1}{k}\left[n \cdot n - \frac{n(n+1)}{2}\right]$$

$$= \frac{1}{k}\left[n^2 - \frac{n^2+n}{2}\right] = \frac{1}{2k}\left[n^2 - n\right] = O(n^2)$$

22. (d)

$n^k < a^n < e^n < n! < n^n$

n^n will take maximum asymptotic time.

$n! = O(n^n)$

$e < n \quad \therefore e^n < n^n$

$k < n \quad \therefore n^k < a^n$

23. (d)

We have to find smallest x such that $N/2 \wedge X\ N$

$$X = \log(N)$$

So, O(log N) is time complexity.

24. (a)

Recurrence relation of function F

$$F(n) = 0 \text{ if } n = 0$$

$$F(n) = F(n-1) + 1,\ n > 0$$

Time complexity = $O(n)$

25. (a)

```
ing count = 0, N; ....O(1)
for (i = 0; i < N * 2; i++) .....O(N)
{
    for(j = 0; j < i/2; i++) .....O(N/3)
    {
        for(k = 0, k < j * j; k++).....O((n * n)/3)
        {
            count ++;
        }
    }
}
```

Here, we using the for loop so that

$O\left(n * \left(\frac{n}{3}\right) * (n * n)/3\right)$ so that $O(n^4)$.

26. (d)

When ever the value of 'sin n' value is -1, then $g(n) = 1$. [sin n value ranges from -1 to $+1$].

Hence $f(n) = O(g(n))$ is violated.

Hence I is false.

And when ever the value of 'sin n' is $+1$, then $g(n) = n^2$

Hence $f(n) = \Omega(g(n))$ is violated. Hence II is false.

27. (a)

In this programme first for loop will run n times and second for loop also run n times, because our array is sorted then second loop will not run and the total time complexity for sorted order will be $O(n)$.

28. (a)

$k = n$

for $i = 0$

while loop will run for ($\log_2 N$) times as in every iteration k becomes half.

Now, k becomes 0 and while loop becomes false.

for $i = 1$

As $k = 0$, while will run 1 times.

for $i = 2$,

Same 1 times.

.....

.....

for $i = n$, 1 time

$T(n) = (\log n) + 1 + 1 + 1 + \ldots..$ (n + times 1)

$T(n) = O(n)$

29. (c)

A has sorting time faster than B, generally but not always because quick sort in worst case running time than merge sort an heap sort. Again bubble sort, insertion sort has $O(n)$ time complexity in best case, where in other case has $O(n^2)$ complexity. So, we cannot say array A sort any array faster than B.

So, option (c) is correct.

30. (a)

$f(n) * \log_2(f(n)^c) = O(g(n) * \log_2 g(n))$.

$\Rightarrow \quad C * f(n) * \log_2(f(n)) = O(g(n) * \log_2 g(n))$.

$\Rightarrow \quad f(n) * \log_2(f(n)) = O(g(n) * \log_2 g(n))$.

Even we can check if it always true or not.

(i) $\frac{1}{n^2} = O\left(\frac{1}{n}\right) \Rightarrow$

True in this (f, g are non- decreasing functions so, i should not include this example, but even if we use, it is true).

(ii) $n = O(n^2) \Rightarrow$

Even true for this.

So, always true.

31. (d)

For layer value of n, put $n = 2^{64}$

(a) $\left(\sqrt{\log n}\right)^{\log^2 n} = 8^{64\times 64}$

$= 2^{3\times 64\times 64} = 2^{12288}$

(b) $(\log n)^{\sqrt{\log n}} = 64^{\sqrt{8}} = 2^{6\sqrt{8}} = 2^{16.97}$

(c) $n^{10} = 2640$

(d) $2^{2^{\sqrt{\log\log n}}} = 2^{2^{\sqrt{8}}} = 2^{7.10}$

So, option (d) is slowest.

32. (c)

In this programme first for loop will run n times and second for loop also run n times, because our array is reverse sorted then second loop will also run and the total time complexity for reverse sorted order will be $O(n^2)$.

33. (d)

This loop iterates as follows:

$i = 2$

$i = 2 \wedge 2$

$i = (2 \wedge 2) \wedge 2$

.....

.....

$i = 2 \wedge (2 \wedge k) <= n$

$k = \log\log n$

So, complexity = $O(\log\log n)$

34. (b)

$$T(n) = 4.T\left(\frac{n}{2}\right) + n^2 + 1$$

$$= \Theta(n^2 \log n)$$

Using Master's theorem

$$n^{\log_b a} = n^{\log_2 4} = 2$$

$$n^2 = n^2$$

$\therefore$ $O(n^2 \log n)$

Case 2 is applied.

So, option (b) is correct.

35. (a)

Recurrence relation is

$$T(n) = T\left(\sqrt{n}\right) + C$$

where, C = Constant

So, time complexity is $O(\log\log n)$.

36. (c)

Let's see the nature of $\frac{e^n}{n+2^n}$

$$\Rightarrow \quad \frac{e^n}{n+2^n} = e^{\log(e^n) - \log(n+2^n)}$$

We find that, $\lim_{x\to\infty} \log(e^n) - \log(n+2^n)$

$$\Rightarrow \quad \lim_{x\to\infty} 0.7n - \log n$$

$$\Rightarrow \quad \lim_{x\to\infty} n * \lim_{x\to\infty}\left(0.7 - \frac{\log n}{n}\right)$$ is infinite as n approaches to infinite.

$$\Rightarrow \quad \frac{e^n}{n*2^n} = \infty$$

$$\Rightarrow \quad e^n = \omega(n*2^n)$$

37. (d)

$\sqrt{n} = n^{1/2} = n^{0.5}$

$2^{\sqrt{\log_2 n}}$

$n^{1.5}$

$n^{1.6}$

10^n

$2^{\sqrt{\log n}} \le n^{0.5} \le n^{1.5} \le n^{1.6} \le 10^n$

38. (b)

In the i^{th} iteration, the inner loop run $O(n/i)$ times, and therefore, the overall complexity is

$O\left(n + \frac{n}{1} + \frac{n}{2} + ... + 1\right)$ and this is $O(n \log n)$.

39. (c)

X = Sum of cubes of {1, 2, 3, ...n}

$$X = \frac{n^2(n+1)^2}{4}$$

1. $X = \theta(n^4)$ True

$$C_1 \cdot (n^4) <= \frac{n^2(n+1)^2}{4} <= C_2 \cdot (n^4)$$

2. $X = \theta(n^5)$ False

$$C_1 \cdot (n^5) <= \frac{n^2(n+1)^2}{4} \le C_2(n^5)$$ not possible

3. $X = O(n^5)$ True $\frac{n^2(n+1)^2}{4} <= C \cdot n^5$

4. $X = \Omega(n^3)$ True $\frac{n^2(n+1)^2}{4} >= C(n^3)$

40. (a)

Let, $T(m, n)$ be the total number of steps.

So, $T(m, 0) = 0$, $T(m, n) = T(n, m \bmod n)$ on average

$$T_n = \frac{1}{n}\sum_{0 \le k \le n} T(k, n)$$

$$T_n = 1 + \frac{1}{n}(T_0\, T_1 + \ldots + T_{n-1})$$

$$T_n = S_n$$

$$S_n = 1 + \frac{1}{n}(S_0\, S_1 + \ldots + S_{n-1})$$

$$S_n = 1 + \frac{1}{n+1}(S_0\, S_1 + \ldots + S_n)$$

$$= 1 + \frac{1}{n+1}(n(S_{n-1}) + S_n)$$

$$= 1 + \frac{1}{n+1} = S_n + \frac{1}{n+1}$$

So, $T_n = \Theta(\log_2 n) + O(1)$

$T = \Theta(\log_2 n)$

41. (b)

We need to calculate upper bound of this series summations.

$$S = 1^k + 2^k + 3^k + \ldots\ldots n^k$$

for $k = 1$, $S = \frac{n*(n+1)}{2} = O(n^2)$

for $k = 2$, $S = \frac{n*(n+1)*(2n+1)}{6} = O(n^3)$

for $k = 3$, $S = \left(\frac{n*(n+1)}{2}\right)^2 = O(n^4)$

So, in this question answer will be $O(n^{k+1})$.

Another solution

Here, inner loop is running for P times which is equal to i^k.

So, total time complexity for all $i (1 \le i <= n)$ will be,

$$T(n) = 1k + 2k + 3k + \ldots\ldots n^k$$
$$= (n+1)k = O(n+1)^k)$$

42. (a)

In this question total possibility is four but only two possibility will give solution and other two possibilities will not give solutions.

$g(n) = O(f(n))$ in case of $0 \le n < 100$

and $0 \le n < 10{,}000$ (it is also not possible)

$g(n) = \Theta(f(n))$ otherwise

From the above two equations the conclusion is $f(n)$ is same as $O(g(m))$ and $g(n)$ and $f(n)$ will be $O(n^2)$.

43. (a, b, d)

Only option (c) is false since $f(n)$ is quadratic function which is obviously greater than $h(n)$ which is linear function of n.

44. (a, c)

Since, both the modules are independent thus both option (a) and (c) would be correct.

45. (c, d)

The time complexity of the above code will be $O(n^3)$.

Thus, (c) and (d) are both true.

46. (c)

$$g(n) = \Omega(n) \qquad g(n) \ge C_1 \cdot n$$
$$f(n) = O(n) \qquad f(n) \le C_2 n$$
$$h(n) = \theta(n) \quad C_3 \cdot n \le h(n) < C_4 \cdot n$$
$$g(n) \cdot f(n) + h(n)$$
$$\ge C \cdot n \qquad \theta(n)$$
$$= \Omega(n)$$

47. (b)

$f_1 = n^4$ (polynomial time)

$f_2 = 4^n$ (ex-polynomial time)

$f_3 = n^{1\,10/37} = n^{3.79}$ (polynomial)

$f_4 = (3.25)^n$ (exponential)

$f_3 < f_1 < f_4 < f_2$

48. (c)

Recurrence relation for the function find:

$$F(n) = 0; \text{ if } n = 0$$
$$= F(n-1) + 1; \text{ if } n > 0$$

Time complexity of $F(n) = O(n)$

49. (c)

for $(i = 1, i < n, i{+}{+})$ -- n time

{

for $(j = 1; j < i * i, j{+}{+})$

$1^2 + 2^2 + 2^3 \ldots\ldots n^2 = \frac{n(n+1)(2n+1)}{6} = O(n^3)$

for $(k = 1; k < j; k{+}{+})$ { $m = m + 1;$ } $\Big]\, O(n)$

Total $= O(n) \times O(n^3) = O(n^4)$

50. (a)

I. $\theta(n^2 \log n)$ II. $\theta(n^3)$
III. $1 + 0 = 1$ IV. $\theta(1)$
V. $\theta(n^{2.2})$

51. (b)

Let $n = 2^m$

$$T(2^m) = T\left(2\left(\frac{m}{2}\right)\right)+1$$

Let, $T(2^m) = S(m)$

Then, $S(m) = 2S\left(\frac{m}{2}\right)+1$

Using Master's theorem

$S(m) = \Theta(m) = \Theta(\log n)$ since $n = 2^m$

Now, $T(n) = T(2^m) = S(m) = \Theta(\log n)$

Therefore, option (b) is correct.

52. (a)

$$T(1) = 1$$
$$T(n) = 2T(n-1) + n \quad n \geq 2$$
$$T(2) = 2T(1) + 2 = 2.1 + 2 = 4$$
$$T(3) = 2T(2) + 3 = 2.4 + 3 = 11$$
$$T(4) = 2T(3) + 4 = 2.11 + 4 = 26$$
$$\vdots$$
$$T(n-1) = 2T(n-2) + n = 2^n - (n-1) - 2$$

So, $T(n) = 2^{n+1} - n - 2$

53. (a)

$$T(n) = \sqrt{2}\cdot T\left(\frac{n}{2}\right)+\sqrt{n} \quad \ldots(i)$$
$$= \sqrt{2}\left[\sqrt{2}\cdot T\left(\frac{n}{4}\right)+\sqrt{\frac{n}{2}}\right]+\sqrt{n}$$
$$= \left(\sqrt{2}\right)^2\cdot T\left(\frac{n}{2^2}\right)+2\sqrt{n} \quad \ldots(ii)$$
$$= \left(\sqrt{2}\right)^3\cdot T\left(\frac{n}{2^3}\right)+3\sqrt{n} \quad \ldots(iii)$$
$$\vdots$$
$$= \left(\sqrt{2}\right)^{2k}\cdot T\left(\frac{n}{2^{2k}}\right)+2k\sqrt{n}$$
$$= \left(\sqrt{2}\right)^{\log n}\cdot 1+\log n\cdot\sqrt{n} \quad [\because \text{Put } 2k = \log n]$$
$$= \sqrt{n}+\log n\cdot\sqrt{n}$$
$$= \sqrt{n}(\log n+1)$$

54. (a)

Since, with every iteration of Binary Search, data set to be searched is divided into half of the original/earlier data set. Also, same constant time is lapsed during each iteration. So, recurrence relation for Binary Search would be:

$T(n) = T(n/2) + k$, where k is a constant

- $T(n)$: Size of earlier data set
- $T(n/2)$: Size of new data set
- k: Time lapsed during each iteration

55. (a)

$$T(n) = T(n-1) + \frac{1}{n}$$
$$\vdots$$
$$T(n) = T(1) + \frac{1}{n} + \frac{1}{n-1} + \ldots\ldots + \frac{1}{2} + \frac{1}{1}$$
$$T(n) = 1 + \log(n)$$
$$\Rightarrow O(\log n)$$

56. (a)

$$T(n) = T\left(\frac{n}{2}\right)+\sqrt{n}$$

Put, $n = 2^K$

$$T(2^K) = T\left(\frac{2^K}{2}\right)+\sqrt{2^K}$$
$$S(K) = S\left(\frac{K}{2}\right)+2^{K/2}$$
$$S(K) = O(2^{K/2})$$

Put, $K = \log_2 n$

So, $T(n) = O(\sqrt{n})$

57. (d)

A *B*-tree of order *M* is a tree having structural property:

1. All leaves are at same depth.
2. Root is either leaf or has children between 2 to *M*.
3. All non-leaf nodes have between $\left\lceil\frac{M}{2}\right\rceil$ to *M*.

58. (b)

$$T(n) = 4T\left(\frac{n}{2}\right) + n^2$$

By applying Master theorem:

$$n^{\log_2 4} = n^2$$

So, $T(n) = n^{\log_2 4} \log n$

$= \Theta(n^2 \log n)$

59. (c)

$$T(n) = 2T\left(\frac{n}{2}\right) + \sqrt{n}$$

By substitution method $T(n) = \Theta(n)$.

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \sqrt{n} \quad ...(1)$$

$$= 2\left[2 \cdot T\left(\frac{n}{4}\right) + \sqrt{\frac{n}{2}}\right] + \sqrt{n}$$

$$= (2)^2 \cdot T\left(\frac{n}{2^2}\right) + \sqrt{n}(1+\sqrt{2}) \quad ...(2)$$

$$= (2)^3 \cdot T\left(\frac{n}{2^3}\right) + \sqrt{n}(1+\sqrt{2}+2) \quad ...(3)$$

$$\vdots$$

$$= (2)^k \cdot T\left(\frac{n}{2^k}\right) + \sqrt{n}\left(\sum_{i=0}^{k-1}(\sqrt{2})^i\right) = \Theta(n)$$

60. (c)

$$T(n) = mT\left(\frac{n}{2}\right) + an^2$$

Since, $m \geq 5$

So by applying Master Theorem:

$$T(n) = O(n^{\log_2 n})$$

61. (a)

$$T(n) = T(n-1) + \frac{1}{n}$$

$$= T(n-2) + \frac{1}{n-1} + \frac{1}{n}$$

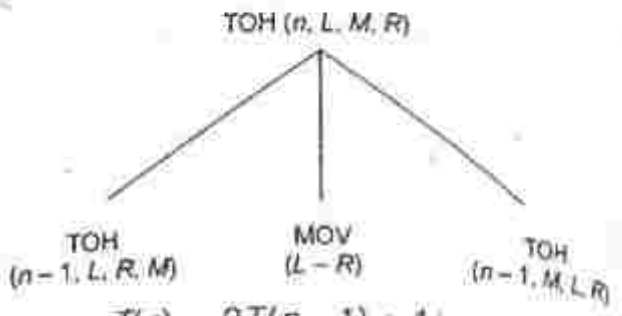
$$= T(n-3) + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}$$

$$= T(n) + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{(n-1)} + \frac{1}{n}$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n-1} + \frac{1}{n}$$

$T(n) = O(\log n)$

62. (d)



$\Rightarrow$ $T(n) = 2T(n-1) + 1$

63. (d)

Since, we have a square root term, considering only perfect squares and those which are multiple of 2 as that can take care of log.

$T(2) = 1$ //Assume

$T(2^2) = T(2) + 1 = 2$

$T(2^{2^2}) = T(4) + 1 = 3$

$T(2^{2^3}) = T(16) + 1 = 4$

$T(2^{2^4}) = T(256) + 1 = 5$

So, we are getting

$T(n) = \log \log n + 1$

$\Rightarrow$ $T(n) = O(\log \log n)$

64. (a)

In recurrence relation, floor, ceil, addition of constant terms etc. are not required for solving as we need asymptotic solution. So,

$$T(n) = T\left(\frac{n}{2} - 1\right) + T\left(n - \frac{n}{2}\right) + \theta(n)$$

$$\Rightarrow \quad T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + \theta(n)$$

$$\Rightarrow \quad T(n) = 2T\left(\frac{n}{2}\right) + \theta(n)$$

$\Rightarrow$ $T(n) = \theta(n \log n)$

[by using 3rd case of Master theorem]

65. (d)

$$T(n) = 7T\left(\frac{n}{2}\right) + an^2$$

Using Master's method: $a = 7$, $b = 2$, $f(n) = an^2$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{an^2}{n^{\log_2 7}} = \frac{an^2}{n^{2.81}} = an^{0.19}$$

So, $U(n) = O(1)$

$T(n) = n^{\log_b a} U(n) = n^{\log_2 7} O(1)$

$T(n) = O(n^{2.81})$

66. (c)

$T(0) = T(1) = 1$

$T(n) = 2T\left(\frac{n}{2}\right) + n$

$T(n)$ be computed as follows:

$$T(n) = n\sum_{i=0}^{\log n}\left(\frac{2}{2}\right)^i = n\sum_{i=0}^{\log n}(1)^i = \Theta(n \log n)$$

[$T(n)$ can also proved by Master Theorem]
If $T(n) = \Theta(n \log n)$ then it is also $O(n \log n)$ and $O(n^2)$ but it is not $\Omega(n^2)$.

67. (d)

$T(n) = T(n-1) + T(n-1) + 1$

[Recursive $(n-1) = T(n-1)$]

$= 2T(n-1) + 1$

$= O(2^n)$ [Using substitution method]

68. (b)

Take $n = 2^m$ then,

$$T(2^m) = T(2^{m/2}) + \log m$$

Solve using Master's theorem

$$S(m) = O(\log^2 m) = T(2^m)$$

Put $m = \log n \Rightarrow T(n) = (\log \log n)^2$

So, option (b) is correct.

69. (d)

$$T(n) = T\left(\frac{n}{2}\right) + 2$$

by substitution method,

$$T\left(\frac{n}{2}\right) = T\left(\frac{n}{4}\right) + 2$$

$$T\left(\frac{n}{4}\right) = T\left(\frac{n}{8}\right) + 2$$

$$T(n) = T\left(\frac{n}{8}\right) + 2 + 2 + 2$$

$$T(n) = T\left(\frac{n}{(2)^3}\right) + 3 * 2$$

$$T(n) = T\left(\frac{n}{(2^k)}\right) + k * 2$$

Let $2k = n$ i.e., $k = \log n$

$$T(n) = T\left(\frac{n}{(2^{\log n})}\right) + \log n * 2$$

$$T(n) = T(1) + 2 * \log n$$

Since, $T(1) = 1$

$$T(n) = 2 * \log_2 n + 1$$

70. (a)

$$T(x) = \sqrt{n}T(\sqrt{n}) + 100n$$

Let $n = 2^k \Rightarrow k = \log n$

$$T(2k) = (2)^{k/2}\, T(2^{k/2}) + 100 \times 2^k$$

(dividing by 2^k)

$$\frac{T(2^k)}{2^k} = \frac{2^{k/2}T(2^{k/2})}{2^k} + 100$$

$$\frac{T(2^k)}{2^k} = \frac{T(2^{k/2})}{2^{k/2}} + 100$$

Let, $y(k) = \frac{T(2^k)}{2^k}$, then ...(i)

$$y(k) = y\left(\frac{k}{2}\right) + 100$$

Now, applying Master theorem $[a = b^k = 1]$

$$y(k) = \log k$$

From (i) we also know that $T(2^k) = 2^k y(k)$, then

$$T(2^k) = 2k \log k = T(n)$$

$$= n \log \log n$$

(because $n = 2^k$ and $k \log n$)

Finally, $T(n) = \theta(n \log \log n)$

71. (c)

Using Master Theorem:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$n^{\log_b a} = n^{\log_2 2} = n^1$

$f(n) = n^2 \log n$

If $n^{\log_b a} > f(n)$ then

$T(n) = O(n^{\log_b a})$

$\Rightarrow n^1 > n^2$

$\Rightarrow T(n) = O(n^1)$

So option (c) is correct.

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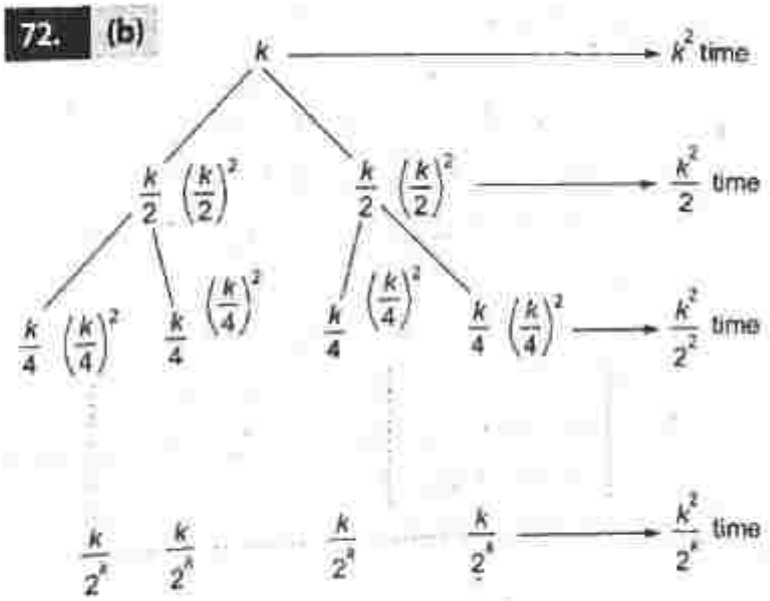
72. (b)



$$T(n) = \frac{k^2}{2^0} + \frac{k^2}{2^1} + \frac{k^2}{2^2} + \cdots \frac{k^2}{2^k}$$

$$T(k) = k^2\left[1 + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^k}\right]$$

$$= k^2 \times O(1) = O(k^2)$$

73. (b)

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \log n$$

$\because$ $\log n = O(n^{\log_b a - \epsilon})$ for some $\varepsilon > 0$

$$T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 2}) = \Theta(n)$$

$\therefore$ $T(n) = \Theta(n)$

So option (b) can be solved using master theorem.

(a) $T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \frac{n}{\log n}$

Try case 1: $\frac{n}{\log n} \neq O(n^{1-\varepsilon})$ for some $\varepsilon > 0$,

because $\frac{1}{\log n} \neq O(n^{-\varepsilon})$

Try case 2: $\frac{n}{\log n} \neq \Theta(n^1)$, because $\frac{n}{\log n} = o(n)$

Try case 3: $\frac{n}{\log n} \neq \Omega(n^{1+\varepsilon})$ for some $\varepsilon > 0$,

because $\frac{n}{\log n} = o(n)$

All cases failed using master theorem for option (a), similarly option (c) is also not applicable.

74. (a)

Tower of Hanoi function works as:

```
TOH (n, x, y, z)
{
    if (n > = 1)
    {
        1. TOH (n, x, z, y)   //...T(n – 1)
        2. move x → y   //...T(1)
        3. TOH (n, z, y, x)   //...T(n – 1)
    }
}
```

Thus, TOH function is recursively called and step 2 is just move which takes O(1) time so as it is asked order of work.

Option (a) is correct.

75. (b)

$$T(n) = 8T\left(\frac{n}{2}\right) + qn \quad \text{if } n > 1$$

$$= p \quad \text{if } n = 1$$

p and q are constants.

Now, according to Master theorem,

$$T(m) = aT\left(\frac{n}{b}\right) + n^c \quad \text{if } n > 1$$

$$= d \quad \text{if } n = 1$$

and if $\log_b a > c$ then,

$$T(n) = \theta(\log_b a)$$

In this recurrence relation,

$$a = 8,\ b = 2,\ c = 1$$

$\therefore$ $\log_b a = \log_2 8 = 3$

Here, $\log_b a > c$

So, order will $T(n) = \theta(n^{\log_b a}) = \theta(n^3)$

76. (a)

For higher values of n, $\frac{n}{2} >> 47$. So, we can ignore 47. Now $T(n)$ will be

$$T(n) = 3T\left(\frac{n}{2}\right) + 2n^2 + 10n - \frac{1}{2}$$

$$= 3T\left(\frac{n}{2}\right) + O(n^2)$$

Apply Master theorem, it is case 3 of Master theorem $T(n) = O(n^2)$

So, option (a) correct.

77. (c)

$$T(n) = T\left(\frac{n}{2}\right) + \frac{n^2}{2} + n$$

$$T\left(\frac{n}{2}\right) = \left[T\left(\frac{n}{2^2}\right) + \left(\frac{n}{2}\right)^2 \frac{1}{2} + \frac{n}{2}\right] + \frac{n^2}{2} + n$$

$$T\left(\frac{n}{2^2}\right) = T\left(\frac{n}{2^3}\right) + \left(\frac{n}{2^2}\right)^2 \frac{1}{2} + \frac{n}{2^2} + \left(\frac{n^2}{2}\right) + \frac{1}{2} + \frac{n}{2} + \frac{n^2}{2} + n$$

$$= T\left(\frac{n}{2^{k+1}}\right) + \left[\left(\frac{n}{2^k}\right)^2 \frac{1}{2} + \ldots + \left(\frac{n}{2}\right)^2 \frac{1}{2} + (n^2)\left(\frac{1}{2}\right)\right] + \left[\frac{n}{2^k} + \frac{n}{2^{k+1}} + \ldots + \frac{n}{2} + \frac{n}{2^0}\right]$$

Terminating condition: $\frac{n}{2^{k+1}} = 1$

$$n = 2k + 1$$
$$k = \log_2 n - 1$$

$$= T(1) + \frac{n^2}{2}\left[\frac{1}{2^{2k}} + \frac{1}{2^{2(k-1)}} + \ldots + \frac{1}{2^2} + \frac{1}{2^0}\right] + n\left[\frac{1}{2^k} + \frac{1}{2^{k-1}} + \ldots + \frac{1}{2^1} + \frac{1}{2^0}\right]$$

$$\Rightarrow 1 + \frac{n^2}{2}\left[\frac{1-\left(\frac{1}{4}\right)^k}{1-\frac{1}{4}}\right] + n\left[\frac{1-\left(\frac{1}{2}\right)^k}{1-\frac{1}{2}}\right]$$

$\Rightarrow O(n^2)$ ignoring other as it is less than 1.

78. (b)

$$T(n) = T(n-1) + \frac{1}{n} \quad \ldots(1)$$

$$= \left[T(n-2) + \frac{1}{n-1}\right] + \frac{1}{n}$$

$$= T(n-2) + \frac{1}{n-1} + \frac{1}{n} \quad \ldots(2)$$

$$= T(n-3) + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n} \quad \ldots(3)$$

$$= T(1) + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n} \quad \ldots(n-1)$$

$$= T(1) - 1 + \left[1 + \frac{1}{2} + \frac{1}{3} + \ldots\ldots + \frac{1}{n}\right]$$

$= T(1) - 1 + O(\log n)$

$= O(1) + O(\log n) = O(\log n)$

79. (a)

A function F is asymptotically larger than function g means for large values all numbers greater than some finite number) $f(n) > g(n)$

$$T(n) = 81T\left(\frac{n}{9}\right) + 10n^2$$

This recurrence will asymptotically grow faster than all the other options given.

By applying Master's theorem

(a) $T(n) = 8T\left(\frac{n}{4}\right) + 100n^2$

Recursive part, $n^{\log_b a} = n^{\log_4 8}$ and $f(n) = 100n^2$

Since, $f(n)$ is polynomially (by a factor of some $n^{\epsilon}, \epsilon > 0$) larger, and $f(n)$ is regular, at per Master theorem.

Case 3. $T(n) = \theta(f(n)) = \theta(n^2)$

(b) $T(n) = 81T\left(\frac{n}{9}\right) + 10n^2$

$\therefore$ $T(n) = O(n^2 \log n)$

Recursive part,

$n^{\log_{10} a} = n^{\log_9 81} = n^2$ and $f(n) = 10n^2$

Since, $f(n) = \theta(n^{\log_b a})$ as per Master theorem case 2,

$T(n) = \theta(f(n) \log n) = \theta(n^2 \log n)$

(c) $T(n) = 16T\left(\frac{n}{4}\right) + 100(n\log n)^{1.99}$

and $f(n) = 100\,(n \log n)^{1.99}$

Since $f(n)$ is polynomially (by a factor of some $n^{\epsilon}, \epsilon > 0$) smaller, as per Master theorem case 1.

$$T(n) = \theta(n^{\log_b a}) = \theta(n^2)$$

(d) $T(n) = 100T\left(\frac{n}{100}\right) + n\log^2 n$

Recursive part, $n^{\log_b a} = n^{\log_{100} 100} = n$

and $f(n) = n \log^2 n$

Here $f(n)$ and $n^{\log_b a}$ are not asymptotically equal and they don't have a polynomial difference. Any polynomial is n is asymptotically larger than any polynomial in log n). So, Master theorem cannot be applied as such. But we can apply the extended Master theorem case 2, which states that

If $f(n) = \theta(n^{\log_b a} \log^k n), k > -1$,

Then, $T(n) = \theta(n^{\log_b a} \log^{k+1} n)$

Here, $k = 2$ so, we get

$$T(n) = \theta(n \log^3 n)$$

80. (b)

$$T(n) = T(n-1) + \frac{1}{n}$$

$$T(n-1) = T(n-2) + \frac{1}{n-1}$$

$$T(n-2) = T(n-3) + \frac{1}{n-2}$$

$$T(n-3) = T(n-4) + \frac{1}{n-3}$$

$$T(n) = T(n-2) + \frac{1}{n-1} + \frac{1}{n}$$

$$= T(n-3) + \frac{1}{(n-2)} + \frac{1}{(n-1)} + \frac{1}{n}$$

$$T(n) = T(n-4) + \frac{1}{(n-3)} + \frac{1}{(n-2)} + \frac{1}{(n-1)} + \frac{1}{n}$$

$$T(n) = T(n-k) + \frac{1}{(n-(k-1))} + \frac{1}{(n-(k-1))}$$

$$+ \frac{1}{(n-(k-3))} + \frac{1}{(n-(k-4))} + \ldots + \frac{1}{n}$$

$$T(n) = T(n-k) + \frac{1}{(n-k+1)} + \frac{1}{(n-k+2)}$$

$$+ \frac{1}{(n-k+3)} + \ldots + \frac{1}{n-1} + \frac{1}{n}$$

$$T(n) = T(n-k) + \frac{1}{n-k+1} + \frac{1}{n-k+2} + \frac{1}{n-k+3} + \ldots$$

$$\ldots + \frac{1}{n-3} + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}$$

Thing that the one condition is given if $(n > 1)$

So, we can substitute $(k = n - 1)$

$$T(n) = T(n-(n-1)) + \frac{1}{n-(n-1)+1} + \ldots + \frac{1}{n}$$

$$T(n) = T(1) + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \ldots + \frac{1}{n}$$

Assume that $T(1) = 1$

$$T(n) = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$$

$T(n) = O(\log n)$

81. (a)

Take base conditions as $T(0) = 1$ and $T(-1) = 0$.

$T(-1) = 0$

$T(1) = 2T(0) - T(-1) = 2$

$T(2) = 2T(1) - T(0)$

$= 3$ ... and so on

So, option (a) is correct.

82. (a)

$$T(n) = T\left(\frac{2n}{3}\right) + 1$$

By using recursive tree method:

$$T(n)$$
$$\downarrow$$
$$T\left(\frac{2n}{3}\right) = O(1)$$
$$\downarrow$$
$$T\left(\frac{2^2 n}{3^2}\right) = O(1)$$
$$\downarrow$$
$$T\left(\frac{2^3 n}{3^3}\right) = O(1)$$
$$\vdots$$
$$T\left(\frac{2^k n}{3^k}\right) = O(1)$$

So, height of tree is $\log_{(3/2)} n$.

So, $T(n)$ = Height × Work at each level

$= \log_{3/2} n \times O(1)$

$= O(\log_{3/2} n)$

83. (b)

According to Master theorem

$a = 7,\ b = 3,\ k = 2$

So, its case of $a < b^k$

So, $O(N^2)$ will be complexity

So, option (b) is correct.

84. (c)

Two cases: $Q(m, 3)$

(i) $m \geq n$: here $m \geq 3$

For $m = 3, Q(3, 3)$

$= Q(3 - 3, 3) + P$

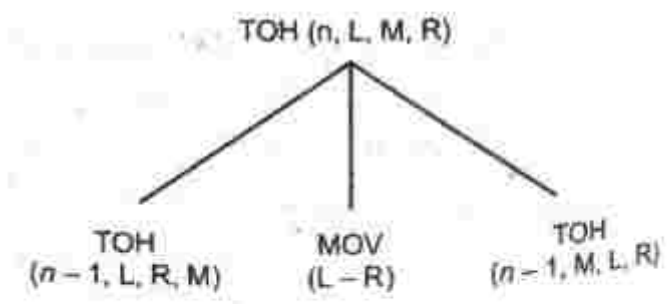
$= Q(0, 3) + P$

$\because$ Since 3 is divide by 3 only 1 time $= 0 + P = P$

So this recursive program generate P, m/n times until $m < n$.

So, $Q(m, 3) = P \times (m \text{ div } 3)$

85. (d)



$\Rightarrow T(n) = 2T(n-1) + 1$

86. (a)

For higher values of n, $\frac{n}{2} >> 47$. So, we can ignore 47. Now $T(n)$ will be

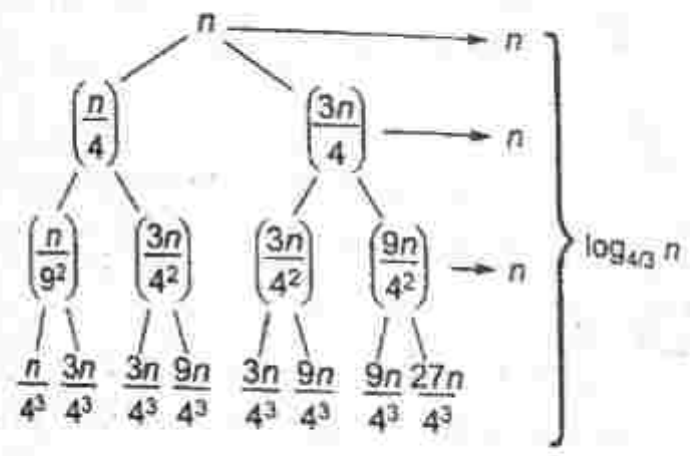
$$T(n) = 3T\left(\frac{n}{2}\right) + 2n^2 + 10n - \frac{1}{2}$$

$$= 3T\left(\frac{n}{2}\right) + O(n^2)$$

Apply Master theorem, it is case 3 of Master theorem $T(n) = O(n^2)$

So, option (a) correct.

87. (d)



$$\Rightarrow \frac{n+3n+3n+9n}{116} = \frac{7n+9n}{16} = \frac{16n}{16} = n$$

$$\Rightarrow n+n+n.....(n * \log_{4/3} n)$$

88. (c)

From Master theorem, case 2

Here, $a = \sqrt{2}$, $b = 2$, $k = 1/2$, $p = 0$

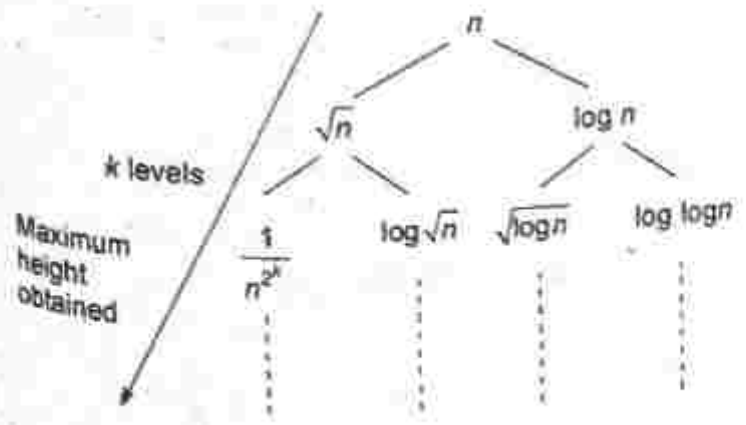
$a = b \wedge k$ which is true.

$p > -1$

So, complexity is theta ($n \wedge$ (log a base to b) $\log \wedge (p+1)n$

$$= \theta(\sqrt{n}(\log n + 1))$$

89. (a)



So, $n^{\frac{1}{2^k}} = 2$

$$\frac{1}{2^k}\log n = 1$$

$$\Rightarrow \log n = 2^k$$

$$k = \log\log n = \theta(\log\log n)$$

90. (a)

$$T(n) = T(\sqrt{n}) + \boxed{T(\log_2 n)} + c$$

$\downarrow$

Smaller with lesser levels

So approximately,

$$T(n) = T(\sqrt{n}) + c$$

Solve using substitution and get log log$n$ levels,

$T(n)$ = $c + c + c +$ ..... log log$n$ times

$= \Omega(\log\log n)$

91. (a, c)

Use substitution method,

$$T(n) = 4\left[4 + \left(\frac{n}{2^2}\right) + \left(\frac{n}{2}\right)^2 \log\frac{n}{2}\right] + n^2\log n$$

$$= 4^2 T\left(\frac{n}{2^2}\right) + n^2\log\frac{n}{2} + n^2\log n$$

In general,

$$T(n) = 4^k T\left(\frac{n}{2^k}\right) + n^2\left[\log n + \log\frac{n}{2} + ... \frac{\log n}{2^{k-1}}\right]$$

$$= \frac{n}{2^k} = 1$$

$$k = \log_2 n$$

$$T(n) = 4^k \cdot T\left(\frac{n}{2^k}\right) + n^2\left[\begin{matrix}\log_2{}^k + \log_2{}^{k-1} \\ + \log_2{}^{k-2} ..... \log_2\end{matrix}\right]$$

$$4\log_2 n\, T(1) + n^2[k + k - 1 + ... 1]$$

$$= n^2 \cdot 1 + n^2\left(\frac{k(k+1)}{2}\right)$$

$$= n^2 + n^2 O((\log n)^2)$$

$$= O(n^2(\log n)^2)$$

It can be solved by extended Master's theorem as well.

92. (b)

Quick sort has two worst cases, when input is in either ascending or descending order, it takes same time $O(n^2)$.

93. (c)

Randomized quick sort worst case time complexity = $O(n^2)$.

94. (c)

Generally, PARTITION process take $O(n)$ time complexity in quick sort.

But here, IMPROVED-PARTITION process take $O(n^2)$ time complexity.

The worst case occurs when the partition process always picks greatest or smallest element as pivot. If we consider above partition strategy where 75th greatest element is selected as pivot, the worst case would occur when the array is already sorted in increasing or decreasing order.

$$T(n) = T(n-1) + O(n^2)$$

The solution of the above recurrence by substitution method is $O(n^3)$.

95. (a)

deletemin( ): Root element is the smallest element. Remove it and swap it with the rightmost leaf node to maintain heap property. Then search for the minimum element out of the remaining elements by looking in the lowermost min (even) level. Let it be at index 'm'. Swap this element at index m with the root. Do this recursively for the subtree rooted at-index m to maintain the min-max property.

deletemax( ): The max element is one out of the two children of root. Find it and then follow the same steps as deletemin( ) operation, this time choosing the max element out of the remaining to be swapped at every stage.

96. (d)

Choosing median element as pivot divide the array into two equal half and time complexity $O(n \log n)$. For 1 and 2 choosing pivot randomly or middle element does not guaranty $O(n \log n)$ time complexity.

97. (b)

Merge sort and Heap sort have $O(n \log n)$ time complexity.

98. (c)

Each comparison will append one item to the existing merge list. In the worst case one needs $m + n - 1$ comparisons which is of order $m + n$.

99. (a)

For this Pseudo code is

(a) Construct a count array such that count[m] = Number of elements with value m //$O(n)$

(b) Convert count to a cumulative count array such that count [m] = Number of elements upto and including m. //$O(k)$

(c) Range Query [a, b] = count [b] – count[$a-1$] //assuming $b >= a$

(d) Total processing time = $O(n + k)$

So, option (a) $O(n + k)$ is correct.

100. (b)

Quick sort will give best case performance if pivot is selected as median of 1st, last and middle element.

101. (b)

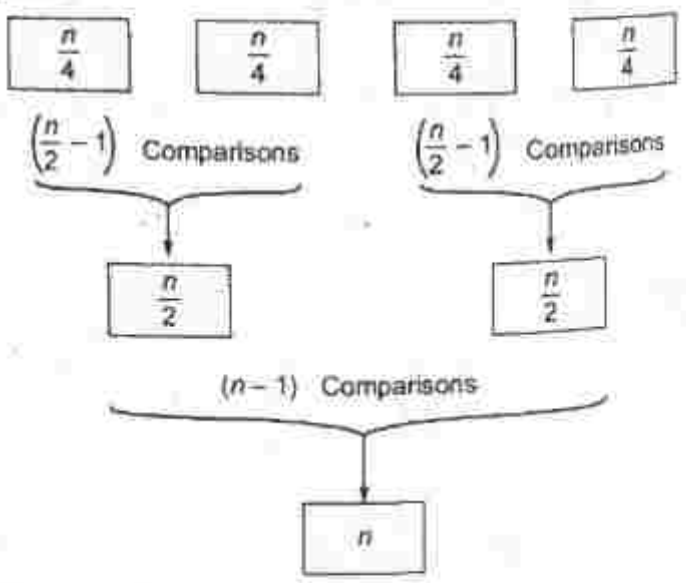
A sorting algorithm is called stable if it keeps element with equal keys in the same relative order in the output as they were in the input.

For example: in the following input the two 4's are indistinguishable. 1, 4a, 3, 4b, 2 and so the output of a stable sorting algorithm must be:

1, 2, 3, 4a, 4b

Bubble sort, merge sort, counting sort, insertion sort are stable sorting algorithms.

102. (c)



Total comparisons = $\left(\frac{n}{2}-1\right)+\left(\frac{n}{2}-1\right)+(n-1)$

$= 2n - 3 = \frac{8}{4}n - 3$

103. (b)

The recurrence tree for merge sort will have height n and $O(n^2)$ work will be done at each level of the recurrence tree (Each level involves n comparisons and a comparison takes $O(n)$ time in worst case). So time complexity of this merge sort will be $O(n^2 \log n)$.

104. (b)

It takes $O(\log n)$ time by using binary search modification.

The idea is to go to the middle element. Check if the difference between middle and next to middle is equal to difference or net, if net then the missing element dies between mid and (mid + 1). If the middle element is equal to $\frac{n}{2}^{th}$ term in arithmetic series (Let n be the number of elements in input array), then missing element lies in right half. Else element lies in left half.

105. (89)

Let X = value N_{C_r} (any value in Pascal's triangle)

Y = number of terms to be computed to determine the value of X.

Then, $Y = 2 * X - 1$

i.e., $X = N_{C_r} = 10_{C_8} = 45$ and $Y = 89$

So, 89 terms computed in worst case.

106. (a)

The n^{th} largest element will be present in one of the leaf nodes and will take $O\left(\frac{n}{2}\right)$ time to find.

Therefore, $O(n)$ is the required complexity.

107. (d)

Each comparison of both the list 'm' and 'n' result one element in sorted array. So in worst case number of comparison will be $m + n - 1$ but for best case min of m and n.

108. (b)

$$T(n) = n + (n+1) + (n+2) + \ldots + (n+k-1)$$
$$= nk + (1 + 2 + \ldots + (k-1))$$
$$= nk + (k-1)\frac{k}{2}$$
$$= O(nk + k^2) = O(k(n+k))$$

109. (b)

We have $\log n$ lists each of size $\frac{n}{\log n}$. And now, if we merge them by 2-ways, then at each level there are : $\frac{\log * n}{\log n} = n$ elements.

And total number of levels are = log (number of sorted list) = log log n

Thus, total time complexity = $n \log \log n$

110. (b)

Merge sort for n letters in string will take $O(n \log n)$ time so for n strings its $O(n^2 \log n)$.

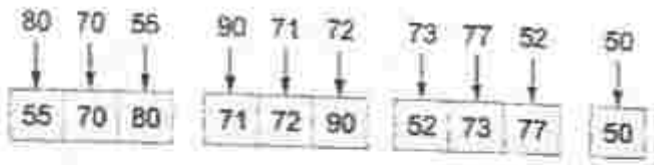
By 2-ways merge question is specifying that sorting is done using merge sort by dividing recursively the array into size of 2 and performing bottom up merging.

So, option (b) is correct.

111. (d)

3 way merge sort will combine 3 sub arrays at a time.

After division elements becomes.



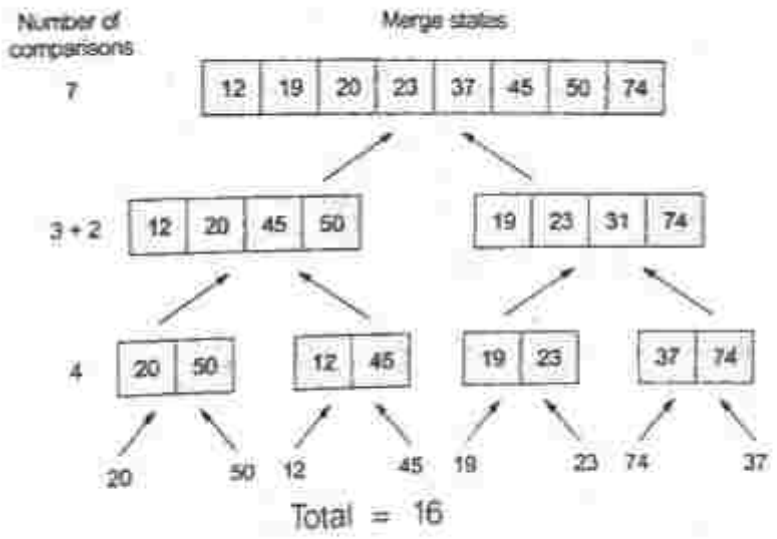
112. (d)

Time complexity of Bubble Sort

Best case = $O(n)$

Worst case = $O(n^2)$

113. (16)



Total = 16

114. (d)

In Binary search we first compare the given element with middle element of the array. If the element matches with middle element, then return middle element index. Otherwise, go to left half of array or right half of array.

115. (a)

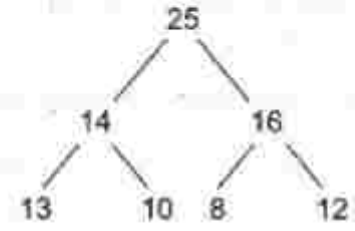
Sorting Algorithm	Worst care complexity (with n inputs)
Merge sort	$O(n \log n)$
Bubble sort	$O(n^2)$
Quick sort	$O(n^2)$
Selection sort	$O(n^2)$

116. (c)

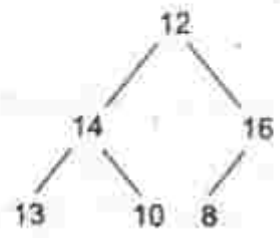
Along with Radix sort two other sorting algo fall in this categories which are counting sort and bucket sort.

117. (c)

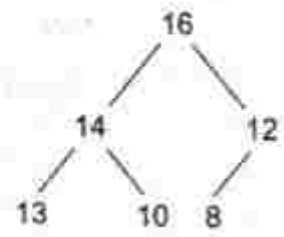
Given max heap is,



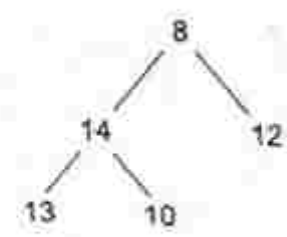
1. Deleting 25,



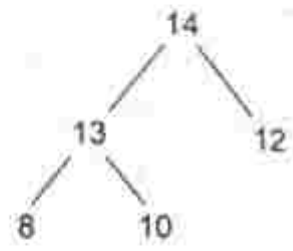
- Apply max heapify on root,



2. Deleting 16,



- Apply max heapify on root,



So the final contents of the array will be 14, 13, 12, 8, 10

Hence option (c) is correct.

118. (c)

To find the first string, algorithm is similar to MaxMin algorithm using divide and conquer.

$$T(n) = 2T\left(\frac{n}{2}\right) + O(n)$$

($O(n)$ is to find the first string from any two subproblems)

$$T(n) = O(n \log n)$$

119. (8)

Input sequence : 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

1st iteration: 8, 9, 7, 6, 5, 4, 3, 2, 1, 0

2nd iteration: 7, 8, 9, 6, 5, 4, 3, 2, 1, 0

3rd iteration: 6, 7, 8, 9, 5, 4, 3, 2, 1, 0

4th iteration: 5, 6, 7, 8, 9, 4, 3, 2, 1, 0

5th iteration: 4, 5, 6, 7, 8, 9, 3, 2, 1, 0

6th iteration: 3, 4, 5, 6, 7, 8, 9, 2, 1, 0

7th iteration: 2, 3, 4, 5, 6, 7, 8, 9,, 1, 0

6th iteration: 1, 2, 3, 4, 5, 6, 7, 8, 9, 0

Hence, answer should be 8th.

120. (c)

If, $n = 2^k$

let at level 1, L_1 and L_2 is merged.

An item in L_1 is compared maximum times when all the items in L_2 is less than item in L_1.

Stay at leaf level k, an item is compared only once when merging.

At level $k-1$, an item is compared twice when merging.

At level 2, it is compared $\frac{n}{2}$ times and after merging we get the sorted array.

Total comparison of an item

$$= \frac{n}{2} + \frac{n}{(2)^2} + \ldots + \frac{n}{(2)^k}$$

where, $$\frac{n}{2^k} = 1 = n^*\left(1-\frac{1}{2^k}\right)$$

$$= n-\frac{n}{2^k} = (n-1)$$

121. (d)

Quick sort takes $O(n^2)$ time to sort the input (ascending order) into ascending order by using divide and conquer.

Insertion sort take $O(n)$ time to sort input (ascending order) into ascending order since need one comparison every time and no swapping.

Merge sort take $O(n \log n)$ and bubble sort can take $O(n)$ time.

122. (b)

Traverse the array element wise and for each element, compare its value with its left adjacent and right adjacent element (if it exists). If the condition is satisfied, print the elements.

123. (720)

We have to choose first element as pivot.

Here 60 is given as first element.

After first pass, the pivot element goes to it's exact location.

Here 60 goes to 6th place.

All the elements less than 60 go to left of 60 and all the elements greater than 60 go to right of 60.

After 1st pass

55 15 7 12 35 60 80 90 95

5! 3!

$\Rightarrow 5! \times 3! \Rightarrow 720$ possible arrangements.

124. (b)

After one iteration the greatest element reaches its correct position i.e., last position, out of all the options only in option (b), the largest element is not present in the last position.

In option (a) and (c) the largest element that is 9 is present in the last position but in case of the (b) option it is not the case, therefore (b) option is not a possible outcome after one iteration of bubble sort. Hence, option (b) is correct.

125. (b)

In the best case quick sort algorithm makes $n \log (n)$ comparisons.

So $1000 \times \log (1000) = 9000$ comparisons, which takes 100 sec.

To sort 100 names a minimum of 100 (log 100) = 600 comparisons are needed.

This takes $100 \times \frac{600}{9000} = 6.7$ sec.

126. (d)

Bubble sort can be improved by using inner loop while required line swapping element while required. Otherwise inner loop should skip.

(a) Doing both way sort is not an efficient solution, as if we do left to right increasing order sorting then right to left decreasing order.

(b) Through quick sort has time complexity ($O(n \log n)$ in average case which is better than bubble sort $O(n^2)$, but that is not needed forgetting better version of bubble sort. Actually here we are using quick sort and then again bubble sort, which is not improving time or space complexity of bubble sort.

(c) Swapping in separate array will increase space complexity of bubble sort.

So, answer will be (d).

127. (b)

The relation $T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + n$

The pivot element is selected in such a way that it will divide the array into $\frac{1}{4}^{th}$ and $\frac{3}{4}^{th}$ always solving this relation give $\Theta(n \log n)$.

128. (a)

The usual $O(n^2)$ implementation of insertion sort to sort an array uses binary search to identify the position, the worst case running time will remain $O(n^2)$. The time required to insert an element x into a binary search tree is bounded by a constant times the number of comparisons made between x and elements already in the tree. Thus we can measure time in terms of the number of comparisons made. In worst case implementation of insertion sort takes $O(n^2)$. So in the worst case adding n elements to a binary search tree could require $O(n^2)$ time. In expected case means element are sorted in increasing order adding n elements to a binary search tree could require $O(n \log n)$ time.

129. (d)

Binary Search: $T(n) = T\left(\frac{n}{2}\right) + 1$[Avg, Worst]

Merge Sort: $T(n) = 2T\left(\frac{n}{2}\right) + kn$[Avg, Best, Worst]

Quick Sort: $T(n) = T(n-1) + n - 1$ [Worst Case]

Max Min: $T(n) = 2T\left(\frac{n}{2}\right) + C$ [Best, Avg, Worst]

130. (b)

The recurrences relation for such an implementation can be given by

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

By using Master theorem, we get

$a = 2$, $b = 2$ and $k = 2$

$$T(n) = \theta(n^2)$$

The most appropriate option is as we do not have $\theta(n^2)$ the most appropriate option is $O(n^2 \log n)$

131. (a)

We can find the middle element of the heap, by simply accessing the $\frac{n}{2}^{th}$ element of the array which can be done in O(1) time. Further, that element is replaced by the last element of the array in O(1) time. Now, perform the heapify operation which will take O(log n) time.

$$\text{Time taken} = O(1) + O(1) + O(\log n)$$
$$= O(\log n)$$

132. (c)

Quick sort is not a stable sorting algorithm the popularity acts as satellite data and the order of satellite data is preserved only by stable sorting algorithms.

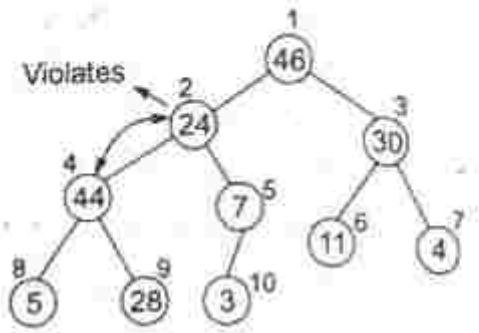
133. (c)

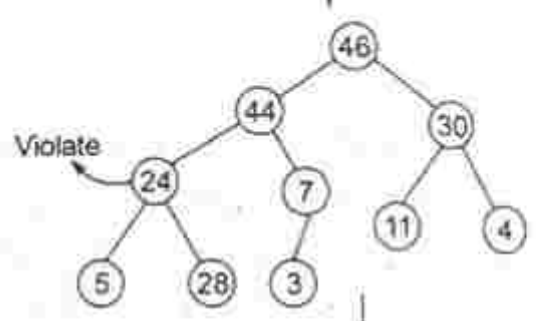
Heap sort is a comparison based sorting algorithm and has time complexity O(n log n) in the average case. Heap sort in an inplace algorithm as it need O(1) of auxiliary space. Heap sort uses heap and operations on heap can change the relative order of items with the same key values.

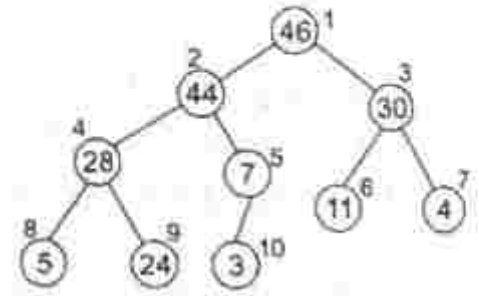
Therefore, heap sort is not a stable sort.

134. (d)

In max-heap element at each node is smaller than or equal to the element at its parent node. On applying the heapify procedure on item at position 2, it will be in position 9 a shown below.







So, it will on 9 position.

135. (a, d)

It is again just memorizing standard results

The expected result is $\theta(n \log n)$, and the worst case result is $\theta(n^2)$.

136. (a, b)

We can use binary search.

137. (b)

Merge sort : $n \log n$

Quick sort : n^*n

Bubble sort : n^*n

Insertion sort : n^*n

Answer is merge sort.

138. (a)

A sorting algo is said to be stable if it preserves the order of equal items.

There are algorithms which are stable naturally like bubble sort and insertion sort. And there are

algorithm which need modification to exhibit this stable sorting property, example of such sorting algo is quick sort.
So, answer is (a) option.

139. (d)

Preorder is root, left, right.

- So, option (b) can't be a valid output. Since, 5 being the root and [312] a left sub-tree and [6487] being a right sub tree. But, $4 < 5$, so not possible.
- In option (a), considering 5 as root, [786] becomes the right sub tree. Further, 7 being the root 6 can't be a part of right sub-tree.
- Similarly, we can shown (c) can't be a valid output.
- So, (d) can be the only possible valid output.

140. (d)

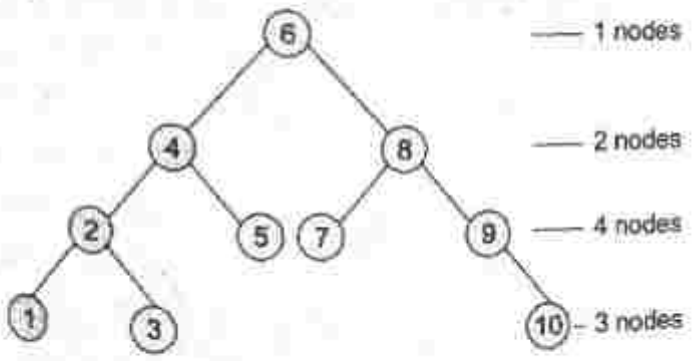
Each comparison of both the list 'm' and 'n' result one element in sorted array. So in worst case number of comparison will be $m + n - 1$ but for best case min of m and n.

141. (c)

The number of comparisons required in the worst case is $O(m + n)$.

142. (d)

Balanced binary search tree look like (since array is sorted):



So average successful time

$$= \sum_{i=1}^{n} \frac{i.(\text{Number of nodes})}{n} \quad [i = \text{level number}]$$

$$= \frac{[1+2\times2+3\times4+4\times3]}{n=10}$$

$$= \frac{[1+4+12+12]}{10} = \frac{29}{10} = 2.9$$

143. (d)

Applying bubble sort on given input:

Step 1	8	22 ↔	7	9	31	5	13
Step 2	8	7	22 ↔	9	31	5	13
Step 3	8	7	9	22	31 ↔	5	13
Step 4	8	7	9	22	5	31 ↔	13
Step 5	8 ↔	7	9	22	5	13	31
Step 6	7	8	9	22 ↔	5	13	31
Step 7	7	8	9	5	22 ↔	13	31
Step 8	7	8	9 ↔	5	13	22	31
Step 9	7	8 ↔	5	9	13	22	31
Step 10	7 ↔	5	8	9	13	22	31
	5	7	8	9	13	22	31

144. (b)

A sorting algorithm is called stable if it keeps elements with equal keys in the same relative order in the output as they were in the input.
For **example** in the following input the two 4's are indistinguishable. 1, 4_a, 3, 4_b, 2 and so the output of a stable sorting algorithm must be:

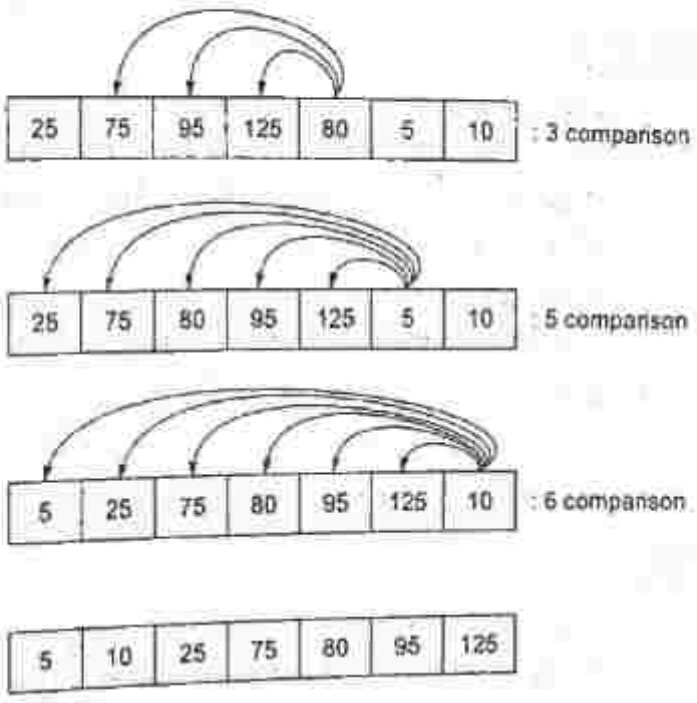
$$1, 2, 3, 4_a, 4_b$$

Bubble sort, merge sort, counting sort, insertion sort are stable sorting algorithms.

145. (17)

$$17 = 3 + 5 + 6 + 3$$

3 comparison, one each for 75, 95, 125



146. (a)

Size sorted list has size 2.
So, number of comparison is between min (2, 2) to $(2 + 2 - 1) = 3$

So, 2, 3, 3 are number of comparisons.
So average number of comparison

$$= \frac{2+3+3}{3} = \frac{8}{3}$$

147. (a)

A sorting algorithm is said to be stable sorting algorithm if two objects/records with equal primary key appears in the same order in the sorted list as in the original unsorted list.

148. (b)

The recurrence tree for merge sort will have height n and $O(n^2)$ work will be done at each level of the recurrence tree (Each level involves n comparisons and a comparison takes $O(n)$ time in worst case). So time complexity of this merge sort will be $O(n^2 \log n)$.

149. (d)

The worst case time complexity of algo given are:
Insertion sort = $\Theta(n^2)$
Merge sort = $\Theta(n \log n)$
Quick sort = $\Theta(n^2)$

150. (a)

Comparison based sorting algorithm has complexity of $O(n \log n)$
Example: Merge sort, Quick sort.

151. (b)

Time complexity of given algorithm is depends on m_1 and m_2 module. How we use, if we use module inside module then the time complexity will be = $O(h(n) * g(n))$.

If we use m_1 and m_2 in parallel, then we can add time complexity of both modules.

Then, $= O(h(n) + g(n))$

$= O(\max(h(n) + g(n)))$

Option (b), (c) and (d) can be possible.

152. (c)

The **minimum depth** occurs for the path that always takes the smaller portion of the split i.e., the nodes that take α proportion of work from the parent node. The first node in the path (after the root) gets α proportion of work (the size of data processed by this node is $n \cdot \alpha$), the second one gets α^2 so on. The recursion bottoms out when the size of data becomes 1. Assuming the recursion ends at level m, we have

$$\Rightarrow \quad \alpha^m \cdot n = 1$$

$$\Rightarrow \quad m = \frac{\log\left(\frac{1}{n}\right)}{\log(\alpha)} = \frac{\log(n)}{\log(\alpha)}$$

Note that similar argument can be used for showing that the maximum depth is

$$= \frac{-\log(n)}{\log(1-\alpha)}$$

153. (d)

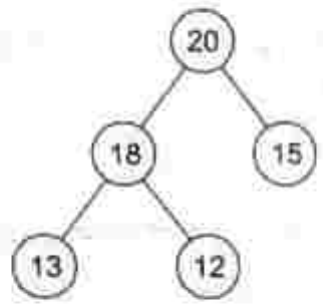
Radix sort complexity is $O(Wn)$ for n keys which are integers of word size (w).

154. (c)

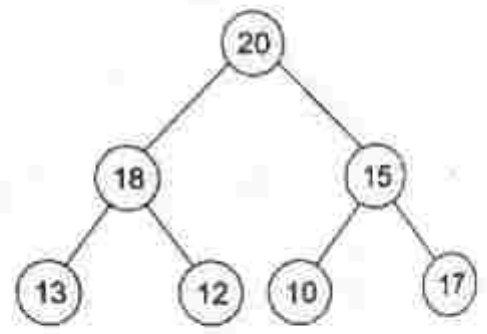
Write each integer in base n, that is each x can be represented as (x_1, x_2) with $x = 1 + x_1 + x_2 * N$. Now you can sort it twice with counting or radix sort, radix sort has time complexity is $O(n)$.

155. (d)

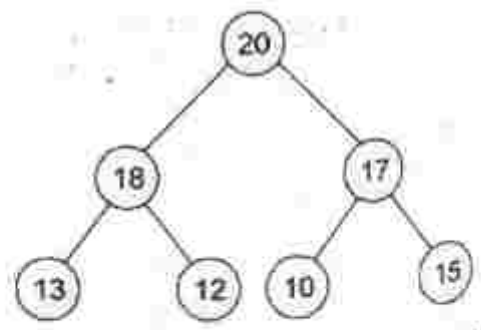
Given level order of max-heap tree:
20, 18, 15, 13, 12



Now, 10 and 15 are inserted, tree will be:



Now there is a need to apply max-heapify. Hence



Level-order traversal is 20, 18, 17, 13, 12, 10, 15

156. (b)

Given array: 10, 4, 7, 23, 67, 12, 5
After 1st pass: 4, 10, 7, 23, 67, 12, 5
After 2nd pass: 4, 7, 10, 23, 67, 12, 5
After 3rd pass: 4, 7, 10, 23, 67, 12, 5
Hence, option (b) is correct.

157. (d)

Lets consider the efficient one, it will be done either by using Dijkstra or Bellman, as we know that Dijkstra is better. So, now in Dijkstra's we keep on choosing the node or vertex with minimum weight, this operation can be mapped to extract minimum operation of min heap which takes O(log n) or O(log V) times and then we are relaxing the edges i.e., O(E log V).
So, it is best option we have.
So, answer is (d) option.

158. (d)

Kruskal's algorithm time complexity
$= O(e \log v)$
$= O(m \log n)$

159. (c)

Heap and priority queue are very neat data structures allowing:

- Add an element to heap with an associated priority.
- Remove the element from the heap or priority queue that has the highest priority, and return it.
- Peak at the element with highest priority without removing it.

A simple way to implement a heap or priority queue data type is to keep a list of elements, and search through the list for the highest priority which gives O(n) time to implement Dijkstra's shortest path algorithm on unweighted graph.

160. (b)

Time complexity:
O($E + V \log E$) using Fibonacci heap.

$$|V| = n$$

$$|E| = \frac{n(n+1)}{2} = \frac{n^2+n}{2}$$

$$T(n) = O\left(\frac{n^2+n}{2} + n\log n\right)$$

$$= O(n^2)$$

OR

Time complexity : O($E + V^2$) using array.
Therefore,

$$T(n) = O\left(n^2 + \frac{n^2-n}{2}\right) = O(n^2)$$

161. (a)

Both dynamic and Greedy algorithm find optimal substructure in the problem but only dynamic programming uses the bottom-up approach, whereas greedy algo uses top-down approach.
So, S_1 is true so, option (a) is correct.

162. (b)

Number of spanning tree possible with n-node $= n^{n-2}$
Given, $n = 7$
Total number of spanning trees $= 7^5$

163. (125)

For a complete graph k_n
Number of spanning tree $= n^{n-2}$
Here, $n = 5$
So, number of spanning tree $= 5^{5-2} = 5^3 = 125$

164. (d)

w be the minimum weight among all edge weights in an undirected connected graph. e is the specific edge of weight w. It may be possible that another edge in the graph having weight w which had been added to minimum spanning tree and when we add e to minimum spanning tree it form a simple circuit. So we can't include e in every minimum spanning tree.

165. (25)

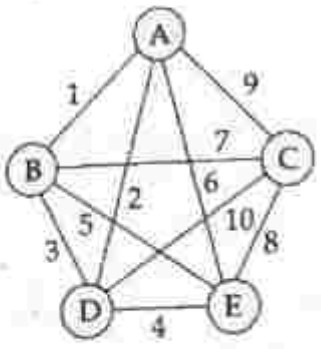
$x \le 11, \quad y \le 6, \quad z \le 8$

166. (c)

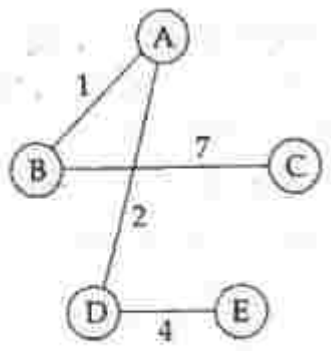
O(m) because the edge weights are sorted already, now you may have check m of the edges in the worst case as intermediate cycles could be formed when adding an edge.
So, option (c) is correct.

167. **(14)**

Consider edges weights are numbered as below:



So minimum cost spanning tree will be:

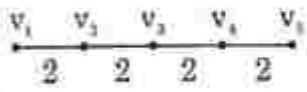


In worst case, cost of MST will be 1 + 2 + 4 + 7 = 14.

168. **(b)**

Given vertex set of G = $\{v_1, v_2, ..., v_n\}$

Weight of an edge = $2|i - j| : (v_i, v_j) \in G$



is MST.

In the case of minimum spanning tree of a graph G we will add upto $(n-1)$ vertices, so the weight of a minimum spanning tree of G

$$= \sum_{i=1}^{n-1} 2|V_i - V_{i-1}|$$

$$= 2\sum_{i=1}^{n-1} |V_i - V_{i-1}|$$

$$= 2\sum_{i=1}^{n-1} |1|$$

$$= 2\,|n-1| = 2(n-1) = 2n-2$$

169. **(b)**

I. If the graph contains a negative weight cycle then no shortest path exist.

II. If shortest path is well defined then it contains at most V-1 edges. Running V-1 iterations of Bellman Ford will find the path.

170. **(c)**

Number of spanning trees of a complete graph k_n is given by Cayley's theorem = n^{n-2}.

171. **(c)**

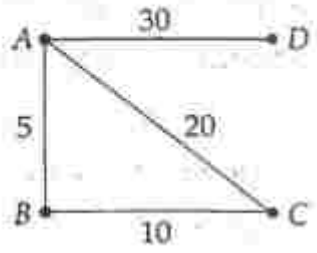
For minimum spanning tree $(n-1)$ edges are enough and graph is complete graph. So each adjacent vertex is connected by an edge of weight 4. So, cost of minimum spanning tree using Prim algo = $(n-1) * 4 = 4n - 4$

172. **(a)**

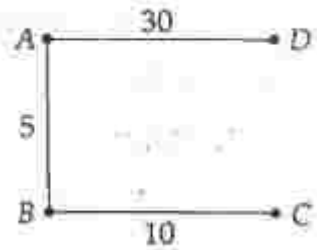
Since edges costs are distinct, so cheapest edge must be present in every minimum spanning tree while expensive edge is may not excluded from every minimum spanning tree.

Statement S_2 is false.

Example:



MST will be:

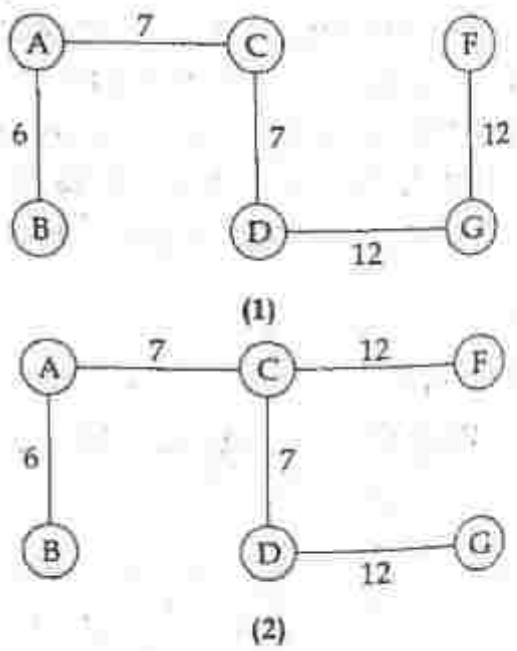


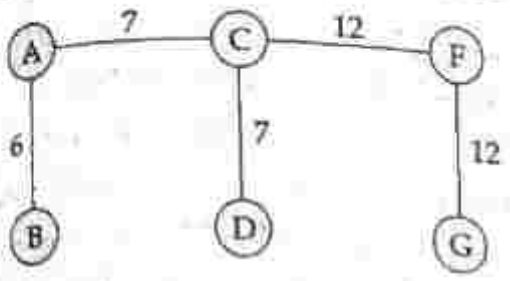
So, most expensive edge is not excluded.

173. **(c)**

Dijkstra's algorithm always terminates after $|E|$ relaxations and $|V| + |E|$ priority queue operations, but may produce incorrect results. So, option (c) is correct.

174. **(3)**





(3)

175. (b)

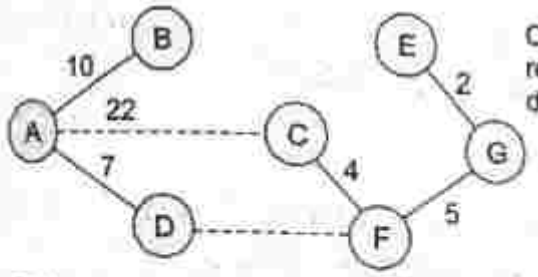
Dijkstra's implementation for single source shortest path

(i) Using binary heap takes
$O((|E|+|V|)\log|V|)$

(ii) Using Fibonacci heap takes
$O(|E|+|V|\log|V|)$

176. (c)

Prim's algorithm having special property which is, while finding minimum cost spanning tree, graph always be connected.



Choice of edges represented by dashed line

Minimum cost spanning tree using Prim's

Now for correct options, we need to check while entry for any edge except for 1st edge only one vertex match from all left side (i.e. visited edges) edges.

(a) (E, G), (C, F) : here for edge (C, F) no vertex match to (E, G). So not possible using Prim's.

(b) (A, D), (A, B), (A, C), (C, F), (G, E) : here for edge (G, E) no vertex match to any edge present left side of it so not possible using Prim's.

(c) (A, B), (A, D), (D, F), (F, G), (G, E), (F, C) not violated rule i.e. always connected in construction of MST.
So True.

177. (b)

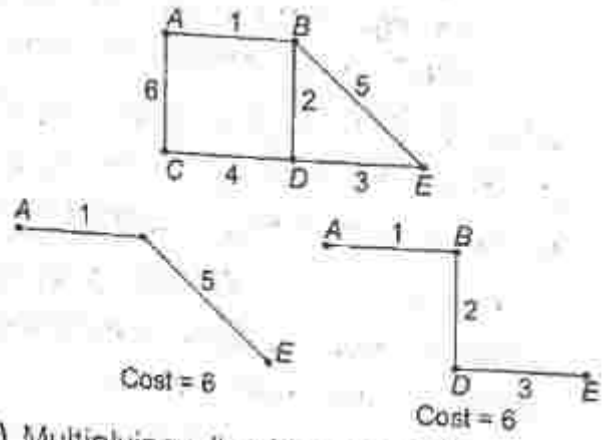
Bellman-Ford shortest path algorithm always finds any negative weighted cycle which is reachable from the source. If the negative weighted cycle is not reachable then, Bellman Ford algorithm not able to find that cycle.

178. (c)

There will be unique minimum cost for any minimum spanning tree in a graph irrespective of algorithm applied, i.e. more than one minimum cost spanning trees are possible but their cost will be same.

179. (b)

(i) Graph can have more than one shortest path between two vertex where all edge weights are distinct.



(ii) Multiplying all edge weight by a positive number should not effect cost of minimum spanning tree.

So option (b) is invalid.

180. (b)

	A	B	C	D	E	F	G	H
	0 Nil	∞	∞	∞	∞	∞	∞	∞
A	–	2 A	15 A	23 A	∞	∞	∞	∞
B	–	–	15 A	23 A	17 B	7 B	∞	∞
F	–	–	13 F	23 A	11 F	–	36 F	y+7 F
Given → E	–	–	13 F	23 A	–	–	36 F	x+11 E

Compare above table with given table

$$x + 11 = 19$$
$$x = 8$$
$$y + 7 \geq x + 11$$
$$y + 7 \geq 8 + 11$$
$$\Rightarrow \quad y \geq 12$$

181. (c)

Sort A by finish time

$S = \{A_1\}$, $J = 1$

For $i = 2$ to n

Do

If $Si \geq Sj$

$S = S \cup \{Ai\}$

$J = i$

Then $\Rightarrow O(n \log n)$

182. **(125)**

According to Cayley's formula for counting spanning trees, for a complete graph k_n.

$T(K_n) = n(\text{pow}(n-2)$ where n = number of vertices

$T(K_5) = 5^{5-2} = 5^3 = 125$

183. **(a)**

1. is true: The Dijkstra's and Bellman Ford algorithms differs in how many times they relax each edge and the order in which they relax edges. Dijkstra's algorithm relaxes each edge exactly once. The Bellman-Ford algorithm relaxes each edge $|V-1|$ times. But both the algorithms will definitely terminate no matter what. Now, on Dijkstra's algo., it always terminates after $|E|$ relaxations and $|V|+|E|$ priority queue operations, but may produce incorrect results in case of negative weight edges.

2. is true: If shortest path is well defined then it cannot include a cycle. Thus, the shortest path contains at most $(V-1)$ edges. Running the usual $(V-1)$ iterations of Bellman-Ford will therefore find that path.

So, option (a) is correct.

184. **(a)**

Adding a constant to every edge weight in a directed graph can change the set of edges that belongs to shortest path tree. Assume unique weights.

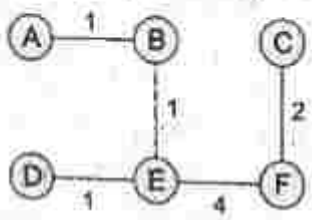
185. **(a)**

Dijkstra algorithm will visit the nodes in the following order: *A, B, C, D, F, E*. So, option (a) is correct.

186. **(a)**

Cost of MST = 9

Cost of maximum spanning tree is 22.



So, option (a) is correct.

187. **(b)**

All pairs shortest path is find using Floyd Warshall algorithm, which is an example of dynamic programming.

- Quick sort and merge sort are examples of "divide and conquer" algorithms.
- MST or minimum weight spanning tree is a tree with a subset of edges such that it connects all vertices and summation of edge weight is minimum Prim's algo and Kruskal's algorithm are used for this, which are examples of greedy approach.
- By using Depth First Search (DFS) one can easily find set of connected components.

188. **(b)**

Number of spanning tree possible with n-node = n^{n-2}.

Given, $n = 7$

Total number of spanning trees = 7^5

189. **(b)**

If we take 33 vertices, then 3 edges are require to connect them in a cyclic graph as in triangle. Similarly, 4 edges require for 4 vertices. And, in general n edges require for n vertices.

190. **(a)**

Bellman-Ford algorithm : $O(nm)$

Kruskal's algorithm : $O(m \log n)$

Floyd-Warshall algorithm : $O(n^3)$

Topological sorting : $O(n+m)$

191. **(b)**

Dijkstra's algorithm solves single source shortest path problem.

Warshall's algorithm finds transitive closure of a given graph.

Prim's algorithm constructs a minimum cost spanning tree for a given weighted graph.

192. **(b)**

The maximum degree of any vertex in simple graph is:

Total edges = $\frac{n(n-1)}{2}$, since number of edges are 'n'.

So, degree will be

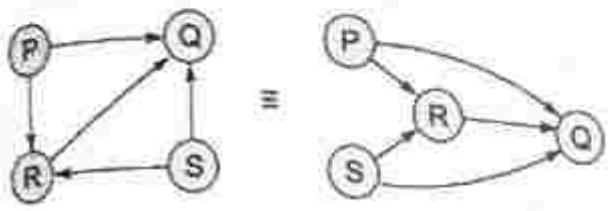
$$\frac{n(n-1)}{2} = e$$

$$\Rightarrow \quad 2e = n(n-1)$$

n = Number of vertex

$(n-1)$ = Degree of each vertex

193. (d)



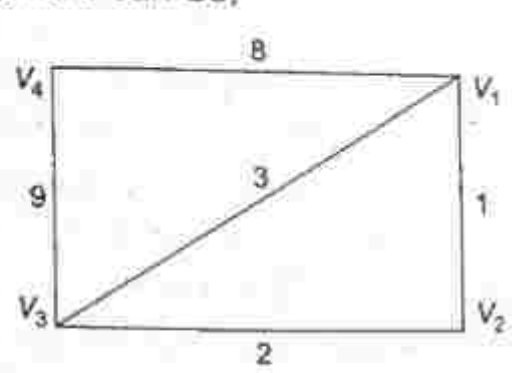
∴ Topological orderings: PSRQ and SPRQ

194. (b)

Bellman-Ford shortest path algorithm always finds any negative weighted cycle which is reachable from the source. If the negative weighted cycle is not reachable then, Bellman Ford algorithm not able to find that cycle.

195. (a)

- Since all the edge weights are unique, hence the minimum spanning tree of the graph will be unique.
- Shortest path between the two vertices need not to be unique. A counter example for the statement can be,



The path from $V_1 \rightarrow V_3$ can be,

I. $V_1 \rightarrow V_2 \rightarrow V_3 : 1 + 2 = 3$

II. $V_1 \rightarrow V_3 : 3$

Hence the path is not unique.

196. (c)

When the edges are already sorted, the problem of finding the Minimum Spanning Tree (MST) reduces to detecting if there is a cycle in the graph or not.

One can take edges one by one and add to MST (if no cycle is formed). Finding a cycle in an undirected graph using DFT or BFT takes $O(v + \varepsilon)$ time.

197. (a)

Bellman-Ford algorithm = $O(|v| \cdot |E|)$

Kruskal's algorithm = $O(|E| \log |E|)$

$= O(|E| \log |v|)$

Floyd-Warshall algorithm = $O(|v|^3)$

Topological sort (DFS) = $O(|v| + |E|)$

198. (a)

A. $V \times O(E \log V) = V \times O(V^2 \log V) = O(V^3 \log V)$

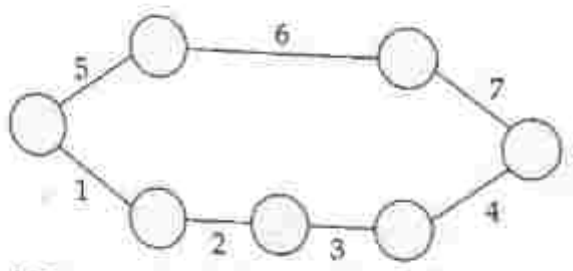
B. $V \times O(EV) = V \cdot O(V^3) = O(V^4)$

C. $O(V^3)$

199. (b, c d)

(a) May give incorrect results for a negative - weighted graph.

(b)



add 100 to all edges, the shortest path changes.

200. (d)

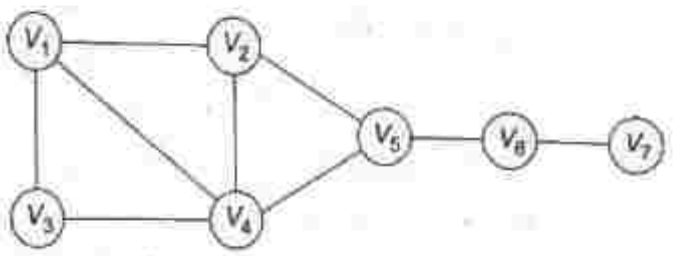
$e_1 + e_2 + e_3 + e_4 = 8$

$^{8+4-1}C_{4-1}$

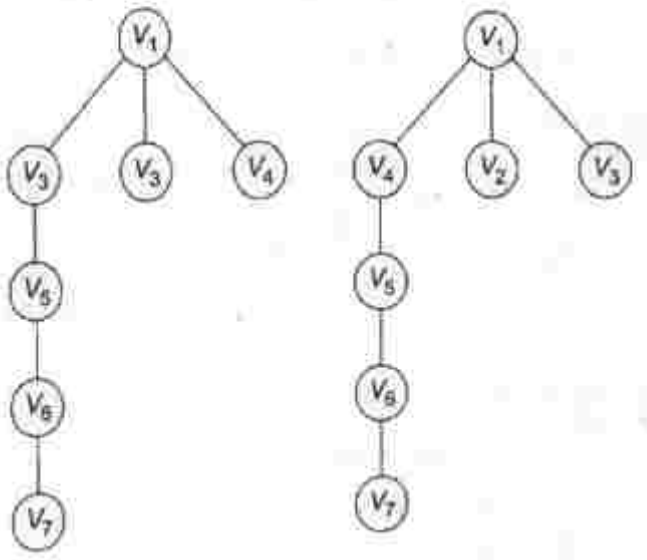
$e_1, e_2, e_3, e_4 \geq 0$

$= {}^{11}C_3 = \frac{11 \times 10 \times 9}{3 \times 2} = 165$

201. (b)



Possible BFS traversals on above graph starting form v_1 is given.



202. (c)

BFS can be used to not only check if a given graph is bipartite, we can also find diameter of graph using BFS.

203. (d)

Number of vertices = n

Degree of each vertex = d

$$\sum_{v \in V} d(v) = 2|e|$$

$$\underbrace{d+d+\ldots+d}_{n \text{ times}} = 2|e|$$

$$n \cdot d = 2|e|$$

$$\Rightarrow \quad |e| = \frac{nd}{2}$$

204. (128)

Since, maximum number of BFS traversal is asked the traversal should start from the root node. Every internal node have two choices to visit the node.

The total number of internal nodes = 7.

So, there are 2^7 ways = 128.

205. (a)

Preorder is same as depth first order.

206. (c)

S_1: A → B → C

Starting with A will get 2 edges

Starting with B will get only 1 edge

Starting with C will get no edge

∴ DFS on directed graph may not give same number of edges.

S_2: Removing all back edges makes the graph as acyclic.

So S_2 is valid statement.

Option (c) is correct.

207. (6)

(A, D); (D, E), (D, H);

(E, C), (E, G); (C, F);

(F, I); (I, B)

The maximum height of tree is 6.

208. (b)

Dijkstra algorithm → $O((E + V) \log V)$

Prim algorithm → $O(E \log V)$

BFS → $O(V + E)$

Floyd-Warshall → $O(V^3)$

Kruskal algorithm → $O(E \log V)$

Bellman Ford → $O(VE)$

So, option (b) is correct.

209. (c)

DFS can be used to find strongly connected components and topological sort.

Hence, option (c) is correct.

210. (64)

As number of undirected graph is = $2^{(n(n-1)/2)}$

211. (d)

In a connected graph, a vertex v is said be an articulation point if by removing that vertex together with its edges the graph become disconnected.

In the given graph there is no articulation point.

212. (d)

Both statement are false because forest is large version of trees. So, there is no bound about number of edges and vertices.

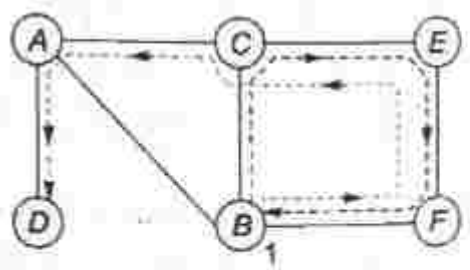
213. (d)

In a graph of strongly connected components, mother vertices are always vertices of source component in component graph. If these exist mother vertex (or vertices), then one of the mother vertices is the last finished vertex in DFS.

1. Do DFS traversal of the given graph. While doing traversal keep track of last finished vertex 'V'. This step takes $O(V + E)$ time.
2. If there exist mother vertex (or vertices), then V must be one (or one of them). Check if V is a mother vertex by doing DFS/BFS from V. This step also takes $O(V + E)$ time.

So, option (d) is correct.

214. (d)



∵ $BCEFAD$ is correct depth first search order.

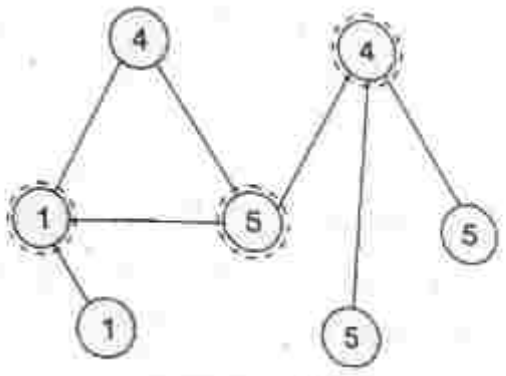
215. (b)

We cannot compute the shortest path in an undirected graph with the help of DFS. BFS is used for it.

So, option (b) is the answer.

216. (d)

An articulation point (or cut vertex) is that vertex removing which (along with its edges) disconnects the graph.



Therefore number of articulation points is 3.

217. (b)

Depth first traversal can be used to detect a cycle in a graph. DFS for a connected graph produces a tree. There is a cycle in a graph only if there is back edge present in the graph. A back edge is an edge that is from a node to itself (self-loop) or one of its ancestor in the tree produced by DFS. For a disconnected graph, we get the DFS forest as output. To detect cycle, we can check for a cycle in individual trees by checking back edges. To detect a backedge, we can keep track of vertices currently in recursion stack of function for DFS traversal. If we search a vertex that is already in the recursion stack, then there is a cycle in the tree. The edge that connects current vertex in the recursion stack is a backedge.

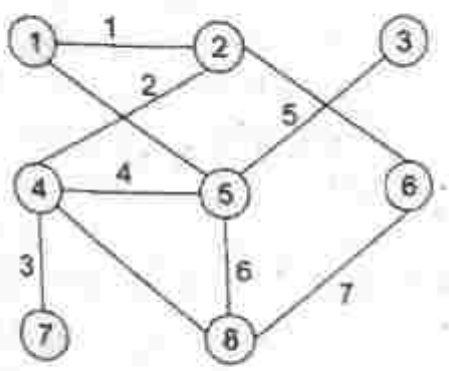
Time complexity of this method is the same as time complexity of DFS traversal which is $O(V+E)$.

So, option (b) is correct.

Option (a) is not correct because it may be guaranteed. So, option (b) is correct.

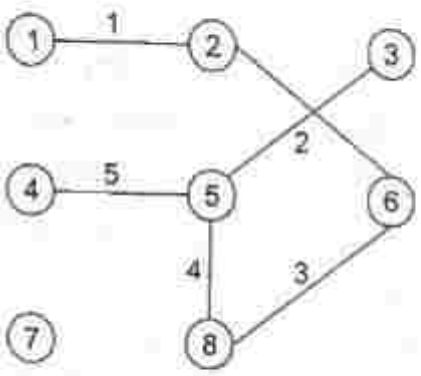
218. (b)

(a)



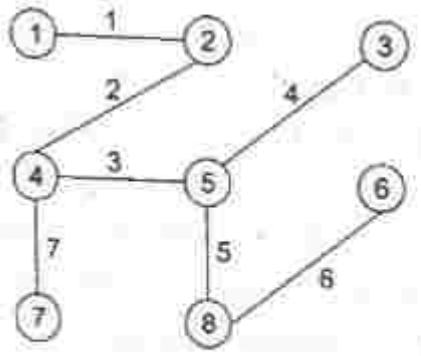
DFS safe sequence

(b)



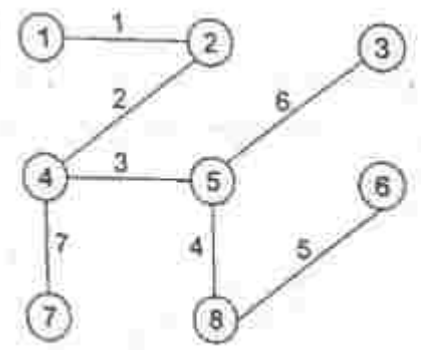
4 to 3 not follow DFS so not DFS sequence because have to go 4 to 7 first then 7 to 3.

(c)



DFS safe sequence

(d)



DFS safe sequence

219. (b)

Connected components of a graph can be computed in linear time by using either breadth-first search or depth-first search.

221. (c)

Statement 1 → True. If DFS finds no back edges, then the graph is acyclic. Removing any back edges found does not change the traversal order of DFS, so running DFS again on modified graph would produce no back edges.

Statement 2 → True. Each needs to keep track of the vertices that have already been visited.

Statement 3 → False. The DFS forest may contain different numbers of trees (and tree edges) depending on the starting vertex and upon the order in which vertices are searched.

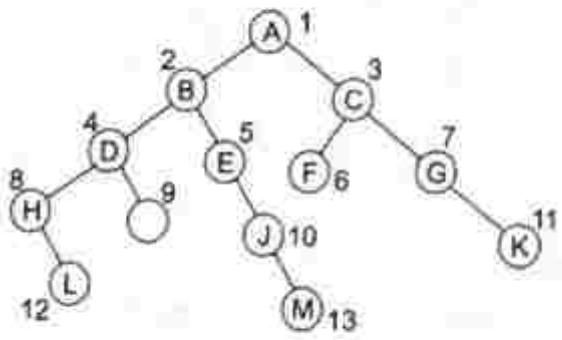
Hence, option (c) is current.

222. (c)

Statement 1: False: If the graph contains a negative weight cycle then no shortest path exists.
Statement 2: False: Even if no two edges have some weight there could be two paths with the same weight.
Statement 3: True: Since all vertices are connected by a path with at most T edges, and since BFS always finds the path with the fewest edges, the BFS tree will have depth atmost T. ADFS tree may have depth up to V-1 (for example in a complete graph).
Statement 4: True: if such an edge existed, if would provide a shorter path to some node than the path found by BFS (interms in number of edges). This cannot happen, as BFS always finds the path with the fewest edges.
Hence, statement 3 and 4 are true.

223. (a)

For BFS algorithm, visiting a node's siblings before its children, while in DFS algorithm, visiting a node's children before its siblings.



Before countering goal node F:
BFS algorithm encounters nodes: ABCDE
DFS algorithm encounters nodes : ABDHLIEJMC
So, option (a) is correct.

224. (c)

If u is visited before v, $d(r, u) \le d(r, v)$ i.e., r to u is having shortest or equal length compared from r to v.

225. (15)

There are 8 possibilities starting with A.
We are missing : : (Starting with A)
$A - C - E - B - D - F$
$A - B - E - C - D - F$ + money more
Starting with $A = 8$, with $C = 4$, with $E = 3$
$\Rightarrow$ Total = 15
and $E - F - A - B - D - C$ is not a topological sort.
So, the number of topological order = 15.

226. (a)

5 strongly component.
$\{s, v, w\}, \{x, z\}, \{q, t, y\}, \{r\}, \{v\}$

227. (d)

In directed graph, no vertex visited twice means acyclic graph then no back edges in directed graph means acyclic graph.
Hence, all statements are true.

228. (c)

There are three strong components:
1. G, A
2. E_1, T, S, E_2
3. C

∴ Option (c) is correct.

229. (a)

Any edge is a minimum cut, so there are $n-1$ minimum cuts.
Anything could happen!
Suppose the graph is a direct chain, adding an edge to make it a cycle can make it reduce a lot of strongly connected components.
$1 \to 2 \to 3 \to 4$ (4 strongly connected components)
Adding $4 \to 1$ will result in just 1 strongly connected components.
On the other hand adding one more edge, say $1 \to 3$, does not decrease it any more.
So the answer is this choice: ...might or might not remain the same (depending on the graph)

232. (a, b)

It is possible to obtain the forest when MSTs are generated using Kruskal algorithm
It is not possible to obtain the forest when MSTs are generated using Prim's algorithm.
Time complexity for Kruskal Algorithm = O(E log E)
Time Complexity for Prim's Algorithm = $O(E \log V)$
= O(E log E)

234. (a, b, c)

ABDEHFGH is not DFS traversal.

235. (a, b)

The algorithm, regardless of correctness, also make progress towards completion by removing a vertex from the unknown set, so it must terminate.

This is true because the graph may not have any negative edge at all.

This is not true given option (a).

This is not true, or else we don't really need the restriction.

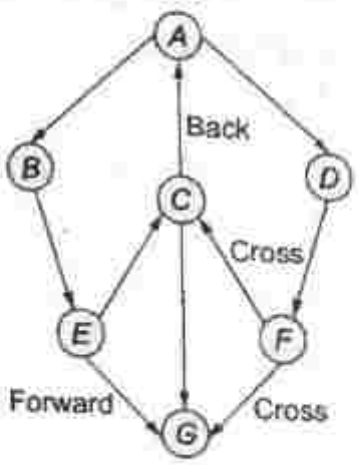
237. (b)

⇒ 2 Cross edges in G,

1 Forward edge in G,

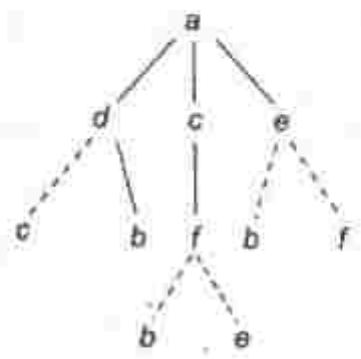
1 Back edge in G

∴ Number of cross edges = 2



238. (b)

Given BFS traversal on G is:



Tree edges are: $\{a, d\}, \{a, c\}, \{a, e\}, \{d, b\}, \{c, f\}$

Cross edges are: $\{d, c\}, \{e, b\}, \{e, f\}, \{f, b\}, \{f, e\}$

⇒ $\{d, b\}$ is not a cross edge.

239. (4)

Definition of backedge → if v is an ancestor of u, then edge (u, v) is a backedge.

- Now, starting from p next unvisited node $\{q, r, s\}$ choosing based on alphabetical order i.e. q.
- Standing at node q next unvisited nodes $\{r, t\}$ choosing based on alphabetical order i.e., r.
- Standing at node r next unvisited nodes $\{v\}$ choosing based on alphabetical order i.e. v.
- Standing at node v next unvisited node $\{s, t\}$ choosing based on alphabetical order i.e. s.
- Standing at node s next unvisited node $\{t\}$ choosing based on alphabetical order i.e., t.

All nodes are traversed.

Back edges are as follows: $(t, v), (s, p), (t, q), (r, p)$

Number of backedges = 4

Hence, 4 is the answer.

240. (a, b, d)

In Kruskal's algorithm, the disjoint-set data structure efficiently identifies the components containing a vertex and adds the new edges. And Kruskal's algorithm always finds the MST for the connected graph.

Prim's algorithm outperforms the Kruskal's algorithm in case of the dense graphs. It is significantly faster if graph has more edges than the Kruskal's algorithm.

241. (b)

Prim's algorithm iterates from one node to another, so it can not be applied for disconnected graph. Kruskal's algorithm can be applied to the disconnected graphs to construct the minimum cost forest. Kruskal's algorithm is comparatively easier and simpler than Prim's algorithm.

242. (a, c)

In the normal execution of Dijkstra's Algorithm, the while loop gets executed V times. The change in the while loop statement causes it to execute only V – 1 times. For similar reasons it will not give correct result.

243. (b, c, d)

In Kruskal's algorithm, the disjoint-set data structure efficiently identifies the components containing a vertex and adds the new edges. And Kruskal's algorithm always finds the MST for the connected graph.

Minimum priority queue is the most commonly used data structure for implementing Dijkstra's Algorithm because the required operations to be performed in Dijkstra's Algorithm match with specialty of a minimum priority queue.

If the total number of edges in all adjacency list is E, then there will be a total of E number of iterations, hence there will be a total of at most E decrease key operations.

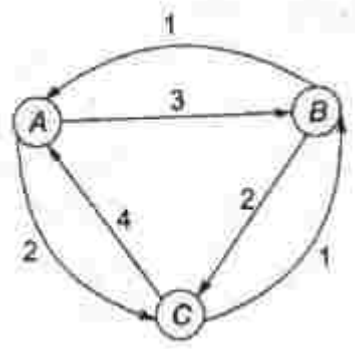
244. (b)

$$\begin{cases} l[i, j], & \text{if } k = 0 \\ \min\left\{d_{i,j}^{k-1},\ d_{i,k}^{k-1} + d_{k,j}^{k-1}\right\}, & \text{if } 1 \le k \le n \end{cases}$$

∴ Option (b) is correct.

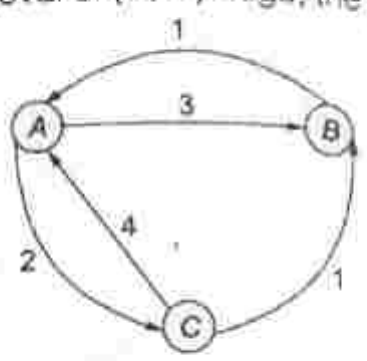
245. (1)

$$D = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 2 \\ 1 & 0 & 2 \\ 4 & 1 & 0 \end{bmatrix} \end{array} \Rightarrow$$



$$D^* = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 2 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} \end{array}$$

After removal of (B, C) edge, the graph is



$$\Rightarrow D_1 = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 2 \\ 1 & 0 & \infty \\ 2 & 1 & 0 \end{bmatrix} \end{array} \Rightarrow D_1 = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 0 & 3 & 1 \\ 1 & 0 & 3 \\ 2 & 1 & 0 \end{bmatrix} \end{array}$$

$\Rightarrow$ B to C shortest path distances are changed

246. (d)

D^i in the Floyd Warshall algorithm represents the shortest paths between all the vertices including the vertices labelled $< = i$.

UNIT 9

Programming and Data Structures

1. Programming Methodology

Q1 Consider the following Python program:

```
def main( ):
    C = "GATECSIT2026"
    p = c

    val_2p = ord(p[2])
    val_6p = ord(p[6])

    index = val_2p – val_6p – 1
    str = c[index:]

    print(len(str))
main( )
```

The output of the program is _____?

(a) 0 (b) 2
(c) 3 (d) 4

Q2 Which of the following statements is/are true?

(a) A Python list can hold complex numbers along with strings.
(b) Lists are stored in a contiguous block of memory like arrays in C.
(c) The append( ) method can add another list as a single element.
(d) List elements can be accessed using both positive and negative indices.

Q3 Consider the following statements:

S_1: When slicing lists, the stop index is exclusive.
S_2: Assigning to a slice can change the size of a list.

Which of the following is true?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of these

Q4 What will the output of the following code?

```
def main( ):
    lst = [10, 15, 20, 25, 30]
    for x in lst:
        if x == 20 or x == 25:
            continue
        print(x, end=",")
main( )
```

(a) 10, 20, 25 (b) 20, 25, 30
(c) 10, 15, 30 (d) 10, 15, 20

Q5 What will be the output of the below code?

```
def modify_tuple(tup):
    result = ( )
    for i in range(len(tup)):
        if i == 0 or i == 2:
            result = result + (tup[i],)
    return result

tup =("MADEEASY", "CSE", "DA", "EC", "ME")
print(modify_tuple(tup))
```

(a) MADECSEDA
(b) MADEEASYDA
(c) MADESYDA
(d) MADECSE

Q6 Consider the following Python code:

```
def fun(p, m):
    m = m + 5
    p[0] = p[0] + m

i = [5]
j = 10
fun(i, j)
print(i[0] + j)
```

What will the output of the above code?

(a) 15 (b) 25
(c) 20 (d) 30

Q7 Consider the following Python code:

```
def madeeasy(x, y):
    x = 0
    ptr = [x]
    y = ptr[0]
    ptr[0] = 1
    x = ptr[0]
    print(f"{x}, {y}")
```

The output of invoking madeeasy (1,1) is

(a) 1, 1 (b) 1, 0
(c) 0, 1 (d) 0, 0

Q.8 Consider the following statements:

S_1: The remove( ) method deletes only the first matching element.

S_2: The clear( ) method empties a list but keeps the list object in memory.

Which of the following is true?

(a) Only S_1 (b) Only S_2

(c) Both S_1 and S_2 (d) None of these

Q.9 Consider the following Python code:

```
k = 12
n = 30

if k > 5 and (n :=4):
    k = 100
else:
    k = 200

print(k)
```

What will be the output of the above code?

(a) 180 (b) 120

(c) 100 (d) 200

Q.10 Which of the following statements is/are true?

(a) The in operator can be used to check whether a value exists in a list.

(b) All elements in a list must be of the same type to avoid a run time error.

(c) Lists are passed by value when used in a function.

(d) Lists can be reversed using slicing with a negative step.

Q.11 What will be the output Python code?

```
def funval(n):
    n = n + 10
    print("Inside function:", n)

a = 5
funval(a)
print("Outside function:", a)
```

(a) 15 ,5 (b) 10, 5

(c) 10, 10 (d) 15, 15

Q.12 Which of the following statements is/are true?

(a) An empty set is defined using { }.

(b) The issubset( ) method checks if all elements of one set exist in another.

(c) The copy( ) method creates a reference to the original set.

(d) Sets are commonly used for membership testing.

Q.13 Consider the following Python code:

```
def func1(x, y, z):
    y[0] = y[0] + 4
    z[0] = x[0] + y[0] + z[0]

def P1( ):
    x = [10]
    y = [3]
    func1(y, x, x)
    print("x = ", x[0])
    print("y = ", y[0])
P1( )
```

What will be the output of the above code?

(a) 27, 2 (b) 10, 3

(d) 31, 3 (d) 28, 2

Q.14 Which of the following statements is/are true?

(a) The list( ) constructor can convert an iterable like a tuple into a list.

(b) Lists are immutable sequences in Python.

(c) List slicing returns a reference to the original list, not a copy.

(d) List comprehension is a method to build lists using loops in a compact form.

Q.15 Consider the following Python code:

```
def incr(i):
    if not hasattr(incr, "count"):
        incr.count = 0
    incr.count = incr.count + i
    return incr.count

def main( ):
    for i in range(5):
        j = incr(i)
    print("Final value of j:", j)

main( )
```

What will be the output of Python code?

(a) 6 (b) 8

(c) 3 (d) 10

Q.16 Consider the following statements:

S_1: Lists allow duplicate values without restriction.

S_2: You can store a list as a value inside a dictionary.

Which of the following is true?

(a) Only S_1

(b) Only S_2

(c) Both S_1 and S_2

(d) None of these

Q.17 What will be the output printed by the following Python program?

```
x = 1
y = 2

for i in range(5):
    x << 1
    y = x + 1

print(f"{x}, {y}")
```

(a) 32, 5 (b) 2, 5
(c) 1, 5 (d) None of these

Q.18 Consider the following statements:

S_1: A set can be created using the set( ) constructor with a list as input.

S_2: Sets maintain the order in which elements are inserted.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.19 Consider the following code:

```
def main( ):

    if not hasattr(main, "i"):
        main.i = 5

    main.i -= 1
    if main.i:
        main( )

    print(main.i, end)

main( )
```

How many zero's are printed in the output?

Q.20 Which of the following statements is/are true?

(a) A list can be part of another list as a nested element.
(b) Using del on a list variable deletes the entire list from memory.
(c) The index( ) method returns the first index of a matching item.
(d) The count( ) method removes duplicates from the list.

Q.21 Consider the following Python code:

```
def pr( ):
    if not hasattr(pr, "i"):
        pr.i = 1
print(chr(65 + pr.i), end= ' ')
pr.i += 1
pr( )
pr( )
pr( )
```

What will be the output of code?

(a) ACC (b) ABD
(c) ACD (d) BCD

Q.22 Which of the following are equivalent to the statements?

$$k = (i << 3) + (j >> 2)$$

Which of the following are equivalent to the above statement?

(a) k = i * 8 + j / 4 (b) k = i * 3 + j * 2
(c) k = i * 3 + j / 2 (d) k = i / 8 + j * 4

Q.23 Consider the following statements:

S_1: A list reference can be reused in multiple variables pointing to the same list.

S_2: The is operator checks whether two lists contain the same elements.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.24 Consider the following Python code:

```
def main( ):
    x = 1
    y = 2
    for i in range(5):
        x = x << 1
        if i % 2 == 0:
            y = y + i
        else:
            y = y - i
    print(x, y)

main( )
```

What will be the output of the above code?

(a) (32, 8) (b) (32, 4)
(c) (34, 3) (d) (36, 6)

Q.25 Consider the following statements:

S_1: The difference between two sets is symmetric.

S_2: The symmetric difference includes only elements unique to each set.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.26 Consider the following code:

```
def main( ):
    tuple = ((10, 20), (35, 12), (32, 50))
    for pair in tuple:
        print(max(pair), end=" ")

main( )
```

What will be the output of the above code?

(a) 20, 12, 32
(b) 20, 35, 50
(c) 35, 32, 10
(d) None of these

Q.27 What will be the output of following Python code?

```
def main( ):
    t = ((1, 2), (3, 1), (0, 5), (4, 0))
    max_pair = ( )
    max_sum = float('–inf')

    for a, b in t:
        s = a + b
        if s > max_sum:
            max sum = s
            max_pair = (a, b)
    print("Sum:", max_sum)

main( )
```

(a) 6 (b) 4
(c) 5 (d) 3

Q.28 What will be the output of the following Python code?

```
def main( ):
    for i in range(6):
        if i % 2 == 0:
            continue
    print(i, end=" ")

main( )
```

(a) 036 (b) 01245
(c) 135 (d) 1354

Q.29 Which of the following statements is/are true?

(a) Tuples are immutable, meaning their elements cannot be changed once defined.
(b) A tuple can contain elements of different types.
(c) Tuples use more memory than lists due to immutability.
(d) You can use indexing to access elements of a tuple.

Q.30 Consider the following Python code:

```
m = 10
m += 1
n = m

n1 = m
m += 1

n –= 1
n1 –= 1

n –= n1

print(n)
```

What will the output of the above code?

Q.31 Consider the following statements:

S_1 : The clear( ) method removes all elements from a set.
S_2 : The set( ) function without arguments returns an empty set.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.32 What will the output of the following code?

```
def main( ):
    1st = [1, 3, 5, 7, 9]
    for val in 1st:
        if val % 3 == 0:
            continue
        print(val, end=" ")

main( )
```

(a) 135 (b) 357
(c) 157 (d) 957

Q.33 Which of the following statements is/are true?

(a) A tuple with one element must end with a comma.
(b) The + operator can concatenate two tuples
(c) The * operator cannot be used with tuples
(d) Tuples are hashable if all their elements are hashable.

Q.34 Consider the following statements:

S_1 : The setdefault( ) method returns a default value if the key is not found in the dictionary
S_2 : A dictionary's values( ) method returns a list of all the values in the dictionary

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.35 What will be the output printed by the following python code?

```
def main( ):
    x = 1
    y = 2
    for i in range(5):
        x <<= 1
        y = x + i
    print(x, y)

main( )
```

(a) 1, 5 (b) 32, 9
(c) 32, 72 (d) 32, 36

Q.36 What will be the output of the following Python code?

```
def main( ):
    values = [2, 5, 8, 11, 14]
    total = 0
    for v in values:
        if v > 10:
            break
        total += v
    print(total)

main( )
```

(a) 10 (b) 15
(c) 20 (d) 25

Q.37 What will be the output of the following code?

```
def main( ):
    data = [4, 7, 10, 13, 16]
    result = 0
    for num in data:
        if num % 2 == 0:
            result += num
        else:
            result -= num
    print(result)

main( )
```

(a) –6 (b) –16
(c) –7 (d) 10

Q.38 Consider the following statements:

S_1: Tuples are immutable, but they can contain mutable objects.

S_2: A tuple with mutable items is still considered hashable.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.39 Consider the Python code:

```
def change_list_val(ref):
    ref[0] = ref [0] + 10
    print("Inside function:", ref[0])

n = [5]
change_list_val(a)
print("Outside function:", n[0])
```

What will be the output of above code?

(a) 10, 5 (b) 10, 10
(c) 15, 5 (d) 15, 15

Q.40 Which of the following statements is/are true?

(a) Dictionaries are mutable data types.
(b) A dictionary key can be a list or a tuple.
(c) You can use a dictionary comprehension to create a new dictionary.
(d) A dictionary cannot store a None value.

Q.41 What will be the output of the following Python code?

```
def main( ):
    vals = [5, 10, 15, 20, 25]
    for i in range(0, len(vals), 2):
        if vals[i] % 10 == 0:
            print(i, end=" ")
        else:
            print(vals[i], end=" ")

main( )
```

(a) 51525 (b) 10515
(c) 20515 (d) 10155

Q.42 Consider the following statements:

S_1 : Keys in a dictionary must be unique.

S_2 : A dictionary can store multiple values for a single key.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.43 What will be the output of the following code?

```
def main( ):
    nums = [2, 4, 8, 16]
    res = 1
    for i in nums:
        if i % 4 == 0:
            res *= 2
        else:
            res += 1
    print(res)

main( )
```

(a) 14 (b) 8
(c) 10 (d) 16

Q.44 Consider the following Python code:

```
var = 5

def foo( ):
    global foo_var
    if 'foo_var' not in globals( ):
        foo_var = 10
    print(foo_var, end=" ")

def main( ):
    global var
    print(var, end=" ")
    foo( )
    print(var)

main( )
```

What will be the output of the above code?

(a) 1015 (b) 5105
(c) 1055 (d) 515

Q.45 Consider the following Python code:

```
def func( ):
    if not hasattr(func, 'i'):
        func.i = 1

    j = 4
    print(func.i, end=" ")
    func.i += 1

    print(j, end=" ")
    j += 1

def main( ):
    func( )
    func( )

main( )
```

What will be the output of the above code?

(a) 1436 (b) 2536
(c) 1424 (d) 1426

Q.46 Which of the following statements is/are true?

(a) Sets are mutable data structures.
(b) A set can contain another set as an element.
(c) The remove( ) method raises an error if the element is not found.
(d) The discard( ) method removes an item without raising an error.

Q.47 Consider the following statements:

S_1: The tuple( ) constructor can be used to convert a list to a tuple.

S_2: Tuples consume more memory than lists because of immutability.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.48 What will be the output of the below code?

```
def string_filter(w):
    S = set( )
    for i in range(len(w)):
        if w[i] not in w[: i]:
            S.add(w[i])
    return ' ' .join(sorted(S))

print(string_filter("Madeeasy"))
```

(a) desy (b) mady
(c) madesy (d) adsye

Q.49 What will be the output of the below code?

```
def process (word):
    S = set( )
    for char in word:
        if char in S:
            S.remove(char)
        else:
            S.add(char)
    return ' ' .join(sorted(S))

print(process("ENGINEERING MADEEASY DA"))
```

(a) MYSA (b) ASYM
(c) AMSY (d) MSYA

Q.50 Consider the following statements:

S_1 : The pop( ) method removes a random key value pair from the dictionary.

S_2 : The update( ) method can be used to merge two dictionaries.

Which of the following is true?

(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q.51 Consider the following Python code:

```
def foo(a, n):
    if n <= 0:
        return 0
    elif a[0] % 3 == 0:
        return a[0] + foo(a[1 :], n – 1)
    else:
        return a[0] – foo(a[1 :], n – 1)
a = [–16, 7, 14, –4, –12, 5]
result = foo(a, len(a))
print(result)
```

The output of this program will be

Q53 What will be the output of the following code?

```
def extract_large_elements(tup):
    result = ( )
    for item in tup:
        if len(item) > 3:
            result = result + (item,)
    return result

tup=("EE", "ME", "MADEASY", "DELHI", "CSEDA")
print(extract_large_elements(tup))
```

(a) MADE EASY CSE DA
(b) MADE EASY CSE
(c) MADE EASY ME
(d) MADE EASY DELHI CSE DA

Q54 Consider the following statements:

S_1: Elements in a tuple cannot be changed, but the entire tuple variable can be reassigned.

S_2: Tuple unpacking can be used to assign multiple variables at once.

Which of the following is true?
(a) S_1 only
(b) S_2 only
(c) Both S_1 and S_2
(d) None of these

Q55 Consider the following Python program:

```
def main( ):
    p = -3
    q = 2
    r = 0
    p += 1
    q += 1
    if p and q:
        s = 1
    else:
        s = int(bool(r))
        r += 1
    t = p + q + s
    s += 1
    print(s, t)
main( )
```

Which of the following output of the above code?
(a) 12 (b) 23
(c) 13 (d) 22

Q55 What will be the output of the following code?

```
def test( ):
    x = 5
    y = 7
    z = 2
    result = (x + y) if x % 2 == 0 else (y – z)
    print(result)
test( )
```

(a) 5 (b) 7
(c) 4 (d) 6

Q56 What will be the output of the following code?

```
def test( ):
    p = 2
    q = 0
    r = (p << 2) and (q or 1)
    print(r)
test( )
```

(a) 0 (b) 1
(c) 2 (d) None of these

Q57 Consider the following statements:

S_1: Parentheses are mandatory to define a tuple.

S_2: The expression 1, is a valid tuple of one element.

Which of the following is true?
(a) S_1 only (b) S_2 only
(c) Both S_1 and S_2 (d) None of these

Q58 What will be the output of the below Python code?

```
a = 300
b = 10
c = 20
if not (a >= 400):
    b = 300
c = 200
print(f"{b}, {c}")
```

(a) 200, 320 (b) 300, 200
(c) 180, 290 (d) 320, 210

Q59 Consider the following Python code:

```
def fun (a, * args, s = '!'):
print(a, s)
for i in args:
    print(i, s)
fun(10, 20, s='+')
```

What is output of the above code?
(a) 10 + 20+ (b) 1020
(c) +10 +20 (d) +10 20+

Q.60 What will be the output of the below Python code?

```
c = 0
d = 5
e = 10

if c > 1 :
    if d > 1 or e > 1:
        a = 100
    else:
        a = 200
else:
    a = 300

print(a)
```

(a) 300 (b) 200
(c) 100 (d) None of these

Q.61 Consider the following statements:

S_1: Tuple slicing returns a new tuple with selected elements.

S_2: The index( ) method can be used to find the position of an item in a tuple.

Which of the following is true?

(a) S_1 only
(b) S_2 only
(c) Both S_1 and S_2
(d) None of these

Q.62 Consider the following code:

```
def foo(tup):
    if len(tup) > 2:
        return tup[1:-1]
    else:
        return( )

tup = ("GATE2025", "GATE2026", "MADEEASY", "DA",
"IES2024")

print(foo(tup))
```

What will be the output of the above code?

(a) GATE 2025 DA
(b) GATE 2026 MADE EASY DA
(c) GATE 2025 MADE EASY
(d) None of these

2. Stack

Q.63 Computes the postfix equivalent of following expression:

$$3 * \log(x + 1) - \frac{a}{2}$$

(a) 3x1 + log * a2/- (b) 31x + log * a2/-
(c) 31x log + * a2/- (d) 31 logx + * a22/-

Q.64 Which of the following statement(s) is/are correct?

(a) In conversion from prefix to postfix using stack data-structure, if operators and operands are pushed and popped exactly once, then the run-time complexity is $O(n)$.
(b) In conversion from prefix to postfix using stack data-structure, if operators and operands are pushed and popped exactly once, then the run-time complexity is $O(n^2)$.
(c) The data structure that can be used for parentheses matching is stack.
(d) The data structure that can be used for parentheses matching is queue.

Q.65 Consider the following infix expression of C.

$g + a - t + e * e / x * a / m - r + e - s + u - l + t$

Find the length of substring in the equivalent postfix expression for the above infix and that substring has only operators, and the number of distinct operators and the length of substring is maximum compared to any other substring.

Q.66 Consider a stack implementation supports, in addition to PUSH and POP, an operation REVERSE, which reverses the order of the elements on the stack. Which of the following represents the minimum stack operations required to implement ENQUEUE and DEQUEUE operations of queue data structure respectively?

(a) 1 and 3
(b) 3 and 1
(c) 2 and 2
(d) Either (a) or (b)

Q.67 Consider the following infix expression which is to be converted to postfix expression using stack.

$$(((P + Q) * (R + S)) / T) + (A * (B + C))$$

The sum of all unique possible heights of stack when converting from infix to postfix is _____

Q.68 Which of the following statement is/are correct?

(a) Postfix operators use value to their right.
(b) Postfix operators use value to their left.
(c) Prefix operators use value to their right.
(d) In postfix expression, operands are followed by operators.

Q.69 Assume stack A has the entries p, q and r (with p on top and r on bottom). Initially stack B is empty. An entry popped out of stack A can be printed immediately or pushed to stack B. A entry popped out of stack B can only be printed. What is the least number of stack permutations of input sequence that start with a particular letter?

Q.70 Assume ↑ is power operator and it has the highest precedence and follows right associativity. What is the output of the following postfix expression evaluated using operand stack?
(**Note:** 2 ↑ 3 = 8)

9 9 1 * 9 1 ↑ / + 9 – 9 +

(a) 10 (b) 19
(c) 18 (d) 81

Q.71 The post fix form of
$A\$B*C-D+E/F/(G+H)$ is
(a) $AB\$C*D-EF/GH+/+$
(b) $AB\$*C-D+EF/GH/+$
(c) $AB\$C+D-EF/GH-/+$
(d) $AB\$C-D*EF/GH/++$

Q.72 Which of the following statements about stacks is/are correct?
(a) Stacks can be implemented using linked lists.
(b) Stacks are first-in, first-out (FIFO) data structures.
(c) New nodes can only be added to the top of the stack.
(d) The last node (at the bottom) of a stack has a null (0) link.

Q.73 Let S be a stack of size $4 \geq 1$ and it is initially empty. Suppose we push the numbers 1, 2, 3, 4 in sequence and then perform 4 pop operations. Let one push operation takes 5 ns; one pop operation takes 5 ns; the time between the end of one such stack operation and the start of the next operation is 2 ns. The stack-life of a particular element $p \geq 1$ is defined as the time elapsed from the end of push (p) to the start of the pop operation that removes p from the stack. The average stack-life of an element of this stack is _____ (in ns).

Q.74 Maximum size of stack required to evaluate given postfix expression is _____

10 5 * 2 6 3 / + –

Q.75 Consider the following statements:
S_1: Two stacks can be implemented by 1 array that grow from each end towards the middle.
S_2: A queue can be implemented using two stacks with worst case time $O(n)$ for each enqueue and dequeue operation.
The number of statements correct are _____.

Q.76 The prefix form of $A - B / (C * D \$ E)$ is
(a) $-/*AB*(C-D)$
(b) $-ABCD*DE\$$
(c) $-A/B*C\$DE$
(d) $-A/BC*\$DE$

Q.77 Which of the following data structure is used for implementing recursion?
(a) Queue (b) Stack
(c) Linked list (d) Heap

Q.78 Assume ↑ is power operator and it has the highest precedence and follows right associativity. What is the output of the following postfix expression evaluated using operand stack?
(**Note:** 2 ↑ 3 = 8)

7 7 1 * 7 1 ↑ / + 7 – 7 +

Q.79 Which of the following statement(s) is/are correct?
(a) Stack overflow refers to adding items to a full stack.
(b) Stack overflow refers to index out of bounds exception.
(c) Stack underflow refers to accessing items from an undefined stack.
(d) Stack underflow refers to removing items from an empty stack.

Q.80 Which of the following permutation can be obtained in the same order using a stack assuming that input is the sequence 5, 6, 7, 8, 9 in that order?
(a) 7, 8, 9, 5, 6
(b) 5, 9, 6, 7, 8
(c) 7, 8, 9, 6, 5
(d) 9, 8, 7, 5, 6

Q.81 The five items : *A*, *B*, *C*, *D*, and *E* are pushed in a stack, one after other starting from *A*. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

(a) *A* (b) *B*
(c) *C* (d) *D*

Q.82 If the sequence of operations Push(1), Push(2), Pop, Push(1), Push(2), Pop, Pop, Pop, Push(2), Pop are performed on a stack, the sequence of popped out values

(a) 2, 2, 1, 1, 2 (b) 2, 2, 1, 2, 2
(c) 2, 1, 2, 2, 1 (d) 2, 1, 2, 2, 2

Q.83 The following postfix expression with single digit operands is evaluated using a stack:

9 3 1 ^ / 6 4 2 + * +

The top two elements of the stack after the first + is evaluated

(a) 4, 2 (b) 6, 6
(c) 4, 6 (d) 2, 4

Q.84 The maximum size of operator stack when converting the following infix to postfix expression is _____.

Assume that ↑ as the highest precedence and follows right associativity.

$a \uparrow b * c * d + e * f \uparrow g$

(a) 4 (b) 2
(c) 5 (d) 3

Q.85 Which of the following is/are the application of stack?

(a) Evaluation of postfix expression.
(b) Conversion of infix to postfix expression.
(c) Balancing symbols.
(d) Line at ticket counter.

Q.86 The total number of push operations performed to evaluate the postfix expression are:

5 4 5 * 8 5 – 2 + / –

Q.87 Consider the following statements:

(i) First-in-first out types of computations are efficiently supported by STACKS.
(ii) Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.
(iii) Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.
(iv) Last-in-first-out type of computations are efficiency supported by QUEUES.

(a) (ii), (iii) are true
(b) (i), (ii) are true
(c) (iii), (iv) are true
(d) (ii), (iv) are true

Q.88 Find postfix expression for the following infix expression?

Assume ↑ as the highest precedence and follows right to left associativity.

Infix: $(a + b) \uparrow (p + q) \uparrow (r * s * t)$

(a) $ab + pq + \uparrow rs * t * \uparrow$
(b) $ab + pq + \uparrow\uparrow rs * t *$
(c) $ab + pq + rs * t * \uparrow\uparrow$
(d) $ab + pq + rst * * \uparrow\uparrow$

Q.89 Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order?

(a) 3, 4, 5, 1, 2 (b) 3, 4, 5, 2, 1
(c) 1, 5, 2, 3, 4 (d) 5, 4, 3, 1, 2

Q.90 What will be the output on evaluating the given postfix-expression:

11 8 + 50 10 / * 2 –

Q.91 Stack *A* has the entries *a*, *b*, *c* (with *a* on top). Stack *B* is empty. An entry popped out of stack *A* is pushed into stack *B*. An entry popped out of stack *B* can only be printed. In this arrangement, which of the following permutations of *a*, *b*, *c* is not possible?

(a) *b a c* (b) *b c a*
(c) *c a b* (d) *a b c*

Q.92 Which of the following statement(s) is/are true?

(a) Two stacks are required for evaluation of prefix expression.
(b) Three stacks are required for evaluation of prefix expression.
(c) Evaluation of infix expression is done based on precedence of operators.
(d) Evaluation of infix expression is independent of precedence of operators.

Q.93 An item that is read as input can be either pushed to a stack and later popped and printed, or printed directly. Which of the following will be the output if the input is the sequence of items – 1, 2, 3, 4, 5?
(a) 3, 4, 5, 1, 2
(b) 3, 4, 5, 2, 1
(c) 1, 5, 2, 3, 4
(d) 5, 4, 3, 1, 2

Q.94 The following sequence of operations is performed on a stack:
PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP
The sequence of values popped out is:
(a) 20, 10, 20, 10, 20
(b) 20, 20, 10, 10, 20
(c) 10, 20, 20, 10, 20
(d) 20, 20, 10, 20, 10

Q.95 The following postfix expression with single digit operands in evaluated using a stack
8 2 3 ^ / 2 3 * + 5 1 * –
Note that ^ is the exponentiation operator. The top two elements of the stack after the first * is evaluated are
(a) 6, 1 (b) 5, 7
(c) 3, 2 (d) 1, 5

Q.96 The best data structure to check whether an arithmetic expression has balanced parentheses is a
(a) Queue
(b) Stack
(c) Tree
(d) List

Q.97 In evaluating the arithmetic expression 2 * 3 – (4 + 5), using stack to evaluate.
Which of the following is stack configuration is not possible

(a)

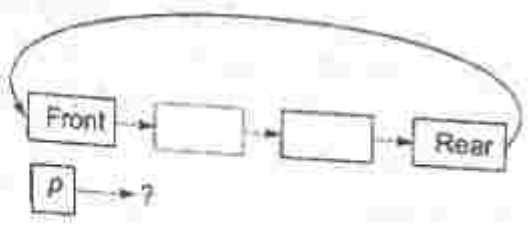
(b)

(c)

(d)

Q.98 Which of the following is essential for converting an infix expression to the post fix form efficiently?
(a) An operator stack
(b) An operand stack
(c) An operand stack and an operator stack
(d) A parse tree

Q.99 A circularly linked list is used to represent a Queue. A single variable *p* is used to access the Queue. To which node should *p* point such that both the operations enQueue and deQueue can be performed in constant time?



(a) Rear node
(b) Front node
(c) Not possible with a single pointer
(d) Node next to front

Q.100 Which of the following statement is/are correct with respect to evaluation of infix expression algorithm?
(a) Operand is pushed on to the stack.
(b) If the precedence of operator is higher, pop two operands and evaluate.
(c) If the precedence of operator is lower, pop two operands and evaluate.
(d) The result is pushed on to the operand stack

3. Queue

Q.101 In breadth first search of graph, which data structure is used?
(a) Stack (b) Queue
(c) Array (d) Linked List

Q.102 In linked list implementation of queue, if only front pointer is maintained, which of the following operation(s) take worst case linear time?
(a) Enqueue
(b) Dequeue
(c) Counting number of elements
(d) Reversing a queue

Q.103 If the number 4, 3, 2, 1 are placed in queue (in that order), and then removed one at a time in what order they will be removed?

Q.104 Suppose the number 1, 2, 3, 4, 5 and 6 arrive in an input stream in that order. Which of the following sequences can be realized as the output of double ended queue?

(i) 1 2 3 4 5 6 (ii) 2 4 3 6 5 1
(iii) 6 5 4 3 2 1 (iv) 1 5 2 4 3 6

(a) (i), (iv) only
(b) (ii), (iii) only
(c) (i) and (ii) only
(d) (i) and (iii) only

Q.105 Suppose you are given an implementation of a queue of integers the operations that can be performed on the queue is

1. empty (*Q*)- returns true if the queue empty, false characters
2. delete (*Q*) - deletes the element at the front of the queue and return its value.
3. insert (*Q*, *i*) - insert the integer *i* at the rear of queue.

Consider the following function:

```
def f(Q)
    if not is_empty(Q):
    i = delete(Q)
    f(Q)
    insert(Q, i)
```

What operation is performed by the above function *f*?

(a) Leaves the queue *Q* unchanged
(b) Reverse the order of element in the queue
(c) Deletes the element at the front of the queue *Q* and inserts it at the rear keeping the other elements in the same order
(d) Empties the queue (*Q*)

Q.106 In an empty queue, rear and front can be initialized as

(a) rear = front = –1
(b) rear = front = 1
(c) rear = 0, front = –1
(d) rear = –1, front = 0

Q.107 In linked list implementation of queue, if both front and rear pointer is maintained, which of the following operation(s) take worst case linear time?

(a) Enqueue
(b) Dequeue
(c) Counting number of elements
(d) Reversing a queue

Q.108 Consider the following statements:

S_1: Implementation of stack using queue, deletion of second element from top of stack time complexity O(n), when insertion take O(1) time.

S_2: In implementation of queue using stack, deletion of second element from front take O(1) time, when insertion take O(n) time.

Which of the following is correct ?

(a) Only S_1 (b) Only S_2
(c) Both S_1 and S_2 (d) None of the above

Q.109 A priority queue is implemented as a Max-heap. Initially it has 5 elements. The level order traversal of the heap is 10, 8, 5, 3, 2. Two new elements '1' and '7' are inserted into the heap in that order. The level order traversal of the heap after the insertion of the elements is

(a) 10, 8, 7, 5, 3, 2, 1
(b) 10, 8, 7, 2, 3, 1, 5
(c) 10, 8, 7, 1, 2, 3, 5
(d) 10, 8, 7, 3, 2, 1, 5

Q.110 Which of the following statement(s) is/are correct?

(a) In a circular queue, We can increment the rear end of the queue as (rear +1) % CAPACITY.
(b) In a circular queue, We can increment the rear end of the queue as (rear % CAPACITY) +1.
(c) Two stacks are required for applying evaluation of infix expression algorithm.
(d) Three stacks are required for applying evaluation of infix expression algorithm.

Q.111 The minimum number of stacks needed to implement a queue is

(a) 3 (b) 1
(c) 2 (d) 4

Q.112 Which of the following is/are true about linked list implementation of queue?

(a) In push operation, if new nodes are inserted at the beginning of linked list, then in pop operation, nodes must be removed from end.
(b) In push operation, if new nodes are inserted at the beginning, then in pop operation, nodes must be removed from the beginning

(c) In push operation, if new nodes are inserted at the end, then in pop operation, nodes must be removed from end.

(d) In push operation, if new nodes are inserted at the end, then in pop operation, nodes must be removed from beginning.

Q.113 Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

(a) A queue cannot be implemented using this stack.

(b) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.

(c) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.

(d) A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

Q.114 What are the applications of dequeue (Double ended queue)?

(a) A-Steal job scheduling algorithm.

(b) Can be used as both stack and queue.

(c) To find the maximum of all sub arrays of size *k*.

(d) To avoid collision in hash tables.

Q.115 Consider the following statements:

(i) First-in-first out types of computations are efficiently supported by STACKS.

(ii) Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.

(iii) Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.

(iv) Last-in-first-out type of computations are efficiency supported by QUEUES.

(a) (ii), (iii) are true

(b) (i), (ii) are true

(c) (iii), (iv) are true

(d) (ii), (iv) are true

Q.116 A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented INSERT (Q, C, K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETEMIN(Q). For a sequence of operations, the keys chosen are in

(a) Non-increasing order

(b) Non-decreasing order

(c) Strictly increasing order

(d) Strictly decreasing order

Q.117 Which of the following statement(s) is/are true?

(a) Minimum number of queues required to implement stack is 1

(b) Minimum number of queues required to implement stack is 2

(c) The properties is associated with a queue is FIFO.

(d) The properties is associated with a queue is LIFO.

Q.118 What can we say about the array representation of a circular queue when it contains only one element?

(a) Front == Rear == None

(b) Front == (Rear + 1)

(c) Front == Rear == –1

(d) Front == Rear and Front is not None

Q.119 A double-ended queue (dequeue) supports adding and removing items from both the ends of the queue. The operations supported by dequeue are AddFront(adding item to front of the queue), AddRear(adding item to the rear of the queue), RemoveFront(removing item from the front of the queue), and RemoveRear(removing item from the rear of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What's the time complexity of performing AddFront( ) and AddRear( ) assuming *m* is the size of the stack and *n* is the number of elements?

(a) $O(m)$ and $O(n)$ (b) $O(1)$ and $O(n)$

(c) $O(n)$ and $O(1)$ (d) $O(n)$ and $O(m)$

Q.120 A queue data-structure can be used for Which of the following

(a) Expression parsing

(b) Recursion

(c) Resource allocation

(d) Semaphores implementation

4. Linked Lists

Q.121 The concatenation of two lists is performed in O(1) time. Which of the following implementations of a list should be used?

(a) Singly linked list
(b) Doubly linked list
(c) Circular doubly linked list
(d) Array implementation of list

Q.122 Consider the following function:

```
class Node:
def _init_(self, value):
    self.value = value
    self.next = None

def find(head):
    P1 = head
    P2 = head
    while P2 and P2.next:
        P1 = P1.next
        P2 = P2.next.next if P2.next else None
    print(P1.value)
```

Assume the node is the structure type with two members: "value" and "next".
Identify the node value printed by the above code if a non-empty linked list header is passed to the function find.

(a) First element of the list
(b) Second element of the list
(c) Middle element of the list
(d) Last element of the list

Q.123 Consider a small circular linked list. Which of the following(s) is/are correct way to detect the presence of cycles in this list effectively?

(a) Keep one node as head and traverse another temp node till the end to check if its 'next points to head.
(b) Have fast and slow pointers with the fast pointer advancing two nodes at a time and slow pointer advancing by one node at a time.
(c) We can apply depth first search in which back edges represents the presence of cycles.
(d) Circular linked list itself represents a cycle. So no new cycles cannot be generated.

Q.124 Given pointer to a node X in a singly linked list. Only one pointer is given, pointer to head node is not given, can we delete the node X from given linked list?

(a) Possible if size of linked list is even
(b) Possible if size of linked list is odd
(c) Possible if X is not first node
(d) Possible if X is not last node. (a) copy the data of next of X to X (b) delete next of X

Q.125 Which of the following operations is performed more efficiently by double linked list than by linear linked list?

(a) Deleting nodes whose location is given
(b) Searching an unsorted list for a given item
(c) Inserting a node after the node with a given location.
(d) Traversing the list to process each node.

Q.126 Consider the following Python code to detect a loop in a linked list.

```
class Node:
    def_init_(self, value):
        self.value = value
        self.next = None

def detect_loop(head):
    p = head
    q = head.next if head else None
    while(A):
        If p == q:
            print("Loop detected")
            return
        p = p.next
        q = q.next.next if q and q.next else None
print("No loop detected")
```

Which of the following options correctly completes the missing statement "A"?

(a) p != None
(b) q != None
(c) (p != None) and (q != None)
(d) (p != None) or (q != None)

Q.127 You are only provided with one pointer pointing to the node which is to be deleted. Which of the following is correct?

(a) Deletion of node is possible if the node to be deleted is the starting node of linked list.
(b) Deletion of node is possible for any arbitrary position of node.
(c) (a) is always true but not (b)
(d) (b) is always true but not (a).

Q.128 Which of the following is/are true about a circular linked list?
(a) Every node has a successor.
(b) Time complexity of inserting a new node at the head of the list is O(1).
(c) Time complexity for deleting the last node is $O(n)$.
(d) We can traverse the whole circular linked list by starting from any point.

Q.129 Let P be a singly linked list. Let Q be the pointer to an intermediate node x in the list. What is the worst-case time complexity of the best-known algorithm to delete the node x from the list?
(a) $O(n)$ (b) $O(\log^2 n)$
(c) $O(\log n)$ (d) $O(1)$

Q.130 Which data structure would be most appropriate to implement a collection of values with the following three characteristics?
- Items are retrieved and removed from the collection in FIFO order.
- There is no prior limit on the number of items in the collection.
- The size of an item is large relative to the storage required for a memory address.

(a) Singly-linked list, with head and tail pointers
(b) Doubly-linked list, with only a head pointer
(c) Binary tree
(d) Hash table

Q.131 In a circularly linked list organization, insertion of a record involves the modification of
(a) no pointer
(b) 1 pointer
(c) 2 pointers
(d) 3 pointers

Q.132 Which of the following operations is performed more efficiently by doubly linked list then by linear linked list? [Note: Using only 1 pointer]
(a) Deleting a node whose location is given
(b) Searching an unsorted list for a given item
(c) Inserting a node after the node with a given location
(d) Traversing the list to process each node

Q.133 A stack is implemented using a circular doubly linked list such that PUSH and POP operations are performed efficiently. Which one of the following statements can be correct?
(a) Both operations can be performed in O(1) time.
(b) The worst case time complexity for both operations will be $\Omega(n)$.
(c) Stack can not be implemented with the given data structure.
(d) None of these

Q.134 A queue is implemented using a singly linked list. The queue has a head pointer and a tail pointer, which point to the starting and the last node of the linked list respectively. Let 'enqueue' be implemented by inserting a new node at the head, and 'dequeue' be implemented by deletion of a node from the tail. Further, let the number of nodes in the queue be n.
Then the time complexity required by the most efficient algorithm for the implementation of enqueue( ) and dequeue( ) operation for this scenario will be:
(a) $O(n), O(n)$
(b) $O(n), O(1)$
(c) $O(1), O(n)$
(d) $O(1), O(1)$

Q.135 Consider the following 'python' function foo1( ) and foo2( ) take a single linked list of integers as a parameters and print the elements of the list in random order. The function is called with linked list containing the integers 7, 6, 5, 4, 3, 2, 1 in given orders:

```
Class Node:
    def_init_(self, data)
        self.data = data
        self.next = None
    def foo2 (head):
        If not head:
            return
        foo2(head.next)
        print(head.data, end = " ")
    def foo1 (head):
        If not head:
            return
        print(head.data, end = " ")
        foo1(head.next)
```

Let the order of the elements printed by foo1( ) be 'x' and the order of elements printed by foo2( ) be 'y'. An element that are at same index in both 'x' and 'y' are _____

Q.136 Which of the following statement(s) is/are true?
(a) Practical application of XOR linked lists are in environments with limited space requirements, such as embedded devices.
(b) XOR lists are not suitable because most garbage collectors will fail to work properly with classes or structures that don't contain literal pointers.
(c) In order to calculate the address of the next node you need to remember the address of the previous node.
(d) XOR lists are much efficient than single, doubly linked lists and arrays.

Q.137 Consider the following statement:
S_1 : Merge sort on linked list take O(n log n) time to sort input of length n.
S_2 : Merge sort on linked list give better space complexity then on array.
S_3 : Inplace merge sort on array will take O(n^2) time.
Which of the following is correct?
(a) Only S_1 (b) Only S_1 and S_2
(c) S_1 , S_2 and S_3 (d) None of these

Q.138 Time required to search for an element in linked list is
(a) O(1) (b) O(log n)
(c) O(n) (d) O(n log n)

Q.139 The concatenation of two lists is to be performed in O(1) time. Which of the following implementations of a list could be used?
(a) Singly linked list
(b) Doubly linked list
(c) Circular doubly linked list
(d) Array implementation of list

Q.140 Identify the functionality of the following code when the function returns the value 1.
[Assume head points to the first node of a non-empty singly linked list.]

```
class Node:
    def _init_(self, value):
        self.value = value
        self.next = None

def find(head):
    P1 = head
    P2 = head
    if head.next:
        P1 = head.next
        P2 = head.next.next if head.next else None

    while P1 and P2:
        if P1 == P2:
            return True
        P1 = P1.next
        P2 = P2.next.next if P2.next else None

    return False
```

What does the function check?
(a) It finds the duplicate element in the list
(b) It finds the middle node in the list
(c) It finds the cycle in the list
(d) It finds the last node in the list

Q.141 In a circular linked list organization, insertion of a record involves modification of
(a) One pointer (b) Two pointers
(c) Multiple pointers (d) No pointer

Q.142 Linked lists are not suitable data structures of which one of the following problems?
(a) Insertion sort
(b) Binary search
(c) Radix sort
(d) Polynomial manipulation

Q.143 In linked list data is stored in
(a) Adjacent location
(b) Different location in memory
(c) Neither (a) nor (b)
(d) None of the above

Q.144 _____ is called self referential structure.
(a) Linked list (b) Stack
(c) Queue (d) Graph

Q.145 Which of the following is not good for linked list?
(a) Sorting (b) Queues
(c) Binary reach (d) None of these

Q.146 Linked lists are not suitable for implementing
(a) Insertion sort
(b) Binary search
(c) Radix sort
(d) Polynomial manipulation

Q.147 In a circularly linked list organization, insertion of a record involves the modification of
(a) No pointer (b) 1 pointer
(c) 2 pointers (d) 3 pointers

Q.148 If we want to find last node of a linked list then the correct coding is

(a) if temp → link is None:
temp = temp.link

(b) if temp.data == Num:
temp = temp.link

(c) while temp link is not None
temp = temp.link

(d) while temp.link! = data:
temp = temp.link

Q.149 What is the output of the following function with head pointing to first node of the following linked list?

1 → 2 → 3 → 4 → 5 → 6

```
class Node:
    def_init_(self, value):
        self.value = value
        self.next = None

def func(head):
    If head is None:
        return
    print(head.value, end=" ") # Print the current
    node value
    if head.next is not None:
        func(head.next.next)    # Skip the next node
        and continue the recursion
    print(head.value, end=" ") # Print the current
    node value after returning from recursion
```

(a) 35135 (b) 12345
(c) 146641 (d) 135531

Q.150 Which of the following statements is/are correct with respect to Singly Linked List (SLL) and Doubly Linked List (DLL)?

(a) Complexity of Insertion and Deletion at known position is O(n) in SLL and O(1) in DLL.
(b) SLL uses lesser memory per node than DLL
(c) DLL has more searching power than SLL
(d) Number of node fields in SLL is more than DLL.

5. Trees

Q.151 Consider the following program:

```
def ht(node):
    If node is None:
        return 0
    return 1 + max(ht(node.left), ht(node.right))
def foo(tree):
    if tree is None:
        return 0
    lh = ht(tree.left)
    rh = ht(tree.right)
    ld = foo(tree.left)
    rd = foo(tree.right)
    return max(1h + rh + 1, max(ld, rd))
```

The value returned by foo( ) when a pointer to the root of a binary tree is passed as its argument is:

(a) Number of leaf nodes in the tree
(b) Difference in the number of nodes in left and right
(c) Diameter (maximum number of nodes on the longest path between two leaf nodes) of the tree
(d) Depth of the tree

Q.152 Consider the following Binary search tree with the following traversals.

Inorder: *A, M, N, O, P, Q, R, S, T, X, Z*
Preorder: *Q, P, A, M, N, O, X, S, R, T, Z*

Find the number of elements in the 4th, 5th and 6th level respectively. Assume root is at level 1.

(a) 3, 2, 2 (b) 2, 2, 2
(c) 3, 1, 1 (d) 3, 2, 1

Q.153 Which of the following statement(s) is/are correct?

(a) Access time of a binary search tree may go worse in terms of time complexity upto O(n).
(b) Access time of a balanced binary search tree may go worse in terms of time complexity upto O(n).
(c) Access time of a binary search tree may go worse in terms of time complexity upto O(log n).
(d) Access time of a Balanced binary search tree may go worse in terms of time complexity upto O(log n).

Q.154 A binary search tree is constructed by inserting the key values 1, 2, 3, 4, 5, 6, 7 in some order specified by a permutation of 1, ..., 7 into an initially empty tree. Which of these permutation will lead to a complete binary search tree?

(a) 1, 2, 3, 4, 5, 6, 7 (b) 4, 6, 5, 7, 1, 2, 3
(c) 4, 2, 6, 1, 3, 5, 7 (d) 4, 1, 5, 3, 6, 2, 7

Q.155 he inorder traversal of some binary tree produced the sequence MFIEJGZ. and post order sequence is MIFJZGE the number of nodes in left subtree is _____?

Q.156 Which is efficient tree structure, considering space complexities?
(a) AVL Tree
(b) Full Binary Tree
(c) Complete Binary Tree
(d) Binary search tree

Q.157 Consider three keys, k_1, k_2, k_3 such that $k_1 < k_2 < k_3$. A binary search tree is constructed with these three keys. Depending on the order in which the keys are inserted, number of binary search trees possible are
(a) 3 (b) 5
(c) 6 (d) 4

Q.158 T is a search tree of order K, its size is N and its height is H. The computation time needed to insert/delete a data item on T is
(a) $O(\log H)$ (b) $O(N)$
(c) $O(K)$ (d) $O(H)$

Q.159 Find the height of a tree for the following given traversals of the tree.
Inorder : *h, d, i, b, e, j, a, f, c, g, k*
Pre-order : *a, b, d, h, i, e, j, c, f, g, k*

Q.160 The number of binary search trees possible with n nodes such that each nodes takes a value from the set {1, 2, 3, 4, 5, 6} root has value 3 and the left child of the root has value 2, is equal to _____.
(a) 5 (b) 6
(c) 4 (d) 7

Q.161 Which of the following traversal is sufficient to construct Binary search tree from given traversal?
I. Preorder
I. Inorder
III. Postorder
(a) Any of the given traversal is sufficient
(b) Either I or III is sufficient
(c) I and III
(d) II and III

Q.162 Consider the following code segment, where both the left and right subtrees of the binary search tree are not null:

```
class Node:
    def_init_(self, data):
        self.left = None
        self.data= data
        self.right = None
def fun(p):
    if p.right and p.right.right is None:
        p.right = p.right.left
    else:
        p = fun(p.right)
    return p
```

What does the function fun do?
(a) Finds the largest node in the binary search tree
(b) Finds the largest node in the binary search tree and deletes it
(c) Finds the smallest node in the binary search tree
(d) Finds the smallest node in the binary search tree and deletes it

Q.163 If a tree has n_1 nodes of degree 1, n_2 nodes of degree 2 and n_3 nodes of degree 3, then which of the following holds?
(a) $n_1 = 2$ (b) $n_1 = n_3 + 2$
(c) $n_1 = n_3 - 1$ (d) $n_1 = n_2 + n_3$

Q.164 Which of the following is/are TRUE (**Note:** this kind of tree once came in gate that's why included as a question)?
(a) 2-3 tree requires less storage than the BST.
(b) Lookup in 2-3 tree is more efficient than in BST.
(c) 2-3 tree is shallower than BST.
(d) 2-3 tree is a balanced tree.

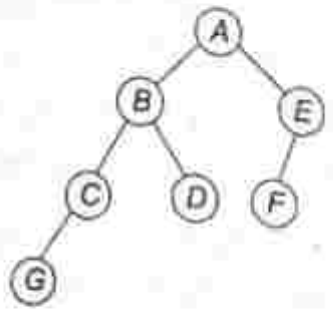
Q.165 Let T be a binary search tree with 120 elements. The smallest possible height of T is _____. Consider root is at height 0.

Q.166 A binary search tree was constructed by inserting following elements into an initially empty binary tree:

50, 27, 16, 88, 34, 65, 52, 77, 93, 4, 12, 29, 44, 92

Preorder and postorder traversals of the resultant binary search tree were stored in arrays A and B respectively. _____ elements have same index location in both the arrays. [Assume arrays A and B start from the same index]

Q.167 A balanced tree is given below:



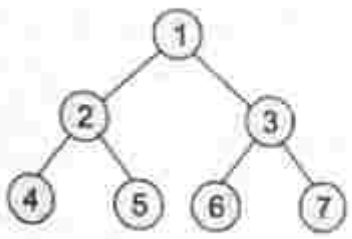
How many nodes will become unbalanced when a node is inserted as a child of node G?

(a) 2 (b) 3
(c) 4 (d) 5

Q.168 In delete operation of binary search tree, we need inorder successor (or predecessor) of a node when a node to be deleted where it has both left and right child. Which of the following is true about inorder successor needed in delete operation?

(a) Inorder successor is always either leaf node or a node with empty right child.
(b) Inorder successor maybe an ancestor of the node.
(c) Inorder successor is always a leaf node.
(d) Inorder successor is always either a leaf node or a node with empty left child.

Q.169 Consider the following tree : If the post order traversal gives $ab - cd^* +$ then the label of the nodes 1, 2, 3, ... will



(a) +, −, *, a, b, c, d
(b) a, −, b, +, c, *, d
(c) a, b, c, d, −, *, +
(d) −, a, b, +, *, c, d

Q.170 A complete binary tree with n non-leaf nodes contains

(a) $\log_2 n$ nodes
(b) $n + 1$ nodes
(c) $2n$ nodes
(d) $2n + 1$ nodes

Q.171 The number of possible binary search trees with 3 nodes is

(a) 12 (b) 13
(c) 5 (d) 15

Q.172 The inorder and preorder traversal of a binary tree are "*dbeafcg*" and "*abdecfg*" respectively. The post order traversal of a binary tree is

(a) *defgbca* (b) *edbfgca*
(c) *edbgfca* (d) *debfgca*

Q.173 A program takes an input a binary tree with 'n' leaf nodes and computes the value of function $g(x)$ for each node 'I'. If the cost of computing $G(x)$ = (Number of leaf nodes in left subtree of I-minimum number of node in right subtree of I), then the worst case time complexity of the program is

(a) $O(n)$ (b) $O(n^2)$
(c) $O(n \log n)$ (d) $O(\log n)$

Q.174 The depth of a complete binary tree with 'n' nodes is (log is to the base two) [**Note:** Assume height of root node is 0]

(a) $\log(n + 1) - 1$ (b) $\log(n)$
(c) $\log(n - 1) + 1$ (d) $\log(n) + 1$

Q.175 Which of the following is/are TRUE about a binary search tree?

(a) The left child is always lesser than its parent.
(b) The right child is always greater than its parent.
(c) The left and right sub-trees should also be binary search trees.
(d) In order sequence gives decreasing order of elements.

Q.176 A strictly binary tree with 12 leaves

(a) has exactly 25 nodes
(b) has atmost 23 nodes
(c) has exactly 23 nodes
(d) has atmost 25 nodes

Q.177 Consider the following code:

```
def find(T1, T2):
    if T1 is None or T2 is None:
        return
    if T1.data == T2.data:
        find(T1.left, T2.left)
        find(T1.right, T2.right)
    else:
        print("GATE")
    exit( )
```

Find when "GATE" will not be printed by the above program? (Assume two binary trees are passed into the function).

(a) If the two trees are same
(b) If the two trees are different
(c) If the two trees have the same levels
(d) If the two trees have different levels

Q.178 Consider the following statements:
1. Infix, Prefix and Postfix notations for expressing sum of A and B are $A + B$, $+AB$, and $AB+$, respectively.
2. AVL tree is a binary tree in which the difference in heights between the left and the right sub tree is not more than one for every node.
3. Stack data structure is used to save and retrieve information in reverse order.
4. Queue data structure is known as LIFO.

Which of the statements given above are correct?
(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2 and 4

Q.179 The number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree is
(a) 0 (b) 1
(c) 2 (d) 3

Q.180 A complete binary tree with the property that the value at each node is atleast as large as the values at its children is known as:
(a) Minheap (b) Binary search tree
(c) AVL Tree (d) Maxheap

Q.181 What is the maximum height of any AVL tree with 20 nodes? Assume that height of tree with single node is 0
(a) 3 (b) 4
(c) 5 (d) 6

Q.182 The following numbers are inserted into an empty Binary search tree in the given order 9, 2, 4, 8, 18, 11, 14. What is the height of the binary search tree (height is the maximum distance of a leaf node from root)? [**Note:** Height of root is 0]
(a) 2 (b) 3
(c) 4 (d) 6

Q.183 Which of the following is/are the advantage of balanced binary search tree, like AVL tree, compared to binary heap?
(a) Insertion takes less time.
(b) Deletion takes less time.
(c) Searching takes less time.
(d) Construction of binary heap takes less time than AVL tree.

Q.184 Which of the following data structure(s) is/are can be used to convert an infix notation to prefix notation. [Choose all the correct options]
(a) Stack (b) Queue
(c) B-Trees (d) Linked-list

Q.185 Which of the following statements is false?
(a) A tree with n nodes has $(n-1)$ edges.
(b) A labeled rooted binary tree can be uniquely constructed given its postorder and preorder traversal results.
(c) A complete binary tree with n internal nodes has $(n+1)$ leaves.
(d) The maximum number of nodes in a binary tree of height h is $(2^{h+1}-1)$.

Q.186 A binary search tree contains the values 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in pre-order and the values are printed out. Which of the following sequences is a valid output?
(a) 53124786 (b) 53126487
(c) 53241678 (d) None of these

Q.187 The number of leaf nodes in a rooted tree of n nodes, with each node having 0 or 3 children is
(a) $\frac{n}{2}$ (b) $\frac{(n-1)}{2}$
(c) $\frac{(n+1)}{2}$ (d) $\left\lceil \frac{(2n+1)}{3} \right\rceil$

Q.188 Consider a B tree of height h, minimum degree $t \geq 2$ that contains any n key, where $n \geq 1$. Which of the following is correct?
(a) $h \geq \log_t \frac{n+1}{2}$ (b) $h \leq \log_t \frac{n+1}{2}$
(c) $h \geq \log_t \frac{n-1}{2}$ (d) $h \leq \log_t \frac{n-1}{2}$

Q.189 Let $T(n)$ be the number of different binary search trees on n distinct elements.
Then $T(n) = \sum_{k=1}^{n} T(k-1)\, T(x)$, where x is
(a) $n-k+1$ (b) $n-k$
(c) $n-k-1$ (d) $n-k-2$

Q.190 Let LASTPOST, LASTIN and LASTPRE denote the last vertex visited in a postorder, inorder and preorder traversal. Respectively, of a complete binary tree. Which of the following is always true?

(a) LASTIN = LASTPOST
(b) LASTIN = LASTPRE
(c) LASTPRE = LASTPOST
(d) None of the above

Q.191 The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

(a) 2 (b) 3
(c) 4 (d) 6

Q.192 Level order traversal of a rooted tree can be done by starting from the root and performing

(a) Preorder traversal
(b) Inorder traversal
(c) Depth first search
(d) Breadth first search

Q.193 In a binary tree, the number of internal nodes of degree 1 is 5, and the number of internal nodes of degree 2 is 10. The number of leaf nodes in the binary tree is

(a) 10 (b) 11
(c) 12 (d) 15

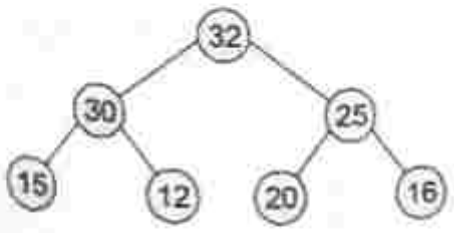
Q.194 How many distinct binary search trees can be created out of 4 distinct keys?

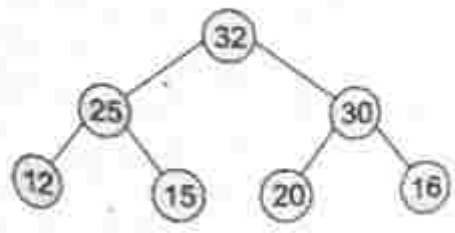
(a) 5 (b) 14
(c) 24 (d) 35

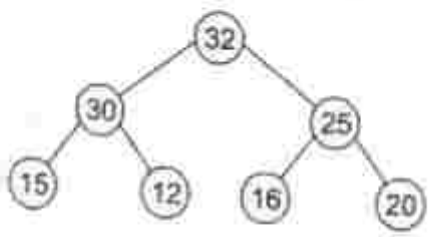
Q.195 In a complete k-ary, every internal node has exactly k children. The number of leaves in such a tree with n internal nodes is

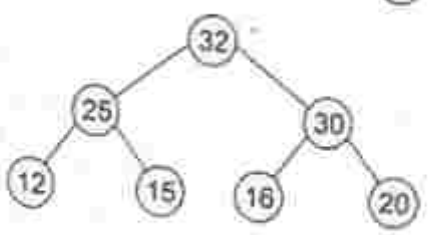
(a) nk (b) $(n-1)k+1$
(c) $n(k-1)+1$ (d) $n(k-1)$

Q.196 The elements 32, 15, 20, 30, 12, 25, 16 are inserted one by one in the given order into a maxHeap. The resultant maxHeap is

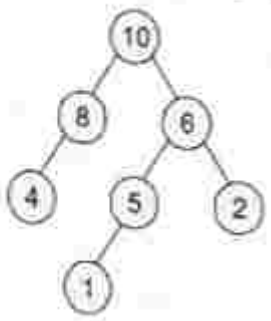
(a)


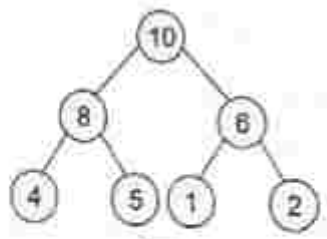
(b)


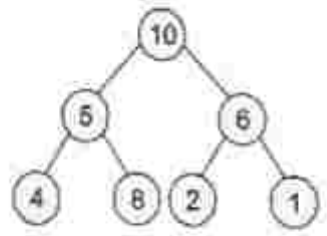
(c)


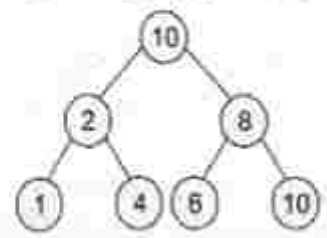
(d)


Q.197 A max-heap is a heap where the value of each parent is greater than or equal to the value of its children. Which of the following is a max-heap?

(a)


(b)


(c)


(d)


Q.198 Two balanced binary trees are given with m and n elements, respectively. They can be merged into a balanced binary search tree in _____ time.

(a) $O(m * n)$ (b) $O(m+n)$
(c) $O(m * \log n)$ (d) $O(m * \log(m+n))$

Q.199 We are given a set of n distinct elements and an unlabeled binary tree with n nodes. In how many ways can we populate the tree with the given set so that it becomes a binary search tree?

(a) 0 (b) 1
(c) $n!$ (d) $\frac{^{2n}C_n}{(n+1)}$

Q.200 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(a) 10, 20, 15, 23, 25, 35, 42, 39, 30
(b) 15, 10, 25, 23, 20, 42, 35, 39, 30
(c) 15, 20, 10, 23, 25, 42, 35, 39, 30
(d) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.201 A complete binary tree of level 5 has how many nodes? (Assume root node at level 0)

(a) 15 (b) 25
(c) 63 (d) 33

Q.202 The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

(a) 63 and 6, respectively
(b) 64 and 5, respectively
(c) 32 and 6, respectively
(d) 31 and 5, respectively

Q.203 How many distinct BSTs can be constructed with 3 distinct keys?

(a) 4 (b) 5
(c) 6 (d) 9

Q.204 Consider the traversal of a tree:
Preorder → ABCEIFJDGHKL
Inorder → EICFJBGDKHLA
Which of the following is correct post order traversal?

(a) EIFJCKGLHDBA (b) FCGKLHDBUAE
(c) FCGKLHDBAEIJ (d) IEJFCGKLHDBA

6. Hashing Techniques

Q.205 Consider a hashing function that resolves collision by quadratic probing. Assume the address space is indexed from 1 to 8. Which of the following locations will never be probed if a collision occurs at position 4?

(a) 4 (b) 5
(c) 8 (d) 2

Q.206 Given keys (x): 3, 23, 1, 0, 15, 31, 4, 7, 11, 43
M (Hash table size) = 11 (location 0 to location 10) and initially hash table is empty.
Find number of collisions occurred in hash table if x mod M hash function is used to map the given keys.

Q.207 A hash table has space for 100 records. What is the probability of collision before the table is 10% full?

(a) 0.45 (b) 0.5
(c) 0.3 (d) 0.34 (approximately)

Q.208 Suppose we used a hash function $H(n)$ to hash 'n' distinct elements (keys) into an array T of length 'm'. What is the expected number of colliding pairs of elements, if we used simple uniform hashing?

(a) $\Theta(n^2)$ (b) $\Theta(m^2)$
(c) $\Theta(n^2/m)$ (d) $\Theta(n^3/m^2)$

Q.209 A hash function f defined as f(key) = key mod 7, with linear probing, insert the keys 37, 38, 72, 48, 98, 11, 56.
Data 11 and 56 will be stored on at the locations _____ , _____ respectively (hash address starts from '0' and location start from '1')

(a) 6, 1 (b) 5, 1
(c) 5, 2 (d) 6, 2

Q.210 A hash function f is defined as f(key) = key mod 7 with chaining, used to insert keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0 to 6. What will be the location of key 11?

(a) 3 (b) 4
(c) 5 (d) 6

Q.211 Given a hash table with 20 slots that stores 2500 elements, the load factor for hash table is ___

Q.212 A hash table with 10 buckets with one slot per bucket is depicted in figure the symbols, S1 and S7 are initially entered using a hashing function with linear probing.
The maximum number of comparisons needed in searching an item that is not present is

0	S7
1	S1
2	
3	S4
4	S2
5	
6	S5
7	
8	S6
9	S3

(a) 4 (b) 5
(c) 6 (d) 3

Q.213 Assume the given hash table uses linear probing and hash function $h(x) = x \bmod 9$.

If keys 22, 11, 17, 13, 16, 12, 20 are inserted into the hash table in the order, find the total number of collisions occurred in the table. (Hash table locations are 0, 1, 2, ...,8)

Q.214 A hash table can store a maximum of 10 records. Currently there are records in locations 1, 3, 4, 7, 8, 9, 10. The probability of a new record going into location 2, with a hash function resolving collisions by linear probing is

(a) 0.6
(b) 0.1
(c) 0.2
(d) 0.5

Q.215 What is the average number of probes required when inserting an element into an open-address hash table with load factor a (assume uniform hashing)?

(a) $\frac{1}{1-\alpha}$ (b) $\frac{1}{\alpha}$

(c) $\frac{1}{1+\alpha}$ (d) $\frac{2}{2-\alpha}$

Q.216 Consider a hash table of size 11. Let $h(k) = k$ mod 11 be the hash function used. A sequence of records with keys:

(33, 68, 62,47, 48, 60, 104, 23, 19, 120, 97) is inserted into an initially empty hash table, the slots of which are indexed from 0 to 10. Which of the following is true?

(a) If collision are resolved by linear probing then maximum number of comparisons in worst case can be 2.
(b) If chaining is used to resolve collision, the average chain length will be 1.09.
(c) Both of these
(d) None of these

Q.217 Consider the hash table of size 12 that uses open addressing with linear probing. Let $h(k) = k$ mod12 be the hash function used. A sequence of records with keys 43, 63, 84, 11, 5, 72, 15, 16 is stored into an initially empty hash table, the bins of which are indexed from zero to 12. The index of the bin into which last element inserted is _____.

Q.218 Which of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for i ranging 0 to 2020?

(a) $h(i) = i^3 \bmod 10$
(b) $h(i) = (11 * i^2) \bmod 10$
(c) $h(i) = (12 * i) \bmod 10$
(d) $h(i) = i^2 \bmod 10$.

Q.219 Consider a hash table N slots. It is given that the collision resolution technique used is chaining. Assuming simple uniform hashing, what is the probability that the last k slots are unfilled after the first 'r' insertions?

(a) $\left(1-\frac{N}{k}\right)^r$ (b) $\left(1-\frac{k}{N}\right)^r$

(c) $\left(1+\frac{N}{k}\right)^{r-1}$ (d) $\left(1-\frac{k}{N}\right)^{r-1}$

Q.220 Which of the following is/are the collision resolution technique?

(a) Separate chaining(b) Linear probing
(c) Quadratic probing(d) Hashing

Q.221 Which of the following is/are the property that you expect a well designed hash function to have?

(a) The hash function should "spread out" most (i.e., "non-pathological") data sets (across the buckets/slots of the hash table).
(b) The hash function should be easy to store (constant space or close to it).
(c) The hash function should be easy to compute (constant time or close to it).
(d) The hash function should "spread out" every data set (across the buckets/slots of the hash table).

Q.222 Advantage of chaining over open addressing is

(a) less space is used
(b) worst case complexity of search operation is less
(c) deletion is easier (d) None of these

Q.223 The characters of the string K R P C S N Y T J M are inserted into a hash table of size of size 10 using hash function

If linear probing is used to resolve collisions, then the following insertion causes collision

(a) T, J (b) J, M
(c) M, P (d) P, T

Q.224 A hash function randomly distributes records one by one in a space that can hold x number of records. The probability that the m^{th} record is the first record to result in collision is
(a) $(x-1)(x-2)...(X-(m-2))(m-1)/X^{m-1}$
(b) $(x-1)(x-2)...(X-(m-1))(m-1)/X^{m-1}$
(c) $(x-1)(x-2)...(X-(m-2))(m-1)/X^{m}$
(d) $(x-1)(x-2)...(X-(m-2))(m-1)/X^{m}$

Q.225 The average search time of hashing, with linear probing will be less if the load factor
(a) is far less than one
(b) equals one
(c) is far greater than one
(d) None of these

Q.226 Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function x mod 10, which of the following statements are true?
1. 9679, 1989, 4199 hash to the same value
2. 1471, 6171 hash to the same value
3. All elements hash to the same value
4. Each element hashes to a different value
(a) 1 only (b) 2 only
(c) 1 and 2 only (d) 3 and 4 only

Q.227 An advantage of chained hash table (external hashing) over the open addressing scheme is
(a) Worst case complexity of search operations is less
(b) Space used is less
(c) Deletion is easier
(d) None of the above

Q.228 Which of the following statement(s) is TRUE?
I. A hash function takes a message of arbitrary length and generates a fixed length code.
II. A hash function takes a message of fixed length and generates a code of variable length.
III. A hash function may give the same hash value for distinct messages.
(a) I only (b) II and III only
(c) I and III only (d) II only

Q.229 Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?
(a) $(97 \times 97 \times 97)/100^3$
(b) $(99 \times 98 \times 97)/100^3$
(c) $(97 \times 96 \times 95)/100^3$
(d) $(97 \times 96 \times 95)/(3! \times 100^3)$

Q.230 The goal of hashing is to produce a search that takes
(a) O(1) time
(b) $O(n^2)$ time
(c) O(log n) time
(d) O(n log n) time

Q.231 An advantage of chained hash table over the open addressing scheme is
(a) Worst case complexity of search operation is less
(b) Space used is less
(c) Deletion easier
(d) None of the above

Q.232 A hash function f defined as f(key) = key mod 7, with linear probing, is used to insert the keys 37, 38, 72, 48, 98, 11, 56, into a table indexed from 0 to 6. What will be the location of key 11?
(a) 3 (b) 4
(c) 5 (d) 6

Q.233 Consider a hash table of size 11 that uses open addressing with linear probing. Let $h(k) = k$ mod 11 be the hash function used. A sequence of records with keys 43 36 92 87 11 4 71 13 14 is inserted into an initially empty hash table, the bins of which are indexed from zero to ten. What is the index of the bin into which the last record is inserted?
(a) 3 (b) 4
(c) 6 (d) 7

Answers Programming and Data Structures

1. (b) 2. (a, c, d) 3. (c) 4. (c) 5. (b) 6. (d) 7. (b)
8. (c) 9. (c) 10. (b, d) 11. (a) 12. (b, d) 13. (d) 14. (a, d)
15. (d) 16. (c) 17. (c) 18. (a) 19. (5) 20. (a, b, c) 21. (d)
22. (a) 23. (a) 24. (b) 25. (b) 26. (b) 27. (c) 28. (c)
29. (a, b, c, d) 30. (0) 31. (c) 32. (c) 33. (a, b, d) 34. (a) 35. (d)
36. (b) 37. (d) 38. (a) 39. (c) 40. (a, c) 41. (a) 42. (a)
43. (d) 44. (b) 45. (c) 46. (a, c, d) 47. (a) 48. (c) 49. (c)
50. (b) 51. (−12) 52. (d) 53. (c) 54. (d) 55. (a) 56. (b)
57. (b) 58. (b) 59. (a) 60. (a) 61. (c) 62. (b) 63. (a)
64. (a, c) 65. (2) 66. (d) 67. (15) 68. (b, c, d) 69. (1) 70. (10)
71. (a) 72. (a, c, d) 73. (23) 74. (4) 75. (1) 76. (c) 77. (b)
78. (8) 79. (a, d) 80. (c) 81. (d) 82. (a) 83. (b) 84. (a)
85. (a, b, c) 86. (11) 87. (a) 88. (c) 89. (b) 90. (93) 91. (c)
92. (a, c) 93. (b) 94. (b) 95. (a) 96. (b) 97. (d) 98. (a)
99. (a) 100. (a, c, d) 101. (b) 102. (a, c, d) 103. (4321) 104. (d) 105. (b)
106. (a) 107. (c, d) 108. (c) 109. (d) 110. (a, c) 111. (c) 112. (a, d)
113. (c) 114. (a, b, c, d) 115. (a) 116. (d) 117. (b, c) 118. (d) 119. (b)
120. (c, d) 121. (c) 122. (c) 123. (b, c) 124. (d) 125. (b) 126. (c)
127. (c) 128. (a, c, d) 129. (a) 130. (a) 131. (c) 132. (a) 133. (a)
134. (c) 135. (4) 136. (a, b, c) 137. (c) 138. (c) 139. (c) 140. (c)
141. (b) 142. (b) 143. (b) 144. (a) 145. (c) 146. (b) 147. (c)
148. (c) 149. (d) 150. (a, b, c) 151. (c) 152. (c) 153. (a, d) 154. (c)
155. (3) 156. (c) 157. (b) 158. (d) 159. (3) 160. (a) 161. (b)
162. (b) 163. (b) 164. (b, c, d) 165. (6) 166. (3) 167. (b) 168. (d)
169. (a) 170. (d) 171. (c) 172. (d) 173. (a) 174. (a) 175. (a, b, c)
176. (c) 177. (a) 178. (a) 179. (d) 180. (d) 181. (c) 182. (b)
183. (c) 184. (a, b, d) 185. (b) 186. (d) 187. (d) 188. (b) 189. (b)
190. (b) 191. (b) 192. (d) 193. (b) 194. (b) 195. (c) 196. (a)
197. (b) 198. (b) 199. (b) 200. (d) 201. (c) 202. (a) 203. (b)
204. (d) 205. (d) 206. (3) 207. (a) 208. (c) 209. (d) 210. (b)
211. (125) 212. (b) 213. (5) 214. (a) 215. (a) 216. (d) 217. (8)
218. (a) 219. (b) 220. (a, b, c) 221. (a, b, c) 222. (c) 223. (b) 224. (a)
225. (a) 226. (c) 227. (c) 228. (c) 229. (c) 230. (a) 231. (b)
232. (c) 233. (d)

Explanations Programming and Data Structures

1. (b)

C	G	A	T	E	C	S	I	T	2	0	2	6
	0	1	2	3	4	5	6	7	8	9	10	11

p	G	A	T	E	C	S	I	T	2	0	2	6
	0	1	2	3	4	5	6	7	8	9	10	11

p[2] = T – ASCII value of T = 84
p[6] = I – ASCII value of T = 73
Index = 84 – 73 – 1 = 10
str = c[index]
str = c[10]
Starting from index 10 onwards so, 2, 6
print(len (2, 6))

So, o/p = 2

2. (a, c, d)

- Python lists can store any type, including mixed types like complex and string.
- Python lists use dynamic arrays, not contiguous memory like C. So, option (b) is incorrect.
- The append( ) methods add the entire list one element at the end.
  So, [1, 2] append[3, 4] becomes [1, 2, [3, 4]].
- Lists support both positive and negative indexing.

3. (c)

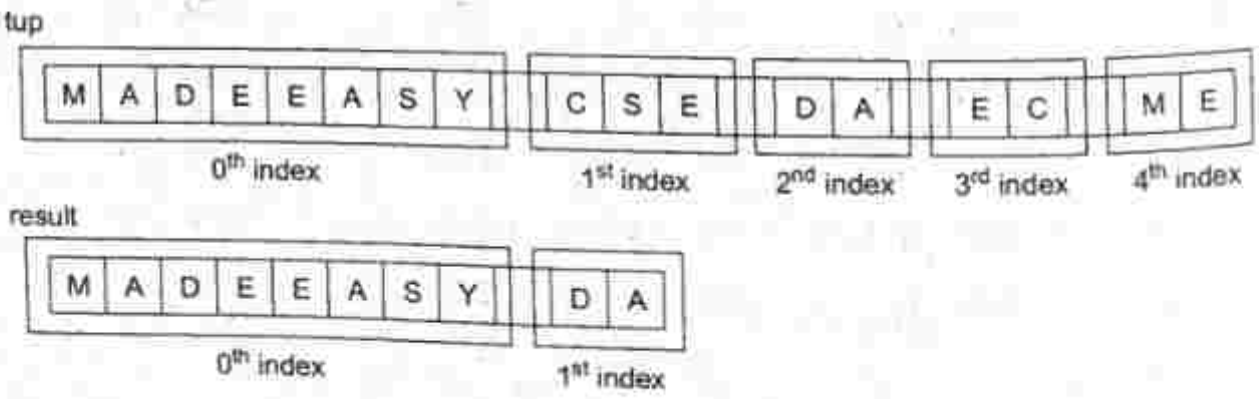
- list slicing, the stop index is execute, meaning the element at the stop position is not included in the resulting list.
- Assigning to a slice can add or remove elements.

4. (c)

1st	10	15	20	25	30
	1	2	3	4	5

x	10	15	30		
	1	2	3	4	5

5. (b)



While executing for loop condition.

Case 1: x = 10

x == 20 or x == 25
10 == 10 or 10 == 25
False False
False

So 10 will be stored in variable x.

Case 2: x = 15

x == 20 or x == 25
15 == 15 or 15 == 25
False False
False

So 15 will be stored in variable x.

Case 3: x = 20

x == 20
20 == 20
True

No required check the second condition.
So 20 could not stored in x.

Case 4: x = 25

x == 20 or x == 25
25 == 20 or 25 == 25
False True
True

Not stored in variable x.

Case 5: x = 30

x == 20 or x == 25
30 == 20 or 30 == 25
False False
False

So 30 will be stored in variable x.

So, o/p = 10, 15, 30

i = 0
0 == 0 or 0 == 2
True
So, Madeeasy add into result.
i = 1
1 == 1 or 1 == 2
False
Skip
i = 2
2 == 0 or 2 == 2
True
So, DA add into result.
i = 3 False Skip
i = 4 False Skip
So, MADEEASYDA.

6. (d)

i [5] 20 (because p and i refers the same list, so, i also becomse 20)
0^{th} index

j [15]

fun(i = [5], m = 10)
↓
fun(p = [5], m = 10)
m = 10 + 5 = 15
p[0] = 5 + 15 = 20
print(i[0] + j) ⇒ 20 + 10 = 30

So, o/p = 30

7. (b)

madeeasy(x = 1, y = 1)
↓
x = 0 [Since x is an integer (immutable), x = 0 only change the local variable inside the function, not the one passed from outside]
y = x 0
ptr = [0] = 0
y = ptr[0] = 0
ptr[0] = 1
ptr = [1]
x = ptr[0]
x = 1

So, o/p = (1, 0)

8. (c)

- remove( ) search from left to right and delete the first occurrence.
- clear( ) removes all contents but doesn't delete the list object.

9. (c)

k [12] 100 n [30] 4

12 > 15 and n := 4 [n := 4 is called largest operator in include in Python 3.8]
True True
True
Assign the value 4 to n, and the same time use this value in the conditon.

So, o/p = 100

10. (b, d)

- The in operator is used to test membership in a list.
  If the element exists, it returns True; otherwise, it returns false. So, option (a) is false.
- Lists are heterogeneous and can store elements of different types.
  Having mixed types in a list does not cause any run time error. So, option (b) is true
- Lists are passed by reference (object reference).
  Changes made to the list inside the function affect the original list. So, option (c) is false.
- Slicing with a negative step like list [:: -1] returns a reversed version.
  This is a simple and common way to reverse a list in Python. So, option (d) is true.

11. (a)

a [5] [The value of a does not charige because integer are immutable and n is a local variable inside the function]

funval(a = 5)
↓
funval(n = 5)
n = n + 10 = 15
print("Inside function", n)
↓
15
print("Outside function", a)
↓
5

So, o/p = (15, 5)

12. (b, d)

- { } creates an empty dictionary, not a set.
  To define an empty set, the correct syntax is set( ).
  So, option (a) is incorrect
- The issubset( ) method is used to test if a set is completely contained within another set.

It returns true if every element in the first set also exists in the second set.
This method is helpful in set theory operations and subset checks. So, option (b) is correct.

- The copy( ) method creates a shallow copy of the set.
  That means the new set is a separate object with the same elements.
  To create a true reference, simple assignment ($a = b$) must be used instead. So, option (c) is incorrect.
- Sets are optimized for fast membership checking using the in keyword.
  They are implemented using hash tables, which allow constant-time lookup.
  This makes sets ideal for scenarios where presence-checking is frequent. So, option (d) is correct.

13. (d)

In P1( )

y [10] 3 14 31

0

y [3] z [10] 31

func1(y = [3], x = [10], x = [10])

↓

func1(x = [3], y = [10], z = [10])

y[0] = 10 + 4 = 14

z[0] = 3 + 14 + 14 = 31

So, o/p = x[0] = 31, y[0] = 3

14. (a, d)

- The list( ) constructor accepts any iterable (like tuples, strings, sets, etc.).
  It returns a new list containing all elements from the given iterable. So, option (a) is true.
- Lists are mutable, meaning elements can be added, removed, or changed.
  Immutable types include tuples and strings, not lists. So, option (b) is false.
- Slicing a list like a[:] creates a shallow copy, not a reference.
  Changes to the sliced list do not affect the original list. So, option (c) is false.
- List comprehensions provide a concise syntax to create new lists using loops.
  They often include an optional condition for filtering elements. So, option (d) is true.

15. (d)

main( )

⇓

i [0 1]

j = incr(i = 0)	count = 0(init) + 0 → count = 0	j = 0
j = incr(i = 1)	count = 0 + 1 = 1	j = 1
j = incr(i = 2)	count = 1 + 2 = 3	j = 3
j = incr(i = 3)	count = 3 + 3 = 6	j = 6
j = incr(i = 4)	count = 6 + 4 = 10	j = 10

So, o/p = j = 10

16. (c)

- Lists can have duplicates.
- Lists can be used as values in dictionaries not keys.

17. (c)

x [1] y [2]

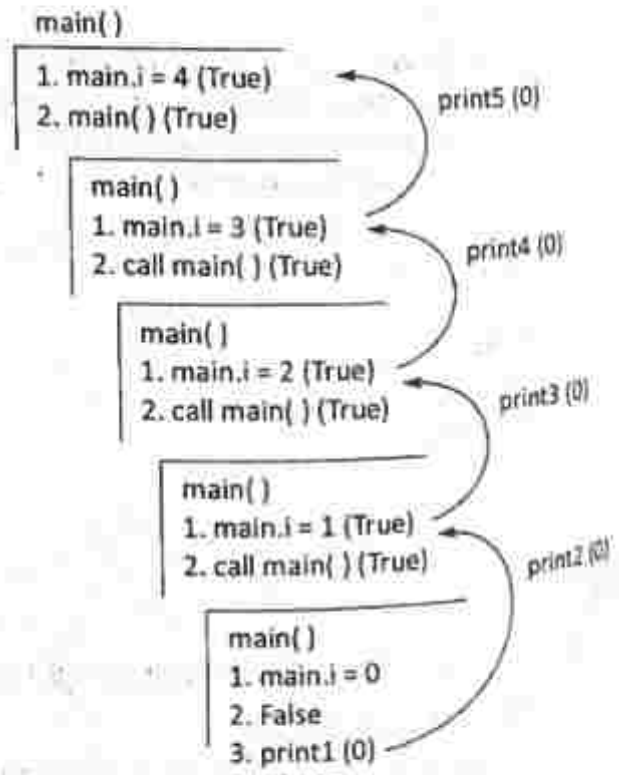
i [0 1 2 3 4]

x << 1 [x << 1 performs a left shift on x but does not update x unless the result is explicitly assigned back]

i = 0	x << 1 = 1	y = 1 + 0 = 1
i = 1	x << 1 = 1	y = 1 + 1 = 2
i = 2	x << 1 = 1	y = 1 + 2 = 3
i = 3	x << 1 = 1	y = 1 + 3 = 4
i = 4	x << 1 = 1	y = 1 + 4 = 5

So, x = 1, y = 5

19. (5)



So, '0' will be prints 5 times.

20. (a, b, c)

- Python supports nested lists, where a list can contain other lists as elements.
  This is commonly used to create 2D structures like matrices.
  Example: nested = [1, 2, [3, 4]] → [3, 4] is a list inside a list. So, option (a) is true.
- The del keyword removes the variable reference from the current namespace.
  If no other references exist, the list object is garbage collected.
  Trying to access the variable afterward will raise a NameError. So, option (b) is true.
- The index( ) method searches the list from left to right for the first match.
  It returns the index of the first occurrence of the given value.
  If the item is not found, it raises a ValueError. So, option (c) is true.
- count( ) does not modify the list or remove duplicates.
  It only returns how many times a specified item occurs in the list.
  So, option (d) is false.

21. (d)

First call

```
pr( )
1. False
2. pr.i = 2
3. chr(65 + 1) = chr(66) = 'B' print = B
4. pr.i = 2
```

Second pr( ) call

```
1. True
2. chr(65 + 2) = chr(67) = 'C' print = C
3. pr.i = 3
```

Third pr( ) call

```
1. True
2. chr(65 + 3) = chr(68) = 'D' print = D
3. pr.i = 4
```

So, o/p = BCD

22. (a)

<< and >> are bit wise operators used to multiply and divide by power of 2 respectively (shift operators)

∴ i << 3 ⇒ i * 8

j >> 2 ⇒ j / 4

23. (a)

- Two variables can point to the same list object.
- is operator checks object identity not element equality.

24. (b)

x [1] y [2]

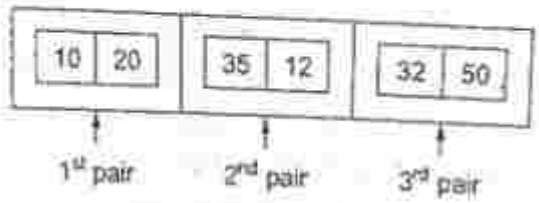
i	Before x = x << 1	After shift	i % 2 == 0	y	Updated y
0	x = 1	x = 2	True	y = y + 0	y = 2
1	x = 2	x = 4	False	y = y − 1	y = 1
2	x = 4	x = 8	True	y = y + 2	y = 3
3	x = 8	x = 16	False	y = y − 3	y = 0
4	x = 16	x = 32	True	y = y + 4	y = 4

So, o/p = x = 32, y = 4

25. (b)

- Set difference is not symmetric (i.e., A − B ≠ B − A).
- Symmetric difference (A ^ B) includes items only in one of the sets.

26. (b)



After 1st for loop iteration.
Take the maximum from the 1st pair.
So, 20 will be printed.
After 2 for loop iteration.
Take the maximum from the 2nd pair.
So, 35 will be printed.
After 3rd for loop iteration.
So, 50 will be printed.

So, output | 20 | 35 | 50 |

27. (c)

Initial values:

max_sum = −∞ (very small number)

max_pair = ( )

Loop iterations:

Iteration	a, b	s = a + b	Compare s > max_sum	Updated max_sum	Updated max_pair
1	(1, 2)	3	3 > −∞ → True	3	(1, 2)
2	(3, 1)	4	4 > 3 → True	4	(3, 1)
3	(0, 5)	5	5 > 4 → True	5	(0, 5)
4	(4, 0)	4	4 > 5 → True	(no change)	(no change)

28. (c)

Iteration	Value of i	i % == 0	Action taken	Output
1	i = 0	0 % 2 = 0	continue → skip	
2	i = 1	1 % 2 = 1	print(1)	1
3	i = 2	2 % 2 = 0	continue → skip	
4	i = 3	3 % 2 = 1	print(3)	13
5	i = 4	4 % 2 = 0	continue → skip	
6	i = 5	5 % 2 = 1	print(5)	135

29. (a, b, c, d)

- Tuples are immutable. Once created, you cannot change, add, or remove their elements. This immutability makes tuples safe and hashable. So, option (a) is true.
- tuples are heterogeneous, they can store any data type.
  Example: (10, "Hello", 3.5) is a valid tuple. Python does not require uniform data types in a tuple. So, option (b) is true.
- Tuples actually use less memory than lists. Because of immutability, Python can optimize memory for tuples better.
  Tuples are more memory efficient and faster in iteration. So, option (c) is true.
- Tuples support both positive and negative indexing, just like lists.
  Example: $t = (10, 20, 30)$; $t[1] \rightarrow 20$, $t[-1] \rightarrow 30$.
  Indexing allows direct access to any element. So, option (d) is true.

30. (0)

(n = n − n1)

12 0 (10 − 10) = 0

11 10 10

m 10 n 11 n1 11

So, n = 0

o/p = 0

32. (c)

1st | 1 | 3 | 5 | 7 | 9 |

For loop execution

val = 1

1 % 3 == 0

1 == 0 [Not divisibly by 3] ⇒ print(1)

val = 3

3 % 3 == 0

0 == 0 (skip)

val = 5

5 % 3 == 0

2 == 0 ⇒ 2 ≠ 0 print(5)

val = 7

7 % 3 == 0

1 ≠ 0 print(7)

val = 9

9 % 3 == 0

0 = 0 (skip)

So, o/p = 1, 5, 7

33. (a, b, d)

- A single-element tuple needs a trailing comma: (3,). So, option (a) is correct.
- Tuples can be concatenated with +. So, option (b) is correct.
- The * operator can be used with tuples. It repeats the tuple a specified number of times.
  Example: (1, 2) * 2 → (1, 2, 1, 2).
  So, option (c) is incorrect.
- Only if all elements inside the tuple are hashable, the tuple itself is hashable. This allows such tuples to be used as dictionary keys or in sets.
  Example: (1, 'a') is hashable, but ([1], 2) is not (because list is unhashable).
  So, option (d) is correct.

34. (a)

- Because setdefault( ) returns a default if the key is missing. So, S_1 is true.
- Values( ) returns a view of values, not a list. So, S_2 is false.

35. (d)

Initial value

x 1 y 2

i	Before x << = 1	After x << = 1 (x * 2)	y = x + i
i = 0	x = 1	x = 2	y = 2 + 0 = 2
i = 1	x = 2	x = 4	y = 4 + 1 = 5
i = 2	x = 4	x = 8	y = 8 + 2 = 10
i = 3	x = 8	x = 16	y = 16 + 3 = 19
i = 4	x = 16	x = 32	y = 32 + 4 = 36

So, o/p = x = 32, y = 36

Note: x << 1 shift all bits of x one position to the left.

36. (b)

Iteration	v	Condition (v > 10)	Action taken	Total after action
1	2	2 > 10 → False	Total = 0 + 2	2
2	5	5 > 10 → False	Total = 2 + 5	7
3	8	8 > 10 → False	Total = 7 + 8	15
4	11	11 > 10 → True	break → loop ends	15
5	14	Not reached	Loop already broke	–

The fifth value (v = 14) is never processed because the loop broke at v = 11 when the condition 10 became true.

37. (d)

data	4	7	10	13	16

result ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 10

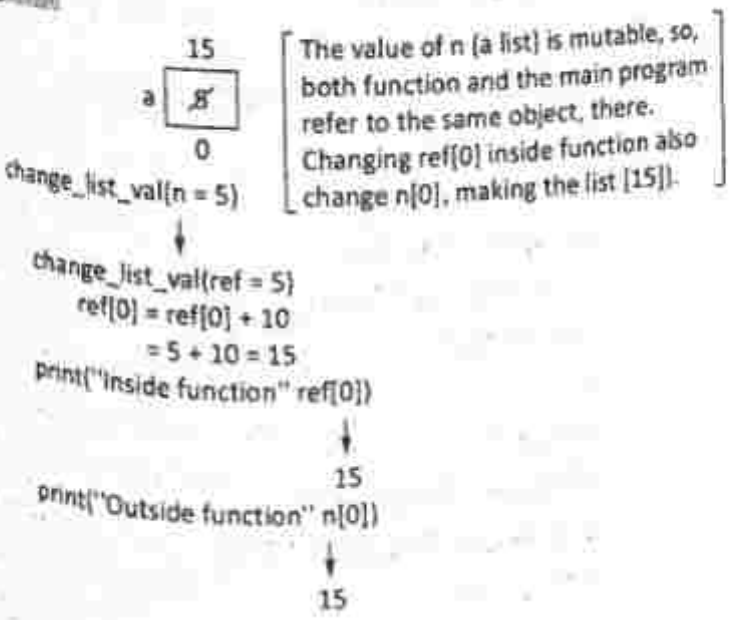
num	4	7	10	13	16

4 % 2 == 0 ⇒ 0 == 0 True
7 % 2 == 0 ⇒ 1 ≠ 0 False
10 % 2 == 0 ⇒ 0 == 0 True
13 % 2 == 0 ⇒ 1 ≠ 0 False
16 % 2 == 0 ⇒ 0 == 0 True

38. (a)

- Tuples themselves cannot be changed after creation.
  However, they can hold mutable elements, like lists or dictionaries.
  Example: $t = (1, [2, 3])$ is valid, and the list inside t can be modified. So, S_1 is correct.
- For a tuple to be hashable, all its elements must also be hashable.
  If it contains a mutable object (like a list), the tuple becomes unhashable. So, S_2 is Incorrect.

39. (c)



40. (a, c)

- Dictionaries are mutable.
- Keys must be immutable (so lists can't be used, but tuples can).
- Dictionary comprehensions allow creating dictionaries.
- Dictionaries can store None as a value.

41. (a)

vals	5	10	15	20	25
	0	1	2	3	4

In the for loop statement 2 elements skip.
It start at i = 0 and skips 2 elements at a time.
Then, i values are

i = 0, 2, 4

i = 0
vals[0] % 10 == 3
5 % 10 == 0
5 ≠ 0 so, print(5)

i = 2
vals[2] % 10 == 0
15 % 10 == 0
5 ≠ 0 print(15)

i = 4
vals[4] % 10 == 0
25 % 10 == 0
5 ≠ 0 print(25)

So, o/p = 51525

42. (a)

- Dictionary keys must be unique.
  So, S_1 is true.
- Each key in a dictionary maps to only one value.
  So, S_2 is false.

43. (d)

Iteration	i value	If condition check (i % 4 == 0)	Action on res	res after action
1	2	2 % 4 = 2 ≠ 0 → False	res = res + 1 = 1 + 1	2
2	4	4 % 4 = 0 → True	res = res * 2 = 2 * 2	4
3	8	8 % 4 = 0 → True	res = res * 2 = 4 * 2	8
4	16	16 % 4 = 0 → True	res = res * 2 = 8 * 2	16

44. (b)

1. var 5
2. main( )
   1. global var 5
   2. print(5)
   - foo( )
     1. True
     2. foo_var 10
     3. print(10)
3. print(var) ⇒ print(5)

So, o/p = 5, 10, 5

45. (c)

main( )
1. func( )
   1. False
   2. func.i = 1
   3. j = 4
   4. print(1)
   5. func.i = ~~X~~ 2
   6. print(4)
   7. j = ~~X~~ 5

main( )
2. func( )
   1. True
   2. j = 4
   3. print(func.i) = 2
   4. func.i += ~~X~~ 3
   5. print(4)
   6. j = ~~X~~ 5

So, o/p = 1424

46. (a, c, d)

- Sets are mutable.
- Sets can't contain other sets.
- remove( ) raises KeyError if element not found.
- discard( ) does nothing if the item isn't present.

47. (a)

- Tuple( ) functions convert any iterable into tuple.
  So, S_1 is correct.
- Tuples actually consumes less memory than list.

48. (c)

String	M	a	d	e	e	a	s	y
	0	1	2	3	4	5	6	7

Set	M	a	d	e	s	y
	0	1	2	3	4	5

i = 0
- w[0] = 'm'
- w[: 0] = " "
- M not in " " True
- Add 'M' to set = {'M'}

i = 1
- w[1] = 'a'
- w[: 1] = "M"
- 'a' not in "M " True
- Add 'a' to set = {'M', 'a'}

i = 2
- w[2] = 'd'
- w[: 2] == "Ma"
- 'd' not in "Ma " True
- Add 'd' to set = {'M', 'a', 'd'}

i = 3
- w[3] = 'e'
- w[: 3] == "Mad"
- 'e' not in "Mad " True
- Add 'e' to set = {'Made'}

i = 4
- w[4] = 'e'
- w[: 4] == "Made"
- 'e' in "Made " True
- Not added (but 'e' is already here)

i = 5
- w[5] = 'a'
- w[: 5] == "Madee"
- 'a' in "Madee " True
- Not added

i = 6
- w[6] = '6'
- w[: 6] == "Madeea"
- 's' is not in "Madeea " True
- Add to set

i = 7
w[7] = 'Y'
w[: 7] == "Madeeas"
'Y' not in "Madeeas"
Add to set "Madeeasy"

51. (c)

String

M	A	D	E	E	A	S	Y	D	A
1	2	3	4	5	6	7	8	9	10

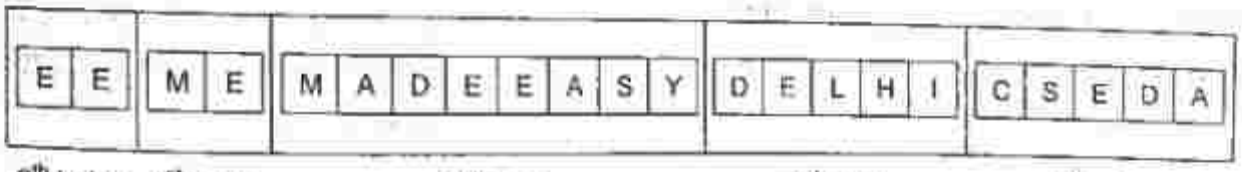
Character-by-character processing table:

i	char	In Set?	Action	Set after step
0	(space)	No	Add	{' '}
1	M	No	Add	{' ', M}
2	A	No	Add	{' ', M, A}
3	D	No	Add	{' ', M, A, D}
4	E	No	Add	{' ', M, A, D, E}
5	E	Yes	Remove	{' ', M, A, D}
6	A	Yes	Remove	{' ', M, D}
7	S	No	Add	{' ', M, D, S}
8	Y	No	Add	{' ', M, D, S, Y}
9	(space)	Yes	Remove	{M, D, S, Y}
10	D	Yes	Remove	{M, S, Y}
11	A	No	Add	{M, S, Y, A}

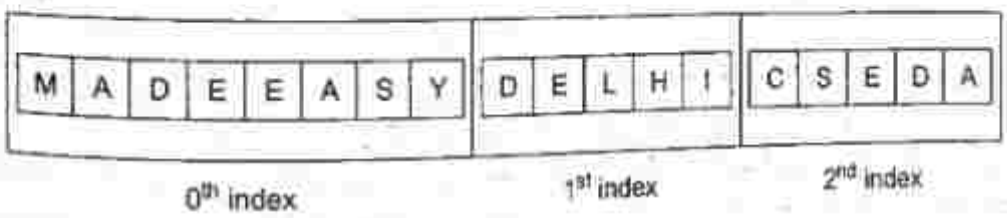
Final set → {M, S, Y, A}
Sorted → "AMSY"

52. (d)

tup



result



For loop execution:
Iteration 1:
len(EE) > 3
2 > 3 False skip
Iteration 2:
len(ME) > 3
2 > 3 False skip
Iteration 3:
len(MADEEASY) > 3
8 > 3 True add to result
Iteration 4:
len(DELHI) > 3
5 > 3 True add to result
Iteration 5:
len(CSEDA) > 3
5 > 3 True add to result
So, o/p : MADE EASY DELHI CSE DA

53. (c)

- A tuple is immutable, but its variable can point to a new object.
- Tuple unpacking assign multiple variables a single line each element of the tuples in assigned to multiple variables.

50. (b)

- Pop( ) removes a specified key-value pair, not random. So, S_1 is false.
- Update( ) merges dictionaries.

51. (–12)

The output will be
= –16 – (7 – (14 – (–4 – (–12 + 5))))
= –12

54. (d)

p [2] q [3] r [0]

if p and q:
−2 and 3 (both non-zero means true)
True

s [2]

t [2] = −2 + 3 + 1 ⇒ 4 − 2 = 2

55. (a)

x [5] y [7] z [2]

5 % 2 == 0
1 == 0 (false) else part will be execute
result = y − 2 = 7 − 2 = 5

56. (b)

p [2] q [0]

8 1

(2 << 2) and (0 or 1) T(both are true)

⇓ ⇓ ⇓

2-shift left by 2 bits return(1) returns the 2nd value
2 × 4 = 8 So, '1' will printed.

57. (b)

- Parentheses are not mandatory; commas define a tuple.
- 1, is a one-element tuple.

58. (b)

a [300] b [20] c [20]

300 200

if not (300 > = 400)

⇓

not (False) ⇒ True

So. o/p = 300, 200

60. (a)

c [0] d [5] e [10]

if c > 1
o > 1 False

[Here outer if false and we directly go to the else block]
No need to check d > 1 or e > 1 because outside if is false.
Never reached

a [300]

61. (c)

- Slicing a tuple like t[1 : 3] returns a new tuple with selected elements.
  Tuples are immutable, so slicing doesn't modify the original.
  Example: (10, 20, 30, 40)[1 : 3] → (20, 30)
  So, S_1 is correct.
- The index( ) method returns the first position of the specified item.
  If the item is not found, it raises a ValueError.
  Example: (1, 2, 3, 2).index(2) → 1. So, S_2 is correct.

62. (b)

tup

G	A	T	E	2	0	2	5	G	A	T	E	2	0	2	6

0th index 1st index

M	A	D	E	E	A	S	Y	D	A	I	E	S	2	0	2	4

2nd index 3rd index 4th index

foo(tup)

↓

len(tup) = 5 > 2 True
return tup[1 : −1]

⇓

Means start from index 1 and include elements upto (but not including) the last element

So, GATE 2026 MADE EASY DA stored in the new tuple.

Input	Postfix Expression	Stack
3	3	
*	3	*
log	3	*log
(	3	*log(
x	3x	*log(
+	3x	*log(+
1	3x1	*log(+
)	3x1+	*log
–	3x+log*	–
a	3x1+log* a	–
/	3x1+log* a	–/
2	3x1+log* a2	–/

Final output → 3x+ log* a2/ –

64. **(a, c)**

Infix to postfix conversion using stack will have run time complexity of $O(n)$.

For every opening brace, push it into the stack, and for every closing brace, pop it off the stack. Do not take action for any other character. In the end, if the stack is empty, then the input has balanced parentheses.

65. **(2)**

$ga + t - ee * x / a * m / + r - e + s - u + l - t +$

∴ Substring: /+ has only operators and two distinct operators.

Any other substring with only operators contains only one length.

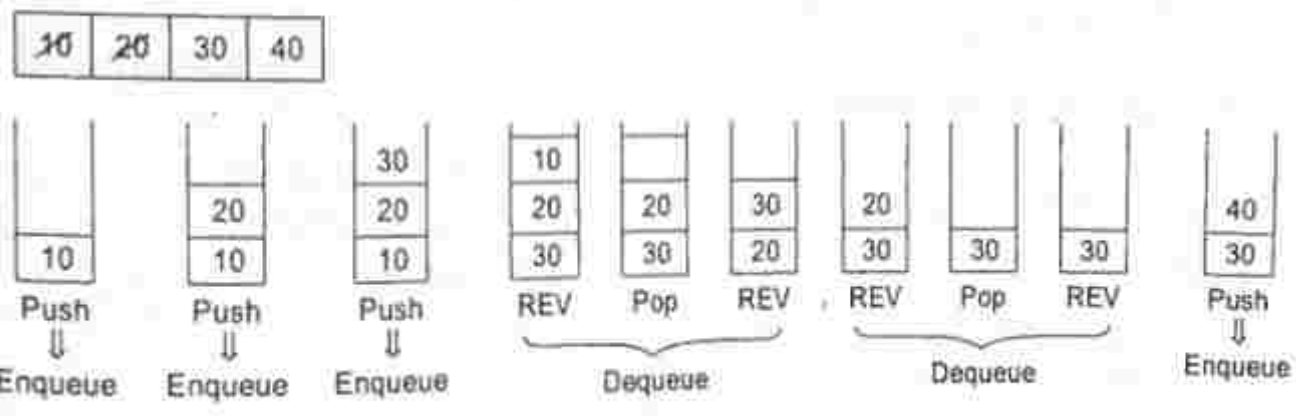
∴ 2 is length of substring.

(d)

Enqueue: PUSH ⇒ 1 operation

Dequeue: REVERSE, POP, REVERSE ⇒ 3 operation

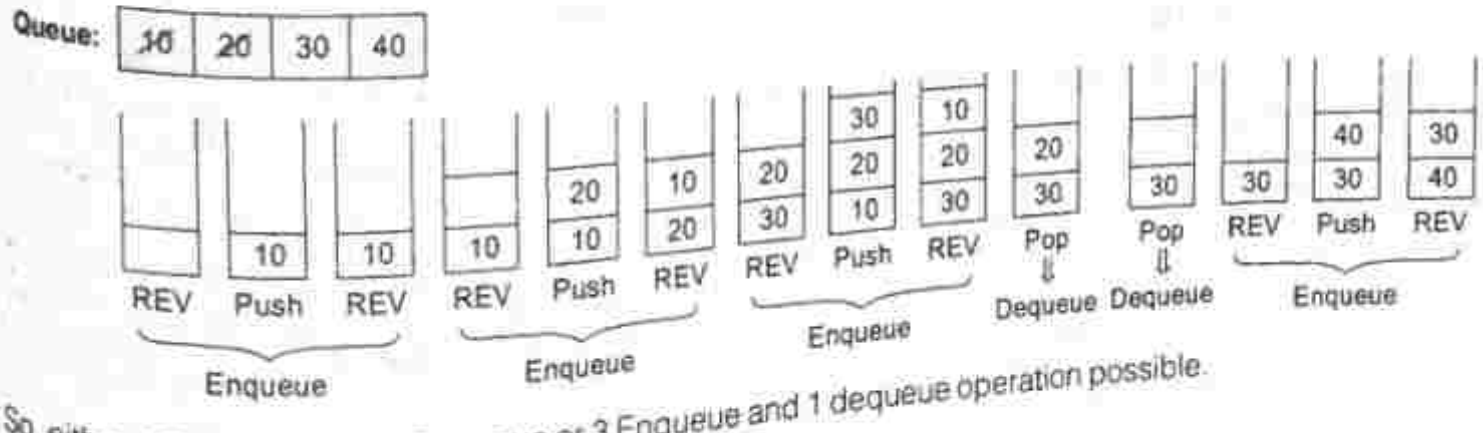
Example: Enqueue (10), Enqueue (20), Enqueue (30)
Dequeue, Dequeue, Enqueue (40)



Enqueue: REV, PUSH, REV

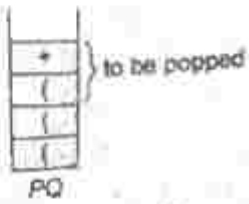
Dequeue: POP

Example: Enqueue (10), Enqueue (20), Enqueue (30)
Dequeue, Dequeue, Enqueue (40)

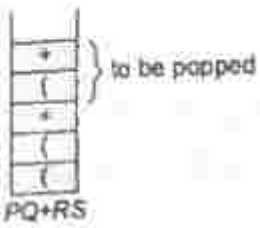


So, either 1 Enqueue and 3 Dequeue or 3 Enqueue and 1 dequeue operation possible.

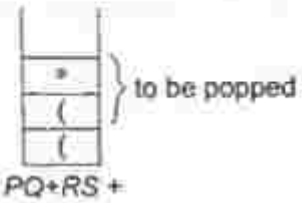
67. (15)



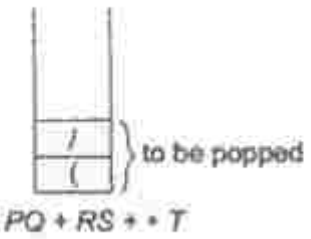
Now ') ' comes in Infix exp, pop up to first occurrence of ' ('



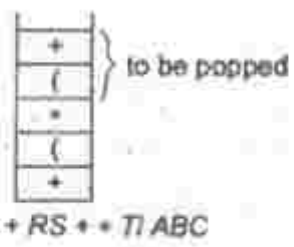
Now ') ' comes in Infix expression pop upto occurrence of ' (' .



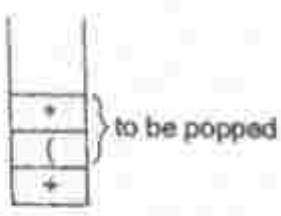
Again ') ' comes pop till first ' (' comes



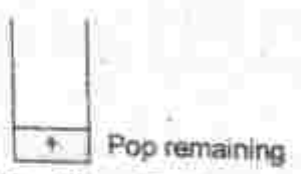
Now ') ' comes pop all elements



Now, ') ' comes, pop upto ' ('



Now ') ' comes pop until ' (' comes maximum possible stack height is 5



$PQ + RS + * T/ABC + *$

$\Rightarrow PQ + RS + *T/ABC + * +$

in the postfix expression.

∴ The sum of all unique possible heights

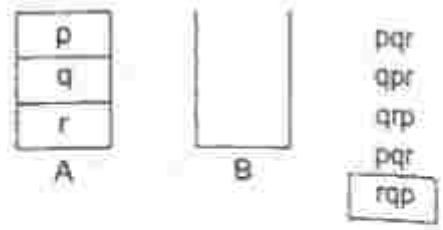
$= 1 + 2 + 3 + 4 + 5 = 15$

68. (b, c, d)

All prefix operators use values to their right and all postfix operators use values to their left.

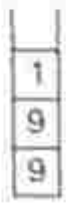
69. (1)

The following are possible stack permutations

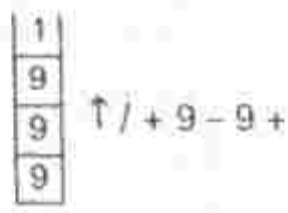


Total stack permutations possible with the given sequence is 5. Here the least number of permutations starting with particular letter is 1 (Starting with *r*).

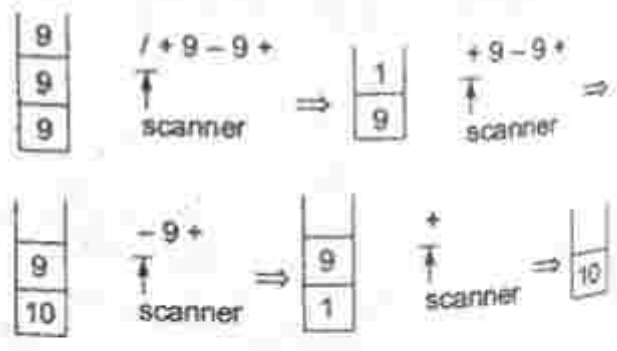
70. (10)



Push the operands one by one when operator comes, POP the top two operands, evaluate the value and then push result on to the stack.



Scanner scans power operator then pop 1 and 9, evaluate 9^1 and then push result back on the stack.



71. (a)

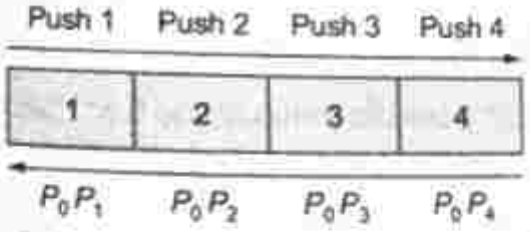
$A\$B^*C-D+E/F/(G+H)$

	Symbol	Postfix expression	Stack top
(1)	A	A	∞
(2)	$	A	$
(3)	B	AB	$
(4)	*	$AB\$$	*
(5)	C	$AB\$C$	*
(6)	–	$AB\$C*$	–
(7)	D	$AB\$C*D$	–
(8)	+	$AB\$C*D-$	+
(9)	E	$AB\$C*D-E$	+
(10)	/	$AB\$C*D-E$	+ /
(11)	F	$AB\$C*D-EF$	+ /
(12)	/	$AB\$C*D-EF/$	+ /
(13)	(	$AB\$C*D-EF/$	+ / (
(14)	G	$AB\$C*D-EF/G$	+ / (
(15)	+	$AB\$C*D-EF/G$	+ / (+
(16)	H	$AB\$C*D-EF/GH$	+ / (+
(17)	)	$AB\$C*D-EF/GH+/+$	

72. (a, c, d)

Since stacks are the LIFO data structure thus option (b) is clearly false.

73. (23)



Time for 1 stack operation = 5 ns

Total number of push = 4

Total number of pop = 4

Stack life of element 1

$= 5\text{ ns} \times 3 + 5\text{ ns} \times 3 + 7 \times 2\text{ ns}$

$= 15\text{ ns} + 15\text{ ns} + 14\text{ ns} = 44\text{ ns}$

Stack life of element 2

$= 5\text{ ns} \times (2 \times 2) + 5 \times 2\text{ ns}$

$= 20\text{ ns} + 10\text{ ns} = 30\text{ ns}$

Stack life element 3

$= 5\text{ ns} \times (1 + 1) + 3 \times 2\text{ ns}$

$= 10\text{ ns} + 6\text{ ns} = 16\text{ ns}$

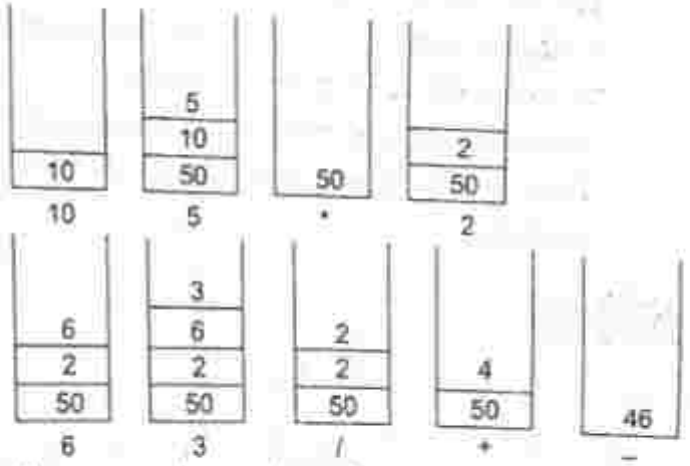
Stack life of element 4

$= 5\text{ ns} \times (0 + 0) + 2\text{ ns} \times 1 = 2\text{ ns}$

∴ Average stack life

$$= \frac{44+30+16+2}{4} = 23\text{ ns}$$

74. (4)



Hence, maximum size of stack needed is 4.

75. (1)

Considering, both the statements:

-

∴ With a single array, two stacks can be implemented.

- Implement a queue using 2 stacks. Denote the two stacks S_1 and S_2. The enqueue operation is simply implemented as a push on S_1 which will be in O(1) time. But, while performing dequeue operation, element of S_1 will be moved to S_2, then pop from S_2 and push to S_1, each element of S_2 which takes $O(n)$ time. It can be vice-versa too i.e., enqueue in $O(n)$ and dequeue O(1) time.

76. (c)

$A - B / (C * D \$ E)$

	Symbol	Postfix expression	Stack top
			)
(1)	)		)
(2)	E	E	$)
(3)	$	E	$)
(4)	D	DE	*)
(5)	*	$\$DE$	*)
(6)	C	$C\$DE$	
(7)	(	$*C\$DE$	/
(8)	/	$*C\$DE$	/
(9)	B	$B*C\$DE$	/B*C$DE-
(10	)	$-$	
(11)	A	$-A/B*C\$DE$	

Hence option (c) is correct.

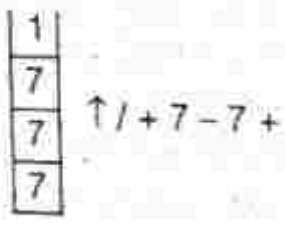
77. (b)

Stack is used to implement recursion stack follows LIFO mechanism (Last in first out). In recursion, same function calls itself and when the last call terminates, it returns value to the parent caller function and parent caller function returns value to its parent caller function and so on.

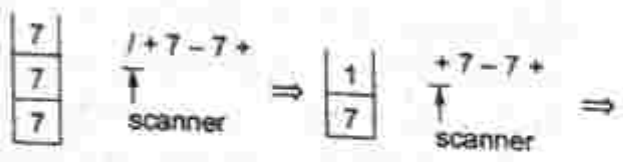
The return pattern of recursion, it also follows LIFO. That is why stack is used for implementing recursion.

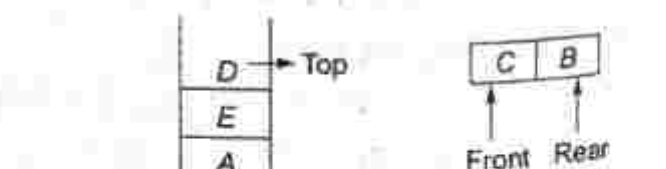
78. (8)

Push the operands one by one when operator comes, POP the top two operands, evaluate the value and then push result on to the stack.



Scanner scans power operator then pop 1 and 7, evaluate 7^1 and then push result back on the stack.





79. (a, d)

Removing items from an empty stack is termed as stack underflow.

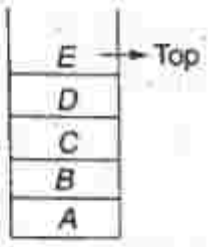
Adding items to a full stack is termed as stack overflow.

80. (c)

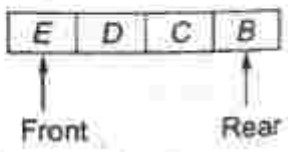
In each possible permutation 6 will become before 5. So only option (c) is correct.

81. (d)

(I) Initially, elements pushed to stack



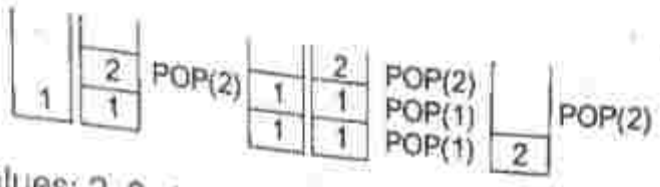
(ii) Pop from tack and insert into queue



(iii) Deque 2 times

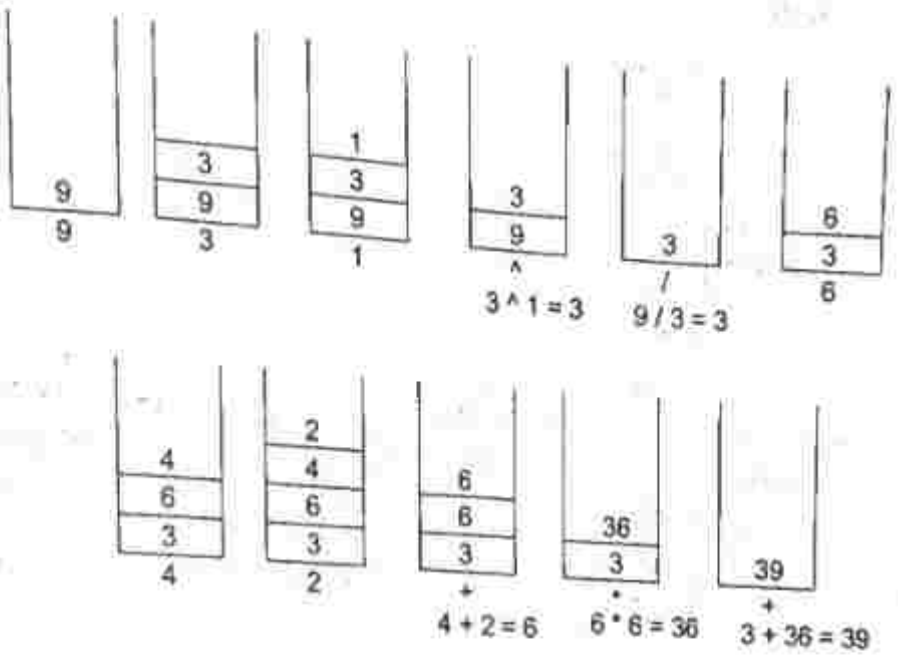
Popped item from stack → D

82. (a)



Sequence of popped out values: 2, 2, 1, 1, 2.

83. (b)



84. (a)

Input	Stack	Output
a		*a*
↑	↑	*a*
b	↑	*ab*
*	↑*	*ab*
c	↑*	*abc*
*	↑*	*abc*
d	↑*	*abc*d*
+	↑+	*abc*d**
e	↑+	*abc*d*e*
*	↑+*	*abc*d*e*
f	↑+*	*abc*d*ef*
↑	↑+*↑	*abc*d*ef*
g	↑+*↑	*abc*d*efg*

Final output: *abc* d* efg* ↑ * + ↑

Maximum size of operator stack ⇒ 4.

85. (a, b, c)

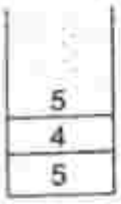
Line at ticket counter is an application of queue whereas conversion of infix to postfix expression, balancing symbols, evaluation of postfix expression are stack applications.

86. **(11)**

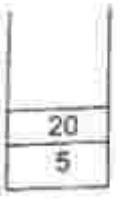
Push (5)

Push (4)

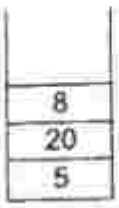
Push (5)



(*) 4 * 5 = 20
Push (20)

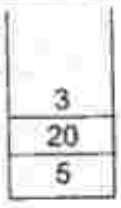


Push (8)

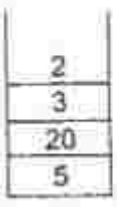


Push (5)

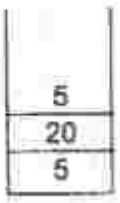
(–) 8 – 5 = 3
Push (3)



Push (2)



(+) 3 + 2 = 5
Push (5)



(/) 20/5 = 4
Push (4)

(–) 5 – 4 = 1
Push (5)

Total push → 11

87. **(a)**

Stack is LIFO system so FIFO not supported by stack hence option statement 1 is incorrect and queue is using FIFO so statement 4 is also wrong.

88. **(c)**

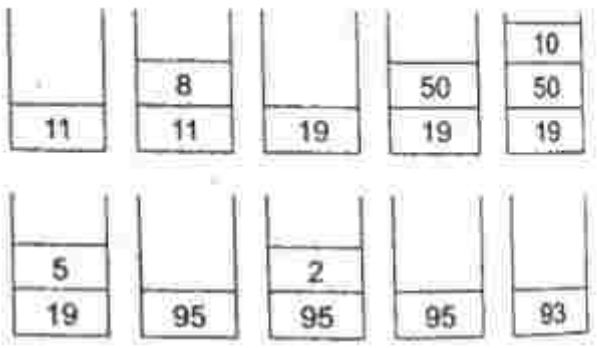
Input	Stack	Postfix
(	(	
a	(	*a*
+	(+	*ab*
)		*ab* +
↑	↑	*ab* +
(	↑(	*ab* +
P	↑(	*ab* + *p*
+	↑(+	*ab* + *p*
q	↑(+	*ab* + *pq*
)	↑	*ab* + *pq* +
↑	↑↑	*ab* + *pq* +
(	↑↑(	*ab* + *pq* +
r	↑↑(	*ab* + *pq* + *r*
*	↑↑(*	*ab* + *pq* + *r*
S	↑↑(*	*ab* + *pq* + *rs*
*	↑↑(*	*ab* + *pq* + *rs**
t	↑↑(*	*ab* + *pq* + *rs* * *t*
)	↑↑	*ab* + *pq* + *rs* * *t* *

Postfix expression: *ab* + *pq* + *rs* * *t* * ↑↑

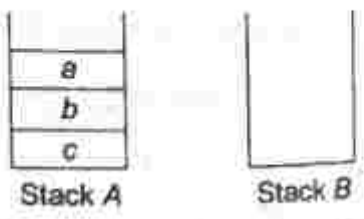
89. **(b)**

PUSH one by one element into Stack and POP accordingly desired output. Only second option can satisfy.

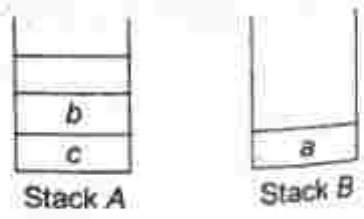
90. **(93)**



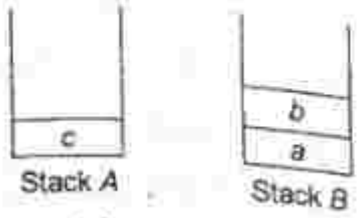
91. **(c)**



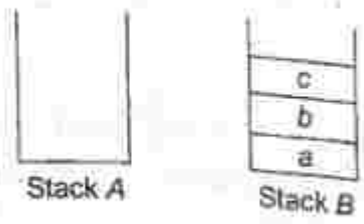
a is popped from stack *A* and pushed to stack *B*



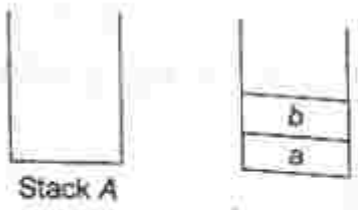
b is popped from stack *A* and pushed to stack *B*



c is also popped from stack A and pushed to stack B



c is popped from stack B and printed



Now, a can't be popped out and printed since top of stack is 'b'.

92. (a, c)

2 stacks are required for evaluation of prefix expression, one for integers and one for characters. During evaluation of infix expression, the operators with higher precedence are evaluated first, followed by operators with lower precedence.

93. (b)

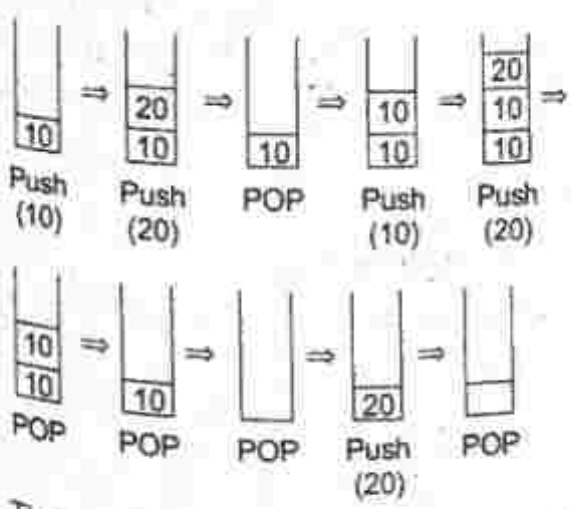
The item can be pushed to stack and later popped and printed, or printed directly. 1, 2, 3, 4, 5 is the input then

(a) is not possible because once pushed 1 is printed after 2.

(c) and (d) are not possible.

Hence, (b) is the output.

94. (b)



∴ The sequence of values popped out is 20, 20, 10, 10, 20.

95. (a)

Given postfix expression is

8 2 3 ^ / 2 3 * + 5 1 * –

Expression	Op1	Op2	Value	top S(RL)
8				8
2				8, 2
3				8, 2, 3
^	2	3	8	8, 8
/	8	8	1	1
2				1, 2
3				1, 2, 3
*	2	3	6	1, 6

So the top two elements of the stack are 6, 1 after the first * is evaluated.

96. (b)

The stack is a data structure in which each push (insert) or pop (delete) operation takes constant $\theta(1)$ time. The best data structure to check whether an arithmetic expression has a balanced parentheses.

Step 1: Start with empty stack of currently open parentheses.

Step 2: Process each char of an expression string.

Step 3: If it is open, push it on the stack.

Step 4: If it is closed, check it against the top of stack element.

Step 4.1: If stack empty or not a matching parentheses then not balanced and return false.

Step 4.2: Otherwise it matches pop the open parentheses then goto step 2.

Step 5: If stack is empty at the end, return true.

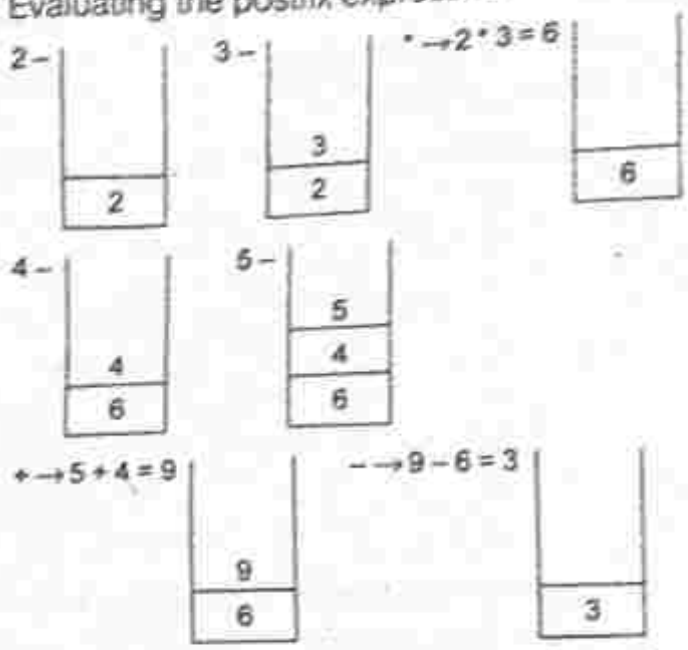
Step 6: If not empty then some parentheses is unmatched, return false.

97. (d)

First converting the infix expression to postfix.

Input	Stack	'Postfix'
2		2
*	*	2
3	*	23
–	–	23*
(	–(	23*
4	–(	23*4
+	–(+	23*4
5	–(+	23*45
)	–	23*45+
Postfix →	23*45+–	

Evaluating the postfix expression–

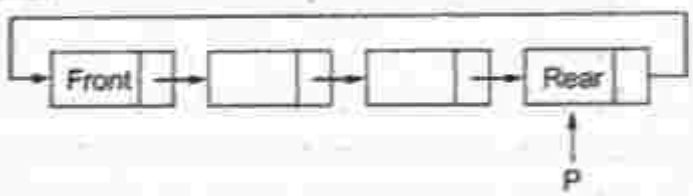


Hence (d) is not possible stack configuration.

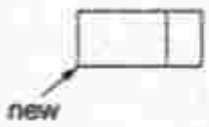
98. (a)

Operands never change its position during expression conversion so only operator stack is needed.

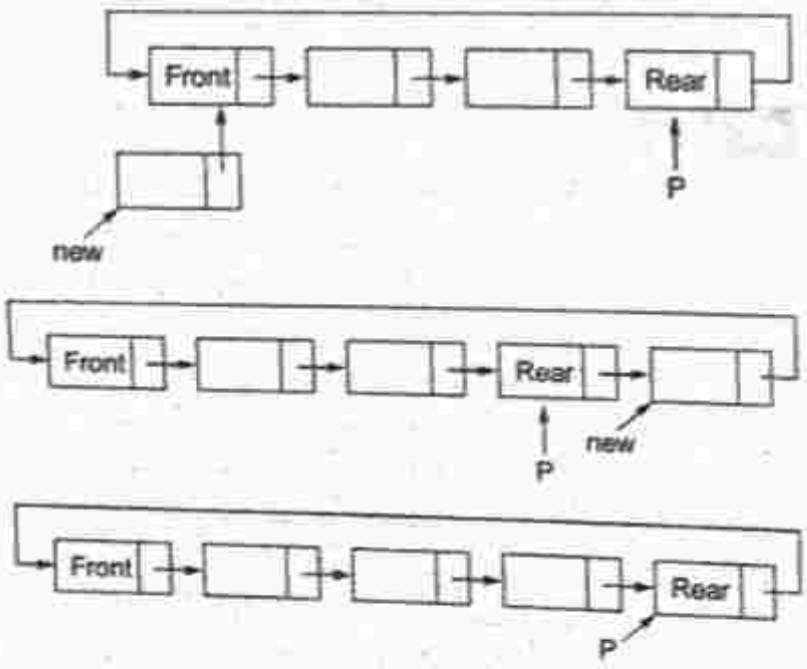
99. (a)



To add a new node:



new → next = Rear → next;
Rear → next = new;
Rear = new;



It will take constant time.

Similarly delete node from Front code will be:

Front = Rear → next;

Rear → next = Front → next;

Free (Front);

It will also take constant time.

100. (a, c, d)

If the precedence of the operator is higher the the stack operator, then it is pushed on to the stack operator.

101. (b)

Queue is suitable structure for BFS.

102. (a, c, d)

Since front pointer is used for deletion then only dequeue operation can be performed in constant time thus other 3 cases will take linear time.

103. (4321)

Queue follows FIFO structure.

104. (d)

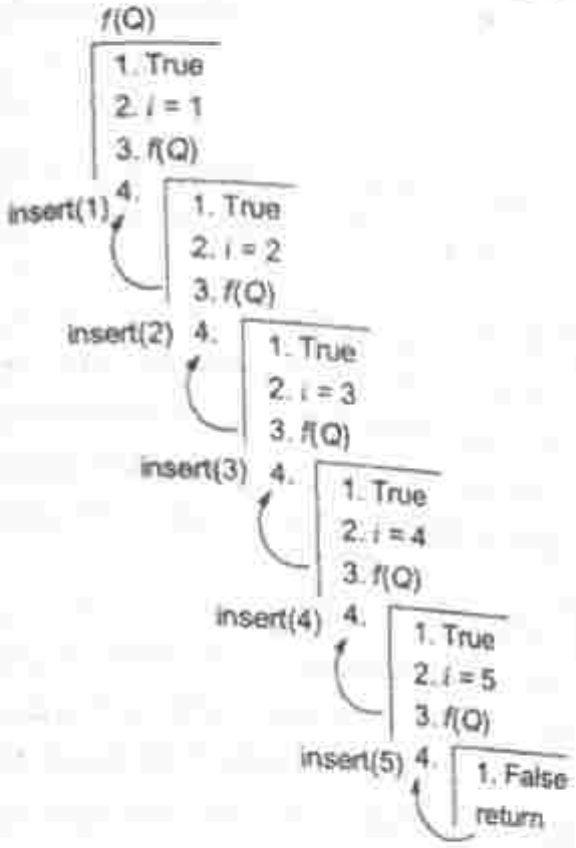
In double ended queue elements can be added and deleted from both the ends.

105. (b)

In this program if statement execute only queue is not empty. This function is called recursively until queue becomes empty after it. Flow of control is transferred to next statement and insert the element in the reverse of the order of elements in the queue.

Let, queue size = 5

```
def f(Q)
    if not is_empty(Q):
    i = delete(Q)
    f(Q)
    insert(Q, i)
```



On returning inserting element back into queue

5	4	3	2	1

106. (a)

In an empty queue rear = front = –1

107. (c, d)

Since front pointer is used for deletion and rear pointer is used for insertion thus only dequeue and enqueue operation can be performed in constant time and other 2 cases will take linear time.

108. (d)

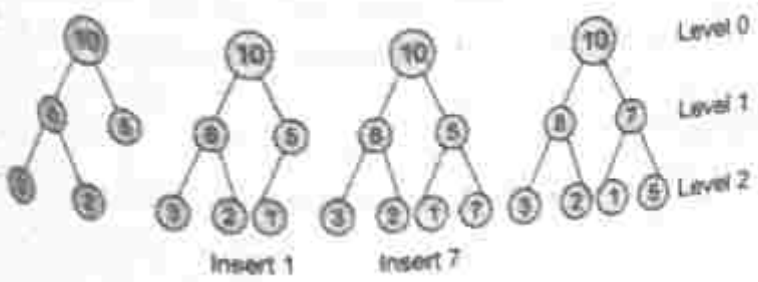
In implementation of stack using queue :

If we do some extra work in inserting the element then deletion of second element from top of stack required queue $(n-1)$ element in another queue then again $(n-1)$ i.e. $(n-2)$ element to another queue so it take $O(n)$ time.

In implementation of using stack; will take $O(1)$ time to delete second element from front will take $O(1)$ time.

109. (d)

Initially the sequence of Max Heap is



So, the sequence of level order traversal is 10, 8, 7, 3, 2, 1, 5

110. (a, c)

Ensures rear takes the values from 0 to (CAPACITY-1).

Two stacks are required for evaluation of infix expression – one for operands and one for operators.

111. (c)

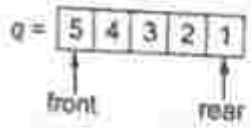
Minimum 2 stacks are required to implement a queue.

112. (a, d)

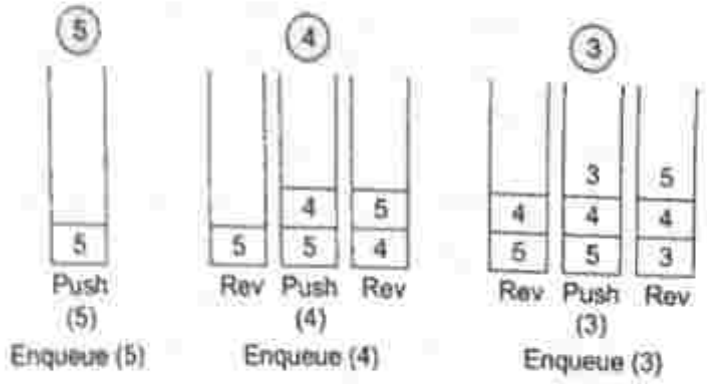
It can be done by both the methods.

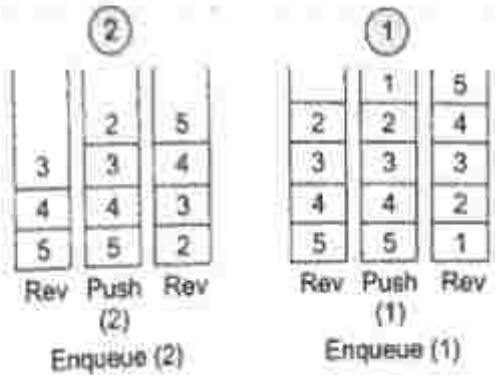
113. (c)

Consider an queue with elements inserted in same order as element present in queue.



Now to implement stack same as queue:





A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.

114. (a, b, c, d)

All of the mentioned can be implemented with a dequeue.

115. (a)

Stack is LIFO system so FIFO not supported by stack hence option statement 1 is incorrect and queue is using FIFO so statement 4 is also wrong.

116. (d)

In stack last element will be deleted first but according to question, it should be work like queue. So last element deleted at last, for that we give lower priority to last element. So all elements should be in strictly decreasing order.

117. (b, c)

Use one queue and one counter to count the number of elements in the queue.

Queue follows First In First Out structure.

118. (d)

If a queue contains an element, then Front == Rear, and neither Front nor Rear can be None or –1, because that would indicate the queue is empty.

Hence, options (a) and (c) are incorrect.

Since it's a circular queue, a bottleneck case arises in the case of two elements where, due to wrap-around, either of the following can be true:

Front == (Fear + 1) % size

Front == (rear – 1) % size

These conditions indicate a full queue, not a one-element queue.

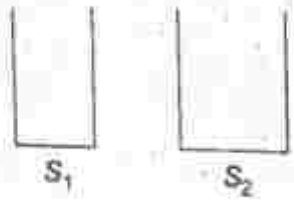
Therefore, what we can confidently say is:

If the circular queue contains only one element, then:

Front == Rear and Front is not None

119. (b)

Use two stacks



Add front( ) → Push to S_1 → O(1)

Remove front( ) → Pop S_1 → O(1)

Add rear( ) → (i) Pop all elements of S_1 to S_2.
(ii) Add new element to S_1.
(iii) Pop all elements to S_2 to S_1.

Remove rear( ) → (i) Pop all elements of S_1 to S_2, except last.
(ii) Pop last element to S_1.
(iii) Pop all elements to S_2 to S_1 → O(n).

If we make some change, then front operations can be O(n) and rear operations can be O(1).

120. (c, d)

Queues can be used for limited resource allocation and semaphores implementation since it exhibits FIFO policy which is very useful in these operations. For other operations, stacks are used.

121. (c)

The concatenation of two lists is to performed on O(1) time by the use of circular doubly linked list. The reason behind this, that it has the pointer in the both direction in front and backward.

122. (c)

- P1 traverses the list one node at a time.
- P2 moves two nodes at a time.
- When P2 reaches the end, P1 will be at the middle node of the list.

123. (b, c)

Advance the pointers in such a way that the fast pointer advances two nodes at a time and slow pointer advances one node at a time and check to see if at any given instant of time if the fast pointer points to slow pointer or if the fast pointer's 'next' points to the slow pointer. This is applicable for smaller lists.

Similarly, DFS can be used to check the presence of cycles.

124. (d)

Struct node * temp = $x \rightarrow$ next;

$x \rightarrow$ data = temp $\rightarrow$ data;

$x \rightarrow$ next = temp $\rightarrow$ next;

free (temp);

125. (b)

In **linear linked list** we can search only in one way but using **double linked list** we can search an unsorted list for a given item in both direction.

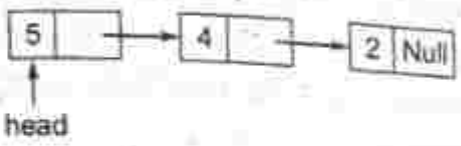
126. (c)

- p is the slow pointer which moves one step at a time.

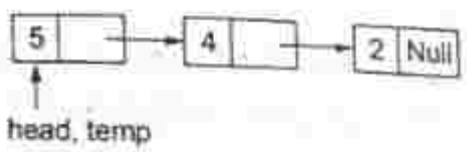
- q is the fast pointer which moves two steps at a time.
- The loop condition while(A) should ensure that both pointers (p and q) are not None. If either pointer becomes None, the list has no loop, and the function should terminate.
- The correct answer is (p != None) and (q != None) because both pointers need to be valid (not None) for further comparison and movement through the list.

127. (c)

If the pointer pointing to first node, it can be deleted.



Let, temp = head



head = temp → next;
free (temp);

But deletion of nodes at other position is not possible always. Suppose pointer is pointing to last node. To delete last node, you need address of second last or the starting address which is not provided.

128. (a, c, d)

Time complexity of inserting a new node at the head of the list is $O(n)$ because you have to traverse through the list to find the tail node.

129. (a)

To delete node x, we should find out the previous node to the x. To find the previous nodes to the x will take $O(n)$.
Therefore it takes $O(n)$.

130. (a)

For FIFO

(i) Elements will be inserted using tail pointer. Elements will be deleted using head pointer.
(ii) In a single linked list, any number of elements can be inserted without declaring prior size.
(iii) An item of larger size can be stored in a linked list using structures.

131. (c)

For insertion/deletion of a record in circularly linked list involves 2 pointer.
But for reversing the linked list, it involves 3 pointer.

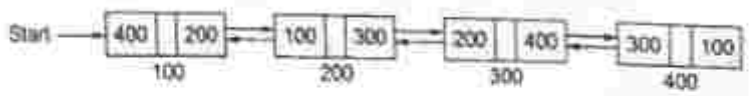
132. (a)

Deleting a new node whose location is given is the operation that is performed more efficiently by doubly linked list than by linear list because we have both the pointers i.e., the pointer to the previous as well as to the next node, since there two pointers need to be edited. But in case of linear list, we don't have pointer to the previous node.

133. (a)

Consider a stack

The characteristic of the stack is both insertions and deletions are performed from one end.
If, it is implemented with a link lists, then both insertions and deletions are needed to be performed from the end.
Since, the linked list is a doubly circular linked list, hence the start node will have address of last node.



So, both the operations can be performed in O(1) time.

134. (c)

Even though insertion can easily be done in O(1), deletion will take $O(n)$ time, since tail points to the last node, and in order to delete the node pointed by tail pointer, we need the address of the second last node for which we need to scan (traverse) the entire list, and thus (c) is the most appropriate choice.

135. (4)

The element printed by foo1( ) be '*x*'
Order of element = 1, 2, 3, 4, 5, 6, 7
The element printed by foo2( ) be '*y*'
Order of element = 7, 6, 5, 4, 3, 2, 1
The element '4' retains its own position.

136. **(a, b, c)**

XOR lists requires same time for most of the operations as arrays would require.

137. **(c)**

- Merge sort on linked list take $O(n \log n)$ time to sort input of length n.
- The merge sort algorithm is recursive, so if requires $O(\log n)$ stack space, for both array and linked list cases. But the array case also allocates an additional $O(n)$ space, which dominates the $O(\log n)$ space required for the stack.
  Hence total stack space required, for merge sort using array is $O(n)$ while for linked list is $O(\log n)$.
- Inplace merge sort on array takes $O(1)$ space and $O(n^2)$ time as number of comparison increase.

138. **(c)**

Linked list allows sequential search, so it takes $O(n)$ time to search an element in linked list of size n.

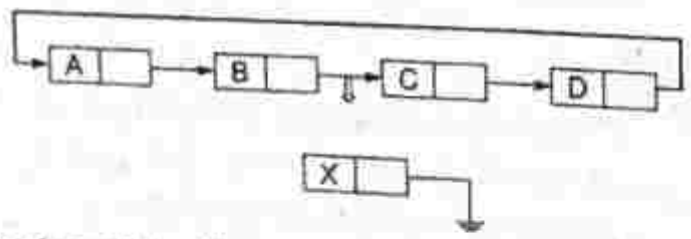
139. **(c)**

In circular doubly linked list, one does not need to traverse the whole list to find the end of the list. The second list can be concatenated at any location. Only fix number of pointers need to be changed. Hence can be done in $O(1)$ time.

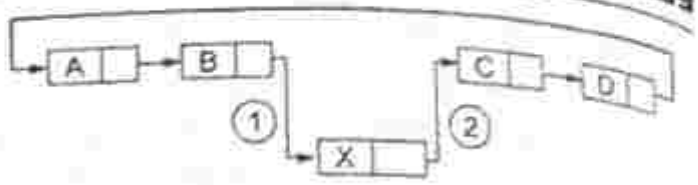
140. **(c)**

- P1 is a slow pointer: it moves one step at a time (P1 = P1.next).
- P2 is a fast pointer: it moves two steps at a time (P2 = P2.next.next), if possible.
- If there is a cycle in the linked list, both pointers (P1 and P2) will eventually meet at some node inside the cycle (i.e., P1 == P2).
- If there is no cycle, P2 will reach the end (None) before P1, and the function will return False.

141. **(b)**



In above linked list a new record X will be inserted in between B and C.



Two pointers are modified to insert X record.

142. **(b)**

Using linked list binary search will take $O(n)$ time. So binary search is inefficient with linked list.

143. **(b)**

In array, contiguous memory allocation is there or data is stored in the adjacent locations.
But, in linked list data is usually stored at different locations in memory. Whenever a new data node is created, its address is stored in the already created linked list and node is appended into the linked list.

144. **(a)**

Self referential data structure is that in which atleast one structure member is present which points to a structure of its own kind. In linked list, every node has a member pointer which points to some another node. Hence, linked list is an example of self-referential structure.

145. **(c)**

Binary search is not good to be implemented by linked list because one can't randomly access data numbers present in link list, which is otherwise possible with arrays.

146. **(b)**

In case of using link list, one cannot randomly access the data or only serial data access is there. But, in case of binary search one needs to jump randomly either to first half or other half, which is not possible with linked list.

147. **(c)**

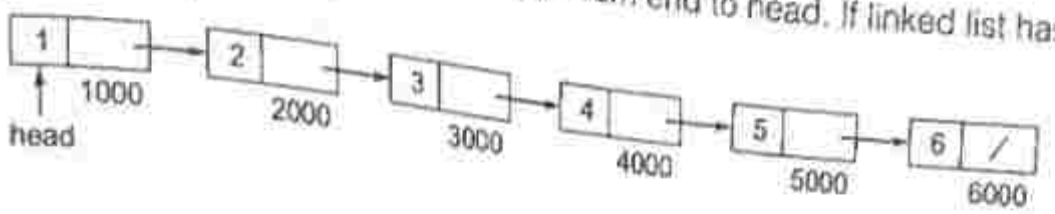
For insertion/deletion of a record in circularly linked list involves 2 pointer.
But for reverse the linked list, it involves 3 pointer.

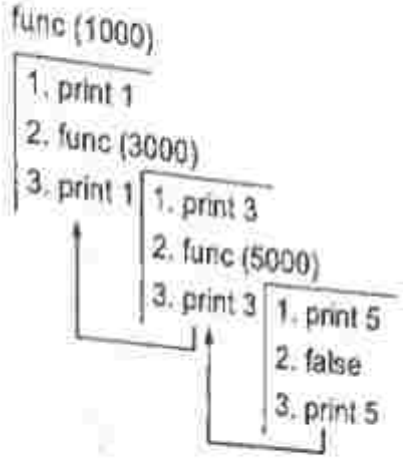
148. **(c)**

By using the given code, we will be able to traverse to the end of the link list.
While ( ) condition will be true for every node except the node before None node, which will help to find last node.

149. (d)

func( ) prints alternate node from head to end and then from end to head. If linked list has even number of nodes, then it skips last node.





Statement-3 → printf(head → data) is called on return.

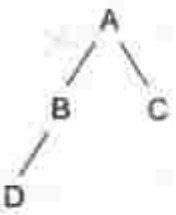
150. (a, b, c)

To insert and delete at known positions requires complete traversal of the list in worst case in SLL, SLL consists of an item and a node field, while DLL has an item and two node fields, hence SLL occupies lesser memory, DLL can be traversed both ways (left and right), while SLL can traverse in only one direction, hence more searching power of DLL.

Node fields in SLL is 2 (data and address of next node) whereas in DLL is 3 (data, address to next node, address to previous node).

151. (c)

Consider the following tree:



The diameter of the tree will be 4, consider from node 'D' to 'C'.

Hence, it calculates the diameter of the tree.

152. (c)

The resultant BST obtained is

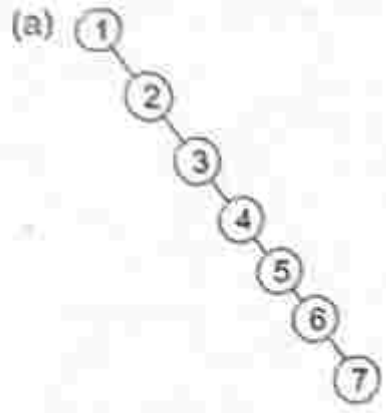
∴ The number of elements in the 4th, 5th and 6th level are. 3, 1, 1

153. (a, d)

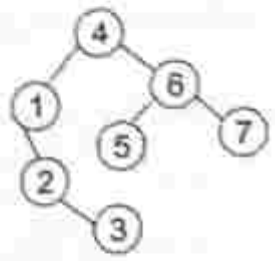
At maximum, BST may need to search all *n* values in the tree in order to access an element, hence, $O(n)$. While Balanced BST will take $\log n$ time to search in worst case.

154. (c)

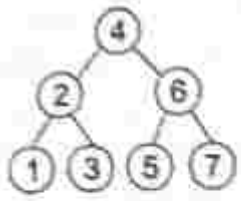
(a)



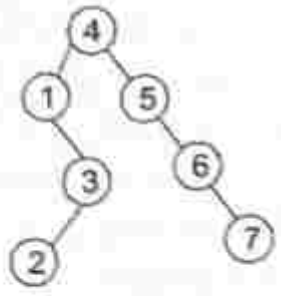
(b)



(c)



(d)



155. (3)

Inorder is given as Left Root Right

Postorder is given as Left Right Root

Inorder (MFI) E (JGZ)

Postorder (MIF) (JZG) E

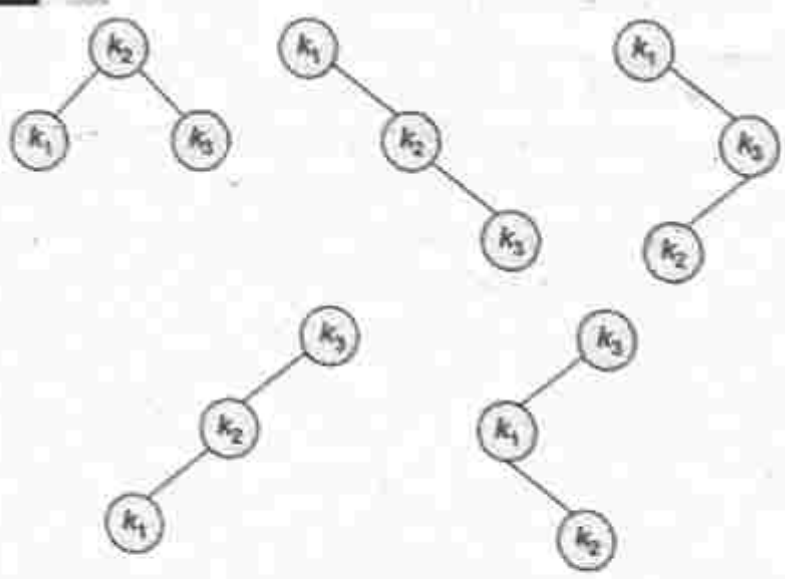
Left Right Root

Left subtree contains 3 nodes M, I, F.

156. (c)

By the method of elimination: Full binary tree looses its nature when operation of insertion and deletion are done. So complete binary tree is better one because for incomplete binary tree, extra storage is required and overhead of NULL node checking takes place.

157. (b)



158. (d)

In this particular case it is BST and search time is proportional to height (H).

159. (3)

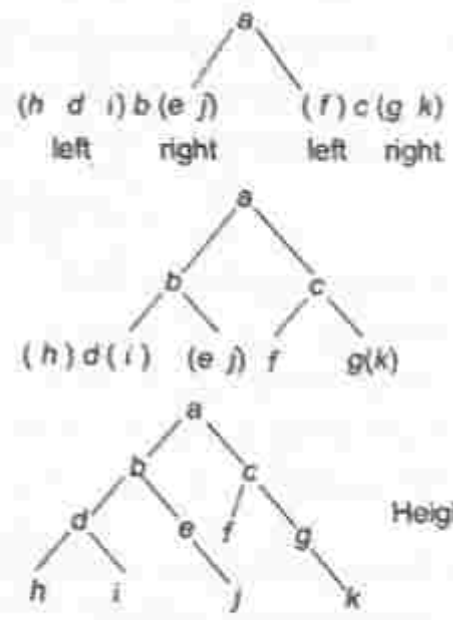
Inorder– Left Root Right

Preorder – Root Left Right

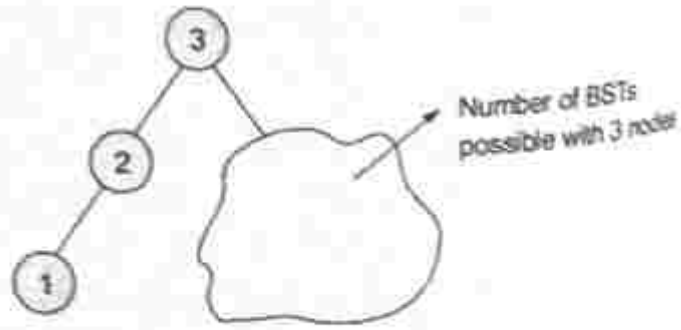
Inorder – (*h, d, i, b, e, j*) (*a*) (*f c g k*)

Preorder – (*a*) (*b, d h i e j*) (*c f g k*)

Root Left Right



160. (a)



$$\text{Number of BSTs} = \frac{^{2N}C_N}{N+1}$$

Putting N as 3,

$$\frac{^6C_3}{4} = \frac{20}{4} = 5$$

Therefore (5) is the answer.

162. (b)

To construct BST either preorder or postorder is sufficient because inorder is always sorted for BST.

163. (b)

Total nodes $(n) = n_1 + n_2 + n_3$

Total edges $(e) = (n_1 + n_2 + n_3) - 1$

$[\because e = n - 1]$

Sum of degrees of vertices $= 2e$

$\Rightarrow 1 * n_1 + 2 * n_2 + 3 * n_3 = 2e$

$\Rightarrow n_1 + 2n_2 + 3n_3 = 2e$

$[\because e = n_1 + n_2 + n_3 - 1]$

$\Rightarrow n_1 + 2n_2 + 3n_3 = 2(n_1 + n_2 + n_3 - 1)$

$\Rightarrow n_1 + 2n_2 + 3n_3 = 2n_1 + 2n_2 + 2n_3 - 2$

$\Rightarrow n_1 = n_3 + 2$

164. (b, c, d)

Search is more efficient in the 2-3 tree than in BST. 2-3 tree is a balanced tree and performs efficient insertion and deletion and it is shallower than BST. But, 2-3 tree requires more storage than the BST.

165. (6)

Complete binary tree can make smallest possible height of Binary Search Tree (T).

$\Rightarrow 2 \le n \le 3 \Rightarrow h = 1$ (smallest)

$\Rightarrow 4 \le n \le 7 \Rightarrow h = 2$

Similarly, $8 \le n \le 15 \Rightarrow h = 3$

$\therefore$ Smallest height possible: [when complete tree]

$$h = \left\lceil \log_2^{(n+1)} \right\rceil - 1$$

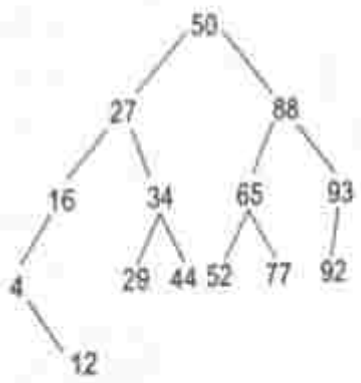
$\therefore$ Largest height possible: [When skew tree]

$$h = n - 1$$

Smallest possible height

$$= \left\lceil \log_2^{(120+1)} \right\rceil - 1 = 7 - 1 = 6$$

166. (3)



Array A:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Preorder:	50	27	16	4	12	34	29	44	88	65	52	77	93	92

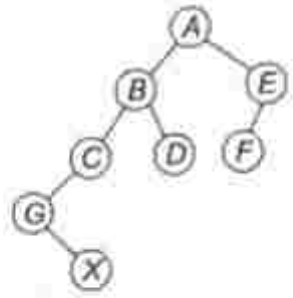
Array B:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Postorder:	12	4	16	29	44	34	27	52	77	65	92	93	88	50

The elements **16, 34, 65** have same location in A and B.

167. (b)

Let a node (x) is inserted

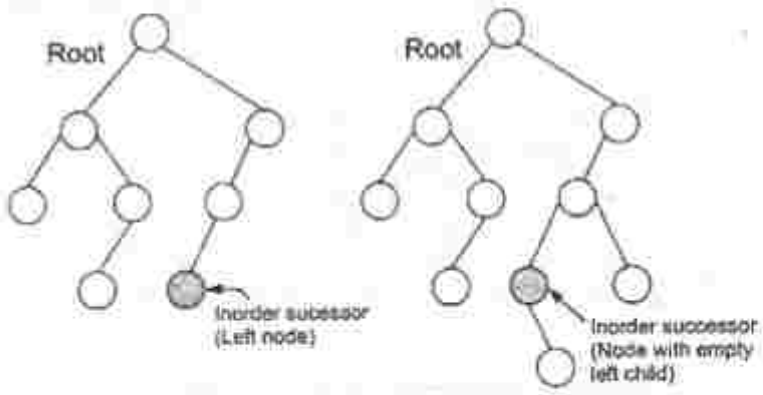


Node A, B, C height = 2

They are unbalanced

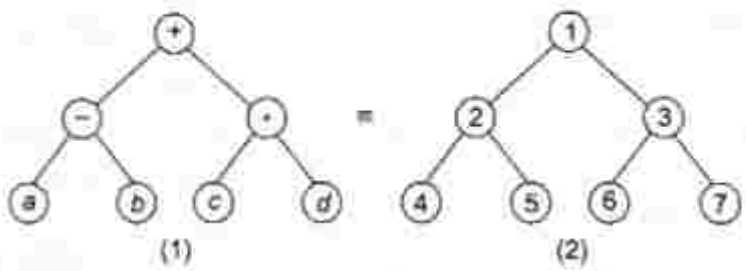
168. (d)

Successor of root element is always the smallest element of the Right subtree. Because it will be the next largest element after the element to be deleted.



169. (a)

Post order = $ab - cd * +$ according to given postorder the binary tree is

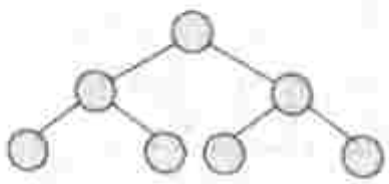


According to tree (1) and (2)

1, 2, 3, 4, 5, 6, 7 ≡ +, −, *, a, b, c, d.

170. (d)

In complete binary tree.



Number of internal nodes = Number of leaf node − 1

So total number of nodes = Internal nodes + Leaf nodes

$$= n + n + 1 = 2n + 1$$

Internal node = 3

Leaf node = 4

Total nodes = 3 + 4 = 7

171. (c)

Number of possible binary search trees with n nodes = $\frac{{}^{2n}C_n}{n+1}$

$\therefore$ For 3 nodes = $\frac{2\times {}^{3}C_{3}}{3+1} = \frac{{}^{6}C_{3}}{4} = 5$

172. (d)

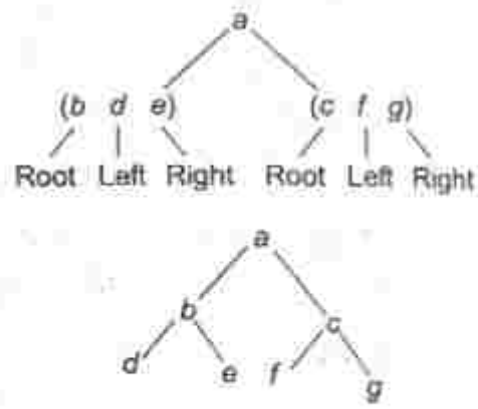
Inorder – Left Root Right

Preorder – Root Left Right

There inorder – $(d\ b\ e)\ a\ (f\ c\ g)$

Preorder – (a) (b, d e) (c f g)

Root Left Right



Postorder – Left Right Root

$-\ d\ e\ b\ f\ g\ c\ a$

173. (a)

$G(x)$ = (Number of leaf nodes in left – Number of leaf nodes in right subtree of I).

To find number of leaf nodes in left and right subtree of node I will take $O(n)$ time. One time visit of each node in left subtree and one time visit of each node in right.

Difference will take constant time.

So, overall time complexity = $O(n)$.

174. (a)

If the depth is d, the number of nodes n will be $2^{(d+1)} - 1$.

So, $n + 1 = 2^{(d+1)}$ or $d = \log(n + 1) - 1$

175. (a, b, c)

In order sequence of binary search trees will always give ascending order of elements.

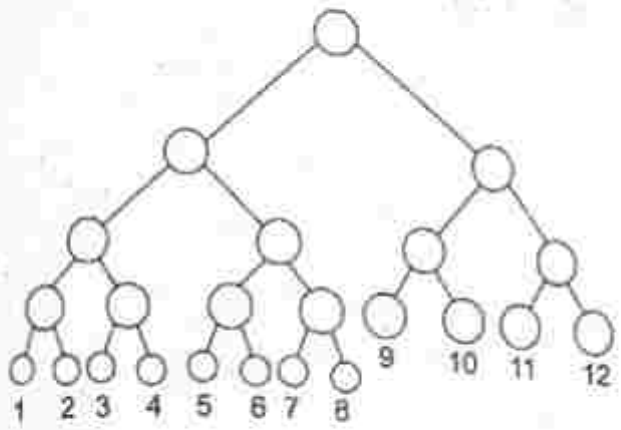
Remaining all are true regarding binary search trees.

176. (c)

If every non-leaf node in a binary tree has non empty left and right subtrees, the tree is termed as strictly binary tree. A strictly binary tree with n leaves always contains $2n - 1$ nodes.

Here, $n = 12$

Total nodes in tree = $2 * 12 - 1 = 23$



177. (a)

In the given code, two binary trees are compared recursively. The function checks whether the data at the current node in both trees is the same.

- If the data matches, it recursively checks the left and right subtrees.
- If there is any mismatch in node data, the function immediately prints "GATE" and exits.

Thus, "GATE" will not be printed only when every corresponding node in both trees contains the same data and structure. That means the two trees are identical.

Hence, the correct option is (a).

178. (a)

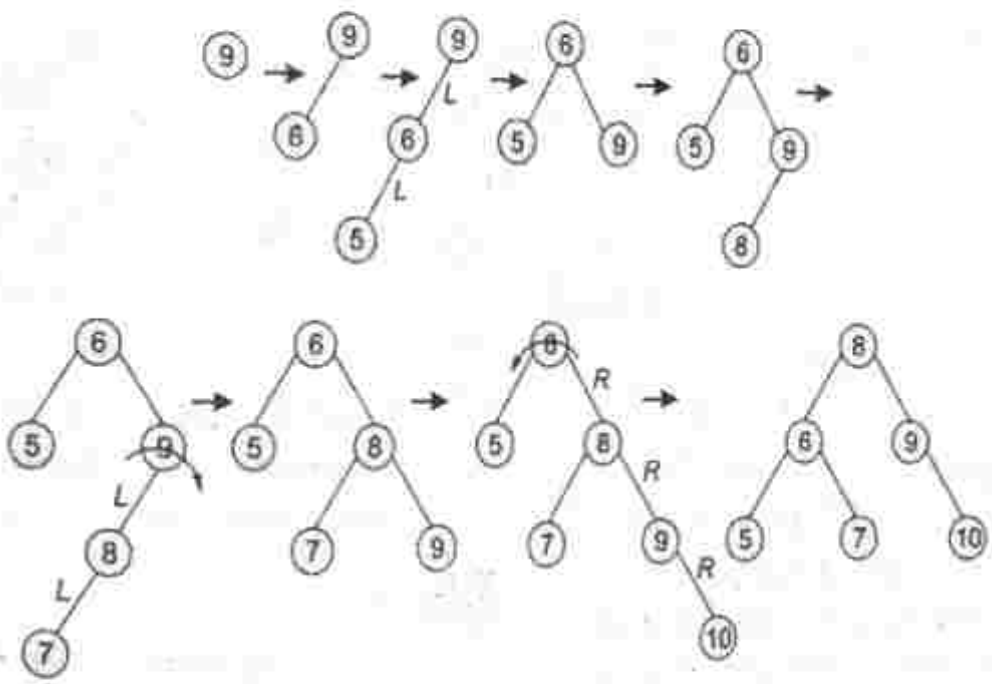
Queue data structure is known as FIFO (First In First Out).

While

Stack is known as LIFO (Last In First Out).

Statements 1, 2 and 3 are correct.

179. (d)

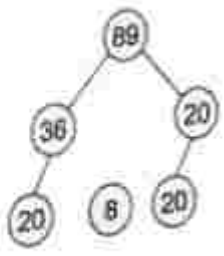


Number of rotations = 3

180. (d)

A max heap is a complete binary tree in which the value in each internal node is greater than equal to the values in children of that node

Example:



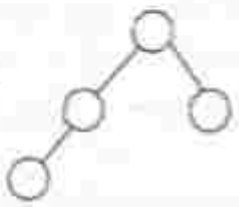
181. (c)

Minimum number of nodes in AVL Tree of height H MNN(H) = MNN (H – 1) + MNN (H – 2) + 1

H = 0 Eg: ○ Minimum 1 node required.

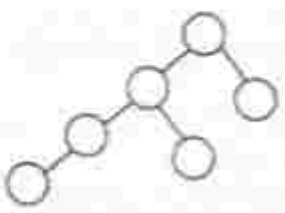
H = 1 Eg: Minimum 2 nodes required.

H = 2 Eg:



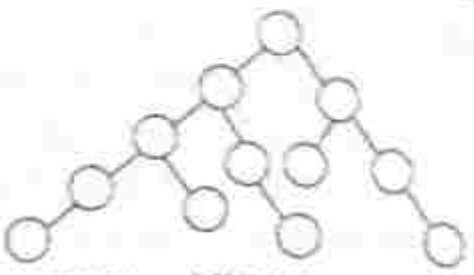
MNN(2) = MNN (1) + MNN (0) H = 2 + 1 H = 4

H = 3 Eg:



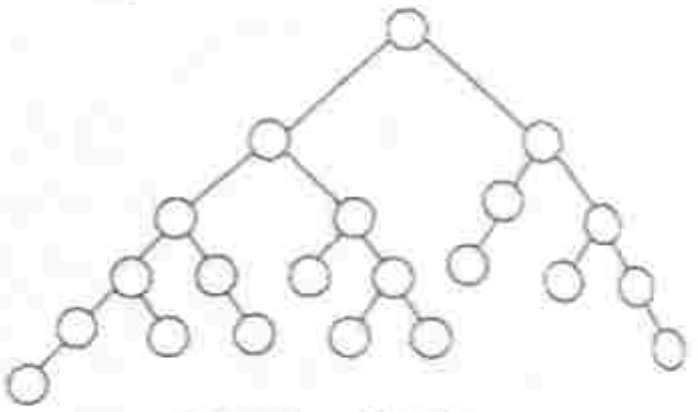
MNN(3) = MNN (2) + MNN (1) H = 4 + 2 H = 7

H = 4 Eg:



MNN(4) = MNN(3) + MNN(2) + 1 = 7 + 4 + 1 = 112

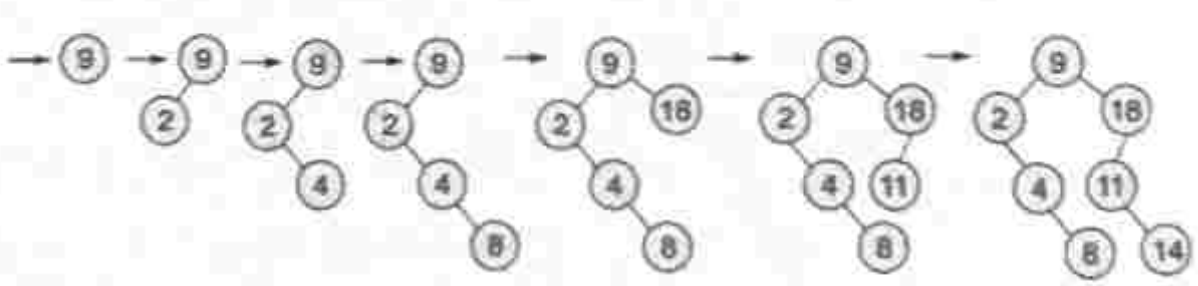
H = 5 Eg:



MNN(5) = MNN(4) + MNN(3) + 1 = 12 + 7 + 1 = 20

So, maximum height of AVL tree with 20 nodes = 5.

182. (b)



Height = 3

183. (c)

Insertion and deletion, in both the binary heap and balanced binary search tree takes O(log *n*). But searching in balanced binary search tree requires O(log *n*) while binary heap takes O(*n*). Construction of balanced binary search tree takes O(*n* log *n*) time while binary heap takes O(*n*).

Note: Option (d) describes the disadvantage of Balanced binary search tree hence can't be chosen.

184. (a, b, d)

First you reverse the given equation and carry out the algorithm of infix to postfix expression. Here, the data structure used is stacks.

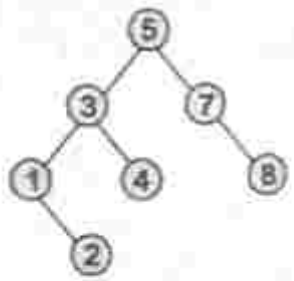
Note: Queue and linked list both can be implemented to perform stack operations with some modifications.

185. (b)

A labelled rooted binary tree can not be constructed uniquely when inorder traversal is given along with post-order or pre-order traversal.

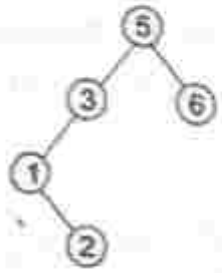
186. (d)

(a) 53124786



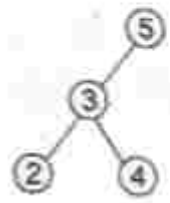
6 can not be child of 8.

(c) 53126487



4 can not be child of 6.

(d) 53241678



1 can not be child of 4 or 5.

188. (d)

In m-ary tree

Total number of nodes = m(internal nodes) + 1

Here, $n = 3i + 1$

Total nodes = Internal nodes + Leaf nodes

$n = i + l,\ i = n - l$

$n = 3(n - l) + 1 \quad [\because n = 3i + 1 \ \& \ i = n - l]$

$n = 3n - 3l + 1$

$3l = 2n + 1$

$$l = \frac{2n+1}{3}$$

Where l is leaf nodes.

189. (b)

Minimum degree = t

Height (h)	Number of keys	Number of pointers
0	1	2
1	$2(t-1)$	$2t$
2	$2(t-1)t$	$2t^2$
⋮		
h	$2(t-1)t^{h-1}$	$2t^h$

For maximum height for n keys, choose minimum degree in tree. And root node can have only one key.

Total keys (n) = $1 + 2(t-1) + 2(t-1)t + \ldots\ldots 2(t-1)\,t^{h-1}$

$$n = 1 + 2(t-1)[1 + t + t^2 + \ldots\ldots + t^{h-1}]$$

$$= 1 + 2(t-1)\left(\frac{t^h - 1}{t-1}\right)$$

$$t^h = \frac{n+1}{2}$$

$$h = \log_t\left(\frac{n+1}{2}\right)$$

Left + Right + 1 = n

So, $k - 1 + x + 1 = n$

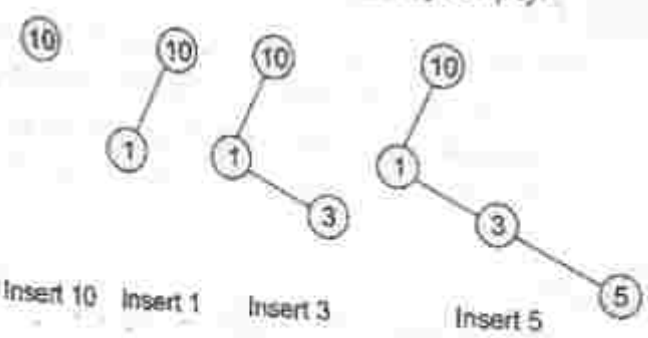
$x = n - k$

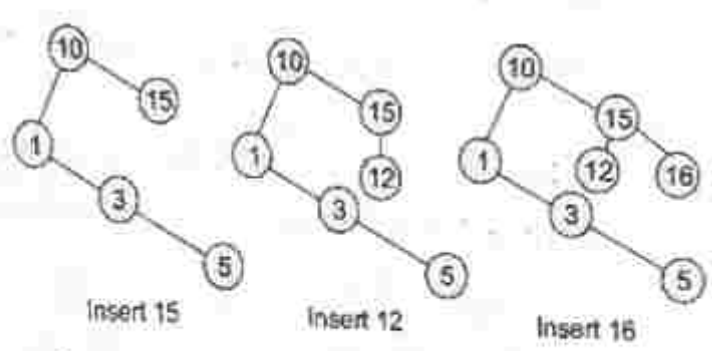
190. (b)

Because of complete binary tree option (b) is always correct.

191. (b)

The binary search tree is initially empty.



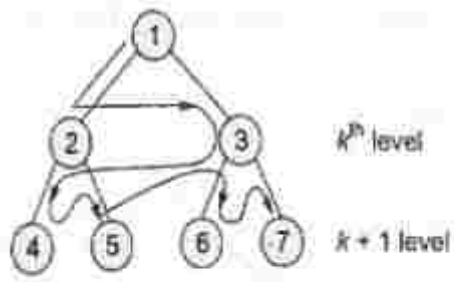


So the maximum height of BST = 3

[Distance from element 5 to root 10 is 3]

192. (d)

Level order traversal of rooted tree can be done by starting from root and visiting all node at k level before visiting any node at $k + 1$ level, which is nothing but breadth first search.



193. (b)

Let number of vertices in binary tree = n

Number of edges = $n - 1$

So, $n - 1 = 1 \times 5 + 2 \times 10$

$n - 1 = 5 + 20$

$\Rightarrow \quad n = 26$

Number of leaf nodes = 26 − 5 − 10 = 11

194. (b)

Total number of distinct binary search tree with n nodes (key)

$$B_n = \frac{1}{n+1}\binom{2n}{n}$$

$$B_4 = \frac{1}{4+1}\binom{8}{4} = \frac{1}{5}\frac{\lfloor 8}{\lfloor 4 \lfloor 4}$$

$$B_4 = \frac{1}{5}\frac{8\times7\times6\times5\times\lfloor 4}{4\times3\times2\times1\times\lfloor 4} = 14$$

195. (c)

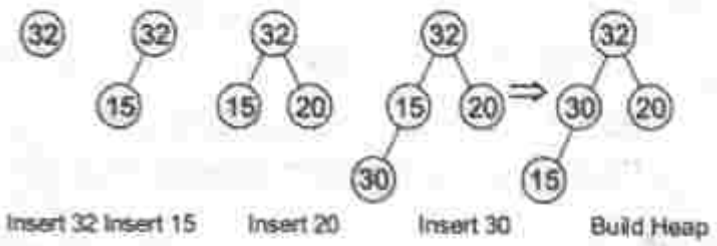
In a complete-k-ary tree. If every internal node has exactly k children then $(k-1)$ key contains no leaves. If there are n internal nodes, then number of leaves is $n(k-1)+1$.

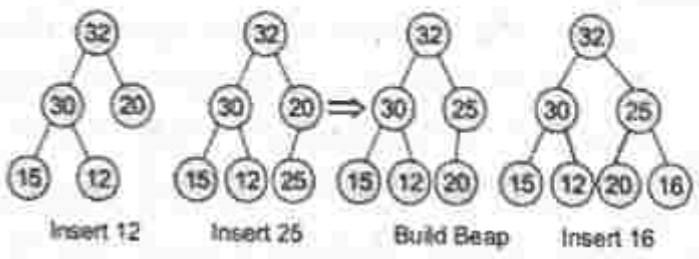
196. (a)

Input sequence 32 ,15, 20, 30, 12, 25, 16

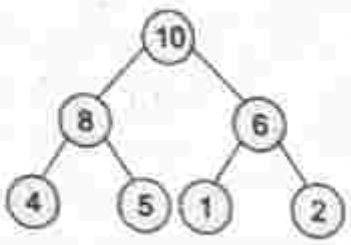
If i is the root then $A[i] \geq A[2i]$ for $1 \leq i \leq \frac{n}{2}$ and

$A[i] \geq A[2i+1]$ for $1 \leq i \leq \frac{n}{2}$ where $2i$ is the left of i and $2i+1$ is the right of i of a max heap.



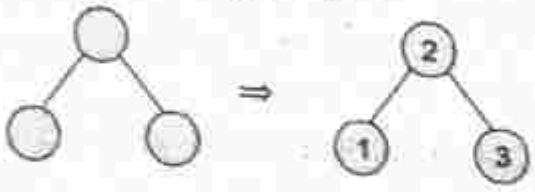


197. (b)



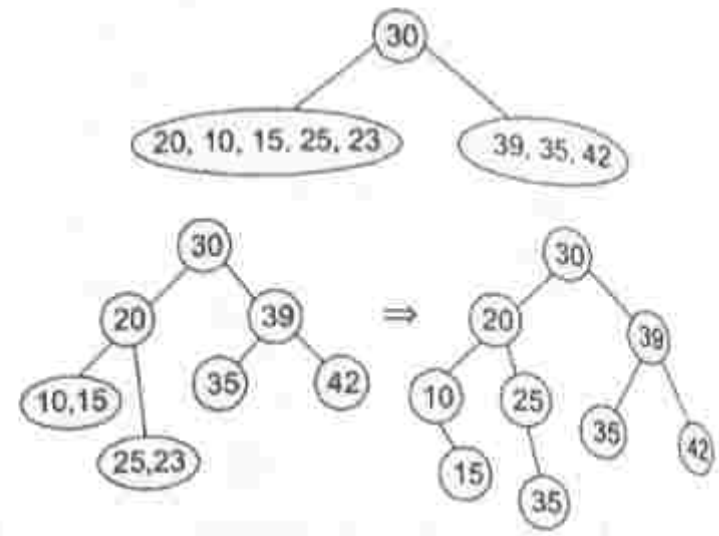
199. (b)

$n = \{1, 2, 3\}$



200. (d)

Binary search tree and preorder is given 30, 20, 10, 15, 25, 23, 39, 35, 42



Postorder traversed of resultant tree is 15, 10, 23, 25, 20, 35, 42, 39, 30

201. (c)

Number of nodes in a complete binary tree of order 'n' is equal to $2^{d+1}-1$, where 'd' is level or height of tree.

Hence for level 5, number of nodes $2^6 - 1 = 63$

202. (a)

Height is defined as path length i.e. (height +1) will give number of levels.

In a binary tree of height $h = 5$:

Maximum number of nodes $= 2^{h+1} - 1$

$= 2^{5+1} - 1 = 64 - 1 = 63$

Minimum number of nodes = 6 i.e. one node at every level.

203. (b)

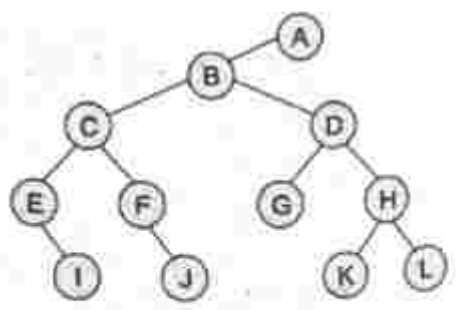
For distinct BST we apply this formula $\frac{C(2n,n)}{n+1}$

$n = 3$ here so $C(6, 3) = 20$ and $\frac{20}{4} = 5$

204. (d)

Read preorder from left and divide inorder in each step, and create tree. Then from tree, create postorder

Tree is



Postorder : IEJFCGKLHDBA

205. (d)

You can verify that the 1st, 3rd, 5th, 7th... probes check at location 5.

The 2nd, 6th, 10th... probes check at location 8.

The 4th, 8th, 12th ... probes check at location 4.

The rest of the address space will never be probed.

206. (3)

$f(\text{key}) = \text{key mod } 11$

$f(3) = 3 \bmod 11 = 3$

$f(23) = 23 \bmod 11 = 1$

$f(1) = 1 \bmod 11 = 1 \rightarrow$ collision

$f(0) = 0 \bmod 11 = 0$

$f(15) = 15 \bmod 11 = 4$

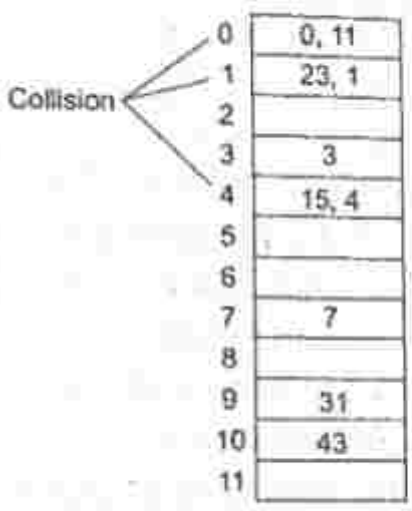
$f(31) = 31 \bmod 11 = 9$

$f(4) = 4 \bmod 11 = 4 \rightarrow$ collision

$f(7) = 7 \bmod 11 = 7$

$f(11) = 11 \bmod 11 = 0 \rightarrow$ collision

$f(43) = 43 \bmod 11 = 10$



Hence, 3 collision occur.

207. (a)

If there is only one record, probability of collision $= \frac{1}{100}$. For 2 records, probability of collision $\frac{2}{100}$.

Similarly, for 9 records, probability of collision $\frac{9}{100}$.

So, required probability is $\frac{1}{100} + \frac{2}{100} + \ldots + \frac{9}{100}$

$$= \frac{45}{100} = 0.45$$

208. (c)

Let $X_{i,j}$ be an indicator random variable equal to 1 if element i and j collide and equal to 0 otherwise. Simple uniform hashing is used i.e., the probability of element i-hashing to slots K is $\frac{1}{m}$.

Therefore the probability that i and j both hashed to same slot $Pr(X_{i,j}) = \frac{1}{m}$.

Hence expected $[X_{i,j}] = \frac{1}{m}$.

The E[no. of collision pairs] $= E\left[\sum_{i=1}^{n}\sum_{j=i+1}^{n} X_{i,j}\right]$

$$= \sum_{i=1}^{n}\sum_{j=i+1}^{n} E[X_{i,j}] = \sum_{i=1}^{n}\sum_{j=i+1}^{n}\frac{1}{m}$$

$$= \frac{n(n+1)}{2m} = \Theta(n^2/m)$$

209. (d)

$f(\text{key}) = \text{key mod } 7$

	Remainder	Location
$f(37) = 37 \bmod 7$	2	3
$f(38) = 38 \bmod 7$	3	4
$f(72) = 72 \bmod 7$	2	(location 2 is not vacant move to next vacant location) (5)
$f(48) = 48 \bmod 7$	6	7
$f(98) = 98 \bmod 7$	0	1
$f(11) = 11 \bmod 7$	4	5
$f(56) = 56 \bmod 7$	0	2

Remainder	0	1	2	3	4	5	6
	98	56	37	38	72	11	48
Locations	1	2	3	4	5	6	7

210. (b)

$f(\text{key}) = \text{key mod } 7$

$f(37) = 37 \bmod 7 = 2$

$f(38) = 38 \bmod 7 = 3$

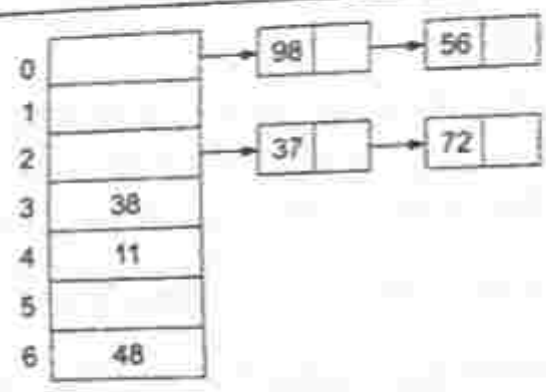
$f(72) = 72 \bmod 7 = 2$

$f(48) = 48 \bmod 7 = 6$

$f(98) = 98 \bmod 7 = 0$

$f(11) = 11 \bmod 7 = 4$

$f(56) = 56 \bmod 7 = 0$



Key 11 is stored at index 4.

211. (125)

$$\text{Load factor } (\alpha) = \frac{\text{Number of elements}}{\text{Number of slots}}$$

$$= \frac{2500}{20} = 125$$

212. (b)

It will be one more than the size of the biggest cluster (which is 4) in this case. This is because, assume a search key hashing onto bin 8. By linear probing the next location for searching is bin 9. Then 0, then 1. If all these resulted in a miss, we try at bin 2 and stop as it is vacant. This logic may not work if deletion operation is done before the search.

213. (5)

$h(x) = x \bmod 9$

$h(22) = 22 \bmod 9 = 4$

$h(11) = 11 \bmod 9 = 2$

$h(17) = 17 \bmod 9 = 8$

$h(13) = 13 \bmod 9 = 4 \rightarrow$ collision

$= (13 + 1) \bmod 9 = 5$

$h(16) = 16 \bmod 9 = 7$

$h(12) = 12 \bmod 9 = 3$

$h(20) = 20 \bmod 9 = 2 \rightarrow 1$ collision

$= (20 + 1) \bmod 9 = 3 \rightarrow 2$ collision

$= (20 + 2) \bmod 9 = 4 \rightarrow 3$ collision

$= (20 + 3) \bmod 9 = 5 \rightarrow 4$ collision

$= (20 + 4) \bmod 9 = 6$

Total 5 collision occur.

0	
1	
2	11
3	12
4	22
5	13
6	20
7	16
8	17

214. (a)

If the new record hashes onto one of the 10 locations 7, 8, 9, 10, 1 or 2, the location 2 will receive the new record. The probability is $\frac{6}{10}$ (as 10 is the total possible number of locations).

215. (a)

Load factor (α) denotes number of filled slots and it is given as

$$\left(\frac{\text{Number of elements in hash table}}{\text{Number of slots}}\right)$$

For open addressing, load factor (α) < 1

Also, $(1 - \alpha)$ given number of free slots, so average number of probes $= \frac{\text{Total slots}}{\text{Free slots}} = \frac{1}{1-\alpha}$

216. (d)

Insertion of keys using linear probing:

$h(x) = x \bmod 11$

$h(33) = 33 \bmod 11 = 0$

$h(68) = 68 \bmod 11 = 2$

$h(62) = 62 \bmod 11 = 7$

$h(47) = 47 \bmod 11 = 3$

$h(48) = 48 \bmod 11 = 4$

$h(60) = 60 \bmod 11 = 5$

$h(104) = 104 \bmod 11 = 5$

$(104 + 1) \bmod 11 = 6$

$h(23) = 23 \bmod 11 = 1$

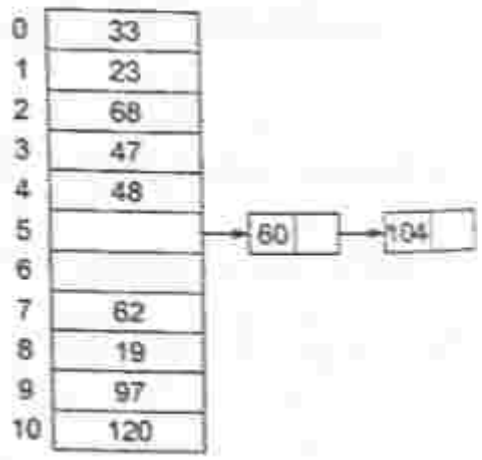
$h(19) = 19 \bmod 11 = 8$

$h(120) = 120 \bmod 11 = 10$

$h(97) = 97 \bmod 11 = 9$

33	0
23	1
68	2
47	3
48	4
60	5
104	6
62	7
19	8
97	9
120	10

Using linear probing, collision = 1



Insertion using chaining

Average chain length

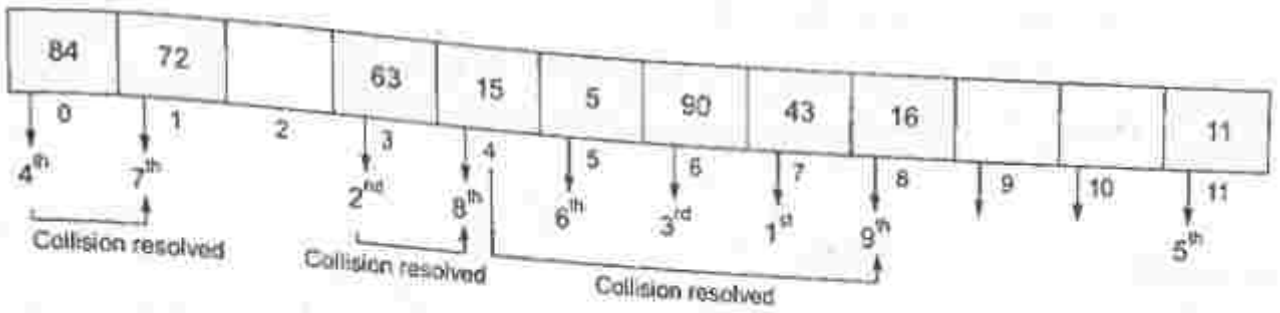
$$= \frac{1+1+1+1+1+2+0+1+1+1+1}{11}$$

$$= \frac{11}{11} = 1$$

217. (8)

Sequence: 43, 63, 84, 11, 5, 72, 15, 16

Hash function ⇒ $h(k) = k \bmod 12$



218. (a)

$h(i) = i^3 \bmod 10$

$h(0) = 0^3 \bmod 10 = 0$
$= h(10) = h(20) = h(30)$

$h(1) = 1^3 \bmod 10 = 1$
$= h(11) = h(22) = h(31)$

$h(2) = 2^3 \bmod 10 = 8$
$= h(12) = h(22) = h(32)$

$h(3) = 3^3 \bmod 10 = 7$
$= h(13) = h(23) = h(33)$

$h(4) = 4^3 \bmod 10 = 4$
$= h(14) = h(24) = h(34)$

$h(5) = 5^3 \bmod 10 = 5$
$= h(15) = h(25) = h(35)$

$h(6) = 6^3 \bmod 10 = 6$
$= h(16) = h(26) = h(36)$

$h(7) = 7^3 \bmod 10 = 3$
$= h(17) = h(27) = h(37)$

$h(8) = 8^3 \bmod 10 = 2$
$= h(18) = h(28) = h(38)$

$h(9) = 9^3 \bmod 10 = 9$
$= h(19) = h(29) = h(39)$

219. (b)

Probability that last k slots are empty after first r iterations

$$= \frac{\overbrace{(N-k)(N-k)(N-k).....(N-k)}^{r \text{ times}}}{N^r}$$

$$= \frac{(N-k)^r}{N^r} = \left[\frac{N-k}{N}\right]^r = \left(1-\frac{k}{N}\right)^r$$

220. (a, b, c)

Hashing is a technique of placing data items in specific locations. Collision may occur in hashing but hashing is not a collision resolution technique.

221. (a, b, c)

Yes: If that is not the case we are screwed-spreading out is the whole point for hashing.

This would be nice, if we take certain size of space to store the hash function itself, it will not be fast to run the hash function either, so yes, we want it to be small.

This would be nice, if it take long time to compute then we might want to use search tree instead.

222. (c)

Deletion is difficult in open addressing. It is because deleting a key from hash take require rearranging of certain keys.

223. (b)

$h(x) = x \bmod 10$

$h(K) = 11 \bmod 10 = 1$

$h(R) = 18 \bmod 10 = 8$

$h(P) = 16 \bmod 10 = 6$

$h(C) = 3 \bmod 10 = 3$

$h(S) = 19 \bmod 10 = 9$

$h(N) = 14 \bmod 10 = 4$

$h(Y) = 25 \bmod 10 = 5$

$h(T) = 20 \bmod 10 = 0$

$h(J) = 10 \bmod 10 = 0 \rightarrow$ Collision 1
$= (10 + 1) \bmod 10 = 1 \rightarrow$ Collision 2
$= (10 + 2) \bmod 10 = 2$

$h(M) = 13 \bmod 10 = 3 \rightarrow$ Collision 1
$(13 + 1) \bmod 10 = 4 \rightarrow$ Collision 2
$(13 + 2) \bmod 10 = 5 \rightarrow$ Collision 3
$(13 + 3) \bmod 10 = 6 \rightarrow$ Collision 4
$(13 + 4) \bmod 10 = 7$

0	T
1	K
2	J
3	C
4	N
5	Y
6	P
7	M
8	R
9	S

224. (a)

Probability for the first record not colliding is x/x.
Probability for the second record not colliding is $x - 1/x$. (This is because one place is already occupied. So favorable number of cases is $x - 1$).
Probability for the third record not colliding is $x - 2/x$.
Probability for the $(m-1)^{th}$ record not colliding is $x - (m-2)/x$.
Now the next (m^{th}) record is resulting in a collision.
Out of the x places, it should hash to one of the $(m-1)$ places already, filled.
So probability is $(m-1)/x$.
The required probability is
$(x/x)\ (x - 1/x)\ (x - 2/x)\ ...(x - (m-2)/x)\ (m - 1/x)$

225. (a)

Load factor is the ratio of number of records that are currently present and total number of records that can be present. If the load factor is less, free space will be more. This means probability of collision is less. So, the search time will be less.

226. (c)

Input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199)

Given Hash function,

$h(a) = x \bmod 10$

$h(a) = h(1)$

$= \{1471, 6171\}$ hash to the same value

$h(a) = h(2) = \{4322\}$

$h(a) = h(3) = \{6173\}$

$h(a) = h(9)$

$= \{9679, 1989, 4199\}$ has to the same value.

$h(0), h(5), h(6), h(7), h(8)$ are empty for given input.
So the statement (i) and (ii) are correct.

227. (c)

Deletion is easier, because chained hash table uses linked list.

228. (c)

A hash function takes a message of arbitrary length and generates a fixed length code by taking mod (some value).
A hash function may give the same hash value for distinct messages.

229. (c)

P(First 3 slots are unfilled after the first 3 insertions)

$= \frac{97}{100} \times \frac{96}{100} \times \frac{95}{100}$ because of uniform hashing

230. (a)

Hasting produces a different key value for every input. So, by using that key value one can search any item in a constant time.
So, complexity of search is O(1).

231. (b)

In open addressing scheme lots of space is wasted while space present inside hash table in order to cooperate with collision. So chained hash table is used to reduce the space needed to store the same set of key, but it will make difficult to deletion of element and searching of element.

232. (c)

0	98
1	56
2	37
3	38
4	72
5	11
6	48

233. (d)

43, 36, 92, 87, 11, 4, 71, 13, 14

$h(k) = k \bmod 11$

87	11	13	36	92	4	71	14			43
0	1	2	3	4	5	6	7	8	9	10

$\therefore$ At index 7, last record 14 is inserted.

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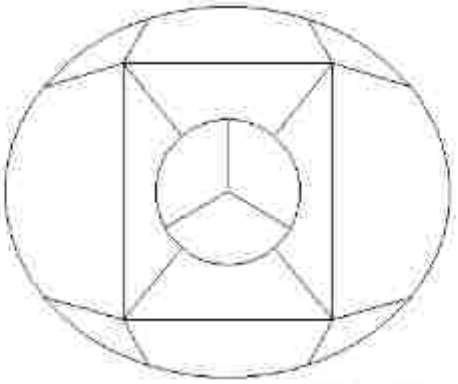
General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

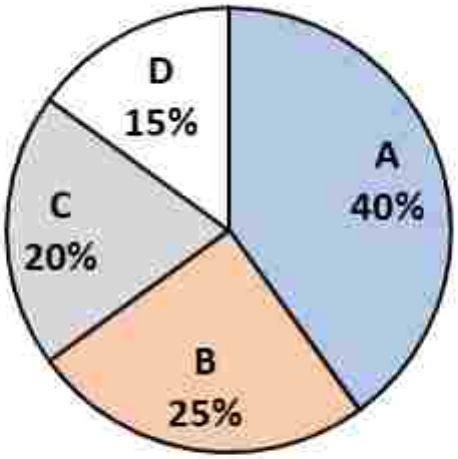
Q.1	If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft]. Which one of the given options is appropriate to fill the blank?
(A)	frown
(B)	fawn
(C)	vein
(D)	vain

GATE 2024

Q.2	The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is 
(A)	4
(B)	3
(C)	5
(D)	6

GATE 2024

Q.3	How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1, 3, 4, 6, 7\}$, such that no digit appears more than once in a number?
(A)	24
(B)	48
(C)	72
(D)	12
Q.4	The sum of the following infinite series is $2+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{8}+\frac{1}{9}+\frac{1}{16}+\frac{1}{27}+\cdots$
(A)	11/3
(B)	7/2
(C)	13/4
(D)	9/2

Q.5	In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid. **Share of valid votes**  Based on the data provided, the total number of valid votes received by the candidates B and C is
(A)	45,000
(B)	49,500
(C)	51,750
(D)	54,000

Q.6 – Q.10 Carry TWO marks Each

Q.6	Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?
(A)	All human beings can digest dairy milk.
(B)	No human being can digest dairy milk.
(C)	Digestion of dairy milk is essential for human beings.
(D)	In human beings, digestion of dairy milk resulted from a mutated gene.
Q.7	The probability of a boy or a girl being born is 1/2. For a family having only three children, what is the probability of having two girls and one boy?
(A)	3/8
(B)	1/8
(C)	1/4
(D)	1/2

Q.8

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund A	Mutual fund B	Mutual fund C
Person 1	₹10,000	₹20,000	₹20,000
Person 2	₹20,000	₹15,000	₹15,000

At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

(A) 7.5%

(B) 10%

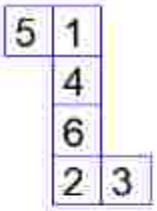
(C) 15%

(D) 20%

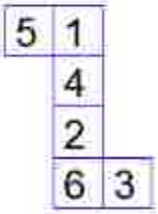
Q.9 Three different views of a dice are shown in the figure below:

The piece of paper that can be folded to make this dice is

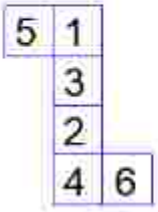
(A)



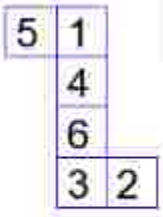
(B)



(C)



(D)



Q.10	Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?
(A)	A rhombus
(B)	A triangle
(C)	An ellipse
(D)	A hexagon

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Q.11 – Q.35 Carry ONE mark Each

Q.11	Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which **ONE** of the following options is **correct**?
(A)	Both (i) and (ii) are true
(B)	(i) is true and (ii) is false
(C)	(ii) is true and (i) is false
(D)	Both (i) and (ii) are false

Q.12	Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails. What is the probability of the event $T \cap S$?
(A)	0
(B)	0.5
(C)	0.25
(D)	1

Q.13	Consider the matrix $\boldsymbol{M} = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$. Which **ONE** of the following statements is **TRUE**?
(A)	The eigenvalues of $\boldsymbol{M}$ are non-negative and real.
(B)	The eigenvalues of $\boldsymbol{M}$ are complex conjugate pairs.
(C)	One eigenvalue of $\boldsymbol{M}$ is positive and real, and another eigenvalue of $\boldsymbol{M}$ is zero.
(D)	One eigenvalue of $\boldsymbol{M}$ is non-negative and real, and another eigenvalue of $\boldsymbol{M}$ is negative and real.

Q.14	Consider performing depth-first search (DFS) on an undirected and unweighted graph G starting at vertex s. For any vertex u in G, $d[u]$ is the length of the shortest path from s to u. Let (u, v) be an edge in G such that $d[u] < d[v]$. If the edge (u, v) is explored first in the direction from u to v during the above DFS, then (u, v) becomes a _____ edge.
(A)	tree
(B)	cross
(C)	back
(D)	gray

Q.15	For any twice differentiable function $f:\mathbb{R}\to\mathbb{R}$, if at some $x^*\in\mathbb{R}$, $f'(x^*)=0$ and $f''(x^*)>0$, then the function f necessarily has a _____ at $x=x^*$. **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	local minimum
(B)	global minimum
(C)	local maximum
(D)	global maximum

Q.16	Match the items in **Column 1** with the items in **Column 2** in the following table: **Column 1** (p) First In First Out (q) Lookup Operation (r) Last In First Out **Column 2** (i) Stacks (ii) Queues (iii) Hash Tables
(A)	(p) – (ii), (q) – (iii), (r) – (i)
(B)	(p) – (ii), (q) – (i), (r) – (iii)
(C)	(p) – (i), (q) – (ii), (r) – (iii)
(D)	(p) – (i), (q) – (iii), (r) – (ii)

Q.17	Consider the dataset with six datapoints: $\{(x_1, y_1), (x_2, y_2), \ldots, (x_6, y_6)\}$, where $x_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $x_2 = \begin{bmatrix}0\\1\end{bmatrix}$, $x_3 = \begin{bmatrix}0\\-1\end{bmatrix}$, $x_4 = \begin{bmatrix}-1\\0\end{bmatrix}$, $x_5 = \begin{bmatrix}2\\2\end{bmatrix}$, $x_6 = \begin{bmatrix}-2\\-2\end{bmatrix}$ and the labels are given by $y_1 = y_2 = y_5 = 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear support vector machine is trained on the above dataset. Which **ONE** of the following sets is a possible set of support vectors?
(A)	$\{x_1, x_2, x_5\}$
(B)	$\{x_3, x_4, x_5\}$
(C)	$\{x_4, x_5\}$
(D)	$\{x_1, x_2, x_3, x_4\}$

Q.18	Match the items in **Column 1** with the items in **Column 2** in the following table: **Column 1**: (p) Principal Component Analysis; (q) Naive Bayes Classification; (r) Logistic Regression **Column 2**: (i) Discriminative Model; (ii) Dimensionality Reduction; (iii) Generative Model
(A)	(p) – (iii), (q) – (i), (r) – (ii)
(B)	(p) – (ii), (q) – (i), (r) – (iii)
(C)	(p) – (ii), (q) – (iii), (r) – (i)
(D)	(p) – (iii), (q) – (ii), (r) – (i)

Q.19	Euclidean distance based k-means clustering algorithm was run on a dataset of 100 points with $k = 3$. If the points $\begin{bmatrix}1\\1\end{bmatrix}$ and $\begin{bmatrix}-1\\1\end{bmatrix}$ are both part of cluster 3, then which **ONE** of the following points is necessarily also part of cluster 3?
(A)	$\begin{bmatrix}0\\0\end{bmatrix}$
(B)	$\begin{bmatrix}0\\2\end{bmatrix}$
(C)	$\begin{bmatrix}2\\0\end{bmatrix}$
(D)	$\begin{bmatrix}0\\1\end{bmatrix}$

Q.20	Given a dataset with K binary-valued attributes (where $K > 2$) for a two-class classification task, the number of parameters to be estimated for learning a naïve Bayes classifier is
(A)	$2^K + 1$
(B)	$2K + 1$
(C)	$2^{K+1} + 1$
(D)	$K^2 + 1$

Q.21	Consider performing uniform hashing on an open address hash table with load factor $\alpha = \frac{n}{m} < 1$, where n elements are stored in the table with m slots. The expected number of probes in an unsuccessful search is at most $\frac{1}{1-\alpha}$. Inserting an element in this hash table requires at most _____ probes, on average.
(A)	$\ln\left(\frac{1}{1-\alpha}\right)$
(B)	$\frac{1}{1-\alpha}$
(C)	$1+\frac{\alpha}{2}$
(D)	$\frac{1}{1+\alpha}$

Q.22	For any binary classification dataset, let $S_B \in \mathbb{R}^{d\times d}$ and $S_W \in \mathbb{R}^{d\times d}$ be the between-class and within-class scatter (covariance) matrices, respectively. The Fisher linear discriminant is defined by $u^* \in \mathbb{R}^d$, that maximizes $$J(u) = \frac{u^T S_B u}{u^T S_W u}$$ If $\lambda = J(u^*)$, S_W is non-singular and $S_B \neq 0$, then (u^*, λ) must satisfy which **ONE** of the following equations? **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	$S_W^{-1} S_B u^* = \lambda u^*$
(B)	$S_W u^* = \lambda S_B u^*$
(C)	$S_B S_W u^* = \lambda u^*$
(D)	$u^{*T} u^* = \lambda^2$

Q.23	Let h_1 and h_2 be two admissible heuristics used in A^* search. Which **ONE** of the following expressions is always an admissible heuristic?
(A)	$h_1 + h_2$
(B)	$h_1 \times h_2$
(C)	$h_1/h_2, (h_2 \neq 0)$
(D)	$\lvert h_1 - h_2 \rvert$

Q.24	Consider five random variables $U, V, W, X,$ and Y whose joint distribution satisfies: $$P(U,V,W,X,Y) = P(U)P(V)P(W\lvert U,V)P(X\lvert W)P(Y\lvert W)$$ Which **ONE** of the following statements is **FALSE**?
(A)	Y is conditionally independent of V given W
(B)	X is conditionally independent of U given W
(C)	U and V are conditionally independent given W
(D)	Y and X are conditionally independent given W

Q.25	Consider the following statement: In adversarial search, $\alpha-\beta$ pruning can be applied to game trees of any depth where α is the **(m)** value choice we have formed so far at any choice point along the path for the MAX player and β is the **(n)** value choice we have formed so far at any choice point along the path for the MIN player. Which **ONE** of the following choices of **(m)** and **(n)** makes the above statement valid?
(A)	**(m)** = highest, **(n)** = highest
(B)	**(m)** = lowest, **(n)** = highest
(C)	**(m)** = highest, **(n)** = lowest
(D)	**(m)** = lowest, **(n)** = lowest

Q.26	Consider a database that includes the following relations: Defender(*name, rating, side, goals*) Forward(*name, rating, assists, goals*) Team(*name, club, price*) Which **ONE** of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where ϕ denotes the empty set?
(A)	$\Pi_{name}(\text{Team}) \setminus \left(\Pi_{name}(\text{Defender}) \cap \Pi_{name}(\text{Forward})\right) = \phi$
(B)	$\left(\Pi_{name}(\text{Defender}) \cap \Pi_{name}(\text{Forward})\right) \setminus \Pi_{name}(\text{Team}) = \phi$
(C)	$\Pi_{name}(\text{Team}) \setminus \left(\Pi_{name}(\text{Defender}) \cup \Pi_{name}(\text{Forward})\right) = \phi$
(D)	$\left(\Pi_{name}(\text{Defender}) \cup \Pi_{name}(\text{Forward})\right) \setminus \Pi_{name}(\text{Team}) = \phi$

Q.27	Let the minimum, maximum, mean and standard deviation values for the attribute *income* of data scientists be ₹46000, ₹170000, ₹96000, and ₹21000, respectively. The z-score normalized *income* value of ₹106000 is closest to which **ONE** of the following options?
(A)	0.217
(B)	0.476
(C)	0.623
(D)	2.304

Q.28	Consider the following tree traversals on a full binary tree: (i) Preorder (ii) Inorder (iii) Postorder Which of the following traversal options **is/are** sufficient to uniquely reconstruct the full binary tree?
(A)	(i) and (ii)
(B)	(ii) and (iii)
(C)	(i) and (iii)
(D)	(ii) only

Q.29	Let x and y be two propositions. Which of the following statements **is a tautology /are tautologies**?
(A)	$(\neg x \wedge y) \Longrightarrow (y \Longrightarrow x)$
(B)	$(x \wedge \neg y) \Longrightarrow (\neg x \Longrightarrow y)$
(C)	$(\neg x \wedge y) \Longrightarrow (\neg x \Longrightarrow y)$
(D)	$(x \wedge \neg y) \Longrightarrow (y \Longrightarrow x)$

Q.30	Consider sorting the following array of integers in ascending order using an in-place Quicksort algorithm that uses the last element as the pivot. 60 | 70 | 80 | 90 | 100 The minimum number of swaps performed during this Quicksort is _____.

Q.31 Consider the following two tables named Raider and Team in a relational database maintained by a Kabaddi league. The attribute *ID* in table Team references the primary key of the Raider table, *ID*.

Raider

ID	*Name*	*Raids*	*RaidPoints*
1	Arjun	200	250
2	Ankush	190	219
3	Sunil	150	200
4	Reza	150	190
5	Pratham	175	220
6	Gopal	193	215

Team

City	*ID*	*BidPoints*
Jaipur	2	200
Patna	3	195
Hyderabad	5	175
Jaipur	1	250
Patna	4	200
Jaipur	6	200

The SQL query described below is executed on this database:

```
SELECT *
FROM Raider, Team
WHERE Raider.ID=Team.ID AND City="Jaipur" AND
RaidPoints > 200;
```

The number of rows returned by this query is _____.

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Q.32 The fundamental operations in a double-ended queue D are:

`insertFirst(e)` – Insert a new element `e` at the beginning of D.
`insertLast(e)` – Insert a new element `e` at the end of D.
`removeFirst()` – Remove and return the first element of D.
`removeLast()` – Remove and return the last element of D.

In an empty double-ended queue, the following operations are performed:

```
insertFirst(10)

insertLast(32)

a ←removeFirst()

insertLast(28)

insertLast(17)

a ←removeFirst()

a ← removeLast()
```

The value of **a** is _____.

Q.33 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = \frac{1}{1+e^{-x}}$.

The value of the derivative of f at x where $f(x) = 0.4$ is _____ (rounded off to **two** decimal places).

Note: $\mathbb{R}$ denotes the set of real numbers.

Q.34	The sample average of 50 data points is 40. The updated sample average after including a new data point taking the value of 142 is _____.

Q.35	Consider the 3×3 matrix $\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$. The determinant of $(\boldsymbol{M}^2 + 12\boldsymbol{M})$ is _____.

Q.36	A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently. What is the expected number of times the die is thrown until **two** consecutive throws of even numbers are seen?
(A)	2
(B)	4
(C)	6
(D)	8

Q.37	Let $f:\mathbb{R} \to \mathbb{R}$ be a function. **Note**: $\mathbb{R}$ denotes the set of real numbers. $$f(x) = \begin{cases} -x, & \text{if } x < -2 \\ ax^2 + bx + c, & \text{if } x \in [-2,2] \\ x, & \text{if } x > 2 \end{cases}$$ Which **ONE** of the following choices gives the values of a, b, c that make the function f continuous and differentiable?
(A)	$a = \frac{1}{4}, b = 0, c = 1$
(B)	$a = \frac{1}{2}, b = 0, c = 0$
(C)	$a = 0, b = 0, c = 0$
(D)	$a = 1, b = 1, c = -4$

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Q.38 Consider the following Python code:

```
def count(child_dict, i):
    if i not in child_dict.keys():
        return 1
    ans = 1
    for j in child_dict[i]:
        ans += count(child_dict, j)
    return ans

child_dict = dict()
child_dict[0] = [1,2]
child_dict[1] = [3,4,5]
child_dict[2] = [6,7,8]
print(count(child_dict,0))
```

Which **ONE** of the following is the output of this code?

(A) 6

(B) 1

(C) 8

(D) 9

Q.39	Consider the function `computeS` (X) whose pseudocode is given below:
	
	`computeS` (X)
	$S[1] \leftarrow 1$
	`for` $i \leftarrow 2$ `to` $length(X)$
	$S[i] \leftarrow 1$
	`if` $X[i-1] \leq X[i]$
	$S[i] \leftarrow S[i] + S[i-1]$
	`end if`
	`end for`
	`return` S
	
	Which **ONE** of the following values is returned by the function `computeS` (X) for $X = [6, 3, 5, 4, 10]$?

(A)	[1, 1, 2, 3, 4]
(B)	[1, 1, 2, 3, 3]
(C)	[1, 1, 2, 1, 2]
(D)	[1, 1, 2, 1, 5]

Q.40	Let $F(n)$ denote the maximum number of comparisons made while searching for an entry in a sorted array of size n using binary search. Which **ONE** of the following options is **TRUE**?
(A)	$F(n) = F(\lfloor n/2 \rfloor) + 1$
(B)	$F(n) = F(\lfloor n/2 \rfloor) + F(\lceil n/2 \rceil)$
(C)	$F(n) = F(\lfloor n/2 \rfloor)$
(D)	$F(n) = F(n-1) + 1$

Q.41	Consider the following Python function:
	
	`def fun(D, s1, s2):`
	`  if s1 < s2:`
	`  D[s1], D[s2] = D[s2], D[s1]`
	`  fun(D, s1+1, s2-1)`
	
	What does this Python function `fun()` do? Select the **ONE** appropriate option below.

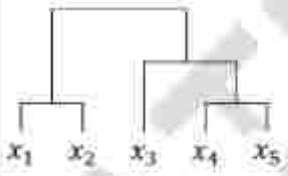
(A)	It finds the smallest element in D from index s1 to s2, both inclusive.
(B)	It performs a merge sort in-place on this list D between indices s1 and s2, both inclusive.
(C)	It reverses the list D between indices s1 and s2, both inclusive.
(D)	It swaps the elements in D at indices s1 and s2, and leaves the remaining elements unchanged.

Q.42 Consider the table below, where the $(i,j)^{th}$ element of the table is the distance between points x_i and x_j. Single linkage clustering is performed on data points, x_1, x_2, x_3, x_4, x_5.

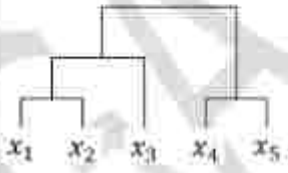
	x_1	x_2	x_3	x_4	x_5
x_1	0	1	4	3	6
x_2	1	0	3	5	3
x_3	4	3	0	2	5
x_4	3	5	2	0	1
x_5	6	3	5	1	0

Which **ONE** of the following is the correct representation of the clusters produced?

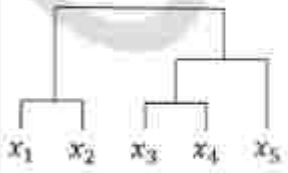
(A)



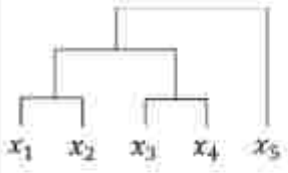
(B)

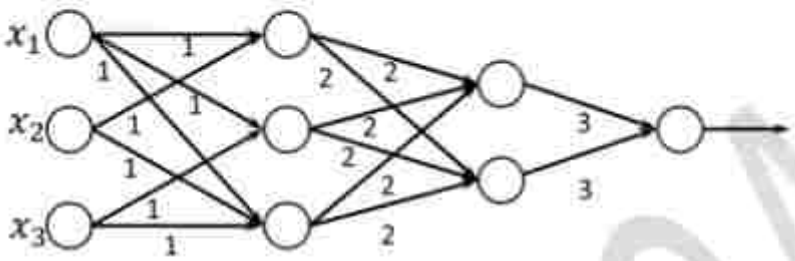
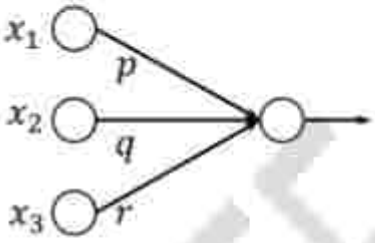


(C)



(D)



Q.43	Consider the two neural networks (NNs) shown in Figures 1 and 2, with $ReLU$ activation ($ReLU(z) = \max\{0, z\}, \forall z \in \mathbb{R}$). $\mathbb{R}$ denotes the set of real numbers. The connections and their corresponding weights are shown in the Figures. The biases at every neuron are set to 0. For what values of p, q, r in Figure 2 are the two NNs equivalent, when x_1, x_2, x_3 are positive? **Figure 1**  **Figure 2** 
(A)	$p = 36, q = 24, r = 24$
(B)	$p = 24, q = 24, r = 36$
(C)	$p = 18, q = 36, r = 24$
(D)	$p = 36, q = 36, r = 36$

Q.44	Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered $n+1$ and $n+2$. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list. Which **ONE** of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?
(A)	BFS expands more states than DFS.
(B)	DFS expands more states than BFS.
(C)	Both BFS and DFS expand equal number of states.
(D)	Both BFS and DFS do not reach the goal state number 6.

Q.45	Consider the following sorting algorithms: (i) Bubble sort (ii) Insertion sort (iii) Selection sort Which **ONE** among the following choices of sorting algorithms sorts the numbers in the array [4, 3, 2, 1, 5] in increasing order after **exactly two** passes over the array?
(A)	(i) only
(B)	(iii) only
(C)	(i) and (iii) only
(D)	(ii) and (iii) only

Q.46	Given the relational schema $R = (U, V, W, X, Y, Z)$ and the set of functional dependencies: $\{U \rightarrow V, U \rightarrow W, WX \rightarrow Y, WX \rightarrow Z, V \rightarrow X\}$ Which of the following functional dependencies can be derived from the above set?
(A)	$VW \rightarrow YZ$
(B)	$WX \rightarrow YZ$
(C)	$VW \rightarrow U$
(D)	$VW \rightarrow Y$

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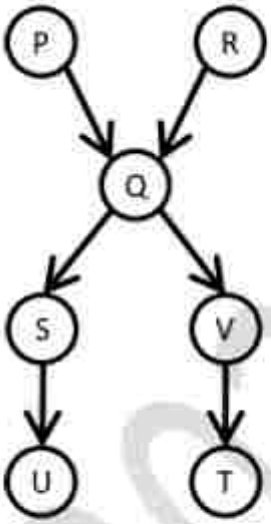
Q.47	Select all choices that are subspaces of $\mathbb{R}^3$. **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(B)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(C)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0\right\}$
(D)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ 5x_1 + 2x_3 + 4 = 0\right\}$

Q.48	Which of the following statements is/are **TRUE**? **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times3}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{Mx} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{Mx} = \boldsymbol{q}$ has infinite solutions.
(B)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times3}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{Mx} = \boldsymbol{p}$ has no solutions and $\boldsymbol{Mx} = \boldsymbol{q}$ has infinite solutions.
(C)	There exist $\boldsymbol{M} \in \mathbb{R}^{2\times3}, \boldsymbol{p} \in \mathbb{R}^2$, and $\boldsymbol{q} \in \mathbb{R}^2$ such that $\boldsymbol{Mx} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{Mx} = \boldsymbol{q}$ has infinite solutions.
(D)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times2}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{Mx} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{Mx} = \boldsymbol{q}$ has no solutions.

Q.49	Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$ and $\boldsymbol{M} \in \mathbb{R}^{3\times3}$ be the matrix corresponding to the projection on to the subspace U. Which of the following statements is/are **TRUE**?
(A)	If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of $\boldsymbol{M}$ is a 1-dimensional subspace.
(B)	If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of $\boldsymbol{M}$ is a 1-dimensional subspace.
(C)	$\boldsymbol{M}^2 = \boldsymbol{M}$
(D)	$\boldsymbol{M}^3 = \boldsymbol{M}$

Q.50	Consider the function $f: \mathbb{R} \to \mathbb{R}$ where $\mathbb{R}$ is the set of all real numbers. $$f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$$ Which of the following statements is/are **TRUE**?
(A)	$x = 0$ is a local maximum of f
(B)	$x = 3$ is a local minimum of f
(C)	$x = -1$ is a local maximum of f
(D)	$x = 0$ is a local minimum of f

Q.51	Consider the directed acyclic graph (DAG) below:  Which of the following is/are valid vertex orderings that can be obtained from a topological sort of the DAG?
(A)	P Q R S T U V
(B)	P R Q V S U T
(C)	P Q R S V U T
(D)	P R Q S V T U

Q.52	Let H, I, L, and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree. Which of the following statements is/are always **TRUE**?
(A)	$L \leq I + 1$
(B)	$H + 1 \leq N \leq 2^{H+1} - 1$
(C)	$H \leq I \leq 2^{H} - 1$
(D)	$H \leq L \leq 2^{H-1}$

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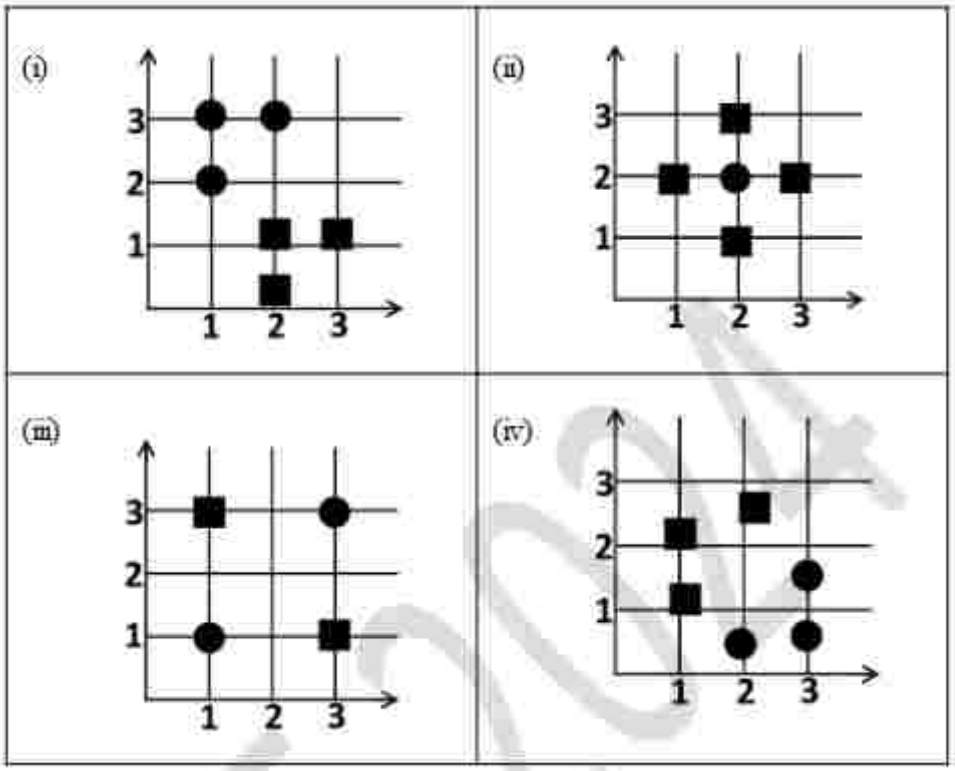
Q.53	Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and squares.  Which of the following is/are **TRUE**?
(A)	(i) is linearly separable.
(B)	(ii) is linearly separable.
(C)	(iii) is linearly separable.
(D)	(iv) is linearly separable.

Q.54	Let *game(ball, rugby)* be true if the ball is used in rugby and false otherwise. Let *shape(ball, round)* be true if the ball is round and false otherwise. Consider the following logical sentences: *s1:* $\forall ball \neg game(ball, rugby) \Longrightarrow shape(ball, round)$ *s2:* $\forall ball \neg shape(ball, round) \Longrightarrow game(ball, rugby)$ *s3:* $\forall ball\ game(ball, rugby) \Longrightarrow \neg shape(ball, round)$ *s4:* $\forall ball\ shape(ball, round) \Longrightarrow \neg game(ball, rugby)$ Which of the following choices is/are logical representations of the assertion, "All balls are round except balls used in rugby"?
(A)	$s1 \wedge s3$
(B)	$s1 \wedge s2$
(C)	$s2 \wedge s3$
(D)	$s3 \wedge s4$

Q.55	An OTT company is maintaining a large disk-based relational database of different movies with the following schema:
	
	`Movie(ID, CustomerRating)`
	`Genre(ID, Name)`
	`Movie_Genre(MovieID, GenreID)`
	
	Consider the following SQL query on the relation database above:
	
	`SELECT *`
	`FROM Movie, Genre, Movie_Genre`
	`WHERE`
	`Movie.CustomerRating > 3.4 AND`
	`Genre.Name = "Comedy" AND`
	`Movie_Genre.MovieID = Movie.ID AND`
	`Movie_Genre.GenreID = Genre.ID;`
	
	This SQL query can be sped up using which of the following indexing options?

A	B^+ tree on all the attributes.
B	Hash index on `Genre.Name` and B^+ tree on the remaining attributes.
C	Hash index on `Movie.CustomerRating` and B^+ tree on the remaining attributes.
D	Hash index on all the attributes.

Q.56	Let X be a random variable uniformly distributed in the interval $[1, 3]$ and Y be a random variable uniformly distributed in the interval $[2, 4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to **three** decimal places).

Q.57	Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by: $$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & otherwise \end{cases}$$ If $5E(X) = Var(X)$, where $E(X)$ and $Var(X)$ indicate the expectation and variance of X, respectively, the value of λ is _____ (rounded off to **one** decimal place).

Q.58	Consider two events T and S. Let $\overline{T}$ denote the complement of the event T. The probability associated with different events are given as follows: $$P(\overline{T}) = 0.6, \quad P(S\|T) = 0.3, \quad P(S\|\overline{T}) = 0.6$$ Then, $P(T\|S)$ is _____ (rounded off to **two** decimal places).

Q.59	Consider a joint probability density function of two random variables X and Y $$f_{X,Y}(x,y) = \begin{cases} 2xy, & 0 < x < 2, \quad 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$ Then, $E[Y\|X = 1.5]$ is _____.

Q.60	Evaluate the following limit: $$\lim_{x \to 0} \frac{\ln((x^2+1)\cos x)}{x^2} = _____.$$

Q.61

Let $\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $\boldsymbol{M} = \boldsymbol{u}\boldsymbol{u}^T$ (where $\boldsymbol{u}^T$ is the transpose of $\boldsymbol{u}$). The value of $\sum_{i=1}^{5} \sigma_i$ is _____.

Q.62

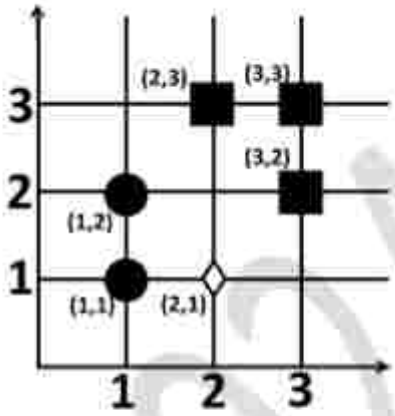
Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute *Pitch* can take one of two values: spin-friendly (represented as *S*) or pace-friendly (represented as *F*). The attribute *Format* can take one of two values: one-day match (represented as *O*) or test match (represented as *T*).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

To develop such a model, the computed InformationGain(C, *Pitch*) with respect to the Target is _____ (rounded off to **two** decimal places).

Table C

Match Number	Pitch	Format	Winner (Target)
1	*S*	*T*	Green
2	*S*	*T*	Blue
3	*F*	*O*	Blue
4	*S*	*O*	Blue
5	*F*	*T*	Green
6	*F*	*O*	Blue
7	*S*	*O*	Green
8	*F*	*T*	Blue
9	*F*	*O*	Blue
10	*S*	*O*	Green

Q.63	Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k-nearest neighbor algorithm for which the diamond ($\diamond$) shaped data point is assigned the label square is _____. 

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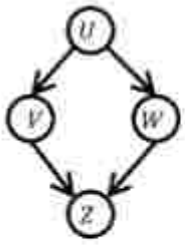
Q.64 Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U=0$	0.5
$U=1$	0.5

	$P(V=0\mid\cdot)$	$P(V=1\mid\cdot)$
$U=0$	0.5	0.5
$U=1$	0.5	0.5

	$P(W=0\mid\cdot)$	$P(W=1\mid\cdot)$
$U=0$	1	0
$U=1$	0	1

		$P(Z=0\mid\cdot)$	$P(Z=1\mid\cdot)$
$V=0$	$W=0$	0.5	0.5
$V=0$	$W=1$	1	0
$V=1$	$W=0$	1	0
$V=1$	$W=1$	0.5	0.5

The value of $P(U=1,V=1,W=1,Z=1)=$ _____ (rounded off to **three** decimal places).

Q.65 Two fair coins are tossed independently. X is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. Y is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise.

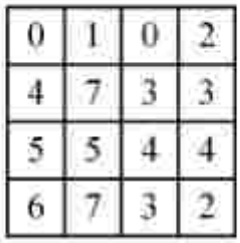
The value of the covariance of X and Y is _____ (rounded off to **three** decimal places).

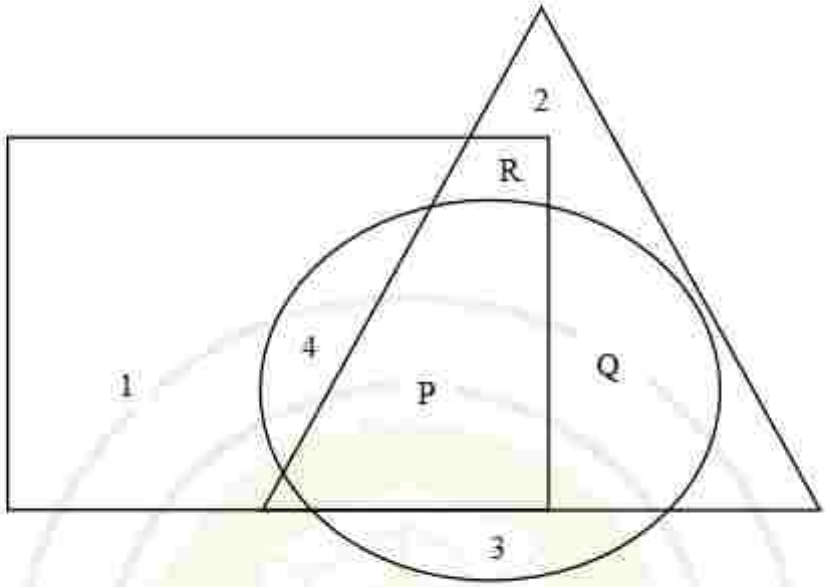
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	Courage : Bravery :: Yearning : _____ Select the most appropriate option to complete the analogy.
(A)	Longing
(B)	Yelling
(C)	Yawning
(D)	Glaring

Q.2	We _____ tennis in the lawn when it suddenly started to rain. Select the most appropriate option to complete the above sentence.
(A)	have been playing
(B)	had been playing
(C)	would have been playing
(D)	could be playing

Q.3	A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:
	
(A)	3
(B)	8
(C)	11
(D)	9

Q.4	In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R ?
	
(A)	P = 6; Q = 5; R = 3
(B)	P = 5; Q = 6; R = 3
(C)	P = 3; Q = 6; R = 6
(D)	P = 5; Q = 3; R = 6

Q.5	A rectangle has a length L and a width W, where L > W. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?
(A)	Perimeter increases by 10%.
(B)	Length of the diagonals increases by 10%.
(C)	Area increases by 10%.
(D)	The rectangle becomes a square.

Q.6 – Q.10 Carry TWO marks Each

Q.6 Column-I has statements made by Shanthala; and, Column-II has responses given by Kanishk.

Column-I		Column-II	
P.	This house is in a mess.	1.	Alright, I won't bring it up during our conversations.
Q.	I am not happy with the marks given to me.	2.	Well, you can easily look it up.
R.	Politics is a subject I avoid talking about.	3.	No problem, let me clear it up for you.
S.	I don't know what this word means.	4.	Don't worry, I will take it up with your teacher.

Identify the option that has the correct match between Column-I and Column-II.

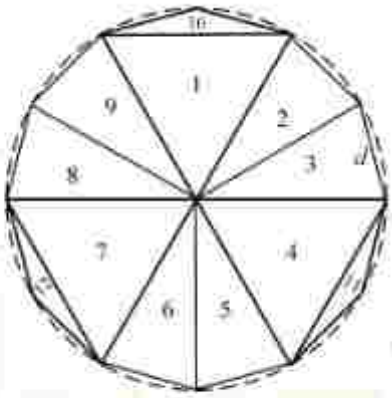
(A) P – 2; Q – 3; R – 1; S – 4

(B) P – 3; Q – 4; R – 1; S – 2

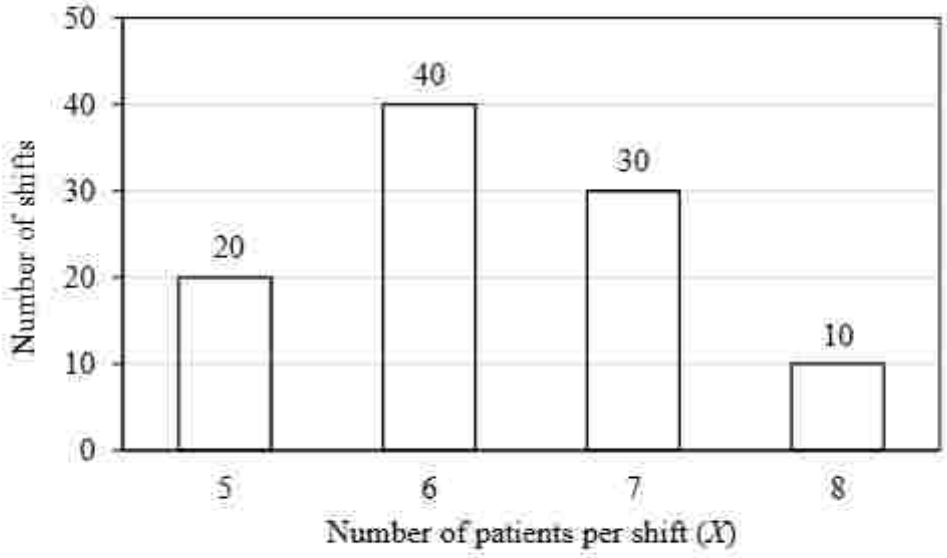
(C) P – 4; Q – 1; R – 2; S – 3

(D) P – 1; Q – 2; R – 4; S – 3

Q.7	Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight?
(A)	$a_1 < a_2 < 50$
(B)	$a_1 < 50 < a_2$
(C)	$50 < a_1 < a_2$
(D)	Either $a_1 = 50$ or $a_2 = 50$

Q.8	A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square. The number of squares that can be formed and the number of triangles required to form each square, respectively, are: Note: The figure shown is representative.
	
(A)	3; 4
(B)	4; 3
(C)	3; 3
(D)	3; 2

Q.9	If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:
(A)	2^{-1}
(B)	2^0
(C)	2^3
(D)	2^{15}

Q.10	The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is ₹ $1000(X - 0.2)$, what is the average amount (in ₹) she has earned per shift in the past 100 shifts? Note: The figure shown is representative.
	
(A)	6,100
(B)	6,300
(C)	6,000
(D)	6,500

Q. 11 – Q. 35 carry one mark each.

Q. 11 Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X|Y]$. Then $E[E[X|Y]]$ equals

(A) $E[X|Y]$

(B) $\dfrac{E[X]}{E[Y]}$

(C) $E[X]$

(D) $E[Y]$

Q. 12 The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order

(A) $O(n)$

(B) $O(n^2)$

(C) $O(n^3)$

(D) $O(n^4)$

Q. 13 The sum of the elements in each row of $A \in \mathbb{R}^{n \times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?

(A) The equation $Bx = 0$ has no solution

(B) The equation $Bx = 0$ has exactly two solutions

(C) The equation $Bx = 0$ has infinitely many solutions

(D) The equation $Bx = 0$ has a unique solution

Q. 14 Let $f(x) = \dfrac{e^x - e^{-x}}{2}, x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a. What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial)

(A) 0

(B) 1

(C) $\dfrac{1}{10!}$

(D) $\dfrac{2}{10!}$

Q. 15 Let p and q be any two propositions. Consider the following propositional statements.

$S_1 : p \to q, \quad S_2 : \neg p \wedge q, \quad S_3 : \neg p \vee q, \quad S_4 : \neg p \vee \neg q,$

where $\wedge$ denotes conjunction (AND operation), $\vee$ denotes disjunction (OR operation), and $\neg$ denotes negation (NOT operation). Which one of the following options is correct? (Note: $\equiv$ denotes logical equivalence)

(A) $S_1 \equiv S_3$

(B) $S_2 \equiv S_3$

(C) $S_2 \equiv S_4$

(D) $S_1 \equiv S_4$

Q. 16 If a relational decomposition is not dependency-preserving, which one of the following relational operators will be executed more frequently in order to maintain the dependencies?

(A) Selection

(B) Projection

(C) Join

(D) Set union

Q. 17 Consider the following three relations:

`Car (model, year, serial, color)`

`Make (maker, model)`

`Own (owner, serial)`

A tuple in `Car` represents a specific car of a given `model`, made in a given `year`, with a `serial` number and a `color`. A tuple in `Make` specifies that a `maker` company makes cars of a certain `model`. A tuple in `Own` specifies that an `owner` owns the car with a given `serial` number. Keys are underlined; (`owner`, `serial`) together form key for `Own`. ($\bowtie$ denotes natural join)

$$\pi_{\texttt{owner}}(\texttt{Own} \bowtie (\sigma_{\texttt{color}=\text{"red"}}(\texttt{Car} \bowtie (\sigma_{\texttt{maker}=\text{"ABC"}}\texttt{Make}))))$$

Which one of the following options describes what the above expression computes?

(A) All owners of a red car, a car made by ABC, or a red car made by ABC

(B) All owners of more than one car, where at least one car is red and made by ABC

(C) All owners of a red car made by ABC

(D) All red cars made by ABC

Q. 18 Consider a hash table of size 10 with indices $\{0, 1, \ldots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10},$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: $1, 4, 5, 6, 14, 15$. The indices where the keys 14 and 15 are stored are, respectively

(A) 2 and 5

(B) 2 and 6

(C) 4 and 5

(D) 4 and 6

Q. 19 Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t, is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \dfrac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

If the median of X is 3, then what is the value of t?

(A) 2

(B) 1

(C) -1

(D) 0

Q. 20 Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4,$$

where $E[X]$ denotes the expectation of random variable X. The values of a, b are:

(A) $a = -2, b = 1$

(B) $a = 2, b = -1$

(C) $a = -2, b = -1$

(D) $a = 1, b = 1$

Q. 21 It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X. What is the value of λ? ($\ln$ denotes natural logarithm)

(A) $\ln 2$

(B) $\ln 4$

(C) $\ln 3$

(D) $\ln 0.25$

Q. 22 Consider designing a linear classifier

$$y = \text{sign}(f(x; w, b)), \quad f(x; w, b) = w^{\mathrm{T}} x + b$$

on a dataset $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)\}$, $x_i \in \mathbb{R}^d$, $y_i \in \{+1, -1\}$, $i = 1, 2, \ldots, N$. Recall that the sign function outputs +1 if the argument is positive, and -1 if the argument is non-positive. The parameters w and b are updated as per the following training algorithm:

$$w_{new} = w_{old} + y_n x_n, \quad b_{new} = b_{old} + y_n$$

whenever $\text{sign}(f(x_n; w_{old}, b_{old})) \neq y_n$. In other words, whenever the classifier wrongly predicts a sample (x_n, y_n) from the dataset, w_{old} gets updated to w_{new}, and likewise b_{old} gets updated to b_{new}. Consider the case $(x_n, +1)$, $f(x_n; w_{old}, b_{old}) < 0$. Then

(A) $f(x_n; w_{new}, b_{new}) > f(x_n; w_{old}, b_{old})$

(B) $f(x_n; w_{new}, b_{new}) < f(x_n; w_{old}, b_{old})$

(C) $f(x_n; w_{new}, b_{new}) = f(x_n; w_{old}, b_{old})$

(D) $y_n f(x_n; w_{old}, b_{old}) > 1$

Q. 23 Consider the following Python declarations of two lists.

```
A=[1,2,3]
B=[4,5,6]
```

Which one of the following statements results in A= $[1, 2, 3, 4, 5, 6]$?

(A) `A.extend(B)`

(B) `A.append(B)`

(C) `A.update(B)`

(D) `A.insert(B)`

Q. 24 Consider two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to (1, \infty)$. Both functions are differentiable at a point c. Which of the following functions is/are ALWAYS differentiable at c? The symbol · denotes product and the symbol $\circ$ denotes composition of functions.

(A) $f \pm g$

(B) $f \cdot g$

(C) $\frac{f}{g}$

(D) $f \circ g + g \circ f$

Q. 25 Which of the following statements is/are correct?

(A) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors

(B) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

(C) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

(D) Orthonormal vectors $\mathbb{R}^n$ are linearly independent

Q. 26 Which of the following statements is/are correct in a Bayesian network?

(A) Variable elimination is an approximate inference algorithm

(B) Gibbs sampling is an exact inference algorithm

(C) Variable elimination is used to determine conditional probabilities

(D) Rejection sampling is an approximate inference algorithm

Q. 27 For which of the following inputs does binary search take time $O(\log n)$ in the worst case?

(A) An array of n integers in any order

(B) A linked list of n integers in any order

(C) An array of n integers in increasing order

(D) A linked list of n integers in increasing order

Q. 28 Let $A = I_n + xx^\top$, where I_n is the $n \times n$ identity matrix and $x \in \mathbb{R}^n$, $x^\top x = 1$. Which of the following options is/are correct?

(A) Rank of A is n

(B) A is invertible

(C) 0 is an eigenvalue of A

(D) A^{-1} has a negative eigenvalue

Q. 29 Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x.

(A) 10

(B) 12

(C) 14

(D) 16

Q. 30 Let C_1 and C_2 be two sets of objects. Let $D(x, y)$ be a measure of dissimilarity between two objects x and y. Consider the following definitions of dissimilarity between C_1 and C_2.

$$\textbf{DIS-1}(C_1, C_2) = \max_{x \in C_1,\, y \in C_2} D(x, y)$$

$$\textbf{DIS-2}(C_1, C_2) = \min_{x \in C_1,\, y \in C_2} D(x, y)$$

Which of the following statements is/are correct?

(A) Single Linkage Clustering uses **DIS-1**

(B) Single Linkage Clustering uses **DIS-2**

(C) Complete Linkage Clustering uses **DIS-2**

(D) Complete Linkage Clustering uses **DIS-1**

Q. 31 There are three boxes containing white balls and black balls.

Box-1 contains 2 black and 1 white balls.

Box-2 contains 1 black and 2 white balls.

Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____(Round off to two decimal places)

Q. 32 $\lim_{t \to +\infty} \sqrt{t^2 + t} - t =$ _____

(Round off to one decimal place)

Q. 33 On a relation named **Loan** of a bank:

Loan		
loan_number	**branch_name**	**amount**
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

the following SQL query is executed.

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
                   FROM Loan L2
                   WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____(Answer in integer)

Q. 34 Given data $\{(-1, 1), (2, -5), (3, 5)\}$ of the form (x, y), we fit a model $y = wx$ using linear least-squares regression. The optimal value of w is _____

(Round off to three decimal places)

Q. 35 The naive Bayes classifier is used to solve a two-class classification problem with class-labels y_1, y_2. Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with

$$P(x|y_1) = \frac{3}{4} \quad \text{and} \quad P(x|y_2) = \frac{1}{4}$$

for a specific feature vector x. The probability of misclassifying x is _____

(Round off to two decimal places)

Q. 36 – Q. 65 carry two marks each.

Q. 36 Let $Y = Z^2$, $Z = \dfrac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2. The variance of Y is

(A) 1

(B) 2

(C) 3

(D) 4

Q. 37 Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?

(A) A is invertible

(B) Determinant of A is 0

(C) The sum of the diagonal elements of A is 1

(D) A and A^2 have the same rank

Q. 38 Let $\{x_1, x_2, \ldots, x_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j)-th element of matrix $A \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = x_i^\top x_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct?

(A) A is invertible

(B) 0 is a singular value of A

(C) Determinant of A is 0

(D) $z^\top A z = 0$ for some non-zero $z \in \mathbb{R}^n$

Q. 39 Consider the cumulative distribution function (CDF) of a random variable X:

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases}$$

The value of $P(X^2 \leq 0.25)$ is

(A) 0.625

(B) 0.25

(C) 0.5

(D) 0.5625

Q. 40 A random variable X is said to be distributed as $Bernoulli(\theta)$, denoted by $X \sim Bernoulli(\theta)$, if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim Bernoulli(\theta)$, $i = 1, 2, \ldots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through Central Limit Theorem, is given by

(Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}}\, dt$)

(A) $\phi(2) - \phi(-2)$

(B) $\phi(1) - \phi(-1)$

(C) $\phi(3) - \phi(-3)$

(D) $\phi(90) - \phi(60)$

Q. 41 For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k + 1,$$

where k is an integer. Let $Y = \lfloor X \rfloor$, where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where ln denotes natural logarithm. For any positive integer ℓ, one can write the probability of the event $Y = \ell$ as follows

$$P(Y = \ell) = q^{\ell}(1 - q)$$

The value of q is

(A) 0.1

(B) 0.01

(C) 0.5

(D) 0.434

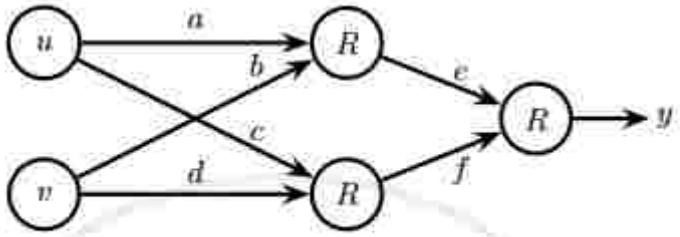
Q. 42 Consider the neural network shown in the figure with

inputs: u, v

weights: a, b, c, d, e, f

output: y

R denotes the ReLU function, $R(x) = \max(0, x)$.



Given $u = 2, v = 3$,

$a = 1, b = 1, c = 1, d = -1, e = 4, f = -1$,

which one of the following is correct?

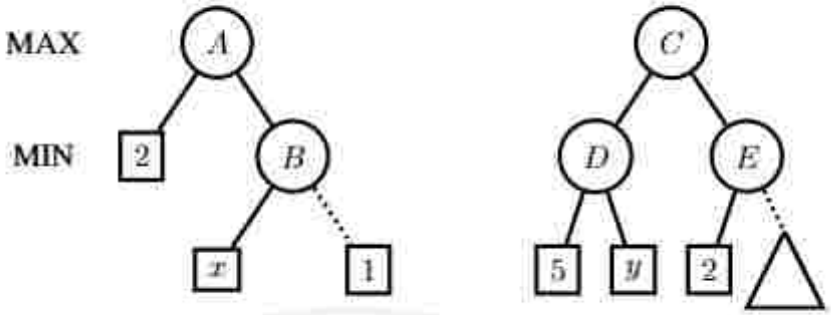
(A) $\dfrac{\partial y}{\partial a} = 8, \dfrac{\partial y}{\partial f} = 0$

(B) $\dfrac{\partial y}{\partial a} = 1, \dfrac{\partial y}{\partial f} = 0$

(C) $\dfrac{\partial y}{\partial a} = 1, \dfrac{\partial y}{\partial f} = -1$

(D) $\dfrac{\partial y}{\partial a} = 2, \dfrac{\partial y}{\partial f} = -1$

Q. 43 Consider game trees Tree-1 and Tree-2 as shown. The first level is a MAX agent and the second level is a MIN agent. The value in the square node is the output of the utility function.



Tree-1 Tree-2

For what ranges of x and y, the right child of node B and the right child of node E will be pruned by alpha-beta pruning algorithm?

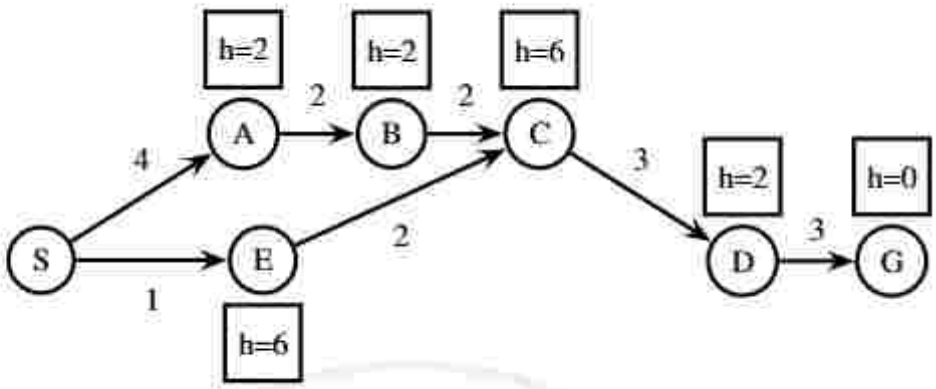
(A) $x \in [1, \infty)$ and $y \in (-\infty, 2]$

(B) $x \in (-\infty, 2]$ and $y \in (-\infty, 5]$

(C) $x \in (-\infty, 2]$ and $y \in [2, \infty)$

(D) $x \in [1, \infty)$ and $y \in (-\infty, 5]$

Q. 44 The state graph shows the action cost along the edges and the heuristic function h associated with each state.



Suppose A^* algorithm is applied on this state graph using priority queue to store the frontier. In what sequence are the nodes expanded?

(A) S,A,E,C,B,D,G

(B) S,E,A,C,B,D,G

(C) S,A,E,B,C,D,G

(D) S,A,B,E,C,D,G

Q. 45 A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

(A) 0

(B) 1

(C) $1 - \left(\frac{5}{6}\right)^{100}$

(D) $\left(\frac{5}{6}\right)^{100}$

Q. 46 Consider a fact table in an OLAP application: `Facts(D1, D2, val)`, where `D1` and `D2` are its dimension attributes and `val` is a dependent attribute. Suppose attribute `D1` takes 3 values and `D2` takes 2 values, and all combinations of these values are present in the table `Facts`. How many tuples are there in the result of the following query?

```
SELECT D1, D2, sum(val)
FROM Facts
GROUP BY CUBE (D1, D2);
```

(A) 1

(B) 6

(C) 9

(D) 12

Q. 47 Consider the following Python code snippet.

```
A={"this","that"}
B={"that","other"}
C={"other","this"}
while "other" in C:
    if "this" in A:
        A,B,C=A-B,B-C,C-A
    if "that" in B:
        A,B,C=C|A,A|B,B|C
```

When the above program is executed, at the end, which of the following sets contains `"this"`?

(A) Only `A`

(B) Only `B`

(C) Only `C`

(D) `A`, `C`

Q. 48 Which of the following statements is/are correct about the rectified linear unit (ReLU) activation function defined as $\text{ReLU}(x) = \max(x, 0)$, where $x \in \mathbb{R}$?

(A) ReLU is continuous everywhere

(B) ReLU is differentiable everywhere

(C) ReLU is not differentiable at $x = 0$

(D) $\text{ReLU}(x) = \text{ReLU}(ax)$, for all $a \in \mathbb{R}$

Q. 49 Consider the function $f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}$, $x \in [-8, 0]$. Which of the following statements is/are correct?

(A) The maximum value of f is attained at $x = -5$

(B) The minimum value of f is attained at $x = -2$

(C) The maximum value of f is $\frac{133}{2}$

(D) The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$

Q. 50 Let x_1, x_2, x_3, x_4, x_5 be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $A = x_1x_1^T + \ldots + x_5x_5^T$. Which of the following statements is/are correct?

(A) Singular values of A are also its eigenvalues

(B) Singular values of A are either 0 or 1

(C) Determinant of A is 1

(D) A is invertible

Q. 51 Let $f : \mathbb{R} \to \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct?

(A) f has a local minima

(B) There does not exist x and y, $x \neq y$, such that $f'(x) = f'(y) = 0$

(C) f has at most one global minimum

(D) f has at most one local minimum

Q. 52 An $n \times n$ matrix A with real entries satisfies the property: $\|Ax\|^2 = \|x\|^2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct?

(A) A must be orthogonal

(B) $A = I$, where I denotes the identity matrix, is the only solution

(C) The eigenvalues of A are either +1 or -1

(D) A has full rank

Q. 53 Consider designing a linear binary classifier $f(x) = \text{sign}(w^\top x + b)$, $x \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1:} \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \text{Class-2:} \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

Hard-margin support vector machine (SVM) formulation is solved to obtain w and b. Which of the following options is/are correct?

(A) $w = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = 1$

(B) The number of support vectors is 3

(C) The margin is $\sqrt{2}$

(D) Training accuracy is 98%

Q. 54 Consider a coin-toss experiment where the probability of head showing up is p. In the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

where n is the total number of independent coin tosses.

Which of the following statements is/are correct?

(A) $E[\widehat{p}] = p$

(B) $E[\widehat{p}] = \frac{p}{n}$

(C) As n increases, variance of $\widehat{p}$ decreases

(D) Variance of $\widehat{p}$ does not depend on n

Q. 55 Consider a two-class problem in $\mathbb{R}^d$ with class labels *red* and *green*. Let μ_{red} and μ_{green} be the means of the two classes. Given test sample $x \in \mathbb{R}^d$, a classifier calculates the squared Euclidean distance (denoted by $\|\cdot\|^2$) between x and the means of the two classes and assigns the class label that the sample x is closest to. That is, the classifier computes

$$f(x) = \|\mu_{red} - x\|^2 - \|\mu_{green} - x\|^2$$

and assigns the label *red* to x if $f(x) < 0$, and *green* otherwise. Which of the following statements is/are correct?

(A) The sample $x = 0$ is assigned the label *green* if $\|\mu_{red}\| < \|\mu_{green}\|$

(B) f is a linear function of x

(C) $f(x) = w^{\top}x + b$, where w and b are functions of μ_{red} and μ_{green}

(D) f is a quadratic polynomial in x

Q. 56 Consider the following two relations, named `Customer` and `Person`, in a database:

```
Person (
        aadhaar CHAR(12) PRIMARY KEY,
        name VARCHAR(32));
Customer (
        name VARCHAR(32),
        email VARCHAR(32) PRIMARY KEY,
        phone CHAR(10),
        aadhaar CHAR(12),
           FOREIGN KEY (aadhaar) REFERENCES Person(aadhaar));
```

Which of the following statements is/are correct?

(A) `aadhaar` is a candidate key in the `Customer` relation

(B) `phone` can be NULL in the `Customer` relation

(C) `aadhaar` is a candidate key in the `Person` relation

(D) `aadhaar` can be NULL in the `Person` relation

Q. 57 Consider a database relation R with attributes ABCDEFG, and having the following functional dependencies:

$A \rightarrow BCEF \qquad E \rightarrow DG \qquad BC \rightarrow A$

Which of the following statements is/are correct?

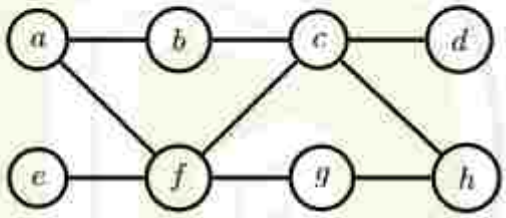
(A) A is the only candidate key of R

(B) A, BC are the candidate keys of R

(C) A, BC, E are the candidate keys of R

(D) Relation R is not in Boyce-Codd Normal Form (BCNF)

Q. 58 Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.



It is given that $a - b - c - d$ is a shortest path between a and d; $e - f - g - h$ is a shortest path between e and h; $a - f - c - h$ is a shortest path between a and h. Which of the following is/are NOT the edges of G?

(A) (b, d)

(B) (b, g)

(C) (b, h)

(D) (e, g)

Q. 59 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then $f(1) - f(0) =$ _____(Answer in integer)

Q. 60 Let $D = \{x^{(1)}, \ldots, x^{(n)}\}$ be a dataset of n observations where each $x^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^{n} x^{(i)} = 0$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \leq i \leq 100$. Let $u \in \mathbb{R}^{100}$ be the direction of maximum variance with $u^\top u = 1$.

The value of

$$\frac{1}{n}\sum_{i=1}^{n}\left(u^\top x^{(i)}\right)^2 = _____$$

(Answer in integer)

Q. 61 A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment.

The expectation of S_{100} denoted by $E[S_{100}] =$ _____

(Round off to one decimal place)

Q. 62 Consider the following tables, **Loan** and **Borrower**, of a bank.

Loan		
loan_num	**branch_name**	**amount**
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower	
customer_name	**loan_num**
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Query: $\pi_{\text{branch_name, customer_name}}(\textbf{Loan} \bowtie \textbf{Borrower}) \div \pi_{\text{branch_name}}(\textbf{Loan})$

where $\bowtie$ denotes natural join.

The number of tuples returned by the above relational algebra query is _____

(Answer in integer)

Q. 63 Consider the following Python code snippet.

```
def f(a,b):
    if (a==0):
        return b
    if (a%2==1):
        return 2*f((a-1)/2,b)
    return b+f(a-1,b)
print(f(15,10))
```

The value printed by the code snippet is _____ (Answer in integer)

Q. 64 Consider the following pseudocode.

```
Create empty stack S
Set x=0, flag=0, sum=0
Push x onto S
while (S is not empty){
  if (flag equals 0){
    Set x = x+1
    Push x onto S}
  if (x equals 8):
    Set flag=1
  if (flag equals 1){
    x = Pop(S)
    if (x is odd):
      Pop(S)
    Set sum = sum + x}
}
Output sum
```

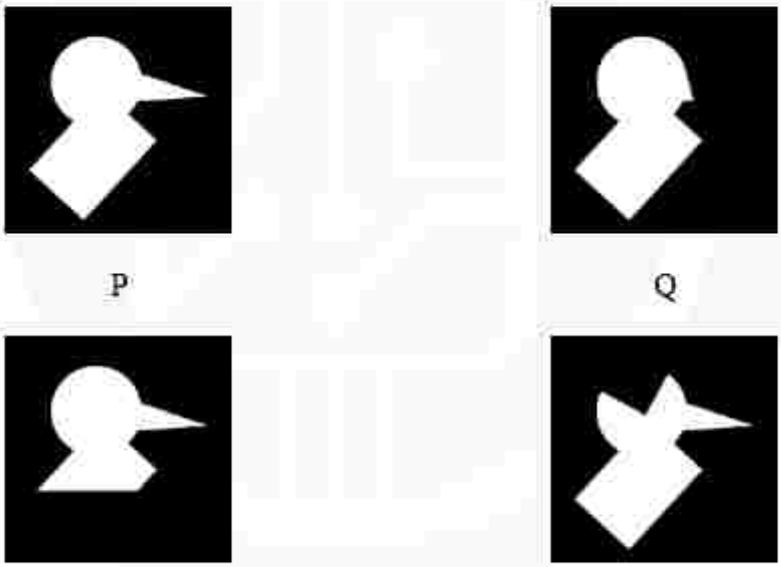
The value of `sum` output by a program executing the above pseudocode is _____
(Answer in integer)

Q. 65 Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \ldots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2, \text{for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in *decreasing* order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____(Answer in integer)

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q. 1	Verbosity : Brevity :: Insolence :_____ Choose the word that best fills the blank.
(A)	Innocence
(B)	Respect
(C)	Solace
(D)	Wealth
Q.2	The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _____
(A)	12
(B)	14
(C)	16
(D)	18

Q.3	The four pieces of a puzzle are shown in the figure below. Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?  P Q R S
(A)	P
(B)	Q
(C)	R
(D)	S

Q.4	Consider two distinct positive real numbers m, n, with $m > n$. Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is _____.
(A)	$x > y$
(B)	$x < y$
(C)	$x = y$
(D)	$x = \log_{10}(y)$

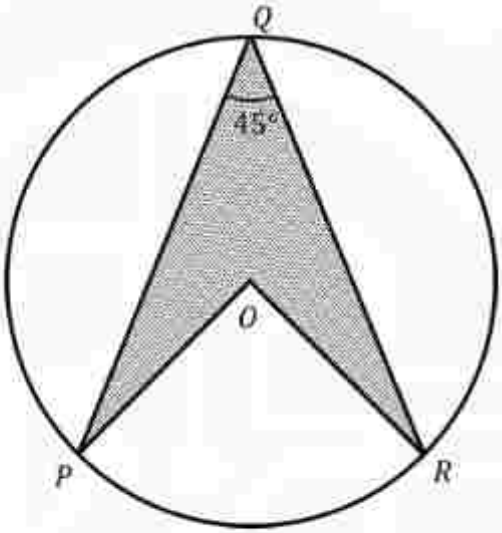
Q.5	'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.' Based only on the above sentence, which one of the following statements is true?
(A)	The actor will certainly sponsor his next movie.
(B)	His latest movie was a commercial success.
(C)	The actor made enough money from his latest movie.
(D)	His latest movie was not commercially successful.

Q.6 – Q.10 Carry TWO marks Each

Q.6	'My friend and I parted __ the door __ the cabin that I had rented __ the night.' Choose the option with the correct sequence of words to fill the blanks.
(A)	at; of; for
(B)	for; at; of
(C)	of; for; in
(D)	in; of; for
Q.7	Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20. Ignoring permutations, the number of ways to pick these five integers is _____
(A)	0
(B)	1
(C)	2
(D)	3

Q.8	Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that: (i) the first game was the only game where two students of the same class played against each other, (ii) the students of Class 5 won more games than the students of Class 4, and (iii) the boys won two games and the girls won one game. The student who did not lose any game is _____.
(A)	Pavan
(B)	Rishi
(C)	Swathi
(D)	Tanvi

Q.9	P, Q, R, S, X, and Y are distinct single-digit whole numbers taking values from 0 to 9. PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number. It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number. The value of Y is _____
(A)	4
(B)	5
(C)	6
(D)	7

Q.10	In the given figure, P, Q, and R are three points on a circle of radius 10 cm with O as its center, $\overline{PQ} = \overline{RQ}$, and $\angle PQR = 45°$. The figure is representative. The area of the shaded region $PQRO$ is _____ cm^2. 
(A)	50
(B)	$25\sqrt{2}$
(C)	$50\sqrt{2}$
(D)	100

Q.11 – Q.35 Carry ONE mark Each

Q.11	For a classification problem, Principal Component Analysis (PCA) has been used to reduce the dimensionality of a feature space from 100 to 10. Which of the following options is true about the angle θ between the first and the tenth principal components?
(A)	$\theta = 0°$
(B)	$\theta = 90°$
(C)	$90° < \theta \leq 180°$
(D)	$0 < \theta < 90°$
Q.12	Consider that you are training a classifier for a 10-class classification problem. Each input is represented as a 512-dimensional vector. There are 1000 samples, out of which first 100 will be used for testing. Let *Leave-One-Out-Cross-Validation* (LOOCV) be used for selection of the classifier model before testing. Which of the following options is the correct number of validation splits that will be generated?
(A)	10
(B)	512
(C)	900
(D)	1000

Q.13	Which of the following algorithms is **NOT** an example of uninformed search?
(A)	Breadth First Search
(B)	Depth First Search
(C)	A* Search
(D)	Depth-limited Search
Q.14	Which of the following statements is **NOT** true? (The names of the predicates are intuitive.)
(A)	$\forall x\ \forall y\ Classmate(x,y) \Rightarrow Classmate(y,x)$
(B)	$\forall x\ Likes(x, Icecream) \Rightarrow \neg\exists x\ \neg Likes(x, Icecream)$
(C)	"*Each king is a person*" is equivalent to $\forall x\ IsKing(x) \wedge IsPerson(x)$
(D)	"*All humans are mortal*" is equivalent to $\forall x\ IsHuman(x) \Rightarrow IsMortal(x)$

Q.15	Consider that the quick sort algorithm is used to sort an array of n distinct randomly ordered elements. In every call, the pivot is chosen as the first element of the current subarray. Let $T(n)$ denote the expected time to sort the array. Assume that the time to partition is linear in the size of the current subarray. Which of the following recurrence relations correctly represents $T(n)$ in this scenario?
(A)	$T(n) = T(1) + T(n-1) + O(n)$
(B)	$T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + O(n)$
(C)	$T(n) = 2T\left(\frac{n}{2}\right) + O(n)$
(D)	$T(n) = \frac{1}{n}\sum_{k=0}^{n-1}[T(k) + T(n-k-1)] + O(n)$

Q.16 Consider the given Python program.

```
def append_to_lst(val, lst=[]):
      lst.append(val)
      return lst
print(append_to_lst(1))
print(append_to_lst(2))
print(append_to_lst(3, []))
```

Which of the following is the correct output of this program?

(A)

```
[1]
[2]
[3]
```

(B)

```
[1]
[1, 2]
[3]
```

(C)

```
[1]
[2]
[1, 2, 3]
```

(D)

```
[1]
[1, 2]
[1, 3]
```

Q.17	Let $R(A,B,C,D,E)$ be a relational schema with functional dependency set $F=\{A \rightarrow BC,\ CD \rightarrow E,\ E \rightarrow A\}$. Which of the following statements is correct?
(A)	AD, ED and CD are the only candidate keys of R.
(B)	AD and ED are the only candidate keys of R.
(C)	A, E and CD are the only candidate keys of R.
(D)	A and CD are the only candidate keys of R.
Q.18	Consider that the visualization of a 3-dimensional data cube is showing *Sales Quantity* for each combination of the attributes *Product Type*, *Month* and *Country*. From this, if we want to further visualize the *Sales Quantity* for each combination of *Product Type*, *Month* and *State*, which of the following OLAP operations should be performed?
(A)	Slicing
(B)	Dicing
(C)	Roll-up
(D)	Drill-down

Q.19	Let M be a randomly chosen non-empty subset of $S = \{1,2,3, \ldots, 2026\}$. Which of the following is the probability that the product of all the elements of M is even?
(A)	$2^{1013}(2^{1013} - 1)/2^{2026}$
(B)	$2^{1013}/2^{2026}$
(C)	$2^{1013}(2^{1013} - 1)/(2^{2026} - 1)$
(D)	$1/(2^{2026} - 1)$
Q.20	Suppose that a computer program provides a non-negative and integer-valued random solution to the equation $n_1 + n_2 + n_3 + n_4 = 20$. Which of the following is the probability that all of n_1, n_2, n_3, n_4 in the provided solution are positive?
(A)	$\binom{19}{3}/\binom{23}{3}$
(B)	$\binom{20}{4}/\binom{24}{4}$
(C)	$\binom{20}{3}/\binom{23}{3}$
(D)	$\binom{19}{4}/\binom{24}{4}$

Q.21	Let $M = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ be a 2×2 matrix, where $\theta = \frac{2\pi}{5}$, and $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Which of the following options is equal to M^{2026} ?
(A)	M^2
(B)	M
(C)	M^{-1}
(D)	I_2
Q.22	Consider a set $S_1 = \{x = (x_1, x_2, x_3)^T \in \mathbb{R}^3 \mid x^T x \leq 16\}$. Let S_2 be another set which is a subspace of $\mathbb{R}^3$ with dimension two. Which of the following gives the area of $S_1 \cap S_2$?
(A)	16π
(B)	4π
(C)	$4\pi^2$
(D)	$16\pi^2$

Q.23 In the following table, the `Task` column lists a few tasks related to machine learning. The `Algorithm` column lists a few algorithms.

Each entry "`t`" from the `Task` column is to be matched with an appropriate entry "`a`" from the `Algorithm` column such that the task "`t`" can be solved using the algorithm "`a`". Denote such a match as `t:a`

`Task`	`Algorithm`
T1 – Clustering	A1 – Markov Chain Monte Carlo
T2 – Classification	A2 – K-Medoid
T3 – Sampling	A3 – Linear Discriminant Analysis
T4 – Feature Extraction	A4 – Naive Bayes

Which of the following options is/are the correct matching(s)?

(A) T1:A4, T2:A3, T3:A1, T4:A2

(B) T1:A2, T2:A4, T3:A1, T4:A3

(C) T1:A3, T2:A4, T3:A1, T4:A2

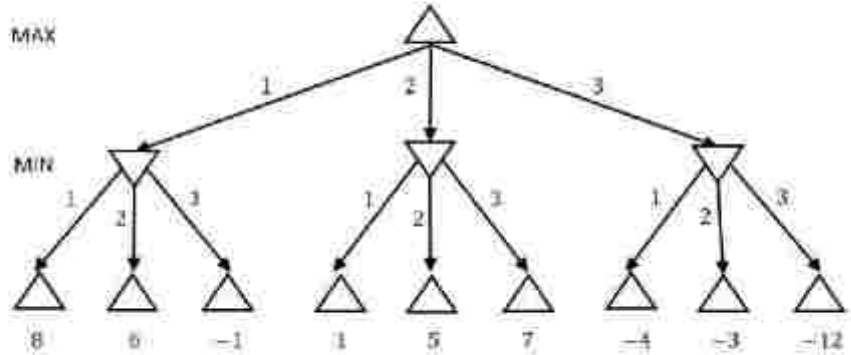
(D) T1:A4, T2:A2, T3:A1, T4:A3

Q.24	Sentence X is said to entail Sentence Y if whenever X is $TRUE$, Y also must hold $TRUE$. Which of the following statements is/are correct if X entails Y?
(A)	$X \Rightarrow Y$
(B)	$X \wedge \neg Y$ is $FALSE$
(C)	if X then Y
(D)	if Y then X

Q.25	You are given the following Pre-order and In-order traversals of a Binary Tree T with nodes E, F, G, P, Q, R, S. Pre-order: P Q S E R F G In-order: S Q E P F R G Which of the following statements is/are true about the Binary Tree T?
(A)	Node P is the root of T
(B)	The Post-order traversal of T is: S E Q F G R P
(C)	Node Q has only one child
(D)	The left subtree of node R contains the node G

Q.26	Consider two relations r and s defined on the relational schemas R(A,B) and S(E,C), respectively. A is the primary key of R and E is a foreign key of S referencing A in R. Which of the following operations will **NEVER** violate the foreign key constraint?
(A)	Inserting records into relation r
(B)	Deleting records from relation s
(C)	Deleting records from relation r
(D)	Inserting records into relation s
Q.27	Let $f(x) = x^3 - 3x^2 + 2$ be a function defined on $(-1, 3]$. Which of the following statements is/are correct?
(A)	$f(x)$ has exactly two roots in $[-0.9, 0]$.
(B)	$f(x)$ has a minimum at 2 only.
(C)	$f(x)$ has maximum at 0 only.
(D)	$f(x)$ has a root at 1.

Q.28	Suppose a random variable Z follows $Normal(\mu = 0, \sigma^2 = 1)$ distribution with probability density function $g(z)$ and cumulative distribution function $G(z)$. Another random variable Y follows t_1 distribution with probability density function $h(y)$ and cumulative distribution function $H(y)$. Let c be the positive real number for which $g(c) = h(c)$. Which of the following statements is/are correct?
(A)	$G(0) = H(0)$
(B)	$G(c) < H(c)$
(C)	$G(-c) < H(-c)$
(D)	$g(0) = h(0)$

Q.29	Consider that for a supervised learning task, the objective function being minimized is $f_w(x) = wx$, where $x \in \mathbb{R}$ is the input and $w \in \mathbb{R}$ is the parameter. Stochastic Gradient Descent with learning rate of 0.10 is used for parameter updates. Suppose that at the end of iteration i, the value of w becomes 10.00. Let $x = 10.00$ be the input for iteration $(i + 1)$. The value of w at the end of iteration $(i + 1)$ is _____. (Rounded off to **two** decimal places)
Q.30	Consider the game tree for a two-player turn-taking minimax game as shown in the figure. The value of a terminal node represents the utility of the game state if the game ends there. The numbers written next to the edges denote the strategies. There are two players MAX and MIN. At any particular state of the game, MAX prefers to move to a state of maximum value. On the other hand, MIN prefers to move to a state of minimum value. Suppose MAX starts the game at the root and has three strategies: 1, 2 and 3. Next, MIN plays and also has three strategies: 1, 2 and 3. The game ends there. Both players always take optimal strategies throughout the game. At the root, the best strategy for MAX is _____. (Answer in **integer**) 
Q.31	Let A be a sorted array containing 1000 distinct integers. You perform a recursive binary search on A to find an element y. Suppose each comparison checks whether the middle element computed during the current recursive step is equal to, less than, or greater than y. The maximum number of comparisons that may have to be performed if y is not an element of A is _____. (Answer in **integer**)

Q.32	In a relational database, a B+ Tree Index is to be constructed for a relation on a key field. In a B+ Tree, a Node Pointer points to a sub-tree and a Data Record Pointer points to a block of database records. Let, Node size = 4096 bytes, Node Pointer size = 10 bytes, Search Key Field size = 11 bytes and Data Record Pointer size = 12 bytes. The maximum number of Node Pointers that can be present in a non-leaf node of the B+ Tree is _____ . (Answer in **integer**)
Q.33	The number of bijections $f(\cdot)$ from the set $S = \{1, 2, 3, 4\}$ to itself such that $f(f(n)) = n$, for all $n \in S$, is _____ . (Answer in **integer**)
Q.34	Let X be an exponentially distributed random variable with mean $\lambda(> 0)$. If $P(X > 5) = 0.35$, then the conditional probability $P(X > 10 \mid X > 5)$ is _____ . (Rounded off to **two** decimal places)
Q.35	The value of $\sum_{i=0}^{\infty} \sum_{j=1}^{\infty} 2^{-i}\, 3^{-j}$ is _____ . (Answer in **integer**)

Q.36 – Q.65 Carry TWO marks Each

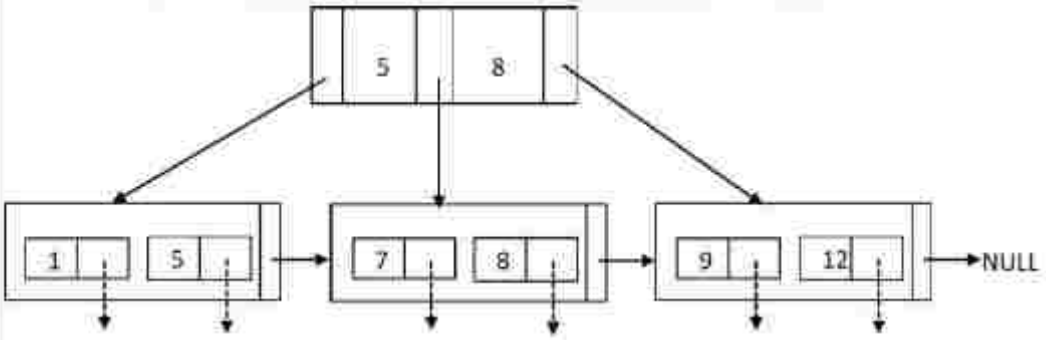
Q.36	Let four points in three-dimensional space be: P1: [2, 3, −1], P2: [3, 1, 1], P3: [5, −2, 3] and P4: [3, 3, 3]. Hierarchical Agglomerative Clustering is used to cluster the above points. If Manhattan Distance is used as the distance metric during clustering, which of the following options indicates the two points that will be merged first?
(A)	$P1, P2$
(B)	$P2, P3$
(C)	$P3, P4$
(D)	$P2, P4$

Q.37	Which of the following statements is true for Ridge Regression?
(A)	The regularizer in the objective function of Ridge Regression is used to guard against scenarios where the model works well for the test data, but poorly for the training data.
(B)	The regularizer of Ridge Regression uses L_1 norm.
(C)	Ridge Regression aims to reduce the number of parameters that have negative values.
(D)	The regularizer of Ridge Regression may increase the bias of the model, but it helps in reducing the variance in predictions.
Q.38	Assume that a Creative (C) person will Succeed (S) if the person is also Disciplined (D), but will not succeed otherwise. Now, consider the following statements: (i) $C \wedge S \Leftrightarrow D$ (ii) $C \Rightarrow (S \Leftrightarrow D)$ (iii) $C \Leftrightarrow ((D \Rightarrow S) \vee \neg S)$ Which of the following options is correct?
(A)	Both (i) and (ii) are TRUE
(B)	Only (ii) is TRUE
(C)	Both (ii) and (iii) are TRUE
(D)	Only (iii) is TRUE

Q.39	A recursive function in Python is given.
	
	`def mystery(n):`
	`  if n <= 0:`
	`  return 1`
	`  else:`
	`  return mystery(n-1) + mystery(n-2)`
	
	Now, consider the following function call:
	
	`mystery(4)`
	
	Assume that a typical runtime stack is used to manage function calls. Each function call is pushed onto the stack and removed only after it finishes execution.
	
	Which of the following options denotes the total number of function calls (i.e., the total number of stack activations), including the initial call, to compute `mystery(4)`?

(A)	5
(B)	9
(C)	15
(D)	17

Q.40	Consider a directed graph $G = (V, E)$, where V is the finite set of vertices and E is the set of directed edges between the vertices. G may contain cycles but there is no self-loop. Further, G may not be strongly connected. Let G^R be the graph obtained by reversing the directions of all the edges in G without changing the set of vertices. Assume that Breadth First Search (BFS) or Depth First Search (DFS) from any given vertex v of a graph visits only the reachable vertices from v in that graph. Which of the following statements must always be true, regardless of the structure of G?
(A)	If u is a reachable vertex in the BFS of G^R from v, then u is also a reachable vertex in the DFS of G from v.
(B)	In G^R, the BFS traversal from v will visit exactly the same set of vertices as the DFS from v in G.
(C)	The order of vertices visited in the BFS of G^R from v is the reverse of the order of vertices visited in the DFS of G from v.
(D)	If u is a reachable vertex in the DFS of G from v, then v is also a reachable vertex in the BFS of G^R from u.

Q.41	Consider a B+ Tree where the maximum number of key values in each leaf node is 2 and the maximum number of pointers in each non-leaf node is 3. Let the content of the B+ Tree be as shown in the figure. Which of the following options denotes the key value(s) stored in the root node after inserting a key value 3 in the given B+ Tree? 
(A)	5
(B)	8
(C)	3 and 5
(D)	3, 5 and 8

Q.42 Consider the given relations X, Y and Z. The relation X has three columns P, Q and R. The relation Y has three columns P, Q and S. The relation Z has two columns P and T.

X

P	Q	R
P1	Q1	R1
P2	Q2	R2
P3	Q3	R2

Y

P	Q	S
P1	Q1	2
P1	Q2	5
P2	Q1	6
P3	Q3	1

Z

P	T
P1	T1
P3	T2
P4	T3
P4	NULL

Consider the relational algebra expression

$$\Pi_{PRS}[(\sigma_{(Q = Q3 \vee R = R2)}[X \bowtie Y]) \bowtie (\sigma_{(S > 1)}[Y \bowtie Z])]$$

where $\bowtie$ denotes natural join operation.

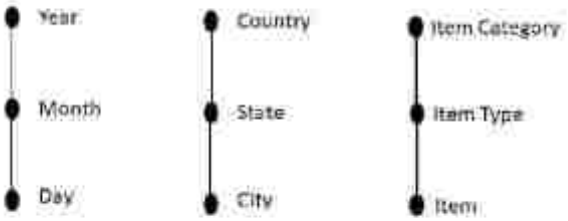
Which of the following options is the correct output for the given expression?

(A) Two rows (P1, R1, 2) and (P1, R1, 5)

(B) Three rows (P1, R1, 2), (P1, R1, 5) and (P2, R2, 6)

(C) One row (P1, R1, 2)

(D) Zero rows

Q.43	Consider the concept hierarchies as shown in the figure. Which of the following options denotes the total number of possible data cuboids from these concept hierarchies? 
(A)	4^3
(B)	2^3
(C)	2!
(D)	4!

Q.44	Let X and Y be two independent random variables. X follows $Bernoulli(p = 0.3)$ distribution and Y follows $Normal(\mu = 0, \sigma^2 = 100)$ distribution. Which of the following options is the variance of $(2X - 1)Y$?
(A)	100
(B)	90
(C)	49
(D)	21
Q.45	Let $L = \lim_{n \to \infty} \sum_{k=0}^{n} \frac{e^{-n} n^k}{k!}$ Which of the following is the value of L?
(A)	0.5
(B)	1.0
(C)	0
(D)	e^{-1}

Q.46	Let $\gamma_1, \gamma_2, \gamma_3$ be the eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{bmatrix}$, where $t \in [-\pi, \pi]$ is in radians. Which one of the following options lists all the possible values of t satisfying $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$?
(A)	$\left\{\frac{\pi}{3}, -\frac{\pi}{4}\right\}$
(B)	$\left\{\frac{\pi}{4}, -\frac{\pi}{3}\right\}$
(C)	$\left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$
(D)	$\left\{\frac{\pi}{3}, -\frac{\pi}{3}\right\}$

Q.47	Consider that 20 stories of Author X and 10 stories of Author Y were kept together without mentioning the names of the authors. A classifier was then asked to predict the author (X or Y) of each of these stories. Let, out of X's stories, 6 were classified as that of Y. On the other hand, out of Y's stories, 2 were classified as that of X. Considering X and Y as two classes, which of the following statements is/are true?
(A)	Accuracy of the classifier is 11/15.
(B)	Precision of Class X is higher than the Precision of Class Y.
(C)	Recall of Class X is higher than the Recall of Class Y.
(D)	Accuracy of the classifier is 14/15.
Q.48	Let $P(x)$ be a predicate. Which of the following statements is/are **NOT** valid in first-order logic?
(A)	$\forall x\, P(x) \Rightarrow \exists x\, P(x)$
(B)	$\exists x\, P(x) \Rightarrow \forall x\, P(x)$
(C)	$\exists x\, P(x) \Leftrightarrow \forall x\, P(x)$
(D)	$\forall x\, P(x) \Rightarrow \exists x\, \neg P(x)$

Q.49	Consider the problem of sorting the given array in ascending order: `P = [1, 2, 3, 5, 4]` Consider two sorting algorithms Bubble Sort (BS) and Insertion Sort (IS). Let N1 be the total number of comparisons done by BS on the elements of P and N2 be the total number of comparisons done by IS on the elements of P. Which of the following options is/are correct?
(A)	$N1 = 10, N2 = 4$
(B)	$N1 > N2$
(C)	IS on P will perform only one swap
(D)	Both BS and IS on P will make at least one unnecessary comparison (i.e., comparing elements that are already in correct order)

Q.50 Consider the given Python program.

```
def outer():
    x = []
    def inner(val):
        x.append(val)
        return x
    return inner
f1 = outer()
f2 = outer()
print(f1(10))    # Line P
print(f1(20))    # Line Q
print(f2(30))    # Line R
print(f1(40))    # Line S
```

Which of the following options is/are correct?

(A) `f1` and `f2` share the same list `x`

(B) Output of `Line Q` is `[10, 20]`

(C) Output of `Line R` is `[10, 20, 30]`

(D) Output of `Line S` is `[10, 20, 40]`

Q.51 Consider a table `Employee(EmpID, TeamID)`, where the column `EmpID` (ID of an employee) is the primary key. The column `TeamID` denotes the team ID of the team of which the employee is a member. `TeamID` is a NOT NULL column.

We want to display the size of the team (denoted as `TeamSize`) in which each employee is a member by using SQL. As an example, the desired output for the given `Employee` table is also shown in tabular form.

Which of the following is/are correct?

`Employee`

EmpID	TeamID
1	8
2	8
3	8
4	7
5	7
6	9

Output

EmpID	TeamSize
1	3
2	3
3	3
4	2
5	2
6	1

(A)

```
SELECT E.EmpID, B.TeamSize
FROM Employee AS E, (SELECT TeamID, COUNT(TeamID) AS
TeamSize FROM Employee GROUP BY TeamID) AS B
WHERE E.TeamID = B.TeamID
```

(B)

```
SELECT A.EmpID, COUNT(B.TeamID) AS TeamSize
FROM Employee AS A, Employee AS B
WHERE A.TeamID = B.TeamID AND A.EmpID = B.EmpID
GROUP BY A.EmpID
```

(C)

```
SELECT B.EmpID, B.TeamSize
FROM (SELECT EmpID, COUNT(TeamID) AS TeamSize
FROM Employee GROUP BY EmpID) AS B
```

(D)

```
SELECT A.EmpID, B.TeamSize
FROM Employee AS A, (SELECT COUNT(TeamID) AS TeamSize FROM
Employee GROUP BY TeamID) AS B
WHERE A.TeamID = B.TeamID
```

Q.52	Let $M = (I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T)$ be a matrix, where $\mathbf{1} = (1,1,1,\ldots,1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n. Which of the following options is/are correct?
(A)	$M^T = M$
(B)	$M^2 = I_n$
(C)	$Trace(M) = n$
(D)	M is a projection matrix
Q.53	Let $X_1, X_2, \ldots, X_n$ be n independent random variables. Each of the random variables follows $Normal(\mu = 0, \sigma^2 = 1)$ distribution. Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Which of the following statements is/are correct?
(A)	$\sum_{i=1}^{n} X_i^2$ follows Chi-square distribution with n degrees of freedom.
(B)	$\sum_{i=1}^{n}(X_i - \bar{X})^2$ follows Chi-square distribution with $(n-1)$ degrees of freedom.
(C)	$X_1^2 + X_n^2$ follows exponential distribution with mean 2.
(D)	$\left(\sqrt{n}\,\bar{X}\right)^2$ follows Chi-square distribution with 2 degrees of freedom.

Q.54	Let X be a discrete valued random variable with cumulative distribution function $F(x)$. Which of the following statements is/are correct?
(A)	$F(x)$ is always a positive function.
(B)	$F(x)$ is a non-decreasing function.
(C)	$F(x)$ has jump discontinuity.
(D)	$F(x)$ is a left continuous function.
Q.55	Consider that Linear Ridge Regression is being used to learn a prediction function $y_{pred} = w^T x$, where $w, x \in \mathbb{R}^2$ and Mean Absolute Error (MAE) is used to measure the prediction error. A weight of 0.20 is associated with the regularizer. At an intermediate step of the training process, assume that the parameter $w = [-3.00,\ 4.00]^T$. In the next step, for the input $x = [1.00,\ 2.00]^T$, the predicted value of y is noted. Let the relation between $x = [x_1,\ x_2]^T$ and the true value of y be $y_{true} = x_1 + x_2$. The value of the overall regularized loss function for this instance is _____. (Rounded off to **two** decimal places)

Q.56	Consider a fully-connected feed-forward multi-layer perceptron. It has 30 neurons in the input layer, followed by two hidden layers and an output layer. The first hidden layer has 4 neurons and the second hidden layer has 3 neurons. The output layer has only one neuron. Assume that no bias parameters are used. The number of learnable parameters in the multi-layer perceptron is _____ . (Answer in **integer**)
Q.57	A clinic specializes in testing for a disease D. The result of the test can be either positive or negative. A study revealed that if a person suffers from the disease D, the test result in that clinic comes out positive 80% of the time, and negative 20% of the time. If a person is not suffering from the disease D, the test comes out positive 10% of the time and negative 90% of the time. It is also known that among the general population, the disease D occurs in 30% of the individuals. If a person tests positive for D in that clinic, the probability that he/she actually suffers from the disease D is _____ . (Rounded off to **two** decimal places)

Q.58	Consider the given Python program.
	
	`def fun(L, i=0):`
	`  if i >= len(L)-1:`
	`  return 0`
	`  if L[i] > L[i+1]:`
	`  L[i+1], L[i] = L[i], L[i+1]`
	`  return 1+fun(L, i+1)`
	`  else:`
	`  return fun(L, i+1)`
	`data = [5, 3, 4, 1, 2]`
	`count = 0`
	`for _ in range(len(data)):`
	`  count += fun(data)`
	`print(count)`
	
	The output of the program is _____ . (Answer in **integer**)

Q.59 Let there be two relations X and Y as shown. X has three columns P, Q and R. Y has two columns P and S.

X

P	Q	R
P1	Q1	R1
P2	Q2	R2
P3	Q3	R2

Y

P	S
P1	10
P1	15
P2	20
P3	1

Consider that the following tuple relational calculus expression is evaluated.

$$\{t | t \in X \wedge \exists z \in X(t[P] = z[P]) \wedge \exists m \in Y\ (m[P] = t[P] \wedge m[S] > 1)\}$$

The number of tuples that will be returned is _____ . (Answer in **integer**)

Q.60 Let `Account` be a relation as shown.

`Account`

AccNo	Balance
A1	5000
A2	5000
A3	10000
A4	15000
A5	18000

Consider the given SQL query.

```
SELECT AccNo FROM Account AS A
WHERE (SELECT COUNT(*) FROM Account AS B
WHERE A.Balance < B.Balance) >= (SELECT COUNT(*)
FROM Account AS C WHERE A.Balance > C.Balance)
```

The number of rows returned by the SQL query is _____ . (Answer in **integer**)

Q.61	Consider an ER model with the entities $E1(A_{11}, A_{12}, A_{13})$ and $E2(A_{21}, A_{22}, A_{23})$, where, A_{11}, A_{12}, A_{13} are the attributes of E1, and A_{21}, A_{22}, A_{23} are the attributes of E2. Let A_{22} be a multi-valued attribute. A_{11} and A_{21} are the primary keys of E1 and E2, respectively. Let R12 be a many-to-many relationship between E1 and E2. Participation of both E1 and E2 in R12 is total. The minimum number of relations required to convert the ER model into relational model (assuming there is no other functional dependency) where each relation is in third normal form (3NF) is _____. (Answer in **integer**)
Q.62	For a given data set $\{x_1, x_2, \ldots, x_n\}$, where $n = 100$, it is known that $$\frac{1}{2000}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2 = 99$$ Let us denote $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ The value of $\frac{1}{99}\sum_{i=1}^{n}(x_i - \bar{x})^2$ is _____. (Answer in **integer**)
Q.63	Let X be a random variable that follows $Uniform(-1,1)$ distribution. The conditional distribution of the random variable Y given $X = x$ is the $Uniform(x^2 - 0.1, x^2 + 0.1)$ distribution. The value of $correlation(X, Y)$ is _____. (Answer in **integer**)
Q.64	Let $A_{5\times5}$ be a matrix such that each of its elements follows $Bernoulli(p = 0.50)$ distribution independently. The probability that the row-sum of the second row and the column-sum of the third column are both equal to 3 is _____. (Rounded off to **two** decimal places)
Q.65	Let $A = \left(I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)$ be a matrix, where $\mathbf{1} = (1,1,1,\ldots,1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n. The value of $\max_S x^T A x$, where $S = \{x \in \mathbb{R}^n \mid x^T x = 1\}$, is _____. (Answer in **integer**)